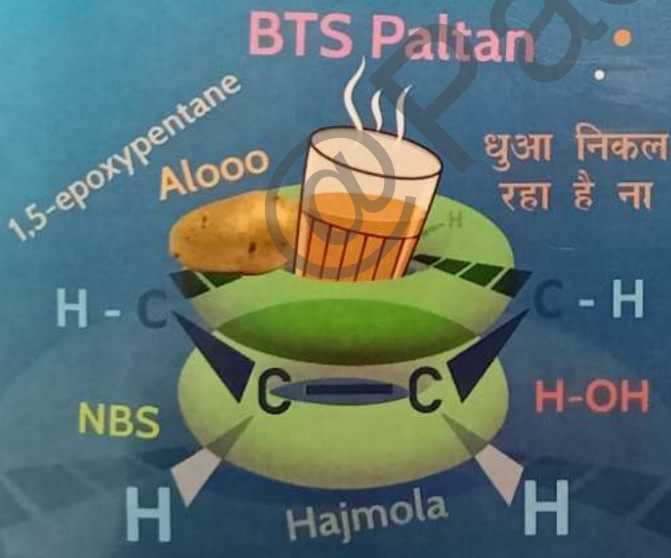
Class-11th + 12th

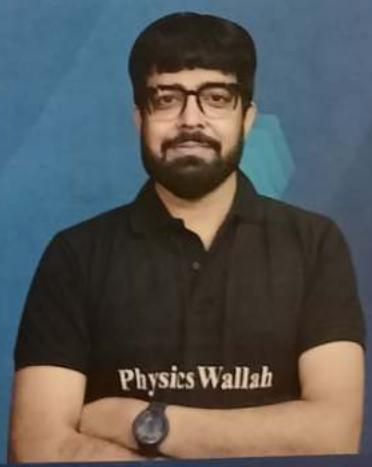
EaJEE NOTES

for

ORGANIC CHEMISTRY

Basic से
Kattar Advanced [illegible]
बाबा के साथ





BACKLOG KILLER ❤
Blue Book

Rohit Agrawal
(RA Baba)

Shubh Karan Choudhary
(SKC Sir)

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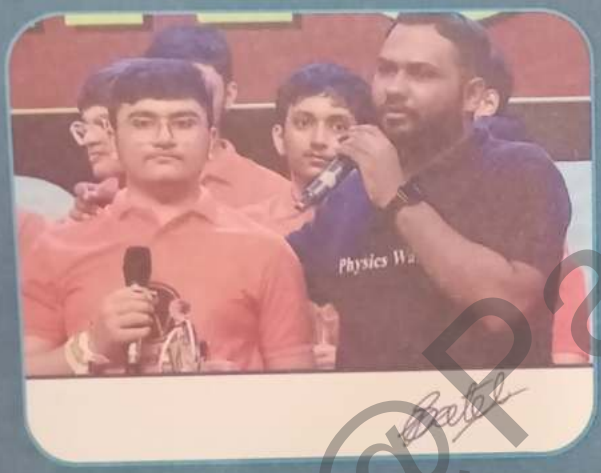
Testimonials

1. I had a great experience learning organic chemistry under Rohit Aggarwal Sir. He teaches the concepts deeply in a way that can applied to all JEE problems and even beyond without much effort. I can attest to the effectiveness and completeness of his notes and in my preparation they were the most important resource for the subject which I followed diligently, and it had a very positive impact in my performance. Additionally, he provided a brilliant selection of problems. The problem sets cover all cases that are relevant for the exam and help in solidifying the concepts even better. The notes along with the problems provide a complete course for JEE organic chemistry. I would recommend it to any serious aspirant who wants to learn organic chemistry in an intuitive and comprehensive way.

Sushant Padha, AIR 52, JEE 2024

Sparsh Patel, AIR 283, JEE 2025.



2. It was a privilege to study organic chemistry under the guidance of Rohit Aggarwal Sir. He teaches organic chemistry at the best level.Thorough practice if his notes and practice questions proved instrumental during the exam. His materials progress systematically from basic principles to advanced topics. The structure and order of his notes and questions offers a wide range of applications, giving a deep understanding of the subject. Each theoretical topic is explained by detailed mechanisms, helping us understand each concept and not simply mug up. I wholeheartedly recommend his resources to any student seeking to excel in the subject and the examination.

3. Baba's class notes are more than sufficient for JEE Advanced.
Maine unke class notes ko chhodkar or kuch bhi extra nahi padha for my organic chemistry.
His class questions are top-level with everything and every concept you need for your exam.

Karan SawAdv. AIR 995, Neet AIR 592, JAT AIR 55

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Kattar Advanced की feel
बाबा के साथ

BACKLOG KILLER ❤

1	2	3	4	5	6
Class Notes of Baba	Reaction Mechanisms Summary	NCERT Book Coverage	PYQs (JEE Main + JEE Adv. & NEET)	Coaching Modules	Reference Book

6 in 1 EaJEE NOTES for Organic Chemistry

"JUNJUNA holders, LUX tera safar hai,
IIT ka dwar hai nahi to PEARS teri manzil."

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Competition Wallah
NCERT Wallah

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Preface

Physics Wallah (PW) strongly believes in conceptual and fun-based learning. PW provides highly exam-oriented content to bring quality and clarity to the students.

A plethora of Competitive Exams Books is available in the market but PW professionals are continuously working to provide the most effective book for Competitive Exams to our students.

Books adopt a multi-faceted approach to master and understanding the concepts by having a rich diversity of questions asked in the examination and equip the students with the knowledge for the competitive exam.

The main objective of the book is to provide a comprehensive theory and a large number of quality problems with varying cognitive levels to facilitate teaching-learning of concepts that are presented through the book.

It has become popular among aspirants because of its easy to understand language and clear illustrations with diagrams, tables, memes and selective previous year questions.

Every effort has been made to ensure that the content is thoroughly reviewed for clarity, ease of understanding, language quality, and comprehensive coverage of all relevant topics and chapters, strictly aligned with the latest syllabus prescribed by NTA for competitive Exams.

Mastering the Physics Wallah (PW) study material curated by the PW team, can help students qualify for the exam with good marks.

In each chapter, for better understanding, theory and examples are according to the latest syllabus for Competitive Exams.

- To strengthen the concept of the topic at the zenith level

This Book Consists of

- Complete Theory and PYQs for JEE Main + Advanced & NEET
- Baba Tea Stall
- Multicolour Handwritten Notes
- Real Classroom Vibes Experience

Special Features of the Book

- More than 3000 practice questions
- Mechanism provided for all reactions
- Special Characteristic of Reactions (BTS)
- Summary provided of all chapters at the end of the book
- Appendix

A step to make our book even better for our students:

Provide your valuable feedback or Report the Error

View Errata (Error Correction from our end)

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About the Author

Mr. Rohit Agrawal (RA Baba Sir)

Ex-Director and Head of Organic Chemistry Department, Nucleus, Kota

Popularly known as RA Baba, he is a name synonymous with excellence in Organic Chemistry. With over 14 years of teaching experience, he has mentored countless aspirants, including top JEE Advanced rankers such as AIR 3, 8, 10, 12, 16, 18, 23, 24, and many others.

A two-time IIT-JEE qualifier himself, he holds a B.Tech degree from NIT Surathkal. He has authored more than 5 books for competitive exams, known for their conceptual clarity and student-focused design.

His mentorship extends to Olympiad achievers like Mrinal Sharan (currently at Oxford University) and Sushant Padha (IIT Bombay) and many more.

Famous for his 12-hour live Organic Chemistry sessions on the JEE Wallah channel, he is known for teaching difficult topics in a fun and simple way that keeps students interested and motivated.

He believes in the philosophy: "A student learns better while enjoying." His intuitive approach and visual tools for reaction mechanisms make complex concepts accessible and enjoyable for learners.

Shubh Karan Chaudhary

With 17+ years of teaching experience in Organic Chemistry, he is a well-known name in JEE and NEET preparation. He has guided top NEET rankers including AIR 1, 2, 3, 5, 6, 9, 11, 15, 20, 21, 25 and many more.

Former #1 Legend Category Educator at Unacademy (5 years)

Held senior leadership roles such as: Vice President of Sarvottam Career Institute Pvt. Ltd., Kota, Director & HOD of Chemistrio Institute Pvt. Ltd., Kota, Former Senior Chemistry Faculty at Allen Career Institute, Kota.

Author of 4 best-selling books, known for their simple and effective explanations.

A registered author with top academic publishers: Pearson India – for NCERT Mantra and Universities Press – for Fundamental Problems in Organic Chemistry.

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This Book is Dedicated To.....

"ये पुस्तक दादा-दादी के स्नेह और
पिता के मार्गदर्शन को समर्पित है-
जिन्होंने मेरे जीवन को अर्थ और दिशा दी।"

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Pravachan

"सभी Lindanes को प्रवचन"

- Nomoskar Lindanes, कैसन बा, आप सभी को BB की बधाई।
- बाबा के इस प्रसाद में, pure basic, NCERT, Short notes और Kattar Advanced का पूरा तड़का हैं।
- इस Book को केवल Reading में Use ना करें और सबसे पहले white pages या कॉपी बना लें।
- कोई भी Mechanism पहले खुद Try करे फिर उसके BTS points को अच्छे से पढ़े वरना सावधानी हटी दुर्घटना घटी।
- Reaction अच्छे से समझने के बाद सारे Question करे चाहे आप JEE के हो या NEET के।
- मैंने PYQ भी कुछ add किये हैं और सारे PYQ आपको QR Code में दे रहा हूँ ताकि COST कम हो जाए।
- Extra Objective & Subjective भी दिए है So समय मिले तो करना, otherwise अगर Book भी करली तो भी NORA का लेवल high रहेगा।
- Last में स्वस्थ रहे, मस्त रहे, व्यस्त रहे और चार लोगों की इज्जत हमेशा करें माता, पिता, गुरु और ..... माता।

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Acknowledgement

Rohit Sir sincerely thanks Alakh Pandey Sir and Prateek Sir, for creating a platform that makes quality education accessible to millions. He also expresses gratitude to Sanyam Badola Sir, for his unwavering support and guidance.

I (Rohit) gratefully acknowledge the encouragement and immense support of my mother Sunita Agrawal, father Bhupendra Agrawal, sister Richa Agrawal and daughter Yati Agrawal. They inspired me all the time during preparation of this book.

I (Shubhkaran) sincerely acknowledge the support of my mother Chandrapati Choudhary, father Baljeet Singh, sister Saroj, brother Vishwanath Choudhary, wife Anandi Choudhary and charming son Kanish Choudhary, Tavish Choudhary.

Our special thanks to guru 'Akshay Choudhary' for shaping this book.

Special thanks to Jyoti Rakheja, Content Faculty at PW Publication, for her dedicated efforts in designing and shaping this book.

I'm also grateful to Deepak Kumar and Rakhi Sharma (Book Compositor) for their continuous efforts.

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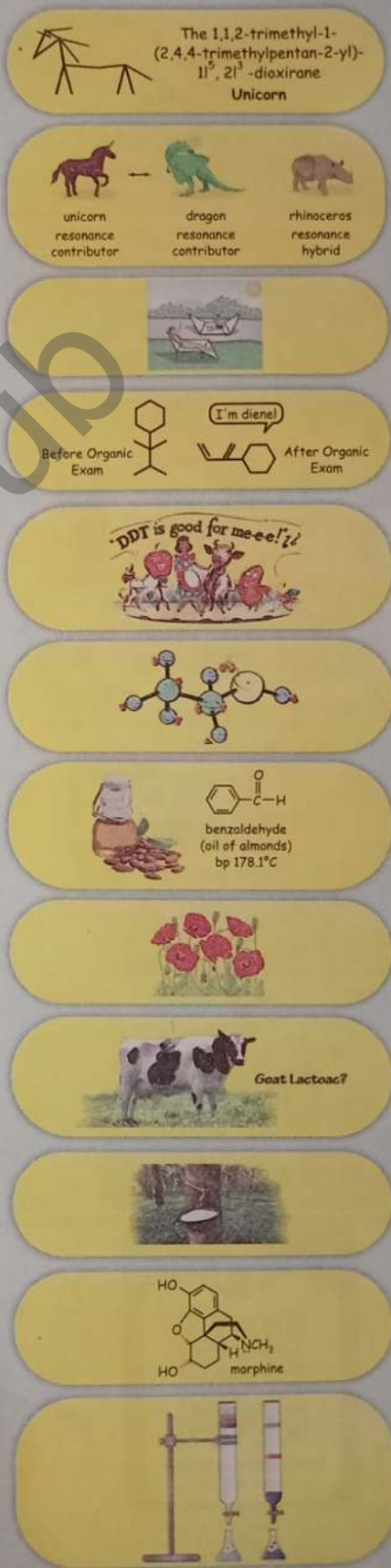

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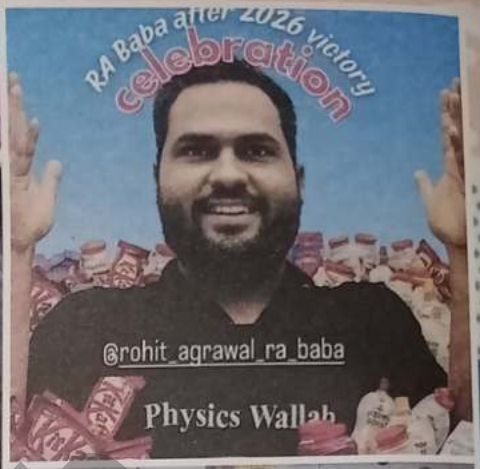

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rohit_agarawal_ra_baba

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1 Nomenclature of Organic Compounds

Definition

1. **Hydrocarbons:** Compound formed by carbon and hydrogen

   ex: CH_4, H_3C-CH_3, etc.

   Alternate π-bond

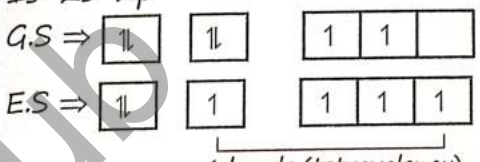
2. **Hydrocarbons derivatives:** Compound formed by carbon, Hydrogen and hetero atoms.

   ex: CH_3Cl, CH_3-OH, CH_3-CH_2-OH etc.

3. **Organic chemistry:** Branch of science which deals with hydrocarbon and its derivative

➢ Carbon

$1s^2\ 2s^2\ 2p^2$



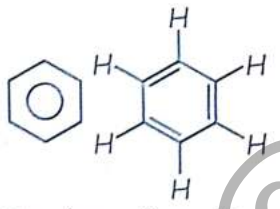
4 bonds (tetravalency)

⇓

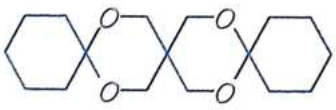
Catenation

Practice Questions

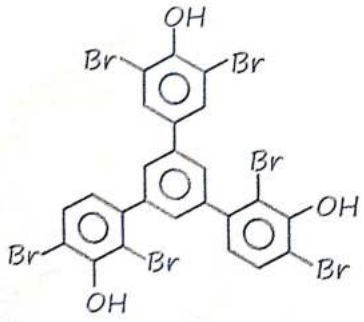
1. No. of CH_2-group present in benzene is:



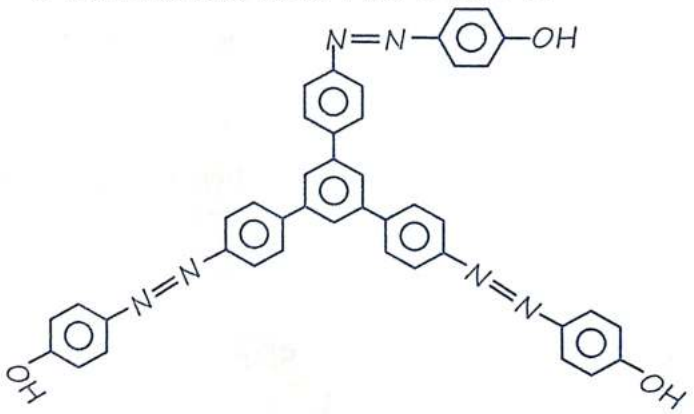
2. Number of $-CH_2$ group and hetero atom present in following compound is:



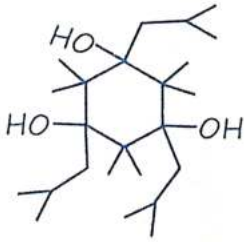
3. No. of hetero atom present in given compound.



4. No. of hetero atom + no. of C ⇒ ??



5. Number of $-CH_3$ group present in following compound is:



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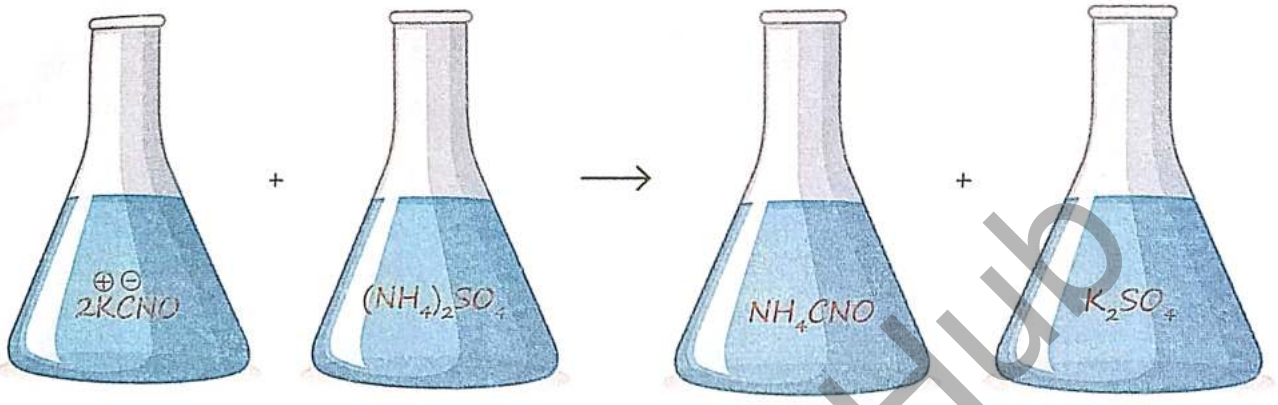
History

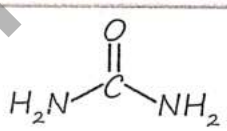
1. Berzelius was a Swedish scientist who in 1815 proposed that organic compounds could only be produced by some special force which must be existing in living organism and could not be prepared in a laboratory.

   This force was called vital force and this theory came to be known as vital force theory

2. Experiment: F. wholer



Reaction

$NH_4CNO \xrightarrow[\text{Rearrangement}]{\Delta}$ 

- Inorganic compound
- Ammonium cyanate
- CH_4ON_2

- Organic compound
- Urea
- CH_4ON_2

Observation

- Organic compounds can be obtained in lab
- Phenomenon of Isomerism

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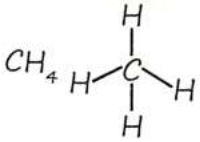
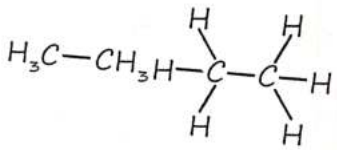
Representation of Organic Compound

Molecular formula

- No. of C, no. of H, no. of hetero atoms
- $C_xH_yO_zN_w$

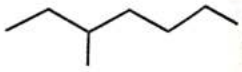
Structural formula

- All bonds are shown
- CH_4 
- H_3C-CH_3 

Condensed formula

- No bonds are shown
- Repeating units and branches as shown in ( ).
- $H_3C-CH_2-\underset{\displaystyle CH_3}{\underset{|}{CH}}-CH_2$
- $-CH_2-CH_2-CH_3$
- $H_3CCH_2CCH(CH_3)(CH_2)_3CH_3$

Bond line formula

- C—H bond is never shown
- C ⇒ represent by.
- 

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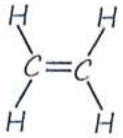
Practice Questions

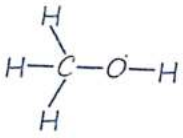
6. Find structural, condensed and bond line formula.

(a) H_3CCH_3

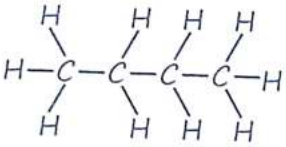
(b) $H-C\equiv C-H$

(c)

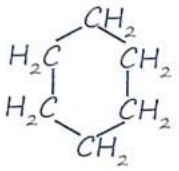
(d) 

(e) 

(f) $H_3C-C\equiv CH$

(g) 

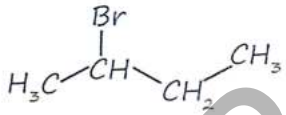
(h)

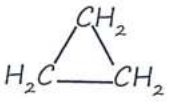
(i) 

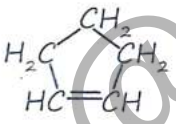
(j)

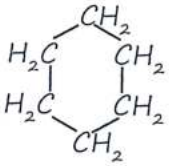
(k)

(l)

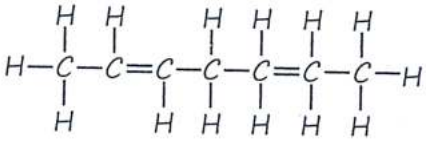
(m) 

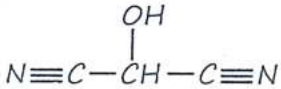
(n) 

(o) 

(p) 

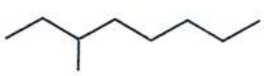
(q) $H_3CCH_2COCH_2CH_3$

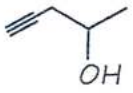
(r) 

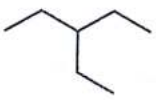
(s) 

(t) $HOCH_2CH_2CH_2CH(CH_3)CH(CH_3)CH_3$

(u)

(v) 

(w) 

(x) 

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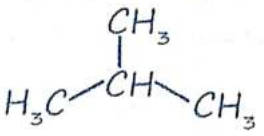
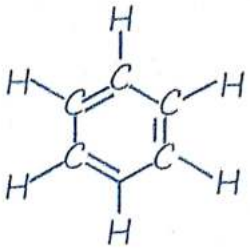
7. Draw the remaining representation of the following compounds?

(i) CH_4

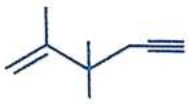
(ii) 

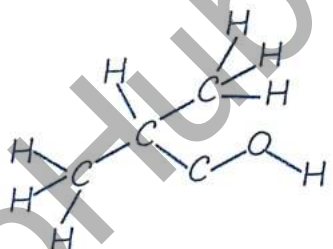
(iii)

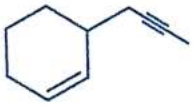
(iv) 

(v) (a)

(b)

(vi) 

(vii) 

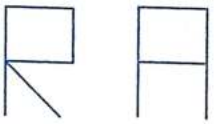
(viii) 

(ix)

(x)

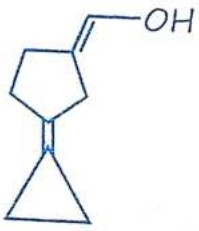
(xi) $(H_3C)_3CCH_2(CH)_2CCH$

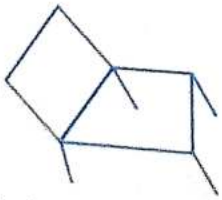
(xii)

(xiii) 

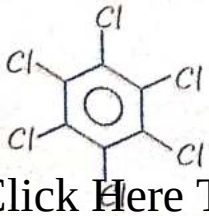
(xiv) $(H_3CCH_2)_2CCHCH_2CCCH\,(OH)\,CH\,(CH_3)\,CH_2CH_3$

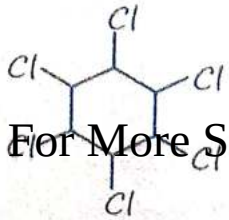
(xv) $(H_3CCH(CH_3))_3CCCCN$

(xvi) 

(xvii) 

(xviii) $(H_3C(OH)CHCHCH)_3CC(OH)CHCH_2CH(CH_2CH_3)_2$

(xix) 

(xx) 

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Hybridization

Hybridisation ⇒ number of σ + number of lone-pair

No. of σ	No. of lone-pair	Hybridisation	Shape
4	0	SP^3	Tetrahedral
3	1	SP^3	Tetrahedral
2	2	SP^3	Tetrahedral
3	0	SP^2	Planer
2	1	SP^2	Planer
2	0	SP	Linear
1	1	SP	Linear

Degree of atom/group

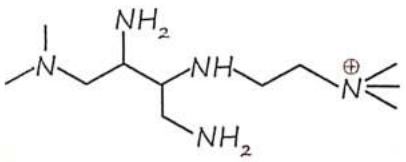
Carbon attached to how many atom.

Ex:

- 4C ⟶ Quaternary (4°)
- 3C ⟶ Tertiary (3°)
- 2C ⟶ Secondary (2°)
- 1C/OC ⟶ Primary (1°)

- Hydrogen: degree of (H) = degree of attached(c)
- 4°H never exist
- degree of (H), (OH), (X), (NO_2) are same as attached degree of (C)

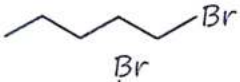
Amine: (N) ⇒ same as (C)

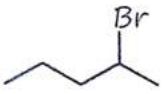
Ex: 

Practice Questions

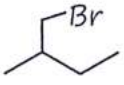
8. Find σ, π and hybridisation of carbon?
   (a) CH_3Cl (b) $(CH_3)_2CO$
   (c) $H_3C\ CN$ (d) $HCONH_2$
   (e) $H_3CCH = CHCN$ (f) CH_3F
   (g) $HCNO$ (h) HCN
   (i) $H_2C{=}CH{-}CH_3$ (j) $HC{\equiv}CCH{=}CHCH_3$

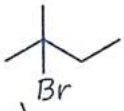
9. Find σ, π and hybridisation of carbon in Qus. 8. Find degree of following halide, alcohol, amine?

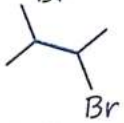
10. 

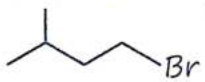
11. 

12.

13. 

14. 

15. 

16. 

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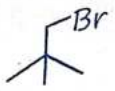
17. 

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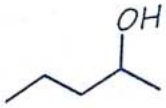
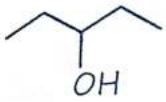
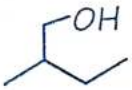
19. 

20. 

21. 

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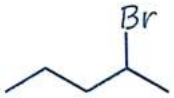
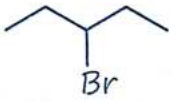
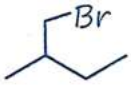
23. 

24. 

25. 

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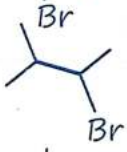
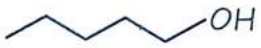
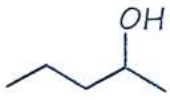
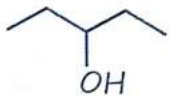
27. 

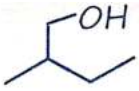
28.

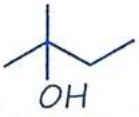
29.

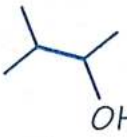
30. 

31. 

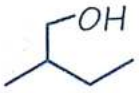
32. 

33. 

34. 

35. 

36.

37. 

38. $(CH_3)_2CHC(Cl)CH_3$

39. $CH_3CH_2CH(CH_3)CH(C_2H_5)Cl$

40. $CH_3CH_2C(CH_3)CH_2I$

41. $(CH_3)_3CCH_2CH\ (Br)C_6H_5$

42. $CH_3C(C_2H_5)CH_2Br$

43. $CH_3C(C_2H_5)_2CH_2Br$

44. $CH_3C(Cl)(C_2H_5)CH_2CH_3$

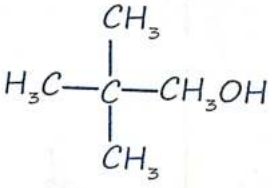
45. $CH_3CH{=}CHC(Br)(CH_3)$

46. $CH_3CH{=}C(Cl),$
$CH_2CH(CH_3C)_2$

47. $p{-}ClC_6H_4CH_2CH(CH_3)_2$

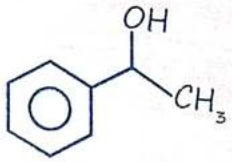
48. $m{-}ClCH_2C_6H_4CH_2C(CH_3)_3$

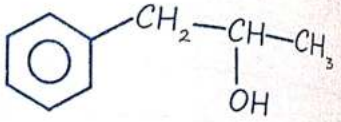
49. $o-Br-C_6H_4CH(CH_3)$
CH_2CH_3

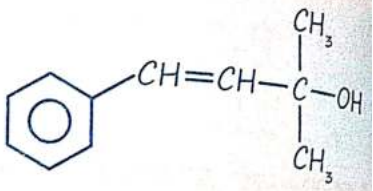
50. 

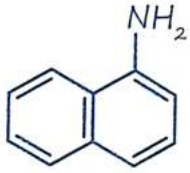
51. $CH_2{=}CH{-}CH_2{-}OH$

52.

53. 

54. 

55. 

56. 

57.

58. $(C_2H_5)_2CHNH_2$

59. $(C_2H_5)_2NH$

60. $(CH_3)_2CHNH_2$

61. $CH_3(CH_2)NH_2$

62. $CH_3NHCH(CH_3)_2$

63. $(CH_3)_3NCH_2$

64. $C_6H_5NHCH_3$

65. $(CH_3CH_2)_2NCH_3$

66. $m{-}BrC_6H_4NH_2$

67. Drawn the structure of alkane having least molecular mass that contains only:

(a) 1° and 2°
(b) 1° and 3°
(c) 1° and 4°
(d) 1°, 2° and 3°C
(e) 1°, 3° and 4°C
(f) 1°, 2°, 3°, and 4°C

68. Draw the structure of cycloalkane having least molecular mass that contains only:

(a) 2°C
(b) 1° and 3°C
(c) 1° and 4°C
(d) 1°, 2° and 3°C
(e) 1°, 3° and 4°C
(f) 1°, 2°, 3° and 4°C

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Degree of Unsaturation/Double Bond Equipment/Index of Hydrogen Deficiency

The presence of p bonds or rings within a molecule is indicated by a quantity called degree of unsaturation.

Definition: Deficiency of 2H atoms with respect to fully saturated acyclic hydrocarbon is equivalent to One DU. It is also known as Hydrogen Deficiency Index (HDI) or Double Bond Equivalent (DBE).

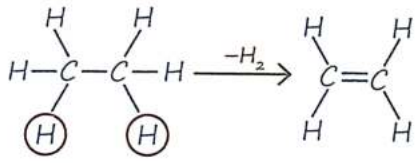
Hydrocrabon

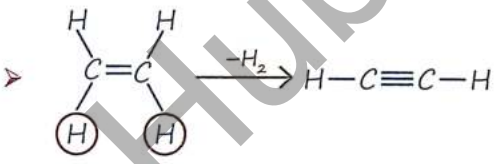
➢ Alkane ⇒ C_nH_{2n+2} ➢ Number of H_2 molecular required to produce open chain alkane is DU/DBE/HDI.

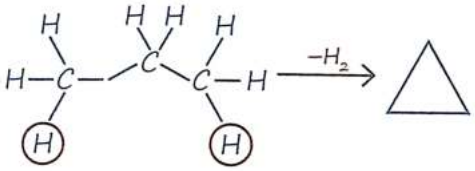
Ex: (a) $C_8H_6 \xrightarrow{+H_2} C_8H_{18}$ (DU)

(b) $C_xHy \Rightarrow \xrightarrow{2x+2-y}$ $x + 1 - y/2$

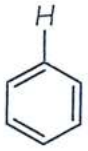
Structures

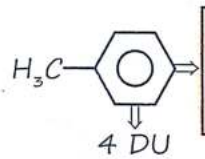
➢ 

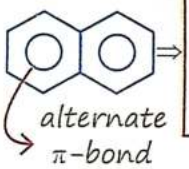
➢ 

➢ 

1 double bond = 1DU
1 Ring = 1DU
1—CN=2DU (—C≡N)
1—Ph=4DU

➢  ⇒ 4DU (3π bond + 1 ring)

➢ 
Phenyl
–Ph
C_6H_5
4 DU
(2π + 1R)

➢ 
⇒ Naphthalene ⇒ 5 double bond
7DU(5π + 2R)
19σ(11C—C, 8C—H
alternate π-bond

Hetero atoms

N	O	X
–H	Ignore	+H

Ex: (a) $C_{12}H_{10}O_2N_4F_2$
$\Downarrow x - 4 + 2$
$C_{12}H_8$
$\rightarrow 26$ (9DU)

(b) $C_{16}H_8O_4N_2FCl_2Br_2I$
$x -2 + 1 + 2 + 2 + 1$
$C_{16}H_{12}$ (11DU)
$\rightarrow 34$

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69. C_3H_6

70. C_4H_8

71. C_5H_6

72. $C_{10}H_{10}$

73. $C_{12}H_6$

74. C_8H_8

75. C_xH_y

76. C_7H_8

77. C_{60}

78. $C_{15}H_{10}$

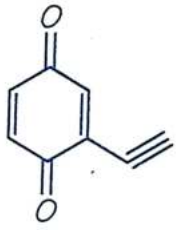
79. C_2H_4DT

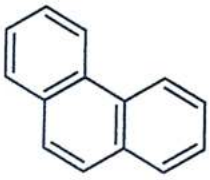
80. $C_6H_5D_2T$

81.

82. Benzene

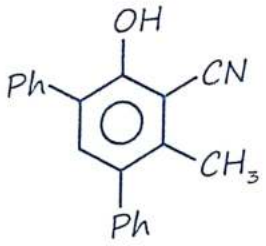
83. Naphthalene

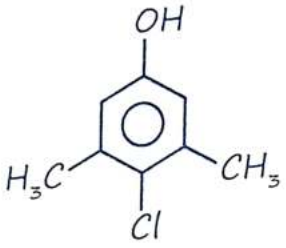
84. 

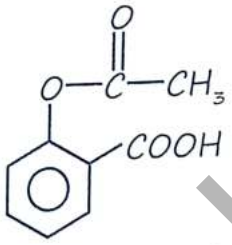
85. 

86.

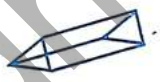
87.

88. 

89. 

90. 

91.

92. 

93. Toluene

94. Anisole

95. C_6H_5OH

96. p—Xylene

97. Ph—Ph

98. Ph_4C

99. Ph_2C_2

100. Ph_3CH

101. Ph_4C_2

102. $(H_2CCH_2)_3C\ C\ C\ C\ C\ N$

103. $HCCC(Ph)_2CH_2$

104.
(Pyrene)

105.

106. C_3H_6O

107. $C_5H_{10}O_2$

108. $C_{10}H_{10}ON_2$

109. C_3H_9N

110. $C_{12}H_{10}O_2F_2Cl_2N_2$

111. $C_{18}H_{10}FCl\ DTO_2$

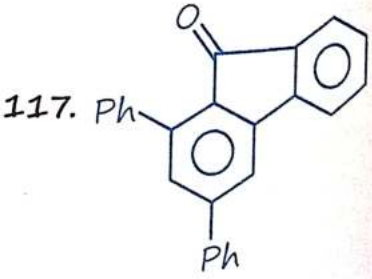
112. $C_{18}H_{10}FCl\ DTO_2$

113. $C_{18}H_{11}Cl$

114. $C_{12}H_{10}O_2F_2Br_2N_2$

115. $C_7H_{10}FN$

116. $C_6H_{12}O_6$

117. 

118. $C_2(CN)_4$

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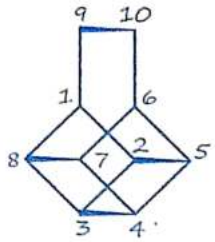
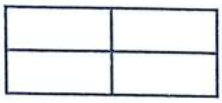
119. Basketane



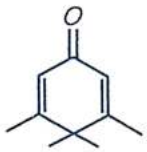
120. Churhane

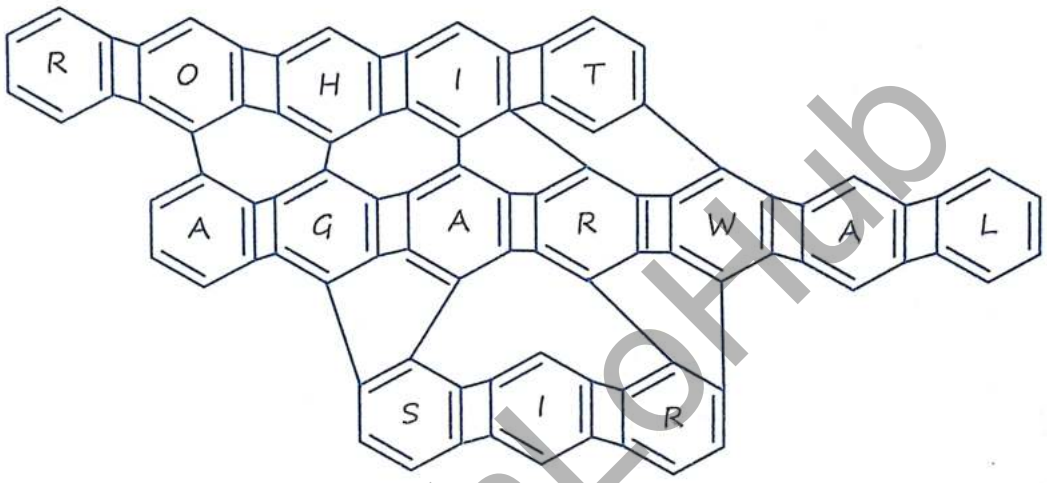
121. Windowane



122. Housane

123. Penguinone



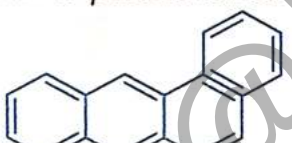


Baba Tea-Stall (BTS)

Carcinogenicity and toxicity

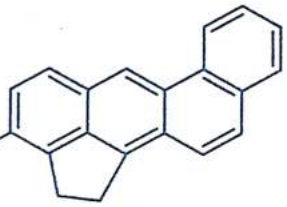
Benzene and poly-nuclear hydrocarbons containing more than two benzene rings fused together are toxic and said to possess cancer producing (carcinogenic) property.

(i)



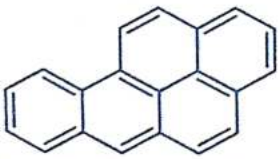
1, 2 Benzanthracene

(ii) H_3C



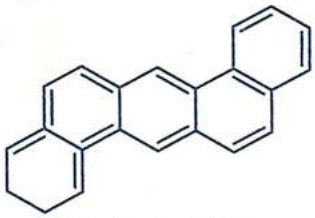
3-methylcholanthrene

(iii)



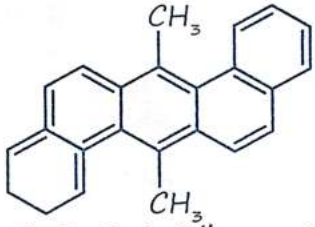
1, 2-Benzpyrene

(iv)



1, 2, 5, 6-Dibenzanthracene

(v)



1, 2, 5, 6-Dibenzanthracene

(vi) $C_{13}HgBrO$

(vii) C_3H_7N

(viii) $C_{10}H_8Cl$

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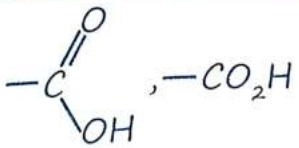
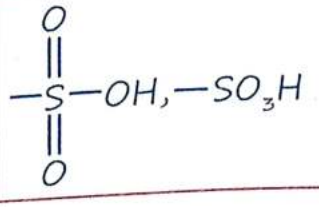
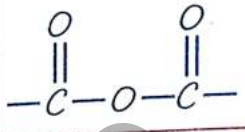
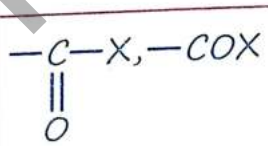
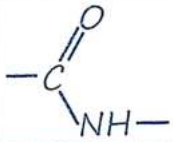
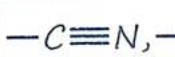
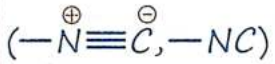
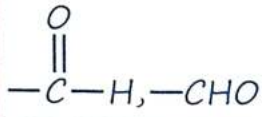
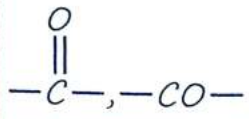
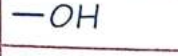
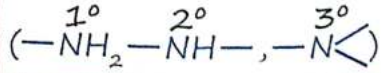
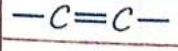
S.N.	Functional Group	
1.	Carboxylic acid	
2.	Sulphonic acid	
3.	Acid anhydride	
4.	Ester	
5.	Acid halide	
6.	Acid amide	
7.	Cyanide	
8.	Isocyanides	
9.	Aldehyde	
10.	Ketone	
11.	Alcohol	
12.	Thiol	
13.	Amines	
14.	Alkene	
15.	Alkyne	
16.	Phenyl	

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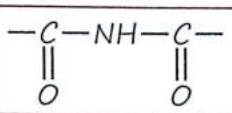
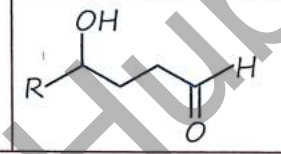
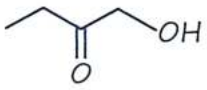
17.	Nitro	$-NO_2$
18.	Nitroso	$-NO$
19.	Thioether	$-S-$
20.	Ether	$-O-$
21.	Halide	$-X$
22.	azo	$-N=N$
23.	Imine	$-CH=NH$
24.	Imide	
25.	enol	$-CH=CH-OH$
26.	enamine	$-CH=CH-NH_2$
27.	Aldol	

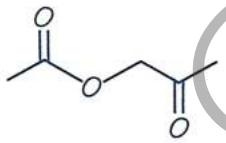
R—OH Points

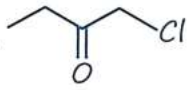
- 1°, 2°, 3° amine are different functional group.
  ⇒ Responsible for any chemical reaction.
- **Alcohol and phenol are different functional group.**
- **1°, 2°, 3° alcohol are same functional group.**
- 1°, 2°, 3° amide are different functional group.

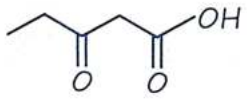
Practice Questions

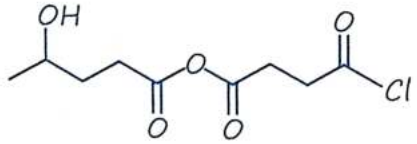
Find different functional group?

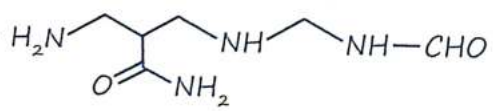
124. 

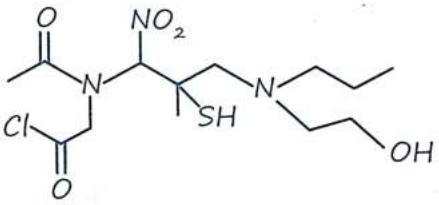
125. 

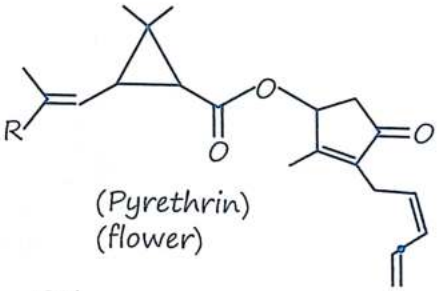
126. 

127. 

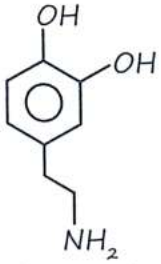
128. 

129. 

130. 

131. 

(Pyrethrin)
(flower)

132. 

(Dopamine) (दिमागी ड्रग)

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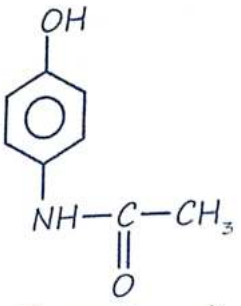
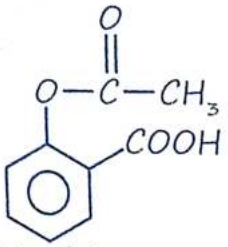
133.



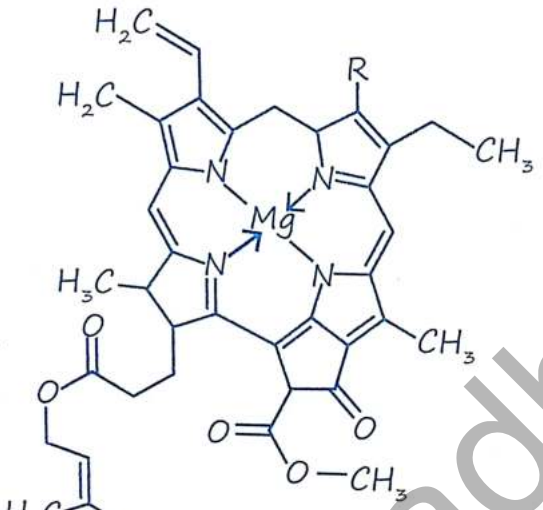
(Paracetamol)

134.

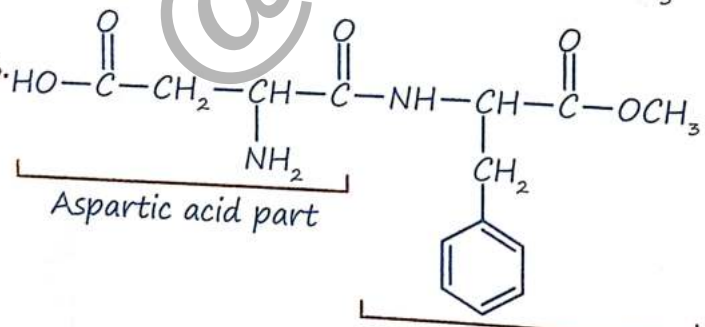


(Aspirin

135.



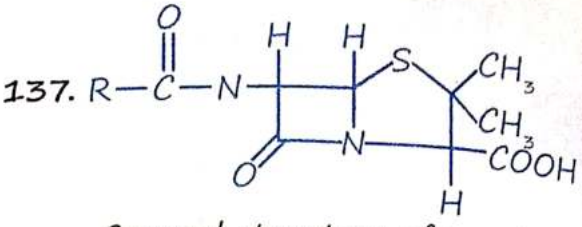
136.



Aspartic acid part

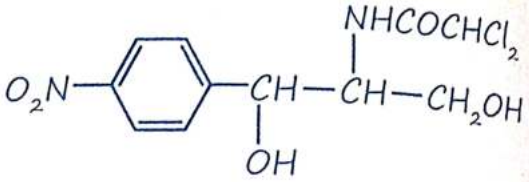
Phenylalanine methyl ester part

137.



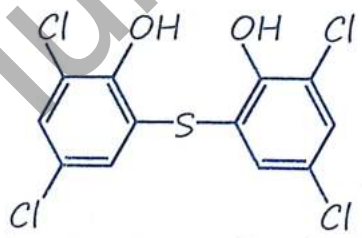
General structure of pencillin

138.



Chloramphenicol

139.



140

Norethindrone

141.

Ethynylestradion (novestrol)

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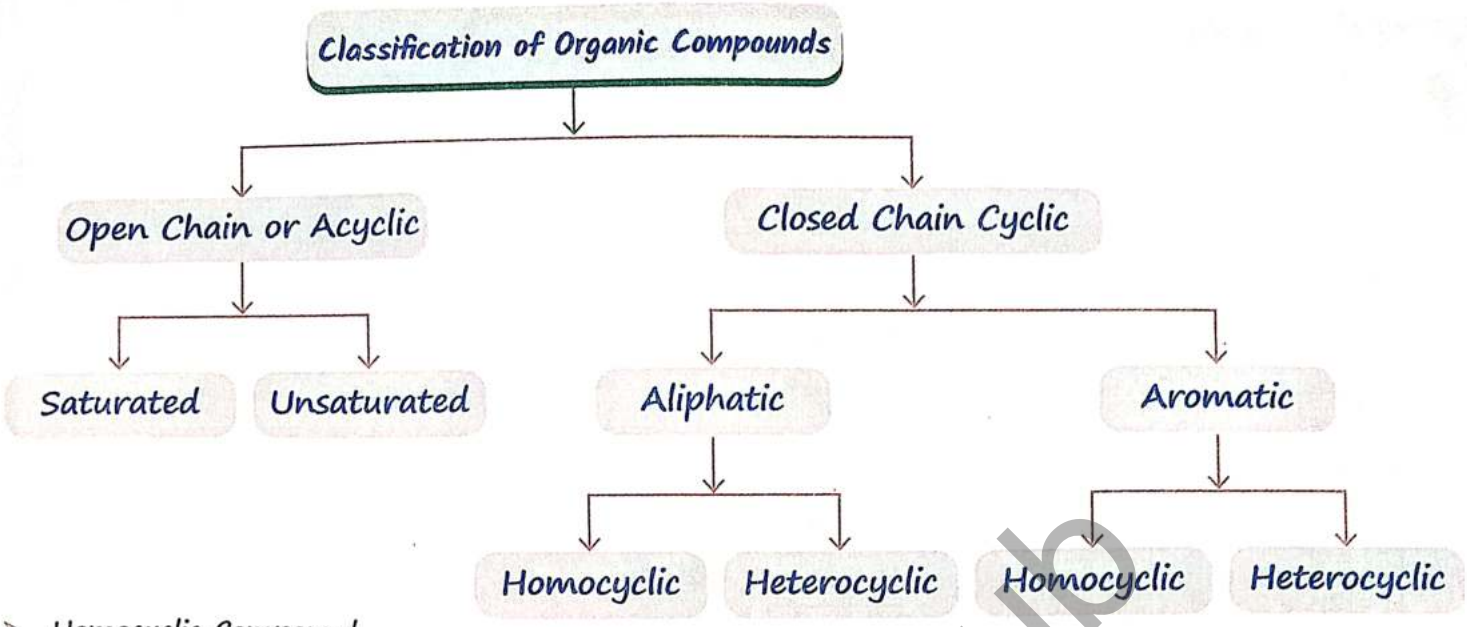
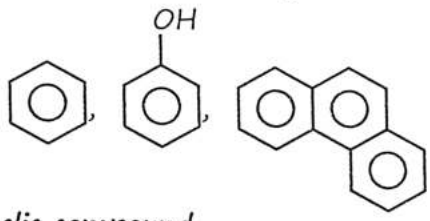
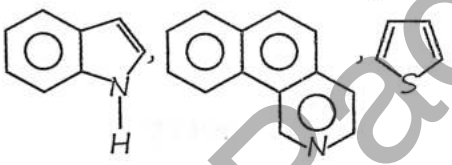


- Homocyclic Compound
  - When all atoms of all rings are same then compound is known as homocyclic compound.

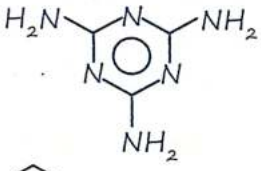
Ex: 

- Hetrocyclic compound.
  - When at least on atom of all ring is different then compound is known as hetrocyclic compound.

Ex: 

Practice Questions

142. Which of the following are heterocyclic compounds?

(a) Benzene (b) Phenol (c) Toluene (d) Melamine 

(e) Aspirin (f) Naphthalene (g) Cumene (h) Pyridine

(i) Paracetamol (j) Pyrrole

(k) Furan

(l) Anthracene

143. Which of the following functional groups are possible for given general formula?

(a) $C_nH_{2n+2}O$ (b) $C_nH_{2n}O$ (c) $C_nH_{2n}O_2$ (d) $C_nH_{2n-2}O_2$

144. For the given DU, which structure are possible?

(a) 1 (b) 2 (c) 3 (d) 4

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General Formula

1. Alkane

   C_nH_{2n+2}

2. Alkene

   $(-C=C-)$

   DU = 1

   $H = 2n \div 2 - 2$

   $= 2$

   C_nH_{2n2}

3. Alkyne

   $(-C\equiv C-)$

   DU = 2

   $H = 2n \div 2 - 4 = 2n - 2$

   C_nH_{2n-2}

4. Alkadiene

   $(-C=C,-C=C-)$

   DU = 2

   $H = 2n \div 2 - 4 = 2n - 2$

   C_nH_{2n-2}

5. Alkenyne

   $(-C=C,-C\equiv C-)$

   DU = 3

   $H = 2n \div 2 - 6 = 2n - 4$

   C_nH_{2n-4}

6. Alcohol

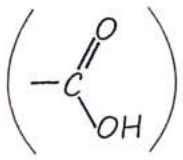
   $(-OH)$

   DU = 0

   $H = 2n + 2$

   $C_nH_{2n+2}O$

7. Carbolic acid

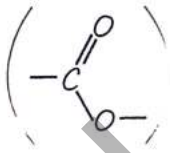
   

   DU = 1

   $H = 2n + 2 - 2 = 2n$

   $C_nH_{2n}O_2$

8. Ester

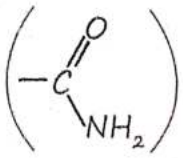
   

   DU = 1

   $H = 2n + 2 - 2 = 2n$

   $C_nH_{2n}O_2$

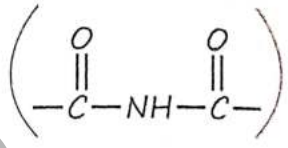
9. Aldehyde
10. acid anhydride
11. ketone
12. Acid amide

13. Amine

    $(-NH_2)$

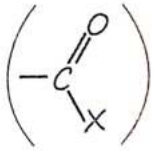
14. Imide

15. Azo

    $(-N=NH)$

16. Acid halide

17. Halide

    $(-X)$

18. Halide

    $(-CH=NH)$

19. Cyanide

    $(-CN)$

Homologous Series

Homologous series may be defined as a series of similarly constituted compounds in which the members possess the same functional group, have similar chemical characteristics and have a regular gradation in their physical properties. The two consecutive members differ in their molecular formula by CH_2.

- Same general formula
- Same functional group
- Same degree of unsaturation
- Same chemical properties
- Different physical properties
- Difference of $-CH_2$ (14gm) in consecutive member.

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Practice Questions

145. Find first 3 member's?

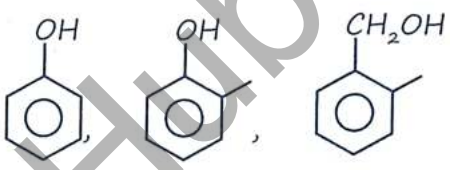
(a) Carboxylic acid
(b) Ester
(c) Aldehyde
(d) Ketone
(e) Acid anhydride
(f) 1° acid amide
(g) Acid halide
(h) Alcohol

146. Which of their following is (are) pairs of same homologous series?

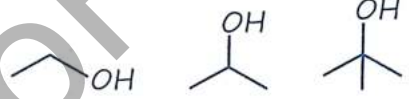
(a) CH_3CHO, CH_2CH_2CHO, $CH_3(CH_2)_5CHO$

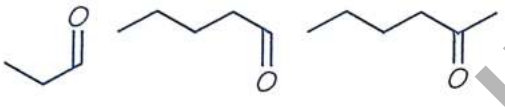
(b) CH_3NH_2, CH_3NHCH_3, $(CH_3)_3N$

(c) $H-OH$, CH_3OH, CH_3CH_2OH

(d) 

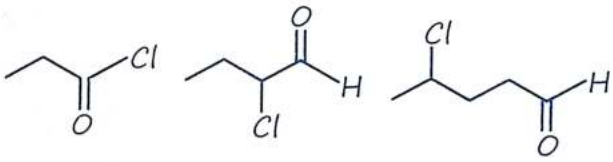
(e) CH_3-CONH_2, $CH_3-CONHCH_3$, $CH_3CON(CH_3)_2$

(f) 

(g) 

(h) C_2H_5Cl, $C_6H_{13}Cl$, $C_9H_{17}Cl$

(i) $H-COOH$, CH_3-COOH, $CH_3-\overset{\overset{O}{\|}}{C}-O-CH_3$

(j) 

(k)

(l) CH_3-CN CH_3-CH_2-CN $CH_3CH_2CH_2NC$

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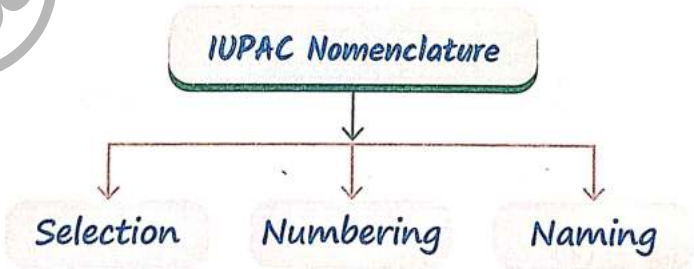
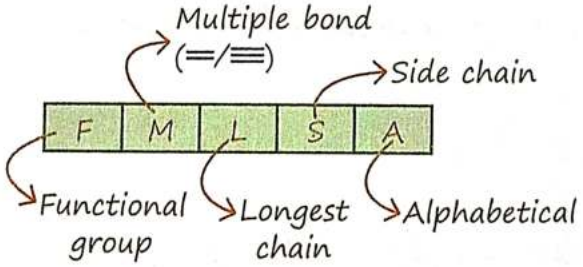


Selection of Parent Chain



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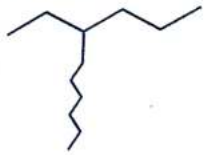
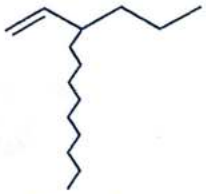
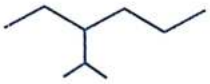
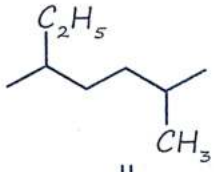
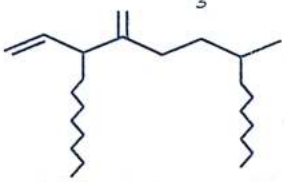
Selection

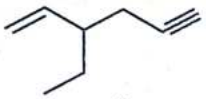
1. 

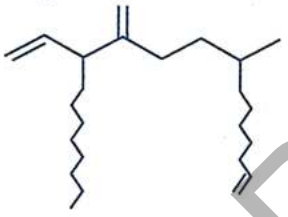
2. 

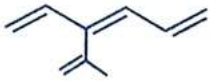
3. 

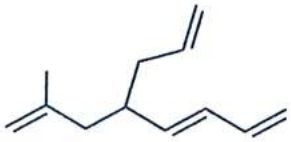
4. 

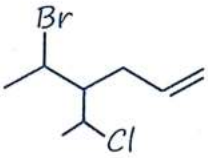
5. 

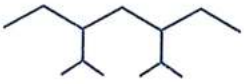
6. 

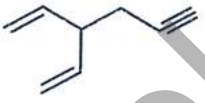
7. 

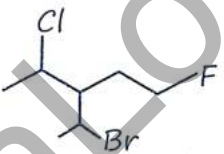
8. 

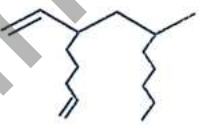
9. 

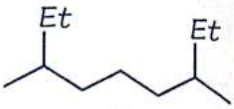
10. 

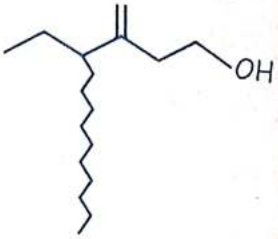
11. 

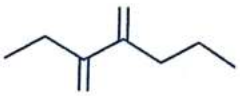
12. 

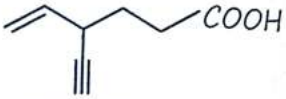
13. 

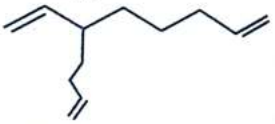
14. 

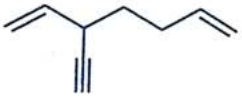
15. 

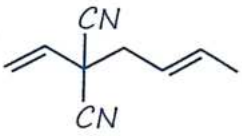
16. 

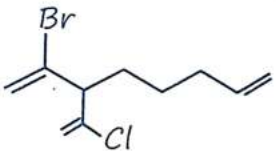
17. 

18. 

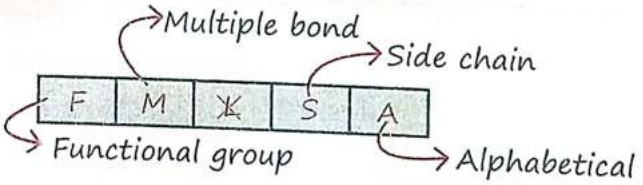
19. 

20. 

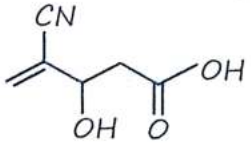
21. 

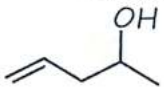
22. 

Numbering of Parent Chain

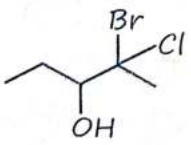


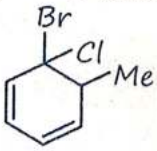
- **Boss rule:** Least number is given to carbon of functional.

EX. (1) 

(2) 

- **Multiple bond rule:** Least number is given to carbon of multiple bond.

EX. (1) 

(2) 

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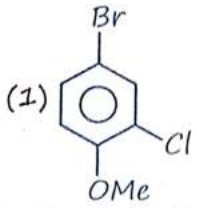
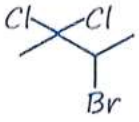
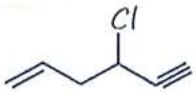
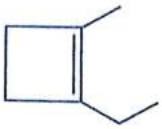
- **Lowest locant or substitutes rule:** If F & M are same then given least numbers to substituents.

EX. (1) 

(2) 

(3) 

- **Alphabetical rule:** If all factors are same then alphabetical order will be considered.

EX. (1) 

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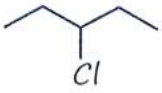
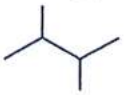
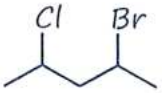
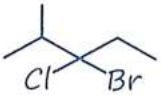
IUPAC

1. 

2. 

3. 

4. 

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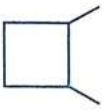
6. 

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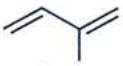
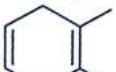
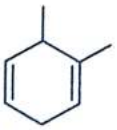
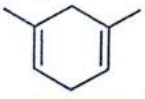
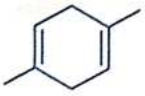
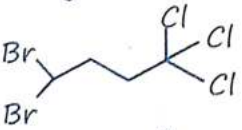
8. 

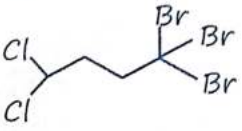
9. 

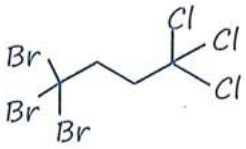
10. 

11. 

12. 

13. 

14. 

15. 

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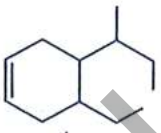
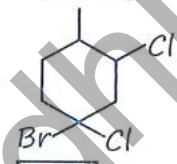
17. 

18. 

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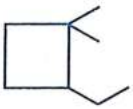
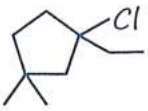
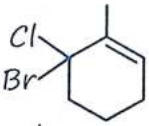
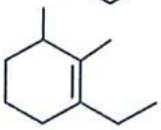
21. 

22. 

23. 

24. 

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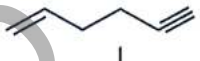
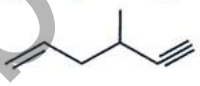
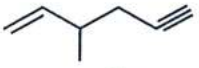
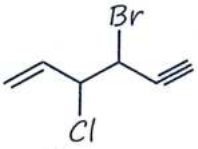
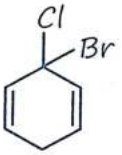
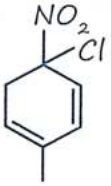
27. 

28. 

29. 

30. 

31. 

32. 

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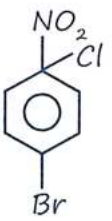
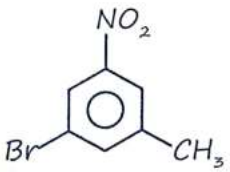
34. 

35. 

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IUPAC Nomenclature

Secondary Prefix	Primary Prefix	Word Root	Primary Suffix	Secondary Suffix
Side Chain	Nature of Compound	No. of (C) in Parent Chain	Nature of Bond	Principal functional group

Word Root: Total carbon in parent chain

- 1C ⇒ meth
- 2C ⇒ Eth
- 3C ⇒ Prop
- 4C ⇒ But
- 5C ⇒ pent
- 6C ⇒ Hex
- 7C ⇒ Hept
- 8C ⇒ Oct
- 9C ⇒ Non
- 10C ⇒ Dec
- 11C ⇒ Undec
- 20C ⇒ Eicosane

Primary suffix: Nature of bonds

- Only single bond ⇒ ane
- Any double bond ⇒ ene
- any triple bond ⇒ yne
- Double and triple bond ⇒ enyne, (P–en–q–yne)

Note

1. Always written with respective position and in alphabetical order.
2. In case of vowel and y, e of suffix will be removed.
3. If di, tri, tetra-is used with MB the 'a' is added to word root.

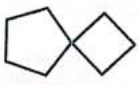
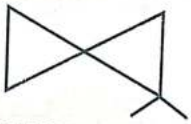
Try Yourself

1.
2.
3. $CH_3-C\equiv CH$
4.
5. $CH_2=CH-CH_2-C\equiv CH$
6. $HC\equiv C-CH_2-CH_2-C\equiv CH$

Primary prefix: Nature of compound.

- Cyclic ⇒
- Bicyclic ⇒
- Spiro ⇒
- Benzene ⇒

Try Yourself

1.
2.
3.
4. 
5. 
6.

Secondary prefix: Side chain in compound

- $-F\Rightarrow$
- $-Cl\Rightarrow$
- $-Br\Rightarrow$
- $-I\Rightarrow$
- $-NO_2\Rightarrow$
- $-NO\Rightarrow$
- $-CH_3\Rightarrow$
- $-CH_2-CH_3\Rightarrow$
- $-CH=CH_2\Rightarrow$
- $-C\equiv CH\Rightarrow$
- $-CH_2-CH=CH_2\Rightarrow$ (Cl on CH_2)
- $-CH(CH_3)-CH_2=CH_2Br$

'yl' is added for carbon chain [Always written with respective position in alphabetical order]

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Practice Questions

147.

148.

149.

150.

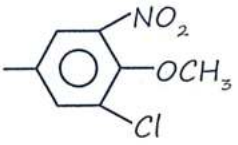
151.

152. $-OCH=CH_2$

153. $-O-\underset{\displaystyle Br}{\underset{|}{CH}}-CH_2-CH=CH_2$

154. $-OH$

155. $-SH$

156. 

Secondary suffix

Functional group	Structure	Prefix	Suffix	Suffix*
Carboxylic	$-COOH$	Carboxy	—oic acid	Carboxylic acid
Sulphonic acids	$-SO_3H$	Sulpho	-Sulphonic acid	-
Carboxylic anhydrides	$-\overset{O}{\overset{\|\|}{C}}-O-\overset{O}{\overset{\|\|}{C}}-$	-	—oic anhydride	—carboxylic anhydride
Carboxylic esters	$-COOR$	Alkoxycarbonyl or Alkanoyl oxy	—oate	—carboxylate
Acid halides	$-COCl$	Halocarbonyl	—oyl halide	—carbonyl halide
Amides	$-CONH_2$	Cyano	—nitrile	—carbonitrile
Aldehydes	$-CHO$	Formyl or oxo	—al	—carbaldehyde
Ketones	$=O$	Oxo or keto	—one	-
Alcohols	$-OH$	hydroxy	—ol	-
Thiols	$-SH$	mercapto	—thiol	-
Amines	$-NH_2$	amine	—amino	-

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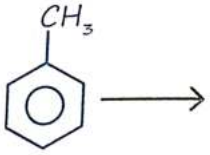
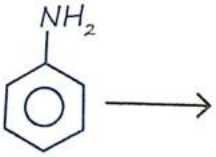
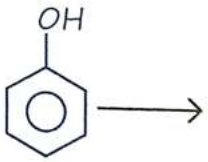
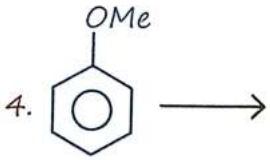
Note:

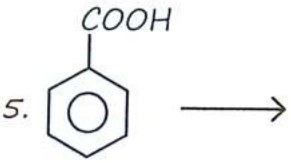
Mono substituted benzene has same common and IUPAC name.

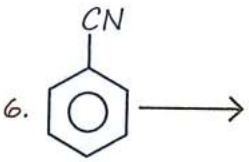
1. 

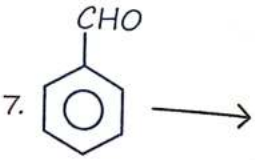
2. 

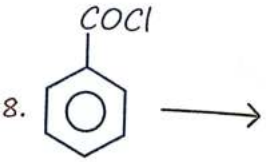
3. 

4. 

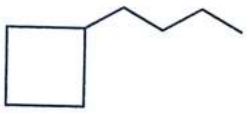
5. 

6. 

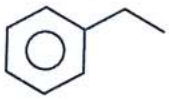
7. 

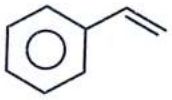
8. 

- di, tri, tetra ... are not used in alphabetical order and only CIN (Cyclo, iso, Neo) is used in alphabetic order.
- If everything is same in open chain and closed chain then closed chain is preferred.

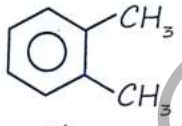
EX: 

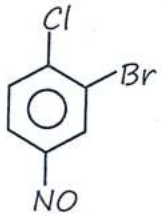
- Generally methyl and ethyl one considered as S.C. in benzene and in other cases benzene will be side chai

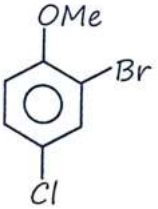
EX: (1)  (2)

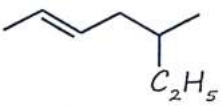
(3) 

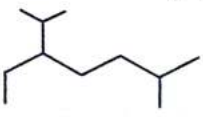
Practice Questions

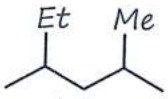
157. 

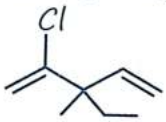
158. 

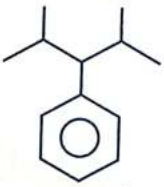
159. 

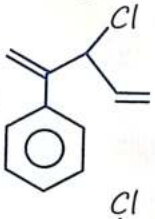
160. 

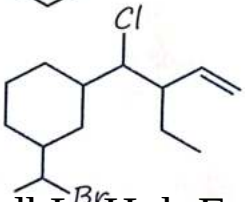
161. 

162. 

163. 

164. 

165. 

166. 

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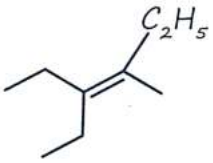
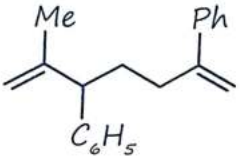
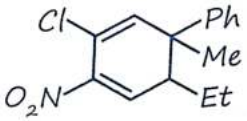
168. 

169. 

170. 

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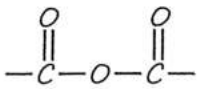
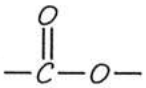
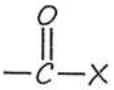
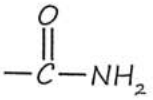
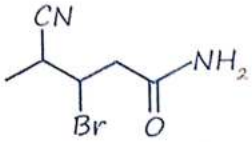
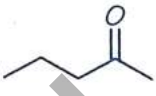
IUPAC of Functional Group

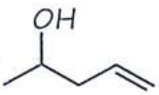
C-Containing
- —COOH
- 
- 
- 
- 
- —CN
- —CHO

EX: (1)

(2) 

NO C-Containing
- $—SO_3H$
- —OH
- $—NH_2$
- —SH
-

EX: (1) 

(2) 

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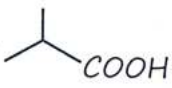
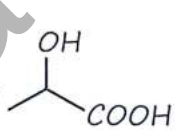
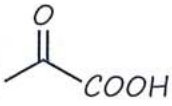
Practice Questions

171. H—COOH

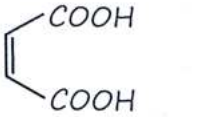
172. CH_3—COOH

173. 

174. 

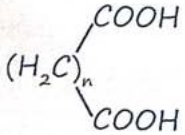
175. 

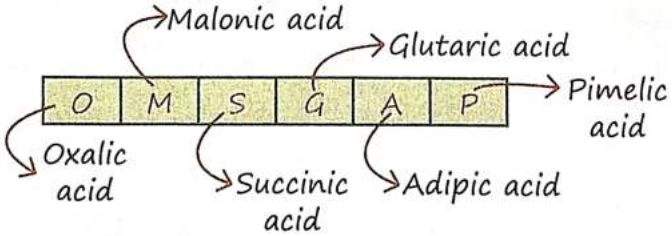
176.

177. 

178. HOOC

Dicarboxylic acid





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179. O

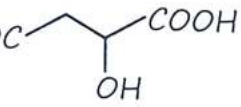
180. M

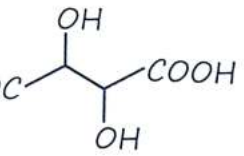
181. O

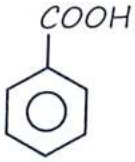
182. G

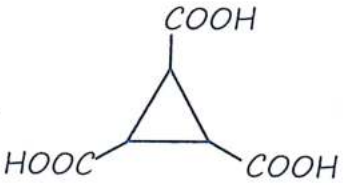
183. A

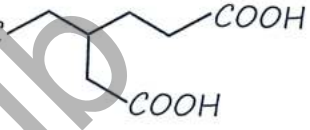
184. P

185. 

186. 

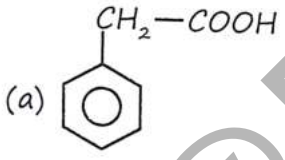
187. 

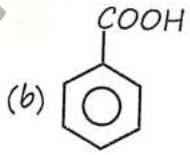
188. 

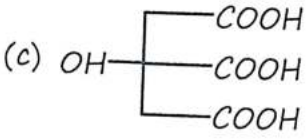
189. 

Note:

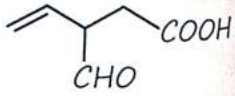
- Carbon of F.G. present in P.C. ⇒
- Carbon of F.G. not present in P.C. and directly attached to P.C. ⇒
- Carbon of F.G. present in side chain ⇒
- If more than two same F.G. are present then rule (2) is applied but if rule (2) is not applicable then select max. F.G. in P.C. and other are treated as S.C.

(a) 

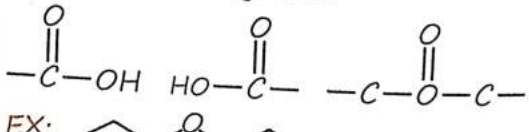
(b) 

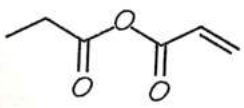
(c) 

1. Carboxylic acid
   Prefix ⇒ Carboxy
   Suffix ⇒ oic acid
   Suffix* ⇒ Carboxylic acid
2. Acid halide
   Prefix ⇒ halocarbonyl
   Suffix ⇒ oyl halide
   Suffix* ⇒ Carbonyl halide
3. Cyanide
   Prefix ⇒ Cyano
   Suffix ⇒ nitrile
   Suffix* ⇒ Carbonitrile
4. Aldehyde
   Prefix ⇒ Formyl, OXO
   Suffix ⇒ all
   Suffix* ⇒ Carbaldehyde
   If (C) of –CHO is present in P.C. ⇒ OXO
   If (C) of –CHO is not present in P.C. ⇒ formyl



5. Acid Anhydride
   Prefix ⇒
   Suffix ⇒ oic anhydride



EX: 

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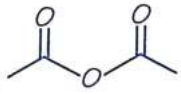
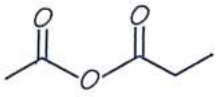
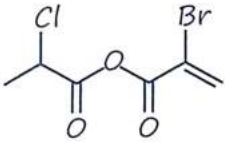
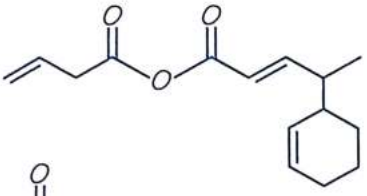
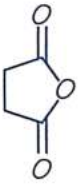
Practice Questions

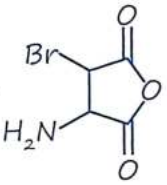
190. 

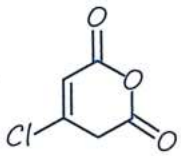
191. 

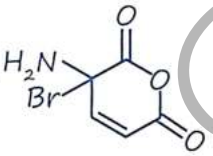
192. 

193. 

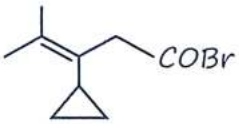
194. 

195. 

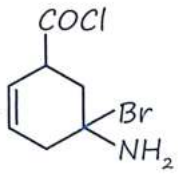
196. 

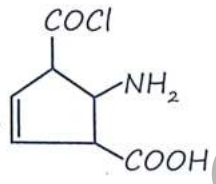
197. 

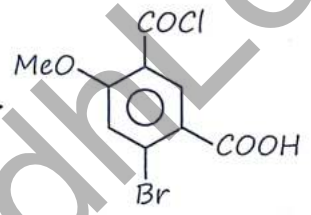
198.

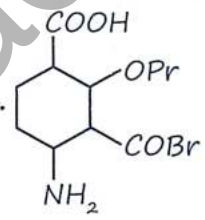
199. 

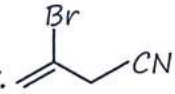
200.

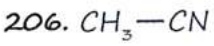
201. 

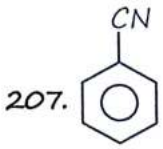
202. 

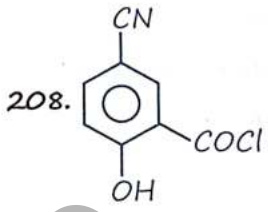
203. 

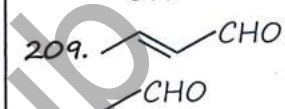
204. 

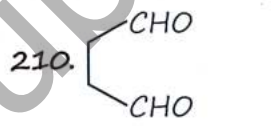
205. 

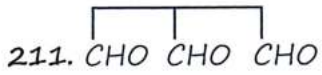
206. 

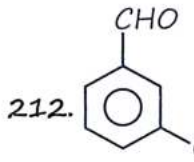
207. 

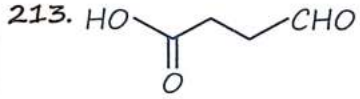
208. 

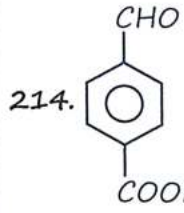
209. 

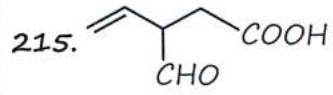
210. 

211. 

212. 

213. 

214. 

215. 

Ketone	Prefix	Keto/oxo	Suffix	one
Alcohol	Prefix	hydroxy	Suffix	ol
Thiol	Prefix	mercapto	Suffix	thiol
Sulphonic acid	Prefix	Sulpho	Suffix	Sulphonic acid
Amine	Prefix	amino	Suffix	amine
NC (Isonitrile)				

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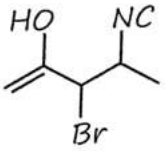
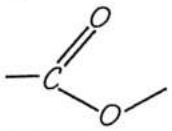
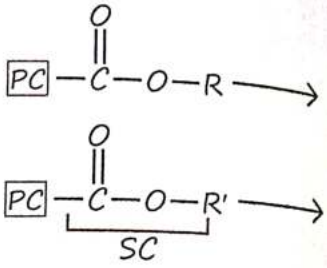
Least number is given to (C) of functional group.

EX: 

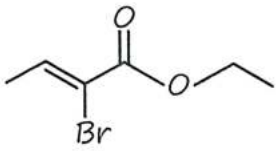
Ester

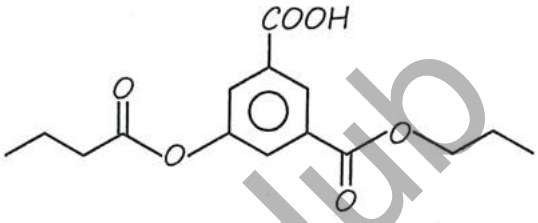




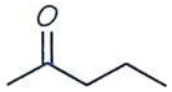
Suffix ⇒ oate, Suffix* ⇒ Carboxylate, Prefix ⇒

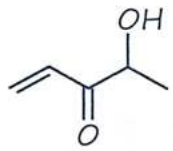
- Group/atom attached to oxygen of ester is always written first.

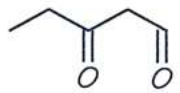
EX: (1) 

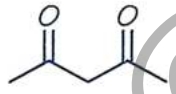
(2) 

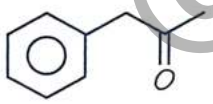
Practice Questions

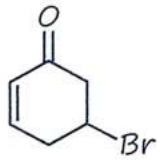
216. 

217. 

218. 

219. 

220. 

221. 

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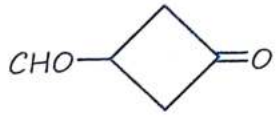
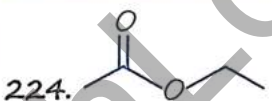
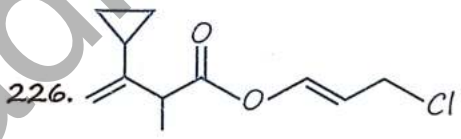
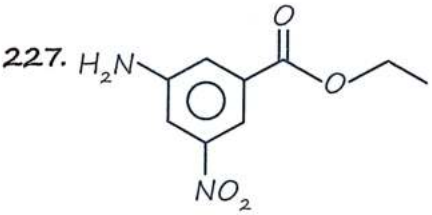
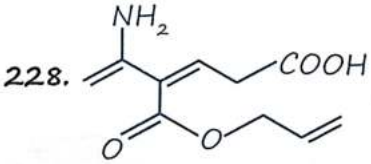
223. 

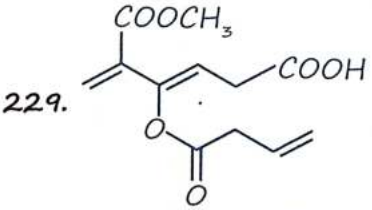
224. 

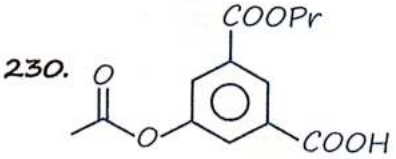
225. 1st member

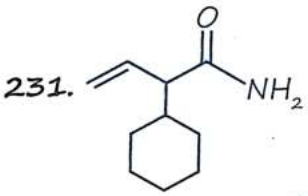
226. 

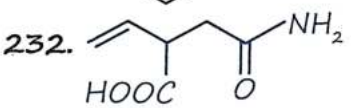
227. 

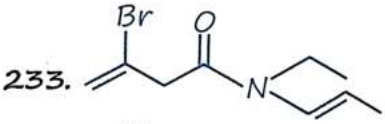
228. 

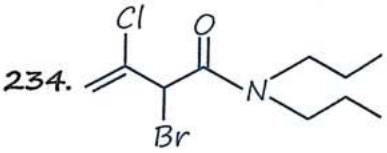
229. 

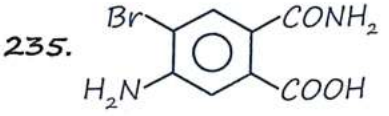
230. 

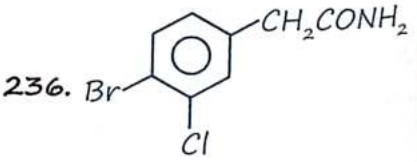
231. 

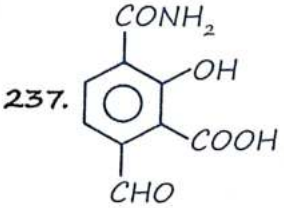
232. 

233. 

234. 

235. 

236. 

237. 

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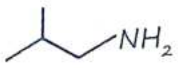
238. 

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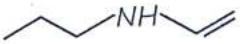
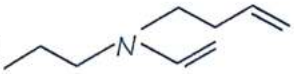
240. 

241. 

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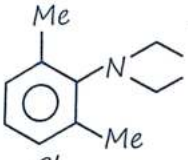
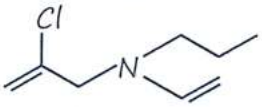
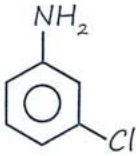
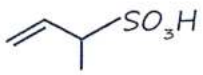
243. 

244. 

245. 

246. 

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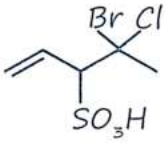
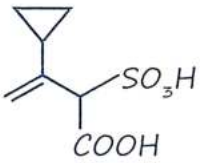
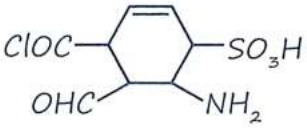
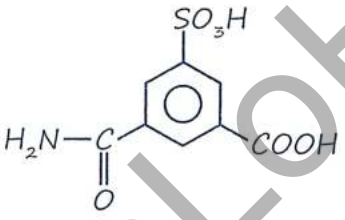
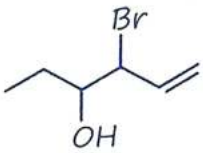
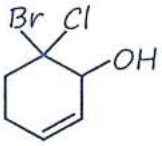
248. 

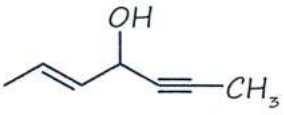
249. 

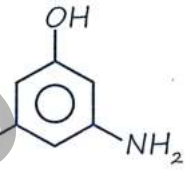
250. 

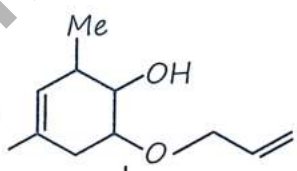
251. 

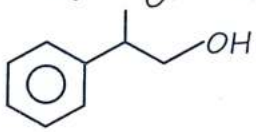
252. 

253. 

254. 

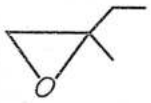
255. 

256. 

257. 

Cyclic Ether

- Cyclic ether ⇒
- Cyclic amide ⇒
- Cyclic ester ⇒

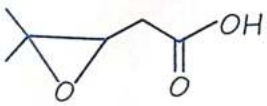
EX: (1)  (2)

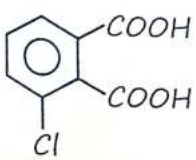
Practice Questions

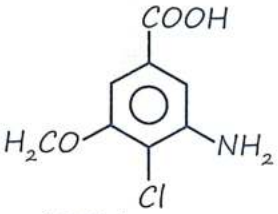
258.

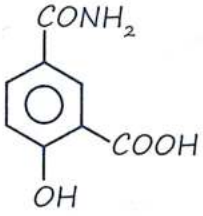
259.

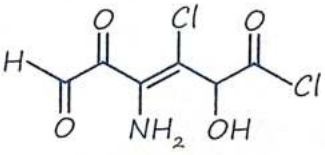
260.

261. 

262. 

263. 

264. 

265. 

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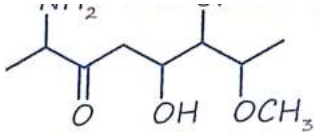
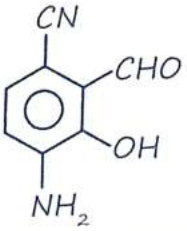
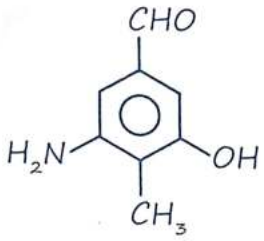
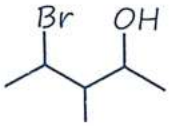
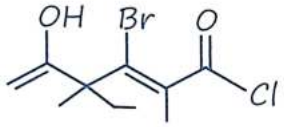
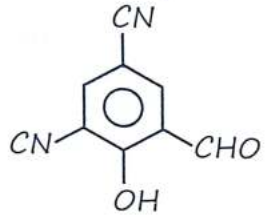
266. 

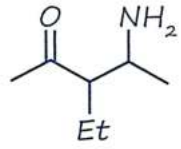
267. 

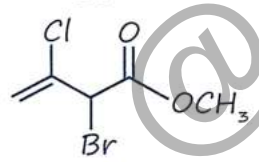
268. 

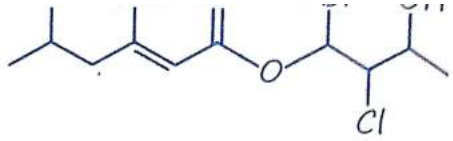
269. 

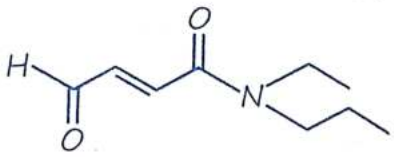
270. 

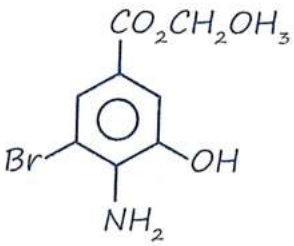
271. 

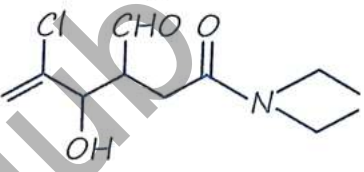
272. 

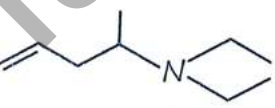
273. 

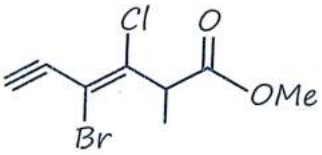
274. 

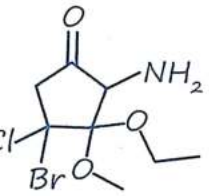
275. 

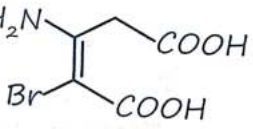
276. 

277. 

278. 

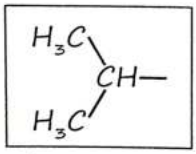
279. 

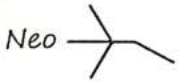
280. 

281. 

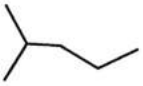
Common Naming

ISO Octane

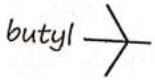
1. ISO 



1. Isohexyl chloride
2. Neohexyl alcohol
3. Isobutene

4. Isohexane 
5. Neohexyl and Isopentyl join

2. Normal (n)
   Secondary (sec)
   Tertiary (tert)

sec. butyl  (tert butyl)

1. n-hexane
2. n-propyl bromide
3. sec. butylchloride
4. Tert-butanol
5. sec. butyl tert butyl keton
6. ditert butyl ether
7. sec. pentyl-n-propyl ketone

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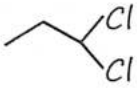
3. Vinyl

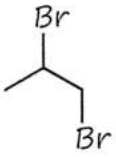
Allyl

1. Vinyl chloride
2. Allylalcohol
3. Allyl and vinyl join
4. Allyl and Iso pentyl
5. No. of vinylic (C)
6. Vinyl and tert butyl join

4. Geminal (Germ)

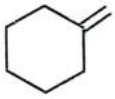
EX: 

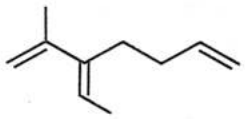
Vicinal (Vic)

EX: 

5. (a) Alkylidene
   1. Ethylidene choride
   2. Ethylene dichloride
   3. Ethylene oxide

   (b) Alkylene

1. 

2. 

3. CH_2Cl_2

6. α, β, γ, δ
   1. β-hydroxy acetic acid
   2. γ-methyl cyclohexane carbaldehyde
   3. methyl cyclohexane carbaldehyde

7. Benzyl, Benzal, Benzo, Phenyl
   1. Benzyl alcohol
   2. Benzyl chloride
   3. Benzal chloride
   4. Benzoic acid
   5. o-substituted benzal chloride

8. (a) Acryl

   C=C—C

   1. Acraldehyde

   $H_2C{=}HC{-}\underset{\underset{O}{\|}}{C}{-}H$

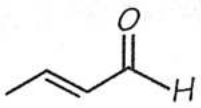
   2. Acrylonitrile/Acrotein

   ($H_2C{=}CH{-}C{\equiv}N$)

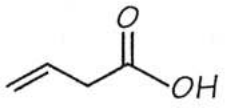
   (b) Croton

   C—C=C—C—

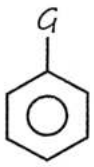
   1. Crotonaldehyde



2. β-crotonic acid



9. o, m, p, IPSO



10. No. of carbon

1C ⟶
2C ⟶
3C ⟶
4C ⟶
5C ⟶

1. Formaldehyde
2. Iodobutyric acid
3. β-Hydroxy valeric acid
4. α-amino butyraldehyde
5. Propionone
6. p-Chlorophenyl acetic acid

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Practice Questions

Write the IUPAC name by joining given radicals:

282. Isopropyl and Isobutyl
283. Isopentyl and Neopentyl
284. Isopropyl and Neopentyl
285. Tertiary butyl and Neohexyl
286. Tert-butyl and Isobutyl
287. Allyl and Allyl
288. Vinyl and Allyl
289. Allyl and Isohexyl
290. Allyl and sec. butyl
291. Disec-butyl

Draw structure and write IUPAC names?

292. Acetone
293. Phenyl acetate
294. β-Hydroxyvaleric acid
295. Isobutyric acid
296. Ethylidene cyclopentane
297. Di tert butyl ketone
298. Sec. butyl aldehyde
299. Acetyl aldehyde
300. Valeric acid

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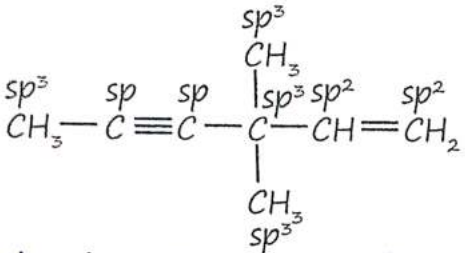
Previous Year Questions

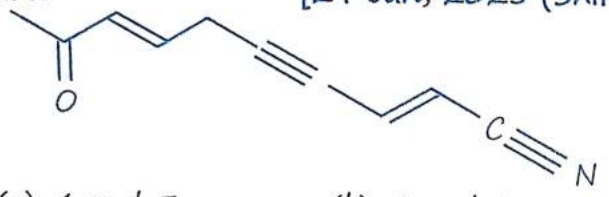
1. In 3,3-dimethylhex-1-ene-4-yne, there are ____ sp^3, ____ sp^2 and ____ sp hybridised carbon atoms respectively:
[02 April, 2025 (Shift-II)]

(a) 4, 2, 2 (b) 3, 3, 2
(c) 2, 4, 2 (d) 2, 2, 4

Sol. (a)

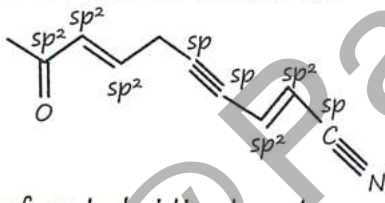


2. In the given structure, number of sp and sp^2 hybridized carbon atoms present respectively are:
[24 Jan, 2025 (Shift-II)]



(a) 4 and 5 (b) 4 and 6
(c) 3 and 5 (d) 3 and 6

Sol. (c) sp and sp^2 hybridized carbon atoms in the structure can be shown as:



No. of sp hybridized carbon-3
No. of sp^2 hybridized carbon-5

3. The sum of sigma (σ) and pi(π) bonds in Hex-1, 3-dien-5-yne is _____.
[29 Jan, 2025 (Shift-I)]

Sol. [15] $CH_2 = CH - CH = CH - C \equiv CH$
Total sigma bonds = 11
Total pi bonds = 4
Total sigma and pi bonds: 11 + 4 = 15
Therefore, the sum of sigma and pi bonds in Hex-1,3-dien-5-yne is 15.

4. Total number of sigma (σ) _____ and pi (π) _____ bonds respectively present in hex-1-en-4-yne are:
[29 Jan, 2025 (Shift-II)]

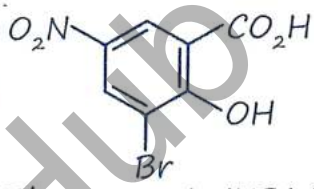
Sol. (d) Structure of hex-1-en-4-yne:
$CH_2=CH-CH_2-C\equiv C-CH_3$
Total Bonds: Sigma (σ) bonds : 13
Pi (π) bonds 3

5. What is the correct IUPAC name of

[03 April, 2025 (Shift-II)]

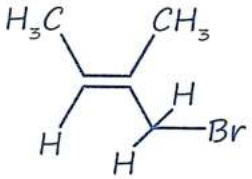
(a) 3-Bromo-2-hydroxy-5-nitrobenzoic acid
(b) 3-Bromo-4-hydroxy-1-nitrobenzoic acid
(c) 2-Hydroxy-3-bromo-5-nitrobenzoic acid
(d) 5-Nitro-3-bromo-2-hydroxybenzoic acid

Sol. (a)



The correct IUPAC name of the given compound is 3-Bromo-2-hydroxy-5-nitro-Benzoic acid.

6. Which of the following is the correct IUPAC name of given organic compound (X) ?

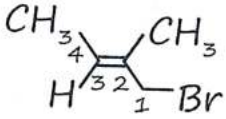


(X)

[07 April, 2025 (Shift-I)]

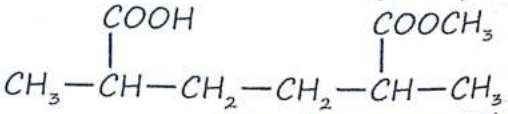
(a) 2-Bromo-2-methylbut-2-ene
(b) 3-Bromo-3-methylprop-2-ene
(c) 1-Bromo-2-methylbut-2-ene
(d) 4-Bromo-3-methylbut-2-ene

Sol. (c)



The IUPAC name of the compound is : 1-Bromo-2-methyl but-2-ene

7. The IUPAC name of the following compound is:



[22 Jan, 2025 (Shift-I)]

(a) 6-Methoxycarbonyl-2, 5-dimethylhexanoic acid.
(b) 2-Carboxy-5-methoxycarbonylhexane.
(c) Methyl-6-carboxy-2, 5-dimethylhexanoate.
(d) Methyl-5-carboxy-2-methylhexanoate.

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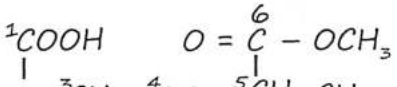
Sol. (a) 

6 - Methoxycarbony l- 2, 5-dimethylhexanoic acid

8. Cyclohexene is _____ type of an organic compound. [27 Jan, 2024 (Shift-I)]

(a) Benzenoid aromatic
(b) Benzenoid non-aromatic
(c) Acyclic
(d) Alicyclic

Sol. (d) is Alicyclic compound

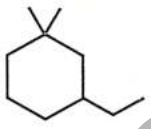
9. IUPAC name of following compound (P) is:

(P)

[27 Jan, 2024 (Shift-I)]

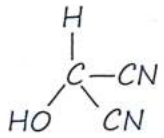
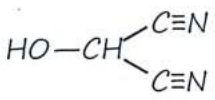
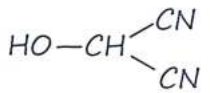
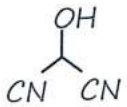
(a) l-Ethyl-5, 5-dimethylcyclohexane
(b) 3-Ethyl-1,1-dimethylcyclohexane
(c) l-Ethyl-3, 3-dimethylcyclohexane
(d) l, l-Dimethyl-3-ethylcyclohexane

Sol. (b) IUPAC name of given compound is

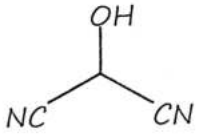


3-ethyl 1, 1-dimethylcyclohexane

10. Bond line formula of $HOCH(CN)_2$ is:
[27 Jan, 2024 (Shift-II)]

(a) 
(b) 
(c) 
(d) 

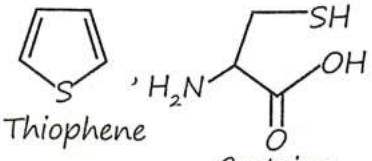
Sol. (d) Bond line formula of compound $CH(OH)(CN)_2$ is

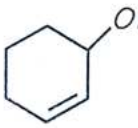


11. Number of compounds among the following which contain sulphur as heteroatom is___.

Furan, Thiophene, Pyridine, Pyrrole, Cysteine, Tyrosine. [29 Jan, 2024 (Shift-I)]

Sol. [2] Thiophene and Cysteine contain sulphur as shown in structures given below

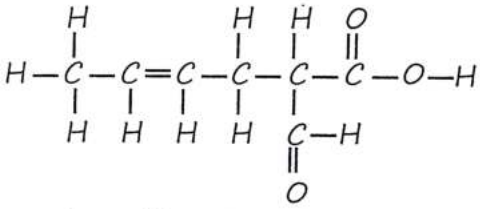

Thiophene Cysteine

12. According to IUPAC system, the compound  is named as
[29 Jan, 2024 (Shift-II)]

(a) Cyclohe x-1-en-2-ol
(b) 1-Hydroxyhex-2-ene
(c) Cyclohex-1-en-3-ol
(d) Cyclohex-2-en-1-ol

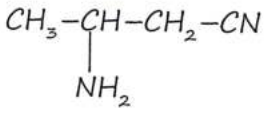
Sol. (d) IUPAC name: Cyclohex-2-en-1-ol

13. The total number of 'Sigma' and Pi bonds in 2 formylhex-4-enoic acid is ___.
[29 Jan, 2024 (Shift-II)]

Sol. [22] 

Total number of bonds are 22

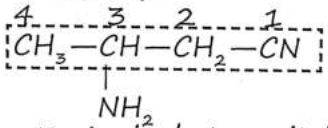
14. IUPAC name of following compound is



[30 Jan, 2024 (Shift-II)]

(a) 2-Aminopentanenitrile
(b) 2-Aminobutanenitrile
(c) 3-Aminobutanenitrile
(d) 3-Aminopropanenitrile

Sol. (c) Priority is given to cyano group as compare to amino group.


3-Aminobutanenitrile

15. Given below are two statements:
Statement-I: IUPAC name of $HO-CH_2-(CH_2)_3-CH_2-COCH_3$ is 7-hydroxyheptan-2-one.
Statement-II: 2-oxoheptan-7-ol is the correct IUPAC name for above compound.

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In the light of the above statements. Choose the **most appropriate answer** from the options given below. [31 Jan, 2024 (Shift-I)]

(a) Statement-I is correct but Statement-II is incorrect.

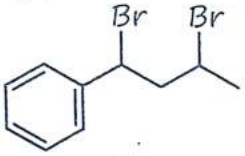
(b) Both Statement-I and Statement-II are incorrect.

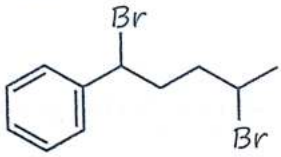
(c) Both Statement-I and Statement-II are correct.

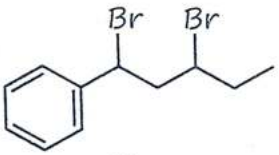
(d) Statement-I is incorrect but Statement-II is correct.

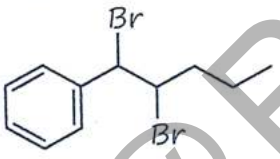
Sol. (a) 7-Hydroxyheptan-2-one is correct IUPAC name of given compound.

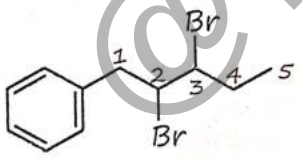
16. **Identify structure of 2, 3-dibromo-1-phenylpentane.** [31 Jan, 2024 (Shift-II)]

(a) 

(b) 

(c) 

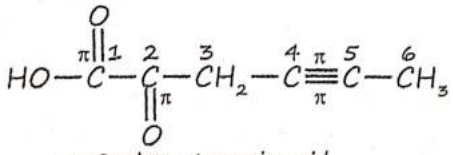
(d) 

Sol. (c) 

2, 3-dibromo -1-phenylpentane

17. **The total number of 'sigma' and 'Pi' bonds in 2-oxohex-4-ynoic acid is___.** [04 April, 2024 (Shift-II)]

Sol. [18] The structure is:



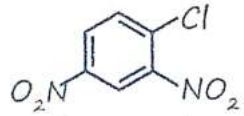
2-Oxohex-4-ynoic acid

Number of σ-bonds = 14

Number of π-bonds = 4

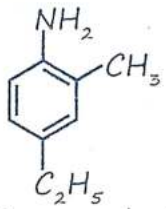
Number of σ & π-bonds = 18

18. **Given below are two statements:**

Statement-I: 

Compund -A

IUPAC name of Compound A is 4-chloro-1, 3-dinitrobenzene:

Statement-II: 

Compund -B

IUPAC name of Compound B is **4-ethyl-2-methylaniline.**

In the light of the above statements, choose the most appropriate answer from the options given below: [08 April, 2024 (Shift-I)]

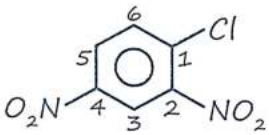
(a) Both Statement-I and Statement-II are correct

(b) Statement-I is incorrect but Statement-II is correct

(c) Statement-I is correct but Statement-II is incorrect

(d) Both Statement-I and Statement-II are incorrect

Sol. (b) Statement I:

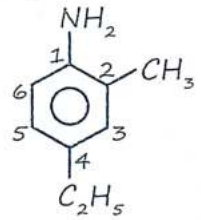


IUPAC name

⇒ 1-Chloro-2, 4-dinitrobenzene

Hence, statement-I is incorrect

Statement-II:

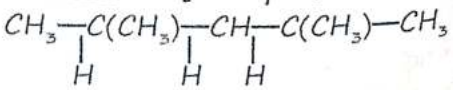


IUPAC name

⇒ 4-Ethyl-2-methylaniline

Hence, statement-II is correct

19. **In the given compound, the number of 2° carbon atom/s is _____.** [08 April, 2024 (Shift-I)]



(a) Three (b) One

(c) Two (d) Four

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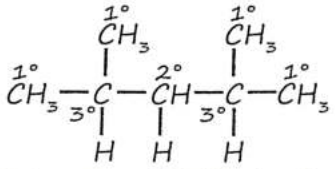
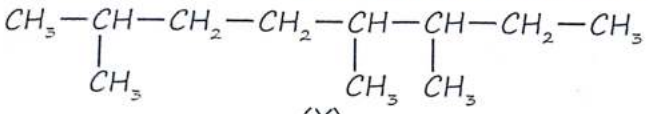
Sol. (b) There is only one 2° carbon atom present in this compound.



20. IUPAC name of following hydrocarbon (X) is:

[08 April, 2024 (Shift-II)]



(X)

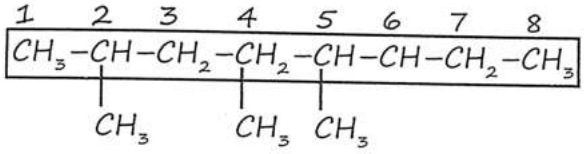
(a) 2-Ethyl-3,6-dimethylheptane

(b) 2-Ethyl-2,6-diethylheptane

(c) 2,5,6-Trimethyloctane

(d) 3,4,7-Trimethyloctane

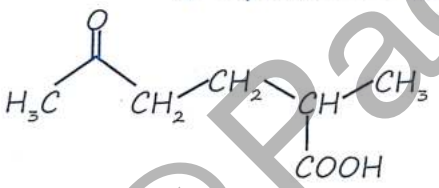
Sol. (c)



IUPAC name is 2,5,6-Trimethyloctane

21. The correct IUPAC nomenclature for the following compound is

[8 April, 2023 (Shift-II)]



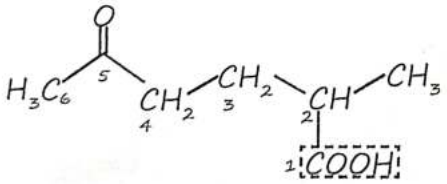
(a) 5-Formyl-2-methylhexanoic acid

(b) 2-Methyl-5-oxohexanoic acid

(c) 2-Formyl-5-methylhexan-6-oic acid

(d) 5-Methyl-2-oxohexan-6-oic acid

Sol. (b) In the given compound, the numbering will start from —COOH group. Double bond O present at 5th position will named as 'oxo' in prefix.

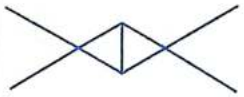
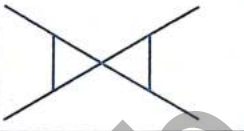
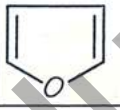


IUPAC name of the above compound is:

2-Methyl-5-oxohexanoic acid

22. Match List-I with List-II.

[28 July, 2022 (Shift-I)]

List-I (Pair of Compounds)			List-II (Isomerism)
(A)		(I)	Spiro compound
(B)		(II)	Aromatic compound
(C)		(III)	Non-planar Heterocyclic compound
(D)		(IV)	Bicyclo compound

Choose the correct answer from the options given below

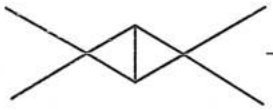
(a) (A)-(II), (B)-(I), (C)-(IV), (D)-(III)

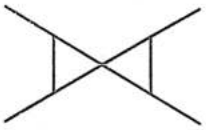
(b) (A)-(IV), (B)-(III), (C)-(I), (D)-(II)

(c) (A)-(III), (B)-(IV), (C)-(I), (D)-(II)

(d) (A)-(IV), (B)-(III), (C)-(II), (D)-(I)

Sol. (c) ⟶ Non-planar heterocyclic compound

 ⟶ Bicyclo compound

 ⟶ Spiro compound

⟶ Aromatic compound

23. The correct decreasing order of priority of functional groups in naming an organic compound as per IUPAC system of nomenclature is

[26 July, 2022 (Shift-II)]

(a) $-COOH > -CONH_2 > -COCl > -CHO$

(b) $-SO_3H > -COCl > -CONH_2 > -CN$

(c) $-COOR > -COCl > -NH_2 > C=O$

(d) $-COOH > -COOR > -CONH_2 > -COCl$

Sol. (b) The decreasing order of priority of functional group is, $-SO_3H > -COCl > -CONH_2 > -CN$

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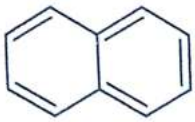
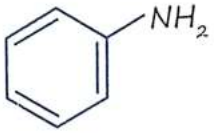
24. Which of the following is not an example of benzenoid compound?

[26 July, 2022 (Shift-II)]

(a)

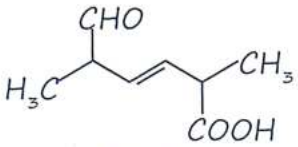
(b)

(c) 

(d) 

Sol. (a,b) The structure consisting benzene ring is called benzenoid compound. Options (a) and (b) do not contain benzene ring, so they are non benzenoid compounds.

25. The IUPAC name for the following compound is:



[2 Sept, 2020 (Shift-I)]

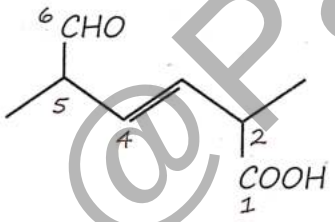
(a) 2, 5-Dimethyl-6-carboxy-hex-3-enal

(b) 6-Formyl-2-methyl-hex-3-enoic acid

(c) 2, 5-Dimethyl-6-oxo-hex-3-enoic acid

(d) 2, 5-Dimethyl-5-carboxy-hex-3-enal

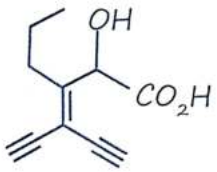
Sol. (c) In the given compound highest priority group is –COOH. Therefore, IUPAC name is given as,

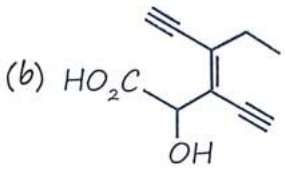


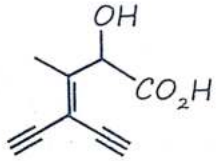
(2, 5-dimethyl-6-oxo-hex-3-enoic acid)

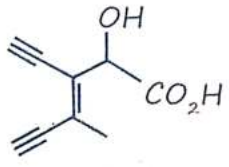
26. Which one of the following structures has the IUPAC name 3-ethynyl-2-hydroxy-4-methylhex-3-en-5-ynoic acid?

C-62.22W-24.61 UA-13.17 (JEE Adv. 2020)

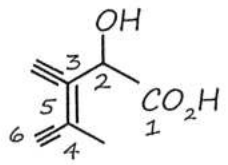
(a) 

(b) 

(c) 

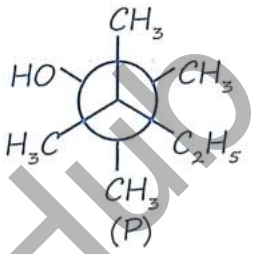
(d) 

Sol. (d) For IUPAC naming, following numbering will be followed

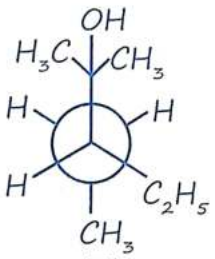


3-ethynyl-2-hydroxy-4-methylhex-3-en-5-ynoic acid

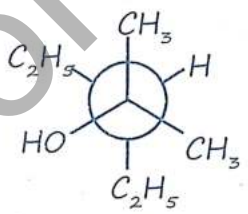
27. Newman projections P, Q, R and S are shown below:



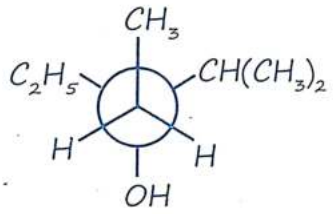
(P)



(Q)



(R)



(S)

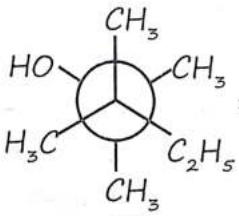
Which one of the following options represents identical molecules?

C-32.04 W-38.77 UA-29.19 (JEE Adv. 2020)

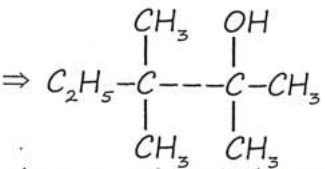
(a) P and Q (b) Q and S

(c) Q and R (d) R and S

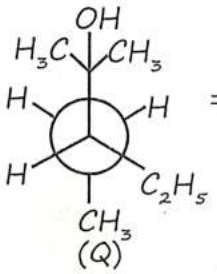
Sol. (c) IUPAC name of P, Q, R and S are:



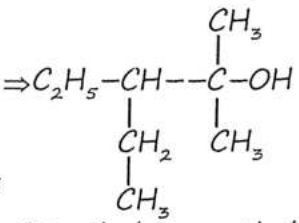
(P)



(2,3,3-trimethylpentan-2-ol)



(Q)



(3-ethyl-2-methylpentan-2-ol)

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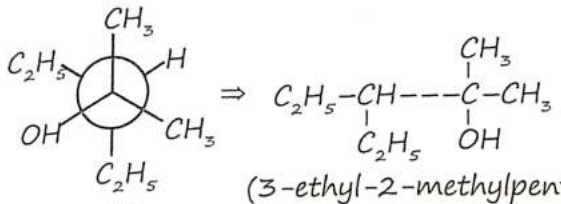
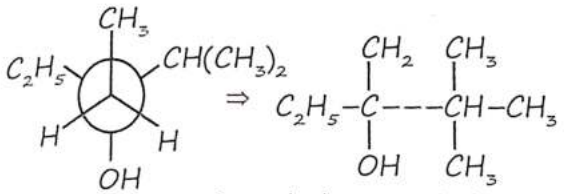


(3-ethyl-2-methylpentan-2-ol)

(R)

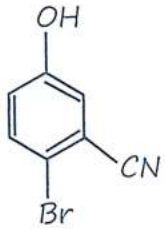


(3-ethyl-2-methylpentan-3-ol)

(S)

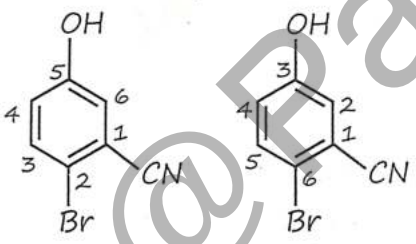
IUPAC name of compounds Q and R are same, thus the correct option (c).

28. **The IUPAC name of the following compound is** (IIT JEE 2009)



(a) 4-bromo-3-cyanophenol
(b) 2 -bromo-5-hydroxybenzonitrile
(c) 2- cyano- 4 -hydroxybromobenzene
(d) 6-bromo-3hydroxybenzonitrile

Sol. (b) –CN has highest priority than –OH.



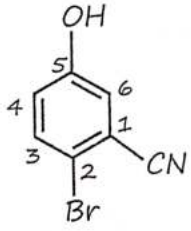
Numbering 1 Numbering 2

From numbering 1 we get sum of locants = 1+2+5=8

And from numbering 2 sum of locant = 1+3+6 = 10

So from the lowest sum rule numbering 1 is correct.

Hence the IUPAC name is 2-bromo-5-hydroxybenzonitrile



29. The IUPAC name of C_6H_5COCl is (IIT JEE 2006)

(a) benzoyl chloride
(b) benzene chloro ketone
(c) benzene carbonyl chloride
(d) chloro phenyl ketone

Sol. (a) C_6H_5COCl is known as benzoyl chloride by IUPA(c)

30. **The IUPAC name of the compound $CH_2 = CH-CH(CH_3)_2$ is** (IIT JEE 1987)

(a) 1 , 1-dimethyl-2-butene
(b) 3 -methyl-1-butene
(c) 2 -vinyl propane
(d) None of these

Sol. (b) $CH_2 = CH-CH(CH_3)_2$

In this structure functional group is alkene. And the longest chain is of 4 carbon. So the IUPAC name is 3 -methyl-1-butene.

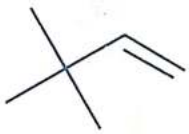
31. **The bond between carbon atom (1) and carbon atom (2) in compound**

$N \equiv C - CH = CH_2$ involves the hybridisation as (IIT JEE 1987)

(a) sp^2 and sp^2 (b) sp^3 and sp
(c) sp and sp^2 (d) sp and sp

Sol. (c) The bond between carbon atom (1) is sp as it is attached to triple bond and carbon atom (2) is sp^2, Since it is attached with a double bond.

32. **The IUPAC name of the compound having the formula is** (IIT JEE 1984)



(a) 3,3,3-trimethyl-1-propene
(b) 1,1,1-trimethyl-2-propene
(c) 3,3 -dimethyl- 1 -butene
(d) 2,2 -dimethyl-3-butene

Sol. (c)

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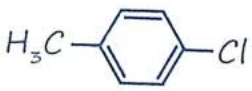
33. The IUPAC name(s) of the following compound is (are)

C-18.6 W-50.63 UA-6.71 PC-24.06

(JEE Adv. 2017)



(a) 4-methylchlorobenzene

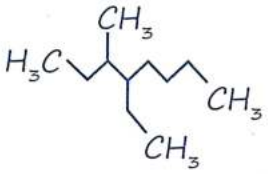
(b) 4-chlorotoluene

(c) 1-chloro-4-methylbenzene

(d) 1-methyl-4-chlorobenzene

Sol. (b, c) 4-chlorotoluene and 1-chloro-4-methyl benzene both are correct answer.

34. Name of the compound given below is:(2003)

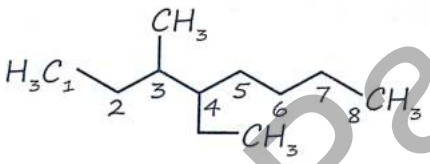


(a) 4-ethyl-3-methyloctane

(b) 3-methyl-4-ethyloctane

(c) 2, 3-diethylheptane

(d) 5-ethyl-6-methyloctane

Sol. (a) 

4-ethyl-3-methyloctane

35. IUPAC name of the following is: (2002)

$CH_2 = CH - CH_2 - CH_2 - C \equiv CH$

(a) 1, 5-hexenyne

(b) 1-hexene-5-yne

(c) 1-hexyne-5-ene

(d) 1, 5-hexynene

Sol. (b) $\overset{1}{C}H_2=\overset{2}{C}H-\overset{3}{C}H_2-\overset{4}{C}H_2-\overset{5}{C}\equiv\overset{6}{C}H$

1-hexene-5-yne.

36. The incorrect IUPAC name is: (2001)

(a) $-CH_3-\underset{\underset{O}{\|}}{C}-\underset{\underset{CH_3}{|}}{CH}-CH_3$,

2-methyl-3-butanone

(b) $CH_3-\underset{\underset{CH_3}{|}}{CH}-\underset{\underset{CH_3-CH_2-CH_3}{|}}{CH}-CH_3$,

2, 3-dimethyl pentane

(c) $CH_3 - C \equiv CCH(CH_3)_2$,

4-methyl-2-pentyne

(d) $CH_3-\underset{\underset{Cl}{|}}{CH}-\underset{\underset{Br}{|}}{CH}-CH_3$,

2-bromo-3-chloro butane

Sol. (a)

37. The IUPAC name of $(CH_3)_2CH - CH_2 - CH_2B$ (19

(a) 1-bromo-3-methylbutane

(b) 2-methyl-3-bromopropane

(c) 1-bromopentane

(d) 2-methyl-4-bromobutane

Sol. (a) $H_3C-\underset{\underset{CH_3}{|}}{\overset{\overset{H}{|}}{C}}-\overset{\overset{H}{|}}{C}-\underset{\underset{H}{|}}{\overset{\overset{H}{|}}{C}}-Br$

1-Bromo-3-methylbutane

38. The IUPAC name for

$CH_3CH = \underset{\underset{NH_2}{|}}{CH}CH_2CHCH_2COOH$ is: (199

(a) 3-amino-5-heptenoic acid

(b) β-amino-δ-heptenoic acid

(c) 5-amino-2-heptenoic acid

(d) 5-amino-hex-2-enecarboxylic acid

Sol. (a) $\overset{7}{C}H_3-\overset{6}{C}H=\overset{5}{C}H-\overset{4}{C}H_2-\underset{\underset{NH_2}{|}}{\overset{3}{C}H}-\overset{2}{C}H_2-\overset{1}{C}OOH$

As – COOH group is highest priority group it is numbered one. So, the IUPAC name is 3-amino-5-heptenoic acid.

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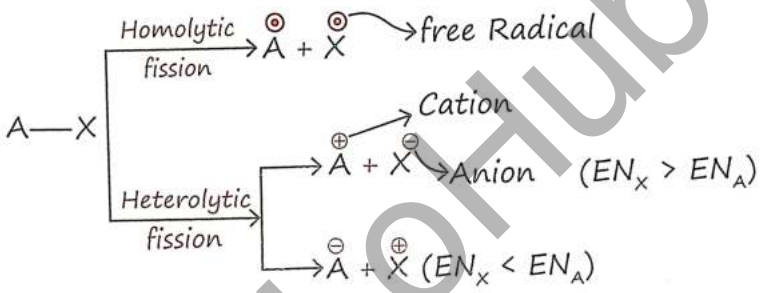
2 General Organic Chemistry (GOC)

Bond Fission

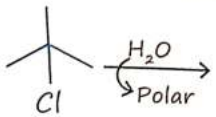
- During any chemical reaction, breaking of bond is known as bond fission.
- Types:

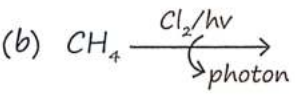


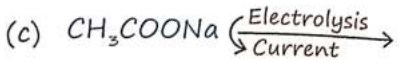
- **Homolytic Fission**
  - During any chemical reaction, if bonded $e^{\ominus}$ pair are cleaved equally then such fission is Known a homolytic fission.
  - EN difference may or may not be zero.
  - Different factors.
- **Factors for Homolytic Fission**
  (1) Sunlight (hv)
  (2) High temp. (Δ)
  (3) Electrolysis
  (4) Metal
  (5) Other free Radical
  (6) Gaseous phase

Practice Questions

1. Which of the following reaction(s) will involve homolytic fission?

(a) 

(b) 

(c) 

(d)

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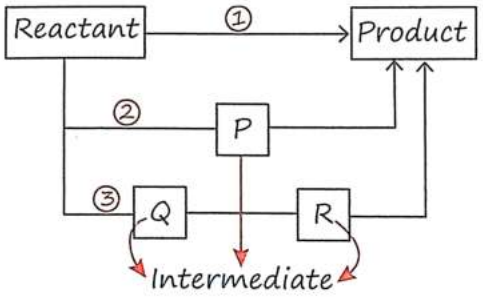
Heterolytic Fission

- During any chemical reaction, if bonded $e^{\ominus}$ pair are cleaved unequally then such fission is known as heterolytic fission.
- Generally occurs in polar solvents.

Intermediate



Examples:

(1) Carbocation ($\overset{\oplus}{C}$) (2) Carbanion ($C^{\ominus}$)

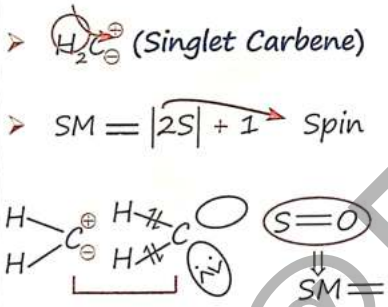
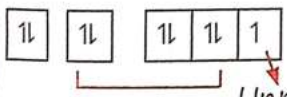
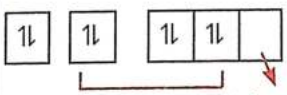
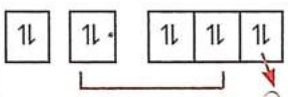
(3) Carbon free radial ($\overset{\ominus}{C}$)

(4) Carbene $\Rightarrow SM = |2S| + 1$

(5) Carbene $\Rightarrow SM = |2S| + 1$

(6) Nitrene ($N^{\oplus}_{\ominus}$) (7) Benzyne

Carbene

➢ $H_2\overset{\ominus}{C}$ (Singlet Carbene) ➢ $SM = \|2S\| + 1$ → Spin  $S = 0$ ⇓ $SM = 1$ (Singlet)	➢ $H_2\overset{\odot}{C}$ (Triplet carbene) ➢ $SM = \|2s\| + 1$ $S = +1$ ➢ $SM = \|2 \times 1\| + 1 = 3$ (Triplet)	C: $1s^2\ 2s^2\ 2p^2$ G.S- 1↿⇂ 1↿⇂ 1 1 E.S- 1↿⇂ 1 1 1 1 4 bonds
✓ $H_3C^{\odot}$  3σ(C–H) Unpaired $e^{\ominus}$ (paramagnetic)	✓ $H_3C^{\oplus}$  3σ(C–H) Vacant orbital(p) (diamagnetic)	✓ $H_3C^{\ominus}$  3σ(C–H) $e^{\ominus}$ pair (diamagnetic)

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Properties	$\oplus CH_3$	$\ominus CH_3$	$\odot CH_3$
Configuration			
LA/LB (Lewis acid/Base)			
U.P. (unpaired)			
l.P. (lone pair)			
$\sqrt{n(n+2)}$B.m M.M. (magnetic moment)			
S.M. $\lvert 2S \rvert + 1$			
Dia/Para			
Hyb. and Shape			
Structure			
Configuration			
LA/LB	Lewis acid	Lewis base	Nothing
U. P.	0	0	1
l. P.	0	1	0
M. M. (magnetic moment)	$\sqrt{n(n+2)}$ B.M. 0	0 B.M.	$\sqrt{1(1+2)}$ $\sqrt{3}$B.M.
S. M.	$\lvert 25 \rvert + 1$ 1(Singlet)	1	$S = \pm y_2 \Rightarrow 2$
Dia/Para	Dia	Dia	Para
Hyb. and Shape	Sp^2, Planner	Sp^3, Tetrahedral	Sp^2, Planer
Structure			

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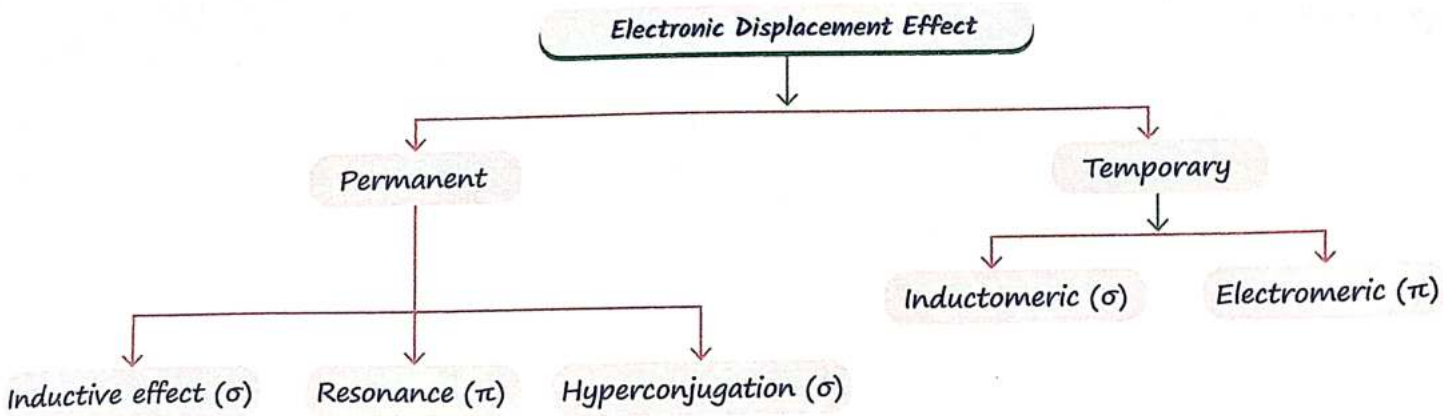
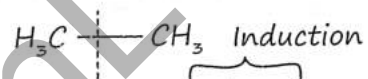


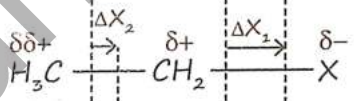
Inductive Effect

- When a covalent bond is formed between atoms of different electro negativity, the electron density is more towards the more electronegative atom of the bond. Such a shift of electron density results in a polar covalent bond.
- This shifting of electron density is known as 'Induction'.
- Due to this induction, adjacent Atom gets effected and developed polarity. This is known as "Inductive Effect".
- Examples

① Ethane:

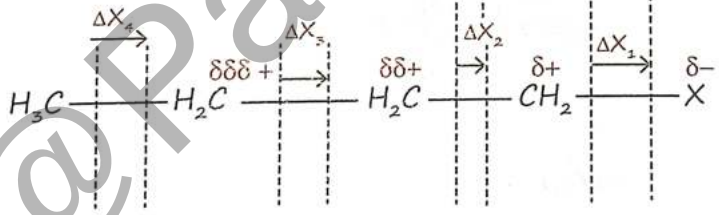


② 1–halo ethane:



$\Delta X_1 >> \Delta X_2 > \Delta X_3 > \Delta X_4 = 0$

③ 1-halo butane:



Properties of Inductive Effect

- It is permanent and partial displacement of sigma electrons.
- It is a permanent effect and weak effect.
- Electrons will never changes its atomic orbital.
- Inductive effect is an additive effect.
- It is a distance dependent effect as distance increases inductive effect decreases.
- Only 10 % of original charge transfers in each bond and $\delta_4 = 0$

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Types of Inductive Effect

C—H ~ Reference

C—A ~ Electron withdrawing group/atom(A) (+I)

C—B ~ Electron Donating group(B) (+I)

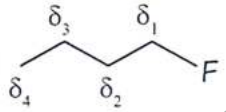
Electron Donating Effect (+I)

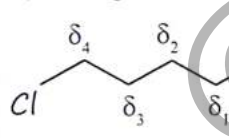
- When bonded pair of electrons are shifted away from any atom or group w. r. t. C — H bond then atom or group shows electron donating effect.

(+ I) → ⊖ve Charge surface } (Real + I)
(+ I) → Alkyl group (C — C, C — H, C — D, C — T)
(+ I) → Less EN

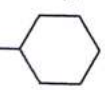
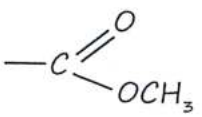
Practice Questions

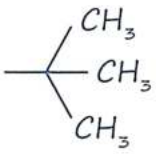
Find the order of charges?

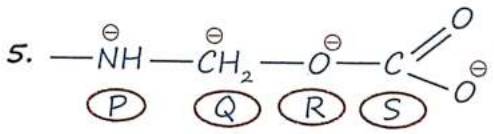
2. 

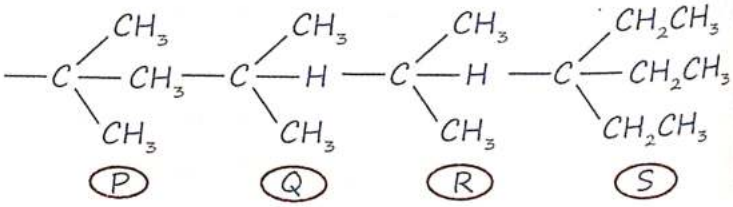
3. 

4. How many of the following Group/atom can acts as +I ?

(a) —OH
(b) $-NH_2$
(c) $-\overset{\ominus}{C}H_2$
(d) $-CH_2OH$
(e) 
(f) $-CH=CH_2$
(g) $-\overset{\ominus}{N}H_2$
(h) 
(i) $-COO^{\ominus}$
(j) —CH=NH
(k) —CN
(l) $-CH_2-CH_3$
(m) —Cl
(n) —CHO

(o) 
(p) $-CT_3$
(q) $-CD_3$
(r) $-\overset{\ominus}{O}$
(s) $-\overset{\ominus}{N}Me_3$

5. 
P Q R S

6. 
P Q R S

7. $-CH_3$ $-SiH_3$
P Q

8. $-CT_3$ $-CD_3$ $-CH_3$
P Q R

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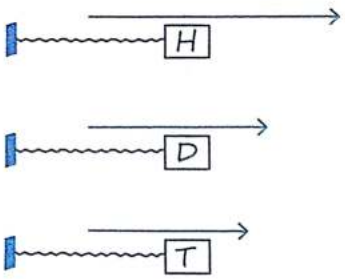
Isotopic Effect



Bond Length ⇒ C—H > C—D > C—T
$e^{\ominus}$ density ⇒ C—T > C—D > C—H

+I effect ⇒

$-\overset{\ominus}{C}H_2 > -\overset{\ominus}{N}H > -\overset{\ominus}{O} > -COO^{\ominus} > -3^{\circ} > -2^{\circ} > -1^{\circ} > -CT_3 > -CD_3 > -CH_3$

(−ve Charged)

Electron Withdrawing Effect (-I)

- When bonded pair of electrons are shifted towards any atom or group w.r.t. C — H bond then atom or group shows electron withdrawing effect.

(+ I) → Positive Charge
(+ I) → Partial positive ($-NO_2 > -CN > -COOH$)
(+ I) → EN (X > O > N > C)

−I effect

$-\overset{\ominus}{N}F_3 > -\overset{\oplus}{N}R_3 > -\overset{\oplus}{N}H_3 > -NO_2 > -CN > -COOH >$

(alky; +ve; Partial + ve)

$-F > -Cl > -Br > -I > -OR > -OH > -NH_2 > -\overset{SP}{C}\equiv CH > -\bigcirc > -CH=CH_2$

(X; O; N; C)

Resonance

- Flow of $\pi e^{\ominus}$ or Delocalisation of $\pi e^{\ominus}$ is known as resonance.
- Condition of Resonance:
  - Continuous P_π/d_π orbital
  - Parallel or nearly parallel orbital
  - flow of $\pi e^{\ominus}$

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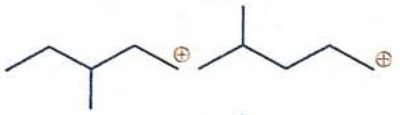
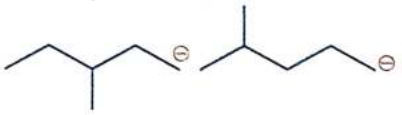
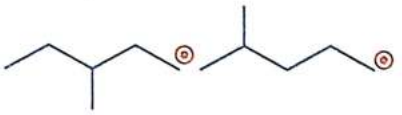
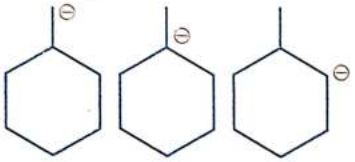
Practice Questions

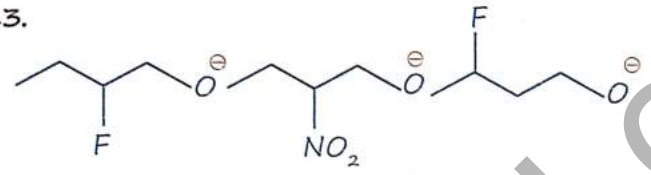
Compare stabilities of ⊕|⊖|⊙:

9. 

10. 

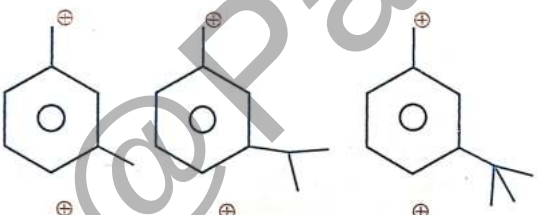
11. 

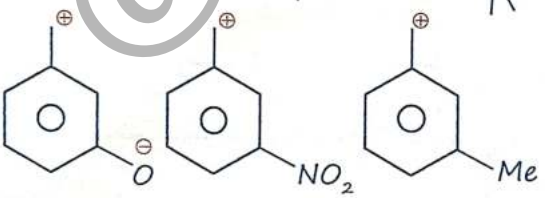
12. 

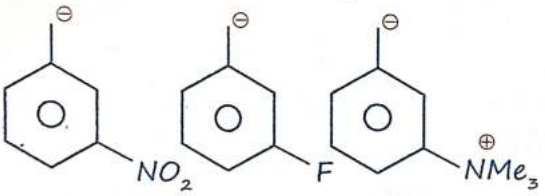
13. 

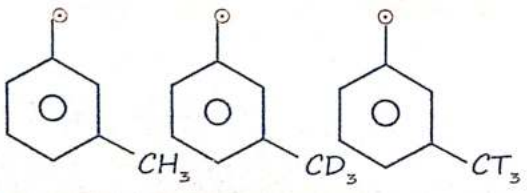
14. $CH_3-\overset{\ominus}{C}OO-CH_2=CH-\overset{\ominus}{C}OO-HC\equiv C-\overset{\ominus}{C}OO$

15. $\overset{\oplus}{N}R_3-CH_2-\overset{\ominus}{O}-F-CH_2-\overset{\ominus}{O}-\overset{\ominus}{O}-O_2N-CH_2-\overset{\ominus}{O}$

16. 

17. 

18. 

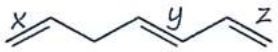
19. 

Q. Draw P_π orbital diagram:

20. $H_2C=CH_2$

21.

22.

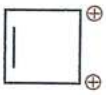
23. 

24.

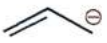
25.

26.

27.

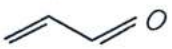
28. 

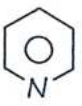
29.

30. 

31.

32.

33. 

34. 

35.

36.

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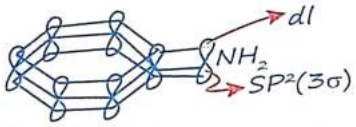
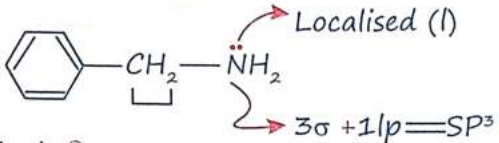
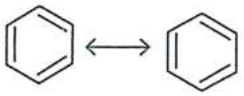
Hybridisation & Delocalisation

(P) 

(Q) 

- If $\pi e^{\ominus}$ undergoes resonance then it is known as delocalised $e^{\ominus}$.
- If lone pair/$e^{\ominus}$ pair undergoes in Reso then that $e^{\ominus}$ pain is not considered in hybridization.

Resonating Structure



- C—C ⇒ 1.54A°, C = C ⇒ 1.33A°
- Kekule structure
- Hypothetical
- Resonating structure

- C = C ⇒ 1.39A°
- Benzene
- Real
- Resonating hybrid

Definitions

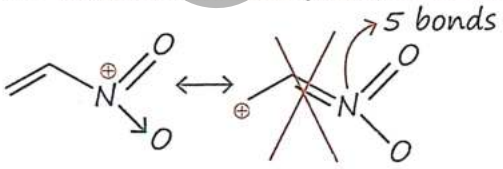
When properties of a compound can't be explained by a single structure then various structures are drawn. These structures are known as resonating structures or canonical structures or contributing structures while real compound is known as resonating hybrid.

Properties of resonance

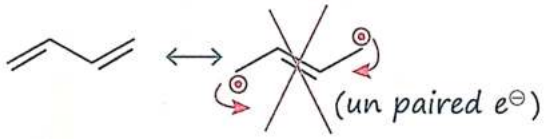
- It is delocalisation of pi electrons.
- It is a permanent effect and strongest effect.
- It is intramolecular phenomenon.
- Generally most stable R.S. contribute most.
- It is a distance independent effect.
- Resonating structures are hypothetical and hybrid is real.

Drawing of R.S.

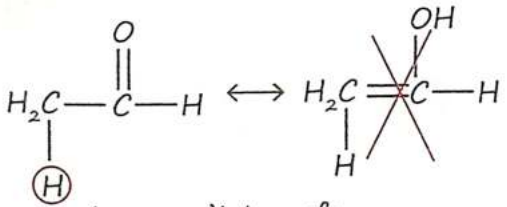
1. All R.S. must be bonafied (valid) structure.



2. All R.S. Must have same unpaired electrons

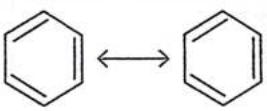


3. Nucleus of an atom must not be changed



atom can't transfer

4. All R.S. must be shown with double headed arrow



5. In case of non polar bond + and – is not valid.

$H_2C{=}CH_2 \longleftrightarrow H_2\overset{\oplus}{C}-\overset{\ominus}{C}H_2$

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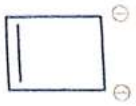
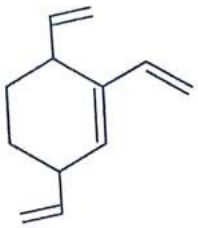
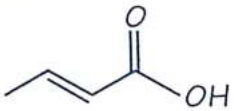
Practice Questions

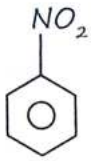
➢ Hybridisation & delocalisation

37.
(Pyran)

38. 

39. 

40. 

41. 

42.

43. $\overset{\ominus}{C}Cl_3$

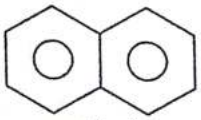
44. $H_2C = C = CH_2$

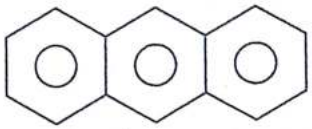
45. CO_2

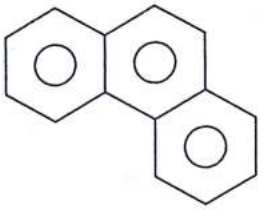
46. $H_2C = C = C = CH_2$

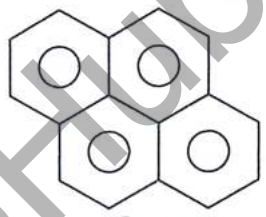
➢ R.S. of Non Polar Compounds

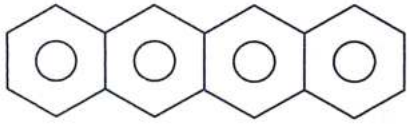
47.
Benzene

48. 
napthalene

49. 
Anthracene

50. 
Phenantheve

51. 
Pyrene

52. 

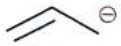
53. Ph_3CH

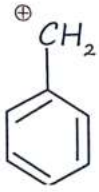
54. Ph_4C

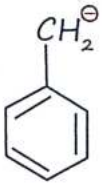
55. $PhCH_2ph$

56. Ph — Ph

57.

58. 

59. 

60. 

61.

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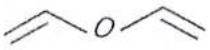
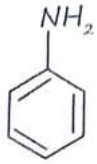
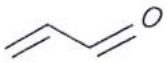
62. 

63. 

64.

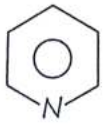
65. 

66. Toluene (Ph — CH_3)

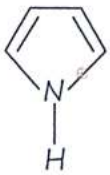
67. Phenol(Ph — OH)

68. Phenoxide (Ph—$\overset{\ominus}{O}$) ion

Pyridine

69. 

Pyrrole

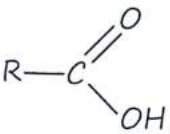
70. 

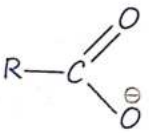
Furan

71.

Thiophene

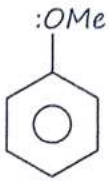
72.

73. 

74. 

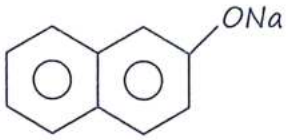
75. Benzaldehyde

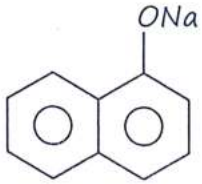
76. Nitrobenzene

77. 

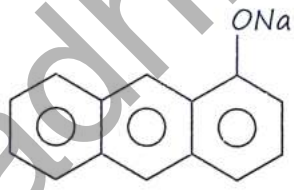
78. Anilinium ion ($Ph—\overset{\oplus}{N}H_3$)

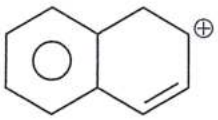
79. Benzonitrile

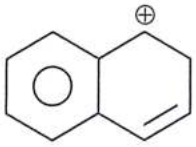
80. 

81. 

82.

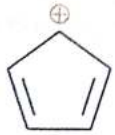
83. 

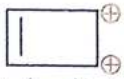
84. 

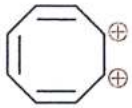
85. 

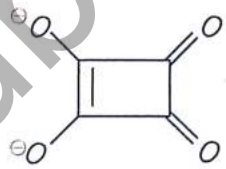
86.
(Only momoion)

87.
(Only momoion)

88. 
(Only momoion)

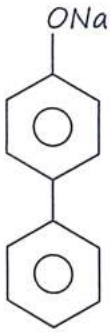
89. 
(Only di-ion)

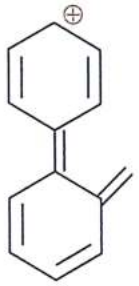
90. 
(Only di-ion)

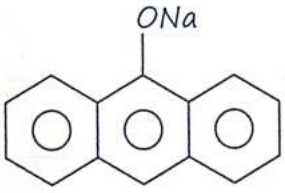
91. 
(Only di-ion)

92. $Ph_3C^{\oplus}$

93. Ph_2CH

94. 

95. 

96. 

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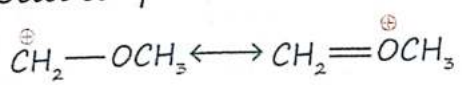
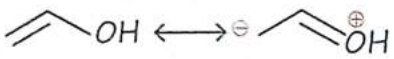
Stability of Resonating Structures

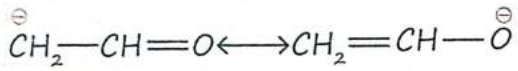
1. Octet completion

eg: 

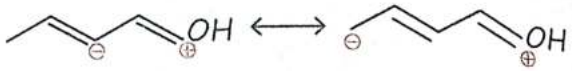
2. Neutral > Ionic

eg: 

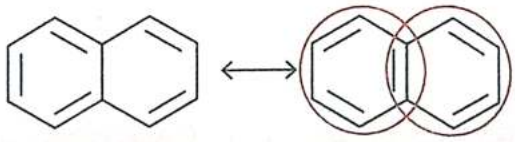
3. EN (-ve more EN) [Inductive effect]

eg: 

4. Opp charges attraction (distance)

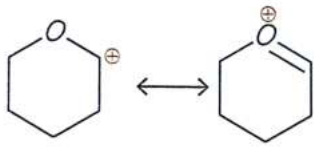
eg: 

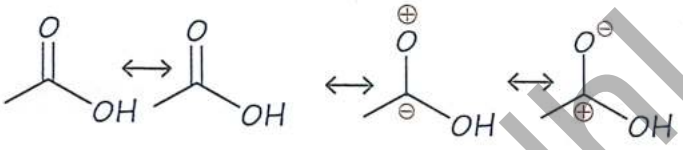
5. Fries rule (More Benzene)

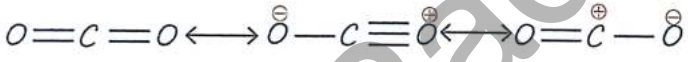
eg: 

Practice Questions

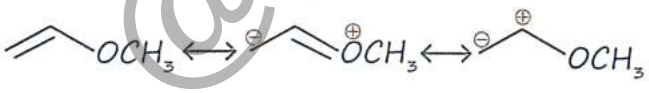
Q. Find stability of given R.S.

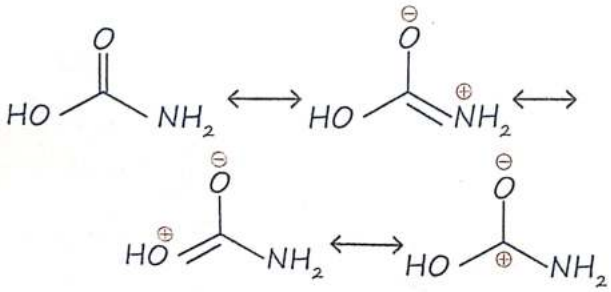
97. 

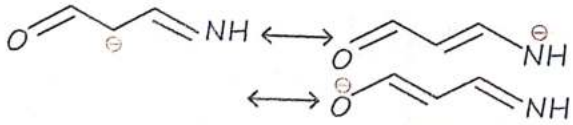
98. 

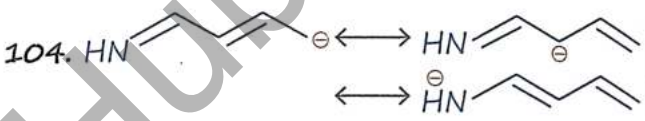
99. 

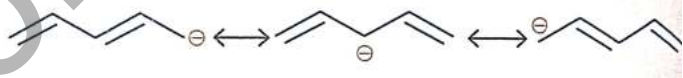
100.

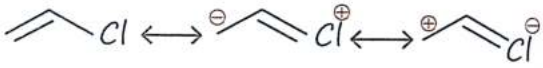
101. 

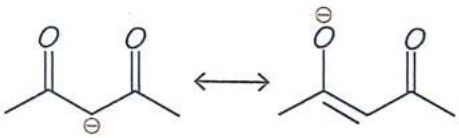
102. 

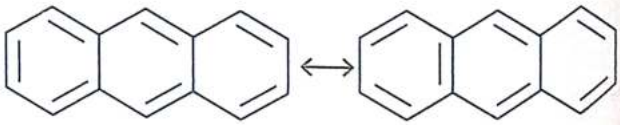
103. 

104. 

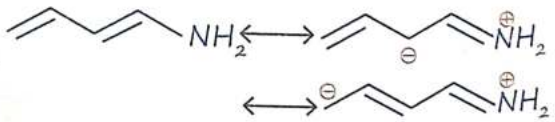
105. 

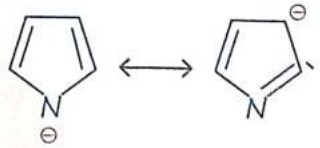
106. 

107. 

108. 

109.

110. 

111. 

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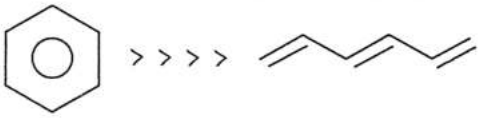
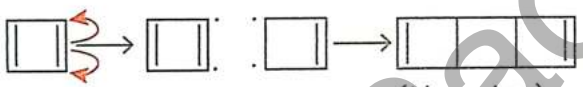
Aromatic Compounds

- Compounds which produces Diamagnetic ring current in presence of external magnetic field are Known as "Aromatic Compounds."
- Generally obtained from fractional distillation of Coal-taar.
- They are more stable than analogous compounds.
- Example 

Anti-Aromatic Compounds

- Compounds which produces paramagnetic ring current in presence of external magnetic field are Known as "Anti-Aromatic Compounds."
- They are less stable than analogous compound.

- They are highly unstable at room temperature so they doesn't exist and remains as Dimer form.



Huckel's Rule

Compound	Cyclic Current	$\pi e^{\ominus}$
Aromatic	✓	$(4n + 2)\pi e^{\ominus}$ $n = 0, 1, 2 \ldots$ $\pi e^{\ominus} = 2, 6, 10 \ldots$
Anti Aromatic	✓	$(4n)\pi e^{\ominus}$ $n = 0, 1, 2, 3 \ldots$ $\pi e^{\ominus} = 4, 8, 12 \ldots$
Non Aromatic	✗	✗

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Rules for aromaticity

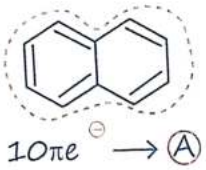
Find loop ⇒ ring current ⇒ apply Huckel's Rule

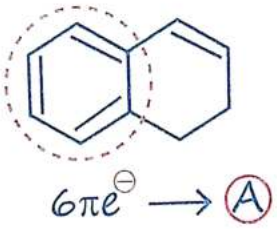
eg: (a)

(b)

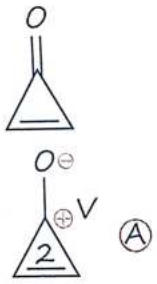
(c)

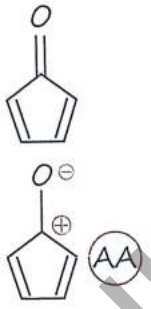
In case of fused rings ⇒ (i) ring current on largest loop ⇒ apply Huckel's Rule.
(ii) If ring current not present on largest loop then go for other small loops.

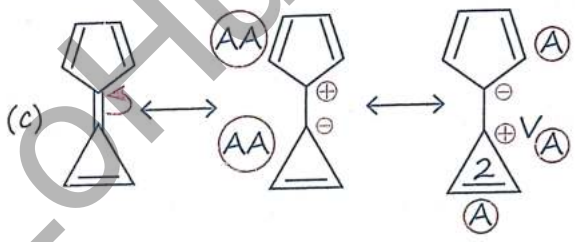
eg: (a) 

(b) 

In case of aromaticity due to ions, compound is known as quasi-aromatic.

eg: (a) 

(b) 

(c) 

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Practice Questions

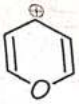
Q. Find Aro, Non or Anti

112.

113.

114.

115.

116. 

117.

118.

119.

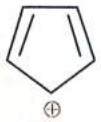
120.

121.

122.

123.

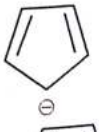
124.

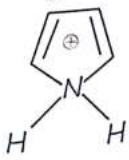
125. 

126.

127.

128.

129. 

130. 

131.

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132.

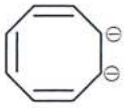
133.

134. Phenol

135.

136.

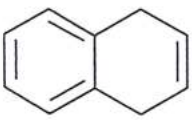
137.

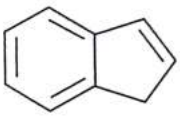
138. 

139. Ph—Ph

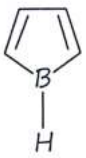
140. Aniline

141.

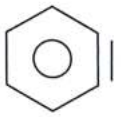
142. 

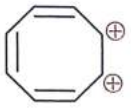
143. 

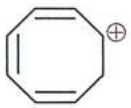
144.

145. 

146.

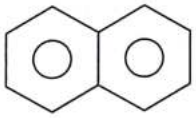
147. 

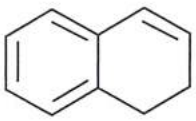
148. 

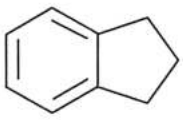
149. 

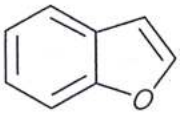
150.

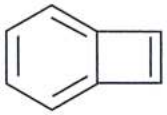
151.

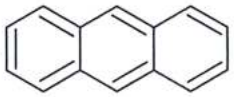
152. 

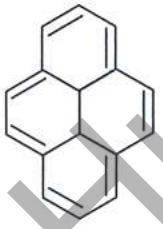
153. 

154. 

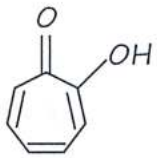
155. 

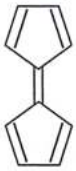
156. 

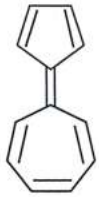
157. 

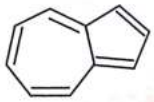
158. 

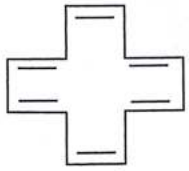
159.

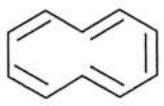
160. 

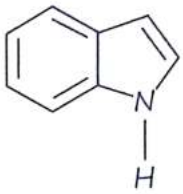
161. 

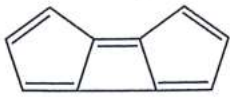
162. 

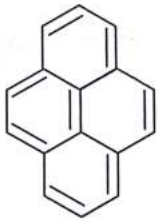
163. 

164. 

165. 

166. 

167. 

168. 

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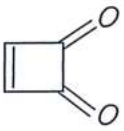
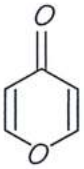
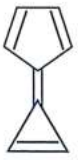
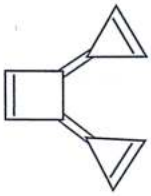
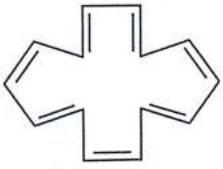
170. 

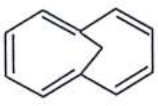
171. 

172. 

173. 

174.

175. 

176. 

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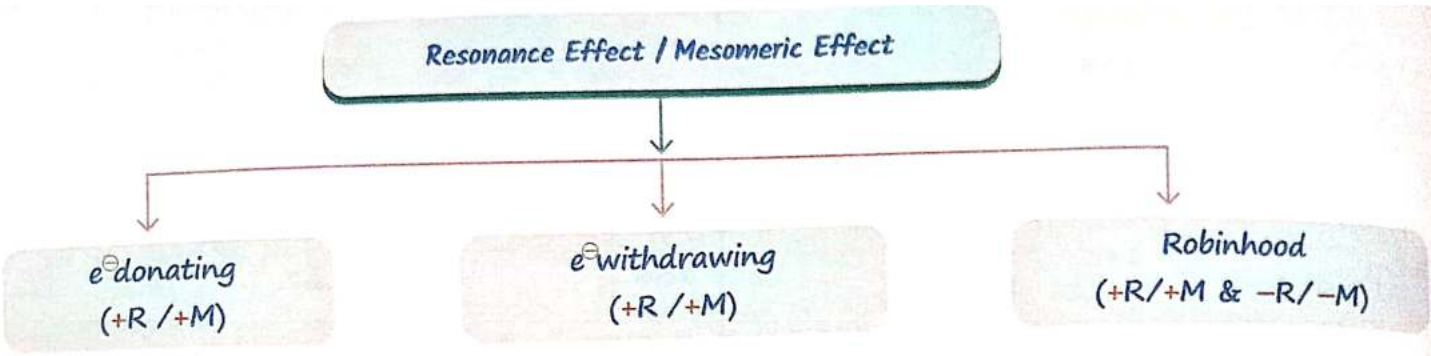
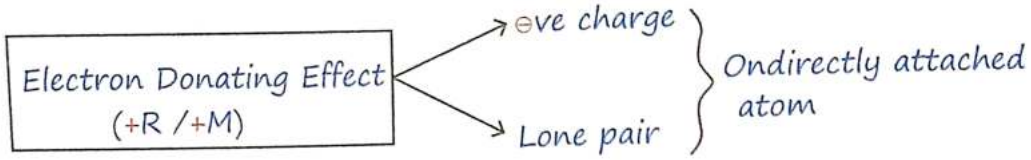
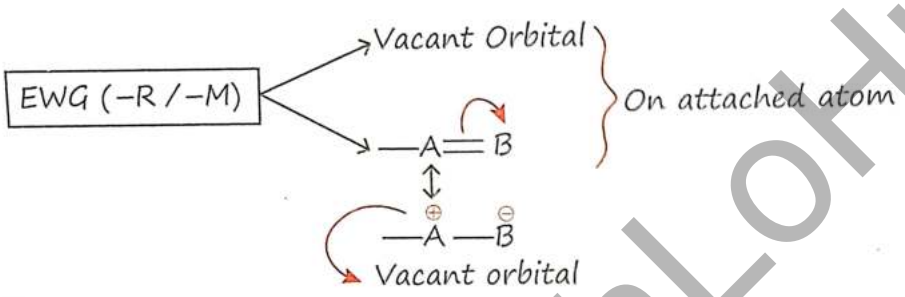


Electron Donating Effect

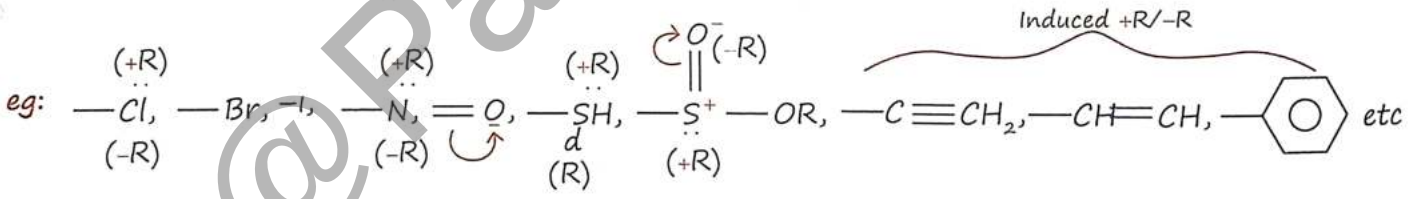


Electron Withdrawing Effect

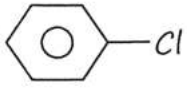


Robinhood Groups

- Atom/groups which shows both +R and −R effect.
- Net effect with neutral system is (−I).



- Find net effect of halogen?

(a) $-\overset{\ominus}{C}H-Cl$ (b) $\overset{\oplus}{C}H_2-Cl$ (c) $CH_2=CH-Cl$ (d)  (f)

Important points of resonance effect

- In case of benzene, effect operates only at ortho and para position only.
- It is strongest effect among hyper & inductive.
- Robinhood groups shows net (−I) effect.
- EDG will activate the benzene for reaction while EWG will deactivate.
- $-NO_2$ shows strongest $e^{\ominus}$ withdrawing effect.

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Practice Questions

177. How many of the following are +R / +M groups?

(a) $-NH_2$

(b) $-NMe_2$

(c) $-OH$

(d) $-\overset{\ominus}{C}H_2$

(e) $-CH_3$

(f) $-C\equiv N$

(g) $-CH=O$

(h)

(i) $-\overset{\ominus}{O}$

(j) $-F$

(k) $-\overset{\ominus}{N}H$

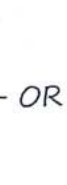
(l) $-\overset{\oplus}{N}H_3$

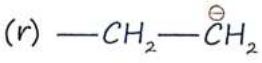
(m) $-Cl$

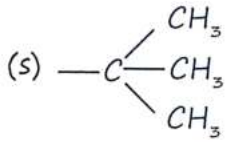
(n) $-N=O$

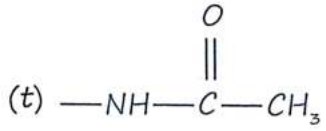
(o) $-SH$

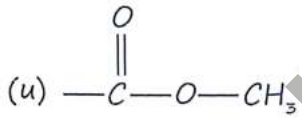
(p)

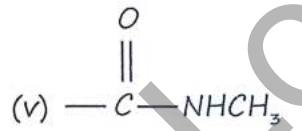
(q) 

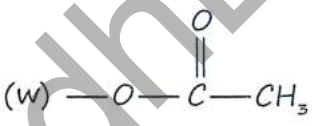
(r) 

(s) 

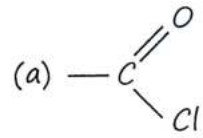
(t) 

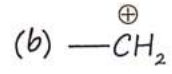
(u) 

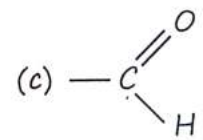
(v) 

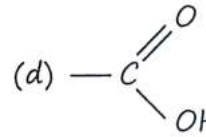
(w) 

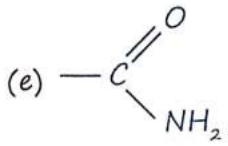
178. How many of the following can be acts as –R / –M group?

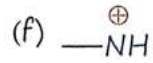
(a) 

(b) 

(c) 

(d) 

(e) 

(f) 

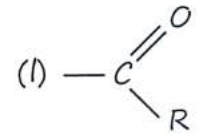
(g) $-F$

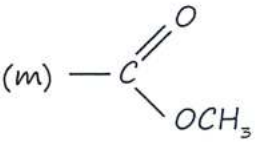
(h) $-CH_3$

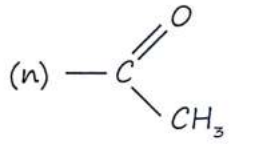
(i) $-Cl$

(j) $-Br$

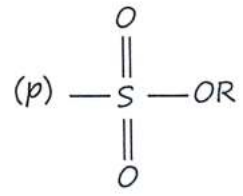
(k) $-CH=NH$

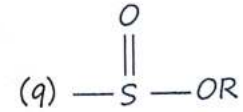
(l) 

(m) 

(n) 

(o) $-SH$

(p) 

(q) 

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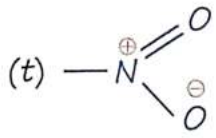
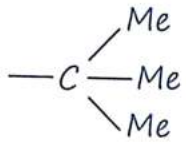
(r) $-N=O$

(s) $-CN$

(t) 

(u) $-\overset{\oplus}{N}R_3$

(v) 

Q. Find permanent Effects

179. $-OH$

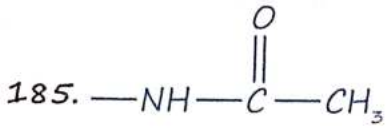
180. $-NH_2$

181. $-\overset{\ominus}{C}H_2$

182. $-CH_2OH$

183. $-NR_2$

184. $-F$

185. 

186. $-Cl$

187. $-CHO$

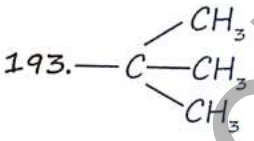
188. $-Br$

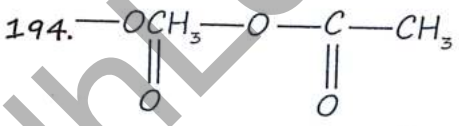
189. $-O^{\ominus}$

190. $-\overset{\ominus}{N}H$

191. $-CH_3$

192. $-CH_2\ CH_3$

193. 

194. 

195. $-COOCH_3$

196. $-CONHCH_3$

197. $-CH=NH$

198. $-COCl$

199. $-CONH_2$

200. $-SH$

201. $-CH=CH_2$

202. $-SO_3H$

203. $-NO_2$

204. $-CN$

205. $-\overset{\oplus}{N}Me_3$

206. $-NO$

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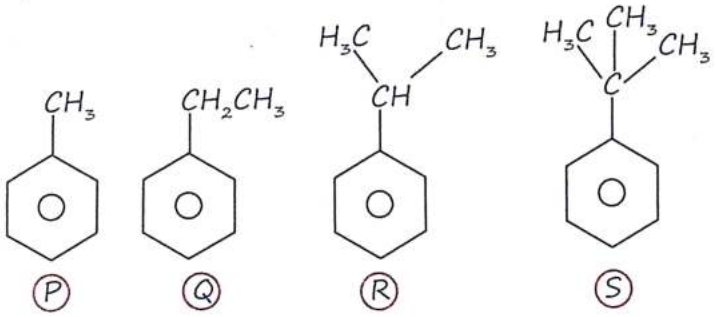
Baker Nathan Effect

- Experiment

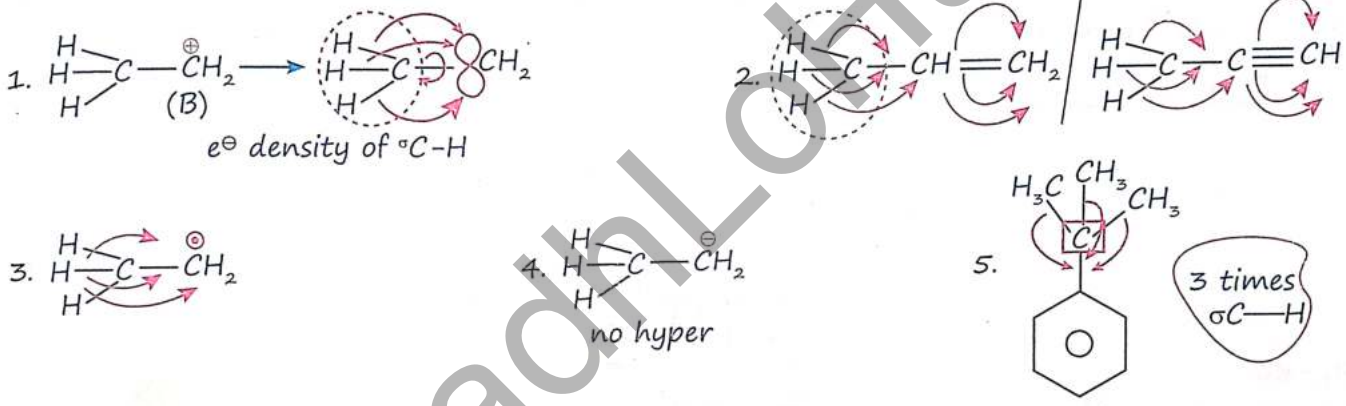


- $E^{\oplus}$ ⟶ S > R > Q > P expected
- $E^{\oplus}$ ⟶ P > Q > R > S Observed
  - P: 3H
  - Q: 2H
  - R: 1H
  - S: OH

Hyperconjugation

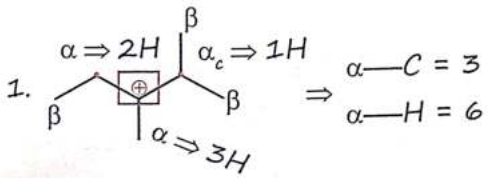
- Delocalisation of σe^{θ} is known as hyperconjugation. (+, B, ene, yne)

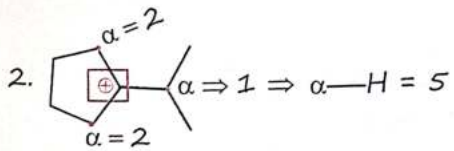
eg:



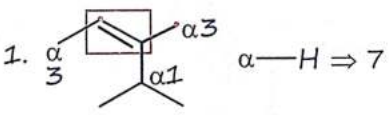
Finding of α-H

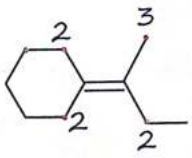
- Cation directly attached C to ⊕
  - B





- Alkene directly attached C to alkene
  Alkyne





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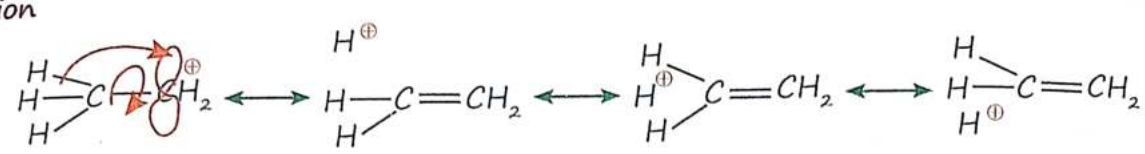
Hyper conjugating structure

➢ Carbocation

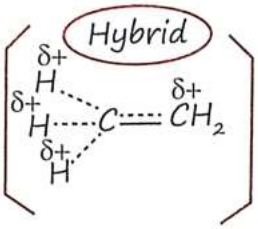


Orbital Overlapping

σ (filled) – p(vacant)

C—H_{sp^3}

Hybrid

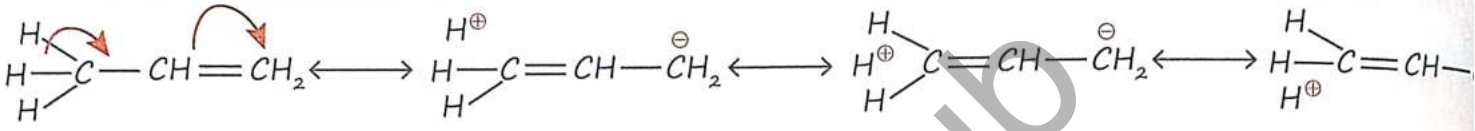


Calculation

Total H.S. involving C—H ⇒ 3

Total H.S. ⇒ 4

NO Bond Resonance

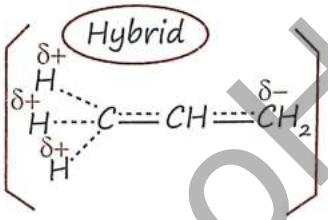


Orbital Overlapping

σ (filled)–π* (vacant)

C—H_{sp^3}

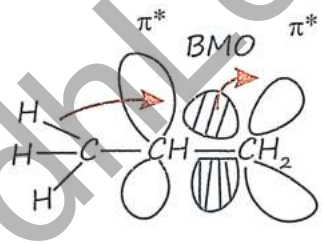
Hybrid



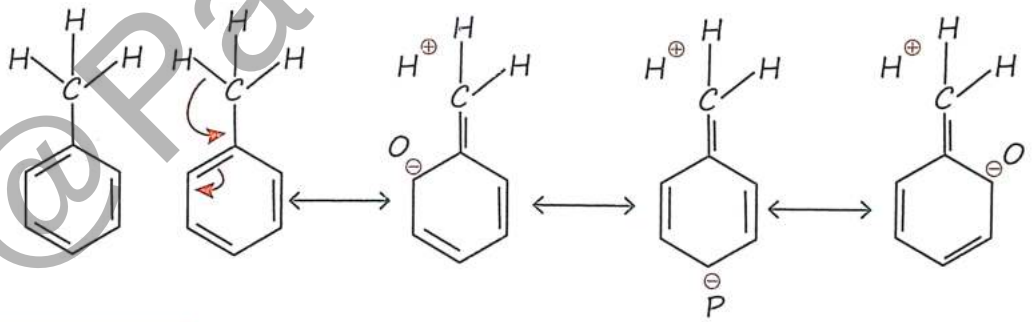
Calculation

Total H.S. involving C—H ⇒ 3

Total H.S. ⇒ 4



Benzene



σ (filled)–π* (vacant)

C—H_{sp^3}

Total H.S. involving C—H ⇒ 9

Total H.S. ⇒ 11

➢ Hyperconjugation effect Operates mainly at o/p positions.
(+, ., B, =, ≡)

→ Total H.S involving C—H bond ⇒ no. of α-H

Total H.S ⇒ no. of α-H + 1

(Benzene)

→ Total H.S. involving C—H bond ⇒ 3 × (no. of α-H)

Total H.S. ⇒ 3 × (no. of α-H) + 1

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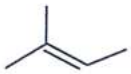
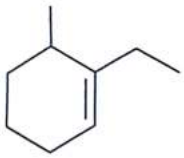
Practice Questions

Find total alpha H present in following species ?

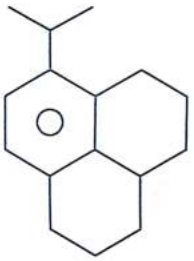
207. 

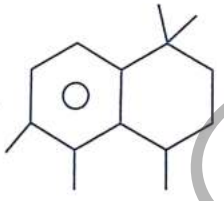
208. $H_3C-C\equiv C-CH_2CH_3$

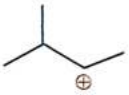
209.

210. 

211.

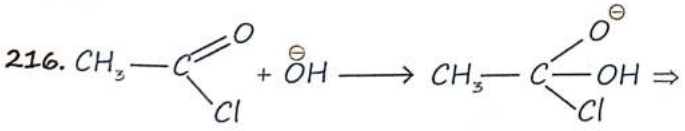
212. 

213. 

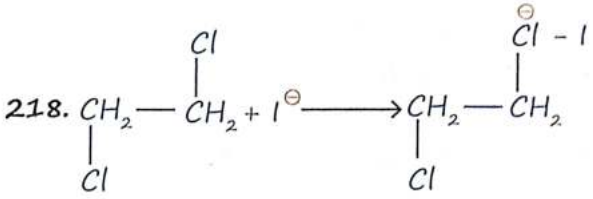
214. 

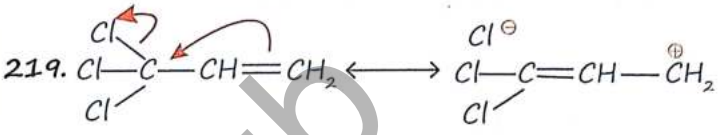
Find vacant orbital in each case?

215. $CH_3^{\oplus} + \overset{\ominus}{O}H \longrightarrow CH_3-OH \Rightarrow$

216. 

217. $CH_3-Cl + \overset{\ominus}{C}N \longrightarrow CH_3-CN + Cl^{\ominus}$

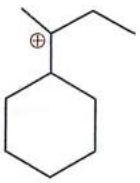
218. 

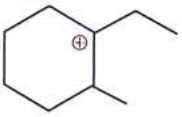
219. 

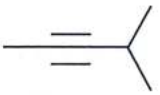
Find total H.S. Involving C—H bond in following compound?

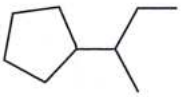
220.

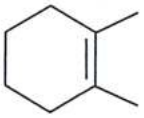
221.

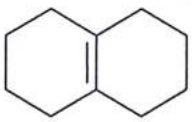
222. 

223. 

224. 

225. 

226. 

227. 

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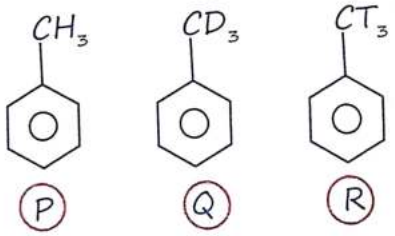
Isotopic effect



- $e^{\ominus}$ density ⇒ P > Q > R (H)

- C—H bond is easy to break so it will show faster hyperconjugation.

Heat of Combustion/Enthalpy of Combustion

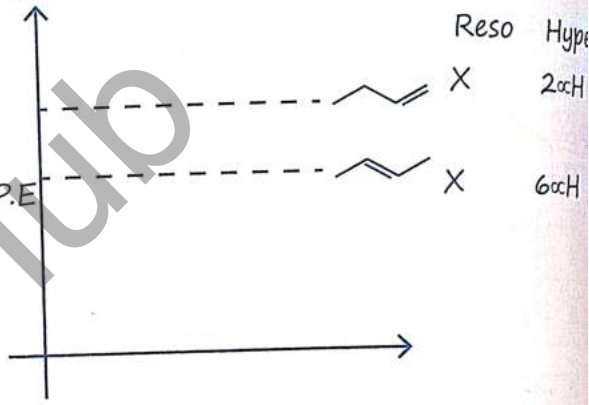
- Enthalpy change due to complete combustion of 1 mole of hydrocarbon.

$$C_xH_y + x + y/4\ O_2 \longrightarrow xCO_2 + y/2\ H_2O \qquad \Delta H_c$$

- HOC $\propto$ no of carbon

$$\propto \frac{1}{\text{Stability}}$$

- HOC per $CH_2 \propto \frac{1}{\text{Stability}}$

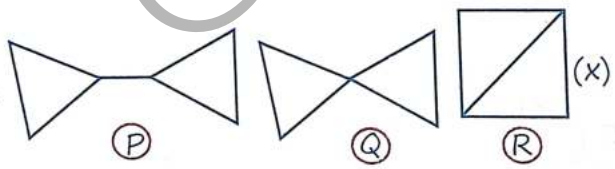


Practice Questions

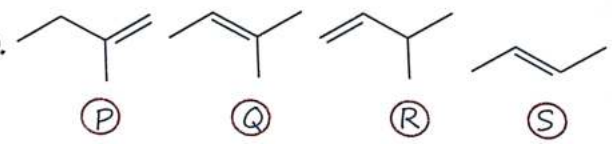
Compare HOC (x) and HOC per C (y) for given compounds.

228.

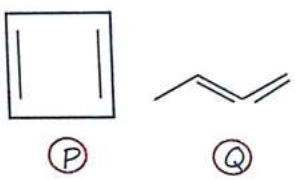
(x, y)

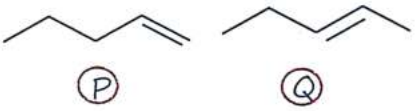
229. 

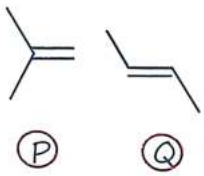
(x)

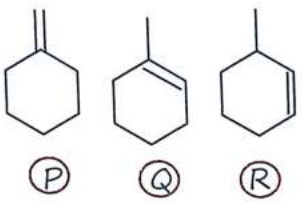
230. 

(x)

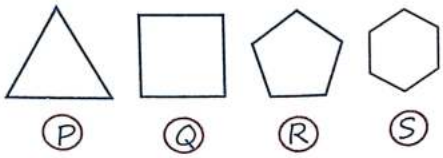
231. 

232. 

233. 

234. 

(x, y)

235. 

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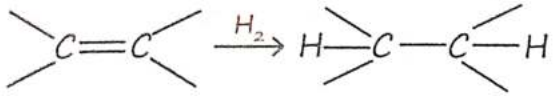
Heat of Hydrogenation (HOH)

- When 1 mole of a compound completely reacts with 1 mole of Hydrogen (H_2) then change in enthalpy is Known as 'Heat of Hydrogenation' [ΔH_H].

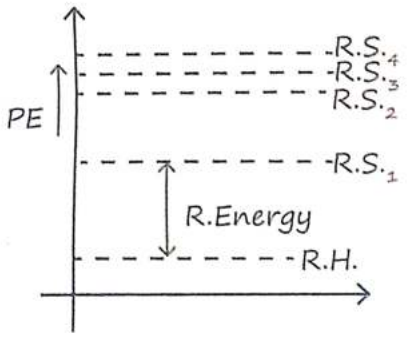
Ex: 

1. HOH
   - No. of π-Bond
   - $\frac{1}{\text{Stability of Compound}}$

2. $\underset{\text{per mole } H_2}{HOH} \longrightarrow \frac{1}{\text{Stability of Compound}}$

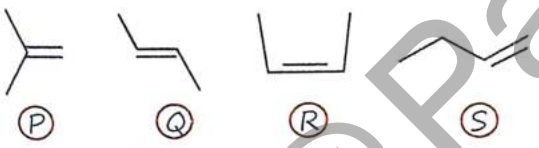
3. Generally heat of Hydrogenation is negative.
4. Catalyst is also added for fast reaction.
5. Sometime Hindrance is also a significant factor.

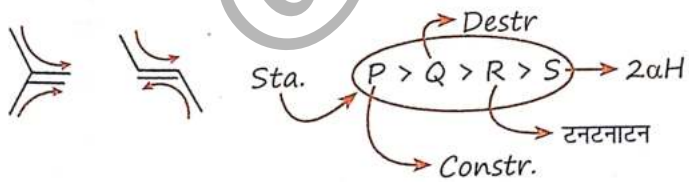
Resonance Stabilization Energy



- Energy difference between most stable R.S & R.H.
- Resonance Energy α Effectiveness of Resonance
- $RE = (\Delta H)_{exp} - (\Delta H)_{obs.}$

Stability of Stability of C_4H_8

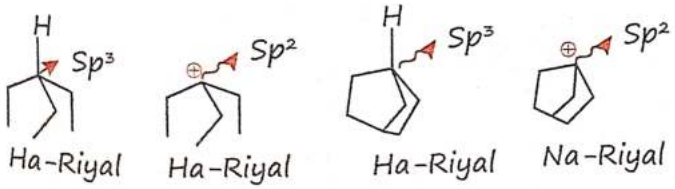




Constructive interference or Extended Hyper
Destructive interference or cross hyper

Bredt's Rule

- When ring ≤ 8, then sp^2 (planarity) is not possible at bridge atom.



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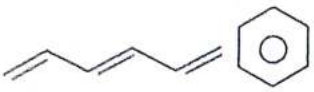
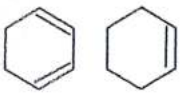
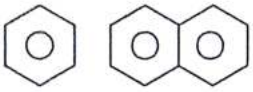
➢ Heat of Hydrogenation (HOH)

236.

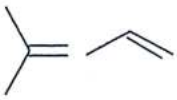
237. 

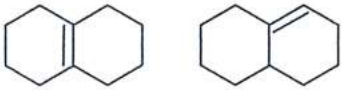
238. 

239. $H_3C—C\equiv CH$ $H_3C—CH=CH_2$

240. 

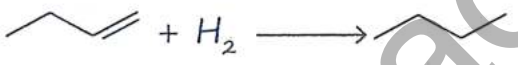
241.

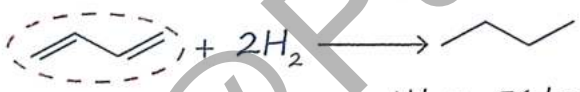
242. $CH_2=CH_2$ 

243. 

➢ Calculation of Enthalpy

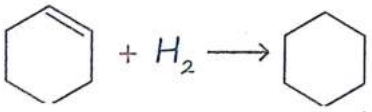
244.

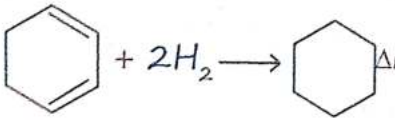
(i)  ΔH_H = -28.6 kcal/mol

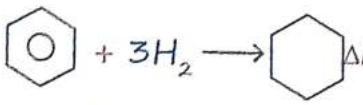
(ii)  ΔH_H = -54 kcal/mol

➢ Find the resonance energy ?

245.

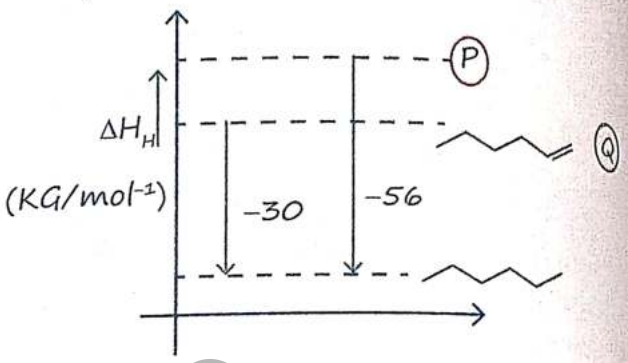
(i)  ΔH_H = -28.6 kcal/mol

(ii)  ΔH_H = -x kcal/mol

(iii)  ΔH_H = (-49.8) kcal/mol

246. Find the resonance stabilization energy of benzene & also find ΔH_H for cyclohexadiene if R.E. is -3 kcal/mol ?

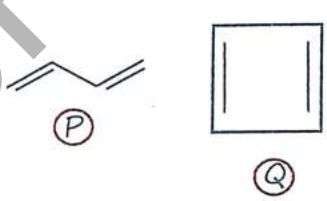
247. If P is formed by addition of 1 pi bond to Q then compound P will be :

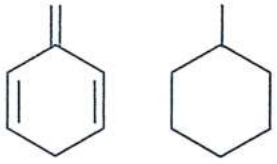


(a) 1,4-Hexadiene (b) 1, 3- Hexadiene

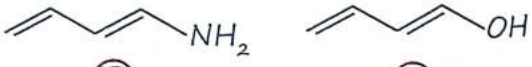
(c) 2,4- Hexadiene (d) 1, 2- Hexadiene

Compare the resonance stabilisation energy for given pair?

248. 

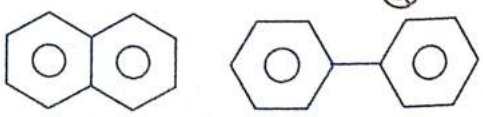
249. 

250.

251. 

252.

253. CH_3COOH $CH_3COO^{\ominus}$

254. 

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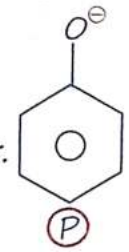
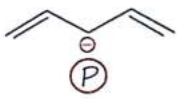
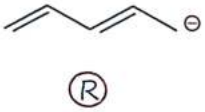
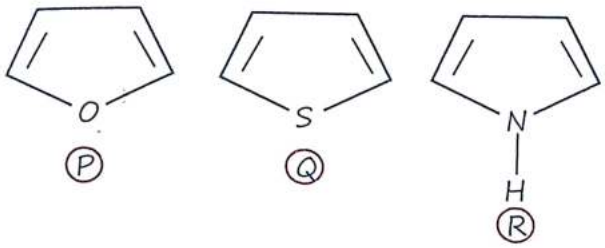
255.  PhCOO⁻

(P) (Q)

256.  

(P) (R)

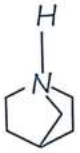
257. 

(P) (Q) (R)

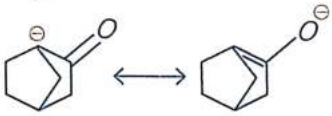
258. Which of the following species or phenomenon are

(a) (b)

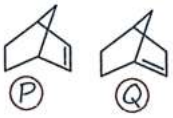
(c) (d)

(e)  (f)

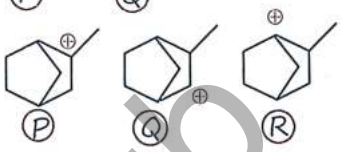
(g) (g)

(i) 

259. Find the Stability of following compounds or species?

(a) 

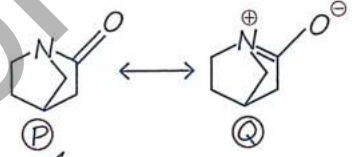
(P) (Q)

(b) 

(P) (Q) (R)

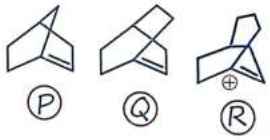
(c)

(P) (Q)

(d) 

(P) (Q)

(e) [No of H.S. involving C—H bond =??]

(f) 

(P) (Q) (R)

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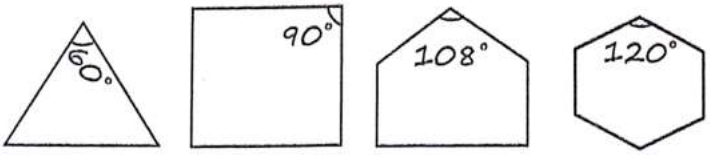
Baeyer's angle strain Theory

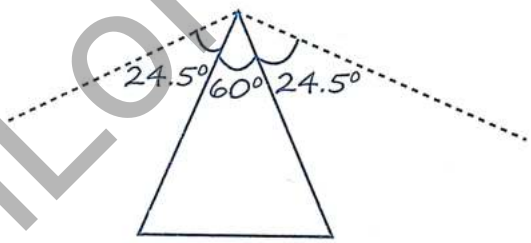


angle strain ⇒ 24.5° 9.5° 0.5° 5.5°

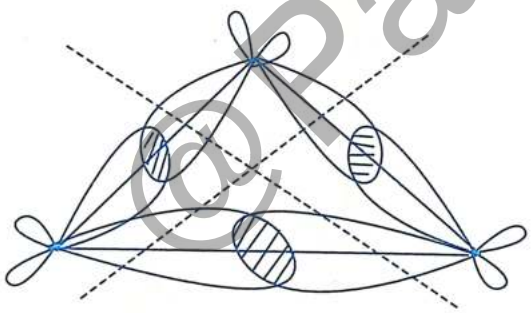
- Stability ⇒ R > S > Q > P
- ~~HOC per CH_2 ⇒ P > Q > S > R~~ (expected)
- HOC per CH_2 ⇒ P > Q > R > S → (**Exp.** Chair, Boat, Twist Boat different conformation)
  (Observed)

किस्सा Cyclopropane का:

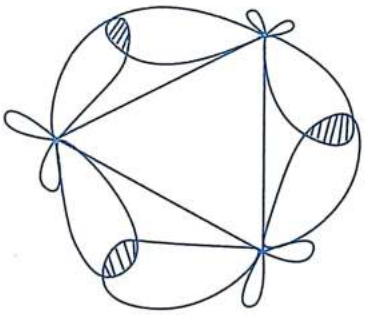
Propane



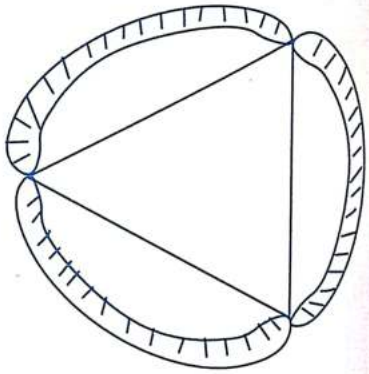
Orbitals



(Expected)



(observed)



(observed)

Observation

(1) Due to high strain in cyclopropane, σ bond behaves like π - bond.

(2) EN of C — H bond increases.

explain: →

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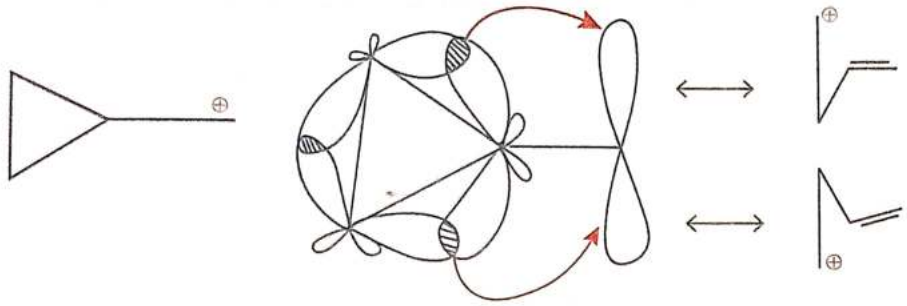
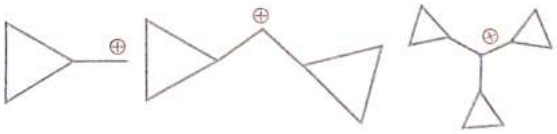
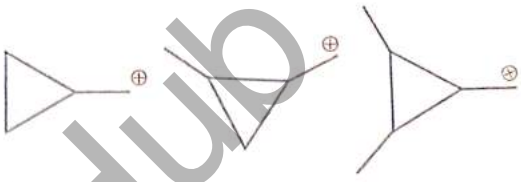
Dancing Resonance/Sigma Resonance

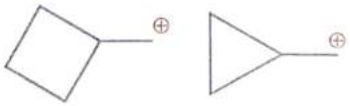


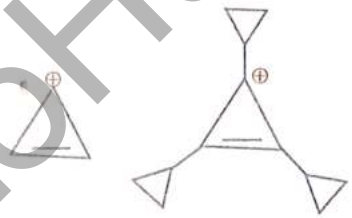
➢ When vacant orbital is present next to cyclopropane then delocalisation is Known as Dancing Resonance.

Find the Stability?

(1) 

(2) 

(3) 

(4) 

Bond length/Bond Order

➢ $C-C$ (P) $\quad C=C$ (Q) $\quad C\equiv C$ (R) $\longrightarrow$ B.L. $\Rightarrow P > Q > R$; B.O. $\Rightarrow R > Q > P$

➢ If % single bond ↑ ⇒ B.L. ↑, B.O. ↓

➢ If % single bond ↓ ⇒ B.L. ↓, B.O. ↑

➢ By Reso & Hyper, single bond develops π-character & double bond develops single character.

Bond Dissociation Energy (BDE)

➢ The amount of energy required to break a bond is same as the amount of energy released when the same bond is formed.

➢ In gaseous state, the energy required for homolytic cleavage of a bond is called Bond Dissociation Energy (BDE) or Bond Strength.

➢ BDE is affected by s-character of the bond and the stability of the radicals formed. Shorter bonds are typically stronger bonds.

Bond Energy ⟶ Gaseous phase ⟶ Homolytic fission ⟶ Free radical

or

BDE

$$B \cdot D \cdot E \propto \frac{1}{\text{Stability of free radical}}$$

eg: $C-\overset{\odot}{C} > C=\overset{\odot}{C} > C\equiv\overset{\odot}{C}$

Stability ↑ EN ↑

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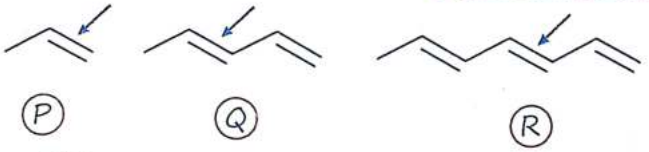
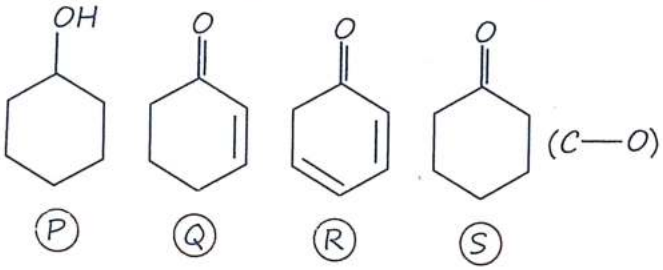
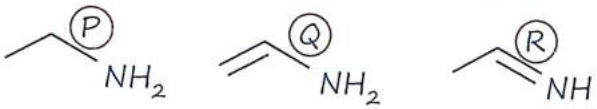
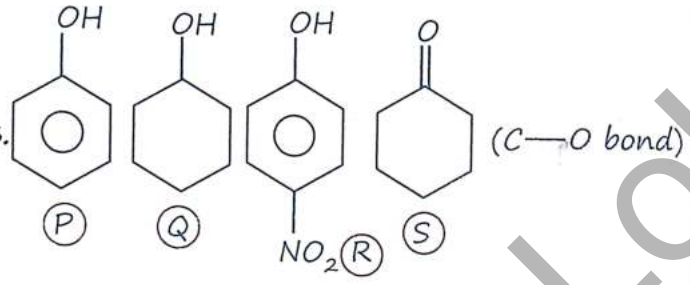
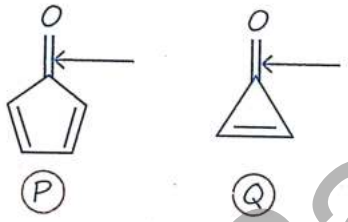
Practice Questions

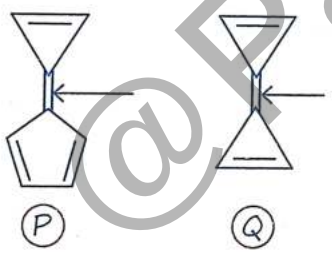
260. 

261. 

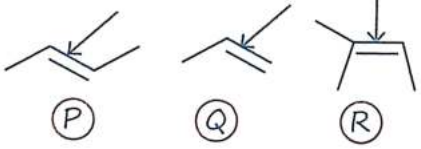
262. 

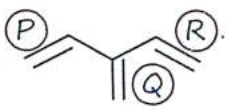
263. 

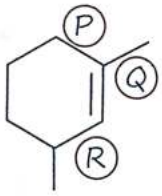
264. 

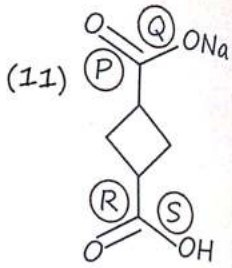
265. 

266. $Ph-Cl$ CH_3Cl $CH_2=CH-Cl$
P Q R

267. 

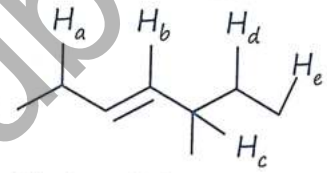
268. 

269. 

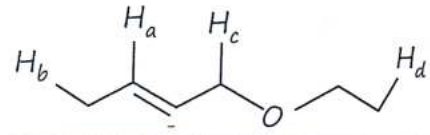
(11) 

270. Find the bond energy of given bonds?

(a) CH_3-CH_2-H
$CH_2=CH-H$
$HC\equiv C-H$

(b) 

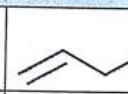
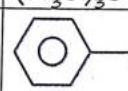
(c) $H_2C=CH-CH=CH_2$
$HC\equiv C-C\equiv CH$
Ph_3C-CPh_3

(d) 

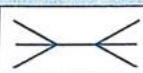
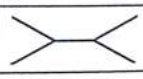
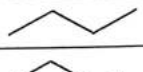
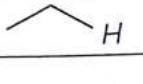
(e)

	Molecule		BDE (Kcal/mol)
(P)	$H-CH(CH_3)_2$	(1)	132
(Q)	$H-CH_2-Ph$	(2)	110
(R)	$H-CH=CH_2$	(3)	95
(S)	$H-C\equiv CH$	(2)	88

(f)

Molecule		BDE (Kcal/mol)	
(P)		(1)	111
(Q)	CH_3-CH_2-H	(2)	92
(R)	$(H_3C)_3C-H$	(3)	89
(S)		(2)	98

(g)

Molecule		BDE (Kcal/mol)	
(P)		(1)	98
(Q)		(2)	88
(R)		(3)	85.5
(S)		(2)	78.5

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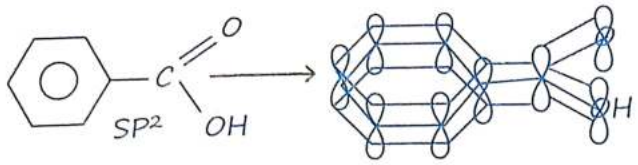
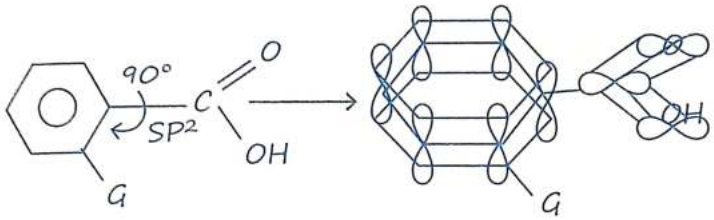
Ortho Effect ⇒ Steric Inhibition of Resonance (SIR effect)

(a)



(b)

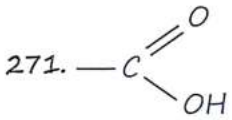


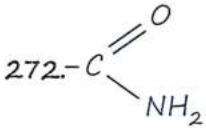
- Due to steric repulsion by group (G) at ortho position, $-Sp^2$ group becomes non planar and resonance will inhibit. This Inhibition of resonance is known as S.I.R. effect.
- **S.I.R. applied by:** All bulky groups except

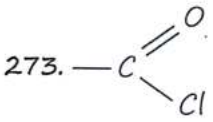
$$\underbrace{-OH, -OR, -NH_2, -H, -F,}\ \text{etc.}$$

Practice Questions

- SIR applied on: [Sp^2 group]

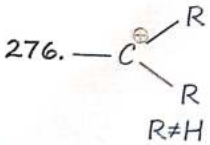
271. 

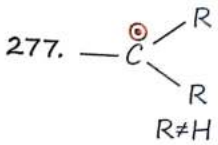
272. 

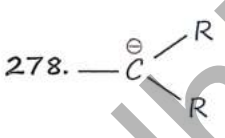
273. 

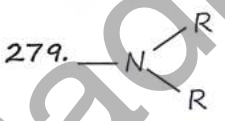
274. $-NO_2$

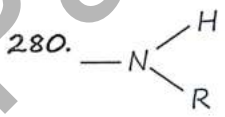
275. $-SO_3H$

276. 

277. 

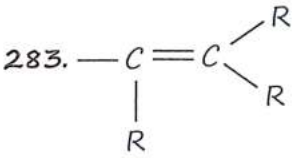
278. 

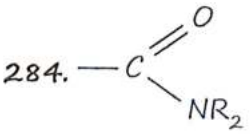
279. 

280. 

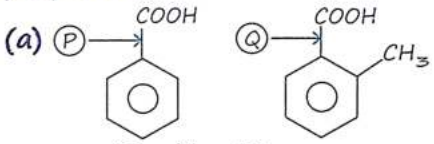
281. $-NH_2$

282. $-C \equiv N$

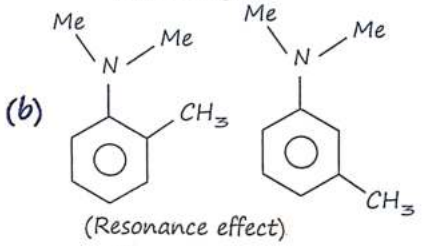
283. 

284. 

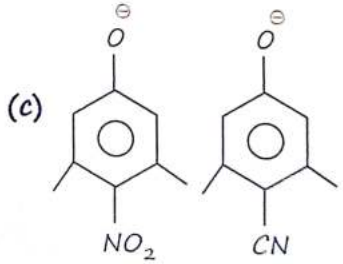
285. **Find the correct order of given properties?**

(a) 

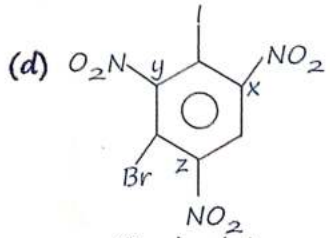
(Bond length)

(b) 

(Resonance effect)

(c) 

(Stability)

(d) 

(Bond order)

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Stability of Intermediates

Carbocation α +R, +H, +I (EDG) (+)	free Radical α +R, −R, +H +I (·)	Carbanion α −R, −I (−)
$\alpha \frac{1}{-R, -I}$	$\alpha \frac{1}{-I}$	$\alpha \frac{1}{+R, +H, +I}$

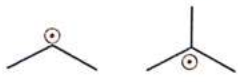
Practice Questions

286. $\overset{\oplus}{C}H_3$ $\overset{\oplus}{C}D_3$ $\overset{\oplus}{C}T_3$

287. $\overset{\odot}{C}H_3$ $\overset{\odot}{C}D_3$ $\overset{\odot}{C}T_3$

288. $\overset{\ominus}{C}H_3$ $\overset{\ominus}{C}D_3$ $\overset{\ominus}{C}T_3$

289. $\overset{\oplus}{C}H_3$

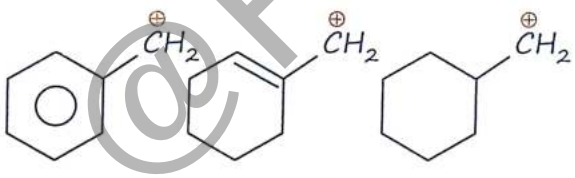
290. $\overset{\odot}{C}H_3$ 

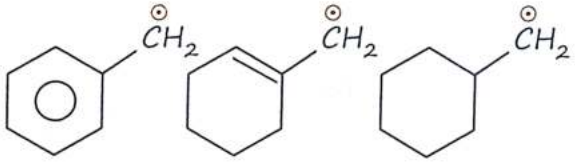
291. $\overset{\ominus}{C}H_3$

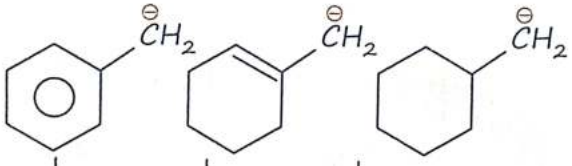
292. $CH_3-\overset{\oplus}{C}H_2$ $H_2C=\overset{\oplus}{C}H$ $HC\equiv\overset{\oplus}{C}$

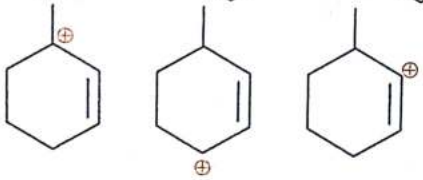
293. $CH_3-\overset{\odot}{C}H_2$ $H_2C=\overset{\odot}{C}H$ $HC\equiv\overset{\odot}{C}$

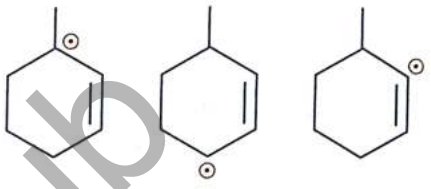
294. $CH_3-\overset{\ominus}{C}H_2$ $H_2C=\overset{\ominus}{C}H$ $HC\equiv\overset{\ominus}{C}$

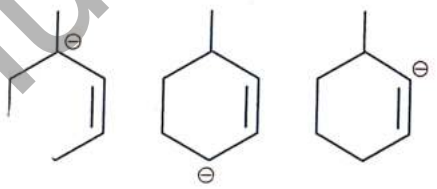
295. 

296. 

297. 

298. 

299. 

300. 

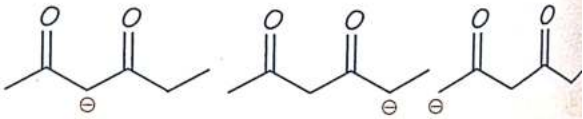
301.

302.

303. $F^{\oplus}$ $Cl^{\oplus}$ $Br^{\oplus}$ $I^{\oplus}$

304. $F^{\odot}$ $Cl^{\odot}$ $Br^{\odot}$ $I^{\odot}$

305. $F^{\ominus}$ $Cl^{\ominus}$ $Br^{\ominus}$ $I^{\ominus}$

306. 

307. $\overset{\oplus}{C}F_3$ $\overset{\oplus}{C}Cl_3$

308. $\overset{\ominus}{C}F_3$ $\overset{\ominus}{C}Cl_3$

309.

310.

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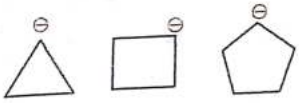
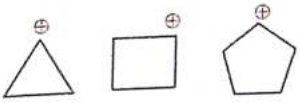
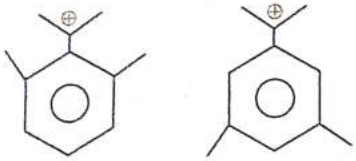
311.



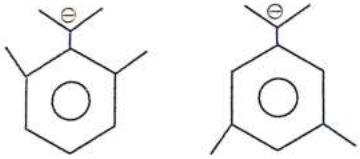
312.



313.

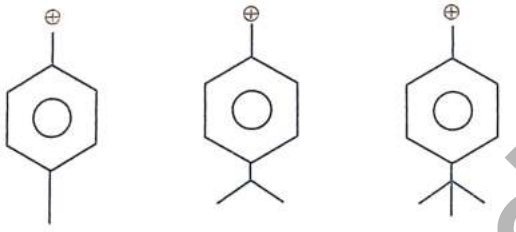


314.

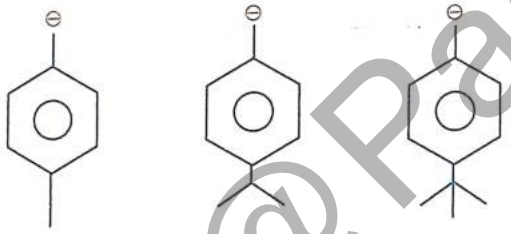


315.

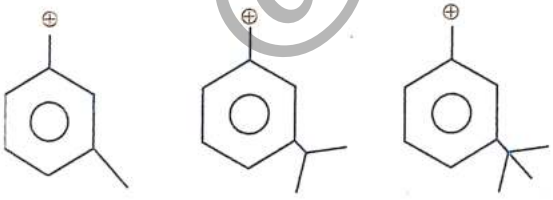
316.



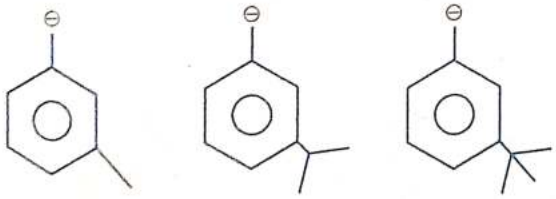
317.



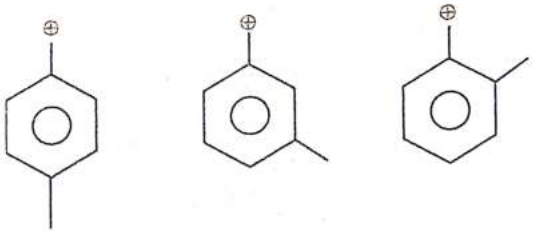
318.



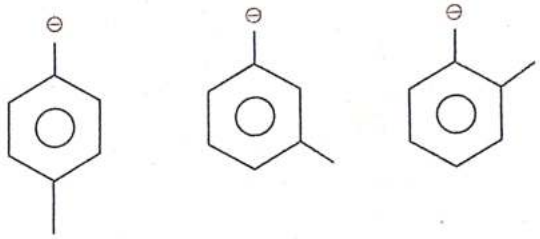
319.



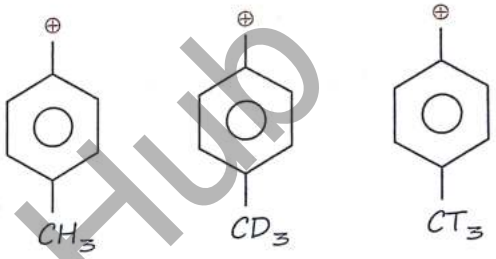
320.



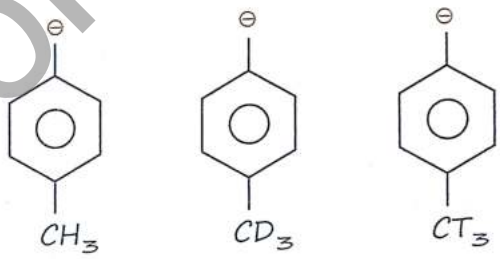
321.



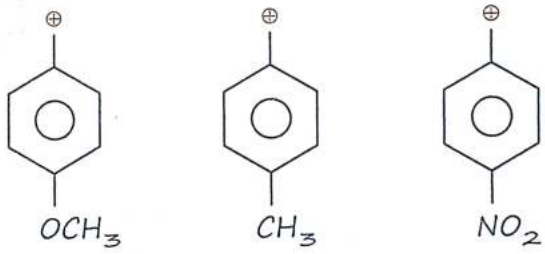
322.



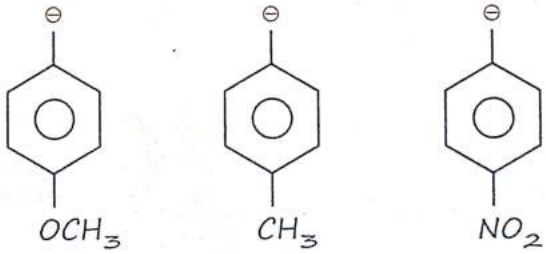
323.



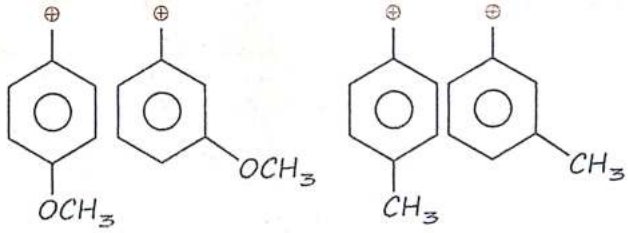
324.



325.



326.



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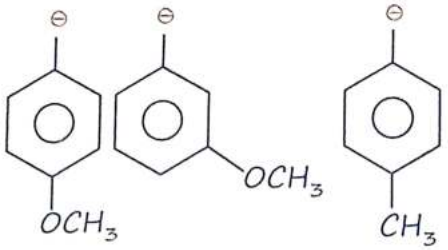
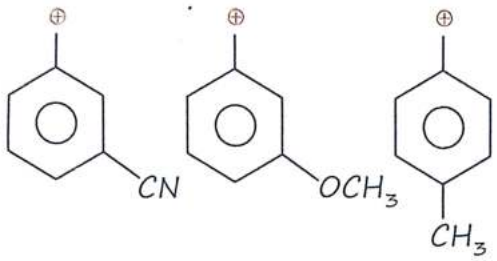
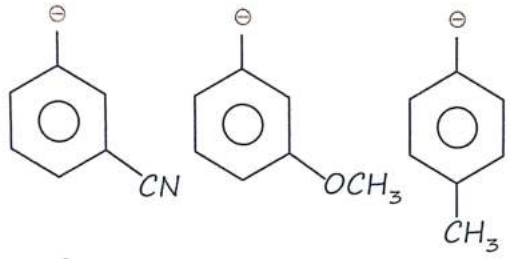
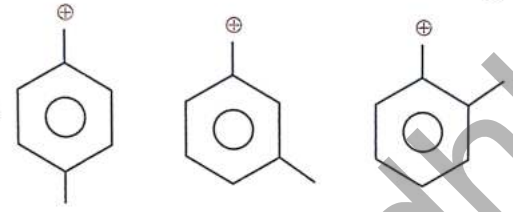
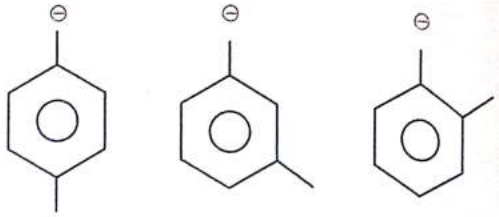
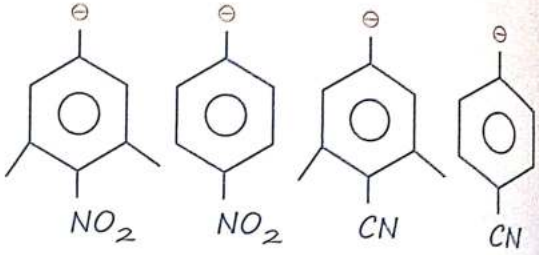
327. 

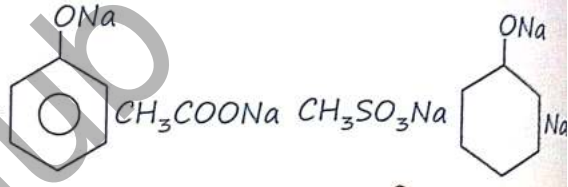
328. 

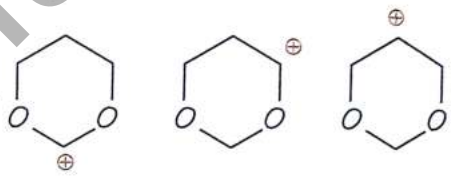
329. 

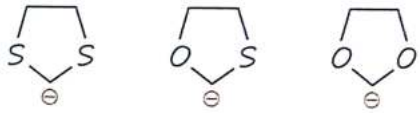
330. 

331. 

332. 

333. 

334. 

335. 

Acidic Strength

Proton Donating tendency

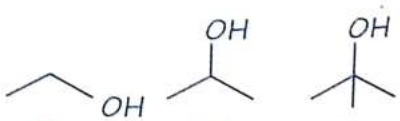
- $HA \underset{}{\overset{K}{\rightleftharpoons}} H^{\oplus} + A^{\ominus}$
  (acid) (Conjugate base)
- $K_a = \frac{[H^{\oplus}][A^{\ominus}]}{[H-A]}$
- $pK_a = -\log(Ka)$

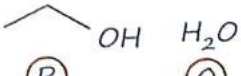
$$A.S. \propto K_a \propto \frac{1}{pK_a}$$
$$\propto \text{Stability of C.B.}(A^{\ominus})$$
$$\propto \text{EWG } (-R, -I)$$
$$\propto \frac{1}{+R, +H, +I}$$
$$\propto \text{H-bonding}$$

Practice Questions

336. HF (P) HCl (Q) HBr (R) HI (S)

337. CH_4 (P) NH_3 (Q) H_2O (R) HF (S)

338. 
(P) (Q) (R)

339.  H_2O (Q) C_6H_5OH (R)
(P)

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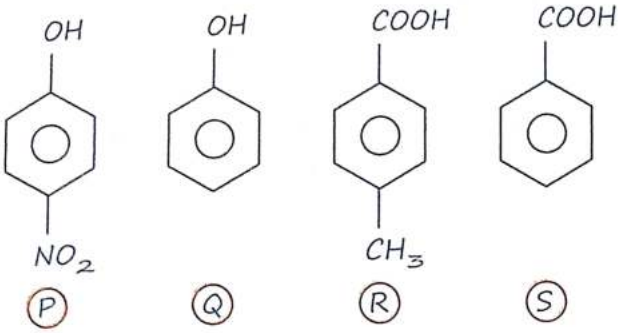
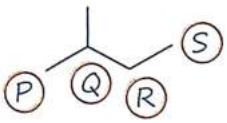
340. $PhOH$ (P) $RCOOH$ (Q) H_2O (R) RSO_3H (S)

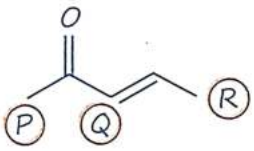
341. 

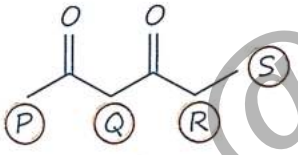
342. $HC \equiv CH$ (P) $CH_2 = CH_2$ (Q) $CH_3 - C \equiv CH$ (R)

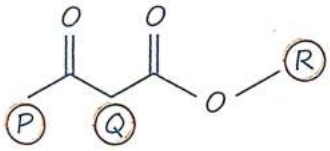
343. $HC \equiv CH$ (P) NH_3 (Q) H_2O (R) HCl (S)

344. 

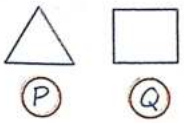
345. $R - COOH$ (P) $R - CONH_2$ (Q)

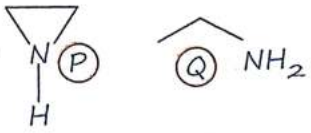
346. 

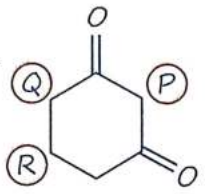
347. 

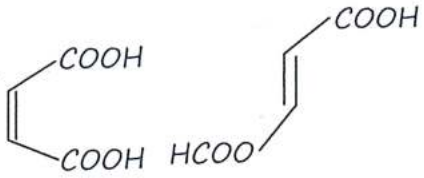
348. 

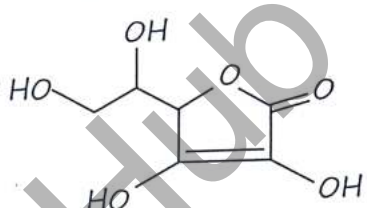
349. CHF_3 (P) $CHCl_3$ (Q)

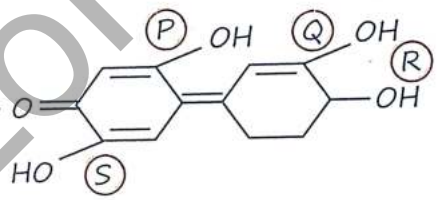
350. 

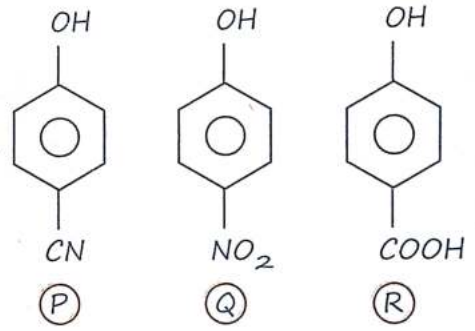
351. 

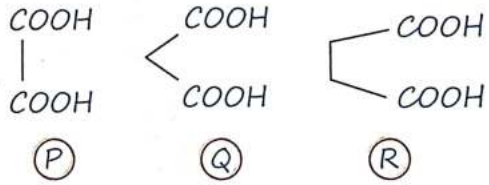
352. 

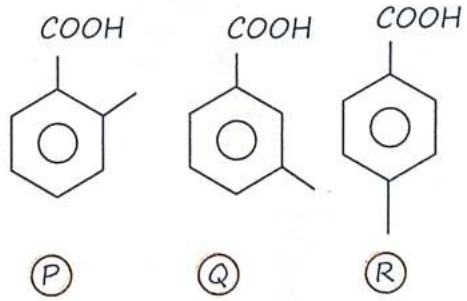
353. 

354. 

355. 

356. 

357. 

358. 

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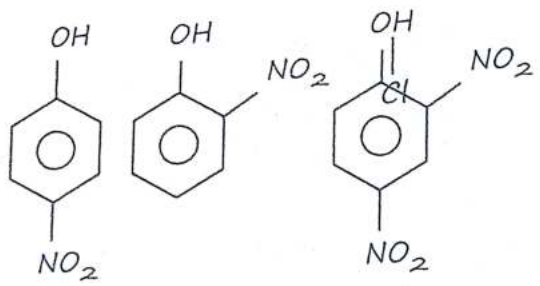
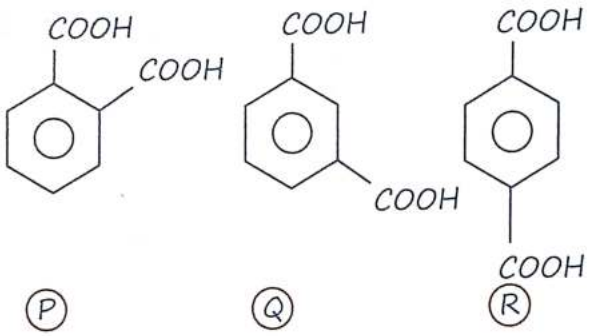
359.



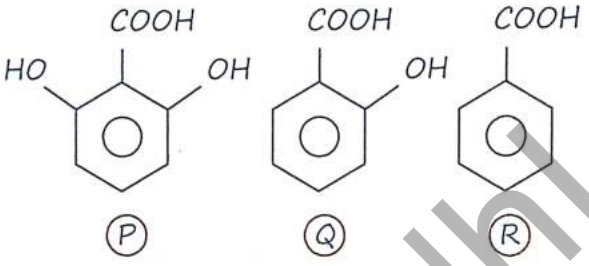
(P) (Q) (R)

360.



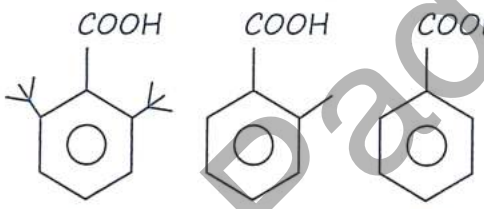
(P) (Q) (R)

361.



(P) (Q) (R)

362.

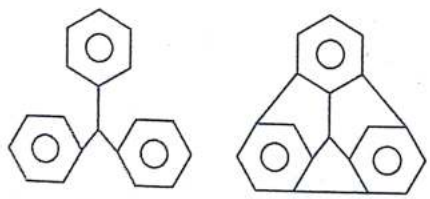


(P) (Q) (R)

363.

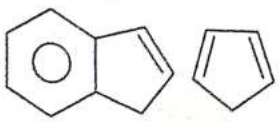
(P) (Q) (R)

364.



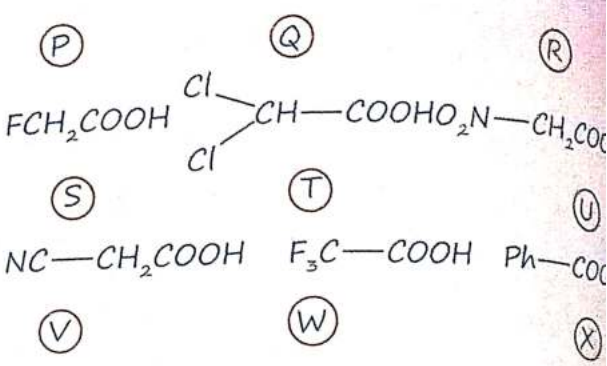
(P) (Q)

365.

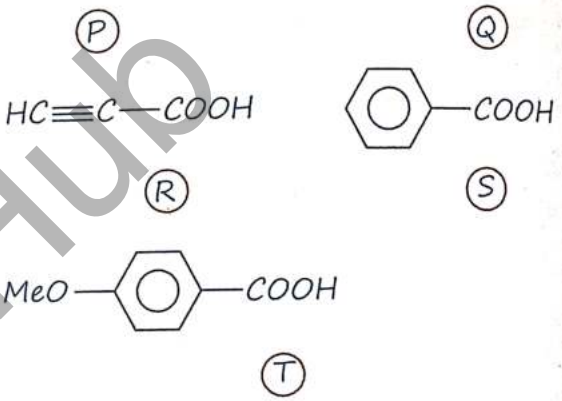


(P) (Q)

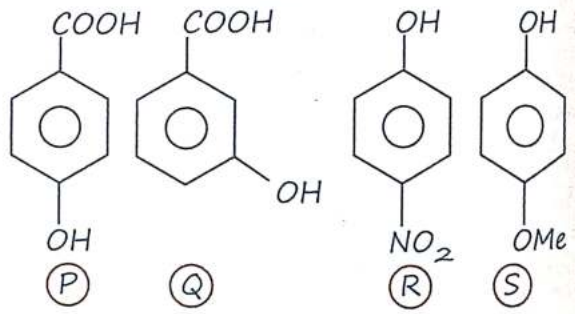
366. HCOOH CH_3COOH $ClH_2C-COOH$



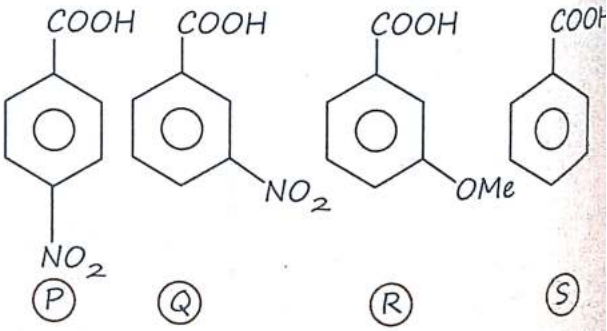
367. H_3CCH_2COOH $CH_2=CHCOOH$



368.



369.



370.

	Acid		pKa
(A)	o-Cresol	(P)	4.19
(B)	Phenol	(Q)	10.2
(C)	Ethanol	(R)	10.0
(D)	Benzoic Acid	(S)	15.9

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(Basic Strength)

➢ $e^{\ominus}$ donating tendency to any $H^{\oplus}$

1. Arrhenius AB ($H^{\oplus} \rightarrow A\overset{\ominus}{O}H \rightarrow B$)
2. Bronsted-Lowry AB ($H^{\oplus} \rightarrow A\,H^{\oplus} \leftarrow B$)
3. Lewis AB ($e^{\ominus}$ pair $\leftarrow$ A $e^{\ominus}$ pair $\rightarrow$ B)

eg: 1. $H_2O \longrightarrow$ B·A, B·B, L·B
2. $NH_3 \longrightarrow$ B·A, B·B, L·B
3. $\overset{\oplus}{H}\overset{\ominus}{SO_4} \longrightarrow$ B·A, B·B, L·B, A·A
4. NaOH $\longrightarrow$ A·B, B·B, L·B

$B^{\ominus} + H^{\oplus} \underset{}{\overset{K_b}{\rightleftharpoons}} H - B$

(Base) (Conjugate acid)

$$K_b = \frac{[H-B]}{[H^{\oplus}][B^{\ominus}]}$$

$$pK_b = -\log K_b$$

- $K_b \propto \frac{1}{pK_a}$

Basic strength
- $e^{\ominus}$ donating tendency
- I > dI
- $e^{\ominus}$ density ↑
- EDG (+R, +H, + α)
- Solvation effect

Practice Questions

371. $\overset{\ominus}{F}$ (P) $\overset{\ominus}{Cl}$ (Q) $\overset{\ominus}{Br}$ (R) $\overset{\ominus}{I}$ (S)

372. $\overset{\ominus}{CH_3}$ (P) $\overset{\ominus}{NH_2}$ (Q) $\overset{\ominus}{OH}$ (R) $\overset{\ominus}{F}$ (S)

373. H_2O (P) NH_3 (Q) $\overset{\ominus}{NH_2}$ (R)

374. NaOH (P) Na (Q) NaSH (R)

375. $R-OH$ (P) $R-NH_2$ (Q)

376. $R-COO^{\ominus}$ (P) $R-O^{\ominus}$ (Q) $Ph-O^{\ominus}$ (R)

377. CH_3NH_2 (P) $CH_2=NH$ (Q) $HC\equiv N$ (R)

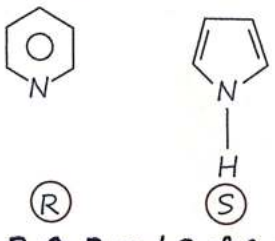
378. $Ph-NH_2$ (P) $Ph-NH-Me$ (Q) $Ph-N(Me)(Me)$ (R)

379. $RO^{\ominus}$ (P) $RS^{\ominus}$ (Q) $Cl^{\ominus}$ (R)

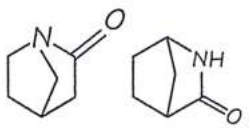
380. $Ph-NH_2$ (P) $Ph-CONH_2$ (Q) $Ph-NH-CH_3$ (R) $Ph-CH_2NH_2$ (S)

381. CH_3NH_2 (P) Me_2NH (Q) Me_3N (R)

382. $Ph-NH_2$ (P) $Ph-CH_2-NH_2$ (Q)

383.  (R) (S)

384. **P, Q, R and S of Q. 382 and 383.**

385.  (P) (Q)

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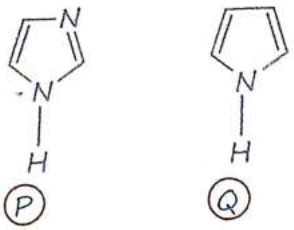
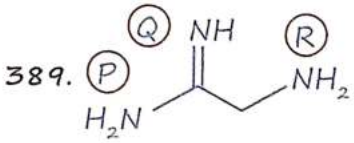
386. 

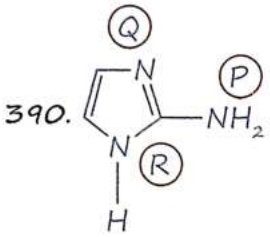
387. Aniline, P–nitroaniline, P–toluidine, Benzylamine

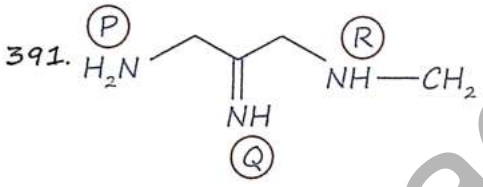
(P) (S)

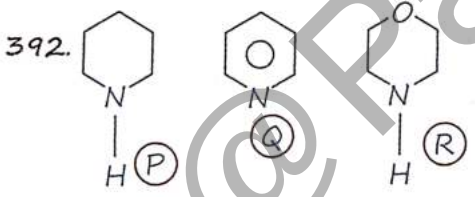
388. $PhCH_2NH_2$ NH_3 CH_3NH_2

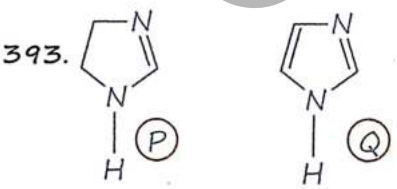
(P) (Q) (S)

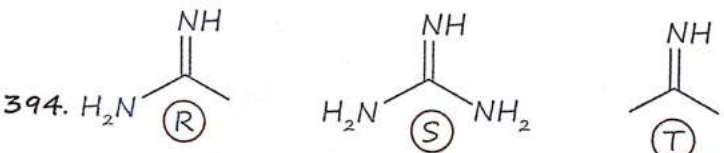
389. 

390. 

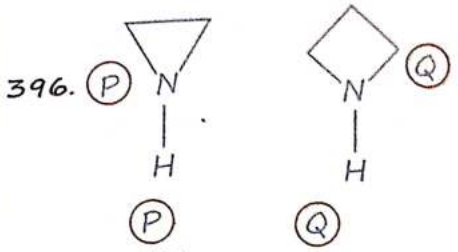
391. 

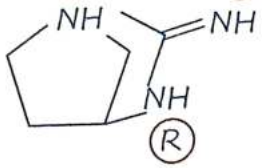
392. 

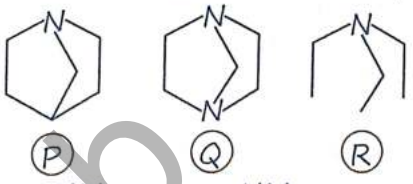
393. 

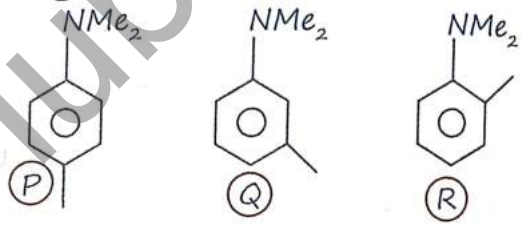
394. 

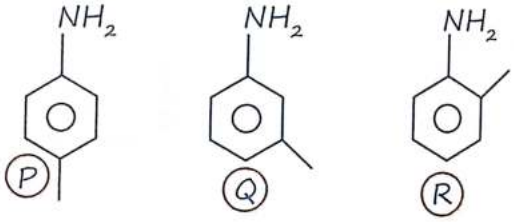
395. P, Q, R, S of Q. 393 and 394.

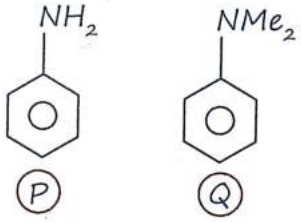
396. 

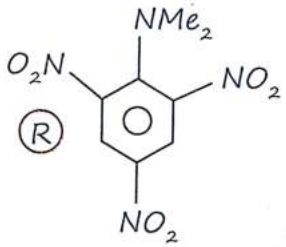
397. 

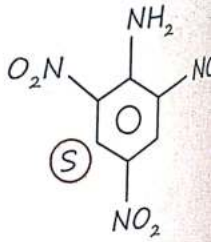
398. 

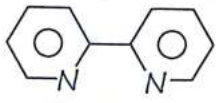
399. 

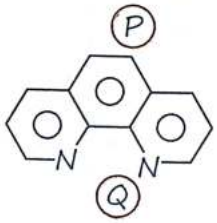
400. 

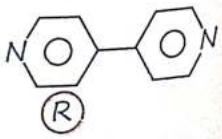
401. 





402. 





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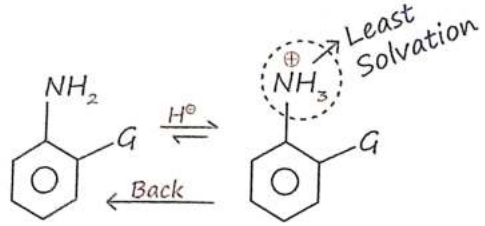
Solvation Effect

1. Ortho effect (SIP Effect)

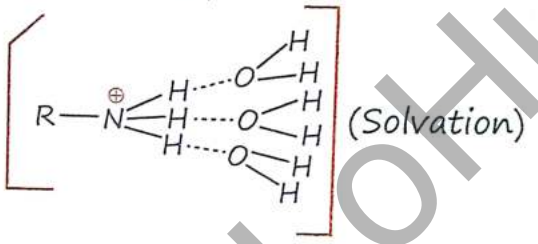


- Due to steric repulsion of (G), ortho substituted aniline becomes least basic among o, m, p isomers.

2. Basicity of 1°, 2°, 3° amines

$R-NH_2$ (1°) $\quad R-NH-R$ (2°) $\quad R-N(R)-R$ (3°)

- Electronic effect ⇒ 3° > 2° > 1° (Inductive)



- Solvation effect ⇒ 1° > 2° > 3° (H-bonding)

# R = Me ⇒ 2° > 1° > 3°
R = Et ⇒ 2° > 3° > 1°

Practice Questions

403. $C_2H_5NH_2$ (P) $\quad C_6H_5NH_2$ (Q) $\quad NH_3$ (R)

$C_6H_5CH_2NH_2$ (S) $\quad (C_2H_5)_2NH_2$ (T) $\quad$ [Aqueous]

404. CH_3NH_2 (P) $\quad (CH_3)_2NH$ (Q) $\quad (CH_3)_3N$ (R)

$C_6H_5NH_2$ (S) $\quad C_6H_5CH_2NH_2$ (T) $\quad [H_2O]$

405. In aqueous phase (write IUPAC name & match)

Compound		pK_b	
(P)	$PhCH_2NH_2$	(1)	9.30
(Q)	Me_2NH	(2)	9.38
(R)	$PhNMe_2$	(3)	4.22
(S)	$PhNH_2$	(4)	3.27
(T)	Me_3N	(5)	4.70
(U)	PhNHMe	(6)	8.92

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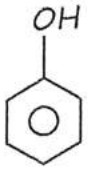
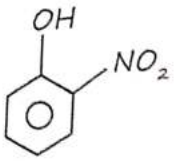
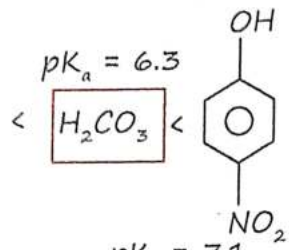
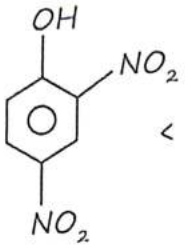
Solubility Table

Alkane < Alkene < $R-NH_2$ < $R-C\equiv CH$ (Alkyne) < $R-OH$ < H_2O <  <

$pK_a = 6.3$ < H_2CO_3 < < < $R-COOH$ < $R-SO_3H$ <

Squaric acid and Inorganic

$pK_a = 7.1$

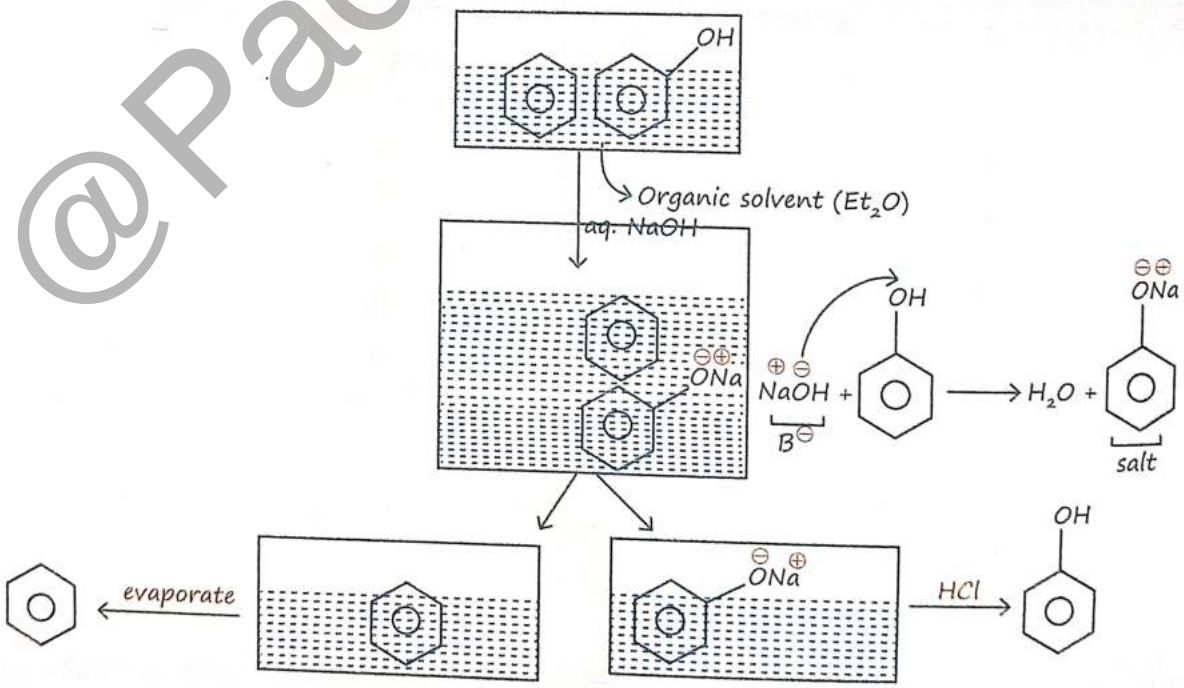
- **Reaction Feasibility**

$$HA + B^{\ominus} \rightleftarrows A^{\ominus} + H-B$$

(Acid) (Base) (C. B.) (C. A.)

- Reaction always moves from SA to WA and SB to WB.
- If Reaction moves in forward direction then HA must be stronger acid than HB.
- If $B^{\ominus}$ is $\overset{\oplus}{Na}\overset{\ominus}{OH}$ ⇒ then HA must be stronger acid than H_2O.
- If $B^{\ominus}$ is $\overset{\oplus}{Na}\overset{\ominus}{HCO_3}$ ⇒ then HA must be stronger acid than H_2CO_3 $H_2CO_3 \rightarrow H_2O$, $H_2CO_3 \rightarrow CO_2$ (Brisk effer.)
- If reaction moves in forward direction then HA is soluble in aq. NaOH/aq. $NaHCO_3$.

Differential Extraction



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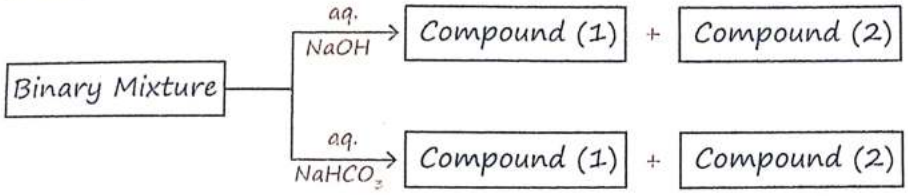
Practice Questions

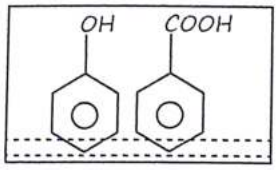
406. Find the pair(s) for given scheme of differential extraction.

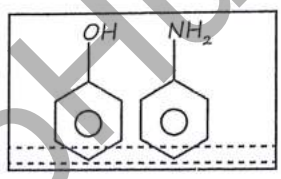


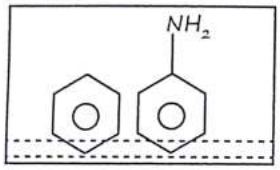
(a) $PhCH_2OH$, C_6H_5OH
(b) CH_3COOH, Cyclohexanol
(c) $C_6H_5CH_2OH$, Salicylic acid
(d) Carbolic acid, Picric acid
(e) C_6H_5OH, $C_6H_5CH_2COOH$
(f) Squaric acid, $PhCH_2OH$
(g) O-Cresol, m-Xylene
(h) Chloroxylenol, Toluene
(i) 2-phenylacetic acid, Anisole
(j) Ethanol, Cyclohexanol

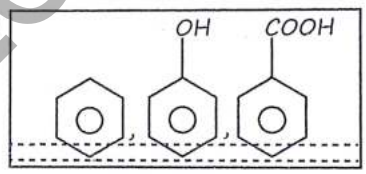
407. Find the reagent(s) for given mixture for separation by scheme of differential extraction.

(a) aq. NaOH
(b) aq. $NaHCO_3$
(c) aq. HCl

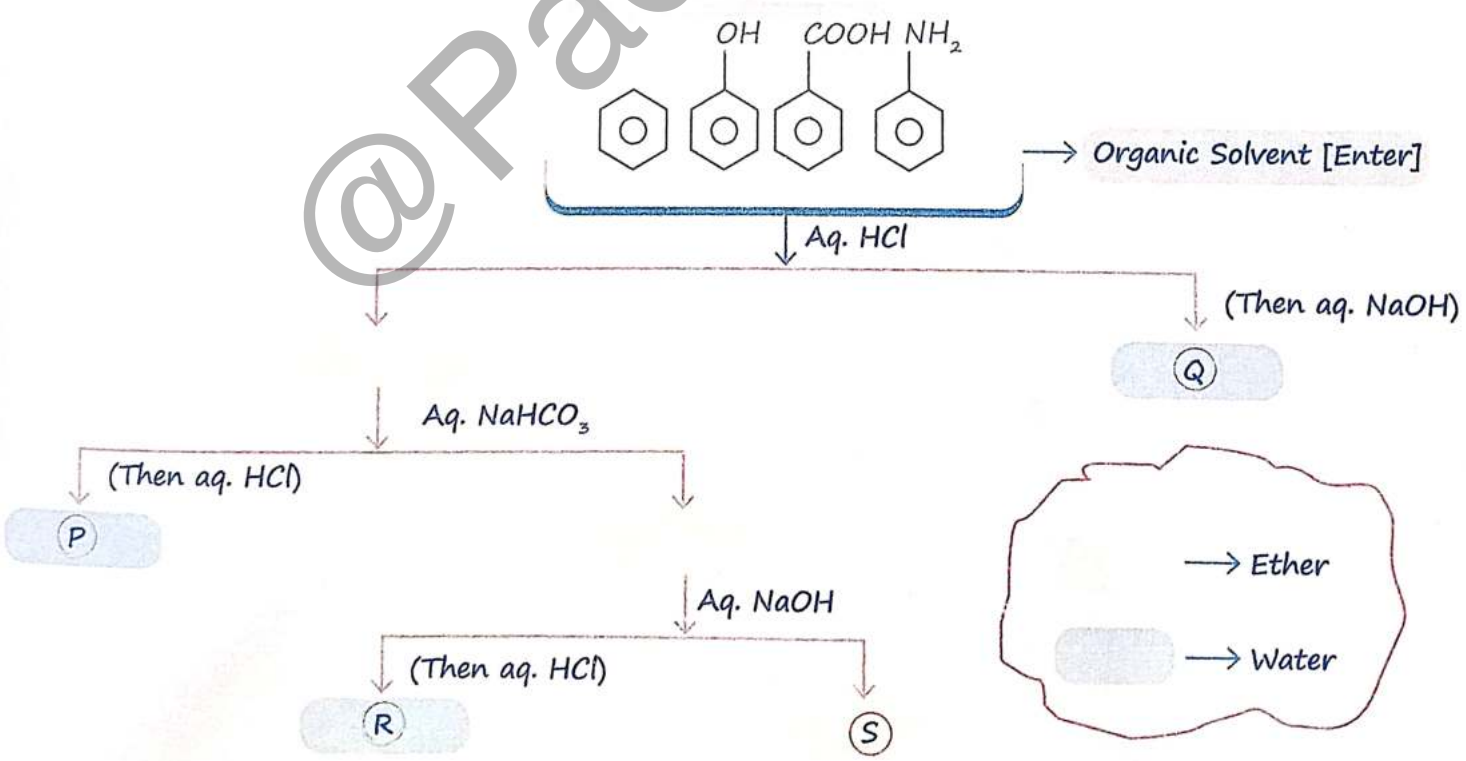
(1) 

(2) 

(3) 

(4) 

408.



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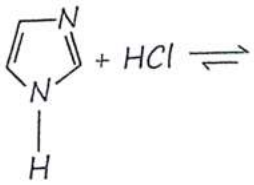
409. How many reactions will move in forward direction dominantly and also find major products?

(A) $CH_3COOH + NaHCO_3 \rightleftharpoons$

(B) $CH_3 - C \equiv CH + NaOH \rightleftharpoons$

(C) $CH_3COOH + CH_3SO_3Na \rightleftharpoons$

(D)  $+ HCl \rightleftharpoons$

410. (A) X is no. of compounds which are soluble in aqueous NaOH

(B) Y is no. of compounds which are soluble in aqueous $NaHCO_3$ and release CO_2 gas

(C) Z is no. of compounds which contains COOH groups Then find the value of X – Y + Z

(1) Phenol	(2) Cyclohexanol	(3) Benzyl alcohol
(4) Benzoic acid	(5) Nitrobenzene	(6) Benzene Sulphonic acid
(7) Aniline	(8) o-Nitro phenol	(9) o-cresol
(10) m-xylene	(11) m-nitro phenol	(12) p-nitro phenol
(13) Picric acid	(14) 3, 4-dinitro phenol	(15) 4-aminophenol
(16) 2-phenyl ethanoic acid	(17) Fumaric acid	(18) Malonic acid
(19) Adipic acid	(20) Salicylic acid	(21) Maleic acid
(22) Tartaric acid	(23) Lactic acid	(24) Cinnamic acid

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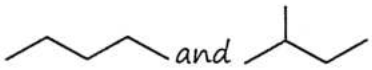
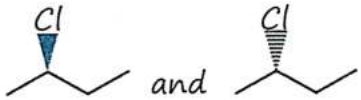
3 Isomerism

Isomerism

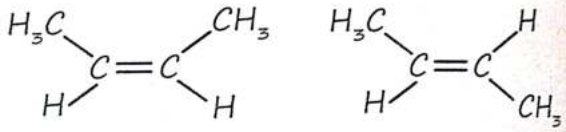
- Compounds having same molecular formula but different
  (a) Connectivity (Structural) or
  (b) Arrangement (Stereo)
  are known as Isomers and Phenomenon is known as Isomerism.
- Example

(1)  and

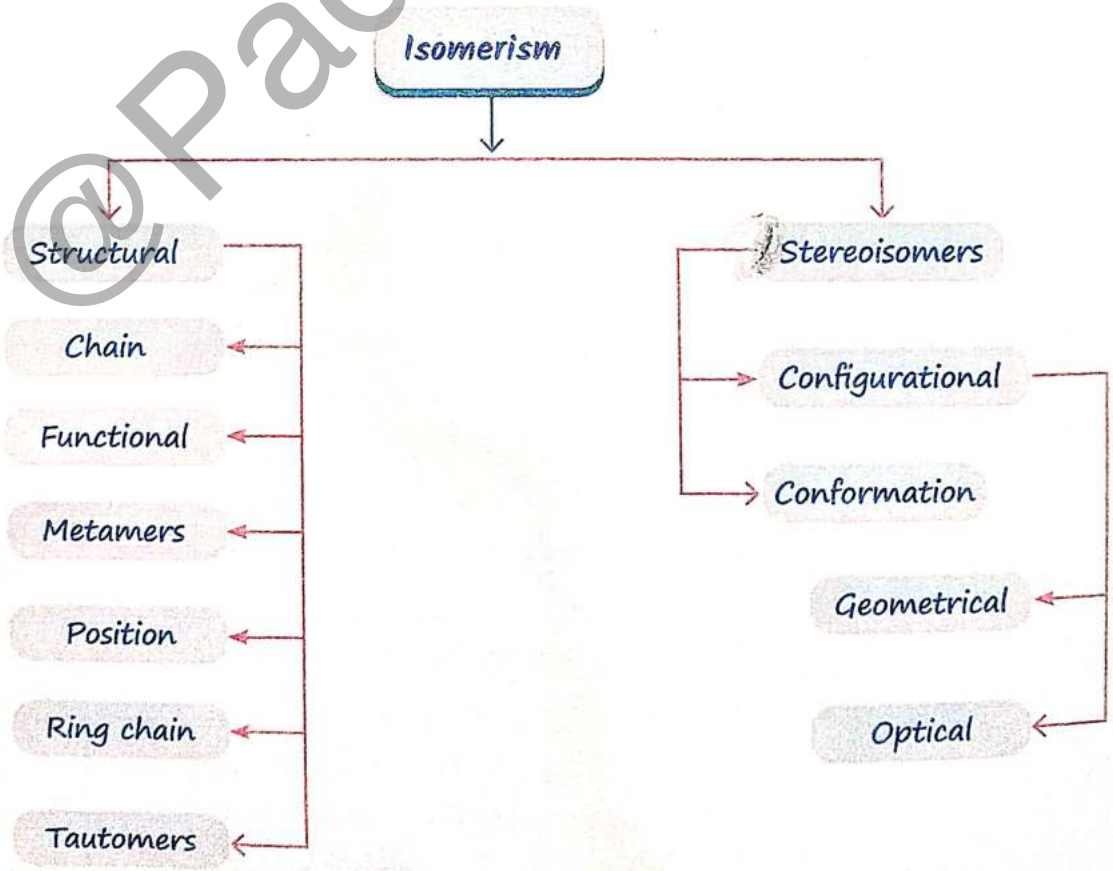
- Connectivity
- Chain Isomer

(2)  and

- Arrangement

(3) 

- Arrangement



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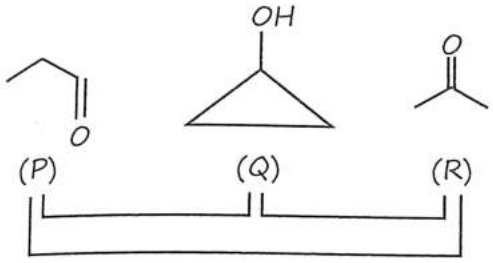
Structural Isomers

(1) Ring chain:
- Same molecular formula
- Different P.C. in open chain or closed chain

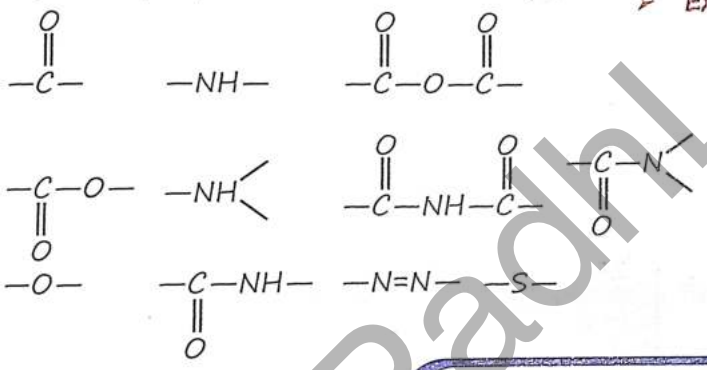
(2) Functional Isomer:
- Same molecular formula
- Different functional group



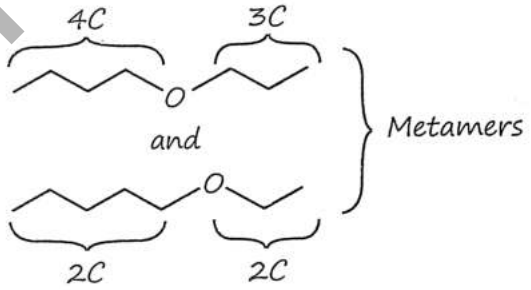
(3) Chain isomer:
- Same molecular formula
- Same functional group
- Different parent chain or side chain

(4) Position Isomers:
- Same molecular formula
- Same functional group and P.C./S.C.
- Different position of M.B./S.C./F.G.

(5) Metamerism:
- When either side of polyvalent functional group has different chain or skeleton then compounds are known as Metamers.
- Polyvalent group:



- Example



Practice Questions

1.
2.
3.
4.
5. CN CN NC
6. COOH COOH
7. OH OH OH

8. CN CN CN CN NC CN
9.
10.
11.
12.

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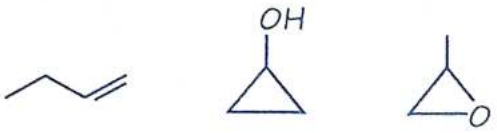
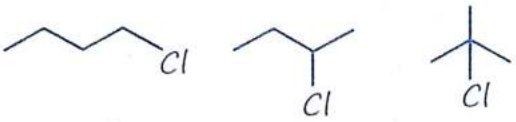
13. 

14. 

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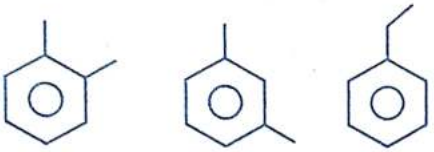
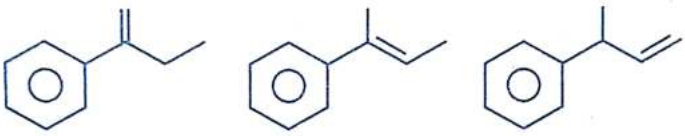
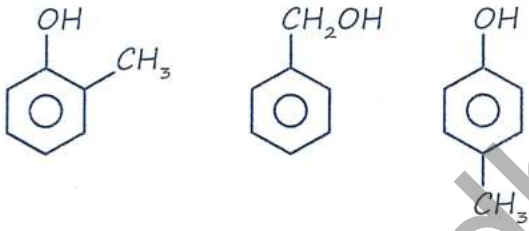
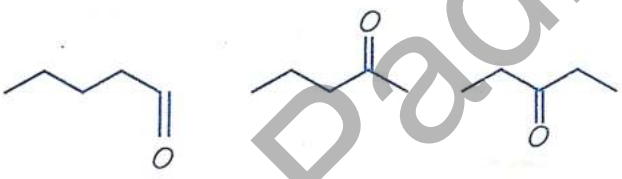
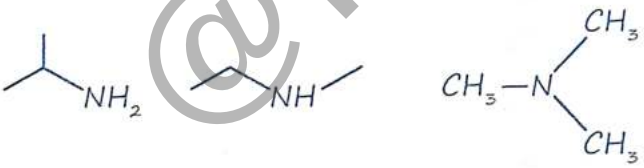
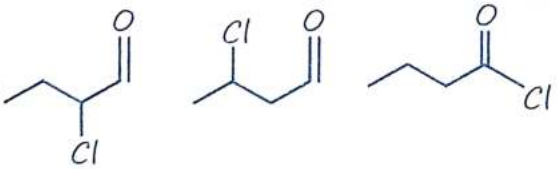
16. 

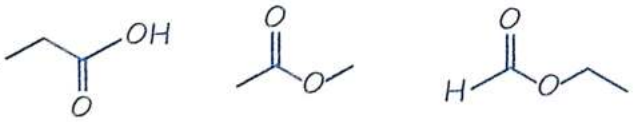
17. 

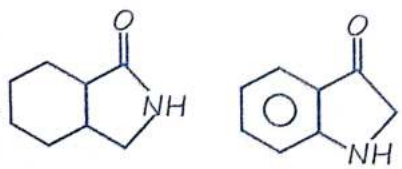
18. 

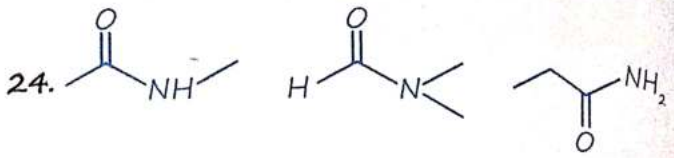
19. 

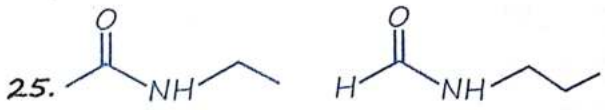
20. 

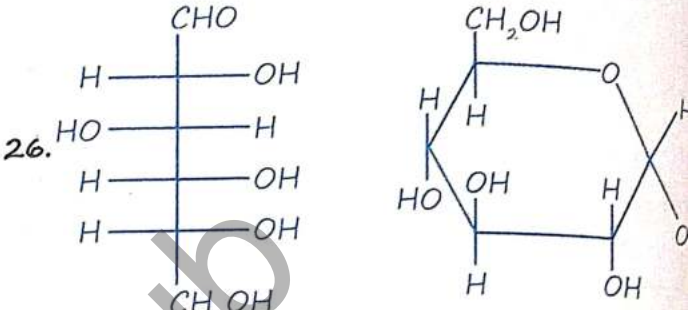
21. 

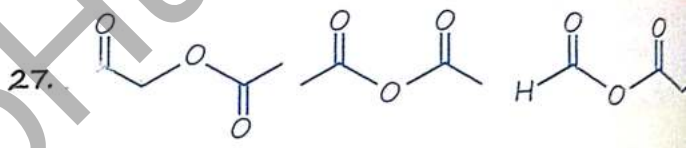
22. 

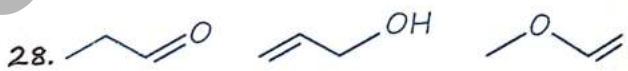
23. 

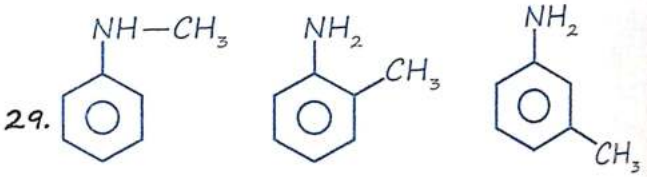
24. 

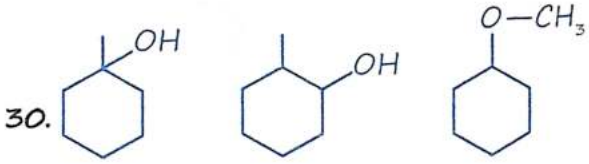
25. 

26. 

27. 

28. 

29. 

30. 

Calculation of Structural Isomers:

31. C_3H_8

32. C_4H_{10}

33. C_5H_{12}

34. C_6H_{14}

35. C_7H_{16}

36. C_3H_7Cl

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37. C_4H_9Cl

38. $C_5H_{11}Cl$

39. $C_6H_{13}Cl$

40. $C_3H_6Cl_2$

41. $C_4H_8Cl_2$

42. C_3H_6ClBr

43. C_4H_8ClBr

44. C_3H_6

45. C_4H_8

46. C_5H_{10}

47. C_6H_{12}

48. C_3H_5Cl

49. C_4H_7Cl

50. $C_3H_4Cl_2$

51. C_3H_4

52. C_4H_6

53. C_5H_8

54. C_3H_8O

55. $C_4H_{10}O$

56. $C_5H_{12}O$

57. $C_6H_{14}O$

58. C_3H_6O

59. C_4H_8O

60. $C_5H_{10}O$

61. $C_6H_{12}O$

62. $C_7H_{14}O$

63. $C_2FClBrI$

64. Tribromo cyclohexane

65. $C_2H_4O_2$ [acid & ester]

66. $C_3H_6O_2$ [acid & ester]

67. $C_4H_8O_2$ [acid & ester]

68. $C_5H_{10}O_2$ [acid & ester]

69. C_8H_{10} [Benzenoid]

70. C_7H_7Cl [Benzenoid]

71. C_7H_8O [Benzenoid]

72. C_7H_9N [Benzenoid]

73. C_3H_9N

74. $C_4H_{11}N$

75. P_1, P_2–dichloro-P_3-methyl Cyclopropane

76. P_1, P_2–dichloro-P_3-methyl Cyclohexane

77. C_6H_5X

78. C_6H_4XY

79. C_6H_3XYZ

80. C_3H_3XYY

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Tautomerism

Desmotropy

- Arises due to oscillation or migration of an atom/group.

(1) $\overset{\ominus}{C}H_2-\overset{O}{\overset{\|}{C}}-H \longleftrightarrow CH_2=\overset{O^{\ominus}}{\overset{|}{C}}-H$

- Hypothetical structures
- Movement of $p.e^{\ominus}$
- Resonance
- No equilibrium

(2) $H_3C-\overset{O}{\overset{\|}{C}}-H \rightleftharpoons CH_2=\overset{OH}{\overset{|}{C}}-H$

- Real structures
- Movement of atom
- Tautomerism
- Dynamic equilibrium

Based on atom migration

(1) **Cationtropy:** When cation transfer

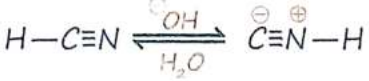
(Prototropy)

(2) **Aniontropy:** Not in JEE

Based on atom between migration

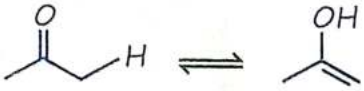
(1) **Diad:** When atom ($H^{\oplus}$) oscillates between two atoms.

- Example

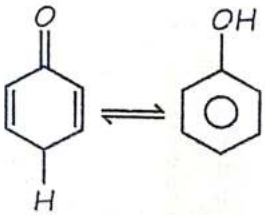


(2) **Triad:** When atom ($H^{\oplus}$) oscillates between three atoms.

- Example



(3) **Space:** When atom ($H^{\oplus}$) oscillates between more than two atoms.



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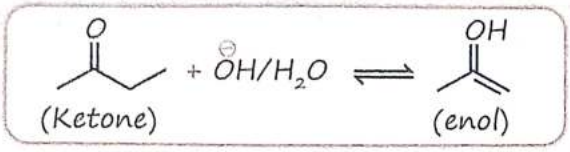
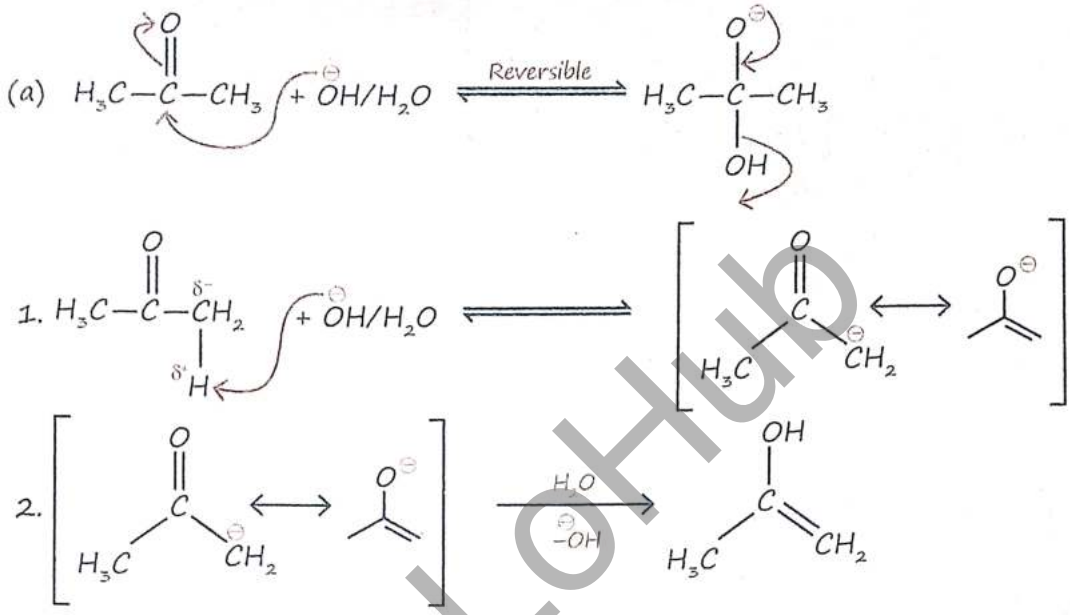
Base catalyst tautomerism

Reaction



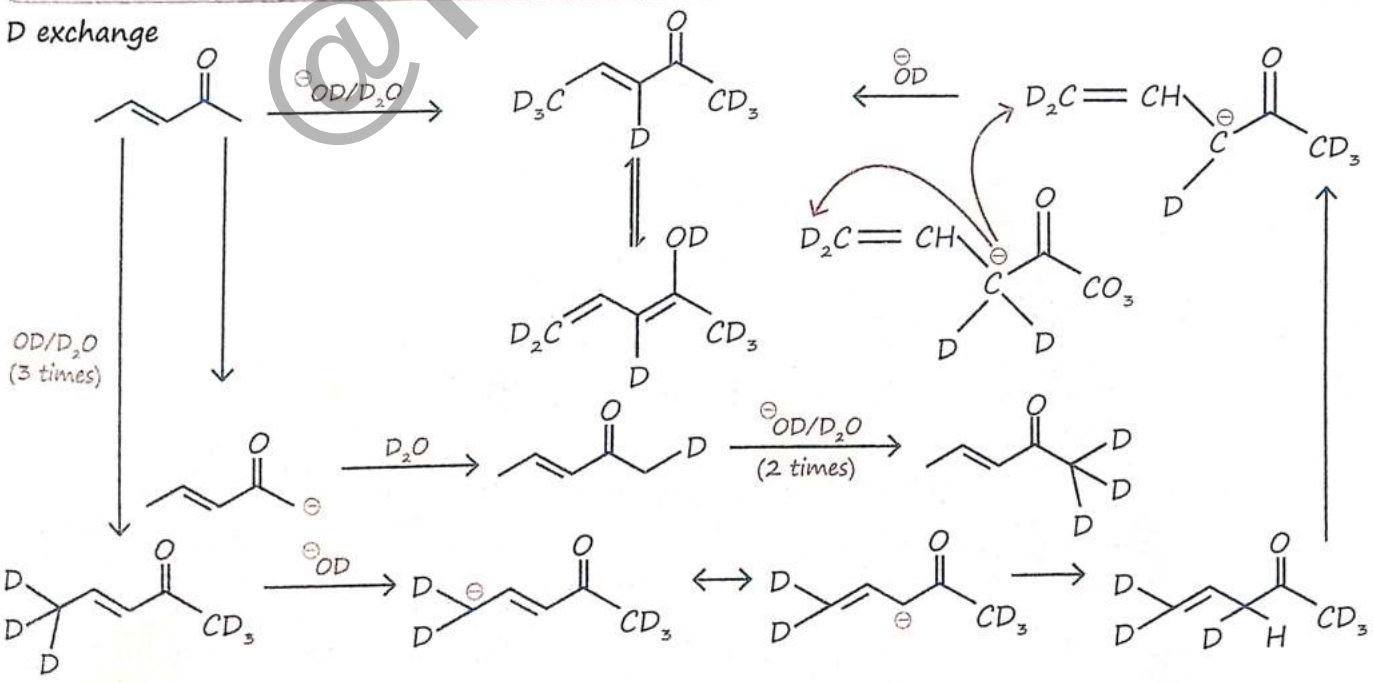
Mechanism



Baba Tea-Stall (BTS)

(i) Also known as Keto-enol tautomerism.
(ii) Keto & enol remains in dynamic equilibrium.
(iii) Slowest step (rds) is formation of carbanion or enolate ion.
(iv) As the stability of −ve ↑↑ ⇒ rate of reaction ↑↑
(v) Instead of $^{\ominus}OH$, we can also use various base.
(vi) $^{\ominus}O$ बनाओ → Reso कराओ → EN atom (O, N, S, P) पर $^{\ominus}$ पहुचाओं।

D exchange



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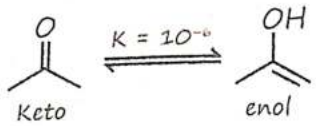
Stability of Keto and Enol

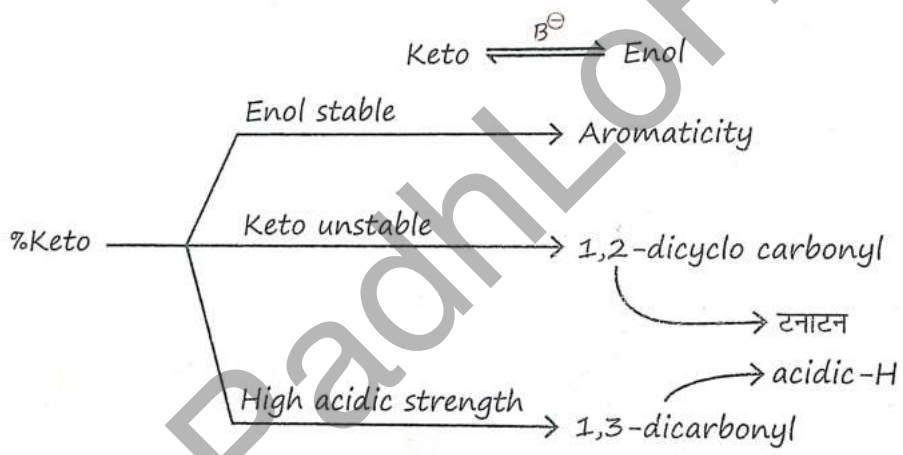
- Generally Keto is more stable than enol form.
- Example:



Bond Strength is KJ mol^{-1}.

Form	–H	π-bond	Sum
Keto	440 [C–H]	720 [C=O]	1160
Enol	500 [O–H]	620 [C=C]	1120

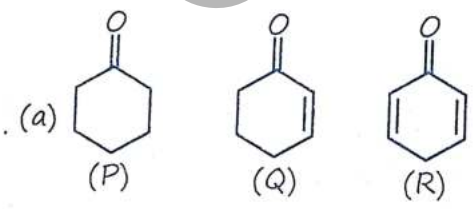
Stability of Enol



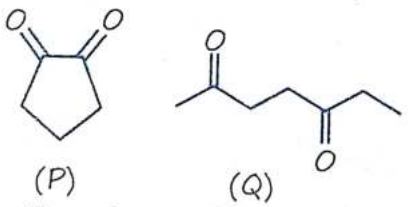
Practice Questions

128. Find the correct option of %enol?

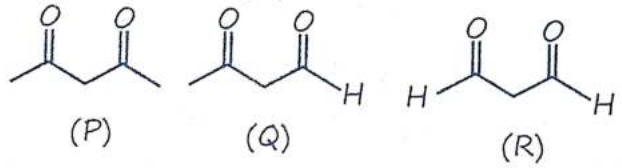
(a) (P) (Q) (R)



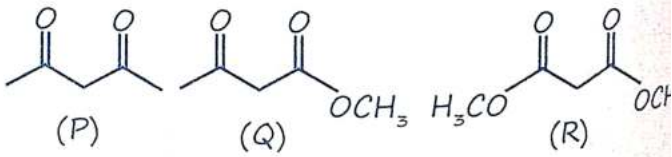
(b) (P) (Q)



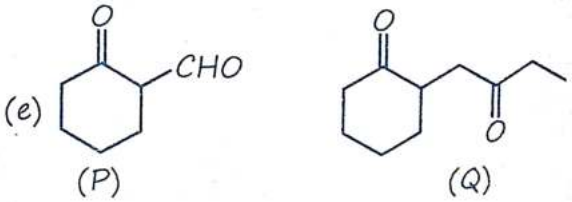
(c) (P) (Q) (R)



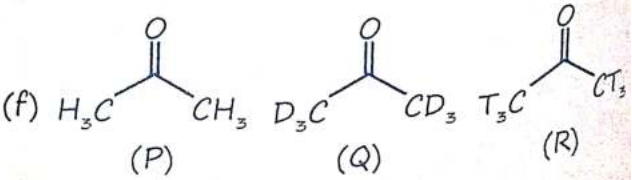
(d) (P) (Q) (R)



(e) (P) (Q)



(f) (P) (Q) (R)



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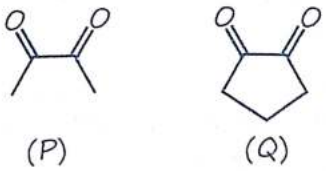
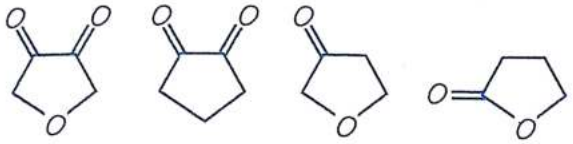
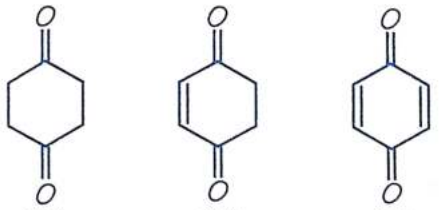
(g) 

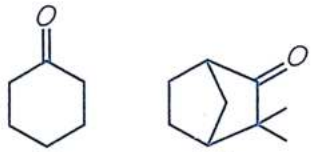
(P) (Q)

(h) 

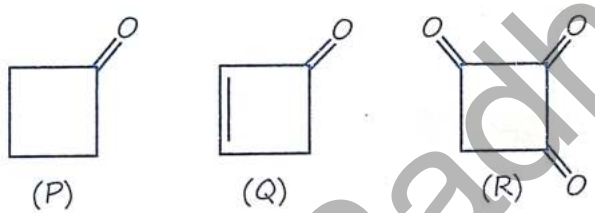
(P) (Q) (R) (S)

(i) 

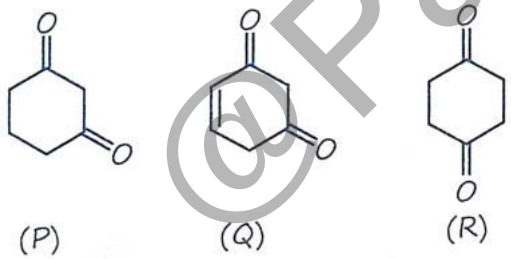
(P) (Q) (R)

(j) 

(P) (Q)

(k) 

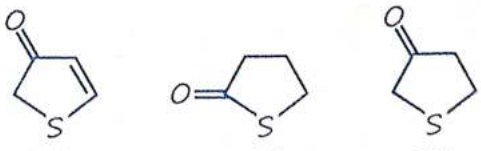
(P) (Q) (R)

(l) 

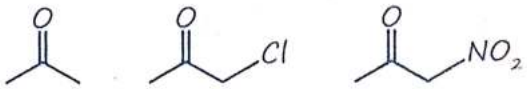
(P) (Q) (R)

(m) CH_3-CHO

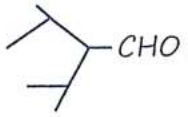
(P) (Q)

(n) 

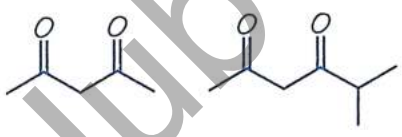
(P) (Q) (R)

(o) 

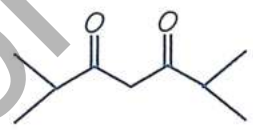
(P) (Q) (R)

(p)  H_3C-CHO

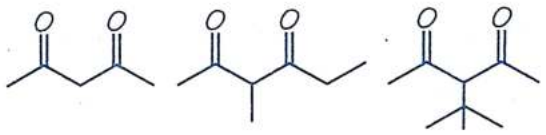
(P) (Q)

(q) 

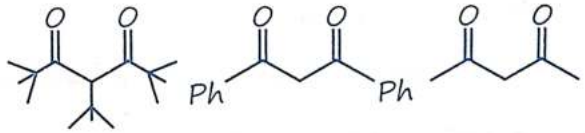
(P) (Q)



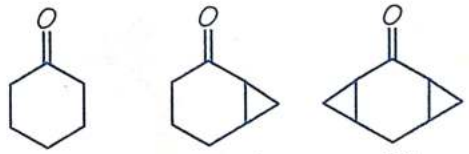
(R)

(r) 

(P) (Q) (R)

(s) 

(P) (Q) (R)

(t) 

(P) (Q) (R)

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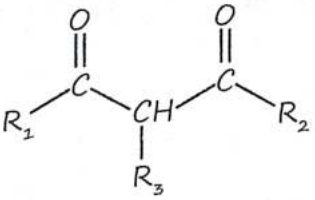
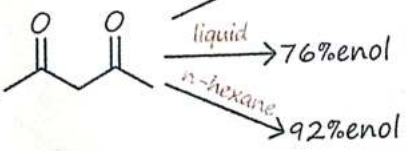
➢ Effect of bulkiness on Enol: 

➢ Effect of Solvent: 

➢ Effect of Temp.: As the temp.↑⇒ %enol↓

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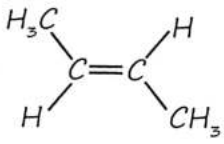
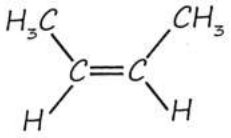
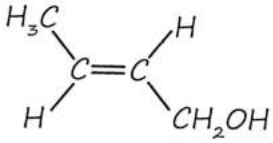
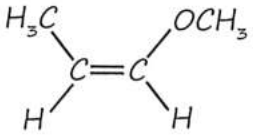
Geometrical Isomerism

Compounds having same molecular formula, same structural formula (connectivity) but different arrangement across restricted rotation system (RRS) are known as Geometrical isomers of phenomenon is known as Geometrical Isomerism.

- Example:

(1)  and  *GI (✓)

(2) and *GI (✗)

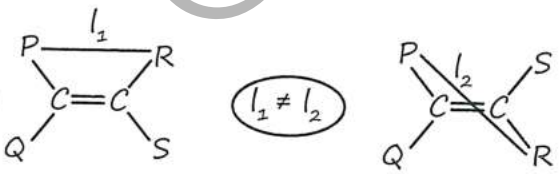
(3)  and  *GI (✗) *Functional (✓)

Properties

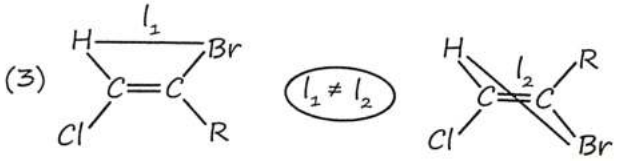
- They are not Interconvertible to each other at room temperature.
- They have different physical and chemical properties.
- They are also known as stereoisomers and cis-Trans Isomers.

Conditions for Geometrical Isomers

- RRS must be present.
- Group/atom at atleast two terminal must be different.
- Geometry must be change.

(1) 

(2)

(3) 

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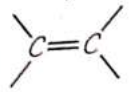
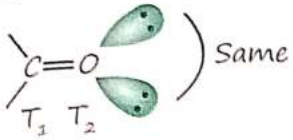
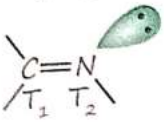
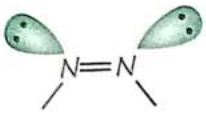
Restricted Rotation System (RRS)

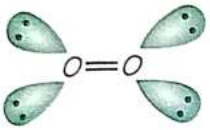
Multiple Bond

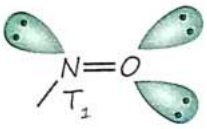
(1) 

(2) 

(3) 

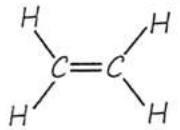
(4) 

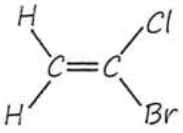
(5) 

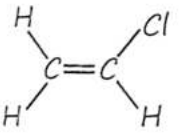
(6) 

(7) $-C\equiv C-$

(8)

➢ 

➢ 

➢ 

Cyclic

(9) Cycloalkane

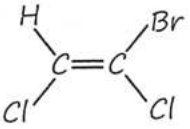
(10) Cycloalkene

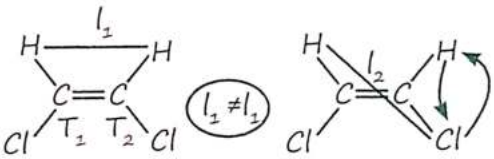
(11) Polyphenyl

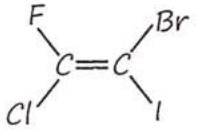
(12) Spiro

(13) Bicyclo

(14) Cumulated ene

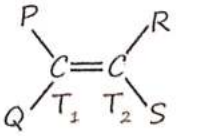
➢ 

➢ 

➢ 

Geometrical Isomerism in Multiple Bond

➢ Atom/Groups at T_1 must be different & atleast T_2 must be different.



$P \neq Q$ & $R \neq S$

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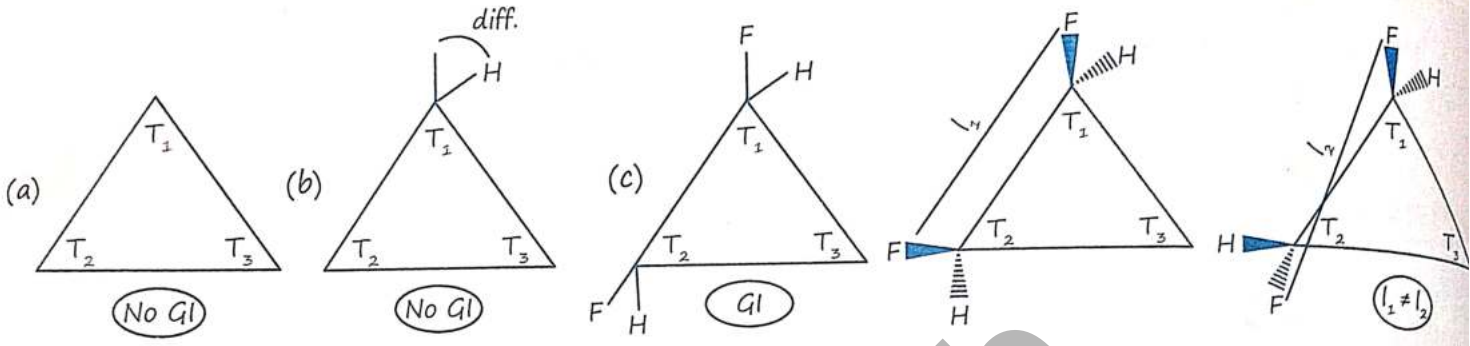
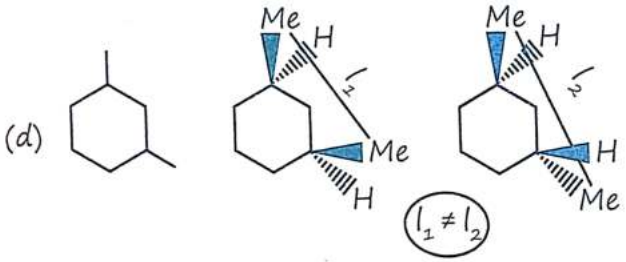
Geometrical Isomerism in Cycloalkane

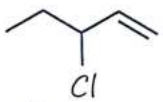
In cyclic compound the rotation about C—C single bond is not possible because of the rigidity caused by the presence of other carbon of the ring which keep them tightly held, thus a disubstituted cyclic compound (having two substitution at the separate ring carbons) might show Geometrical Isomerism.



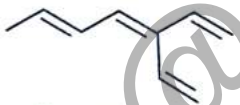


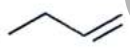
Practice Questions

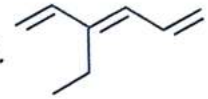
129.

130. 

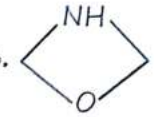
131.

132. 

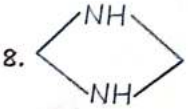
133. 

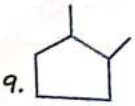
134. 

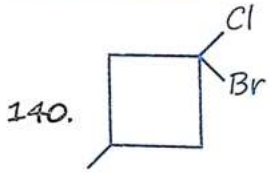
135.

136. 

137. N_2H_4

138. 

139. 

140. 

141. ether

142. 2-Bromopropene

143. 1-Bromo but-1ene

144. 1,3-butadiene

145. Penta-1,3-diene

146. 1,2-dichloro propene

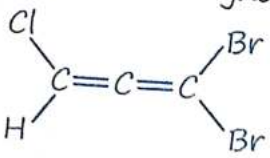
147. 2-methyl-1-butene

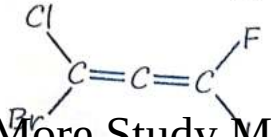
148. 1,1-dichloro propene

149. 2-methyl-2-butene

150. Pent-1-en-3-yne

151. Pent-3-en-1-yne

152. 

153. 

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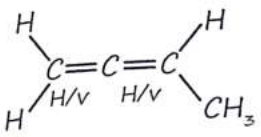
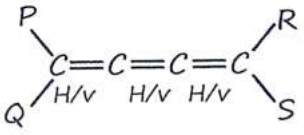
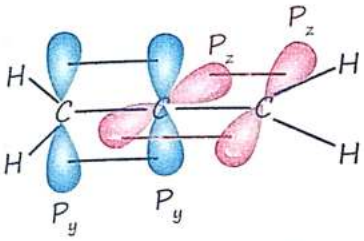
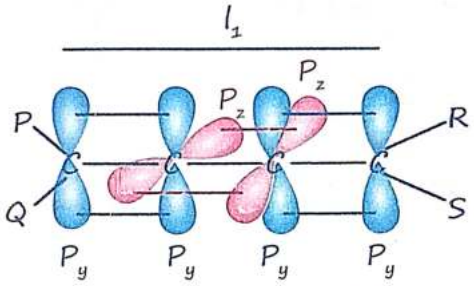
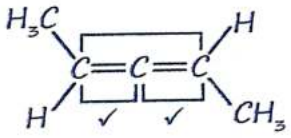
Geometrical Isomerism in Cumulated Ene

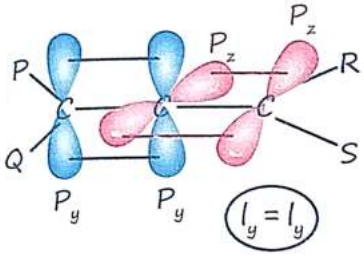




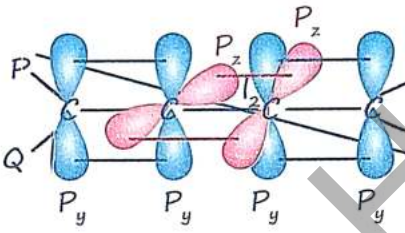








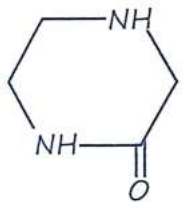
$l_y = l_y$



$l_1 = l_2$

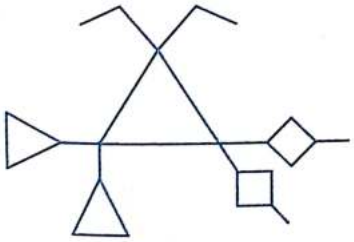
Practice Questions

154.



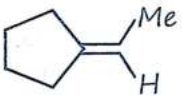
155.

156.



157.

158.



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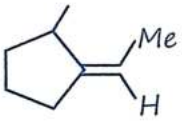

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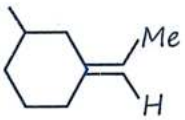

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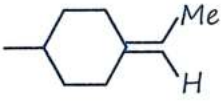
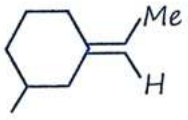


162.



163. $H_3C-CCl=C=CCl-CH_3$

164. $H_3C-HC=C=C=CH-CH_3$

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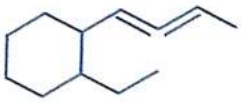
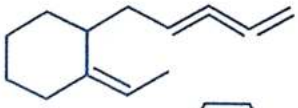
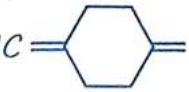
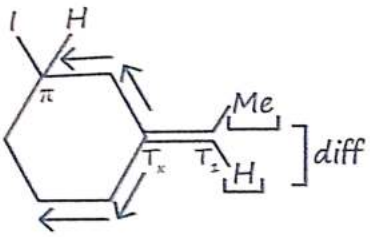
165. $Cl-CCl=C=C=C<^{Cl}_{Cl}$

166. $Me-CCl=C=C=C=C=CHCl$

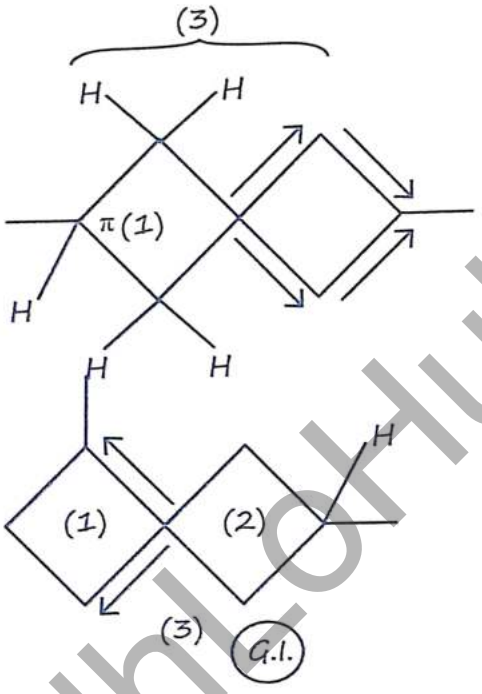
167. 

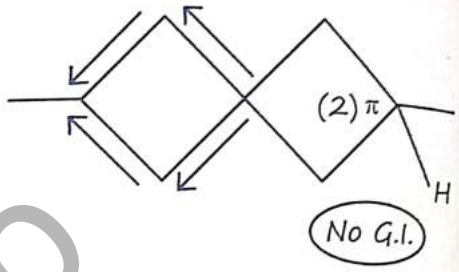
168. 

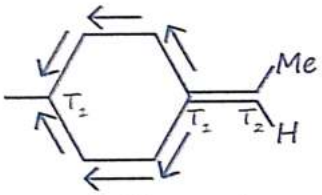
169. H_3C-HC  $CCl-CH_3$



RRS ⇒ double & Ring

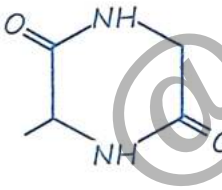


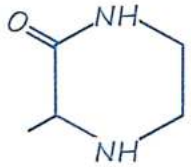


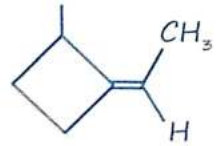


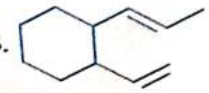
RRS ⇒ double & Ring

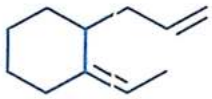
Practice Questions

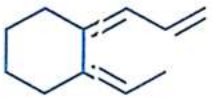
170. 

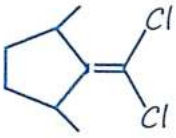
171. 

172. 

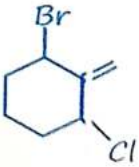
173. 

174. 

175. 

176. 

177.

178. 

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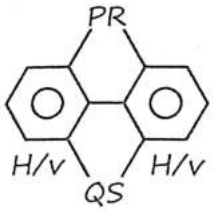
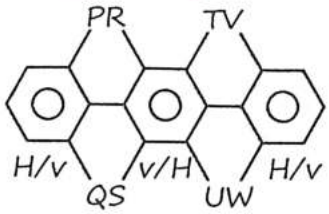
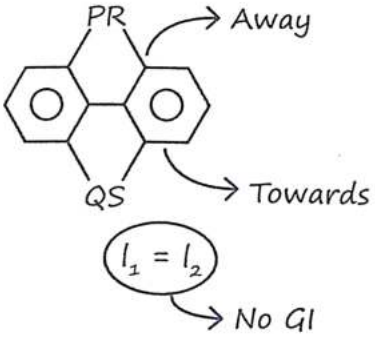
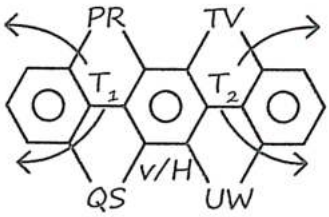
Geometrical Isomerism in Polyphenyl Rings







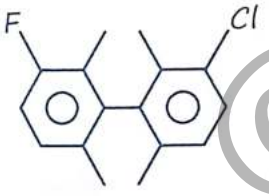


$P \neq Q$ & $V \neq W$

Practice Questions

179.

180.



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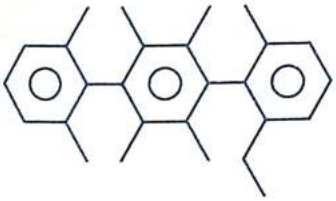

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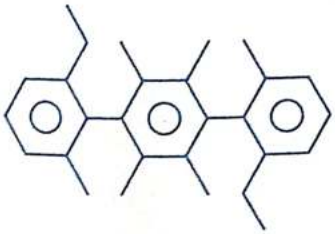

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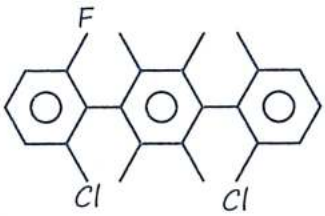

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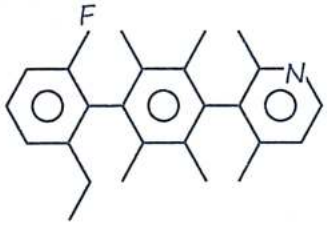

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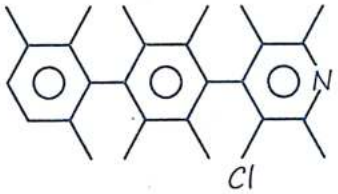

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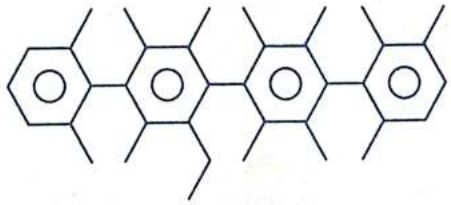

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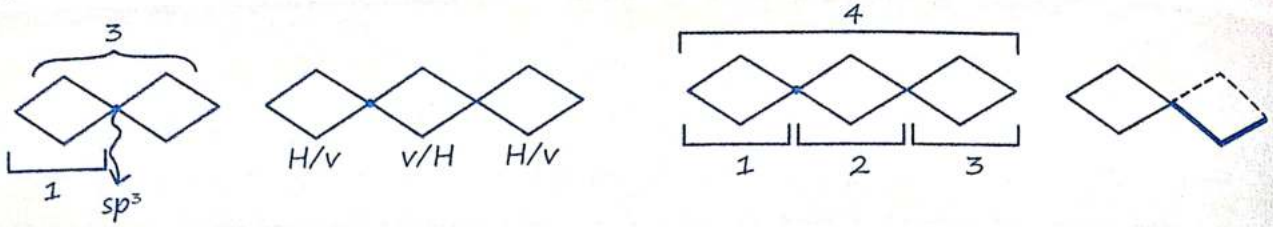
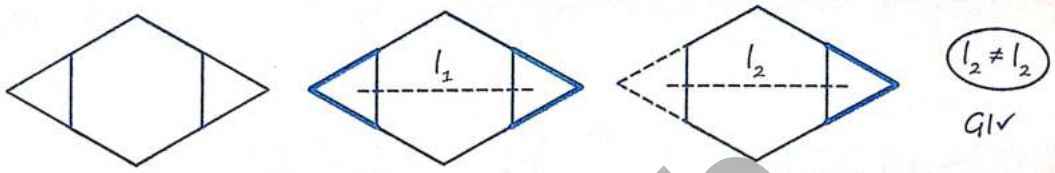
Geometrical Isomerism in Spiro



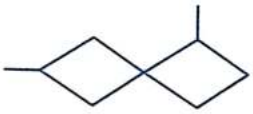
Geometrical Isomerism in Bicyclo



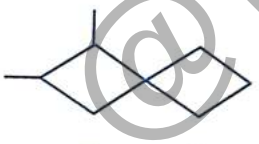
Practice Questions

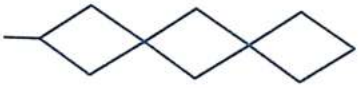
187.

188.

189. 

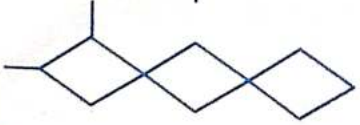
190.

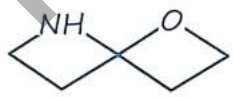
191. 

192. 

193.

194.

195. 

196. 

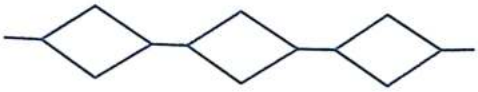
197.

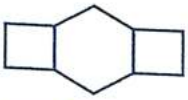
198.

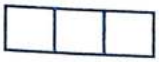
199.

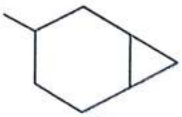
200.

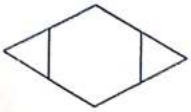
201.

202. 

203. 

204. 

205. 

206. 

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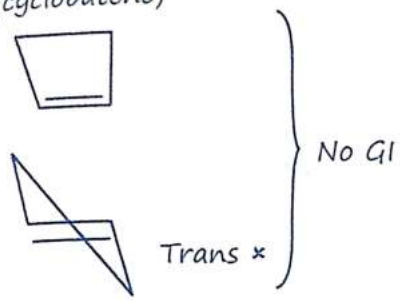
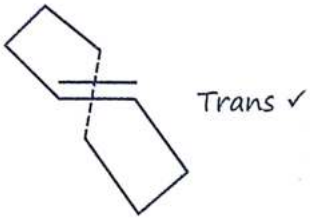
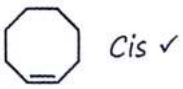
Geometrical Isomerism in Cycloalkene

- ring ≥ 8 ⇒ cycloalkene can show GI.
- 8 ring ≤ 11 ⇒ Cis > Trans (stable)
- ring ≥ 12 ⇒ Trans > Cis (stable)

(cyclobutene)

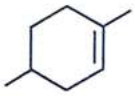


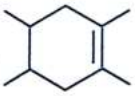


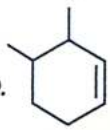


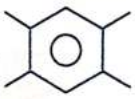
Practice Questions

207.

208. 

209. 

210. 

211. 

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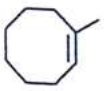
213. 

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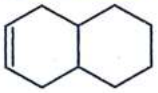
215. 

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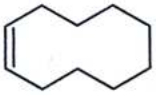
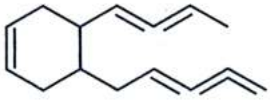
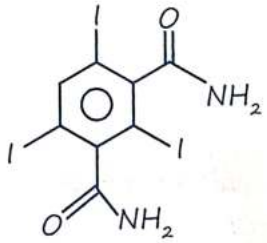
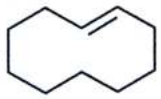
217. 

218. 

219. 

220. 

221. Stability of 217 & 220.

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Nomenclature of Geometrical Isomers

- Cis-trans
- E-Z
- Syn-anti

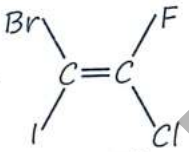
E-Z Nomenclature

If Highest priority gnp/atom acc. to CIP rule on T_1 and T_2 are on:

Same side	→ Z
Opposite side	→ E

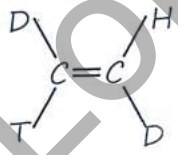
Cahn-Ingold-Prelog Rule

Rule 1: Atomic no. of directly attached atom



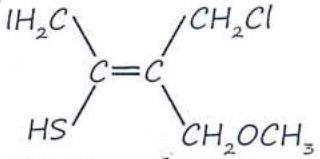
EX: $-SH$ (1) $-F$ (2) $-Cl$ (3) $-OH$ (4) $-CH_3$ (5) $-NH_2$ (6)

Rule 2: Atomic mass of directly attached atom



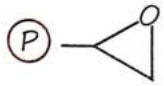
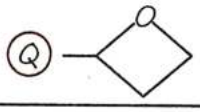
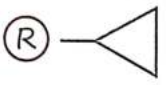
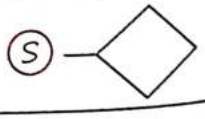
EX: $-{}^{35}Cl$ (1) $-H$ (2) $-D$ (3) $-T$ (4) $-{}^{37}Cl$ (5)

Rule 3: If at no. mass and directly attached atom is same then move to next atom.



EX: $-CH_2Cl$ (1) $-CH_2CH_2Br$ (2) $-\underset{\underset{CH_3}{|}}{CH}-CH_2Cl$ (3) $-OCH_3$ (4) $-CH_2CH_2I$ (5)

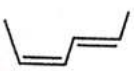
Rule 4: In case of double bond and ring. dummy is used.

(P) $-CH_2=CH_2$	(Q) $-CH_2=O$	(R) $-C\equiv N$	(S) $-COOH$
(P) 	(Q) 	(R) 	(S) 

Rule 5: If both dummy and real present then real is more preferred.

EX: $-CH=CH_2$ (1) $-\underset{\underset{CH_3}{|}}{CH}-CH_2CH_3$ (2) $-CH=O$ (3) $-\underset{\underset{OH}{|}}{CH}-O-CH_3$ (4)

Rule 6:

Ex: 

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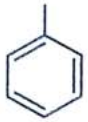
Q. Assign priority number to the following group as per Cahn, ingold. prelog sequence rule

(i) $-CH_2OH$, $-CH_3$, $-CH_2CH_2OH$ $-H$

(ii) $-Cl$, $-Br$, $-CH=CH_2$ $-CH_3$

(iii) $-\overset{\overset{O}{\|}}{C}-H$, $-OH$, $-CH_3$ $-CH_2OH$

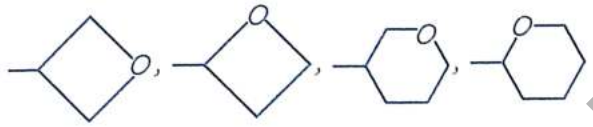
(iv) $-CH(CH_3)_2$, $-CH_2CH_2Br$, $-Cl$ $-CH_2CH_2CH_2Br$

(v) $-CH=O$, $-CH_2CH_3$  $-CH_3$

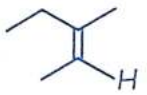
(vi) $-CH=CH_2$, $-C\equiv CH$ $CH_2-CH(CH_3)_2$ $-C(CH_3)_2-CH(CH_3)_2$

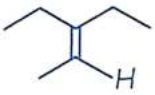
(vii) $-CH_2CH_2CH_2I$, $-CH_2-CH(Br)-CH_3$ $-CH(Cl)-CH_2-CH_3$ $-F$

(viii) $-CN, -OH, Cl, -D, -H, -SO_3H, -SH, -F, Br, I, -NH_2$

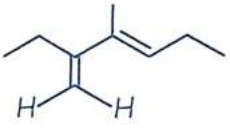
(ix) 

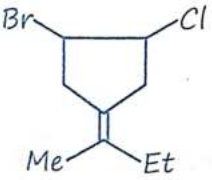
Practice Questions

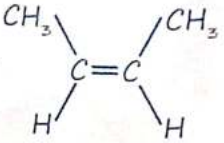
222. 

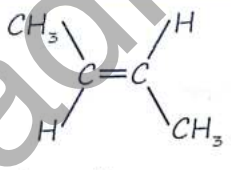
223. 

224.

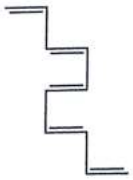
225. 

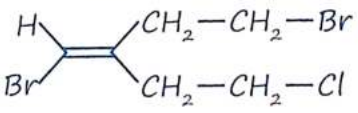
226. 

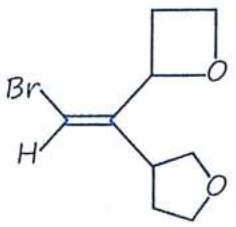
227. 

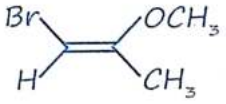
228. 

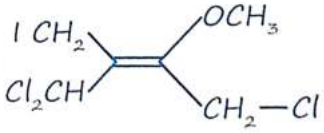
229.

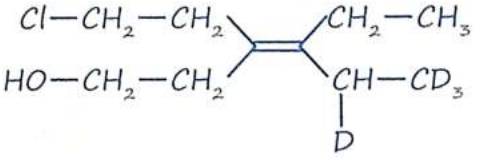
230. 

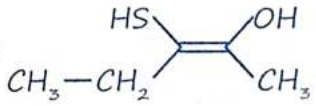
231. 

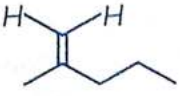
232. 

233. 

234. 

235. 

236. 

237. 

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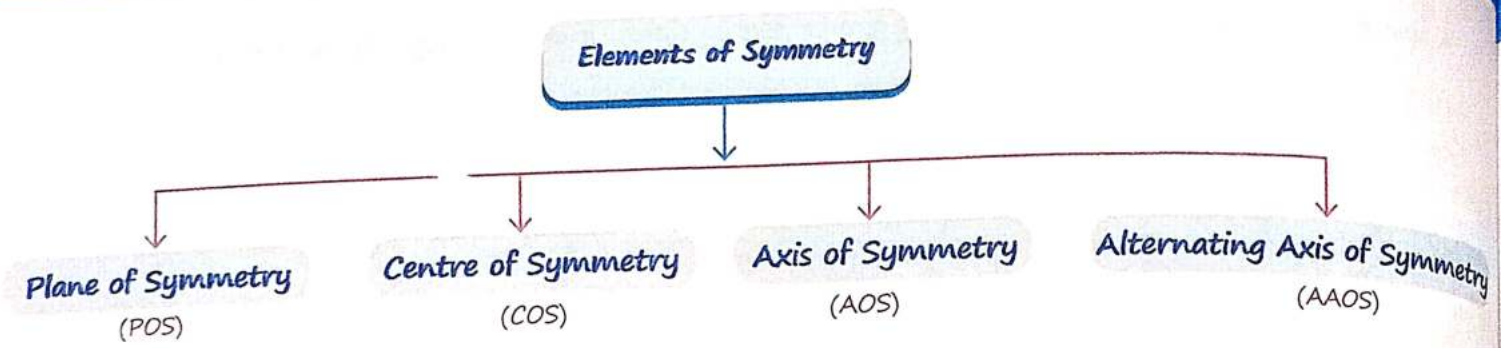
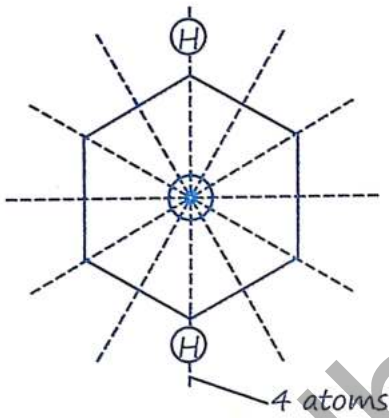


Plane of Symmetry

- Any imaginary plane across a molecule can be divided into mirror images of each other is known as 'plane of symmetry'.
- It is also known as 'POS' or 'σ_n' or σ plane of symmetry.
- Example:

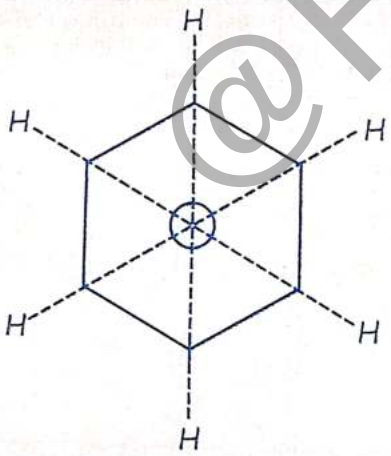


7 POS (6 + 1) ⟶ Molecular plane (slice)

Centre of Symmetry

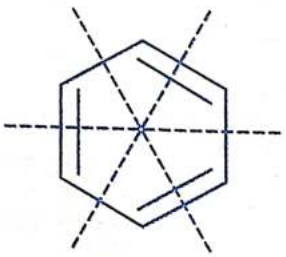
- The point from which the identical atom exist on the opposite side from this centre at equal distance.
- It is also known as COS or centre of inversion or C_i.

(a)



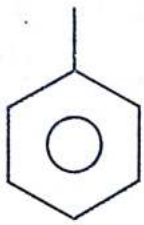
COS ✓

(b)



- COS ×
- POS (3 + 1 = 4)

(c)



- COS ×
- POS ✓

(1 + 1 = 2)

Axis of symmetry

An n-fold axis of symmetry (C_n) means that rotating the molecule by 360°/n degrees around that axis results in a molecule that is indistinguishable from the original.

Alternating axis of symmetry

An alternating axis of symmetry, also known as an improper axis of rotation (S_n), is a symmetry element in molecules. It involves a combination of rotation and reflection.

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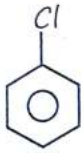
Practice Questions

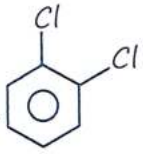
238.

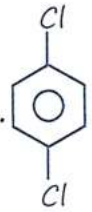
239.

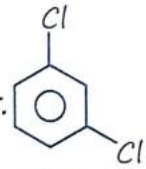
240.

241.

242. 

243. 

244. 

245. 

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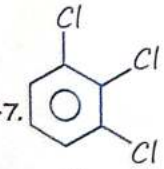
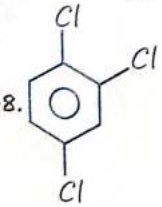
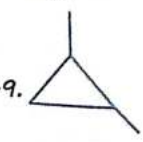
247. 

248. 

249. 

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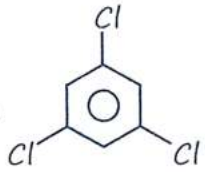
251. 

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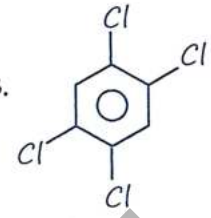
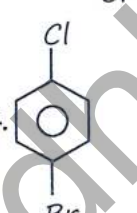
253. 

254. 

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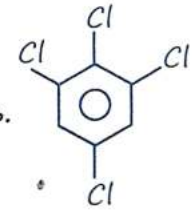
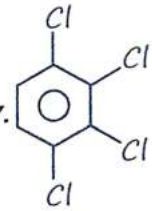
256. 

257. 

258. CH_4

259. CH_3Cl

260. CH_2Cl_2

261. $CHCl_3$

262. CCl_4

263. CH_2ClBr

264. BF_3

265. SO_3

266.

267. Human hand

268. Chair

269. Fan

270. Kids tricycle

271. Hockey stick

272. Screw

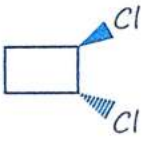
273. Nail

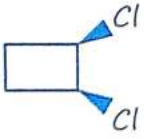
274. Wine Glass

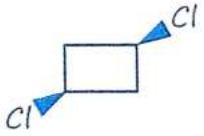
275. Spelling

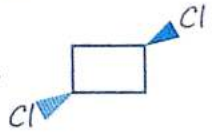
276. Prism

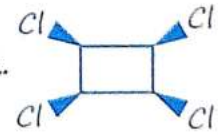
277. Coffee Mug

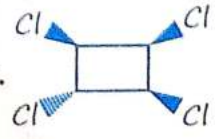
278. 

279. 

280. 

281. 

282. 

283. 

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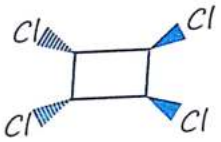
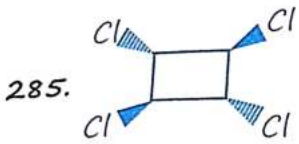
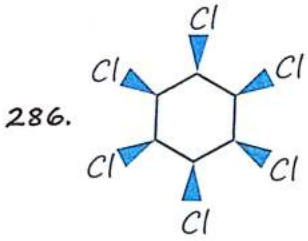
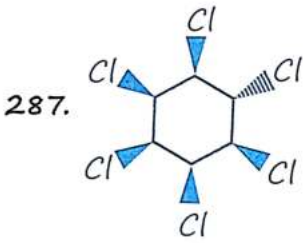
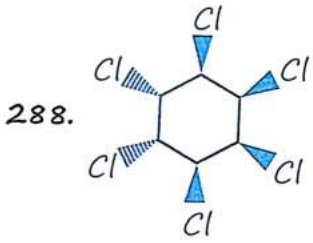
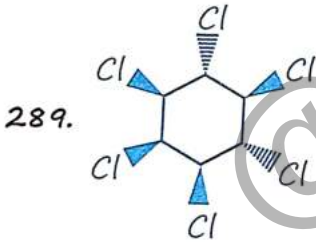
284. 

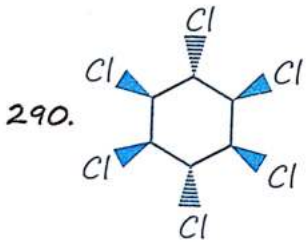
285. 

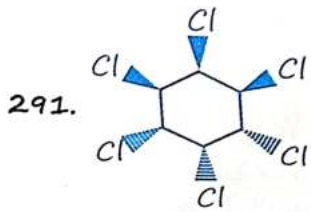
286. 

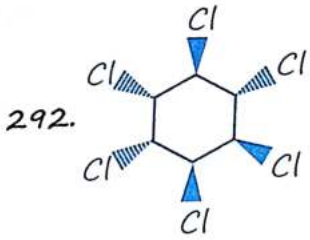
287. 

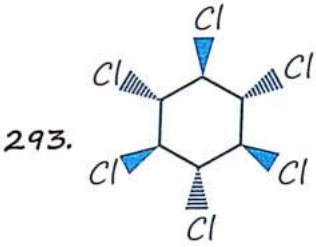
288. 

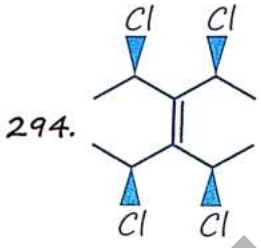
289. 

290. 

291. 

292. 

293. 

294. 

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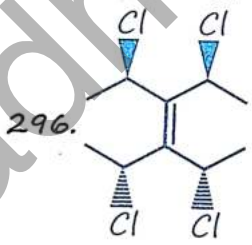
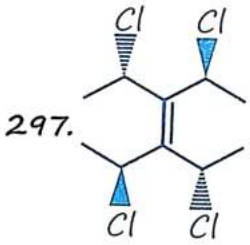
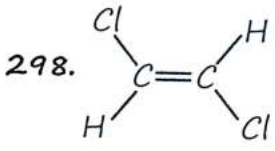
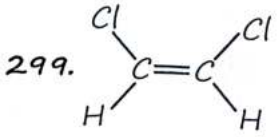
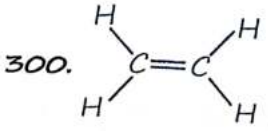
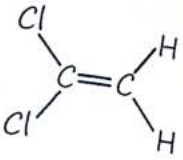
296. 

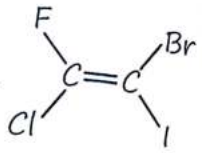
297. 

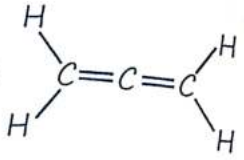
298. 

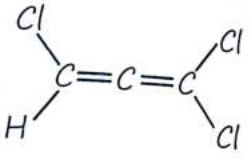
299. 

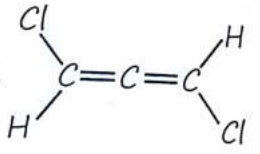
300. 

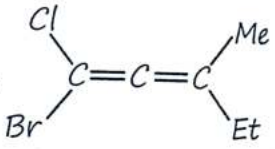
301. 

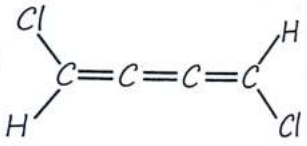
302. 

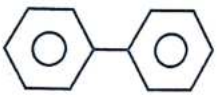
303. 

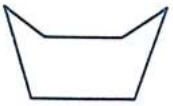
304. 

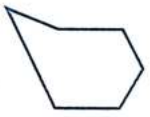
305. 

306. 

307. 

308. 

309. 

310. 

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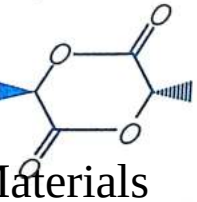
312. 

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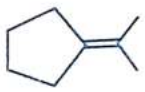
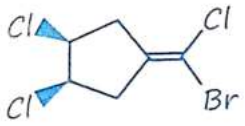
313. 

314. 

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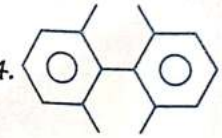
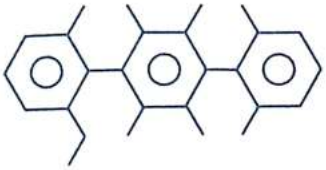
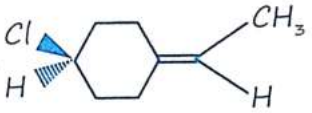
324. 

325. 

326. 

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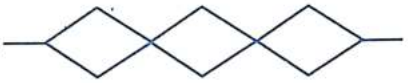
328. 

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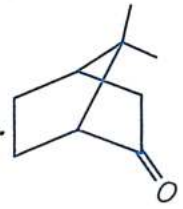
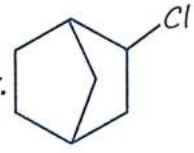
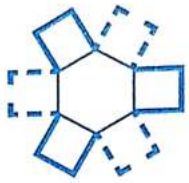
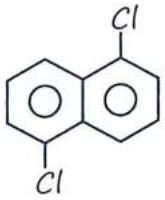
334. 

335. 

336. 

337. 

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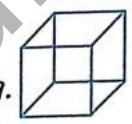
339. 

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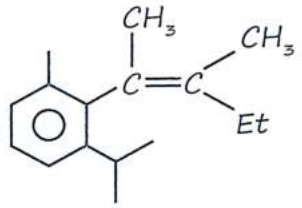
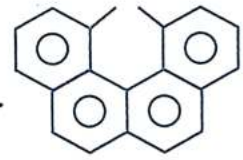
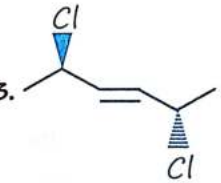
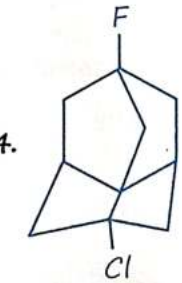
341. 

342. 

343. 

344. 

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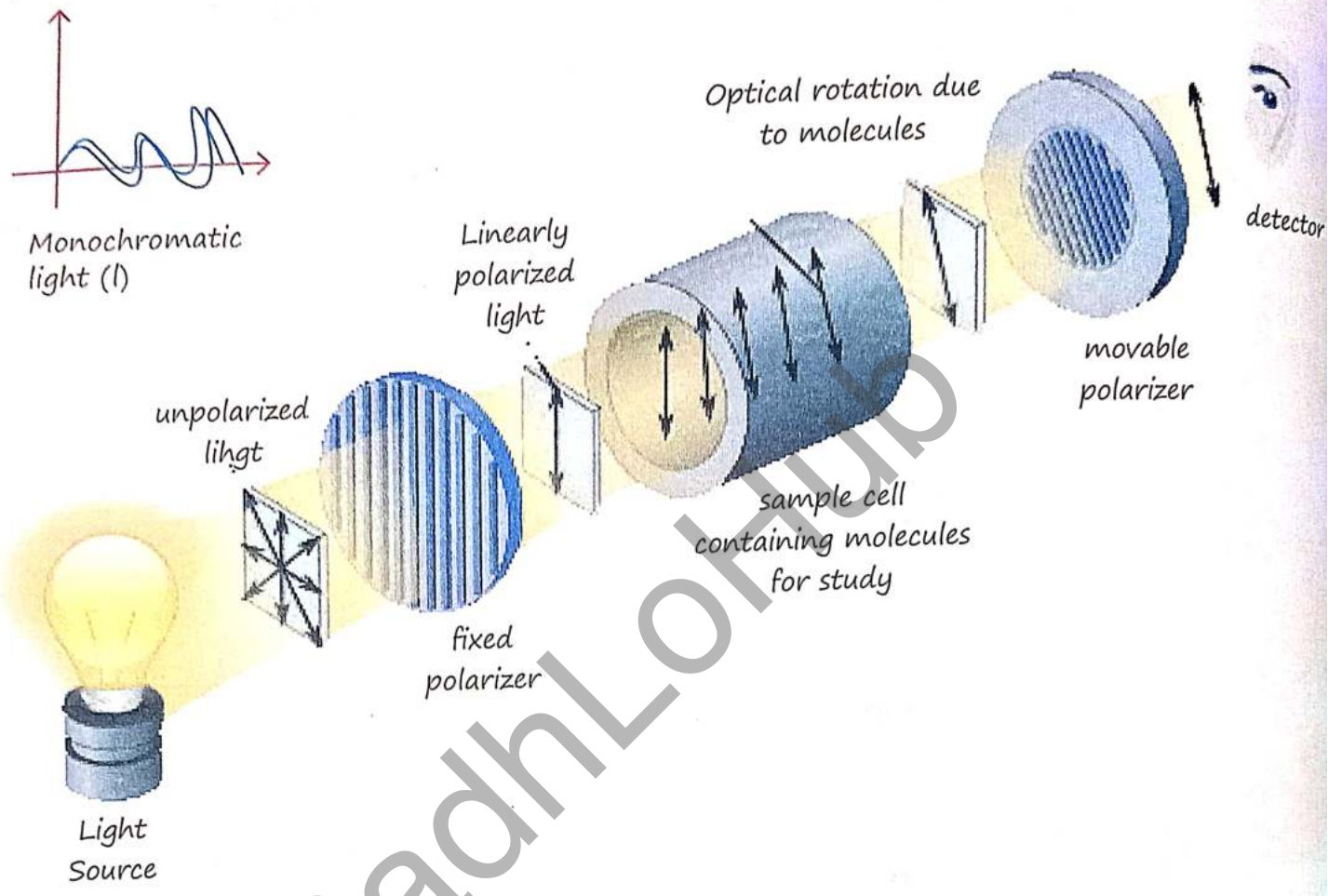
Optical Experiment



Observation

PPL → Bright zone → Optically inactive → 0°
PPL → Bright zone → Optically inactive → ±nπ°
PPL → Dark zone → Optically active → +θ° or C.W or Dextro rotary (d)
PPL → Dark zone → Optically active → θ° or A.C.W. or Laevoratatory (l)

Factors

(1) Concentration of compound
(2) Length of the tube
(3) Solvent
(4) Light source (λ)
(5) Temperature condition

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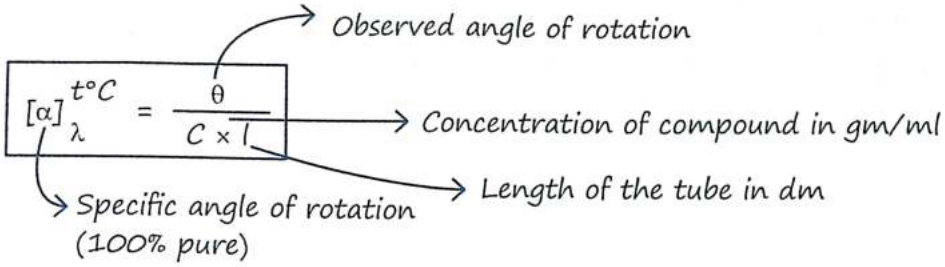
Calculation



No.	%d	%l	Enantiomeric mixture (Racemic)	Enantiomeric excess (Purify)	Net rotation (+ or −)	Observed angle
1	100					2Cl°
2		100				
3	60					
4			60		−	
5		50				
6				80	+	
7	33.33					
8					−	12°
9		45				
10			30		+	

Practice Questions

345. Specific rotation of pure (−)-2-butanol is 21°. After mixing rotation will be +12°. Find %d & optical purity.

346. A sample of + & − has rotation −130° & optical purity is 80% then find α.

347. A solution of unknown compound 3 g of compound in 20 ml solution when placed in tube of 2 dm, it rotated the ppl 1.8° in anticlockwise direction. Find specific rotation of the compound.

348. Calculate the specific angle of rotation when 0.5 M solution of (+) 2-butanol is taken in sample tube of length 10 cm . Angle of rotation observed is +20°.

349. 5M solution of (+) 2-chlorobutane whenever taken in sample tube of length 100 mm then observed angle of rotation is +10°. Then Calculate
   (a) $[\alpha]^{t°C}\lambda$
   (b) $[\alpha]^{t°C}\lambda$ and α_{obs} if length is halved
   (c) $[\alpha]^{t°}$ and λ and α_{obs} if length is double and dilution is doubled.

350. When 1 gm/ml of an optically active compound is taken in a sample tube of length 10 cm and α_{obs} = +357.75° when same the sample is taken in tube of length 20 cm , observed angle is +355.50°. Calculate Specific angle of rotation.

351. Optically pure 2-butanol has a specific rotation of +13.52 degrees. A synthesized and purified sample of 2-butanol has the observed specific rotation of +6.76 degrees. The correct statement based on this observation is
   (a) the sample is completely racemized
   (b) 25% of the sample is racemic
   (c) 50% of the sample is racemic
   (d) 6.76% of the sample is racemic

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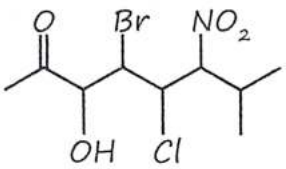
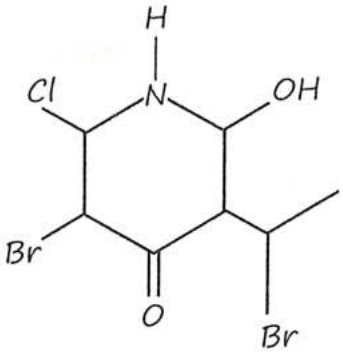
Chiral Centre

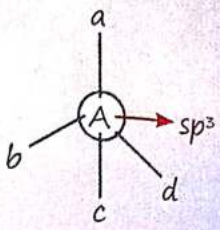
- If all four atom/group around a sp^3 centre are different then that sp^3 centre is known as chiral cent
- Example:

(1)



(2)



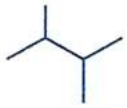


$a \pm b \pm c \pm d$

Chiral centre or Asymmetric centre or Stereocentre

Practice Questions

352.



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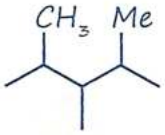

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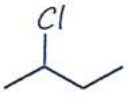

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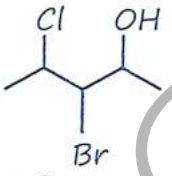

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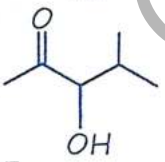

357.

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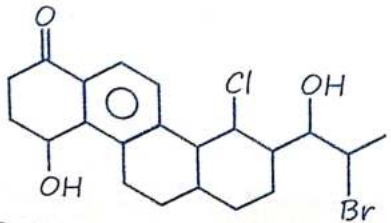

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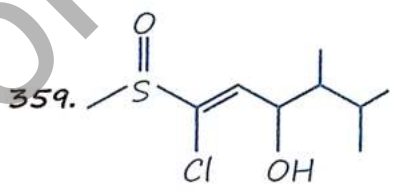

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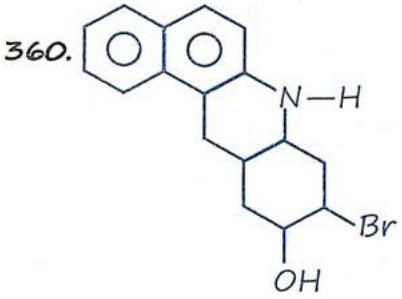

361.

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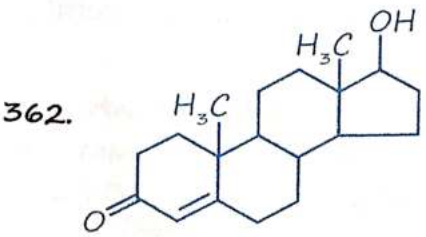

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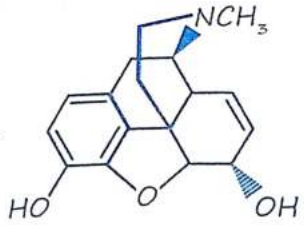
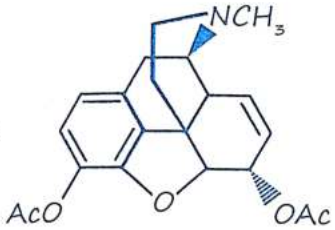
363.



Morphine

364.



Heroin

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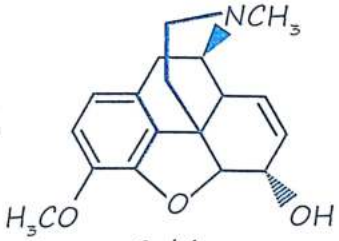

Codeine

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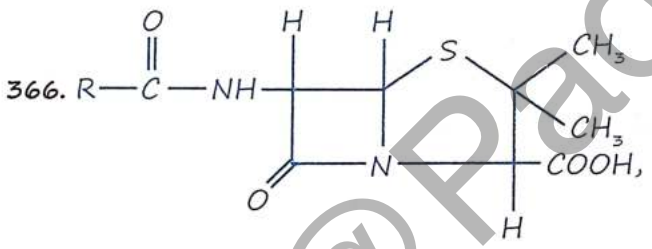
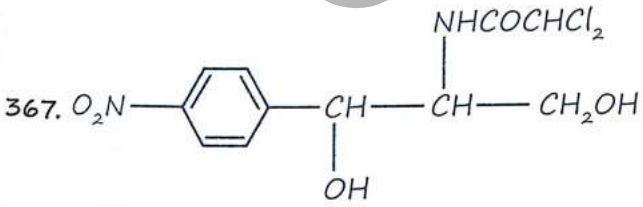


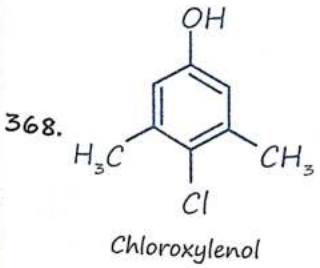
General Structure of Pencillin

367.



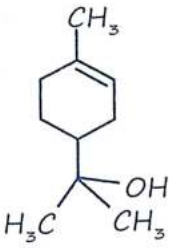
Chloramphenicol

368.



Chloroxylenol

369.



Terpineol

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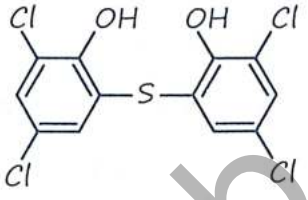
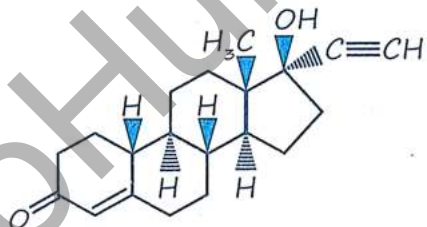


Bithionol

371.



Norethindrone

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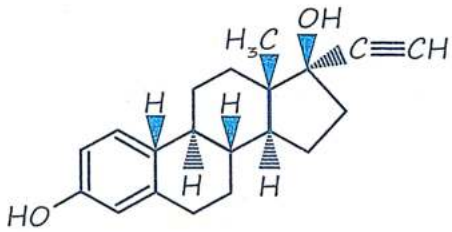

Ethynylestradiol (novestrol)

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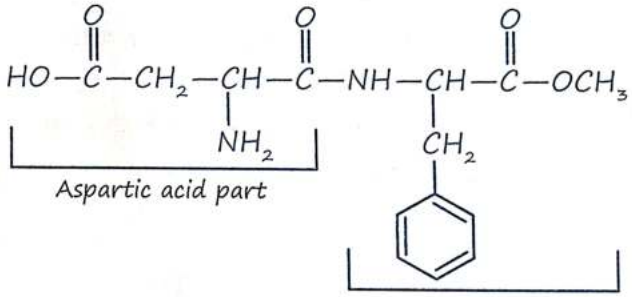
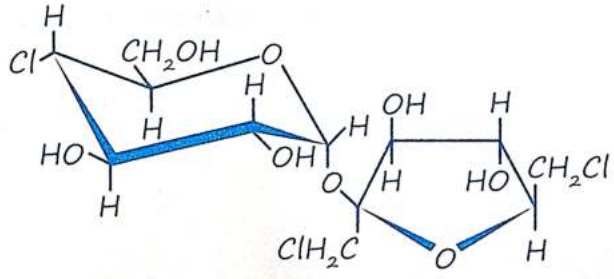


Aspartic acid part

Phenylalanine methyl ester part

374.



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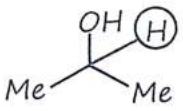
Important Terminology

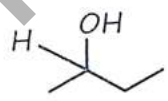
1. Optically active
2. Dextro (d)
3. Laevo (l)
4. Asymmetric
5. Disymmetric
6. Optically inactive
7. Optical Isomerism
8. Enantiomer
9. Diastereomer
10. Homomer
11. Meso
12. Racemic mixture
13. Stereo centre
14. Stereogenic centre
15. Pseudo centre

Optically Active Compound

- Compounds which rotates the plane polarised light (ppl) are known as 'Optically active' compound.
- They are also known as 'Chiral compound' or 'Resolvable compound'.
- For a compound to O.A. ⇒
  - POS → ×
  - COS → ×
  - AAOS → ×
  - AOS → ✓/×
  - Chiral Centre → ✓/×

  } 'AAOS' is must condition for chirality
- Mirror Image of an O.A. compound is different.
- Example:

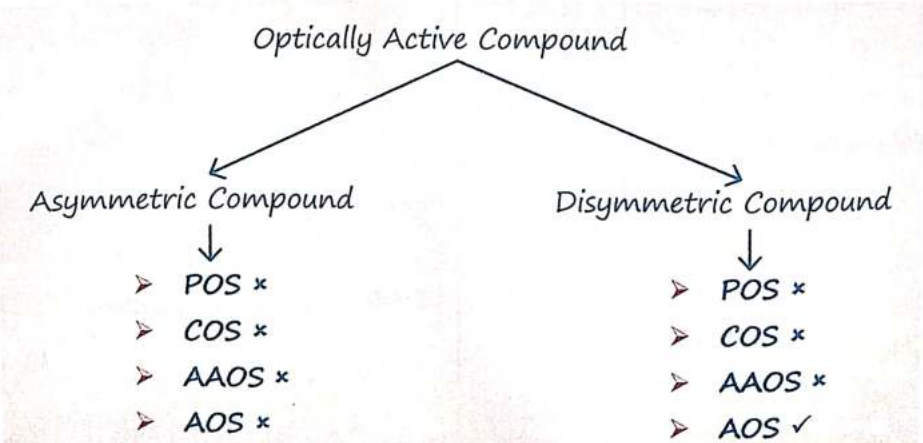
(a) 

(Achiral) (b) 

POS, COS — Chiral/O.A.

Dextrorotatory & Laevorotatory Compound

- Compounds which rotates the plane polarised light (PPL) in clockwise direction are known as 'dextrorotatory compound' (d).
- Compounds which rotates the plane polarised light (PPL) in anticlockwise direction are known as 'Laevorotatory compound' (l).
- d & D are not same & l & L are also not same.
- If one compound is 'd' then its mirror image will be 'l'.



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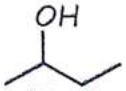
Optically Inactive Compound

- Compounds which not rotates the plane polarised light (PPL) are known as 'Optically inactive' compound.
- They are known as 'Achiral compound' or 'Non-Resolvable Compound'.
- For a compound to O.I. ⇒
  - POS →
  - COS →
  - AAOS →
  - AOS → ✓/×
  - Chiral Centre → ✓/×

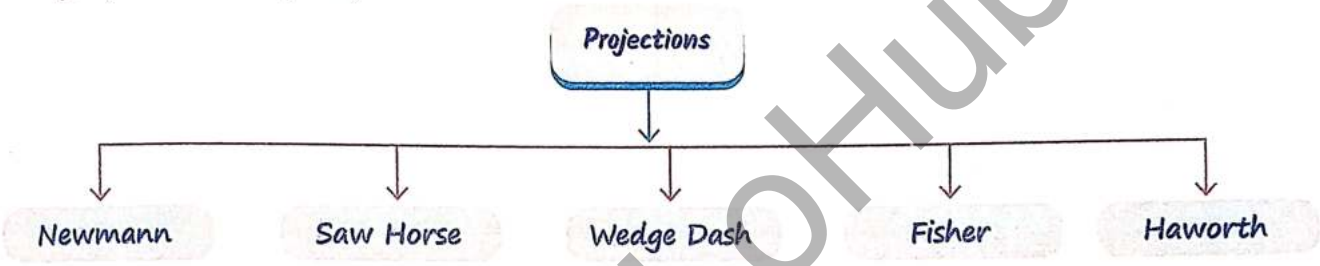
  } ⇒ Atleast one must be present
- Mirror Image of an O.I. compound is same.
- Example:

(a)

(O.I.)

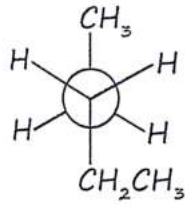
(b) 

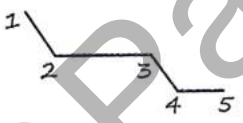
(O.A.)



(I) Newmann Projection

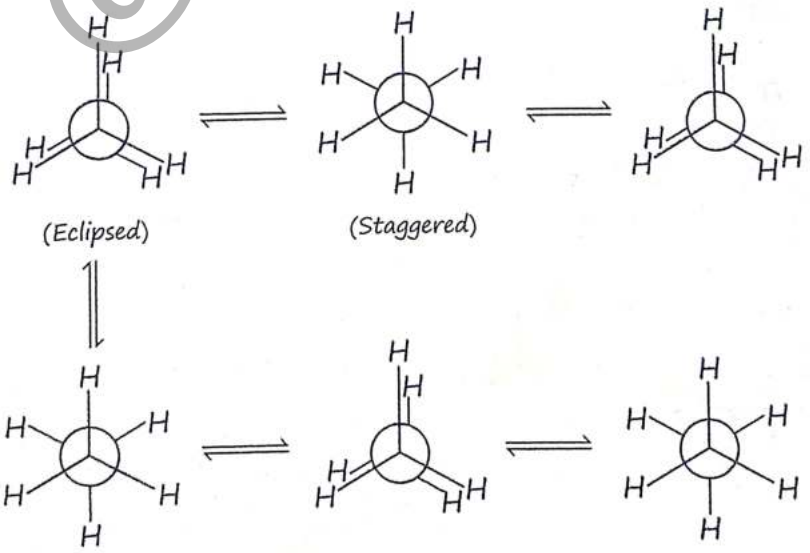
• ⟶ Front Carbon
O ⟶ Rear Carbon

(1) 



Pentane

(2) H_3C-CH_3



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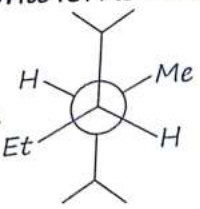
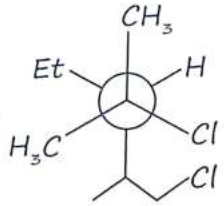
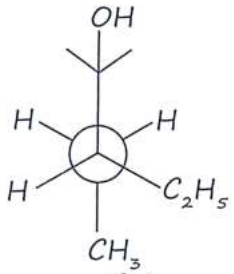
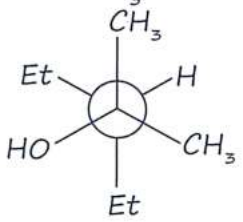
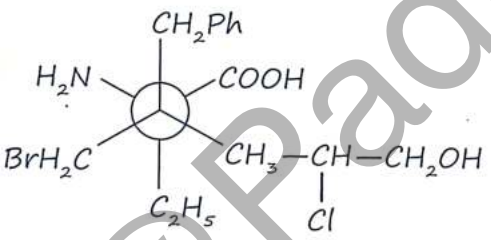
Q. Write IUPAC name of given compounds (375–382):

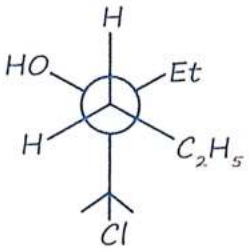
375. 

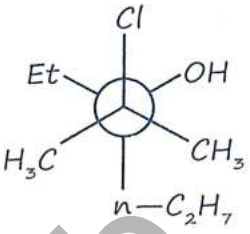
376. 

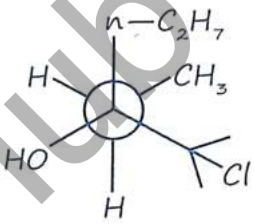
377. 

378. 

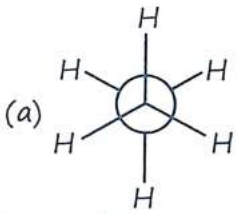
379. 

380. 

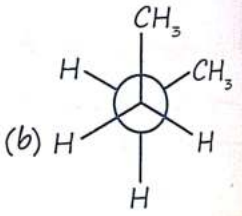
381. 

382. 

383. By the given rotation to following newmann projection which conformation is obtained?

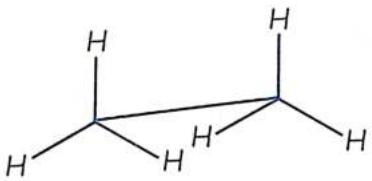
(a) 

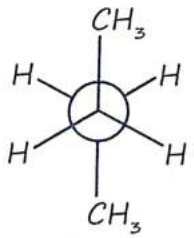
[120°clockwise: Front
240° Anticlockwise: Rear]

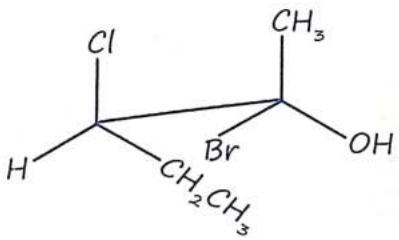
(b) 

[120°clockwise: Front
60° Anticlockwise: Rear]

(2) Saw Horse Projection

- Side view of newmann: 

- Example: 



⇒ IUPAC name??

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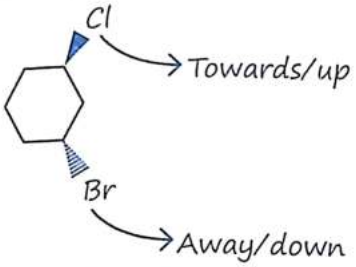
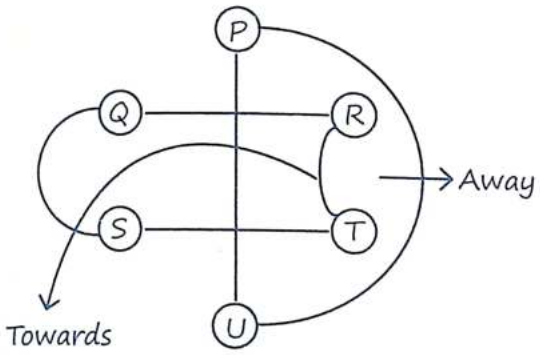
(3) Wedge-Dash Projection

- ◀——▶ up/towards the viewer
- ▬——▶ down/away from viewer
- Example:

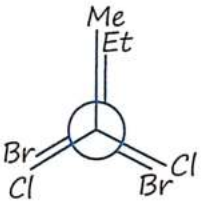


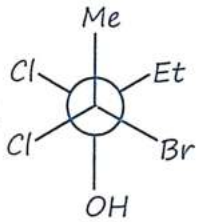
(4) Fisher Projection

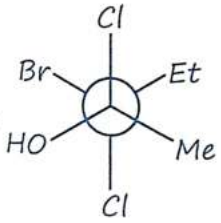


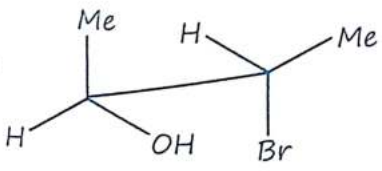
- Horizontal line ⇒ Towards
- Vertical line ⇒ Away
- Always drawn in eclipsed form.
- Always arrange newmann like
- Golden rule ⇒ follow IUPAC rules and carbon must be in a line.

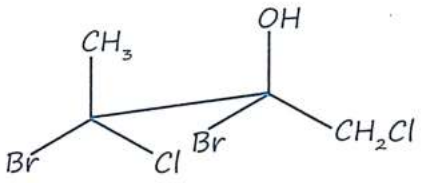
Practice Questions

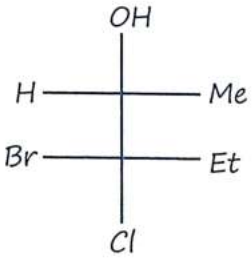
384. 

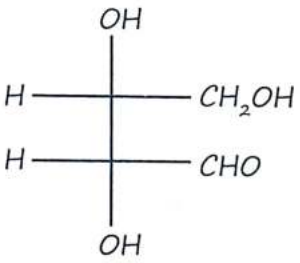
385. 

386. 

387. 

388. 

389. 

390. 

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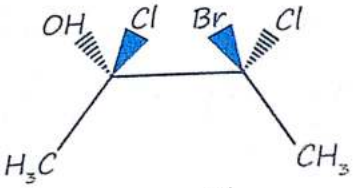
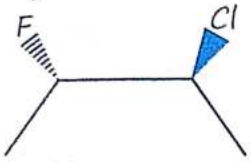
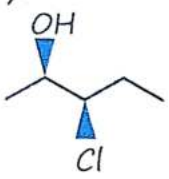
391.



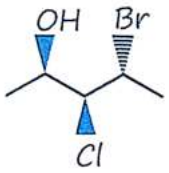
392.



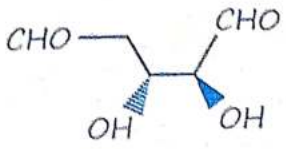
393.



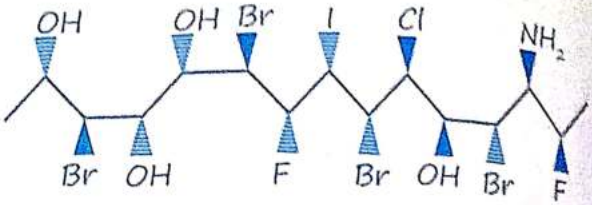
394.



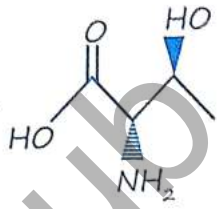
395.



396.



397.



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Nomenclature of Optical Isomers

- Absolute
- Relative
  - D/L
  - Threo/Erythro

D/L Nomenclature

- At last chiral center from top, if $-OH/-NH_2$ is on

Right side ⇒ D
left sede ⇒ L

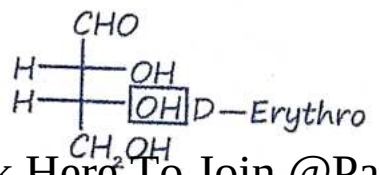
- If both $-OH$ and $-NH_2$ are present in compound then ($-NH_2$) will decide D/L

Threo and Erythro Nomenclature

- When two same group/atom are on

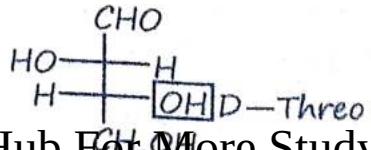
Same side ⇒ E
Opp side ⇒ T

Ex: (1)



D–Erythro

(2)



D–Threo

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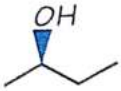
R,S Nomenclature

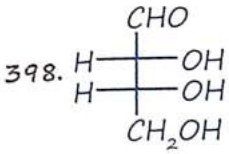
- First decide (1) (2) (3) according to C.I.P. rule.
- Then move (1) → (2) → (3)

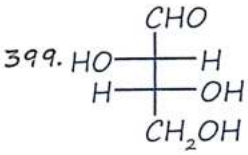
Ex: 

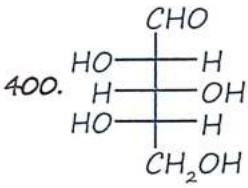
- If 4th priority is away then:
- If 4th priority is towards then:

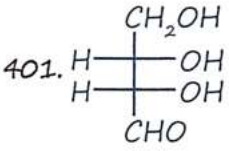
Practice Questions

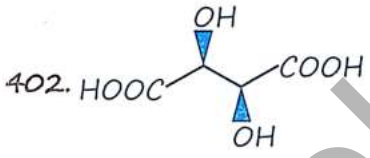
Q. Find D/L, R/S nomenclature?

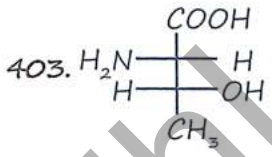
398. 

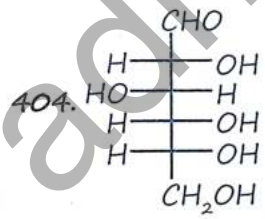
399. 

400. 

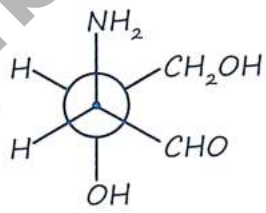
401. 

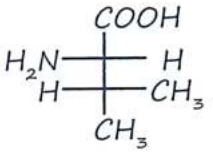
402. 

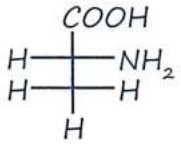
403. 

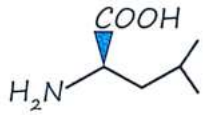
404. 

405. Find L—Glucose

406. 

407. 

408. 

409. 

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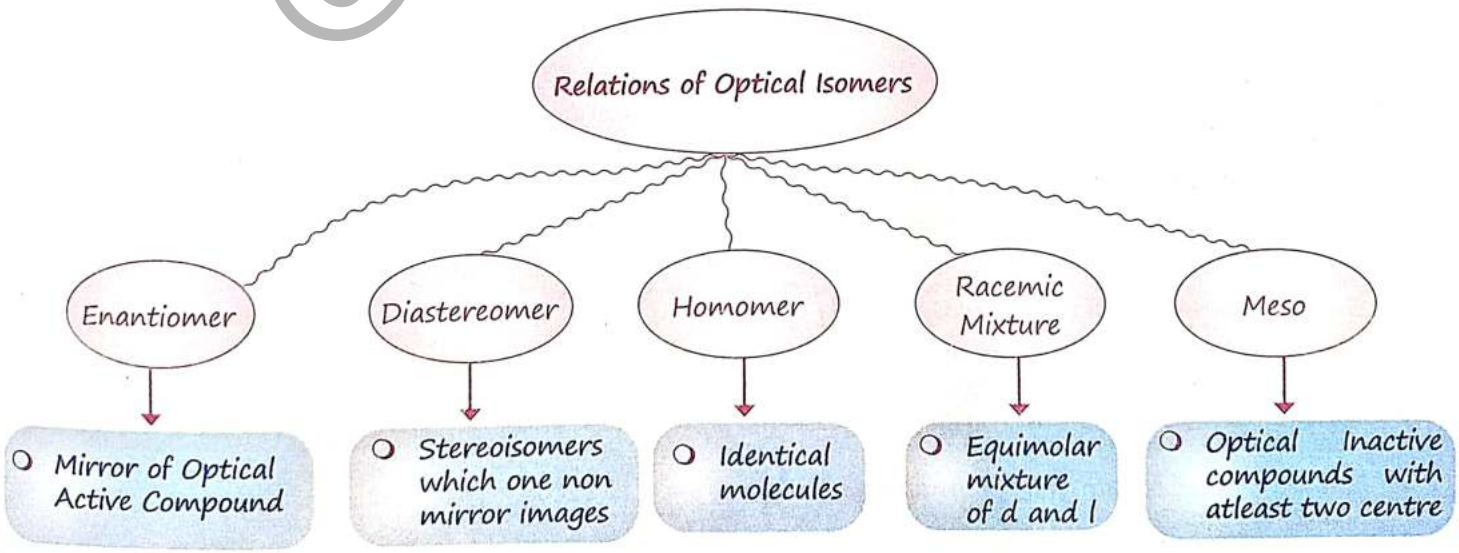

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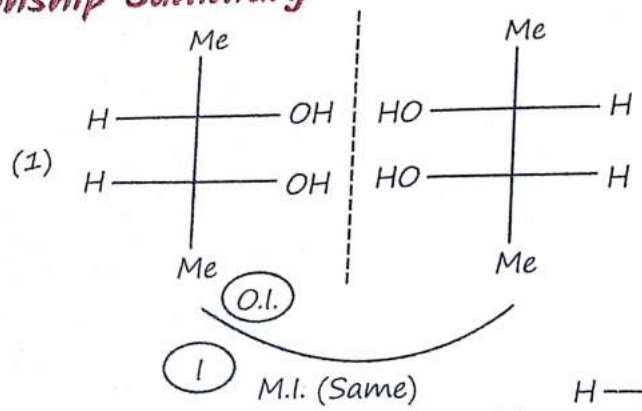
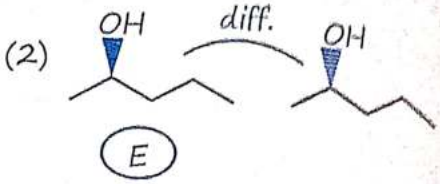
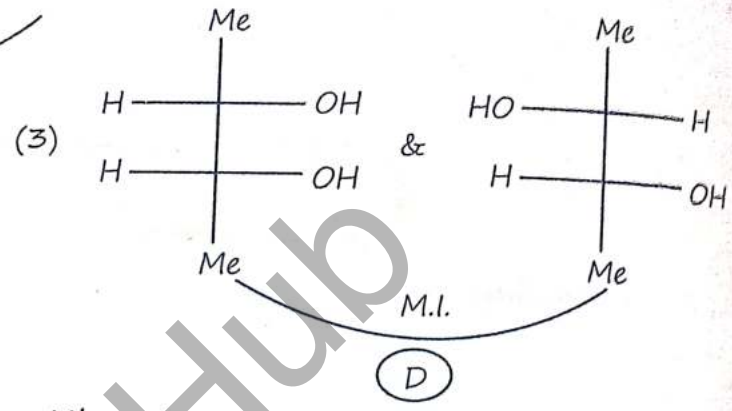
Relationship Summary



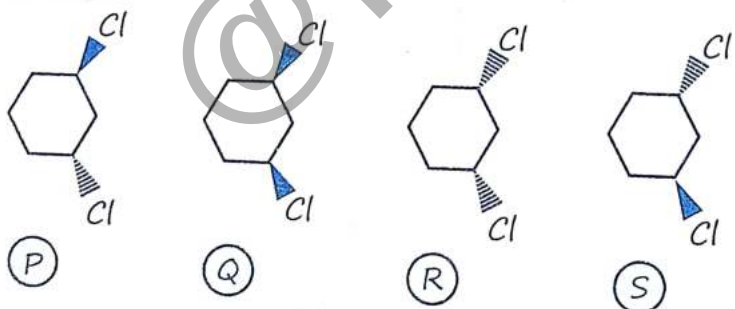




Mirror Images

- Yes
  - Different → Enantiomer
  - Same → Identical or Homomer
- No → Diastereomer

- Example:



- P – S ⇒
- P – Q ⇒
  - P – Q
  - P – R
  - R – S
- Q – S ⇒
- 50% P + 50% S ⇒
- Q ⇒

- Non-Super imposable (different) mirror images ⇒ (E)
- Super imposable (different) mirror images ⇒ (I)/(H)
- Non-super imposable non mirror images ⇒ (D)
- Super imposable (same) non mirror images ⇒ (I)/(H)

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Enantiomer

1. Mirror images of optically active compound
2. They are Non superimposable mirror images of each others.
3. They have same physical properties like boiling point, melting point, solubility, refractive index etc.
4. Only difference in optical rotation towards PPL
5. They can't be separated by fractional distillation and crystallisation.

Diastereo isomerism

1. Stereoisomers which are non mirror images of each other
2. They are Non superimposable non mirror images of each others.
3. They have different physical properties like boiling point, melting point, solubility, refractive index etc.
4. They can be optically active or inactive.
5. They can be separated by fractional distillation and crystallisation.

Flomomer

1. Mirror images of optically inactive compound or compound obtained by just rotation.
2. They are superimposable on each others.
3. They have same physical properties like boiling point, melting point, solubility, refractive index etc.
4. They can be optically active or inactive.
5. They can't be separated by fractional distillation and crystallisation.

Meso Compound

1. Optically inactive compound with atleast 2 chiral centre.
2. Internal compensation.
3. They shows optical isomerism.

Racemic Mixture

1. Equimolar mixture of Enantiomers
2. External compensation.
3. They shows zero rotation towards PPL.

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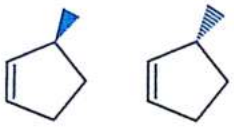
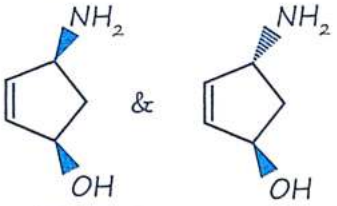
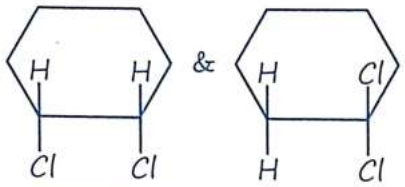
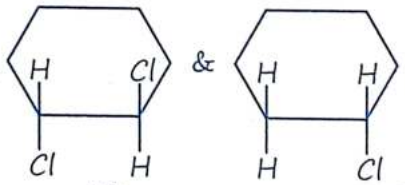
Practice Questions

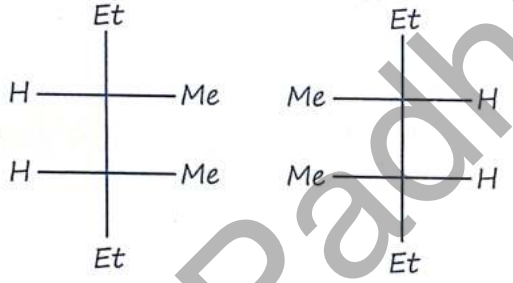
Find relation between given pairs of compounds:

410. 

411. 

412. 

413. 

414. 

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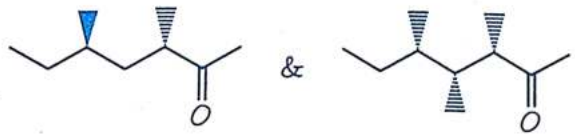
416. 

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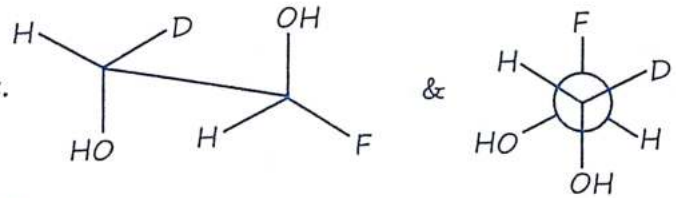
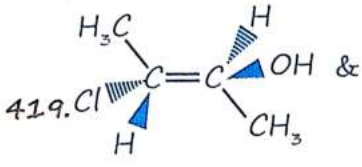
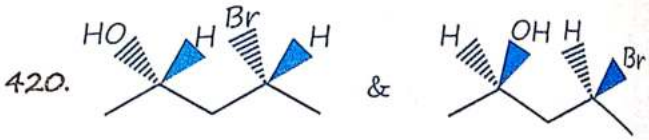
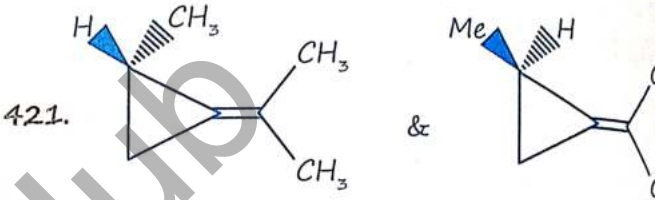
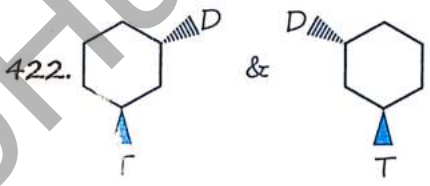
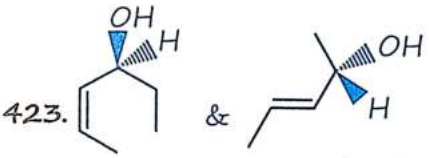
418. 

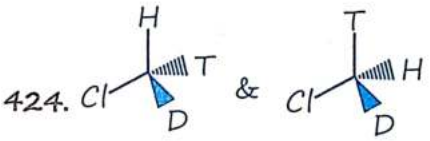
419. 

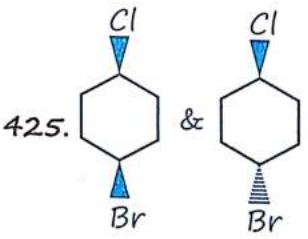
420. 

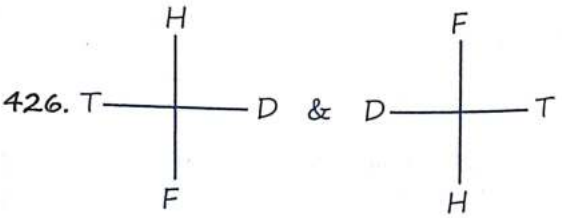
421. 

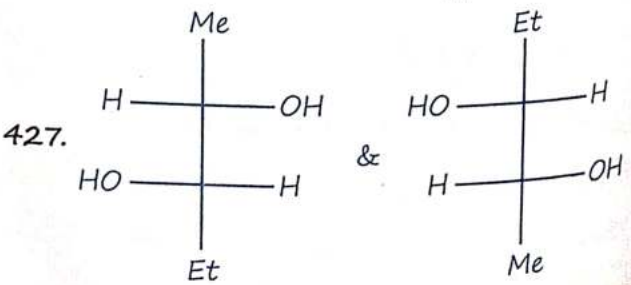
422. 

423. 

424. 

425. 

426. 

427. 

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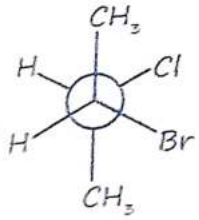
2-Bromo-3-Chloro Butane:

428.



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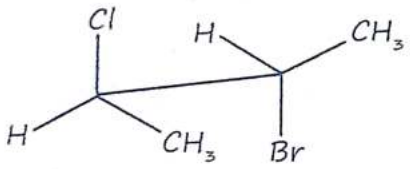

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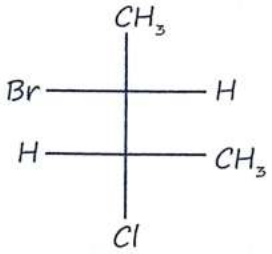

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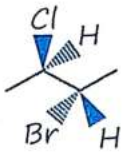

(S)

Cumulated ene:

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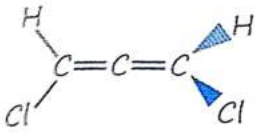

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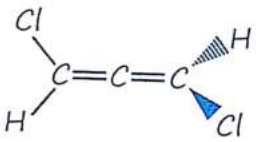

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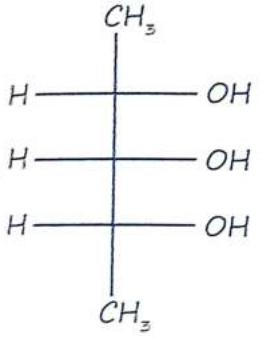

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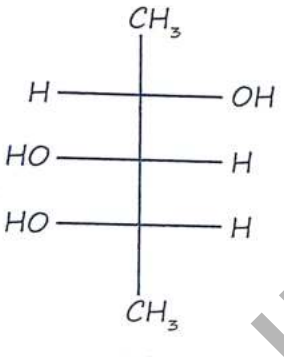

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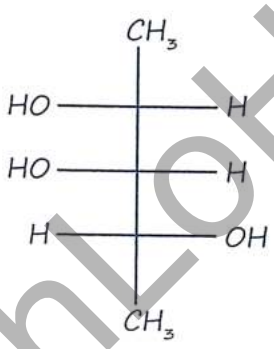

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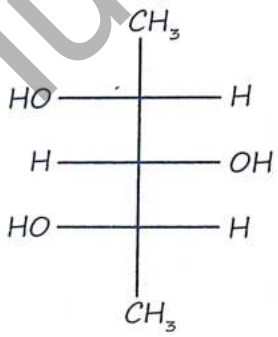

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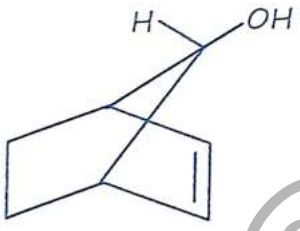

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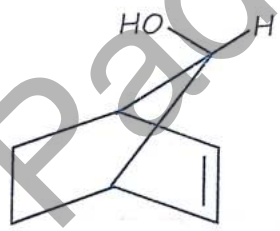

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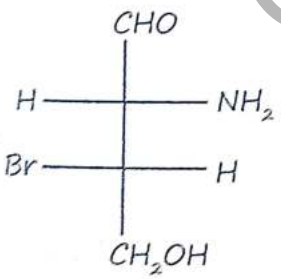

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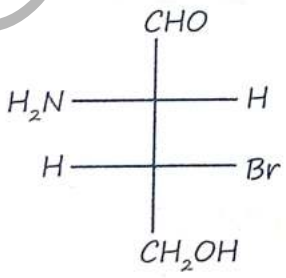

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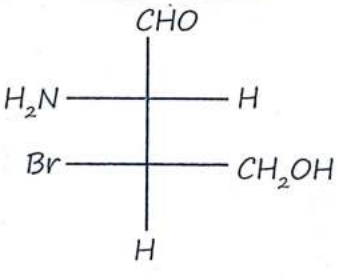

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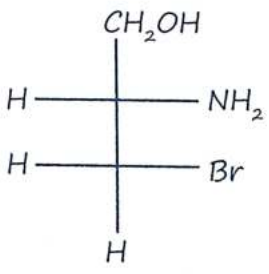

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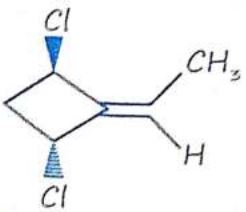

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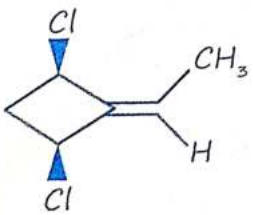

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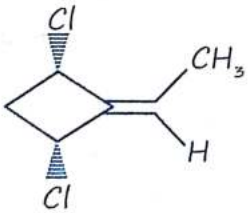

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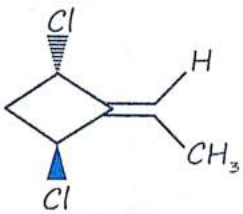

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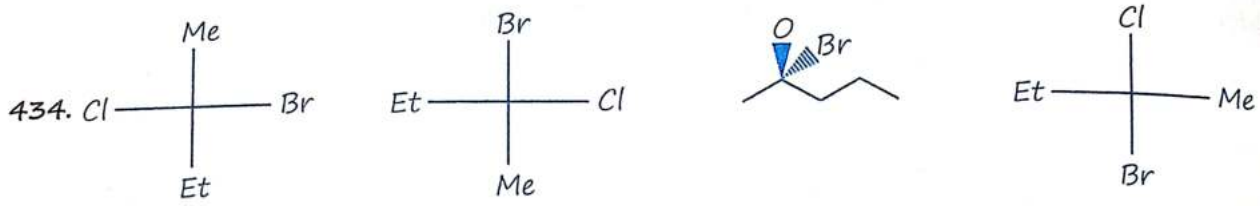
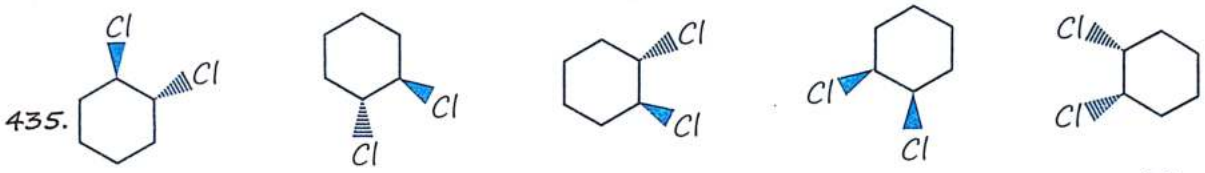
434. 

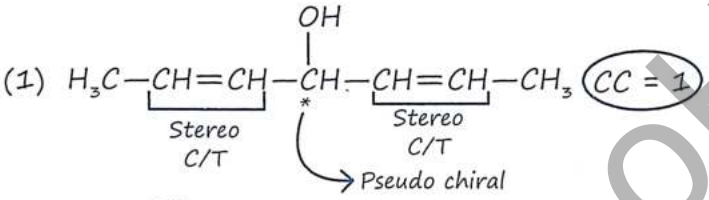
(P) (Q) (R) (S)

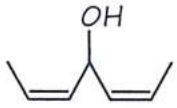
435. 

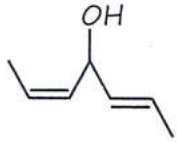
(P) (Q) (R) (S) (T)

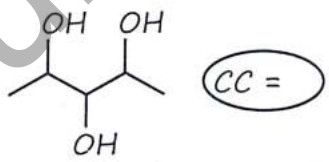
Pseudo Chiral Centre

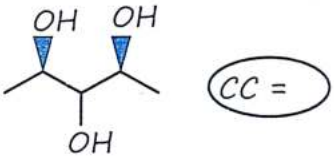
A centre which depends on stereochemistry of surrounding is known as pseudo chiral chentre.

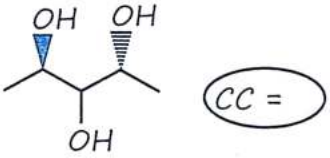
(1) 

(2) 

(3) 

(4) 

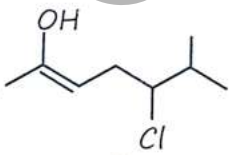
(5) 

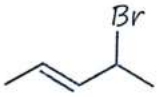
(6) 

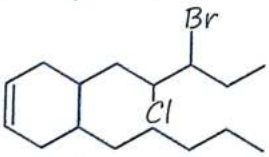
Calculation of Stereoisomers

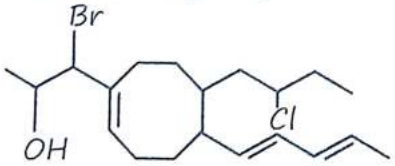
2^n → Units which shows stereoisomerism (OI, GI)

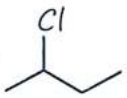
Practice Questions

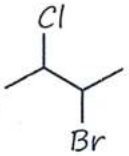
436. 

437. 

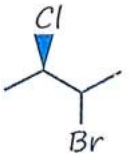
438. 

439. 

440. 

441. 

442.

443. 

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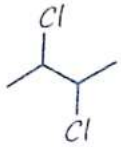
444. $CH_3-CH=CH-CH_3$

445. $CH_3-CH=CH=CH-CH_2-CH_3$

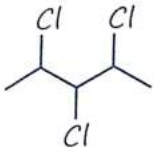
446. $CH_3-CH=CH-CH-\underset{\displaystyle OH}{\underset{|}{CH}}-CH_3$

447. $CH_3-CH=CH-CH=CH-CH_3$

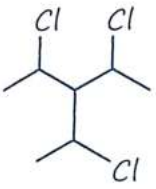
448. $H_3C-CH=CH-CH=CH-CH=CH-CH_3$

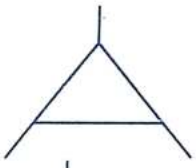
449. 

450. $H_3C-\underset{\displaystyle OH}{\underset{|}{CH}}-CH=CH-\underset{\displaystyle OH}{\underset{|}{CH}}-CH_3$

451. 

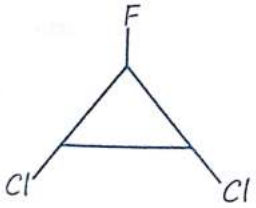
452. $(H_3C-CH=CH)_2-\underset{\displaystyle OH}{\underset{|}{CH}}$

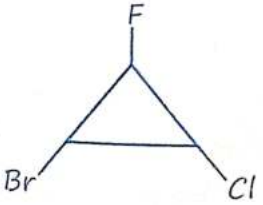
453. 

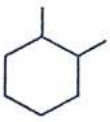
454. 

455.

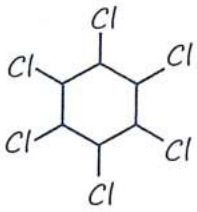
456.

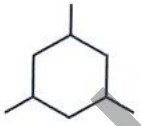
457. 

458. 

459. 

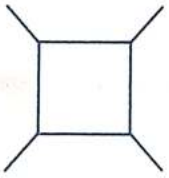
460.

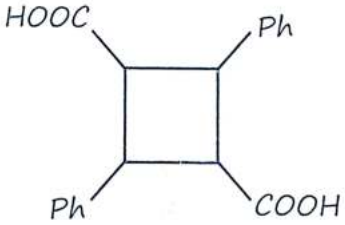
461. 

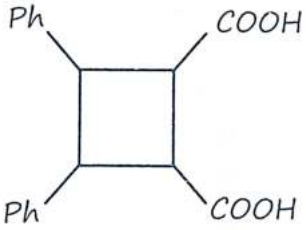
462. 

463.

464.

465. 

466. 

467. 

468.

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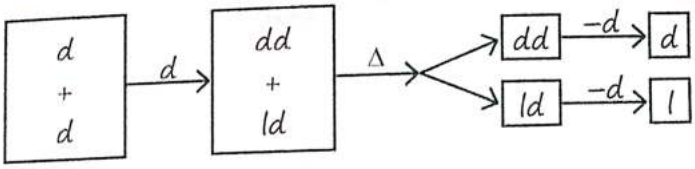
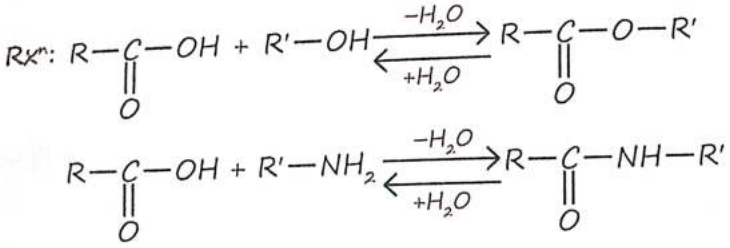
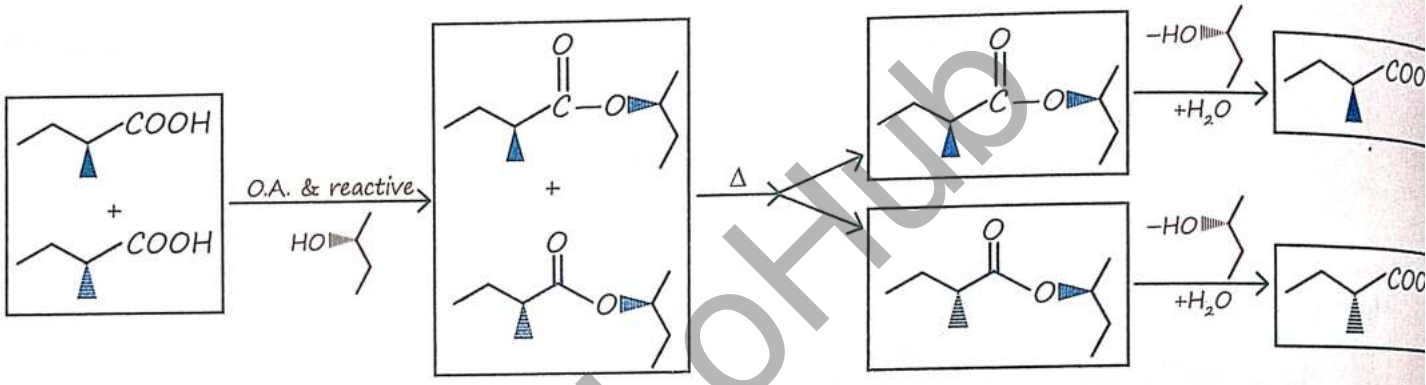
Resolution of Enantiomers

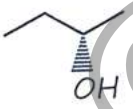






Practice Questions

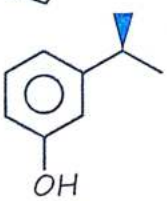
Q. Which of the following can be used to resolve enantiomeric mixture of recemic 2-methyl butanoic acid?

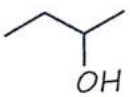
469. 

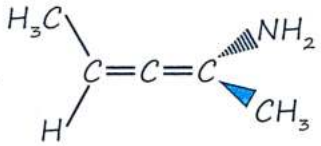
470.

471.

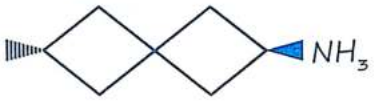
472.

473. 

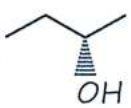
474. 

475. 

476.

477. 

Q. Which of the following can be used to reso enantiomeric mixture of racemic amino a valine?

478. 

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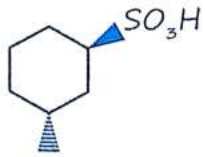
480. 

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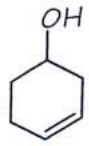
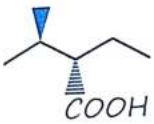
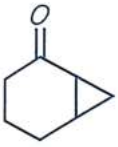
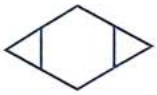
481. 

482. 

483.

➢ Find Stereoisomerism

484. 

485. 

486.

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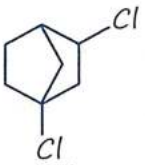
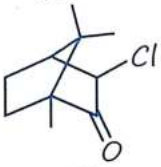
488.

489. 

490. 

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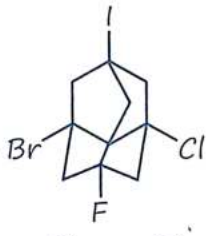
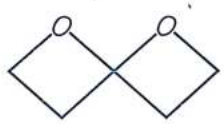
492. 

493. 

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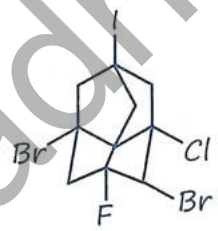
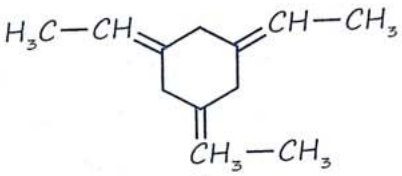
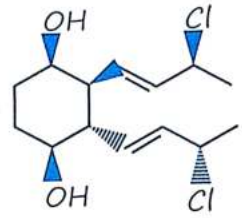
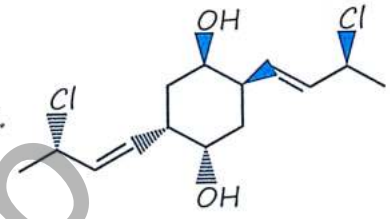
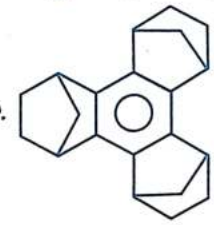
495. 

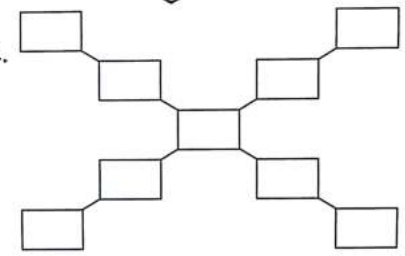
496. 

497. 

498. 

499.

500. 

501. 

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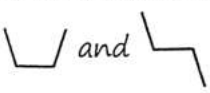
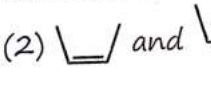
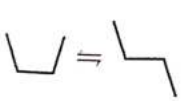
Conformational Isomerism

Compound having same molecular formula, different arrangement across single bond is known as conformational isomers and phenomenon is known as conformational isomerism.

Ex: (1)  and (2)  and

➢ 

➢ They are also known as rotamers

➢ The are not true isomers because they are infinite.

➢ They have same physical and chemical properties.

➢ They are inter convertible to each other.

➢ Potential energy graph of conformations is known as conformational analysis.

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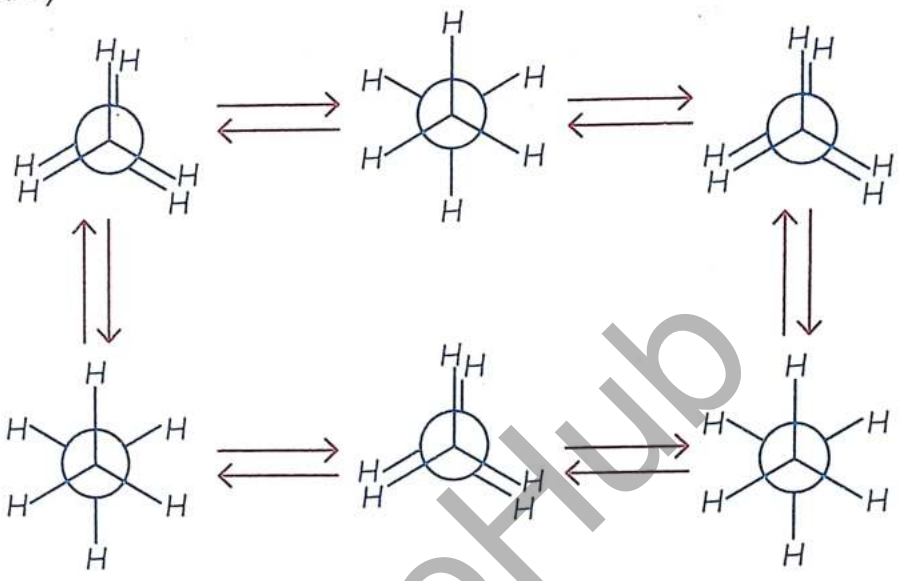
Q. What is the difference of conformations and conformer?

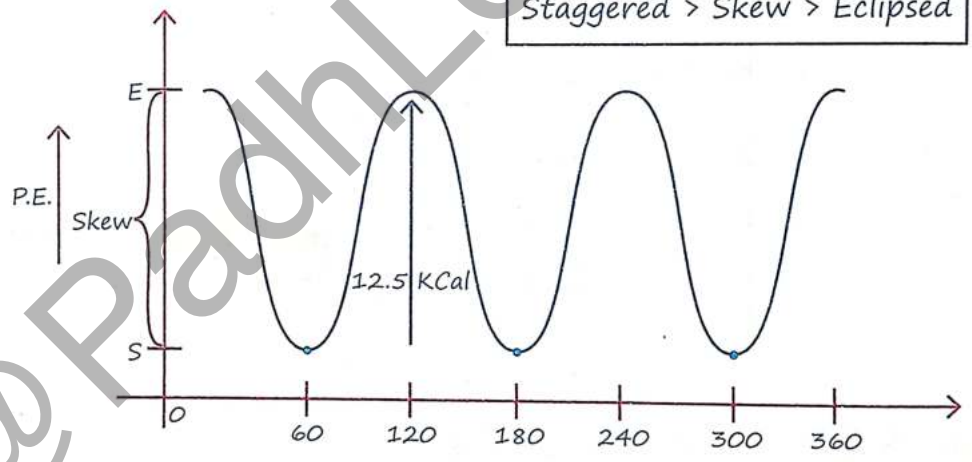
- All arrangements across σ-bond are know as conformations and they are infinite.
- Arrangements at minima in potential energy analysis is known as conformer.
- Stable conformations (absolute or relative minima) are conformer.

Conformations

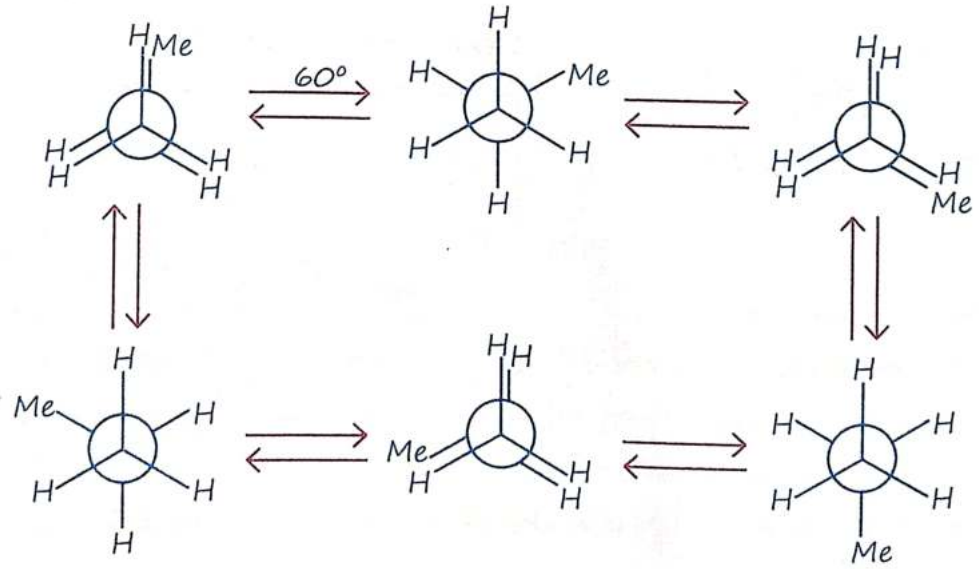
1. $CH_3—CH_3$ (ethane)



Staggered > Skew > Eclipsed



2. $CH_3—CH_2—CH_3$ (Propane)



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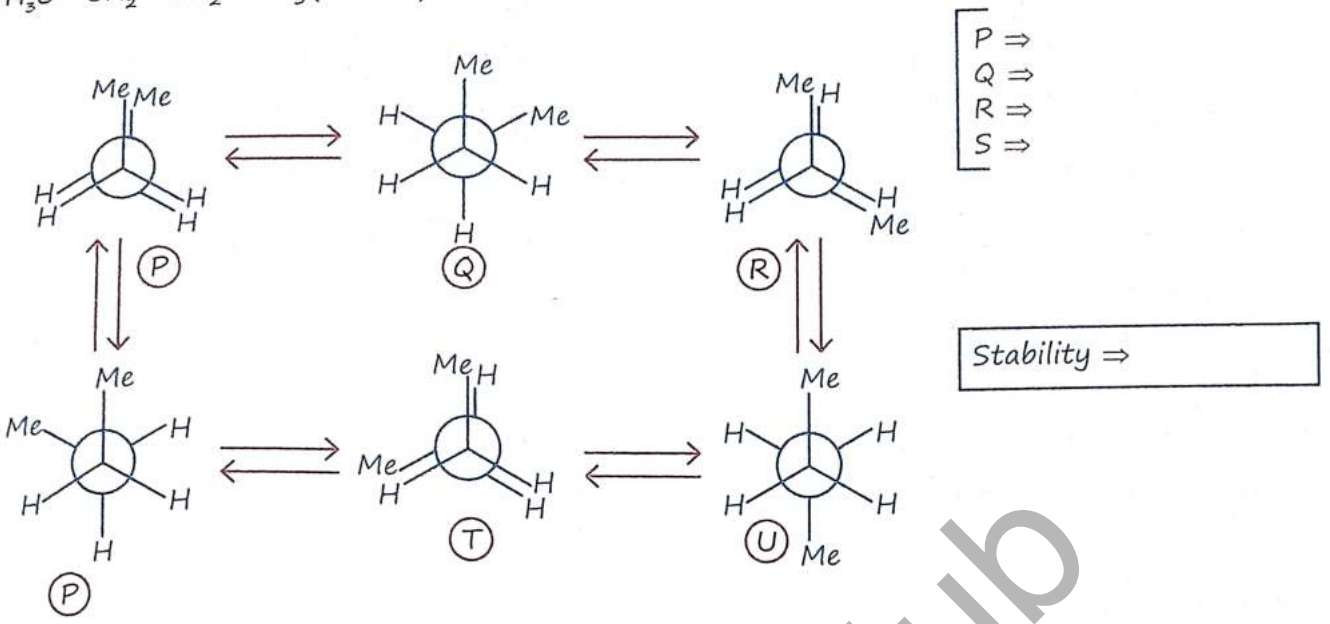
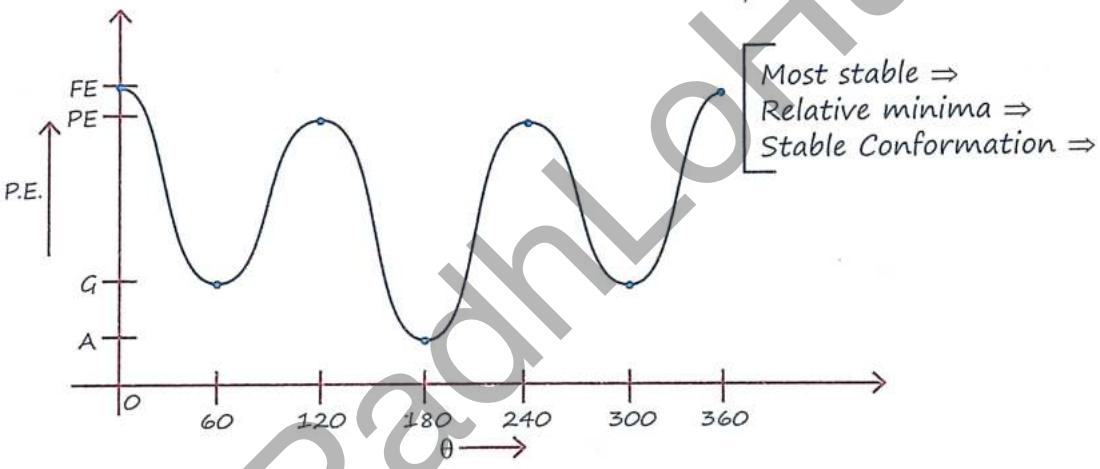
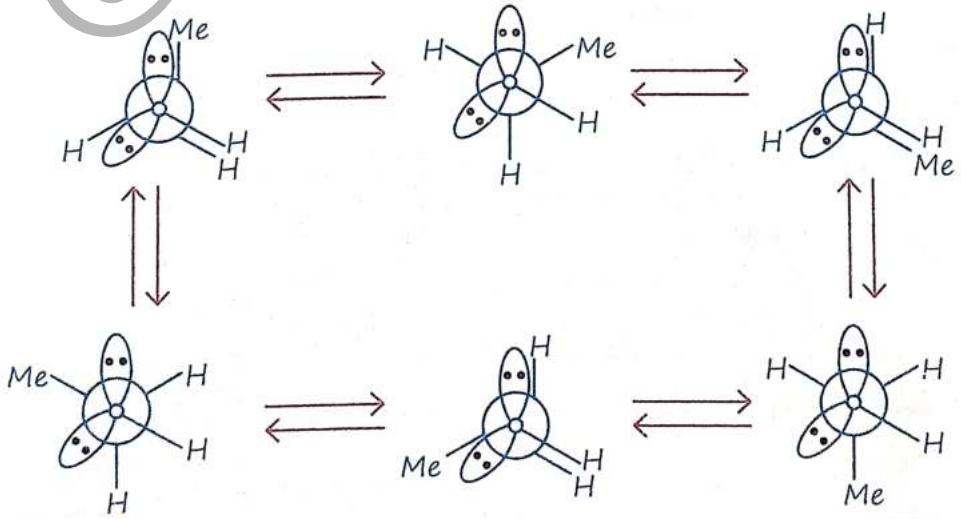
3. $H_3C-CH_2-CH_2-CH_3$ (Butane)





4. CH_3-CH_2-OH



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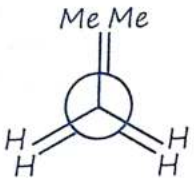
Important terminology

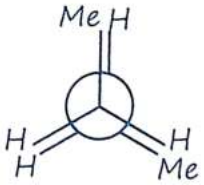
1. Eclipsed conformation: When groups/atoms at front atom is 0°/120°/240° to rear atom.

Eg:

(a) Fully eclipsed: When bulkiest groups/atoms at front atom is 0° to rear atom.

Eg: 

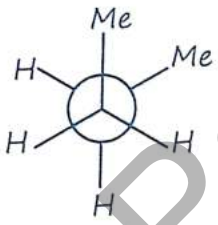
(b) Partially eclipsed: When bulkiest groups/atoms at front atom is 120°/240° to rear atom.

Eg: 

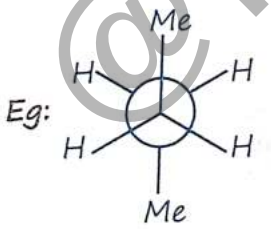
2. **Staggered Conformations:** When group/atoms at front is 60°/180°/300° to rear atom.

Eg:

(a) Gauche: When bulkiest groups/atoms at front is 60°/300° to rear atom.

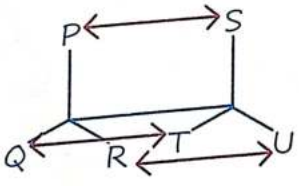
Eg: 

(b) Anti: When bulkiest groups/atoms at front atom is 180° to rear atom.

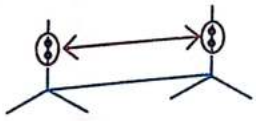
Eg: 

Strain:

(a) Vanderwal/Steric strain: Strain due to repulsion b/w groups/atom



(b) Torsional strain: Strain due to repulsion b/w bonded e⊙pair



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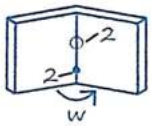
Dihedral Angle/Torsional Angle: Angle between two intersecting planes or Half planes.



Q. Find the dihedral angle of given conformational?

1. Eclipsed form of ethane
2. Staggered form of ethane
3. gauche form of n-butane
4. Anti form of n-butane

Skew conformations:

$0° < \theta < 60° \Rightarrow$ skew conformation

Stability $\Rightarrow$ Staggered > Skew > Eclipsed

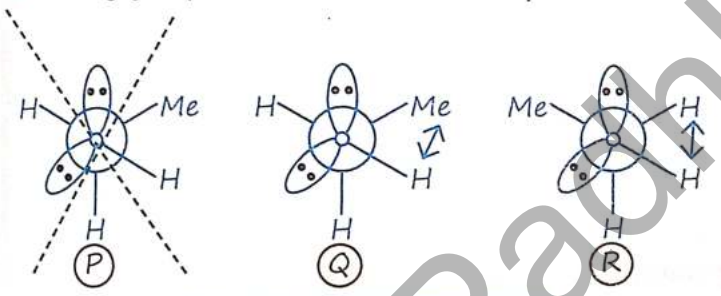
Stability of gauche and anti

Generally anti is more stable than gauche but gauche will be more stable if:

(a) Electrostatic attraction
(b) H—Bonding
(c) Gauche effect

Gauche effect

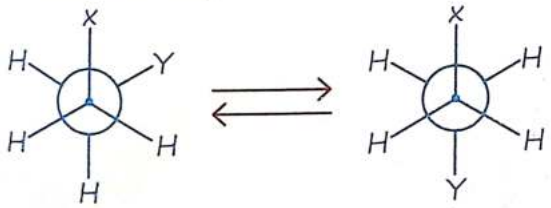
➢ Bulky group remains closer to lone pair.



Stability $\Rightarrow$

Practice Questions

Q. Find the stable conformation of given pairs (x,y) and also find Keq?



502. [CH_3, CH_3]
503. [CH_3, OH]
504. [CH_3, H]
505. [OH, OH]
506. [OH, F]
507. [OH, COOH]
508. [OH, NH_2]
509. [NH_2, COOH]
510. [COOH, COOH]
511. [COOH, COOH] at $P^H = 10$
512. [Cl, Cl]
513. [CH_3, Cl]
514. [H, H]
515. [$\overset{\oplus}{N}Me_3$, $\overset{\ominus}{C}OO$]
516. [$\overset{\oplus}{N}Me_3$, $\overset{\ominus}{O}$]
517. [:, CH_3]

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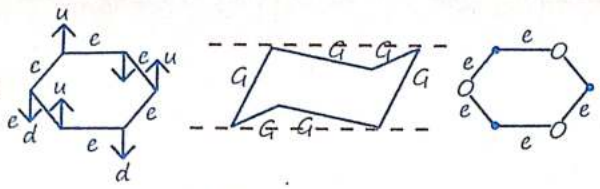
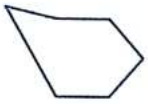
Cyclohexane

➢ Chair:



• → up/down
o → down/up

➢ Half Boat:



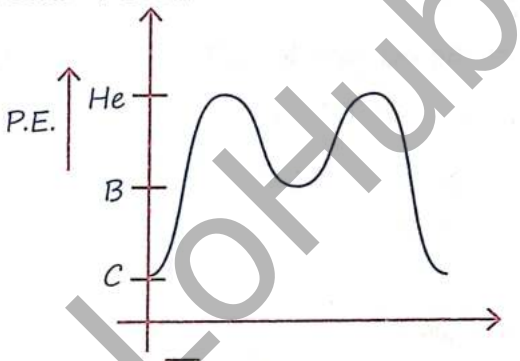
4e, 2G
Highly unstable
1 POS
O-Inactive

➢ Boat:

2e, 4G
Stable than H.b
2 POS
O-Inactive

Flipping about(σ)

Chair ⇌ Half boat ⇌ Boat ⇌ Half boat ⇌ Chair



➢ Half Chair:

3e, 3G
Highly unstable
POS, COS
O-Active

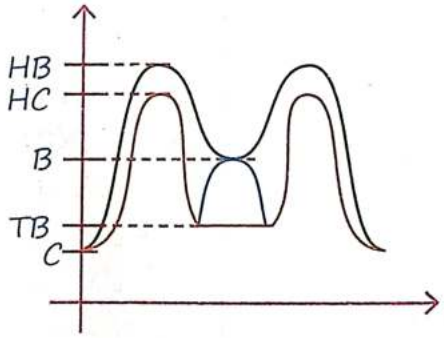
➢ Twist Boat:

6G
Stable but less than chair
POS, COS
O-Active

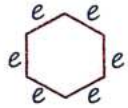
Flipping about axis

1. $C \rightleftharpoons H.C_1 \rightleftharpoons TB_1 \rightleftharpoons TB_2 \rightleftharpoons HC_2 \rightleftharpoons C$
2. $TB \rightleftharpoons B \rightleftharpoons TB$

Draw graph of path-1, path-2, path-3



➢ Flexible ⇒

➢  ⇒ Not exist

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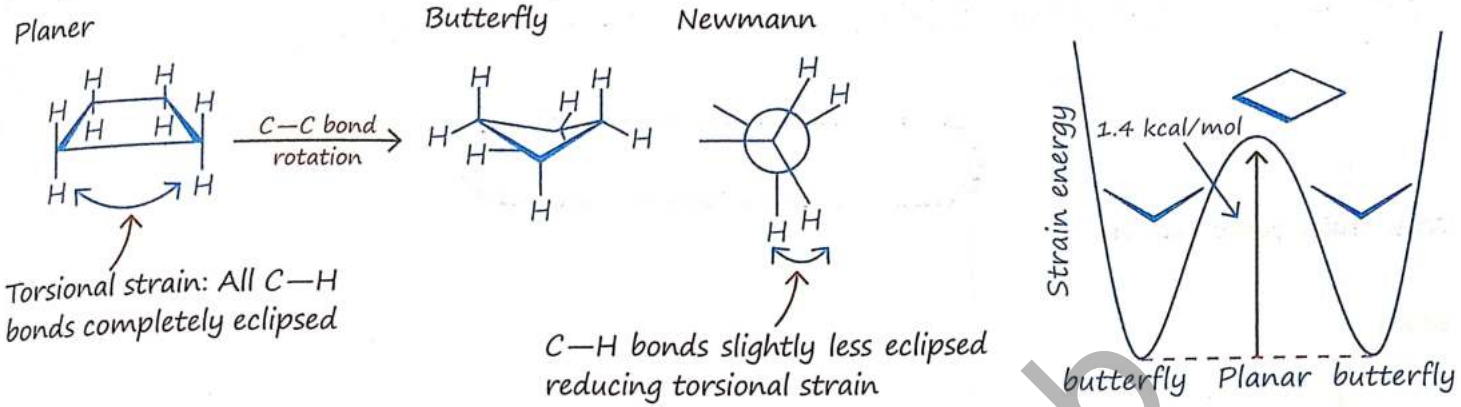
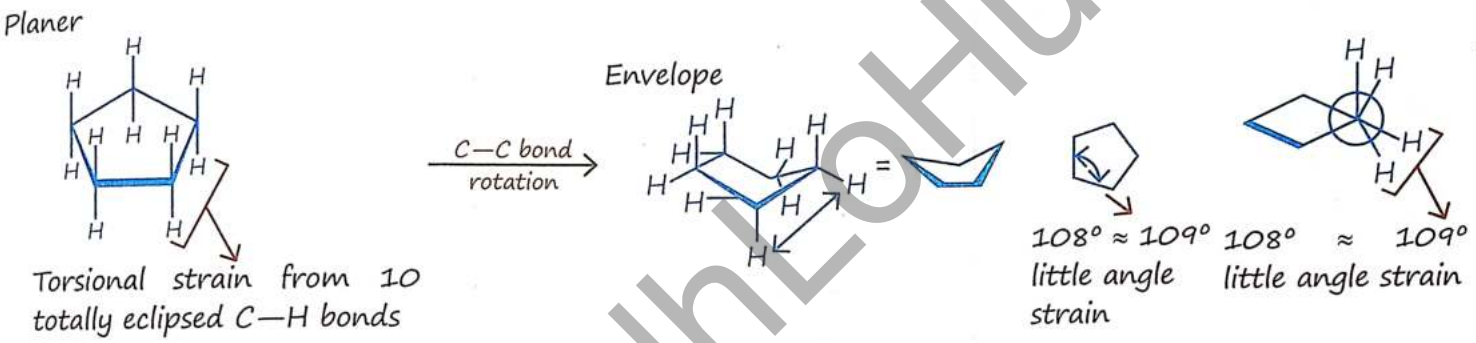
Cyclic Conformation

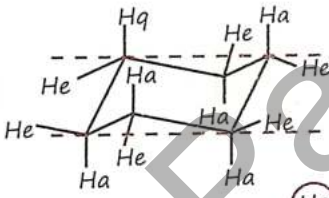
- All cycloalkanes are non-planar except cyclopropane.
- Cyclobutane



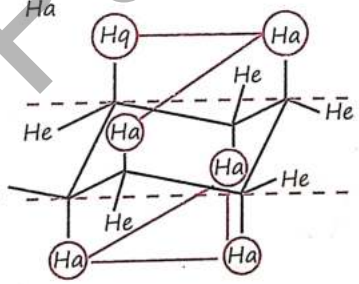
- Cyclopentane envelope form



- Bond in Chair:

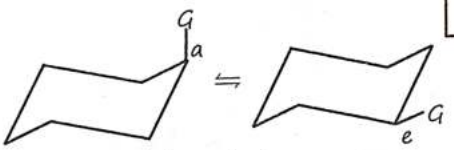


- Stability of bonds in chair:

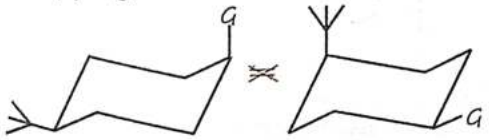


- Generally equitorial bonds are more stable than axial due to flag pole repulsion
- Axial bond can be more stable if:
  - (a) H—bonding
  - (b) Ionic attraction
  - (c) lane pair

- Flipping of Cyclohexane:



Due to flipping, axial becomes equatorial and vice-versa



Due to (t-Bu), flipping is not possible

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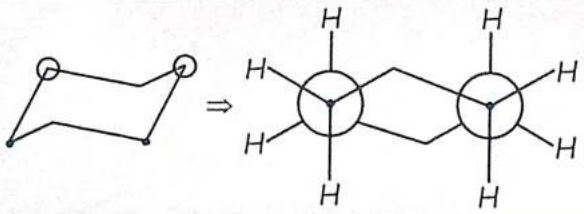
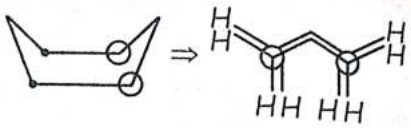
Newmann of chair and Boat

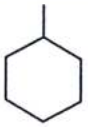




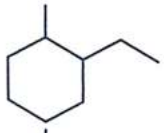
Practice Questions

Draw stable conformation?

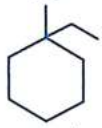
518.



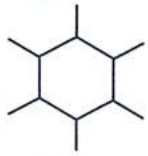
519.



520.



521.



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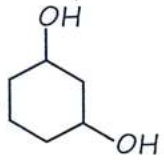

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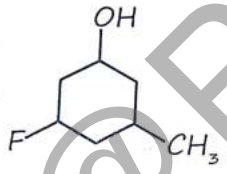

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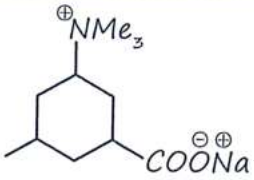
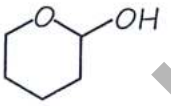
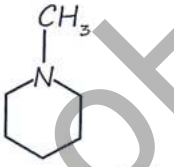


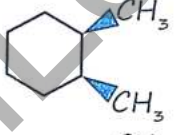
525.



526.



527.



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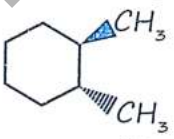

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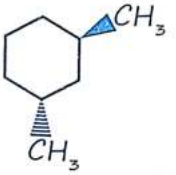
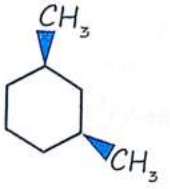
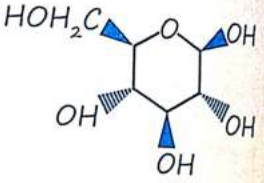


530.

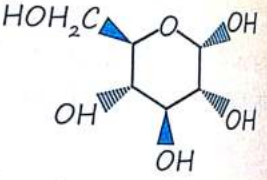


531.

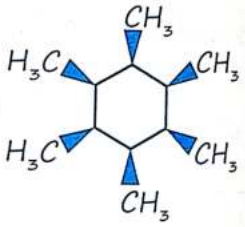
532.



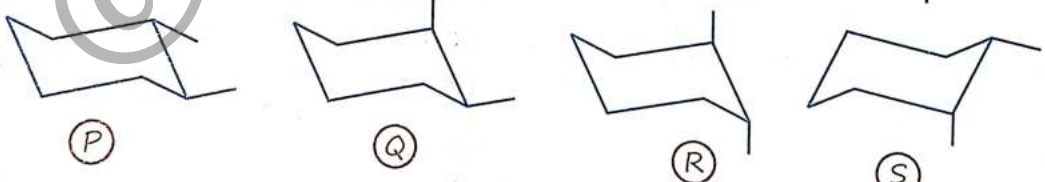
533.



534.

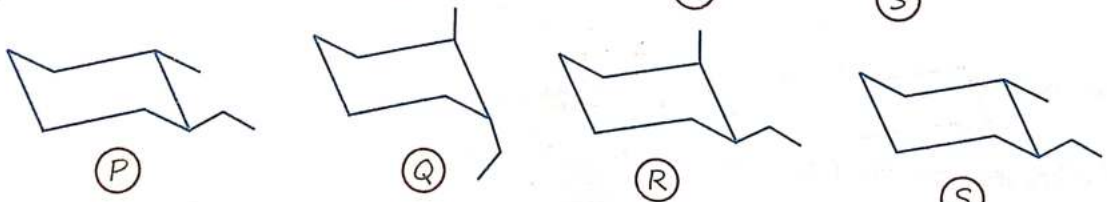


535.



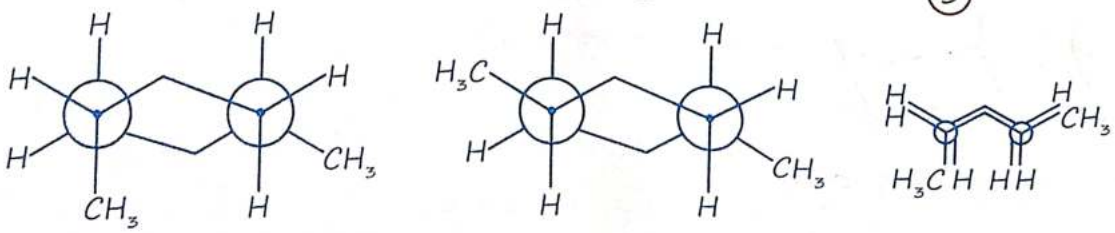
(P) (Q) (R) (S)

536.



(P) (Q) (R) (S)

537.



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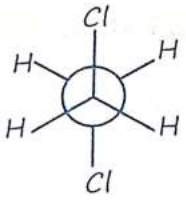
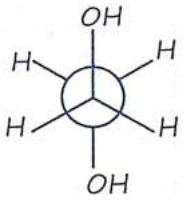
Dipole Moment

$\mu_{net}^2 = \sum x_i \mu_i^2$

Eg: (1)



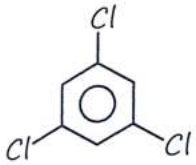
(2)



Practice Questions

538.

539.



540.

541.

542.

543.

544.

545.

at $P^H = 10$

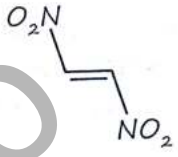
546.

547.

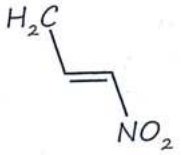
548.

549.

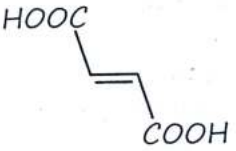
550.



551.

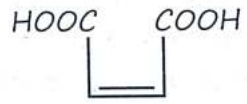


552.

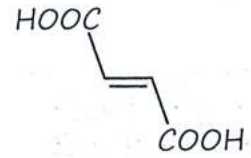


at $P^H = 11$

553.



554.



555.

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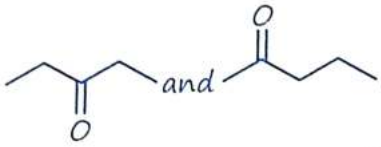
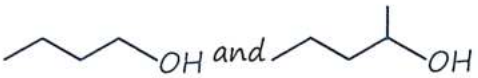
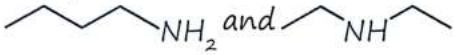
1. Identify the correct statements from the followings

(A)  are metamers

(B) are functional isomers

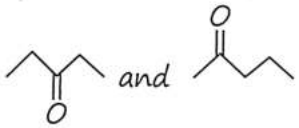
(C)  are position isomers

(D)  are homologous

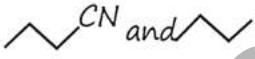
Choose the correct answer from the options given below. [03 April, 2025 (Shift-I)]

(a) C & D only (b) B & C only
(c) A & B only (d) A, B & C only

Sol. (c)

(a)  = Metamer

(same functional group, but different carbon chains or branches).

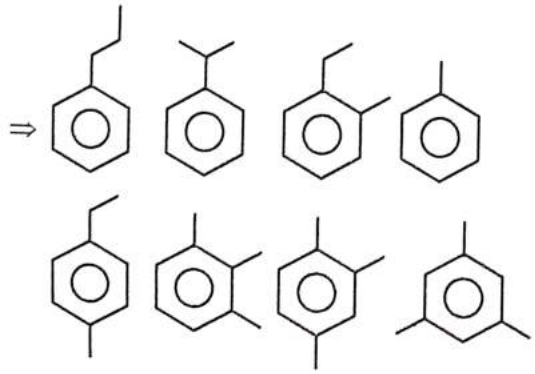
(b)  = Functional group isomer

(same molecular formula, but different functional groups).

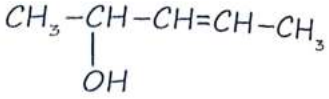
2. The total number of structural isomers possible for the substituted benzene derivatives with the molecular formula C_9H_{12} is _____.

[03 April, 2025 (Shift-II)]

Sol. [8] There are eight structural isomers that are possible for the substituted benzene derivatives with the molecular formula, C_9H_{12} which are given below:

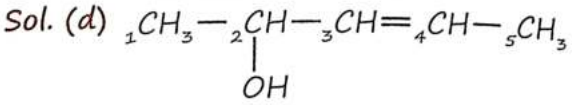


3. How many different stereoisomers are possible for the given molecule?



[22 Jan, 2025 (Shift-I)]

(a) 1 (b) 2
(c) 3 (d) 4

Sol. (d) 

The carbon at position 2 is a chiral center, hence, contributes to 2 stereoisomers (R/S configurations).

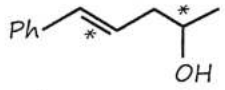
The C = C double bond at position 3 allows for cis-trans isomerism.

Therefore, it has total of 4 stereoisomers

4. The possible number of stereoisomers for 5-phenylpent-4-en-2-ol is _____.

[24 Jan, 2025 (Shift-II)]

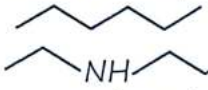
Sol. [4] In 5-phenylpent-4-en-2-ol, there are One chiral center (giving 2 stereoisomers), and One C=C double bond with E/Z isomerism (giving 2 stereoisomers).



No of stereoisomers $2^n = 2^2 = 4$

5. Given below are two statements:

Statement (I): and are isomeric compounds.

Statement (II):  and are functional group isomers.

In the light of the above statements, choose the correct answer from the options given below:

[28 Jan, 2025 (Shift-II)]

(a) Statement I is false but Statement II is true
(b) Statement I is true but Statement II is false
(c) Both Statement I and Statement II are true
(d) Both Statement I and Statement II are false

Sol. (c) Both Statement I and Statement II are true.

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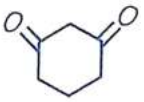
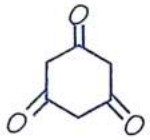
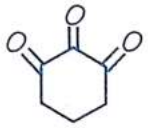
6. Highest enol content will be shown by:
[27 Jan, 2024 (Shift-I)]

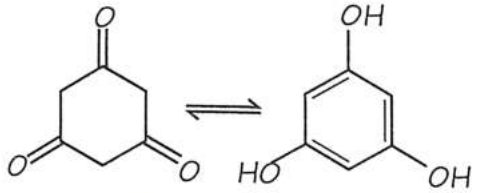
(a) 

(b) 

(c)

(d) 

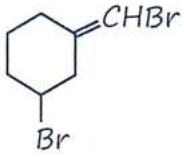
Sol. (b) Keto-enol structure

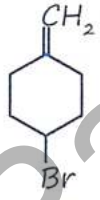


7. Which one of the following will show geometrical isomerism?
[29 Jan, 2024 (Shift-II)]

(a)

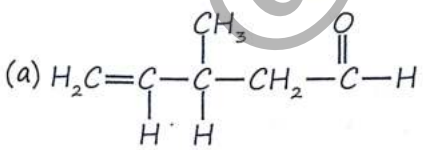
(b)

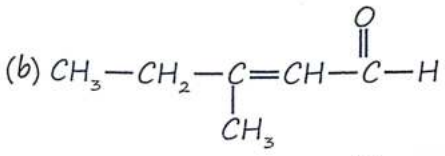
(c) 

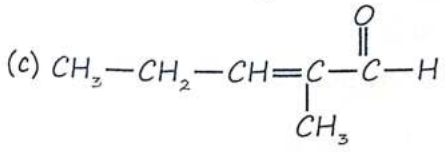
(d) 

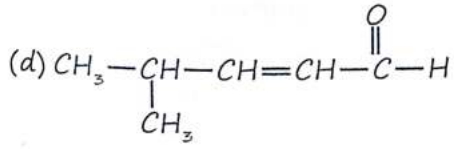
Sol. (c) Due to unsymmetrical structure it

8. Structure of 4-Methylpent-2-enal is
[30 Jan, 2024 (Shift-I)]

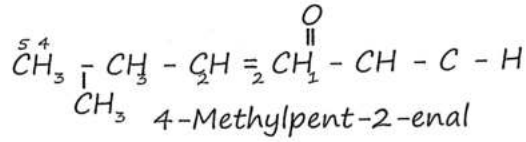
(a) 

(b) 

(c) 

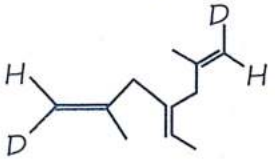
(d) 

Sol. (d) The given compound has the following structure:

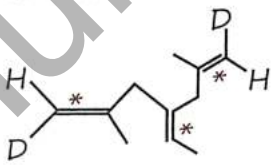


4-Methylpent-2-enal

9. Number of geometrical isomers possible for the given structure is/are _____.
[30 Jan, 2024 (Shift-II)]

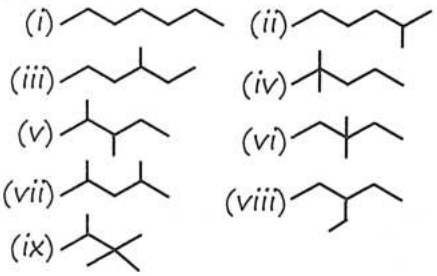


Sol. [4] Three stereocenters, symmetrical
Total Geometric isomers 4.
EE, ZZ and two isomers – EZ



10. The number of different chain isomers for C_7H_{16} is _____.
[04 April, 2024 (Shift-I)]

Sol. [9] There are nine chain isomers present for C_7H_{16} which are shown below:



11. Match List-I with List-II.
[05 April, 2024 (Shift-II)]

	List-I (Pair of Compounds)	List-II (Isomerism)
(A)	n-propanol and Isopropanol	(I) Metamerism
(B)	Methoxypropane and ethoxyethane	(II) Chain Isomerism
(C)	Propanone and propanal	(III) Position Isomerism
(D)	Neopentane and Isopentane	(IV) Functional Isomerism

(a) (A)–(II), (B)–(I), (C)–(IV), (D)–(III)
(b) (A)–(III), (B)–(I), (C)–(II), (D)–(IV)
(c) (A)–(I), (B)–(III), (C)–(IV), (D)–(II)
(d) (A)–(III), (B)–(I), (C)–(IV), (D)–(II)

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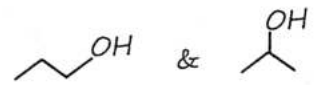
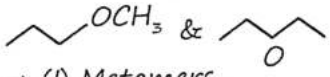
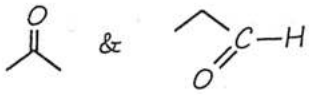
Sol. (d) (A)  ⇒ (III) Position Isomers

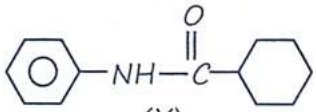
(B)  ⇒ (I) Metamers

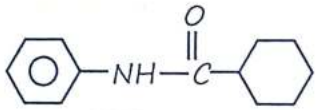
(C)  ⇒ (IV) Functional Isomers

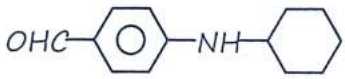
(D)
neopentane isopentane

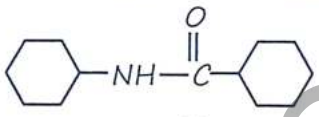
⇒ (II) Chain Isomers

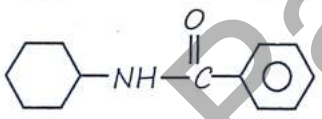
12. Which of the following is metamer of the given compound (X)? [06 April, 2024 (Shift-I)]


(X)

(a) 

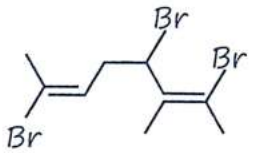
(b) 

(c) 

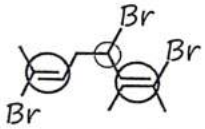
(d) 

Sol. (d) Isomers which have the same molecular formula, same functional group, but different aryl/alkyl group on either side of functional group are called metamers.

13. Total number of stereo isomers possible for the given structure: [09 April, 2024 (Shift-II)]



(a) 8 (b) 2
(c) 4 (d) 3

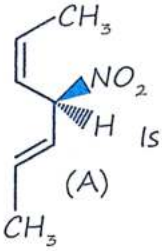
Sol. (a) 

In the given molecule, there are three stereo centre, hence

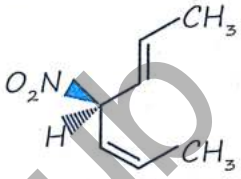
Number of stereoisomers = $2^n = 2^3 = 8$
(where, n = number of stereoisomers)

14. Given below are two statements. [29 July, 2022 (Shift-II)]

Statement-I: The compound


(A) Is optically active

Statement-II:


Is mirror image of above compund A.

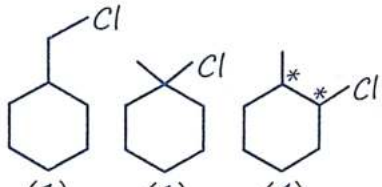
In the light of the above statement, choose the most appropriate answer from the options given below.

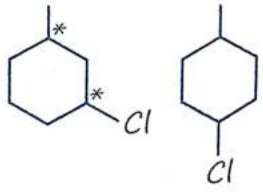
(a) Both statement-I and Statement-II are correct
(b) Both statement-I and Statement-II are incorrect
(c) Statement-I is correct but Statement-II is incorrect
(d) Statement-I is incorrect but Statement-II is correct

Sol. (c) No mirror image of each other is there and also they have same configuration.

15. Total number of isomers (including steroisomers) obtained on monochlorination of methylcyclohexane is _____. [26 July, 2022 (Shift-II)]

Sol. [12] Monochlorinated isomers of methylcyclohexame are given as,


(1) (1) (4)


(4) (2)

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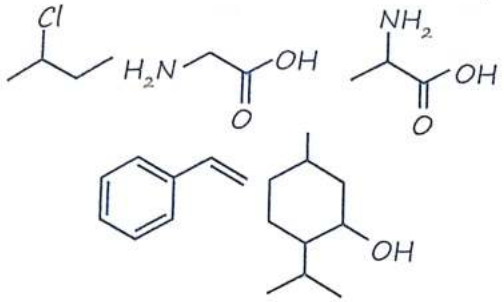
No. of isomers:

Total no. of isomers = 1 + 1 + 4 + 4 + 2 = 12

16. Observe structure of the following compounds

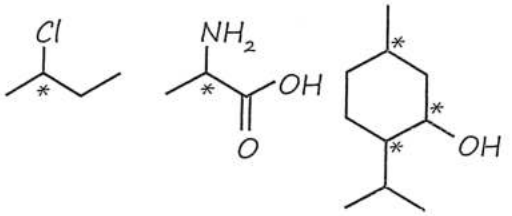


The total number of structures/compounds which possess asymmetric atoms is____.

[29 June, 2022 (Shift-I)]

Sol. [3] Asymmetric carbons are sp^3-hybridised carbon with four different valencies.

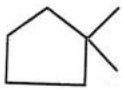
Number of compounds containing asymmetric carbon are:



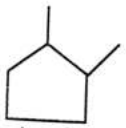
17. Total number of possible stereoisomers of dimethyl cyclopentane is_____.

[27 June, 2022 (Shift-I)]

Sol. [6] The possible stereoisomers of dimethyl cyclopentane are:

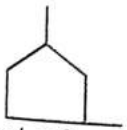


will have only one form and has no stereo-isomer.



(Total-3forms)

will have only two forms, one is cis and another is trans, but trans form has again two forms i.e dextro and laevo.



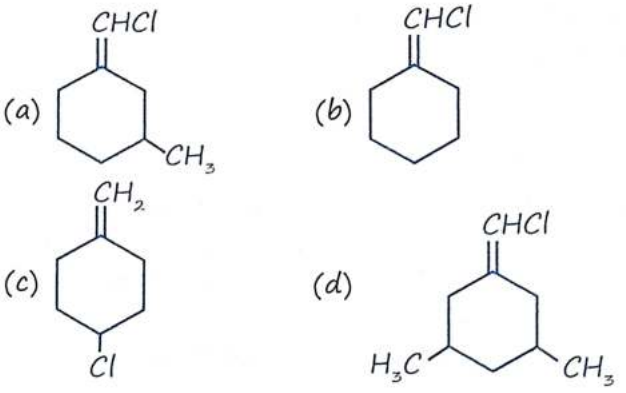
(Total-3forms)

will have two forms of geometrical: cis and trans form, trans forms have again two form dextro and laevo.

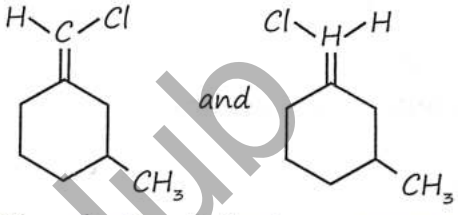
Hence, the total number of possible stereoisomers are 6.

18. Among the following compounds, geometrical isomerism is exhibited by:

[5 Sept, 2020 (Shift-II)]



Sol. (a) Compound A exhibits-cis and- trans isomers.

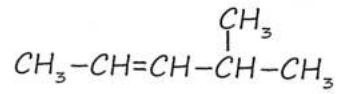


19. Which of the following compounds shows geometrical isomerism?

[6 Sept, 2020 (Shift-I)]

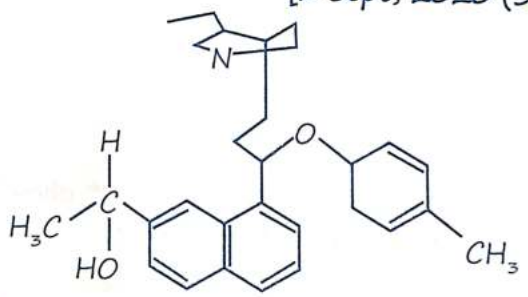
(a) 2 – methylpent – 2 – ene
(b) 4 – methylpent – 2 – ene
(c) 4 – methylpent – 1 – ene
(d) 2 – methylpent – 1 – ene

Sol. (b) 4-methylpent-2-ene will show geometrical isomerism.

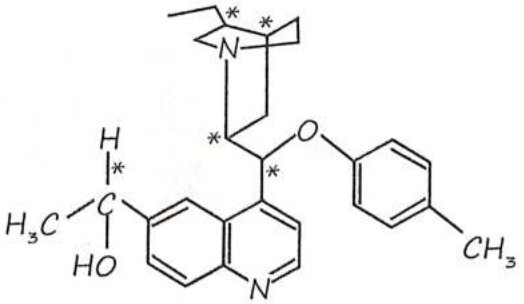


20. The number of chiral carbons present in the molecule given below is _____.

[2 Sept, 2020 (Shift-I)]



Sol. [5] In the given figure starred carbons are chiral in nature.



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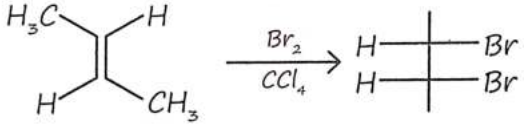
21. The number of stereoisomers obtained by bromination of trans-2-butene is (IIT JEE 2007)

(a) 1 (b) 2
(c) 3 (d) 4

Sol. (a) Bromination of Trans-2-butene

Here addition Br is anti addition which make Meso compound so there is only one stereoisomer .

Here we can see symmetry.



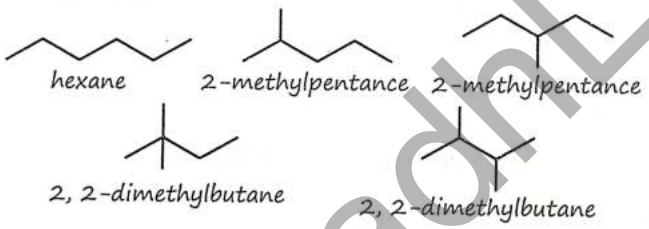
trans-2-butene

22. The number of isomers of C_6H_{14} is (IIT JEE 1987, 2007)

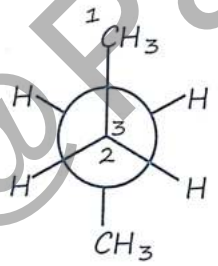
(a) 4 (b) 5
(c) 6 (d) 7

Sol. (b) There are 5 isomers of C_6H_{14}.

These isomers are positional and chain isomer.



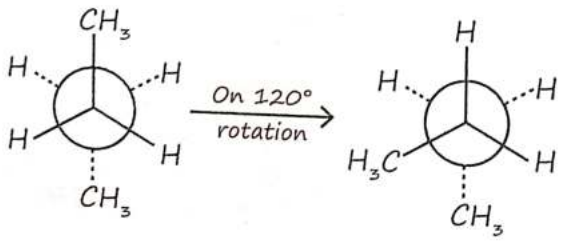
23. (IIT JEE 2004)



C_2 is rotated anti-clockwise 120° about C_2-C_3 bond. The resulting conformer is

(a) partially eclipsed (b) eclipsed
(c) gauche (d) staggered

Sol. (c)



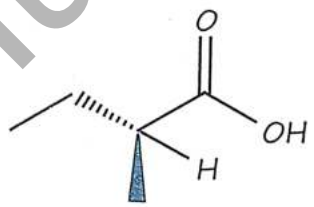
After 120 degree rotation, CH_3 groups which are bulky and on C_2 and C_3 are at dihedral angle of 60 degree. The conformer is gauche.

24. Which of the following compounds exhibits stereoisomerism? (IIT JEE 2002)

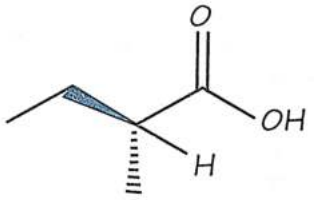
(a) 2 -methylbutene-1
(b) 3 -methylbutyne-1
(c) 3 -methylbutanoic acid
(d) 2 -methylbutanoic acid

Sol. (d) In 2 -methylbutanoic acid, there is chiral carbon (which is connected with 4 different groups). In other option, there are no chiral carbon. So from the given ones only 2-methylbutanoic acid exhibit stereoisomerism.

Following are stereoisomer of it



(S)-2-Methylbutanoic acid



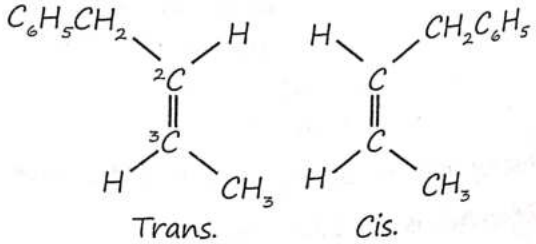
(R)-2-Methylbutanoic acid

25. Which of the following compounds will exhibit geometrical isomerism? (IIT JEE 2001)

(a) 1 -phenyl-2-butene
(b) 3-phenyl-1-butene
(c) 2-phenyl-1-butene
(d) 1,1 -diphenyl-1-propene

Sol. (a) We can see that 1 -phenyl-2-butene exhibit isomerism

It contains two different groups on carbon with double bond.



In 3-phenyl-1-butene the 1 st carbon

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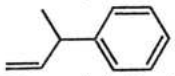
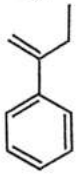
which have double bond is connected with two same groups which are both H. So it will not exhibits geometrical isomerism .

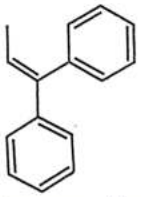


3-Phenyl-1-butene

In 2-phenyl-1-butene the 1 st carbon which have double bond is connected with two same groups which are both H. So it will not exhibits geometrical isomerism



In 1,1 -diphenyl-1-propene the 1 st carbon which have double bond is connected with two same groups which are both phenyl So it will not exhibits geometrical isomerism

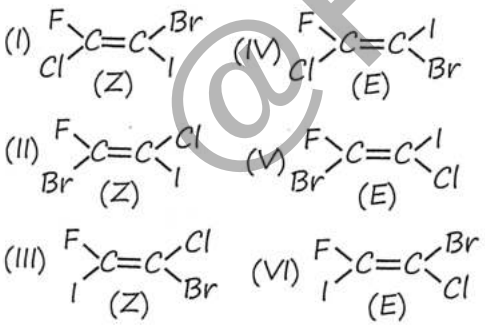


26. The number of isomers for the compound with molecular formula C2 BrClFI is (IIT JEE 2001)

(a) 3 (b) 4
(c) 5 (d) 6

Sol. (d) There are 6 isomers for compound with molecular formula C_2 BrClFI .

The following are the structure of them ,



27. The optically active tartaric acid is named as D-(+)-tartaric acid because it has a positive. (IIT JEE 1999)

(a) Optical rotation and is derived from D-glucose.
(b) pH in organic solvent.
(c) optical rotation and is derived from D-(+)-glyceraldehydes.
(d) Optical rotation when substituted by deuterium.

Sol. (c) D is used for the arrangement of -OH group in right side at second last carbon atoms as in glyceraldehyde.

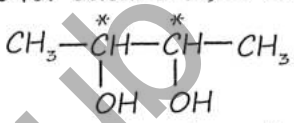
(+) sign is used to represent the rotation in right side.

Hence,The optically active tartaric acid is named as D-(+)-tartaric acid because it has a positive optical rotation and is derived from D-(+)-glyceraldehydes

28. How many optically active stereoisomers are possible for butane-2, 3-diol? (IIT JEE 1997)

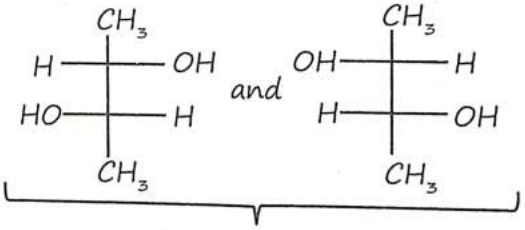
(a) 1 (b) 2
(c) 3 (d) 4

Sol. (b) Two optically active stereoisomers are possible for butane-2, 3-diol.

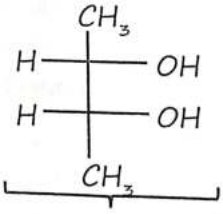


Here, 2 asymmeteric carbon and plane of symmetry so optically active stereoisomers $= 2^{n-1} = 2^{2-1} = 2$; n = no. of asymmetric carbons

butane-2, 3-diol have 3 stereoisomers from which 2 are d and l which are otically active, but 1 is meso compound which is optically inactive because of internal compensation.



Optically active



meso-compound
(optically inactive)

29. Isomers which can be interconverted through rotation around a single bond are (IIT JEE 1992)

(a) conformers (b) diastereomers
(c) enantiomers (d) positional isomers

Sol. (a) Conformers are the spatial arrngement of the atoms of the molecular structure.

It is type of stereoisomerism in which isomers are interconverted by rotation about c-c sigma bond.

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30. If two compounds have the same empirical formula but different molecular formulae, they must have (IIT JEE 1987)

(a) different percentage composition
(b) different molecular weight
(c) same velocity
(d) same vapour density

Sol. (b) If two compounds have the same empirical formula but different molecular formulae, they must have different molecular weight.

31. An isomer of ethanol is (IIT JEE 1986)

(a) methanol (b) diethyl ether
(c) acetone (d) dimethyl ether

Sol. (d) Isomers are the compounds with the same molecular formula but different arrangements.

Ethanol = $C_2H_5OH = C_2H_6O$

Methanol = $CH_3OH = CH_4O$

Acetone = $CH_3COCH_3 = C_3H_6O$

Diethyl ether = $C_2H_5OC_2H_5 = C_4H_{10}O$

Dimethyl ether = $CH_3OCH_3 = C_2H_6O$

We can see that ethanol and dimethyl ether have same molecular formula

Hence, the isomer of ethanol is dimethyl ether.

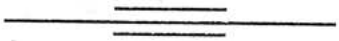
32. Which of the following compounds will exhibit cis-trans (geometrical) isomerism? (IIT JEE 1983)

(a) 2 -butene (b) 2 -butyne
(c) 2-butanol (d) butanal

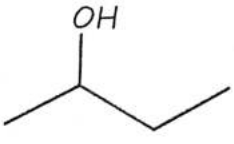
Sol. (a) 2-butene exhibits geometrical isomerism in the following manner .

Cis-2-butene trans-2-butene

In 2 -butyne, there is triple bond, does not show geometrical isomerism .

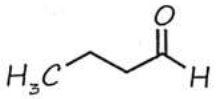


In 2-butanol, there is no double bond, so it can't exhibits cis-trans isomerism.



In butanal, there is double bond not between C=C but that is between C=O

So it will not exhibits cis- trans isomerism



33. The compound which is not isomeric with diethyl ether is. (IIT JEE 1981)

(a) n-propyl methyl ether
(b) butane-1-ol
(c) 2 -methyl propane- 2-ol
(d) butanone

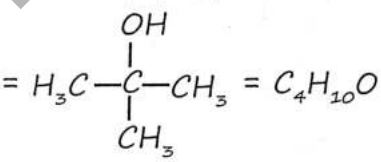
Sol. (d) Isomers are the compounds with the same molecular formula but different arrangements.

Diethyl ether = $C_2H_5OC_2H_5 = C_4H_{10}O$

n-propyl methyl ether = $C_3H_7OCH_3$ = $C_4H_{10}O$

butane-1-ol = $C_4H_9OH = C_4H_{10}O$

2-methyl propane-2-ol



butanone = $CH_3CH_2COCH_3 = C_4H_8O$

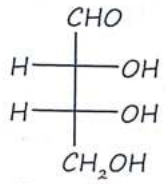
We can see that butanone is not isomer with diethyl ether.

34. The correct combination of names for isomeri alcohols with molecular formula $C_4H_{10}O$ is/are C-27.68 W-69.29 UA-3.03 (JEE Adv. 201

(a) tert-butanol and 2-methylpropan-2-ol
(b) tert-butanol and 1,1-dimethylethan-1-
(c) n-butanol and butan-1-ol
(d) iso-butyl alcohol and 2-methylpropan-1-

Sol. (a,c,d)

35. The Fischer projection of D-erythrose is shown below:



D-erythrose

D-erythrose and its isomers are listed as P, Q, R, and S in Column-I. Choose the correct relationship of P. Q. R, and S with D-erythrose from Column-II.

C-24.79 W-43.7 UA-31.51 (JEE Adv. 2020)

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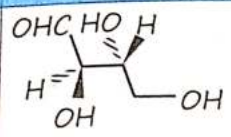
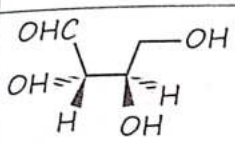
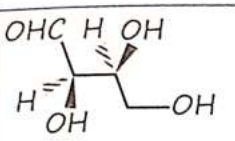
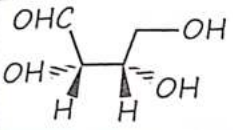
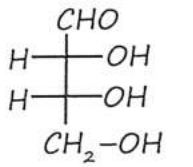
	Column-I		Column-II
P.		1.	Diastereomer
Q.		2.	Identical
R.		3.	Enantiomer
S.			

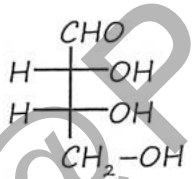
Codes:

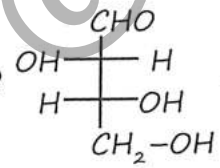
	P	Q	R	S		P	Q	R	S
(a)	2	3	2	2	(b)	3	1	1	2
(c)	2	1	1	3	(d)	2	3	3	1

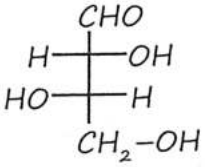
Sol. (c)

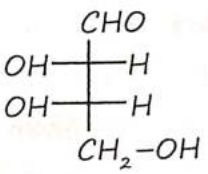


D-Erythrose

Compound P  identical

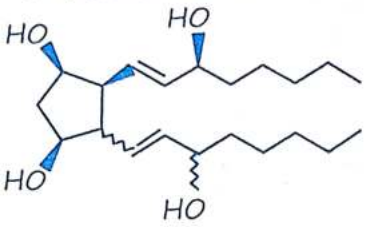
Compound Q  diastereomer

Compound R  diastereomer

Compound S  enantiomer

36. For the given compound X, the total number of optically active stereoisomers is.....

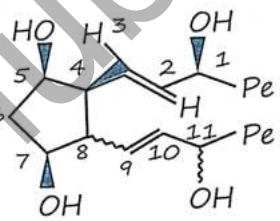
C-4.01 W-84.88 UA-11.1 ((JEE Adv. 2018)



This type of bond indicates that the configuration at the specific carbon and the geometry of the double bond is fixed

This type of bond indicates that the configuration at the specific carbon and the geometry of the double bond is not fixed

Sol. [7] The given compound X is:



Position in space and geometry (cis-trans configuration) is fixed.

Position in space and geometry is not fixed

If the bottom double bond $C_9 = C_{10}$ is in the trans position then there can be different possibilities with the C_8-C_9 and $C_{10}-OH$. Similarly, there will be additional 4-different possibilities when the double bond $C_9=C_{10}$ is in the cis configuration. Now, out of these eight different possibilities, only 1 configuration has a POS. This makes it an optically inactive compound. We can also solve this problem by considering the two chiral centers at C_8 and C_{11}.

Number of stereoisomers::= $2^n = 2^2 = 4$

Now, each of these 4 stereoisomers can be in cis or trans configuration about $C^9=C^{10}$ bond. This makes the total number of stereoisomers = $4 \times 2 = 8$

$\therefore$ Total number of optically active stereoisomers = $8-1=7$.

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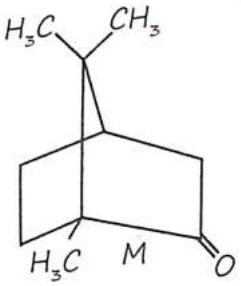
37. The total number of stereoisomers that can exist for M is (JEE Adv. 2015)

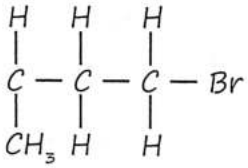


Sol. [2] The given molecule cannot show the geometrical isomerism, so its mirror image will be its only other stereoisomer.

38. The IUPAC name of $(CH_3)_2CH - CH_2 - CH_2Br$ is: (1996)

(a) 1-bromo-3-methylbutane
(b) 2-methyl-3-bromopropane
(c) 1-bromopentane
(d) 2-methyl-4-bromobutane

Sol. (a)

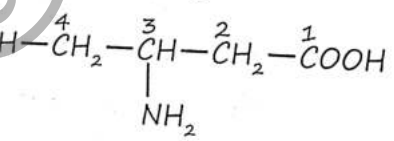


1-Bromo-3-methylbutane

39. The IUPAC name for $CH_3CH = \underset{\underset{CH_2}{|}}{C}HCH_2CHCH_2COOH$ is: (1995)

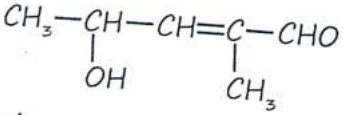
(a) 3-amino-5-heptenoic acid
(b) β-amino-δ-heptenoic acid
(c) 5-amino-2-heptenoic acid
(d) 5-amino-hex-2-enecarboxylic acid

Sol. (a)



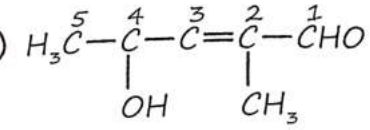
As – COOH group is highest priority group, it is numbered one. So, the IUPAC name is 3-amino-5-heptenoic acid.

40. The IUPAC name of: (1992)



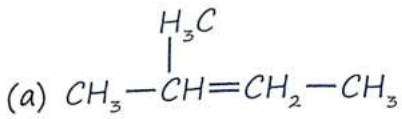
(a) 4-hydroxy-1-methylpentanal
(b) 4-hydroxy-2-methylpent-2-en-1-al
(c) 2-hydroxy-4-methylpent-3-en-5-al
(d) 2-hydroxy-3-methylpent-2-en-5-al

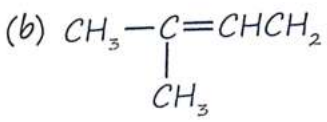
Sol. (b)

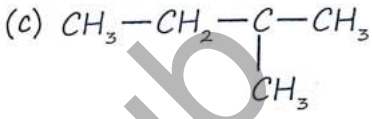


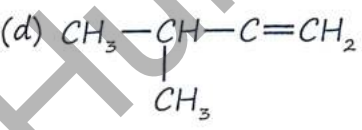
4-hydroxy-2-methyl pent-2-ene-1-al

41. 2-Methyl-2-butene will be represented as: (1992)

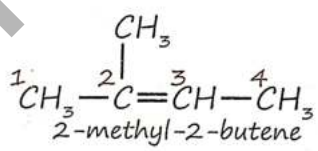
(a) 

(b) 

(c) 

(d) 

Sol. (b)

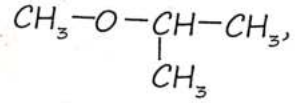


2-methyl-2-butene

42. The compound which shows metamerism is: (2021)

(a) C_3H_8O (b) C_3H_6O
(c) $C_4H_{10}O$ (d) C_5H_{12ss}

Sol. (c) A type of isomerism in which compounds having the same molecular formula but different alkyl groups on either side of functional groups is called metamerism. Among the given options, compounds with molecular formula $C_4H_{10}O$ can be ethers and may exhibit metamerism. i.e.,



(methoxy propane)

and $CH_3-CH_2-O-CH_2-CH_3$. These are
(ethoxy ethane)

metamers because they have different structure of alkyl chains around the functional group.

43. Which among the given molecules can exhibit tautomerism? (2016-II)

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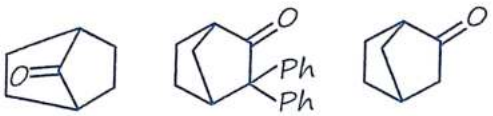
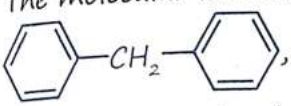


I II II

(a) III only (b) Both I and III
(c) Both I and II (d) Both II and III

Sol. (a) α-H at bridge carbon never participate in tautomerism. Thus, only (III) shows tautomerism.

44. The molecular formula of diphenyl methane,  , is $C_{13}H_{12}$.

How many structural isomers are possible when one of the hydrogens is replaced by a chlorine atom? (2004)

(a) 4 (b) 8
(c) 7 (d) 6

Sol. (a) Only 4 structural isomers are possible for diphenyl methane when one of the hydrogens is replaced by a chlorine atom.

45. Geometrical isomers are different in: (2002)

(a) Position of functional group
(b) Position of atoms
(c) Spatial arrangement of atoms
(d) Length of carbon chain

Sol. (c) Geometrical isomers are those isomers which posses the same molecular & structural formula but differ in the arrangement of atom.

46. Tautomerism is exhibited by: (1997)

(a) R_3CNO_2 (b) RCH_2NO_2
(c) $(CH_3)_3CNO$ (d) $(CH_3)_2NH$.

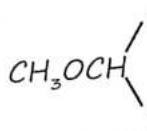
Sol. (b) It is a special type of functional isomerism, in which both the isomers are represented by one and the same substance and are always present in equilibrium. It is exhibited by nitroalkane (RCH_2NO_2) and isonitroalkane.

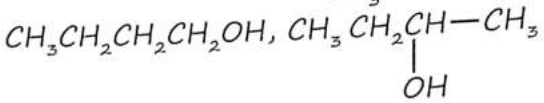
47. The number of isomers in $C_4H_{10}O$ will be: (1996)

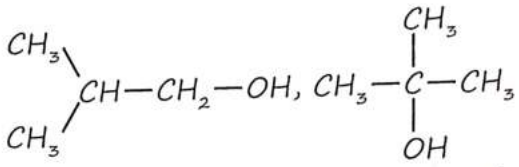
(a) 7 (b) 8
(c) 5 (d) 6

Sol. (a) There are 7 isomers in $C_4H_{10}O$. Out of these 4 are alcohols and 3 are ethers.

$CH_3CH_2 - O - CH_2CH_3$, $CH_3OCH_2CH_2CH_3$







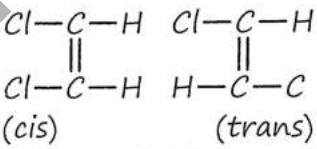
48. Isomers of a substance must have the same: (1991)

(a) Structural formula
(b) Physical properties
(c) Chemical properties
(d) Molecular formula

Sol. (d) Isomers must have same molecular formula but different structural formula.

49. Which one of the following can exhibit cis-trans isomerism? (1989)

(a) $CH_3-CHCl-COOH$ (b) $H-C\equiv C-Cl$
(c) $ClCH=CHCl$ (d) $ClCH_2-CH_2Cl$

Sol. (c) 

(cis) (trans)

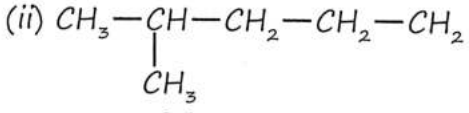
1, 2 -dichloroethene exhibits cis-trans (geometrical) isomerism.

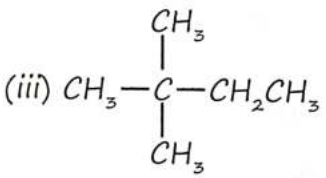
50. How many chain isomers could be obtained from the alkane C_6H_{14}? (1988)

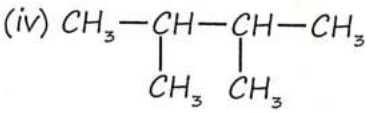
(a) Four (b) Five
(c) Six (d) Seven

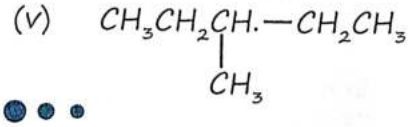
Sol. (b) 5-Chain isomers are obtained from alkane C_6H_{14}.

(i) $CH_3CH_2CH_2CH_2CH_2CH_3$

(ii) 

(iii) 

(iv) 

(v) 

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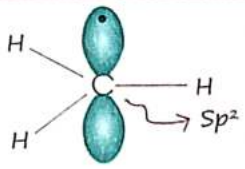
4 Hydrocarbons

Free Radical

A free radical is a molecule or atom with an unpaired electron, making it highly reactive.

Properties

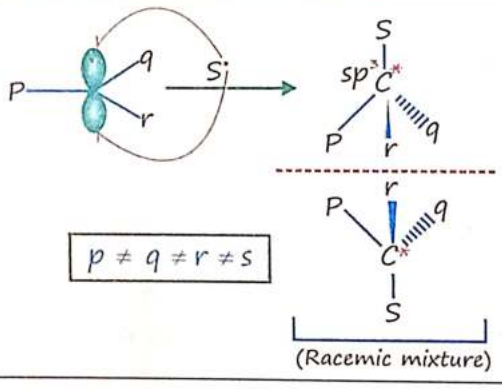
- Octet incomplete
- Highly unstable
- Paramagnetic
- 1 U.P. (U.P. : unpaired electron)
- $\sqrt{1(1+2)}$ B.M $\Rightarrow \sqrt{3}$ B.M.
- 0 l. P. (l.p. : lone pair)
- 3 B.P. (B.P. : Bond Pair)
- sp^2, Planar
- 0 A.E. for Rxⁿ (A.E. (Activation energy)



Factors

- Sunlight (hv, uv rays)
- High temperature (Δ)
- Metal
- Electrolysis
- Gaseous phase
- Another radical

Attack by Radical



$p \neq q \neq r \neq s$

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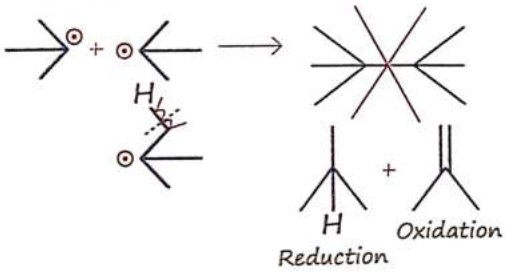
Reaction By Free Radical

Combination

$R^{\odot} + R^{\odot} \longrightarrow R - R$

$R^{\odot} + R'^{\odot} \longrightarrow R - R' + R - R$
(Major) $+ R' - R'$

Disproportionation



Rearrangement

Just as free radicals initiate powerful chains, your small steps today start unstoppable progress.

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Wurtz Reaction

The Wurtz reaction is an organic chemical reaction used to couple two alkyl halides to form a higher alkane. It involves the reaction of alkyl halides with sodium metal in dry ether.

Reaction

$$2R - X + 2Na \xrightarrow{\text{dry } Et_2O} R - R + 2NaX$$

Mechanism 1

Involves homolytic cleavage of R–X by sodium metal, forming alkyl radicals (R•), which then couple to form R–R.

1. $R\overset{\delta+}{-}X\overset{\delta-}{} + Na \xrightarrow{rds} R^{\odot} + Na^{\oplus}X^{\ominus}$ ($e^{\ominus}$)

2. $\dot{R} + \dot{R} \longrightarrow R - R$

Mechanism 2

Sodium donates electrons to R–X forming R–Na+, which acts as a nucleophile and displaces halide from another R–X to form R–R.

1. $R\overset{\delta+}{-}X\overset{\delta-}{} + 2Na \xrightarrow{rds} {}^{\ominus\oplus}RNa + NaX$ ($2e^{\ominus}$)

(rds = Rate determining step)

2. $RNa + R - X \longrightarrow R - R + NaX$

Wurtz: Two alkyl halides walk into a room, sodium says 'Let's team up!' and bam – a new alkane is born!

Baba Tea-Stall (BTS)

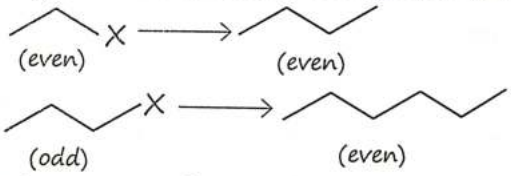
(i) Both Carbanion and free intermediate obtained

(ii) As the stability of (•) ↑ ⇒ Rate of reaction ↑

(iii) Rate of reaction for different R – X is :

R – I > R – Br > R – Cl > R – F

(iv) Symmetrical alkane obtained as major product



(v) In case of 3° – X, major product is disproportionation reaction.

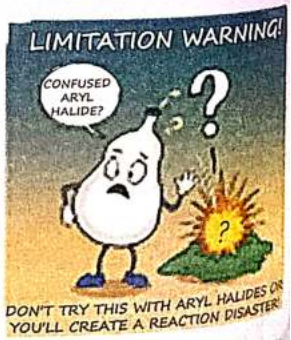
Wurtz ka khandaan

$R - X + Ar - X \xrightarrow[\Delta]{Na} R - Ar$ (Wurtz-Fittig Reaction)

- Wurtz Reaction: $R - X \xrightarrow[\text{Dry } Et_2O]{Na} R - R + Nax$
- Fittig Reaction: $Ar - X \xrightarrow{Na}$ No reaction
- $Ar - X \xrightarrow[\Delta]{Na} Ar - Ar + NaX$

Ulmann Reaction: $Ar - X \xrightarrow[\Delta]{Cu} Ar - Ar$

Frankland Reaction: $R - X \xrightarrow{Zn} R - R$



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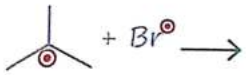
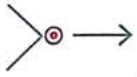
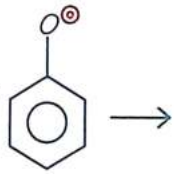
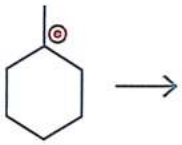
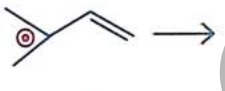
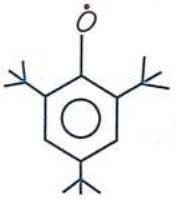
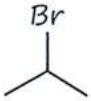
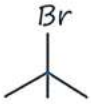
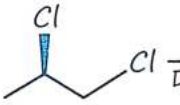
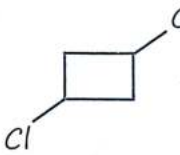
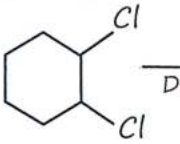
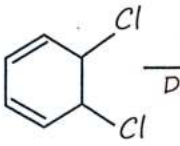
Practice Questions

Free Radical

1. $\overset{\odot}{C}H_3 \longrightarrow$
2. $\overset{\odot}{C}H_3 + \overset{\odot}{Cl} \longrightarrow$
3. $\overset{\odot}{Cl} \longrightarrow$
4.  $+ \overset{\odot}{Br} \longrightarrow$
5.  $\longrightarrow$
6. $+ \overset{\odot}{C}H_3 \longrightarrow$
7. $+ \overset{\odot}{Cl} \longrightarrow$
8. $+ \overset{\odot}{Br} \longrightarrow$
9.  $\longrightarrow$
10. $\longrightarrow$
11.  $\longrightarrow$
12.  $\longrightarrow$
13.  $\longrightarrow$
14. $\longrightarrow$

Wurtz Reaction

15. $CH_3Cl \xrightarrow[dryEt_2O]{Na}$
16. $PhCH_2Br \xrightarrow[dryEt_2O]{Na}$
17. $\xrightarrow[Dry\ Et_2O]{Na}$
18.  $\xrightarrow[Dry\ Et_2O]{Na}$
19.  $\xrightarrow[Dry\ Et_2O]{Na}$
20. $\xrightarrow[Dry\ Et_2O]{Na}$
21. $\xrightarrow[Dry\ Et_2O]{Na}$
22. $\xrightarrow[Dry\ Et_2O]{Na}$
23.  $\xrightarrow[Dry\ Et_2O]{Na}$
24.  $\xrightarrow[Dry\ Et_2O]{Na}$
25.  $\xrightarrow[Dry\ Et_2O]{Na}$
26. $\xrightarrow[Dry\ Et_2O]{Na}$
27.  $\xrightarrow[Dry\ Et_2O]{Na}$

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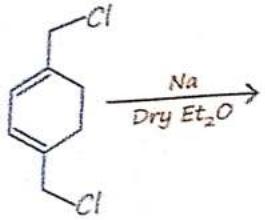
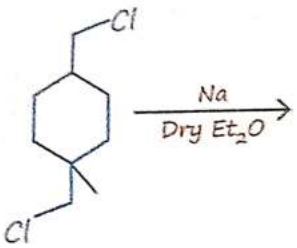
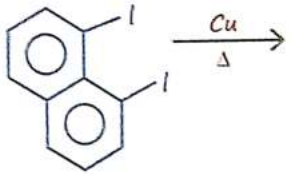
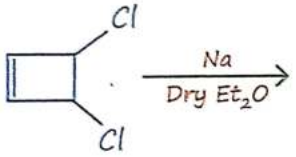
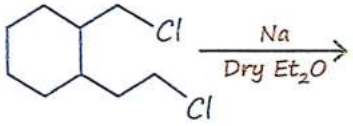
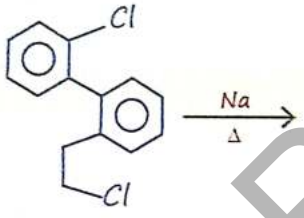
28. 

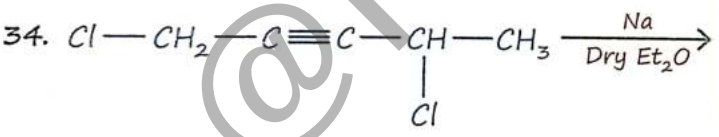
29. 

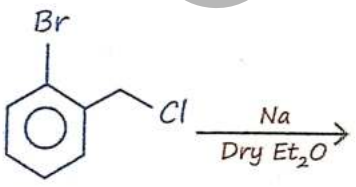
30. 

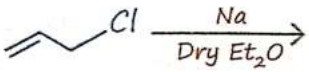
31. 

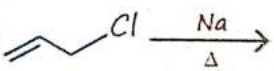
32. 

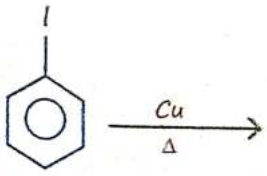
33. 

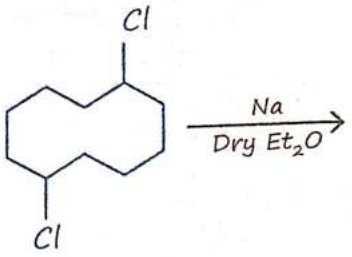
34. 

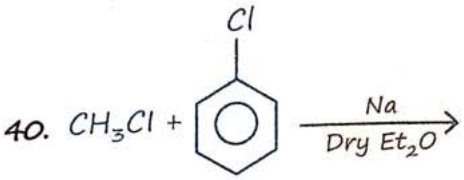
35. 

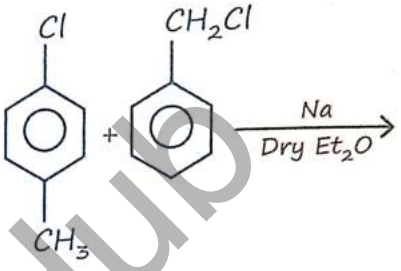
36. 

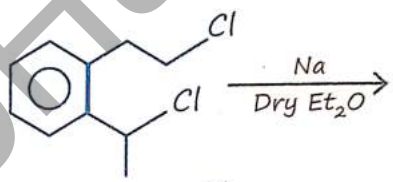
37. 

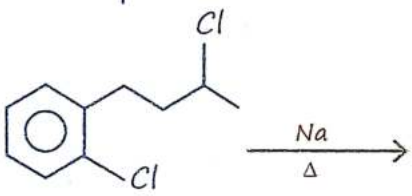
38. 

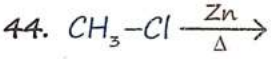
39. 

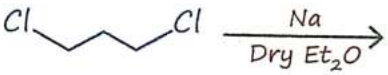
40. 

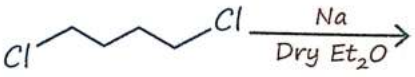
41. 

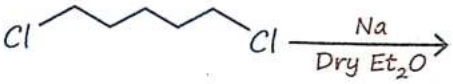
42. 

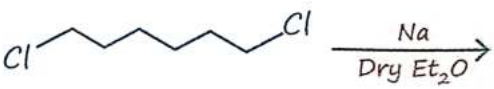
43. 

44. 

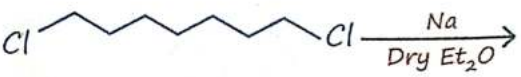
45. 

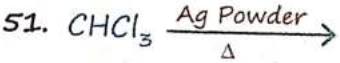
46. 

47. 

48. 

49. **Compare rate of 45 – 47**

50. 

51. 

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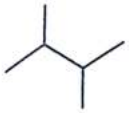
52. $CH_3 - CHCl_2 \xrightarrow[\Delta]{\text{Ag Powder}}$

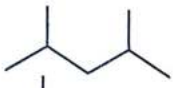
53. $CH_3 - CCl_3 \xrightarrow[\Delta]{\text{Ag Powder}}$

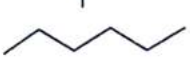
54. How many of the following compounds can be obtained in good yield by wurtz reaction using single reactant?

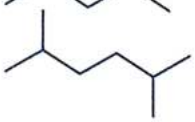
(a)

(b) CH_4

(c) 

(d) 

(e) 

(f) 

(g)

(h)

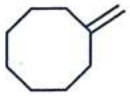
(i)

(j)

(k)

(l)

(m)

(n) 

(o)

Previous Year Questions

1. Which of the following alkane cannot be made in good yield by Wurtz reaction? (NEET, 2020)

(a) 2, 3-Dimethylbutane

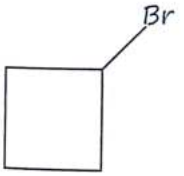
(b) n-Heptane

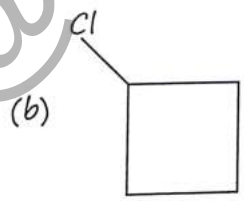
(c) n-Butane

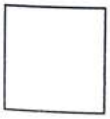
(d) n-Hexane

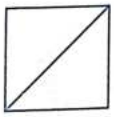
Sol. (b)

2. 1-bromo-3-chlorocyclobutane when treated with two equivalents of Na, in the presence of ether which of the following will be formed? (IIT JEE 2005)

(a) 

(b) 

(c) 

(d) 

Sol. (d)

3. Hydrocarbon (A) reacts with bromine by substitution to form an alkyl bromide which by Wurtz reaction is converted to gaseous hydrocarbon containing less than four carbon atoms. (A) is (NEET, 2018)

(a) $CH \equiv CH$ (b) $CH_2 = CH_2$

(c) CH_4 (d) $CH_3 - CH_3$

Sol. (c)

4. Hydrocarbon (A) reacts with bromine by substitution to form an alkyl bromide which by Wurtz reaction is converted to gaseous hydrocarbon containing less than four carbon atoms. (A) is (2018)

(a) $CH \equiv CH$ (b) $CH_2 = CH_2$

(c) CH_4 (d) $CH_3 - CH_3$

Sol. (c) Wurtz reaction gives symmetrical alkane with even no. of carbon atoms and since hydrocarbon containing less than four carbon atoms need to be obtained, therefore, the hydrocarbon A would be CH_4. The reaction takes place as:

$$CH_4 \xrightarrow[\text{Substitution reaction}]{Br_2/hv} CH_3Br \xrightarrow[\text{Wurtz reaction}]{Na/\text{dry ether}} CH_3 - CH_3$$

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Reaction with Metal and Acid

When an alkyl halide reacts with zinc and hydrochloric acid (HCl), the halide is reduced, resulting in the formation of an alkane (R–H).

Reaction

$$R-X \xrightarrow[HCl]{Zn} R-H \quad (x = Cl, Br, I)$$

Mechanism

$$R-X \xrightarrow[2e^{\ominus}]{Zn} R^{-} \xrightarrow[HCl]{\overset{\delta+}{}\ \overset{\delta-}{}} R-H$$

Other Example

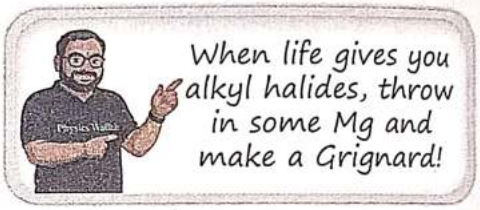
1. $\overset{\delta-}{R}-\overset{\delta+}{Mg}-X+H-OH \rightarrow R-H$
2. $\overset{\delta-}{R_2}Zn^{2+}+H^{+} \rightarrow R-H$
3. $\overset{\delta-}{R_2}\,CuLi \xrightarrow{H_2O} R-H$

Formation of Grignard Reagent

When an alkyl halide (R–X) reacts with magnesium (Mg) in dry ether, it forms a Grignard reagent (R–Mg–X), which is an organometallic compound.

Reaction

$$R-X+Mg \xrightarrow{Et_2O} R-Mg+X$$



Mechanism

$$\overset{\delta+}{R}-\overset{\delta-}{X}+Mg \xrightarrow{Et_2O} R^{-}+Mg^{2+}+X^{-} \rightarrow \underset{\substack{\text{Grignard reagent}\\ \text{(more Covalent)}}}{R-MgX}$$

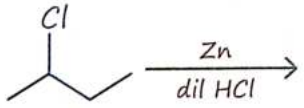
Baba Tea-Stall (BTS)

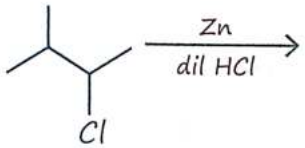
(i) Carbanion and free radical Intermediate
(ii) Organometallic compound obtained
(iii) R–Mg–X known as Grignard reagent

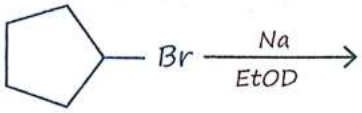
Practice Questions

55. $CH_3Cl \xrightarrow[dil\ HCl]{Zn}$

56. $Et-Br \xrightarrow[dil\ HCl]{Zn}$

57. 

58. 

59. 

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60. $PhCH_2Br$ → (Na/Et_2O) / ($Na/EtOH$)

61. $PhCH_2Br$ → ($Zn/dil.HCl$) / (Zn/Δ)

62. → (Na/Et_2O) / ($Zn/dil.HCl$)

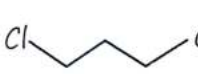
63. $CH_3Cl \xrightarrow[Et_2O]{Mg}$

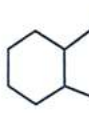
64. $CH_3-CH_2Br \xrightarrow[Et_2O]{Mg}$

65. $PhCH_2Cl \xrightarrow[Et_2O]{Mg}$

66. $CH_3F \xrightarrow[Et_2O]{Mg}$

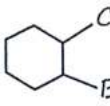
67. $\xrightarrow[Et_2O]{Mg}$

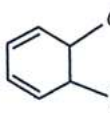
68.  $\xrightarrow[Et_2O]{Mg(excess)}$

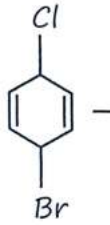
69.  $\xrightarrow[Et_2O]{Mg}$

70. $\xrightarrow[Et_2O,\ \Delta]{Mg}$

71. $\xrightarrow[Et_2O,\ \Delta]{Mg}$

72.  $\xrightarrow[Et_2O]{Mg(excess)}$

73.  $\xrightarrow[Et_2O]{Mg}$

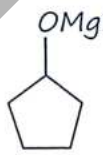
74.  $\xrightarrow[Et_2O]{Mg}$

75. $\xrightarrow[Et_2O,\ \Delta]{Mg}$

76. 14 $\xrightarrow[Et_2O\ \Delta]{Mg}$

Q. Which of the following is (are) Grignard reagents?

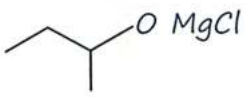
77.

78. 

79. $(CH_3)_2CuLi$

80. CH_3MgCl

81. $PhMgI$

82. 

Previous Year Questions

5. Grignard reagent is prepared by the reaction between: (1994)

(a) Magnesium and alkane
(b) Magnesium and aromatic hydrocarbon
(c) Zinc and alkyl halide
(d) Magnesium and alkyl halide

Sol. (c) $R-X \xrightarrow[HCl]{Zn} R-H$

Grignard reagent is prepared by heating an alkyl halide with dry magnesium powder in dry ether.

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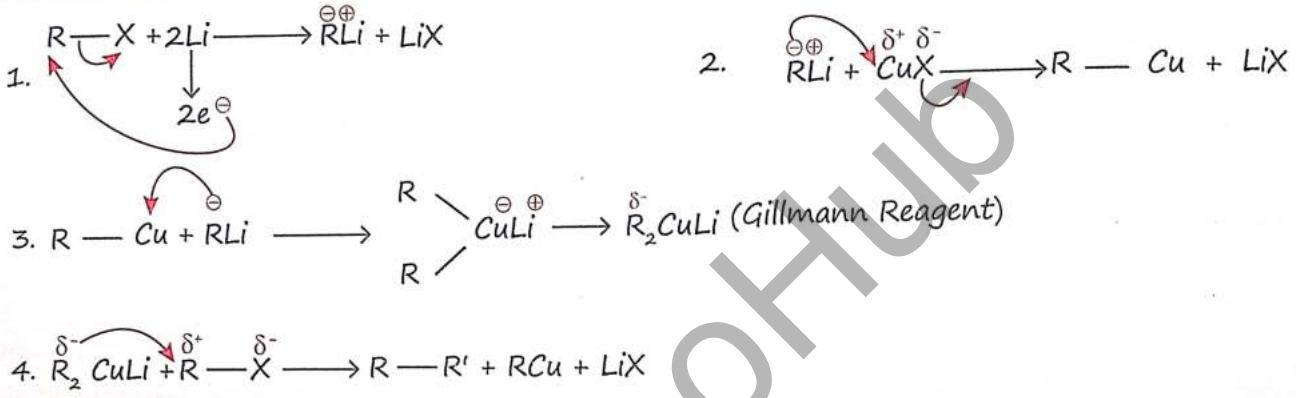
Corey House Synthesis

The Corry-House synthesis involves reacting an alkyl halide (R-X) with lithium (Li) to form an alkyl lithium (R-Li), which then reacts with copper halide (CuX) to form an organocuprate (R-Cu-Li). The organocuprate then reacts with another alkyl halide (R'-X) to form a new carbon-carbon bond (R-R').

Reaction

$$R-X \xrightarrow[3.\ R'-X]{1.\ Li \\ 2.\ CuX} R-R'$$

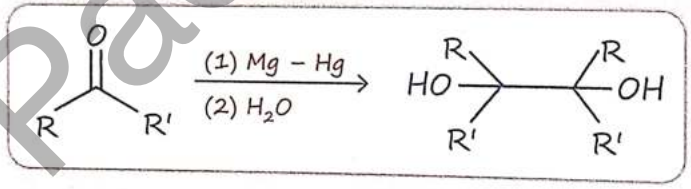
Mechanism



Pinacol Formation

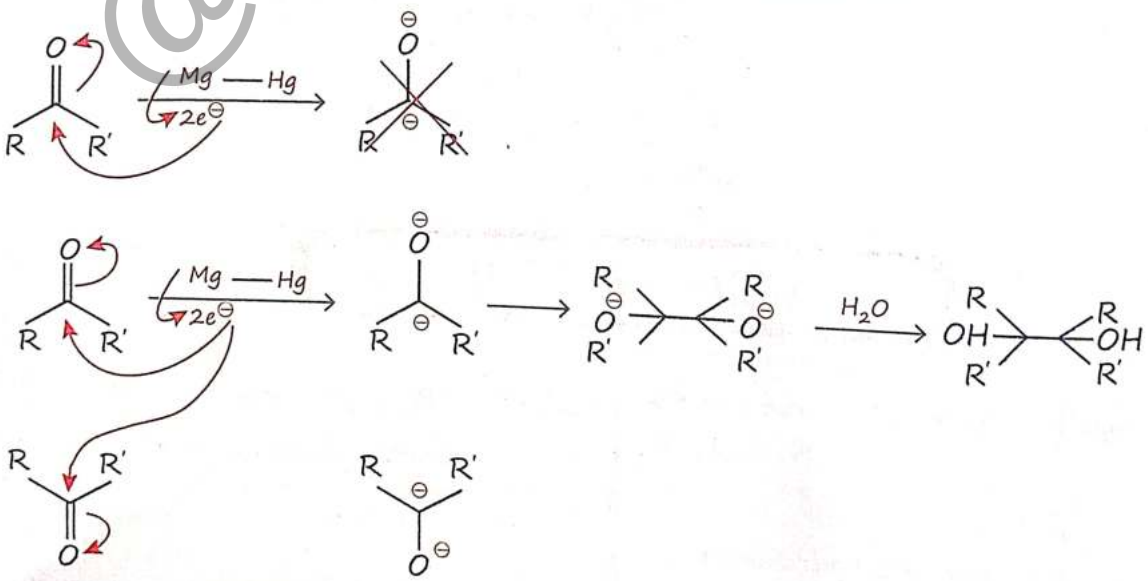
Pinacol formation is a reduction reaction where the carbonyl group (-C=O) is reduced to a hydroxyl group (-OH), leading to the formation of a vicinal diol (pinacol).

Reaction



- If R = R' ⇒ 1 pinacol
- If R ≠ R' ⇒ 3 pinacol

Mechanism



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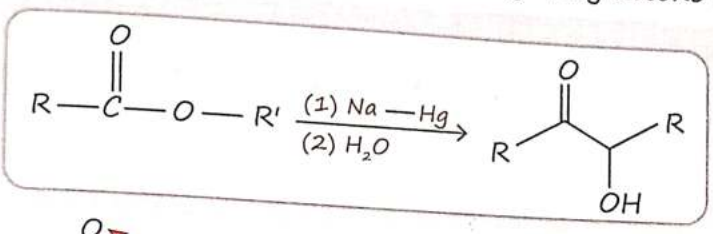
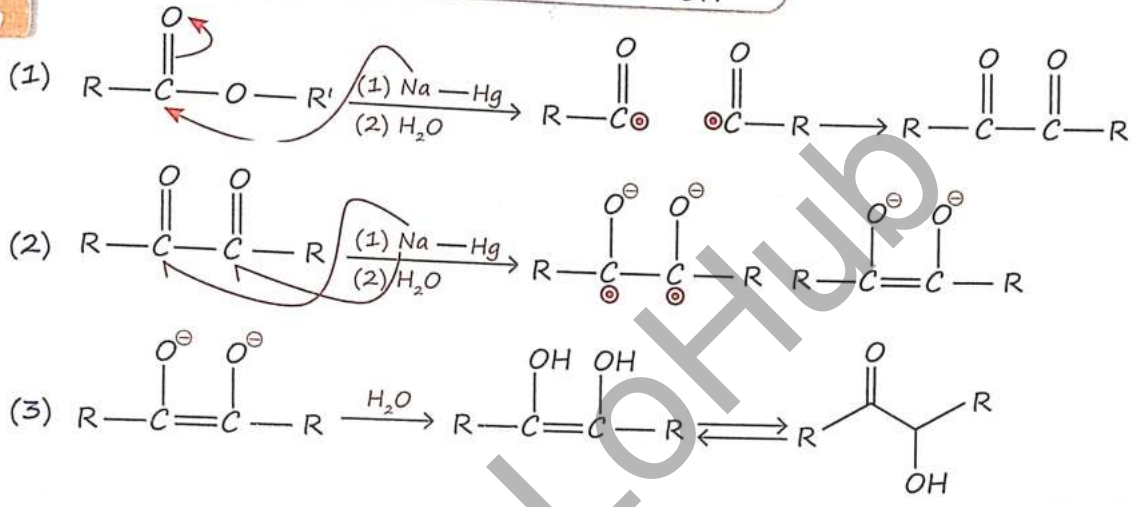
Acyloin Condensation

The Acyloin Condensation involves the reaction of two molecules of aldehydes or ketones in the presence of a $Na-Hg/H_2O$, resulting in the formation of an α-hydroxy ketone (acyloin).

Reaction



Mechanism



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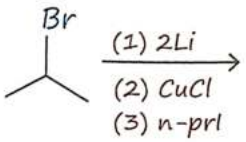
Practice Questions

➤ **Corey House Synthesis**

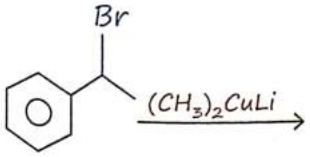
83. $CH_3Br \xrightarrow[(2)\,CuCl\ (3)\,EtBr]{(1)\,2Li}$

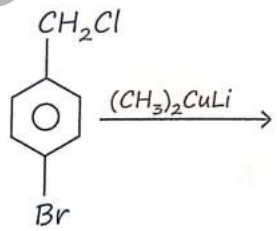
84. $PhCH_2Cl \xrightarrow[(2)\,CuCl\ (3)\,CH_3Cl]{(1)\,2Li}$

85. 

86. $P \xrightarrow[(2)\,CuCl\ (3)\,EtBr]{(1)\,2Li} \text{Iso-hexane}$

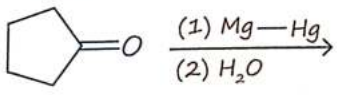
87. $CH_3CH_2Cl \xrightarrow{(C_2H_5)_2CuLi}$

88. 

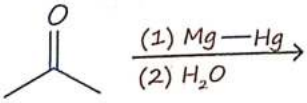
89. 

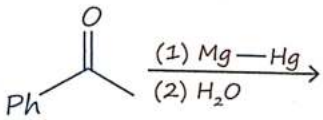
90. $PhCH_2CH_2Br \xrightarrow{(i\text{-}Pr)_2CuLi}$

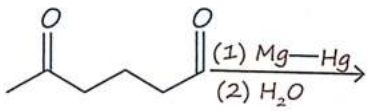
➤ **Pinacol Formation**

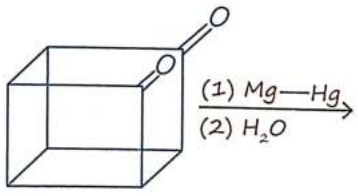
91. 

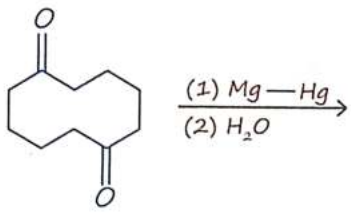
92. $CH_3-CHO \xrightarrow[(2)\,H_2O]{(1)\,Mg-Hg}$

93. 

94. 

95. 

96. 

97. 

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Kolbe's Electrolysis

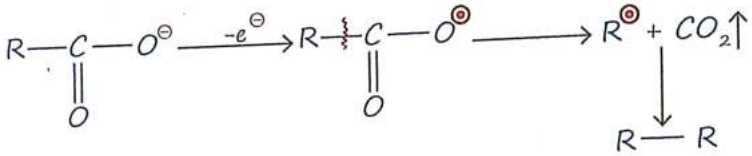
Kolbe's electrolytic method: An aqueous solution of sodium or potassium salt of a carboxylic acid on electrolysis gives alkane containing even number of carbon atoms at the anode.

Reaction

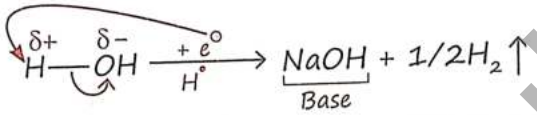
$$R-COOH + NaOH \xrightarrow{Electrolysis} R-COO^{\ominus}Na^{\oplus} + H_2O \longrightarrow R-R + CO_2 + H_2 + NaOH$$

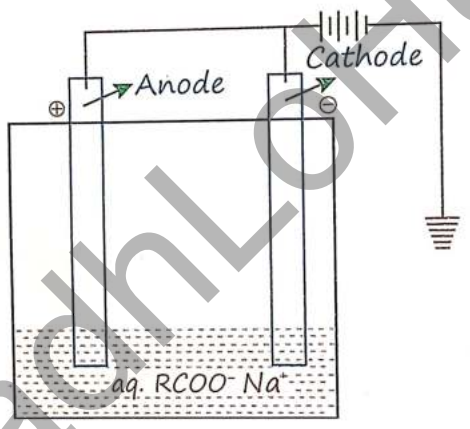
Mechanism

Anode:



Cathode:





Baba Tea-Stall (BTS)

(i) Free radical Obtained as intermediate
(ii) CO_2 and $R-R$ at anode, H_2 at cathode obtained.
(iii) pH of the medium increases (due to NaOH)
(iv) CH_4 can never be obtained
(v) In case of 3°–COOH, ester (RCOOR) will be obtained.

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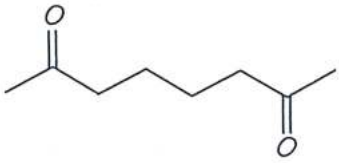
Practice Questions

➢ Acyloin Condensation

98. $CH_3COOCH_3 \xrightarrow[(2)H_2O]{(1)Na-Hg}$

99. $PhCOOCH_3 \xrightarrow[(2)H_2O]{(1)Na-Hg}$

100. $CH_3COOPh \xrightarrow[(2)H_2O]{(1)Na-Hg}$

101.  $\xrightarrow[(2)\ H_2O]{(1)\ Na-Hg}$

➢ Kolbe's Electrolysis

102. $CH_3COOH \xrightarrow[Electrolysis]{NaOH,}$

103. $\xrightarrow{Electrolysis}$

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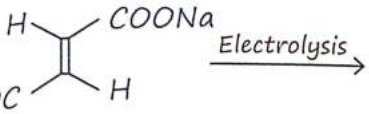
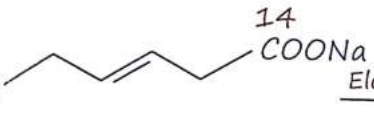
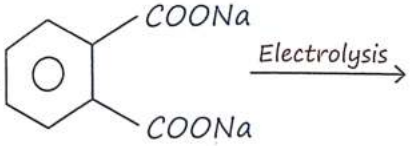
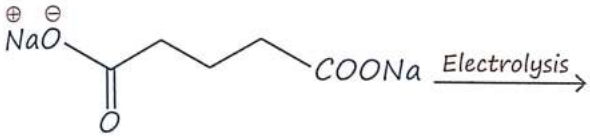
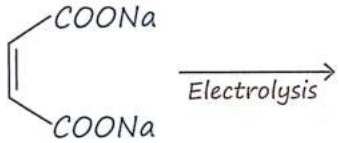
104. $CH_2-COONa$ / $CH_2-COONa$ $\xrightarrow{\text{Electrolyis}}$

105.  $\xrightarrow{\text{Electrolysis}}$

106.  $\xrightarrow{\text{Electrolysis}}$

107.  $\xrightarrow{\text{Electrolysis}}$

108.  $\xrightarrow{\text{Electrolysis}}$

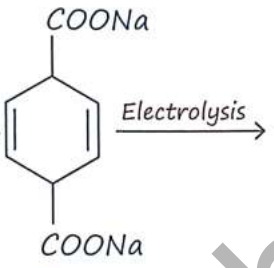
109.  $\xrightarrow{\text{Electrolysis}}$

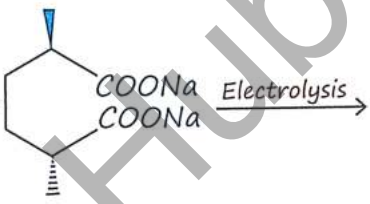
110. $CH_3COONa + C_2H_5COONa \xrightarrow[\text{Electrolysis}]{NaOH}$

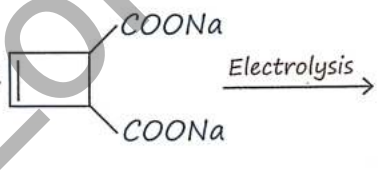
111. $CH_3COONa + C_2H_5COONa \xrightarrow{\text{Electrolysis}}$

Hydrogen act both anode and cathode.

112. $CH_3\overset{14}{C}OONa \xrightarrow{\text{Electrolysis}}$

113.  $\xrightarrow{\text{Electrolysis}}$

114.  $\xrightarrow{\text{Electrolysis}}$

115.  $\xrightarrow{\text{Electrolysis}}$

116. aq. $NaCl \xrightarrow{\text{Electrolysis}}$

117. $P \xrightarrow[\text{Electrolysis}]{NaOH,} Q \xleftarrow[Et_2O]{Na} CH_3Cl$

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Previous Year Questions

6. Number of alkanes obtained on electrolysis of a mixture of CH_3COONa and C_2H_5COONa is____.

[31 Jan, 2024 (Shift-I)]

Sol. [3] $CH_3COONa \longrightarrow \dot{C}H_3$

$C_2H_5COONa \longrightarrow \dot{C}_2H_5$

$2\dot{C}_2H_5 \longrightarrow CH_3-CH_2-CH_2-CH_3$

$2\dot{C}H_3 \longrightarrow CH_3-CH_3$

$\dot{C}H_3 + \dot{C}_2H_5 \longrightarrow CH_3-CH_2-CH_3$

7. Which of the following reaction is expected to readily give a hydrocarbon product in good yields? (1997)

(a) $CH_3CH_3 \xrightarrow[h\nu]{Cl_2}$

(b) $(CH_3)_2CHCl \xrightarrow{C_2H_5OH}$

(c) $RCOOK \xrightarrow[\text{Oxidation}]{\text{Electrolysis}}$

(d) $RCOOAg \xrightarrow{I_2}$

Sol. (c) When an aqueous solution of sodium or potassium salt of carboxylic acid is electrolysed, then an alkane containing even number of carbon atoms is formed at the anode. The reaction occurs as:

$RCOOK \xrightarrow{\text{Electrolysis}} RH + CO_2 + KOH$

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Photohalogenation

Halogenation takes place either at higher temperature (573-773 K) or in the presence of diffused sunlight or ultraviolet light. Lower alkanes do not undergo nitration and sulphonation reactions. These reactions in which hydrogen atoms of alkanes are substituted are known as substitution reactions

Reaction

$$R - H + X_2 \xrightarrow[\text{or hv or sunligh}]{\text{UV rays}} R - X + H - X$$

Mechanism

Initiation: $X_2 \xrightarrow{hv} 2X^{\odot}$

Propagation:

$R - H + X^{\odot} \xrightarrow{rds} R - X + H^{\odot}$

$R^{\odot} + H - X$

$R^{\odot} + X - X \longrightarrow R - X + X^{\odot}$

Termination: (1) $R^{\odot} + R^{\odot} \longrightarrow R - R$

(2) $X^{\odot} + X^{\odot} \longrightarrow X_2$

(3) $R^{\odot} + X^{\odot} \longrightarrow X - X$

Baba Tea-Stall (BTS)

(i) Example of Substitution Reaction
(ii) Free radical intermediate
(iii) UV rays/hv is used to initiate the reaction
(iv) Rate of reaction for different X_2 is $F_2 > Cl_2 > Br_2 > I_2$
(v) Kinetic isotopic effect ($R - H > R - D$)
(vi) RDS is formation of $R^{\odot}$

Types of Photohalogenation

Fluorination

- Highly Exothermic and explosive

$$CH_4 + 2F_2 \longrightarrow C_{black} + 4HF$$

Iodination

- Least reactive and reversible

$CH_4 + I_2 \rightleftharpoons CH_3 - I + H - I$

O.A. (HIO_3, HNO_3) I_2

Chlorination

- More reactive
- Less Selective

⇓

1:3.8:4.5 → 3° − H

1° − H, 2° − H

Bromination

- Less Reactive
- More selective

⇓

1:80:1600 → 3° − H or Reso/BB/Aro

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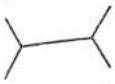
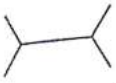
Practice Questions

118. $\xrightarrow{Cl_2/h\nu}$

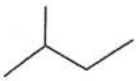
119. $\xrightarrow{Br_2/h\nu}$

120.  $\xrightarrow{Cl_2/h\nu}$

121.  $\xrightarrow{Br_2/h\nu}$

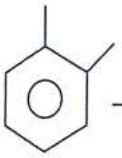
122. Toluene $\xrightarrow{Cl_2/h\nu}$

123. Toluene $\xrightarrow{Br_2/h\nu}$

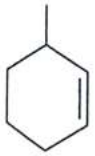
124.  $\xrightarrow[\Delta]{Br_2}$

125. $\xrightarrow{Br_2/h\nu}$

126. $\xrightarrow{Cl_2/500°C}$

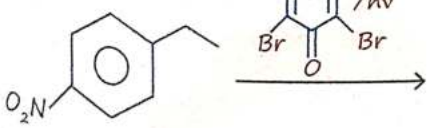
127.  $\xrightarrow[(2eq)]{Br_2/h\nu}$

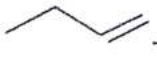
128. $\xrightarrow[(excess)]{Cl_2/h\nu}$

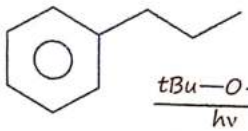
129.  $\xrightarrow{Cl_2/h\nu}$

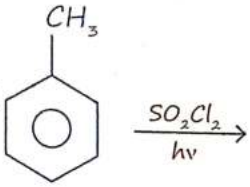
130. $\xrightarrow{Br_2/h\nu}$

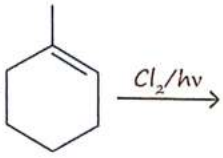
131. $(CH_3)_2CHCH_2CH_2CH_3 \xrightarrow{Br_2/h\nu}$

132. 

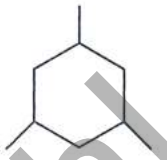
133.  $\xrightarrow{Cl_2/\Delta}$

134.  $\xrightarrow[h\nu]{tBu-O-Cl}$

135.  $\xrightarrow[h\nu]{SO_2Cl_2}$

136.  $\xrightarrow{Cl_2/h\nu}$

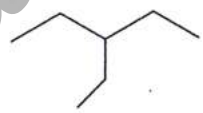
137. Find % yield of each monochlorinated product

(a) 

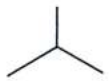
[1 : 3.8 : 4.5]

(b)

[1 : 4 : 6]

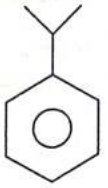
(c) 

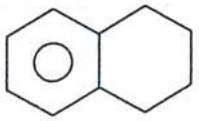
[1 : 3.8 : 4.5]

(d) 

[1 : 3.8 : 4.5]

138. $\xrightarrow{O_2/h\nu}$

139.  $\xrightarrow{O_2/h\nu}$

140.  $\xrightarrow{O_2/h\nu}$

141. 2—butene $\xrightarrow{tBu-OCl}$

142. After monochlonination reaction on given compound, find

(a) Structures obtained

(b) Total products

(c) Optically active

(d) Meso

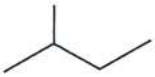
(e) Fractions

143.

144.

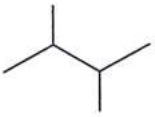
145.

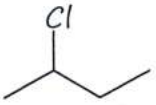
146.

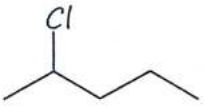
147. 

148.

149.

150. 

151. 

152. 

153.

154.

155.

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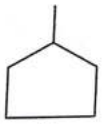
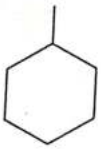
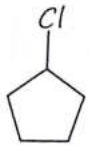
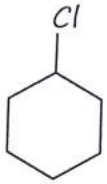
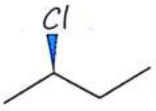
156. 

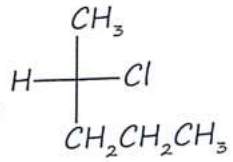
157. 

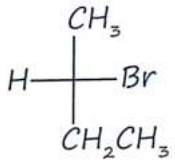
158. 

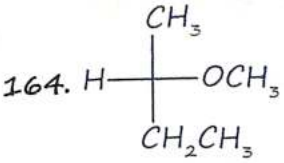
159. 

160.

161. 

162. 

163. 

164. 

165.

166. $CH_3COOH \xrightarrow[\text{Electrolysis}]{NaOH} P \xrightarrow{Br_2/hv} Q$

- $Q \xrightarrow{Na/Et_2O} R \xrightarrow{Br_2/hv}$ All possible bromoderivative
- $Q \xrightarrow{Zn,\ \Delta} S \xrightarrow{Cl_2/hv} V \xrightarrow[(2)\ D_2O]{(1)\ Mg/THF} Y$
- $Q \xrightarrow{Zn/HCl} T \xrightarrow[(Bright)]{Cl_2/hv} W$
- $Q \xrightarrow{(PhCH_2)_2CuLi} U \xrightarrow{Cl_2/hv} X \xrightarrow[(2)\ CH_3COOH]{(1)\ Mg/THF} Z$

167. $CH_3-Cl \xrightarrow[(2)Br_2/hv]{(1)Na/Et_2O} (P) \xrightarrow[(2)Cl_2/hv]{(1)Zn,\Delta} (Q) \xrightarrow[(2)Br_2/hv\ (2\ eq.)]{(1)Na/D.E} (R) \xrightarrow[(2)NBS]{(1)Zn\ dust} (S)$

168. $\xrightarrow[\text{Electrolysis}]{NaOH} (P) \xrightarrow[hv]{Cl_2(excess)} (Q)$ [Find all possible stereoisomers for Q]

169. $CH_3Cl \xrightarrow[Dry(CH_3)_2O]{Na} (P) \xrightarrow[500°C]{Cl_2} (Q) \xrightarrow[Et_2O]{Na} (R)$ [xgm]

(101gm) (60%) (50%) (80%)

Previous Year Questions

8. Number of bromo derivatives obtained on treating ethane with excess of Br_2, in diffused sunlight is _____ [6 April, 2023 (Shift-I)]

Sol. [9]

9. The maximum number of isomers (including stereoisomers) that are possible on mono-chlorination of the following compound, is $CH_3CH_2CH(CH_3)CH_2CH_3$ (IIT JEE 2011)

Sol. [8]

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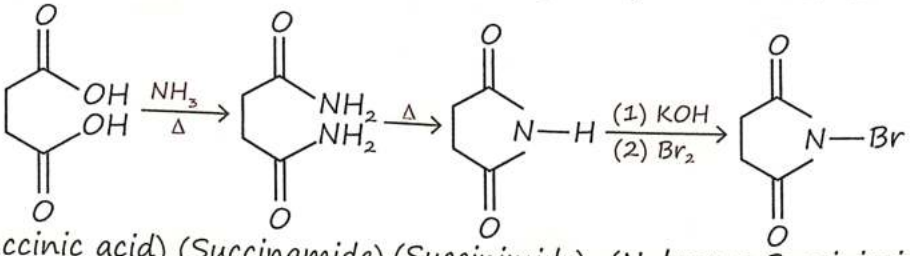
N-bromo succinimide (NBS)

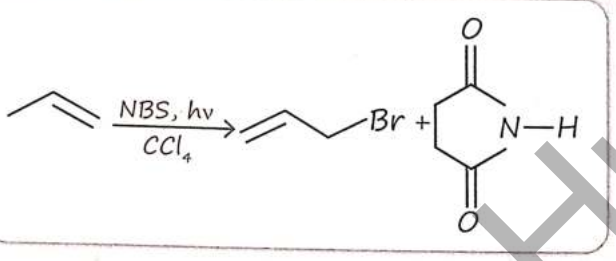
N-Bromosuccinimide (NBS) is a mild brominating agent used mainly for allylic and benzylic bromination via a radical mechanism, and for α-bromination of carbonyl compounds under acidic conditions.

➢ Formation



(Succinic acid) (Succinamide) (Succinimide) (N-bromo Succinimide)

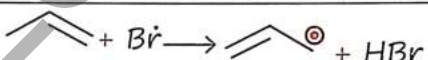
Reaction

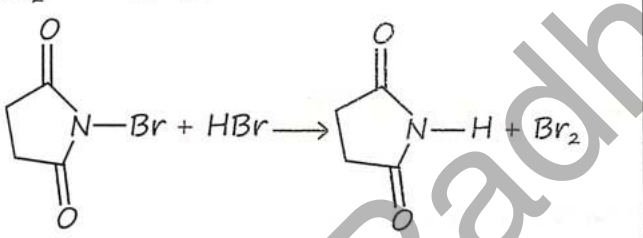


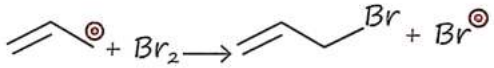
Mechanism 1.

Impurity of Br_2

1. $Br_2 \xrightarrow{h\nu} 2Br^{\odot}$

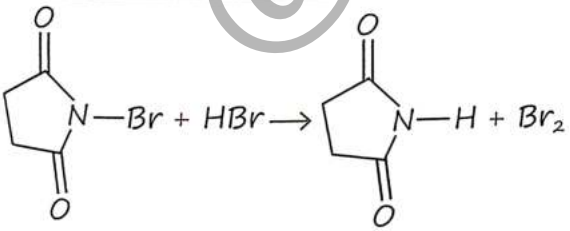
2. 

3. 

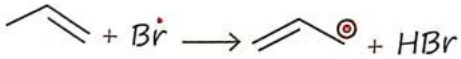
4. 

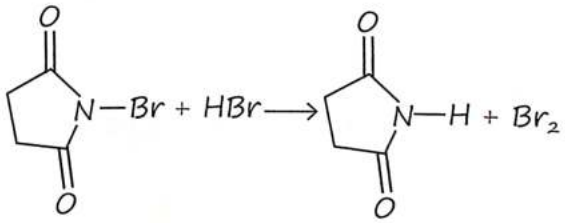
Mechanism 2.

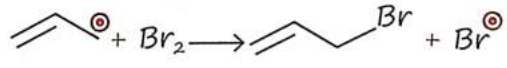
Impurity of HBr

1. 

2. $Br_2 \xrightarrow{h\nu} 2Br^{\odot}$

3. 

4. 

5. 

Baba Tea-Stall (BTS)

(i) NBS is inert in nature
(ii) trace amount of impurity Br_2 or HBr is added
(iii) Reacts only with allylic ($C=C-\dot{C}$) and benzylic ($Ph-\dot{C}H_2$) positions.

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Practice Questions

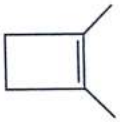
170. When following compound treated with NBS/hv then find:

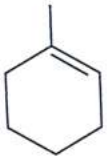
(a) Structure of monobrominated product

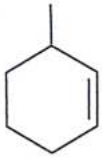
(b) Total monobrominated product

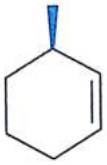
(c) optically active product

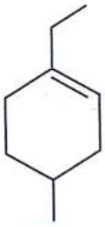
(d) Fractions obtained.

171. 

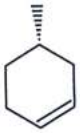
172. 

173. 

174. 

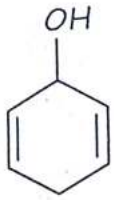
175. 

176.

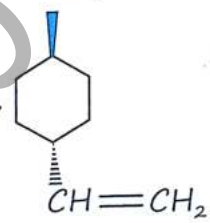
177. 

178.

179.

180. 

181.

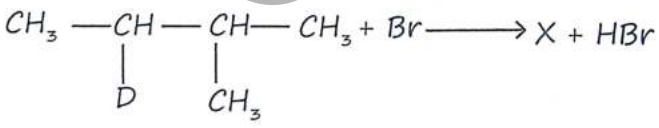
182. 

183.

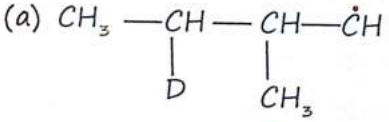
184.

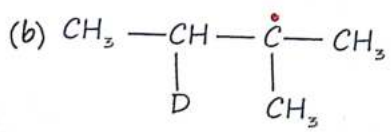
Previous Year Questions

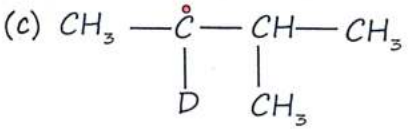
10. Consider the following reaction

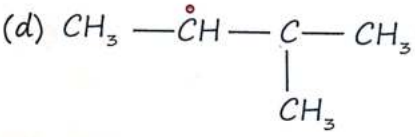


Identify the structure of the major product X (IIT JEE 2002)

(a) 

(b) 

(c) 

(d) 

Sol. (b)

11. Which of the following is a free radical substitution reaction? (2020-Covid)

(a) Acetylene with HBr

(b) Methane with Br_2/hv

(c) Propene with $HBr/(C_6H_5COO)_2$

(d) Benzene with $Br_2/AlCl_3$

Sol. (b)

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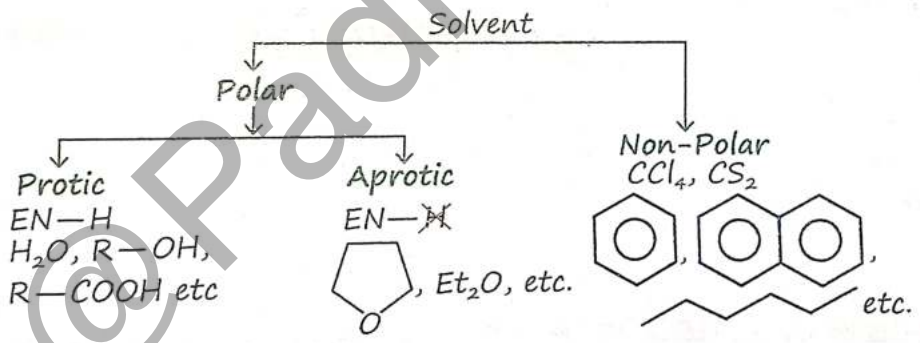
Attacking Reagents

In chemistry, an attacking reagent is a molecule or ion that initiates a chemical reaction by interacting with a substrate molecule or an intermediate. It acts as an electron donor or electron acceptor, leading to the formation of a new chemical bond.

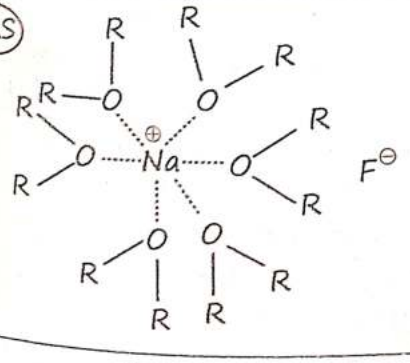
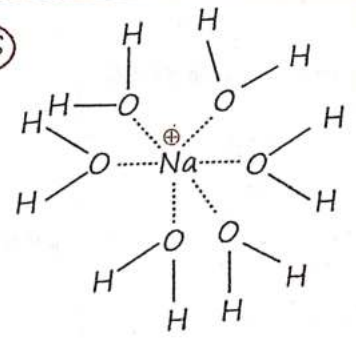
Attacking Reagents

Electrophile	Nucleophile	Solvent
➢ Electrons loving Or ➢ Electron deficient species Or ➢ electron seeking species	➢ Nucleus loving Or ➢ electron rich species Or ➢ Electron donating species.	

Nucleophilicity	Basicity
➢ Electron donating tendency to any $E^{\oplus}$	➢ Electron donating tendency to H^{+}
$Nu^{\ominus} + E^{\oplus} \rightleftharpoons E - Nu$	$B^{\ominus} + H^{\oplus} \rightleftharpoons B - H$
➢ Kinetic property	➢ Thermodynamics properties



Nucleophilicity in different solvent

➢ NaF NaCl NaBr NaI	➢ NaF NaCl NaBr NaI	➢ NaF NaCl NaBr NaI
(PAS) 	(PPS) 	(NPS)
$F^{\ominus} > Cl^{\ominus} > E > I^{\ominus}$	NaI > NaBr > NaCl > NaF	NaI > NaBr > NaCl > NaF

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Electrophile	Nucleophile
1. Positive changed incomplete octet $\overset{\oplus}{C}H_3$, $\overset{\oplus}{F}$, $\overset{\oplus}{Cl}$, $\overset{\oplus}{Br}$, $\overset{\oplus}{I}$, $\overset{\oplus}{NO}$, $\overset{\oplus}{NO_2}$, $\overset{\oplus}{OH}$, $H^{\oplus}$, $D^{\oplus}$, $T^{\oplus}$, $R-\overset{\oplus}{C}=O$ etc.	1. Negatively Changed & complete octet $\overset{\ominus}{C}H_3$, $Cl^{\ominus}$, $F^{\ominus}$, $Br^{\ominus}$, $I^{\ominus}$, $\overset{\ominus}{O}H$, $\overset{\ominus}{S}H$, $\overset{\ominus}{C}N$, $N_3^{\ominus}$, $RCOO^{\ominus}$, $H^{\ominus}$, $D^{\ominus}$, $\overset{\ominus}{N}O_2$ $\left(\overset{\ominus}{O}-\overset{\ominus}{N}=O\right)$
2. Neutral & incomplete octet $AlCl_3$, $AlBr_3$, BF_3, BBr_3, Fe etc.	2. Neutral & Complete octet (lone pair) H_2O, NH_3, NR_3, $R-\ddot{N}H_2$, $H_2N-\ddot{N}H_2$
3. Neutral & Complete octet (π^*, σ^*) FeX_3, sbF_5, $ZnCl_2$, $R-C(=O)-H$, $R-C(=O)-X$, CO_2, SO_3, Br_2, I_2, $I-Cl$ etc.	$Ph-NH-NH_2$, $H_2\ddot{N}-NH-C(=O)-NH_2$
	3. π – electrons Alkene, Alkyne, Napthalene etc.

Practice Questions

185. Which of the following statements are correct for nucleophile?
(a) All negatively charged species are nucleophiles
(b) Nucleophiles are Lewis bases
(c) Alkenes, alkynes, benzene and pyridine are nucleophiles
(d) All are correct

186. Consider the species.
(I) $\overset{\ominus}{O}H$ (II) $CH_3-\overset{\ominus}{O}$
(III) $\overset{\ominus}{C}H_3$ (IV) $\overset{\ominus}{N}H_2$
Arrange these nucleophilic species in their decreasing order of nucleophilicity:
(a) III > IV > I > II (b) II > I > III > IV
(c) I > II > III > IV (d) III > I > II > IV

187. The nucleophilicity of CH_3^-, NH_2^-, OH^- and F^- decrease in which order?
(a) $CH_3^- > NH_2^- > OH^- > F^-$
(b) $OH^- > NH_2^- > CH_3^- > F^-$
(c) $NH_2^- > OH^- > CH_3^- > F^-$
(d) $CH_3^- > OH^- > F^- > NH_2^-$

188. Arrange the following nucleophiles in the order of their nucleophilic strength:
(a) $OH^- > CH_3COO^- > OCH_3^- > C_6H_5O^-$
(b) $CH_3COO^- < C_6H_5O^- < OCH_3^- < OH^-$
(c) $C_6H_5O^- > CH_3COO^- < CH_3O^- < OH^-$.
(d) $CH_3COO^- < C_6H_5O^- < OH^- < CH_3O^-$

189. Correct order of leaving group tendency is :
(a) $I^- > Br^- > Cl^- > F^-$ (b) $F^- > Cl^- > Br^- > I^-$
(c) $Cl^- > F^- > Br^- > I^-$ (d) $I^- > Cl^- > Br^- > F^-$

190. Match the following

Column – I	Column – II
(a) I^-	(P) A good leaving group
(b) F^-	(Q) Very weak base
(c)	(R) A good nucleophile in polar protic solvent
(d) OH^-	(S) A poor leaving group

191. The most reactive nucleophile among the following is
(a) CH_3O^- (b) $C_6H_5O^-$
(c) $(CH_3)_2CHO^-$ (d) $(CH_3)_2CO^-$

192. Which of the following is not an electrophile?
(a) CH_2CH_2 (b) BF_3 (c) $^+NO_2$ (d) Fe^{+3}

193. Which is the MOST basic nucleophile in the following series?
(a) F^- (b) CH_3CH_2OH
(c) H_2O (d) $CH_3CH_2O^-$

194. How are basicity and leaving group ability related?
(a) They are not related to each other
(b) Good leaving groups are strong bases
(c) Good leaving groups are weak bases
(d) Leaving group $\propto$ Basic strength

195. Correct order of leaving group character is :
(a) $CCl_3COO^{(-)} > CH_3COO^{(-)} > CH_3O^{(-)} > PhO^{(-)}$
(b) $OTs^{(-)} > CH_3SO_3^{(-)} > PhO^{(-)} > OH^{(-)}$
(c) $CCl_3COO^{(-)} > PhO^{(-)} > CH_2=CH-O^{(-)} > CH_3O^{(-)}$
(d) $HCOO^{(-)} > PhO^{(-)} > CH_3COO^{(-)} > CH_3O^{(-)}$

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Types of Reaction

Addition Rxn

- $A=B \xrightarrow{X-Y} \underset{\underset{X}{|}}{\overset{\overset{|}{A}}{}}-\underset{\underset{Y}{|}}{\overset{\overset{|}{B}}{}}$
- DU ↓
- $E^{\oplus}, Na^{\ominus}$, fR Possible

Elimination Rxn

- $\underset{\underset{X}{|}}{\overset{\overset{|}{A}}{}}-\underset{\underset{Y}{|}}{\overset{\overset{|}{B}}{}} \xrightarrow{-XY} A=B$
- DU ↑

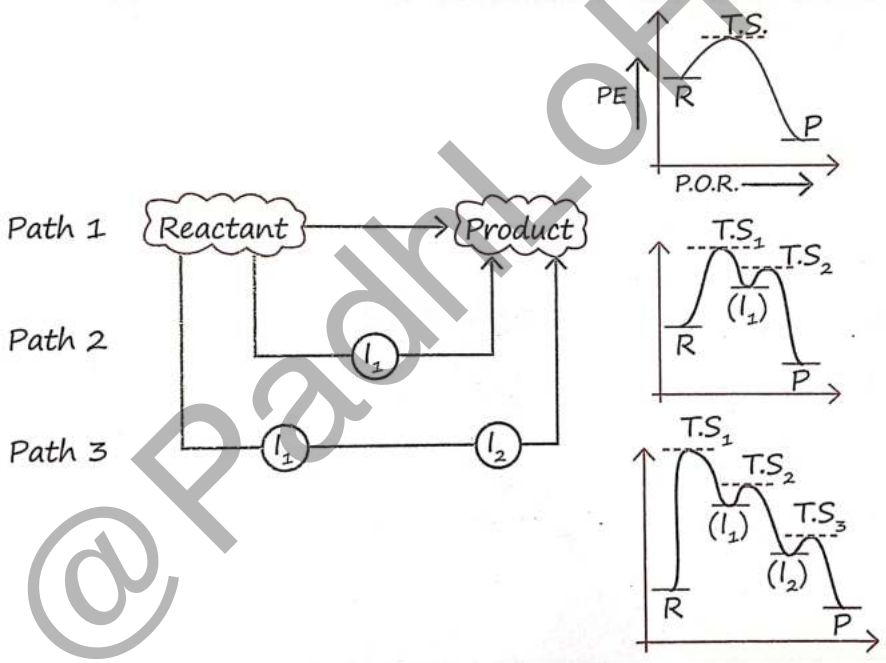
Substitution Rxn

- $A-B-X \xrightarrow{Y} A-B-Y$
  - $E^{\oplus}$ (Electrophilic Sub)
  - $Nu^{\ominus}$ (Nucleophilic Sub)
- DU Same
- $E^{\oplus}$, $Nu^{\ominus}$, fR Possible

Rearrangement Rxn

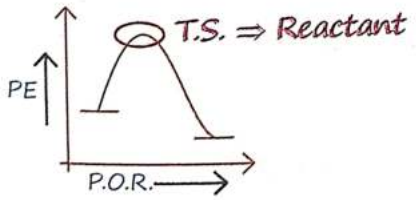
- $A-B-C \longrightarrow B-A-C$
- DU Same

Basics of Rxn



Hammond's Postulate

- Reaction is exothermic if T.S. will look like Reactant
- Reaction is endothermic if T.S. will look like Product



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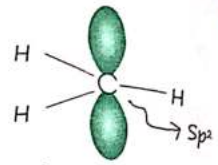
Basics of Carbocation

A species having a carbon atom possessing sextext of electrons and a positive charge is called a carbocation (earlier called carbonium ion).

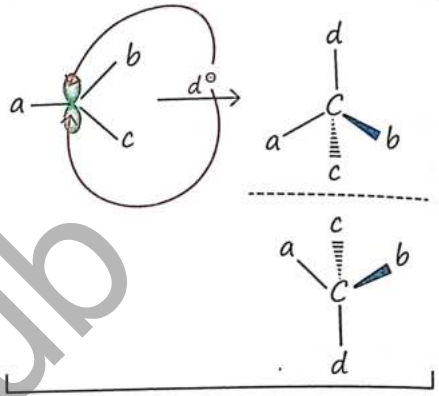


Trigonal Planar

Properties

- $\overset{\oplus}{C}H_3$
- , sp^2
- Magnetic moment = $\sqrt{n(n+2)}$ B.M. = 0 M.M.
- SM = $|2s|+1$ = 1 SM, singlet,
- Vacant Pπ orbital
- Octet incomplete
- EDG (+R, +H, +I)
- 3 Bond Pair
- 0 Unpaired electron

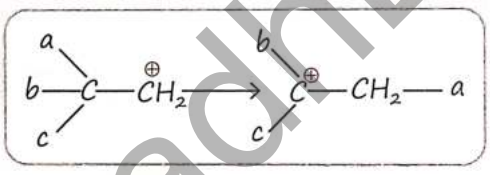
Reaction of Carbocation



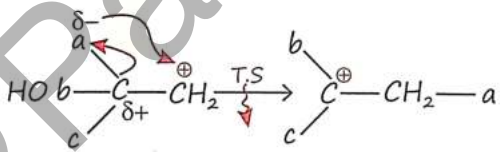
(Racemic mixture)
If $a \neq b \neq c \neq d$

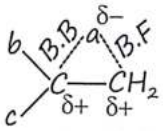
Rearrangement of Carbocation

Reaction



Mechanism





Baba Tea-Stall (BTS)

(i) 3 Member cyclic T.S involved
(ii) 3C – $2e^-$ bond
(iii) 1, 2- shift involved
(iv) Carbocation Rearrangement
(v) Stability increases
(vi) In case of rearrangement, carbocation gets stability in each step.
(vii) During rearrangement, Configuration doesn't change
(viii) In case of B. B. following migratory aptitude is applied. H > Ar > R

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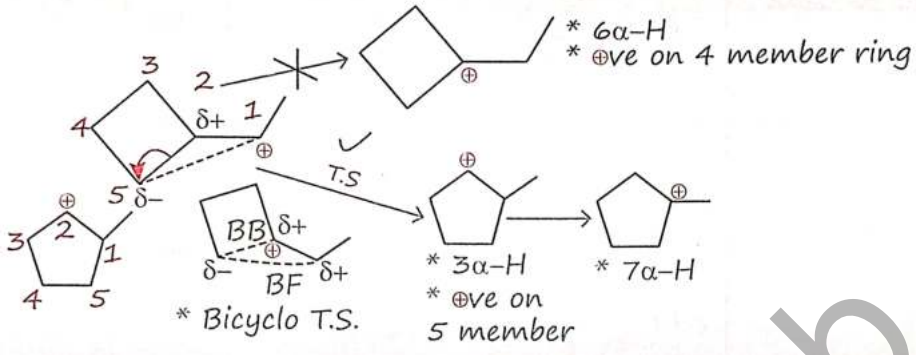
Ring Expansion

Rearrangement of carbocation is a process in which a carbocation shifts its structure (via hydride shift or alkyl shift) to form a more stable carbocation, usually moving from a less stable (1° or 2°) to a more stable (2° or 3° or resonance-stabilized) form.

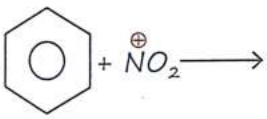


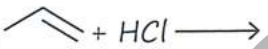
Practice Questions

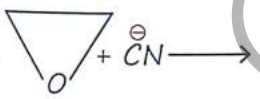
➢ Attacking Reagents

196. Find the E^+ and Nu^- in the following reactions.

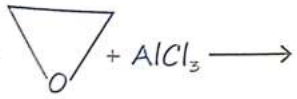
197. $CH_3 - Cl + H_2O \longrightarrow$

198.  $+ \overset{\oplus}{N}O_2 \longrightarrow$

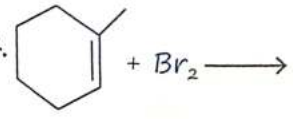
199.  $+ HCl \longrightarrow$

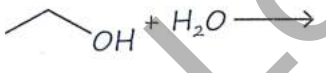
200.  $+ \overset{\ominus}{C}N \longrightarrow$

201. $CH_3-\overset{\overset{O}{\|}}{C}-Cl + CH_3O^{\ominus} \longrightarrow$

202.  $+ AlCl_3 \longrightarrow$

203. $CH_3CH_2OH + PCl_3 \longrightarrow$

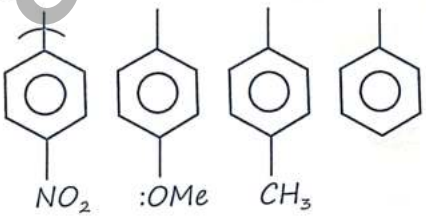
204.  $+ Br_2 \longrightarrow$

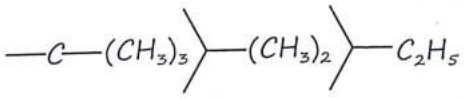
205. 

206.

➢ Carbocation

207. $-H > -D > -T$

208. 

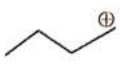
209. 

210. $\overset{\oplus}{C}H_3$

211. $CH_3-\overset{\oplus}{C}H_2$

212.

213.

214. 

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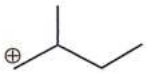
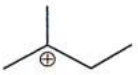
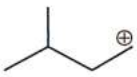
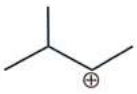
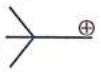
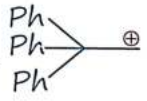
216. 

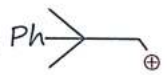
217. 

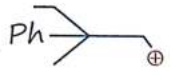
218. 

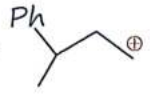
219. 

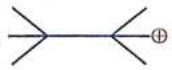
220. 

221. 

222. 

223. 

224. 

225. 

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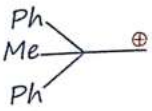
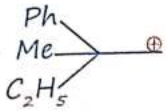
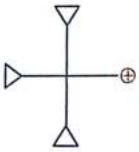
226. 

227. 

228. 

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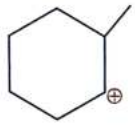
230. 

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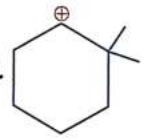
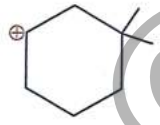
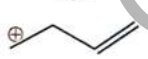
233. 

234. 

235. 

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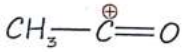
237. 

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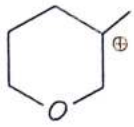
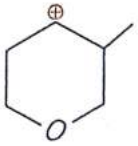
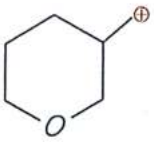
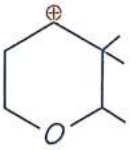
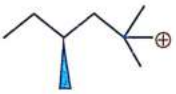
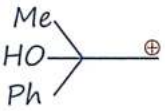
240. 

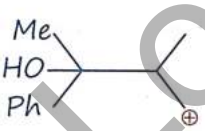
241. 

242. 

243. 

244. 

245. 

246. 

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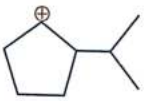
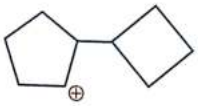
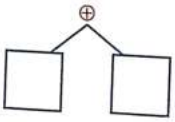
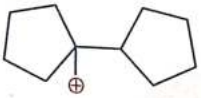
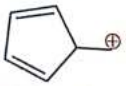
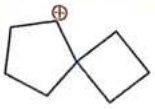
249. 

250. 

251. 

252. 

253. 

254. 

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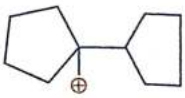
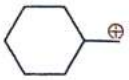
256. 

257. 

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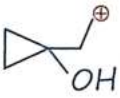
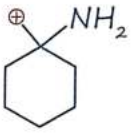
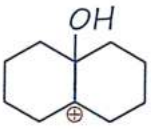
260. 

261. 

262. 

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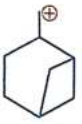
267. 

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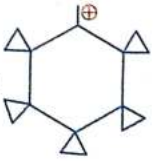
269. 

270.

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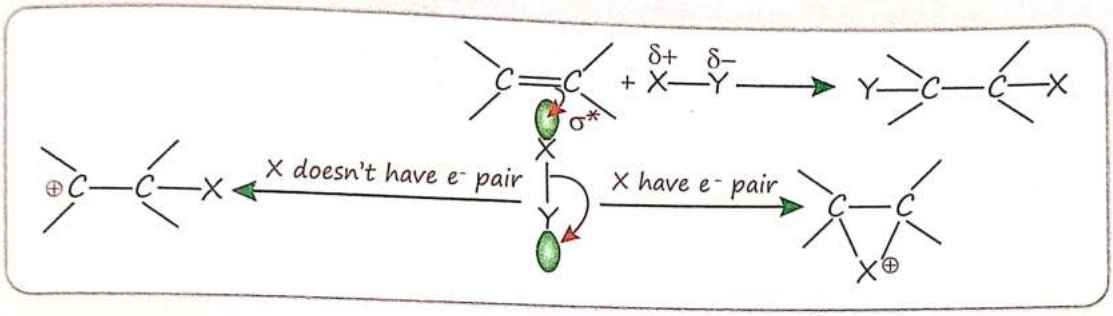
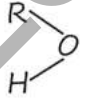
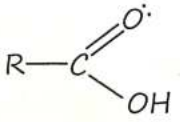
Electrophilic Addition Reaction

Electrophilic addition reaction is a reaction where an electrophile attacks a double or triple bond in an unsaturated compound, leading to the addition of atoms or groups across the π bond.

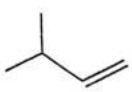
Reaction

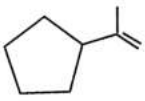


Reagent	$E^{\oplus}$	$Nu^{\ominus}$	Path
H-Cl/H-Br/H-I	$H^{\oplus}$	$X^{\ominus}$ (Hydrohalogenation)	1(cc)
D-Cl	$D^{\oplus}$	$Cl^{\ominus}$	1(cc)
$H^{\oplus}/H_2O$ or $H_3O^{\oplus}$ dilute H_2SO_4	$H^{\oplus}$	H_2O (Hydration)	1(cc)
$ROH/H^{\oplus}$	$H^{\oplus}$		1(cc)
$RCOOH/H^{\oplus}$	$H^{\oplus}$		1(cc)

Practice Questions

271. $H_3C-CH=CH_2$ → HX, H_2SO_4(Cold), H^+/H_2O, ROH/H^+, RCOOH/H^+, D-Cl

272.  → HX, H_2SO_4(Cold), H^+/H_2O, ROH/H^+, RCOOH/H^+, D-Cl

273.  → HX, H_2SO_4(Cold), H^+/H_2O, ROH/H^+, RCOOH/H^+, D-Cl

274. → HX, H_2SO_4(Cold), H^+/H_2O, ROH/H^+, RCOOH/H^+, D-Cl

275. → HX, H_2SO_4(Cold), H^+/H_2O, ROH/H^+, RCOOH/H^+, D-Cl

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Hydrohalogenation

Hydrohalogenation is an electrophilic addition reaction in which a hydrogen halide (HX) adds across th double bond of an alkene, forming an alkyl halide.

Reaction

$$H_2C{=}CH_2 \xrightarrow[CCl_4]{H\text{-}X} H_3C{-}CH_2{-}X$$

Mechanism

$$H_2C{=}CH_2 \xrightarrow[rds]{H\text{-}X} H_2\overset{\oplus}{C}{-}CH_3$$

$$H_2\overset{\oplus}{C}{-}CH_3 \xrightarrow{X^{\ominus}} CH_3{-}CH_2{-}X$$

Baba Tea-Stall (BTS)

(i) Carbocation obtained as intermediate
(ii) Rearrangement may be possibles.
(iii) Formation of carbocation is rds
(iv) Both Syn and anti addition possible
(v) X and Y may or may not be vicinal

(vi) Rate of reaction:
Rate of H—X ⟶ HI > HBr > HCl > HF
Rate of Electron addition →
Nucleophilicity (e^- density) of alkene
Stability of carbocation

$$\frac{1}{\text{Stability of alkene}}$$

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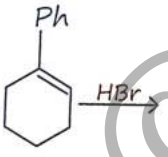
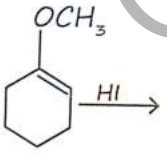
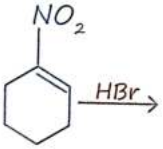
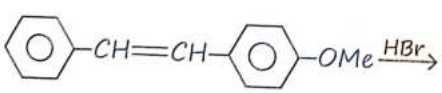
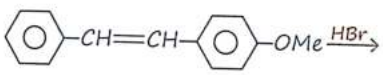
Practice Questions

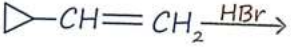
276. 

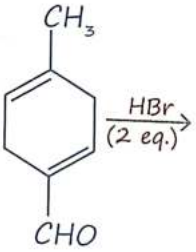
277. 

278. 

279. 

280. 

281. 

282. 

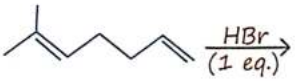
283. $CH_2{=}CH{-}Cl \xrightarrow{HCl}$

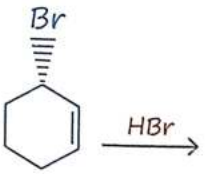
284. $HC{\equiv}CH \xrightarrow[(excess)]{HCl}$

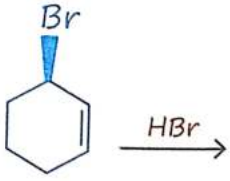
285. $CH_3{-}C{\equiv}CH \xrightarrow[DI]{DCl}$

286. $CH_2{=}C{=}CH_2 \xrightarrow[(excess)]{HCl}$

287.

288. 

289. 

290. 

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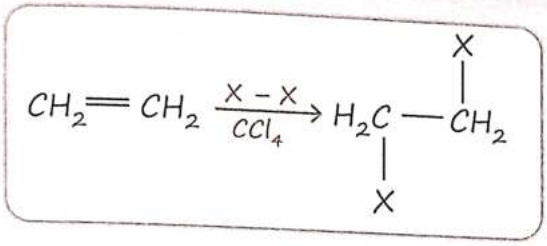
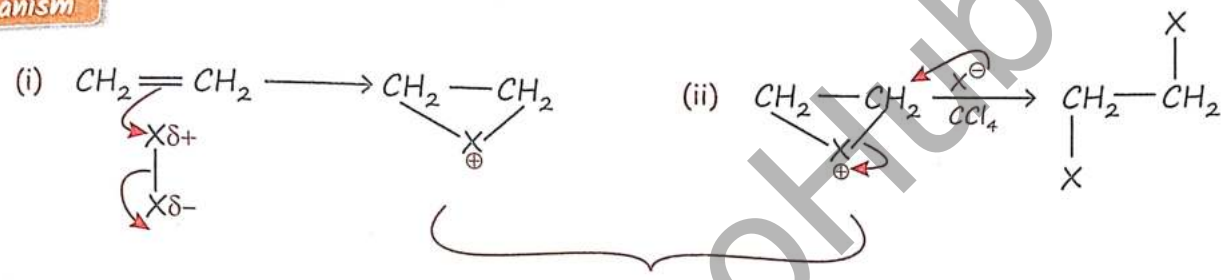
Addition of X – X

Halogens like bromine or chlorine add up to alkene to form vicinal dihalides. However, iodine does not show addition reaction under normal conditions. The reddish orange colour of bromine solution in carbon tetrachloride is discharged when bromine adds up to an unsaturation site. This reaction is used as a test for unsaturation. Addition of halogens to alkenes is an example of electrophilic addition reaction involving cyclic halonium ion formation.

Reaction



Mechanism



Baba Tea-Stall (BTS)

- Non classical cation (NCC) intermediate
- No rearrangement
- Both X and X must be vicinal
- Involves 'Anti addition'
- Rate of reaction ⇒ $[F_2 > Cl_2 > Br_2 > I_2]$

Path – 2

Reagent	$E^{\oplus}$	$Nu^{\ominus}$	Path
X_2/CCl_4	$X^{\oplus}$	$X^{\ominus}$	2(NCC)
Br_2/H_2O or HOBr Reddish brown	$Br^{\oplus}$	$Br^{\ominus}$, $H_2O/OH^{\ominus}$ (major)	2(NCC)
Br_2/H_2O in NaCl Brine solution	$Br^{\oplus}$	$Br^{\ominus}$, H_2O, $Cl^{\ominus}$	2(NCC)
NOCl [Tilden reagent]	$[\overset{\oplus}{N}O=O \longleftrightarrow NO\equiv\overset{\oplus}{O}]$	$Cl^{\ominus}$	2(NCC)
IN_3	$I^{\oplus}$	N_3	2(NCC)

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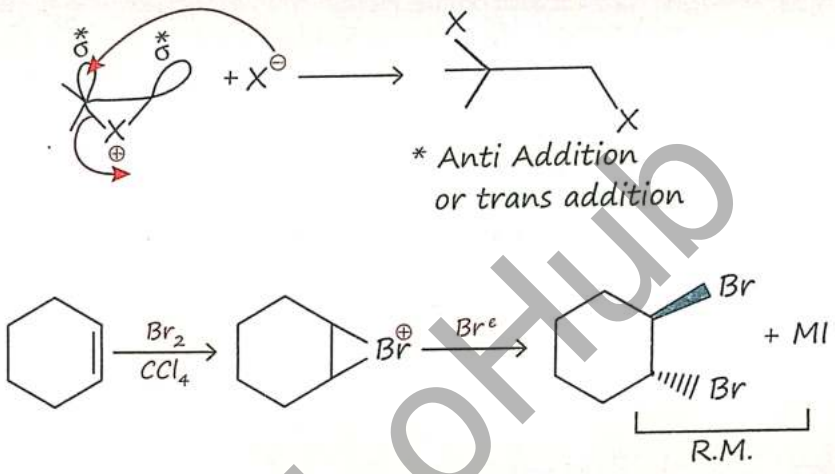
NCC Opening

- NCC opens from side at which carbocation is more stable.
- In other cases, where only ring present then it will be open from less hindered site.

Stereochemistry

Stereochemistry of addition of X_2 (halogenation) involves anti addition, where the two halogen atoms add to opposite faces of the double bond, leading to the formation of trans (anti) dihalides.

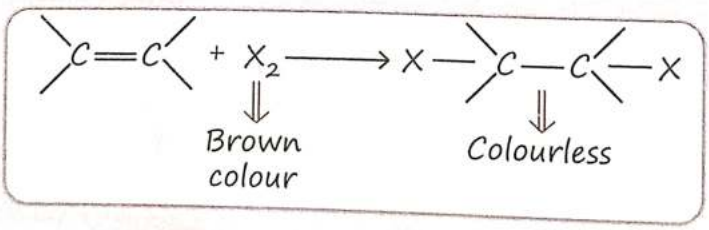


Types of alkene	Type of addition	Type of product
Cis (+)	Syn (+)	Erythro (same side) (+)
Trans (−)	Anti (−)	Erythro (same side) (÷)
Cis (+)	Anti (−)	Threo (−)
Trans (−)	Syn (+)	Threo (−)

Use in POC

1. Used to detect unsaturation in compound.
2. Reddish brown colour of Br_2 becomes colorless

Reaction



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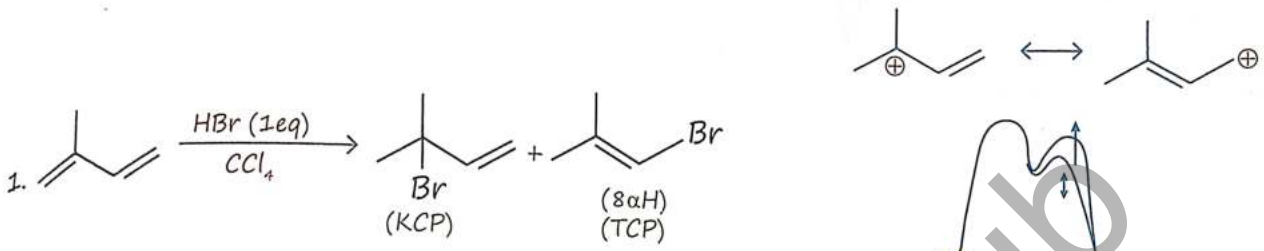
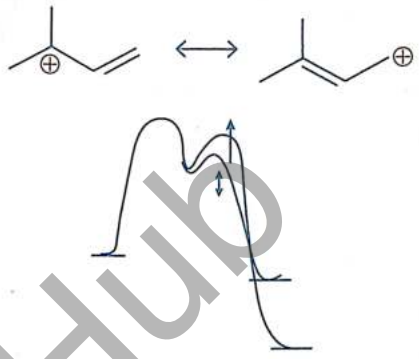
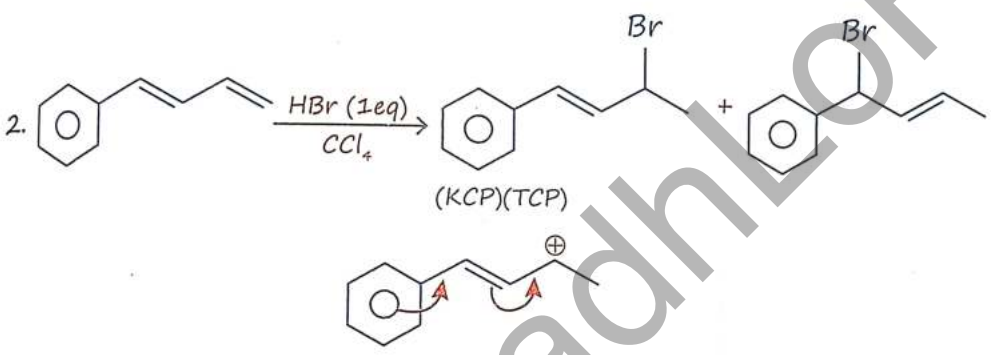
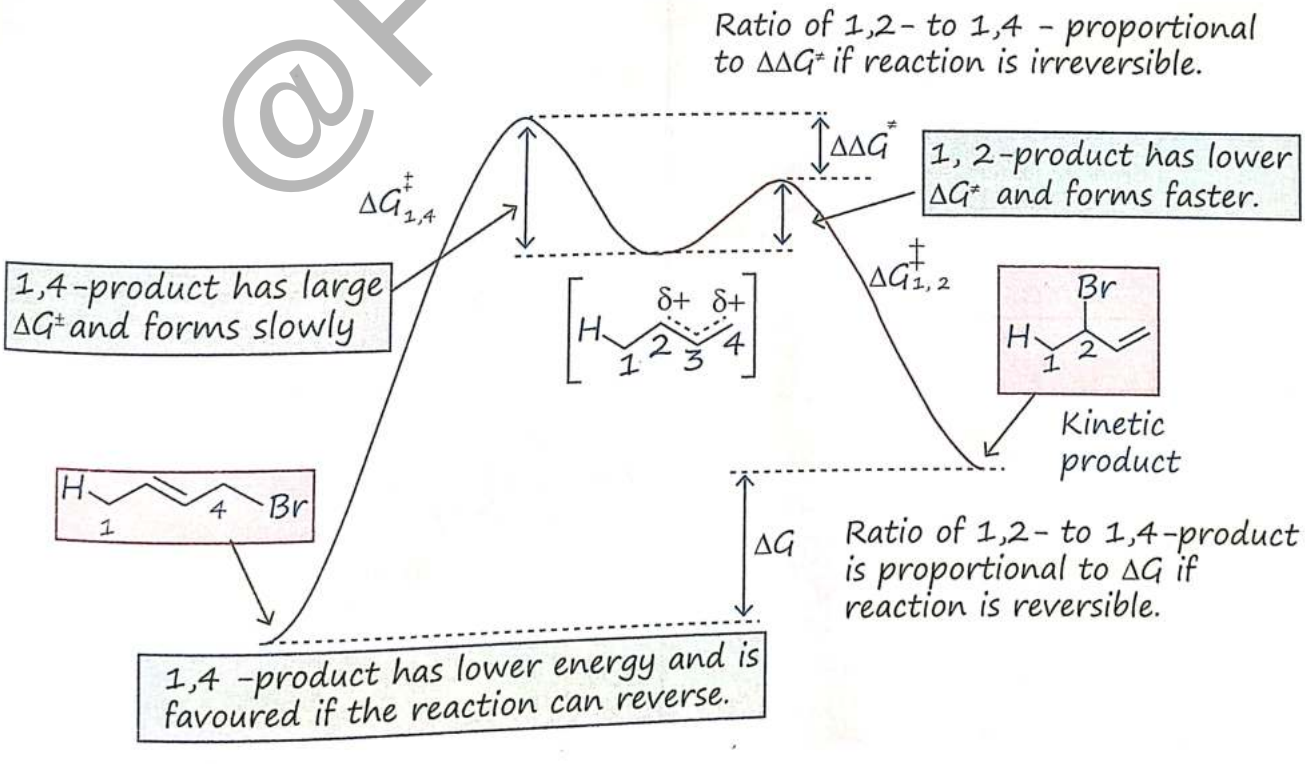
KCP and TCP

KCP	KCP
▫ Kinetically controlled product	▫ Thermodynamically controlled product
▫ 1, 2-product is assumed to be KCP.	▫ Can be 1, 2 and 1, 4
▫ Rate (fast)	▫ Stability (product stable)
▫ Favours at low temperature	▫ Favours at high temperature.









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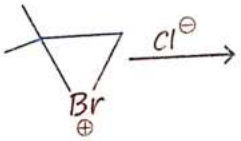
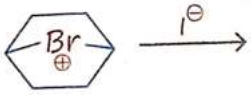
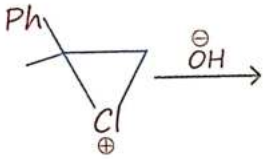
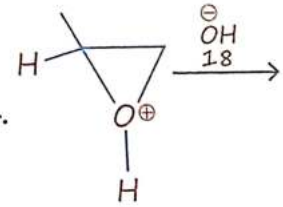
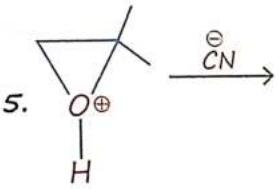
Practice Questions

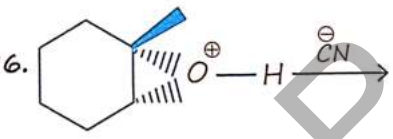
291. 

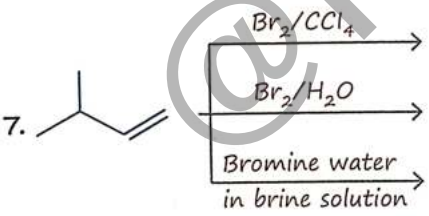
292. 

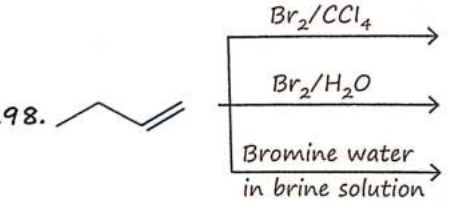
293. 

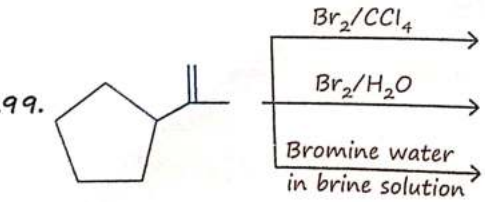
294. 

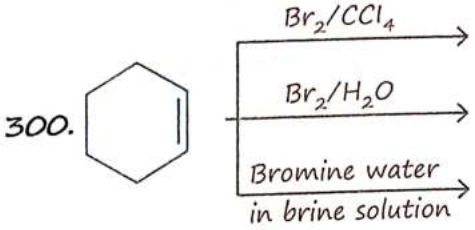
295. 

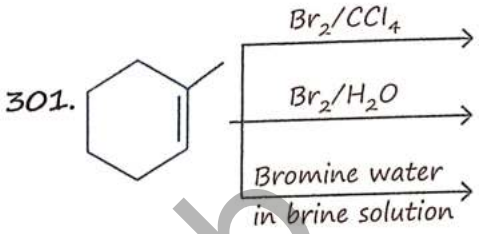
296. 

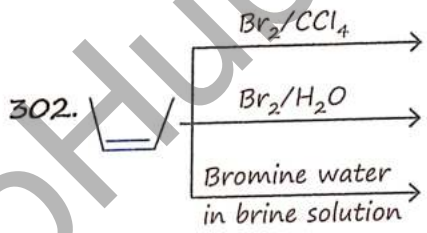
297. 

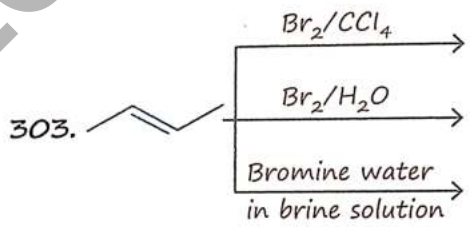
298. 

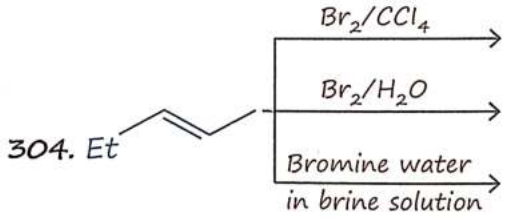
299. 

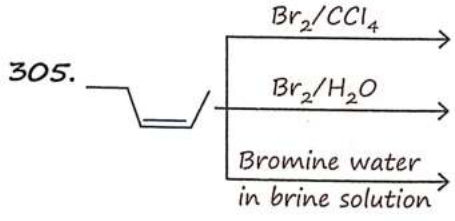
300. 

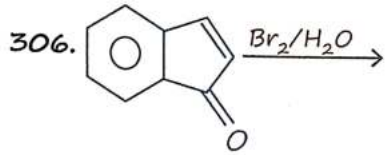
301. 

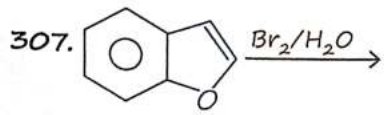
302. 

303. 

304. 

305. 

306. 

307. 

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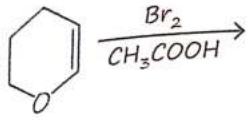
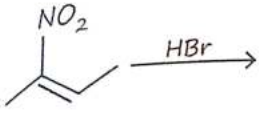
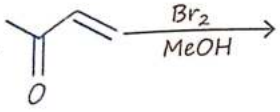
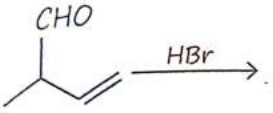
308. 

309. 

310. 

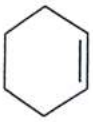
311. 

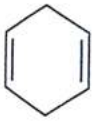
Q. How many of given compounds will decolourise Bromine solution?

312.

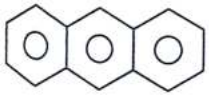
313. $HC \equiv CH$

314.

315. 

316. 

317.

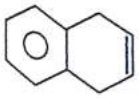
318. 

319. $CH_3-C \equiv CH$

320.

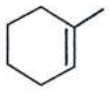
321.

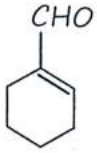
322.

323. 

- Markovnikov and AntiMarkovnikon
- In $E^{\oplus}$ addition reaction, $Nu^{\ominus}$ add to the lesser H containing carbon.

324. $O_2N-CH=CH_2 \xrightarrow[CCl_4]{HBr}$

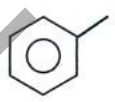
325.  $\xrightarrow[H_2SO_4]{dilute}$

326.  $\xrightarrow[CCl_4]{HBr}$

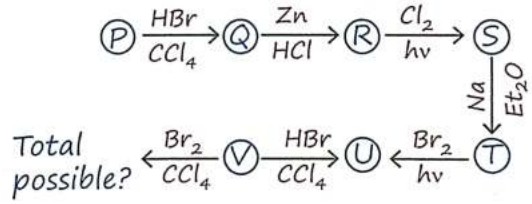
327. $HC \equiv CH \xrightarrow[(excess)]{Br_2/CCl_4}$

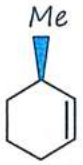
328. $H_3C-\underset{(8\ gm)}{C} \equiv CH \xrightarrow[(excess)]{Br_2/CCl_4}$ (x gm) Mass of product

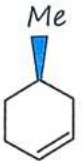
329. $H_3CCH_2C \equiv CH \xrightarrow[(excess)]{Br_2/CCl_4}$

330.  $\xrightarrow[(excess)]{Br_2/H_2O}$

331. P can be obtained by electrolysis of sodium succinate.



332.  $\xrightarrow{Br_2/CCl_4}$

333.  $\xrightarrow{Br_2/CCl_4}$

334. $\xrightarrow{Br_2/CCl_4}$

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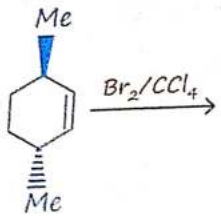
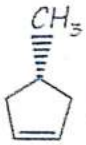
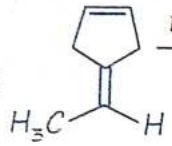
335. 

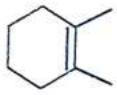
Br_2/CCl_4

336. 

Br_2/CCl_4

337. 

Br_2/CCl_4

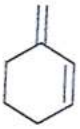
338. 
Br_2/CCl_4

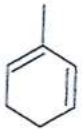
339. Cyclooctene $\xrightarrow{Br_2/CCl_4}$

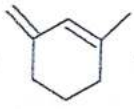
340. C_5H_{10} $\xrightarrow{Br_2/CCl_4}$

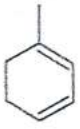
341. C_4H_8 $\xrightarrow{Br_2/CCl_4}$

342.
HBr (1eq.)

343. 
HBr (1eq.)

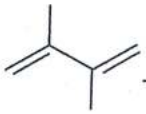
344. 
HBr (1eq.)

345. 
HBr (1eq.)

346. 
HBr (1eq.)

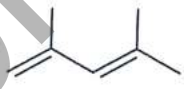
347.
HBr (1eq.)

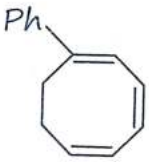
348.
HBr (1eq.)

349. 
HBr (1eq.)

350.
HBr (1eq.)

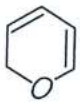
351.
HBr (1eq.)

352. 
HBr (1eq.)

353. 

HBr (1.eq.)

354. $CH_2=CH-C\equiv CH$ $\xrightarrow[(1eq.)]{HBr}$

355. 
HBr (1eq.)

356.
Br_2 (1eq.)

357.
Br_2 (1eq.)

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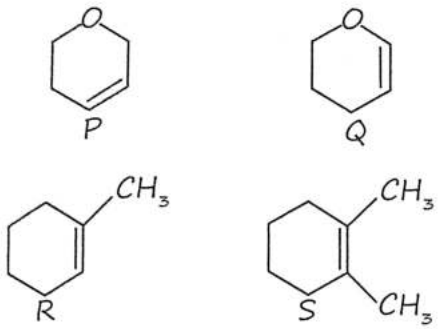
12. Following are the four molecules "P", "Q", "R" and "S".

Which one among the four molecules will react with $H-Br_{(aq)}$ at the fastest rate?



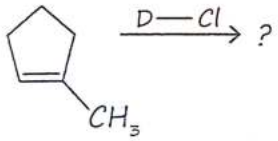
[24 Jan, 2025 (Shift-I)]

(a) Q (b) R

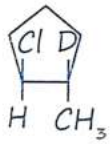
(c) P (d) S

Sol. (a)

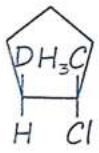
13. Major product of the following reaction is

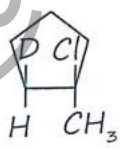


[31 Jan, 2024 (Shift-II)]

(a) 

(b)

(c) 

(d) 

Sol. (c, d)

14. The one giving maximum number of isomeric alkenes on dehydrohalogenation reaction is (excluding rearrangement)

[29 Jan, 2023 (Shift-II)]

(a) 1-Bromo-2-methylbutane

(b) 2-Bromopropane

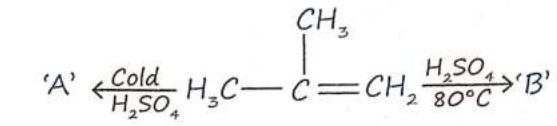
(c) 2-Bromopentane

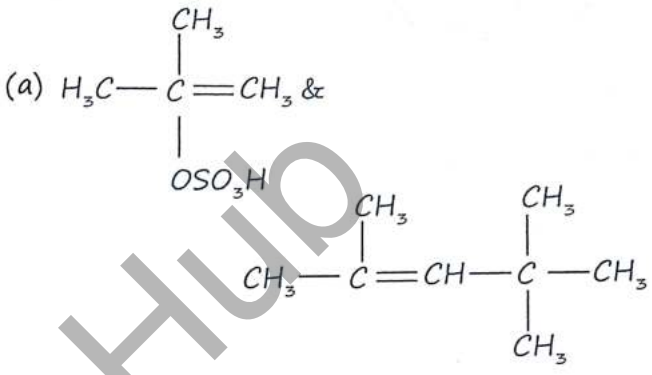
(d) 2-Bromo-3,3-dimethylpentane

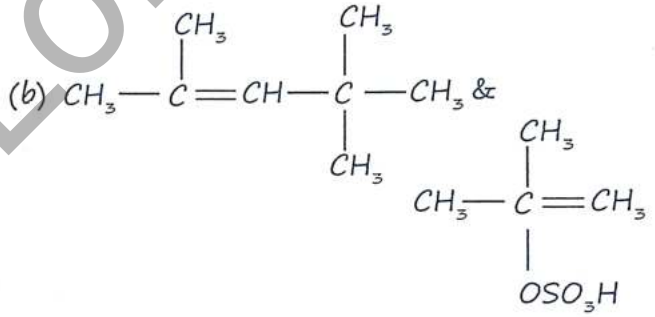
Sol. (c)

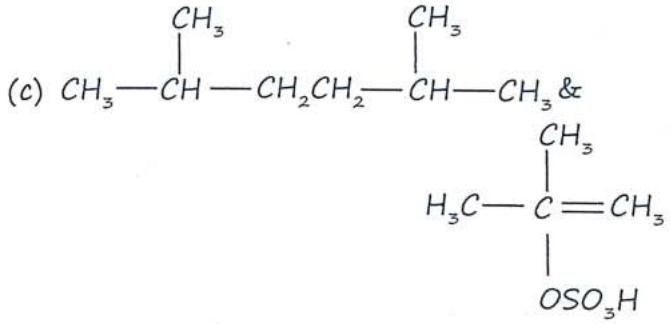
15. The major products 'A' and 'B', respectively, are

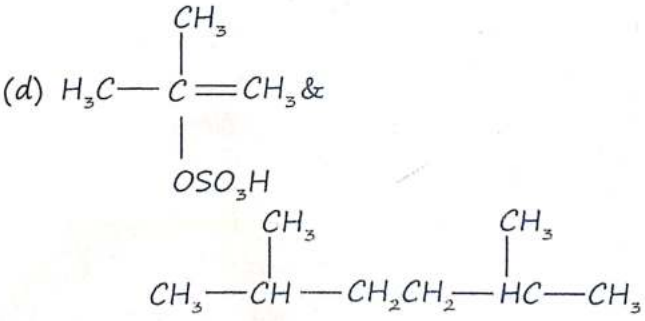
[30 Jan, 2023 (Shift-I)]



(a) 

(b) 

(c) 

(d) 

Sol. (a)

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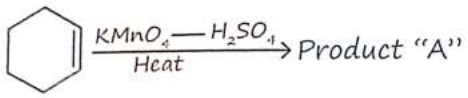
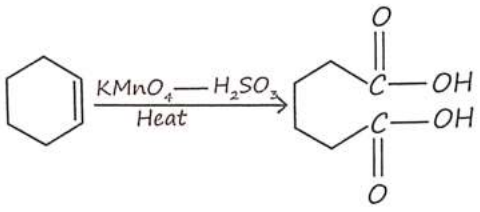
16. Consider the given chemical reaction:



Product "A" is [05 April, 2024 (Shift-II)]

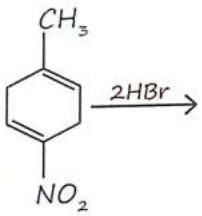
(a) picric acid
(b) oxalic acid
(c) acetic acid
(d) adipic acid

Sol. (d)

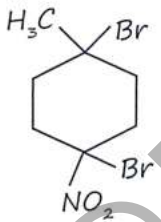


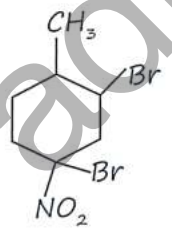
Adipic Acid (A)

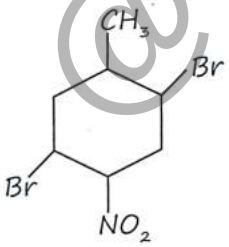
17. The major product of the following reaction is

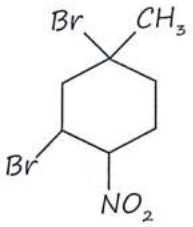


[6 Sept, 2020 (Shift-I)]

(a)


(b)


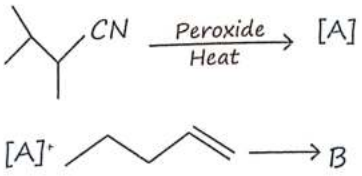
(c)


(d)


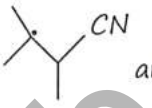
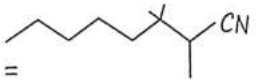
Sol. (d) $-NO_2$ is an electron withdrawing group while

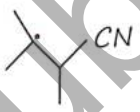
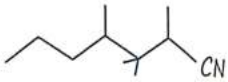
$-CH_3$ is an electron releasing group.

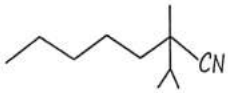
18. The major products A and B in the following reactions are:

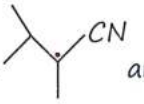
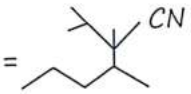


[8 Jan, 2020 (Shift-I)]

(a) A =  and B = 

(b) A =  and B = 

(c) A = and B = 

(d) A =  and B = 

Sol. (d)

19. Statement-I: Addition of bromine to trans-2-butene yields meso-2, 3-dibromo butane.

Statement-II: Bromine addition to an alkene is an electrophilic addition. (IIT JEE 2001)

(a) Statement-I and II are true and statement-II is correct explanation of statement-I

(b) Statement-I and II are true and statement-II is NOT correct explanation of statement-I

(c) Statement-I is correct, Statement-II is wrong

(d) Statement-I is wrong, Statement-II is correct

Sol. (b)

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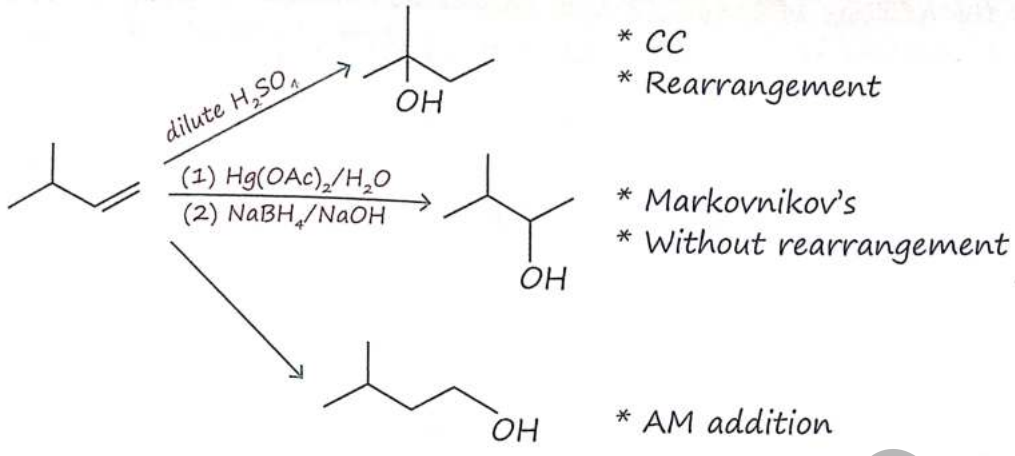
Hydration of Alkene



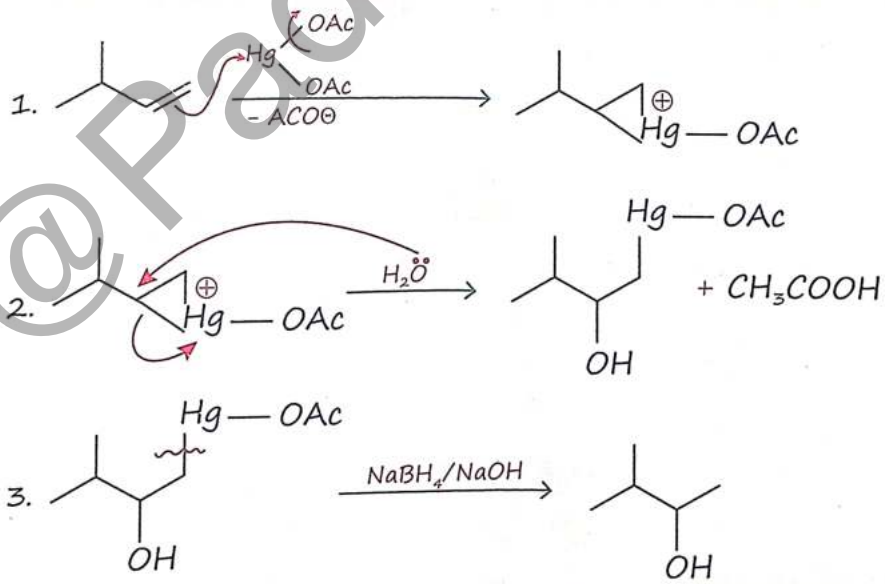
* CC
* Rearrangement
* Markovnikov's
* Without rearrangement
* AM addition

Oxymercuration and Demercuration OMDM

Hydration of alkyne is the addition of water (H_2O) across the triple bond, usually in the presence of $HgSO_4$ and H_2SO_4, leading to the formation of an enol, which rapidly tautomerises to a ketone (Markovnikov product).

Reaction

Mechanism



Baba Tea-Stall (BTS)

- Markonikov's addition.
- ON is anti/trans addition.
- DM involves free radical.
- Both syn and anti addition possible overall.
- Metallic Hg(O) is obtained.

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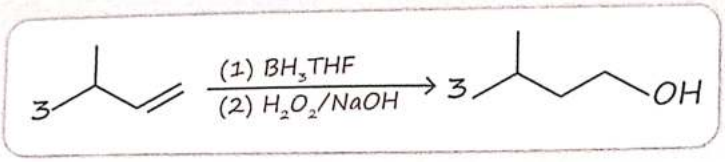
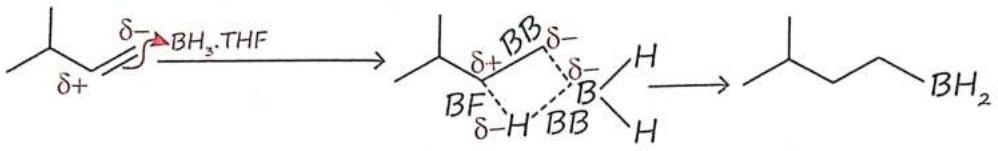
Hydroboration Oxidation (HBO)

Hydroboration is the addition of borane (BH_3) to an alkene, followed by oxidation (with $H_2O_2/NaOH$) resulting in an anti-Markovnikov alcohol with syn addition (both H and OH add to the same face).

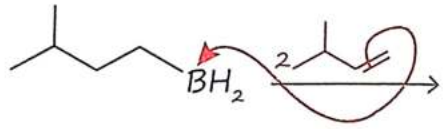
Reaction

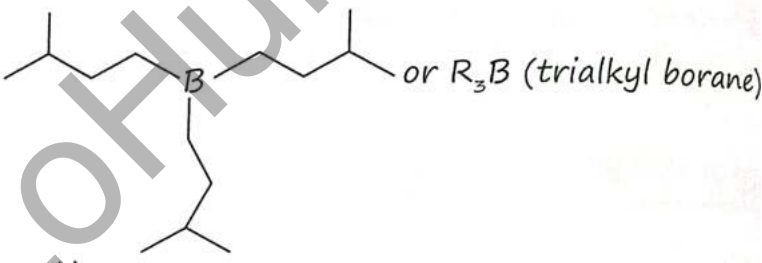


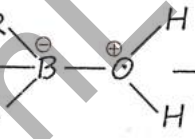
Mechanism



1. * 4MCTS
   * Synaddⁿ

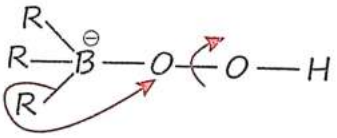
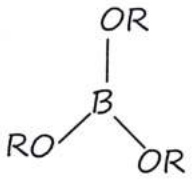
2. 

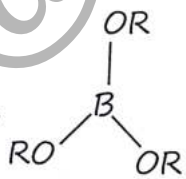


(a) $R_3B + H_2\ddot{O} \longrightarrow$  $\longrightarrow H_3BO_3 + 3R-H$ (hydroboration reduction)

(b) $R_3B + Cl-Cl \longrightarrow$

(c) $R_3B + H_2N-Cl \longrightarrow$

(d) 1. $R_3B + H_2O_2 \xrightarrow{NaOH}$  $\longrightarrow$  $+ 3\overset{\ominus}{O}H$

2.  $+ 3\overset{\ominus}{O}H \longrightarrow 3R-OH$

Baba Tea-Stall (BTS)

- Example of 'syn|cis' addition.
- Antimarkovnikov's addition.
- Involves 4MCTS.
- In case of acid/water, reduction occurs.
- Generally 3 moles of alkene required for 1 mole BH_3.
- 9-BBN | Sia_2BH is used for selective on less crowded terminal alkenes.

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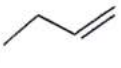
Practice Questions

Oxymercuration and Demercuration OMDM

358. 

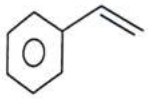
(1) $Hg(OAc)_2/H_2O$
(2) $NaBH_4/NaOH$

(1) $Hg(OAc)_2/H_2O$
(2) $NaBH_4/NaOD$

(1) $Hg(OAc)_2/D_2O$
(2) $NaBD_4/NaOD$

(1) $Hg(OAc)_2/D_2O$
(2) $NaBH_4/NaOH$

(1) $Hg(OAc)_2/CH_3OH$
(2) $NaBH_4/NaOH$
Dilute H_2SO_4

359. 

(1) $Hg(OAc)_2/H_2O$

(1) $Hg(OAc)_2/H_2O$
(2) $NaBH_4/NaOD$

(1) $Hg(OAc)_2/D_2O$
(2) $NaBD_4/NaOD$

(1) $Hg(OAc)_2/D_2O$
(2) $NaBH_4/NaOH$

(1) $Hg(OAc)_2/CH_3OH$
(2) $NaBH_4/NaOH$
Dilute H_2SO_4

360.

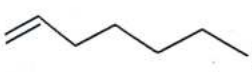
(1) $Hg(OAc)_2/H_2O$

(1) $Hg(OAc)_2/H_2O$
(2) $NaBH_4/NaOD$

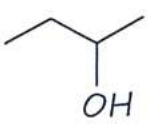
(1) $Hg(OAc)_2/D_2O$
(2) $NaBD_4/NaOD$

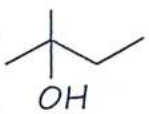
(1) $Hg(OAc)_2/D_2O$
(2) $NaBH_4/NaOH$

(1) $Hg(OAc)_2/CH_3OH$
(2) $NaBH_4/NaOH$
Dilute H_2SO_4

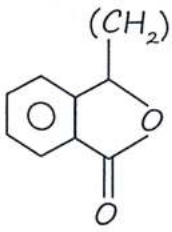
361. 

1. $Hg(OAc)_2/H_2O$
2. $NaBH_4/NaOH$

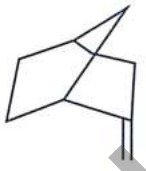
362. Alkene $\xrightarrow[H^{\oplus}/H_2O]{OMDM}$ 

363. Alkene $\xrightarrow[H^{\oplus}/H_2O]{OMDM}$ 

364. Alkene $\xrightarrow[H^{\oplus}/H_2O]{OMDM}$

365. 

(1) $Hg(OAc)_2/H_2O$
(2) $NaBH_4/NaOH$

366. 

(1) $Hg(OAc)_2/H_2O$
(2) $NaBH_4/NaOH$

367.

(1) $Hg(O_2CCF_3)_2/(CH_3)_2CH—OH$
(2) $NaBH_4/NaOH$

368.

(1) $Hg(O_2CCF_3)_2/EtOH$
(2) $NaBH_4/NaOH$

Hydroboration Oxidation (HBO)

369.

(1) B_2H_6-THF
(2) $H_2O_2/NaOH$

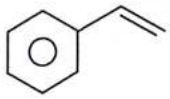
(1) B_2H_6-THF
(2) $D_2O_2/NaOH$

(1) B_2D_6-THF
(2) $H_2O_2/NaOH$

(1) B_2H_6-THF
(2) H_2O

(1) B_2H_6-THF
(2) CH_3COOD

B_2H_6-THF
(1 eq.)

370. 

(1) B_2H_6-THF
(2) $H_2O_2/NaOH$

(1) B_2H_6-THF
(2) $D_2O_2/NaOH$

(1) B_2D_6-THF
(2) $H_2O_2/NaOH$

(1) B_2H_6-THF
(2) H_2O

(1) B_2H_6-THF
(2) CH_3COOD

B_2H_6-THF
(1 eq.)

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371.

(1) B_2H_6-THF
(2) H_2O_2/NaOH

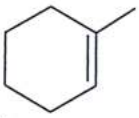
(1) B_2H_6-THF
(2) D_2O_2/NaOH

(1) B_2D_6-THF
(2) H_2O_2/NaOH

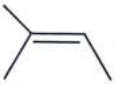
(1) B_2H_6-THF
(2) H_2O

(1) B_2H_6-THF
(2) CH_3COOD

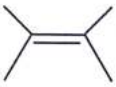
B_2H_6-THF
(1 eq.)

372. 

(1) B_2H_6/THF
(2) H_2O_2/NaOH

373. 

(1) B_2H_6.THF
(2) H_2O_2/NaOH

374. 

(1) B_2H_6.THF
(2) H_2O_2/NaOH

375. $HC \equiv CH$ (1) B_2H_6.THF, (2) H_2O_2/NaOH

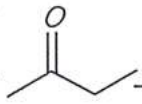
376. $H_3C - C \equiv CH$ (1) B_2H_6.THF, (2) H_2O_2/NaOH

377.

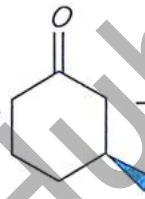
(1) B_2H_6.THF
(2) H_2O

378.

(1) B_2H_6.THF
(2) D_2O

379. 

(1) B_2H_6.THF
(2) H_2O

380. 

(1) B_2H_6.THF
(2) H_2O

Previous Year Questions

20. The product (A) formed in the following reaction sequence is [24 Jan, 2025 (Shift-I)]

$CH_3-C\equiv CH \xrightarrow[(ii)\ HCN]{(i)\ Hg^{2+}H_2SO_4} \xrightarrow[(ii)\ H_2/Ni]{} (A)$ Product

(a) $CH_3-CH_2-\underset{NH_2}{\underset{|}{CH}}-CH_2-OH$

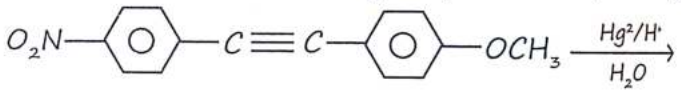
(b) $CH_3-CH_2-\underset{OH}{\underset{|}{CH}}-CH_2-NH_2$

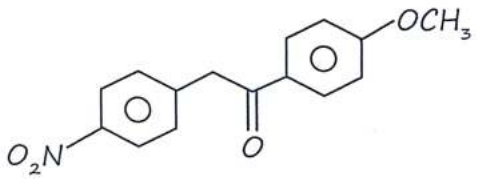
(c) $CH_3-\underset{CH_3}{\underset{|}{\overset{OH}{\overset{|}{C}}}}-CH_2-NH_2$

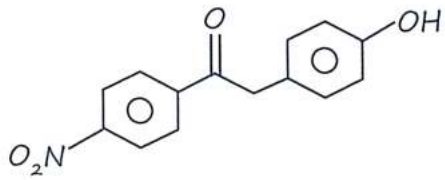
(d) $CH_3-\underset{CH_3}{\underset{|}{\overset{NH_2}{\overset{|}{C}}}}-CH_2-OH$

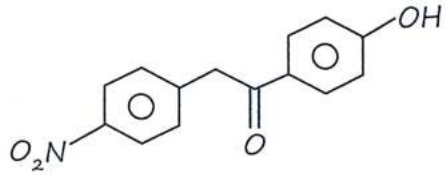
Sol. (c)

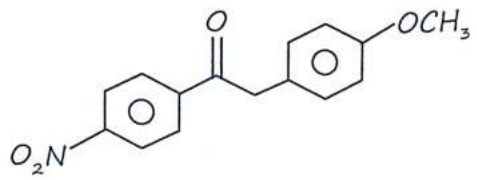
21. The major product obtained from the following reaction is [6 Sept, 2020 (Shift-I)]



(a) 

(b) 

(c) 

(d) 

Sol. (a)

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Passage

An acyclic hydrocarbon P, having molecular formula C_6H_{10}, gave acetone as the only organic product through the following sequence of reactions, in which Q is an intermediate organic compound

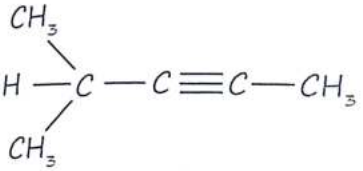
$$P\ (C_6H_{10}) \xrightarrow[\text{(ii) } NaBH_4/\text{ethanol}]{\text{(i) dil.}H_2SO_4/HgSO_4} \xrightarrow[\text{(iii) dil.acid}]{}$$

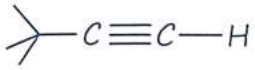
$$Q \xrightarrow[\text{(catalytic amount}(-H_2O))]{\text{(i) dil.}H_2SO_4} \xrightarrow[\text{(iii) } Zn/H_2O]{\text{(ii) } O_3/\text{ethanol}} \underset{H_3C}{\overset{O}{\overset{\|}{C}}}\underset{CH_3}{}$$

22. The structure of compound P is (IIT JEE 2011)

(a) $CH_3CH_2CH_2CH_2-C\equiv C-H$

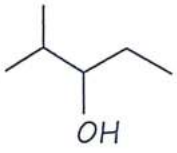
(b) $H_3CH_2C-C\equiv C-CH_2CH_3$

(c) 

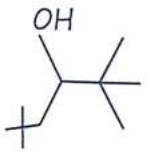
(d) 

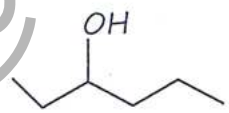
Sol. (d)

23. The structure of the compound Q is (IIT JEE 2011)

(a) 

(b)

(c) 

(d) 

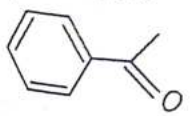
Sol. (b)

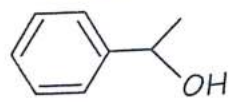
24. $C_8H_6 \xrightarrow[H_2]{Pd-BaSO_4} C_8H_8 \xrightarrow[\text{(ii) } H_2O_2, NaOh, H_2O]{\text{(i) } B_2H_6} X$

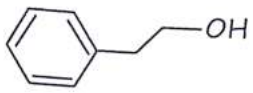
$$\downarrow \begin{matrix} H_2O \\ HgSO_4, H_2SO_4 \end{matrix}$$

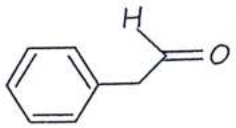
$$C_8H_6 \xrightarrow[\text{(ii) } H^+, \text{Heat}]{\text{(i) } BtMgBr, H_2O} Y$$

Compound X is (JEE Adv. 2015)

(a) 

(b) 

(c) 

(d) 

Sol. (c)

25. Which reagent converts propene to 1-propanol? (2000)

(a) H_2O, H_2SO_4

(b) B_2H_6, H_2O_2, OH^-

(c) $Hg(OAc)_2$, $NaBH_4/H_2O$

(d) Aq. KOH

Sol. (b)

$$\underset{\text{Propene}}{H_3C-CH=CH_2} \xrightarrow{B_2H_6} H_3C-\underset{H}{\overset{H}{C}}=CH_2 \xrightarrow{C_3H_6} (C_3H_7)_2BH \xrightarrow{C_3H_6} (C_3H_7)_3B$$

$$\underset{\text{1-propanol}}{H_3C-CH_2-CH_2-OH} \xrightarrow{OH^\ominus, H_2O_2}$$

26. Predict the correct intermediate and product in the following reaction (2017-Delhi)

$$H_3C-C\equiv CH \xrightarrow[H_2SO_4]{H_2O_2H_2SO_4} \underset{(A)}{\text{Intermediate}} \rightarrow \underset{(B)}{\text{Product}}$$

(a) A: $H_3C-\underset{SO_4}{\overset{|}{C}}=CH_2$ B: $H_3C-\underset{O}{\overset{\|}{C}}=CH_3$

(b) A: $H_3C-\underset{OH}{\overset{|}{C}}=CH_2$ B: $H_3C-\underset{O}{\overset{\|}{C}}=CH_3$

(c) A: $H_3C-\underset{OH}{\overset{|}{C}}=CH_2$ B: $H_3C-\underset{SO_4}{\overset{|}{C}}=CH_2$

(d) A: $H_3C-\underset{O}{\overset{|}{C}}-CH_3$ B: $H_3C-C\equiv CH$

Sol. (b)

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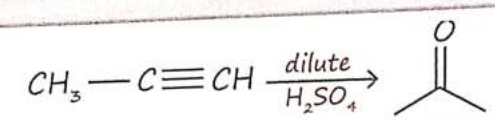
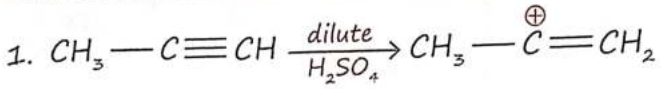
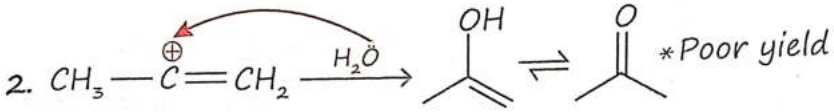
Hydration of Alkyne

Reaction



Mechanism

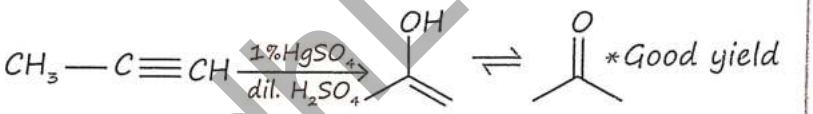
1. 
*Highly unstable

2. 
*Poor yield

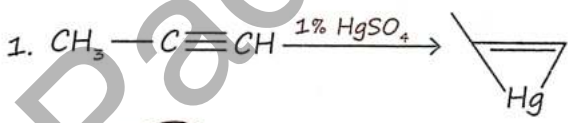
Kucherov's Reaction

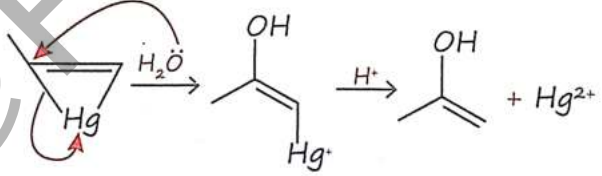
Kucherov reaction is the acid-catalyzed hydration of alkynes using $HgSO_4$ and H_2SO_4, where water add across the triple bond to form an enol, which quickly tautomerizes to a ketone (follows Markovnikov's rule)

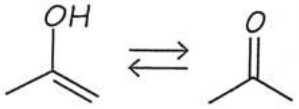
Reaction


*Good yield

Mechanism

1. 

2. 

3. 

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Baba Tea-Stall (BTS)

- Example of electrophilic addition reaction.
- NCC type intermediate.
- Involves tautomerism
- Mixed carbonyl are obtained.
- Involves S_E2 mechanism.

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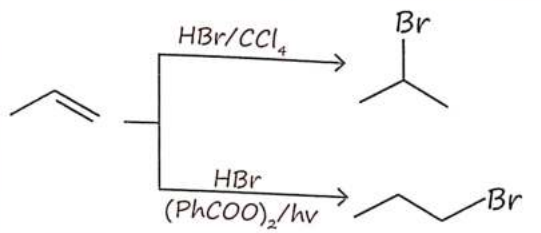
Kharasch Effect/Peroxide Effect

The Kharasch effect refers to the anti-Markovnikov addition of HBr to alkenes in the presence of peroxides, due to a free radical mechanism.

Reaction



- Markovonikov addⁿ
- Carbocation

- AM Addition
- Free radical

Mechanism

Initiation

1. $Ph-C(=O)-O-O-C(=O)-Ph \xrightarrow{h\nu} 2PhCOO^{\odot} \xrightarrow{-CO_2\uparrow} 2Ph^{\odot}$

2. $Ph^{\odot} + HBr \longrightarrow$ $+ Br^{\odot}$

Propagation

3.

4.

Baba Tea-Stall (BTS)

(i) Free radical intermediate obtained.
(ii) Example of free radical addition reaction.
(iii) Involves antimarkovnikov's addition.
(iv) Only HBr shows peroxide effect.

H–X	ΔH_1(kJ/mol) (X + Alkene)	ΔH_2(kJ/mol) (Alkyl radical + HX)
HCl	–67 (exo)	+12.6 (endo)
HBr	–25.1 (exo)	–50.2 (exo)
HI	+46 (endo)	–117.1 (exo)

Practice Questions

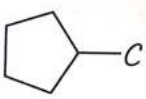
- Kucherov's Reaction

Q. Draw major product:

381. $CH_3CH_2C \equiv CH \xrightarrow[dil.\ H_2SO_4]{HgSO_4}$

382. $HC \equiv CH \xrightarrow[dil.\ H_2SO_4]{HgSO_4}$

383. Phenyl acetylene $\xrightarrow[dil.\ H_2SO_4]{HgSO_4}$

384.  $\xrightarrow[dil.\ H_2SO_4]{HgSO_4}$

385. Propyne $\xrightarrow[H^{\oplus}/H_2O]{Hg^{+2}}$

386. $CH_3(CH_2)_3C \equiv CH \xrightarrow[H_2SO_4]{Hg^{2+}}$

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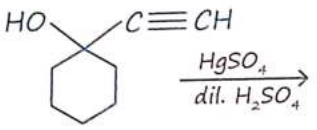
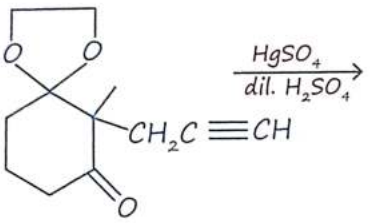
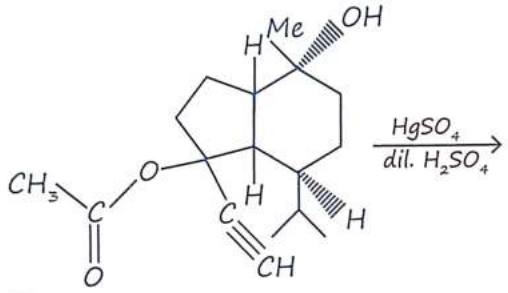
387. 

388. 

389. 

390. How many of the following compounds can be obtained in Kucherov's reaction?

(a) CH_3-CHO (b) CH_3CH_2CHO

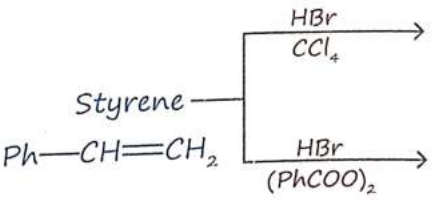
(c) $PhCH_2CHO$ (d) $PhCHO$

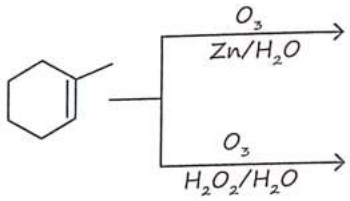
(e) $CH_3CH_2CH_2CHO$ (f) acetone

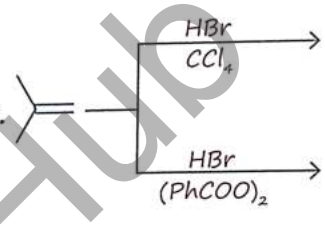
(g) Acetophenone (h) Benzophenone

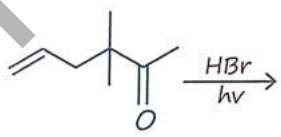
(i) Butanone (j) Phenyl acetone

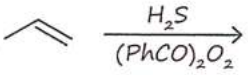
➢ Kharasch Effect/ Peroxide Effect

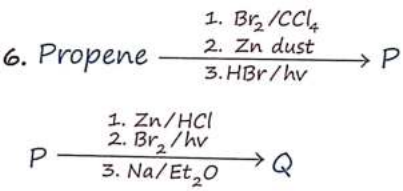
391. 

392. 

393. 

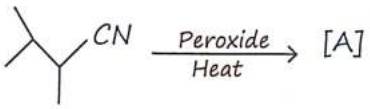
394. 

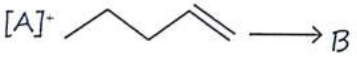
395. 

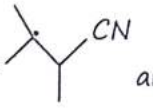
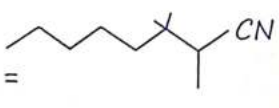
396. 

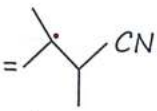
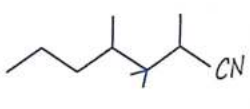
Previous Year Questions

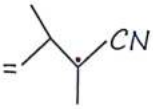
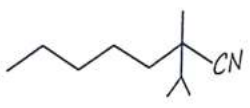
27. The major products A and B in the following reactions are: [8 Jan, 2020 (Shift-I)]

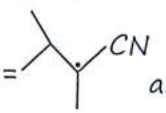
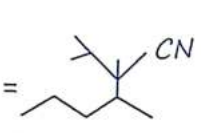




(a) A =  and B = 

(b) A =  and B = 

(c) A =  and B = 

(d) A =  and B = 

Sol. (d)

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Ozonolysis

Ozonolysis of alkenes involves the addition of ozone molecule to alkene to form ozonide, and then cleavage of the ozonide by Zn-H2O to smaller molecules. This reaction is highly useful in detecting the position of the double bond in alkenes or other unsaturated compounds.

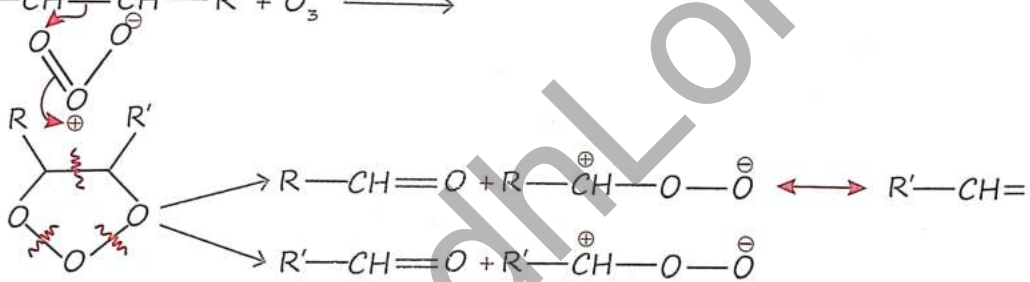
Reaction

1. $R-CH=CH-R' \xrightarrow[\text{2. } Zn/H_2O \text{ or } PPh_3 \text{ or } SMe_2]{\text{1. } O_3} R-CH=O + R'-CH=O + ZnO$
(Reductive ozonolysis)

2. $R-CH=CH-R' \xrightarrow[\text{2. } H_2O_2/H_2O]{\text{1. } O_3} R-COOH + R'-COOH$
(Oxidative ozonolysis)

Mechanism of Ozonolysis

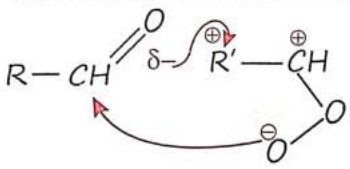
1. $R-CH=\overset{\delta+}{C}H-R' + O_3 \longrightarrow$

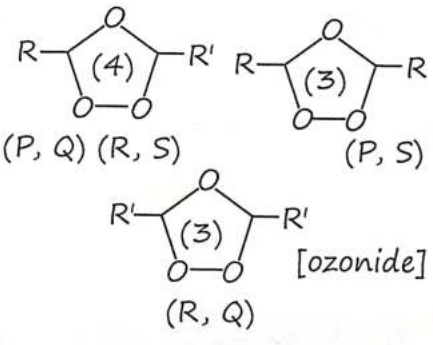


(Molo Ozonide)

2. $R-CH=O$ (P) + $R-\overset{\oplus}{C}H=O-\overset{\ominus}{O}$ (Q)

$R-CH=O$ (R) + $R-\overset{\oplus}{C}H=O-\overset{\ominus}{O}$ (S)





(P, Q) (R, S) (P, S) [ozonide] (R, Q)

Practice Questions

397. $CH_2=CH_2$ → (1. O_3, 2. Zn/H_2O) / (1. O_3, 2. H_2O_2/H_2O)

398. Propene ⟶

399. Isobutene ⟶

400. Isopentene ⟶

401. 2,3-dimethyl 2-butene ⟶

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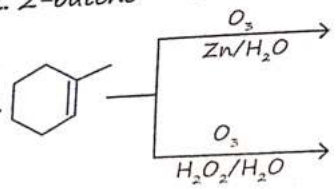
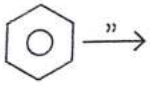
402. 2-butene $\xrightarrow{\text{»}}$

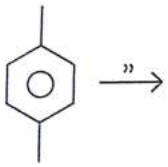
403. 

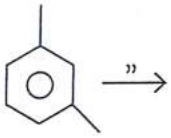
404. Methylidene Cyclohexane $\xrightarrow{\text{»}}$

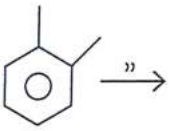
405. Styrene $\xrightarrow{\text{»}}$

406. 1,5-dimethyl cyclopentene $\xrightarrow{\text{»}}$

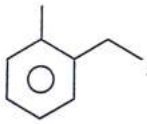
407.  $\xrightarrow{\text{»}}$

408.  $\xrightarrow{\text{»}}$

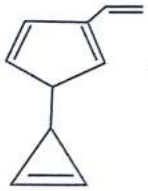
409.  $\xrightarrow{\text{»}}$

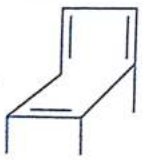
410.  $\xrightarrow{\text{»}}$

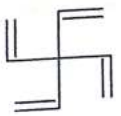
411. Toluene $\xrightarrow{\text{»}}$

412.  $\xrightarrow{\text{»}}$

413. Allene →

414.  $\xrightarrow{\text{»}}$

415.  $\xrightarrow{\text{»}}$

416.  $\xrightarrow{\text{»}}$

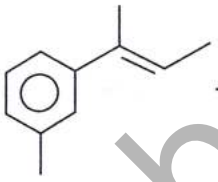
417. $\xrightarrow{\text{»}}$

418. Ethyne $\xrightarrow{\text{»}}$

419. Propyne $\xrightarrow{\text{»}}$

420. 2-Butyne $\xrightarrow{\text{»}}$

421. Phenyl acetylene $\xrightarrow{\text{»}}$

422.  $\xrightarrow{\text{»}}$

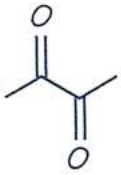
423. P $\xrightarrow[Zn/H_2O]{O_3}$ 4 – methyl – 5 – oxo Hexanal

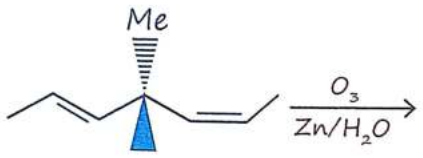
424. P $\xrightarrow[Zn/H_2O]{O_3}$ 2 mole of methyl glyoxal + glyoxal

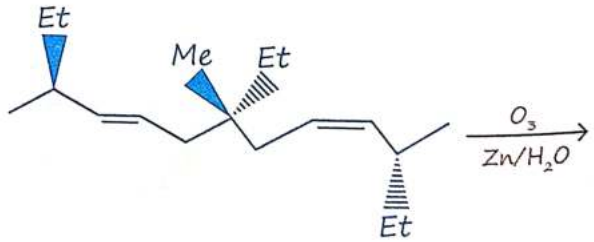
425. P $\xrightarrow[Zn/H_2O]{O_3}$ methyl glyoxal + 2-methyl butane – 1,2 – dial

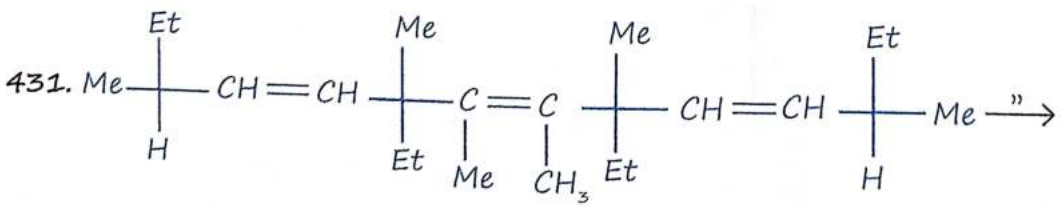
426. P $\xrightarrow[Zn/H_2O]{O_3}$ only acetaldehyde

427. P $\xrightarrow[Zn/H_2O]{O_3}$ Acetaldehyde + Acetone

428. P $\xrightarrow[Zn/H_2O]{O_3}$ CH_3-CHO +  H + HCHO

429. 

430. 

431. 

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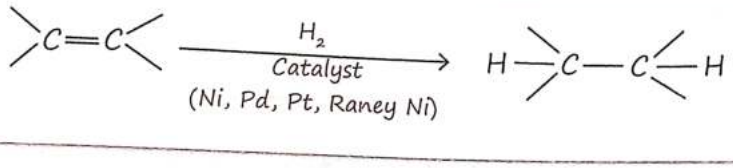
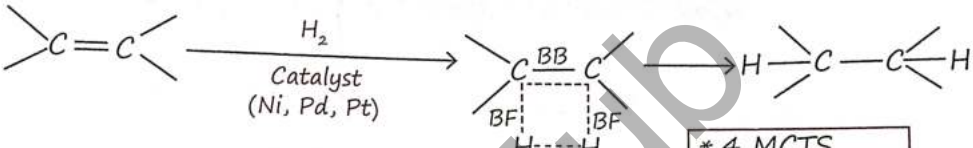
Catalytic Hydrogenation

Dihydrogen gas adds to alkenes and alkynes in the presence of finely divided catalysts like platinum, palladium or nickel to form alkanes. This process is called hydrogenation. These metals adsorb dihydrogen gas on their surfaces and activate the hydrogen – hydrogen bond. Platinum and palladium catalyse the reaction at room temperature but relatively higher temperature and pressure are required with nickel catalysts.

Reaction



Mechanism



* 4 MCTS
* Syn addition
* Adsorption

Baba Tea-Stall (BTS)

- It is an example of addition reaction and redox reaction.
- An example of syn addition or cis-addition.
- Example of surface phenomenon and involves 4 MCTS.
- $Pd/BaSO_4$ or $CaCO_3$ known as lindlar's catalyst and quinoline is added as poison.
- In case of conjugated diene, 1,4 addition takes place due to 6 MCTS.

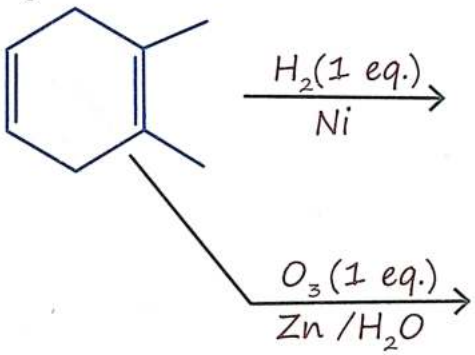
Hydrocarbon

$$\text{Rate} \propto \text{adsorption}$$

$$\propto \frac{1}{\text{Steric hindrance}}$$

$$\propto \frac{1}{\text{Stability of compound}}$$

Eg.



Functional Group	Δ with Ni
C-S bond > R-COCl > Ar – NO_2 > Alkyne > Alkene > Aldehyde > Ketone > R–CN > > Acid and its derivative	Pd | $BaSO_4$ or $CaCO_3$ → Lindlar's catalyst (Alkyne, RCOCl) Quinoline|s → Poison

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Practice Questions

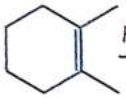
432. $CH_2 = CH_2 \xrightarrow{H_2/Ni}$

433. $\xrightarrow{H_2/Ni}$

434. $\xrightarrow{H_2/Ni}$

435. $\xrightarrow{H_2/Ni}$

436. Rate of 432-435.

437.  $\xrightarrow{H_2/Ni}$

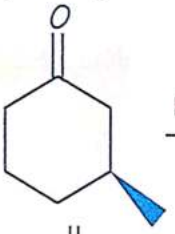
438. $CH_3 - C \equiv CH \xrightarrow{H_2/Ni}$

439. $CH_3 - C \equiv C - CH_3 \xrightarrow{H_2/Ni}$

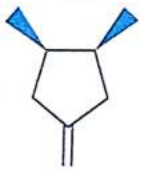
440. $CH_3 - C \equiv N \xrightarrow{H_2/Ni}$

441. $CH_3 - CHO \xrightarrow{H_2/Ni}$

442. $\xrightarrow{H_2/Ni}$

443.  $\xrightarrow{H_2/Ni}$

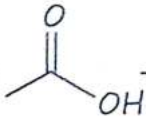
444. $\xrightarrow{H_2/Ni}$

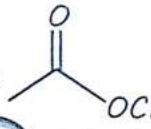
445.  $\xrightarrow{H_2/Ni}$

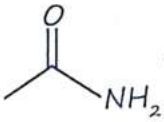
446. $R - COCl \xrightarrow{H_2/Ni}$

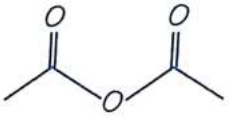
447. $R - NO_2 \xrightarrow{H_2/Ni}$

448. $R - CH = NH \xrightarrow{H_2/Ni}$

449.  $\xrightarrow[\Delta]{H_2/Ni}$

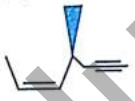
450.  $\xrightarrow{H_2/Ni}$

451.  $\xrightarrow{H_2/Ni}$

452.  $\xrightarrow{H_2/Ni}$

453. $+ H_2 \xrightarrow[\Delta]{Ni}$

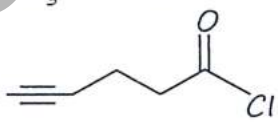
(a) 1 : 3 (b) 1 : 1

454.  $\xrightarrow[BaSO_4]{H_2/Pd}$

455. $CH_3 - C \equiv CH \xrightarrow[BaSO_4]{H_2|Pd}$

456. $CH_3 - C \equiv C - CH_3 \xrightarrow{H_2/Ni}$

457. $CH_3 - COCl \xrightarrow{H_2/Ni}$

458.  $\xrightarrow{H_2/Ni}$

459. $CH_3 - CH = CH_2 \xrightarrow{H_2/Ni}$

460. $\xrightarrow[(1\ eq.)]{H_2/Ni}$

461. $\xrightarrow[(1\ eq.)]{H_2/Ni}$

462. Alkene $\xrightarrow{H_2/Ni}$ Butane

463. Alkene $\xrightarrow{H_2/Ni}$ 3-MethylPentane

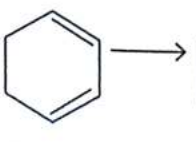
464. Alkene $\xrightarrow{H_2/Ni}$ Methylcyclohexane

465. Alkene $\xrightarrow{H_2/Ni}$ 1-Ethyl-4-Methylcyclohexane

466. Alkene $\xrightarrow[(2\ mol)]{H_2/Ni}$ Pentane

467. C_5H_{10} [O.A.] $\xrightarrow{H_2/Ni}$ Optically inactive product

468. C_6H_{12} [O.A.] $\xrightarrow{H_2/Ni}$ Optically inactive product

469.  Use only H_2/Ni and D_2/Ni

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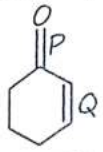
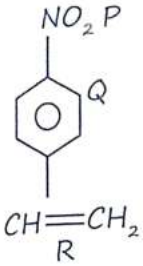
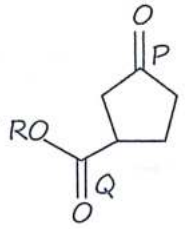
470. Simplest O.A. alkyne $\xrightarrow[\text{(excess)}]{H_2|Ni}$ P $\xrightarrow[h\nu]{Cl_2}$

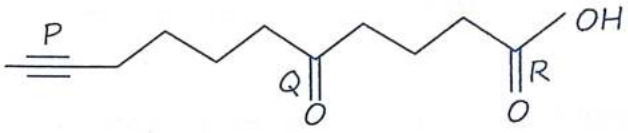
Total monochlorinated products

Q. Which of the following group will react faster with H_2/Ni?

471. 

472. 

473. 

474. 

Previous Year Questions

28. A hydrocarbon 'X' with formula C_6H_8 uses two moles of H_2 on catalytic hydrogenation of its one mole. On ozonolysis, 'X' yields two moles of methane dicarbaldehyde. The hydrocarbon 'X' is

[31 Jan, 2023 (Shift-II)]

(a) hexa-1, 3, 5-triene

(b) 1-methylcyclopenta-1, 4-diene

(c) cyclohexa-1, 3-diene

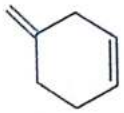
(d) cyclohexa-1, 4-diene

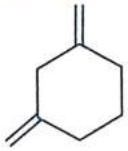
Sol. (d)

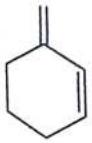
29. An unsaturated hydrocarbon X absorbs two hydrogen molecules on catalytic hydrogenation, and also gives following reaction.

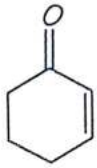
$X \xrightarrow[ZN/H_2O]{O_3} A \xrightarrow{[Ag(NH_3)_2]^+} B$ (3-oxo- hexanedicarboxylic acid)

X will be [8 Jan, 2020 (Shift-II)]

(a) 

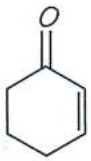
(b) 

(c) 

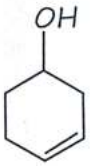
(d) 

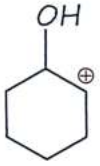
Sol. (a)

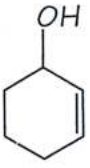
30. The correct structure of the product A formed in the reaction: (2016 - II)

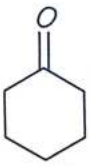


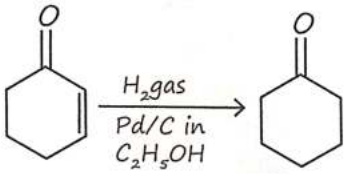
A is

(a) 

(b) 

(c) 

(d) 

Sol. (d) 

H_2 gas in pd/C causes simple reduction reaction and results in the formation of alkanes. It adds hydrogen to the double bond.

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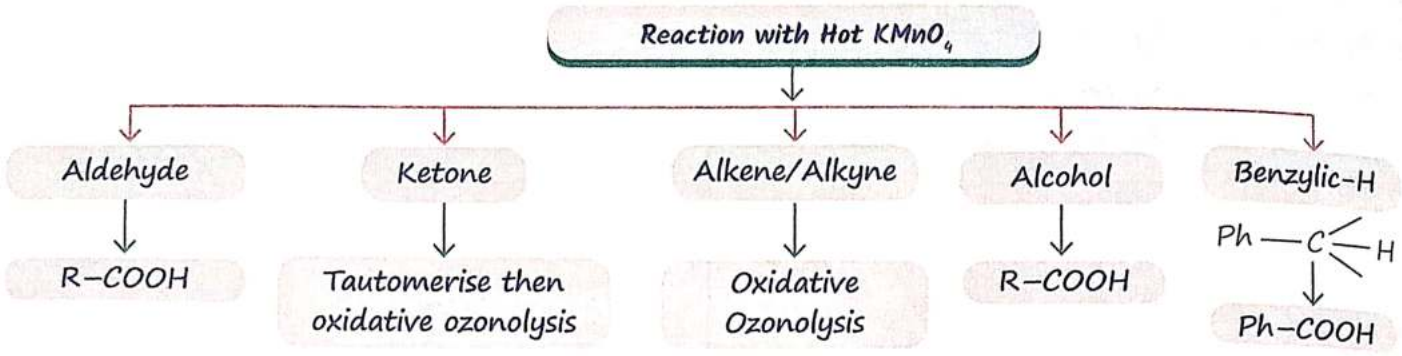
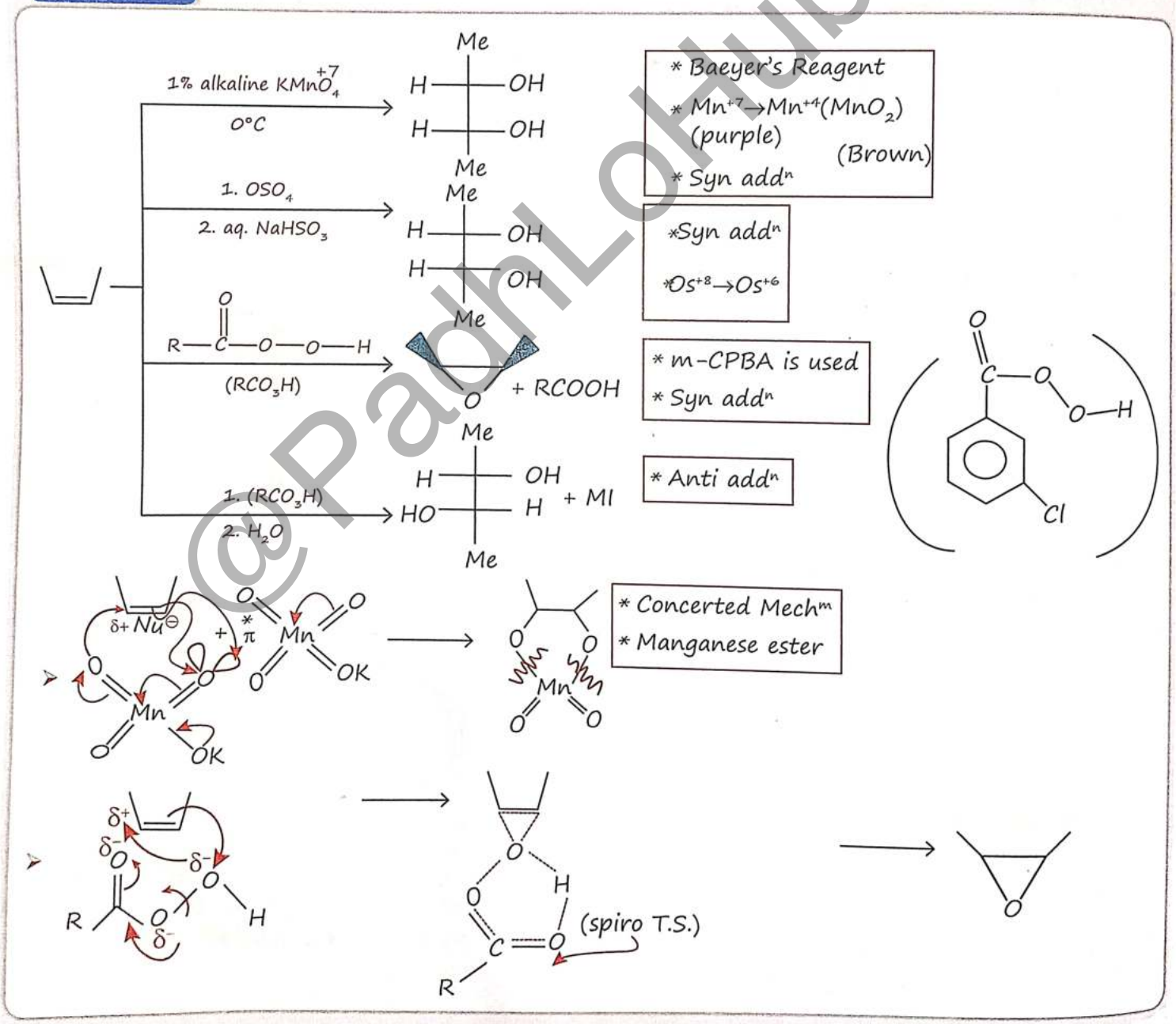


If benzylic carbon has any —H present or $=$, $\equiv$ present then it will convert to benzoic acid.

Oxidation of Alkene/alkyne

Reaction



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Practice Questions

475. $CH_2=CH_2$ → $KMnO_4$, 0°C / 1. OsO_4, 2. $NaHSO_3$ / 1. m-CPBA, 2. H_2O / m-CPBA

476. → $KMnO_4$, 0°C / 1. OsO_4, 2. $NaHSO_3$ / 1. m-CPBA, 2. H_2O / m-CPBA

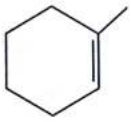
477. → $KMnO_4$, 0°C / 1. OsO_4, 2. $NaHSO_3$ / 1. m-CPBA, 2. H_2O / m-CPBA

478. → $KMnO_4$, 0°C / 1. OsO_4, 2. $NaHSO_3$ / 1. m-CPBA, 2. H_2O / m-CPBA

479. → $KMnO_4$, 0°C / 1. OsO_4, 2. $NaHSO_3$ / 1. m-CPBA, 2. H_2O / m-CPBA

480. → $KMnO_4$, 0°C / 1. OsO_4, 2. $NaHSO_3$ / 1. m-CPBA, 2. H_2O / m-CPBA

481. → $KMnO_4$, 0°C / 1. OsO_4, 2. $NaHSO_3$ / 1. m-CPBA, 2. H_2O / m-CPBA

482.  → $KMnO_4$, 0°C / 1. OsO_4, 2. $NaHSO_3$ / 1. m-CPBA, 2. H_2O / m-CPBA

483. $CH_3-C\equiv C-CH_3 \xrightarrow[1\%\ KMnO_4]{Cold}$

484. $CH_3-C\equiv CH \xrightarrow[1\%\ KMnO_4]{Cold}$

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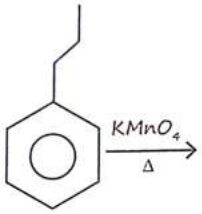
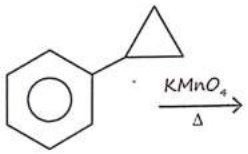
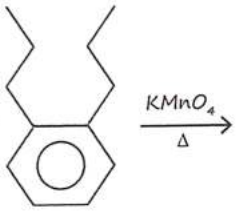
485. $HC \equiv CH \xrightarrow[1\%\ KMnO_4]{Cold}$

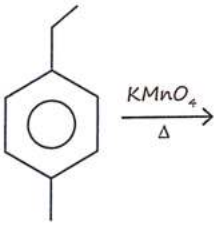
486. $\xrightarrow[1\%\ KMnO_4]{Cold}$ Find moles of $KMNO_4$ for complete reaction

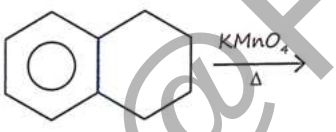
(6 mole)

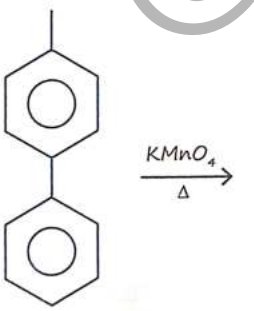
487. 

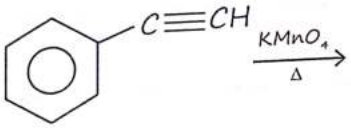
488. 

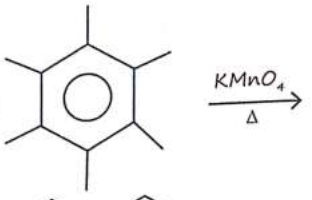
489. 

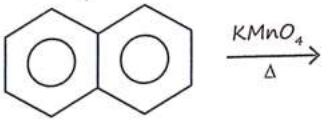
490. 

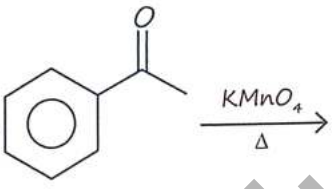
491. 

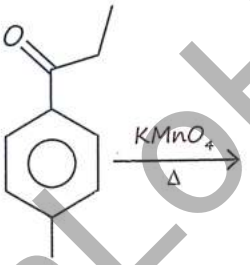
492. 

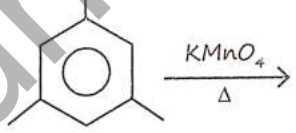
493. 

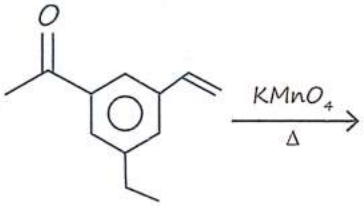
494. 

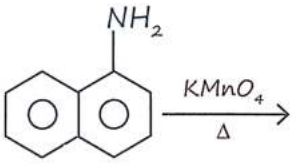
495. 

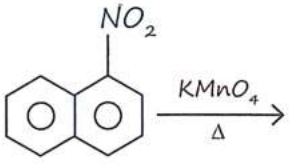
496. 

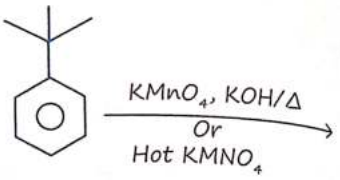
497. 

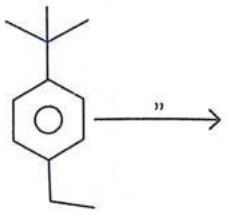
498. 

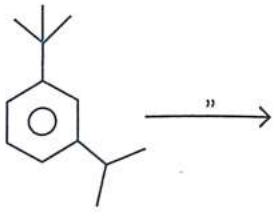
499. 

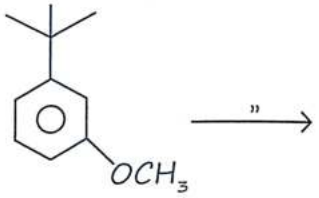
500. 

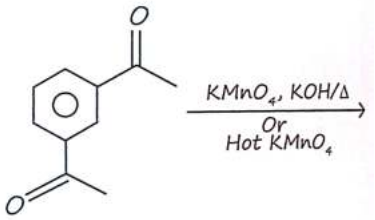
501. 

502. 

503. 

504. 

505. 

506. 

Q. How many ozonide will form with following compound?

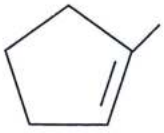
507. Ethene

508. Propene

509. 2-Butene

510. 2-pentene

511. Isopentene

512. 

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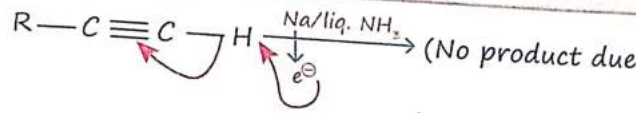
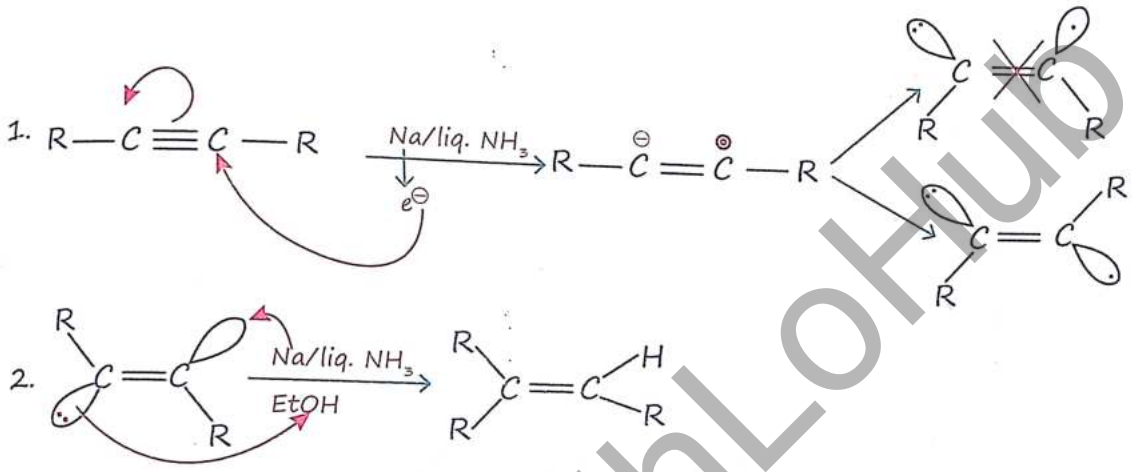
Birch Reduction

Birch reduction of alkynes involves the use of Na (or Li) in liquid NH_3 to reduce an alkyne to a trans-alkene (anti addition of hydrogen).

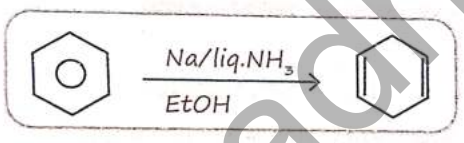
Reaction



Mechanism



Reaction



Mechanism

Baba Tea-Stall (B.T.S.)

(i) Na/liq-NH_3 is a blue colour solution.
(ii) Both free radical and carbanion.
(iii) R–OH is used as proton source.
(iv) Terminal alkyne will not give alkene.
(v) In case of substituted benzene, more stable alkene is major product.

Practice Questions

513. Propyne $\xrightarrow{Na/liq.NH_3}$

514. 2 – Butyne $\xrightarrow{Na/liq.NH_3}$

515. Hepta – 1,5 – diyne $\xrightarrow{Na/liq.NH_3}$

516. $\xrightarrow{Na/liq.NH_3}$

517. $HC\equiv CH \xrightarrow[2.\ CH_3-I]{1.\ Na/liq.NH_3} P$

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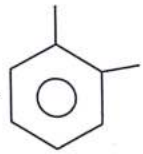
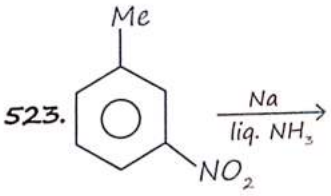
518. P $\xrightarrow[2.\ CH_3-I]{1.\ Na/liq.\ NH_3}$ Q

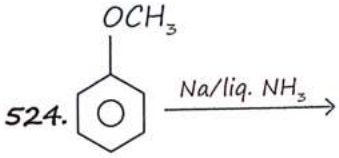
519. $\xrightarrow{Na/liq.\ NH_3}$

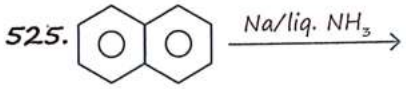
520. $\xrightarrow{Na/liq.\ NH_3}$

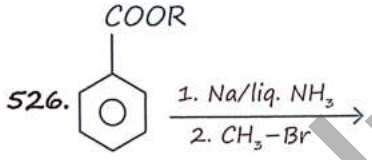
521. $\xrightarrow[liq.\ NH_3]{Na}$

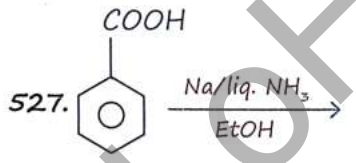
522.  $\xrightarrow[liq.\ NH_3]{Na}$

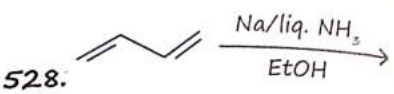
523. 

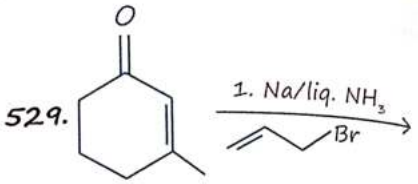
524. 

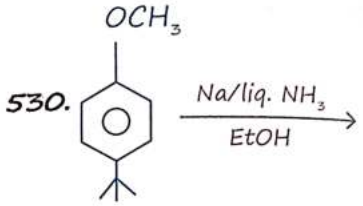
525. 

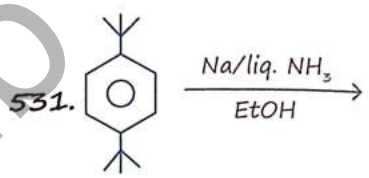
526. 

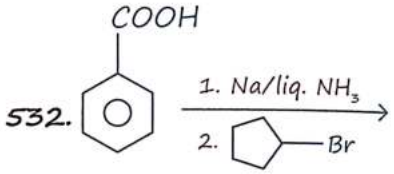
527. 

528. 

529. 

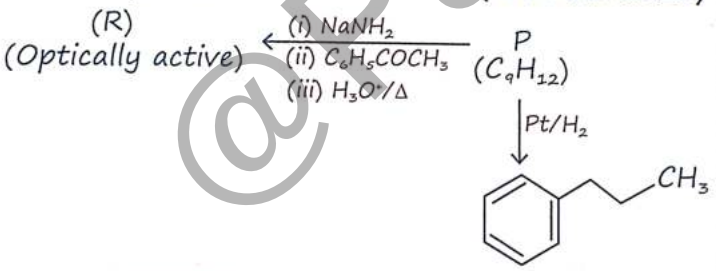
530. 

531. 

532. 

Previous Year Questions

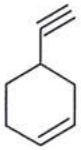
31. Consider the following transformations of a compound P. (JEE Adv. 2020)



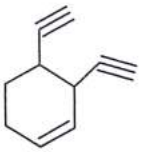
$\xrightarrow[(ii)\ KMnO_4/H_2SO_4/\Delta]{(i)\ X\ (reagent)}$ Q $(C_8H_{12}O_6)$

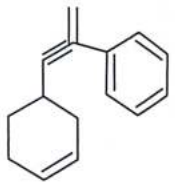
(Optically active acid)

Choose the correct option(s):

(a) P is 

(b) X is Pd-C/quinoline/H_2

(c) P is 

(d) R is 

Sol. (b, c)

32. In the reaction: (2016 - I)

$H-C\equiv CH \xrightarrow[(2)\ CH_3CH_2Br]{(1)\ NaNH_2/liq.NH_3} X \xrightarrow[(2)\ CH_3CH_2Br]{(1)\ NaNH_2/liq.NH_3} Y$

(a) X = 1-Butyne ; Y = 2-Hexyne

(b) X = 1-Butyne ; Y = 3-Hexyne

(c) X = 2-Butyne ; Y = 3-Hexyne

(d) X = 2-Butyne ; Y = 2-Hexyne

Sol. (b)

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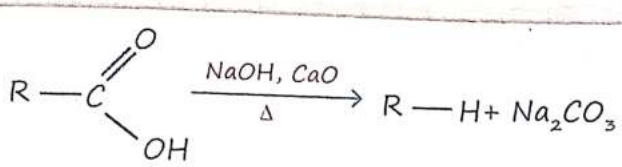
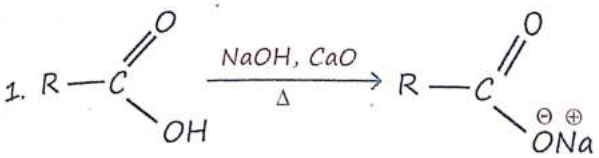
Sodalime Decarboxylation

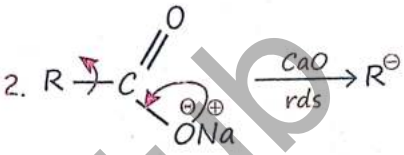
Sodium salts of carboxylic acids on heating with soda lime (mixture of sodium hydroxide and calcium oxide) give alkanes containing one carbon atom less than the carboxylic acid. This process of elimination of carbon dioxide from a carboxylic acid is known as decarboxylation.

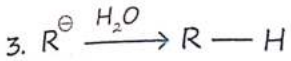
Reaction



Mechanism







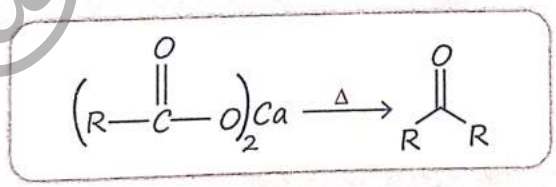
Baba Tea-Stall (BTS)

(i) 3 : 1 of NaOH and CaO taken respectively.
(ii) Carbanion is obtained as intermediate.
(iii) Stability of (–) increase ⇒ Rate increase
(iv) Oakwood Degradation.

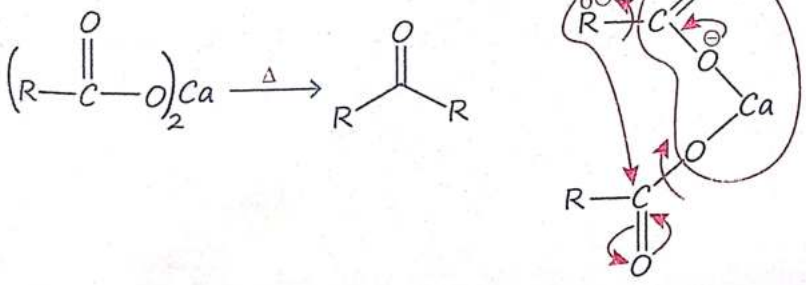
Heating or distillation of calcium salt

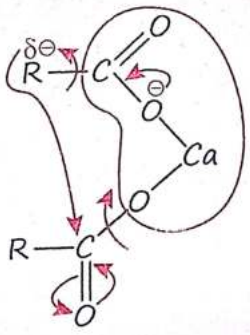
Heating with heating salt refers to dry distillation or thermal decomposition of organic compounds using inorganic salts (like anhydrous $CaCl_2$ or $KHSO_4$) that provide high, uniform heat and sometimes act as dehydrating agents.

Reaction



Mechanism





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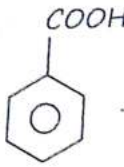
Practice Questions

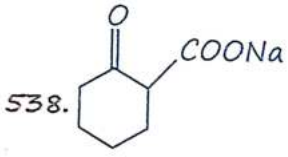
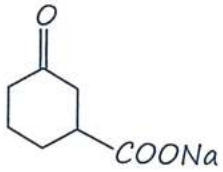
533.  $\xrightarrow[CaO,\Delta]{NaOH}$

534. Phenyl Acetic acid $\xrightarrow[CaO,\Delta]{NaOH}$

535. $\xrightarrow[CaO,\Delta]{NaOH}$

536. o – Toluic acid $\xrightarrow[CaO,\Delta]{NaOH}$

537. $CH_3COOH \xrightarrow[CaO,\Delta]{NaOH}$

538.  

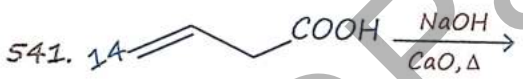
Rate?

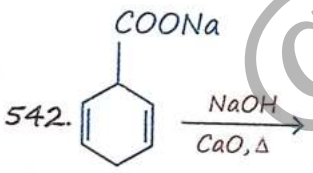
539. Propanoic acid, Propenoic acid, Propynic acid

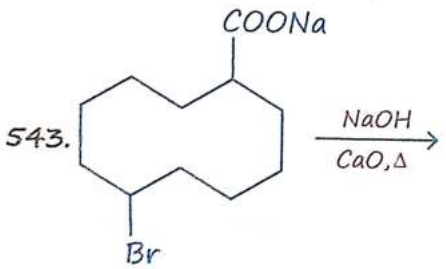
(a) (b) (c)

540. PhCOONa, $PhCH_2COONa$

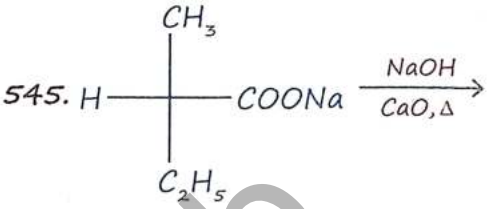
(a) (b)

541.  $\xrightarrow[CaO,\Delta]{NaOH}$

542.  $\xrightarrow[CaO,\Delta]{NaOH}$

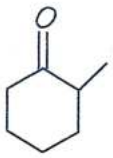
543.  $\xrightarrow[CaO,\Delta]{NaOH}$

544. $\xrightarrow[CaO,\Delta]{NaOH}$

545. 

546. How many monocarboxylic acid will produce Toluene on reaction with NaOH/CaO, Δ?

547. How many monocarboxylic acid will produce given compound on reaction with NaOH/CaO, Δ?



548. $(CH_3COO)_2Ca \xrightarrow{\Delta}$

549. $(PhCOO)_2Ca \xrightarrow{\text{dry distillation}}$

550. $EtCOOH \xrightarrow{Ca(OH)_2}$

551. $\xrightarrow[2.\ Ca(OH)_2,\Delta]{1.\ O_3/H_2O}$

552. $(PhCOO)_2Ca + (CH_3COO)_2Ca \xrightarrow{\Delta}$

553. $PhCH_2COONa \xrightarrow[\Delta]{CaCl_2}$

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Formation of Hydrocarbon

Hydrolysis of Carbides

Hydrolysis of carbide involves the reaction of metal carbides with water to produce hydrocarbons:

$$Al_4C_3 \xrightarrow{H_2O} 4Al(OH)_3 + 3CH_4$$

$$Mg_2C_3 \xrightarrow{4H_2O} 2Mg(OH)_2 + C_3H_4 \quad (CH_3-C\equiv CH)$$

$$CaC_2 \xrightarrow{2H_2O} Ca(OH)_2 + C_2H_2 \quad (HC\equiv CH)$$

$$Be_2C \xrightarrow{4H_2O} 2Be(OH)_2 + CH_4$$

Aromatisation

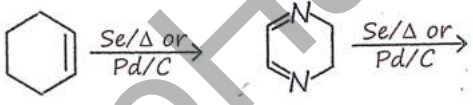
n-Alkanes having six or more carbon atoms on heating to 773K at 10-20 atmospheric pressure in the presence of oxides of vanadium, molybdenum or chromium supported over alumina get dehydrogenated and cyclised to benzene and its homologue. This reaction is known as aromatisation or reforming.

(a) Red hot Fe tube

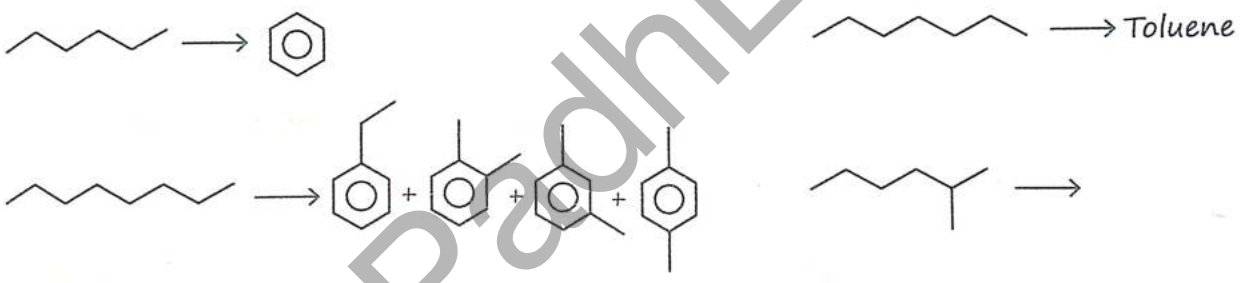
$$3HC\equiv CH \xrightarrow[\text{Fe tube}]{\text{Red hot}}$$

$$3H_3C-C\equiv CH \xrightarrow[\text{Fe tube}]{\text{Red hot}}$$

(b) By Se/Δ or Pd/C

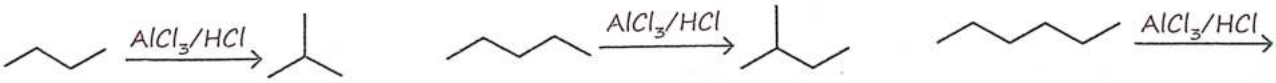


(c) By $Cr_2O_3/V_2O_5/Mo_2O_3$ at 773 K, 10-20 atm



Isomerisation

n-Alkanes on heating in the presence of anhydrous aluminium chloride and hydrogen chloride gas isomerise to branched chain alkanes. Major products are given below.



Reaction with Soda Amide ($NaNH_2$)

Reaction

$$H_3C-C\equiv CH \xrightarrow[CH_3-I]{NaNH_2 \text{ then}}$$

$$H_3C-C\equiv CCH_3 \xrightarrow[\Delta]{NaNH_2} H_3C-CH_2-C\equiv CH$$

$$H_3C-C\equiv CCH_3 \xleftarrow[\Delta]{\text{Alc. KOH}} H_3C-CH_2-C\equiv CH$$

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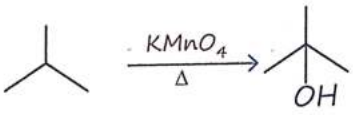
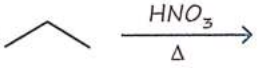
Reactions of Hydrocarbon

(a) Combustion: $C_xH_y + \left(x+\frac{y}{4}\right)O_2 \longrightarrow xCO_2 + y/2\ H_2O$

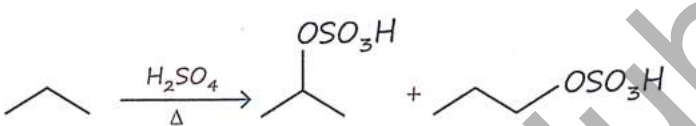
(b) By $KMnO_4$, (only with Δ3°-H):



(c) Nitration:



(d) Sulphonation:



(e) Incomplete combustion: $CH_4 + O_2 \longrightarrow C_{(s)} + 2H_2O(l)$

(f) Reaction with Terminal Alkyne:

1. $HC \equiv CH \xrightarrow[\overset{\ominus}{O}H]{Ag_2O} CH \equiv \overset{\ominus}{C}\overset{\oplus}{Cl}$ (White ppt)

2. $HC \equiv CH \xrightarrow[NH_4OH]{Cu_2Cl_2} CH \equiv \overset{\ominus}{C}\overset{\oplus}{Ag}$ (Red ppt)

3. $HC \equiv CH \xrightarrow[NH_4Cl]{Cu_2Cl_2} HC \equiv C - CH = CH_2$

(g) Controlled Oxidation

1. $CH_4 + O_2 \xrightarrow[\Delta(523\ K)]{Cu} CH_3 - OH$

2. $CH_4 + O_2 \xrightarrow{Mo_2O_3} CH_2(OH)(OH) \longrightarrow HCHO$

3. $CH_3 - CH_3 + O_2 \xrightarrow[\Delta]{Mn(CH_3COO)_2} CH_3 - COOH$

(h) Reaction with Steam

$CH + H_2O \longrightarrow CO + 3H_2O$

(i) Pyrolysis (Cracking)

1. $C_6H_{14} \xrightarrow{773\ K} C_6H_{12} + H_2$
   $C_6H_{14} \xrightarrow{773\ K} C_4H_8 + C_2H_6$
   $C_6H_{14} \xrightarrow{773\ K} C_3H_6 + C_2H_4 + CH_4$

2. $C_{12}H_{26} \xrightarrow{773\ K} C_7H_{16} + C_5H_{10} + Others$

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Practice Questions

554. $HC_3-C\equiv CH \xrightarrow{NaOH}$

555. CaC_2 (15 mole) $\xrightarrow{H_2O}$ P (50%) $\xrightarrow[\text{then } CH_3-I]{NaNH_2}$ Q 100% $\xrightarrow[\text{Fe tube}]{\text{Red hot}}$

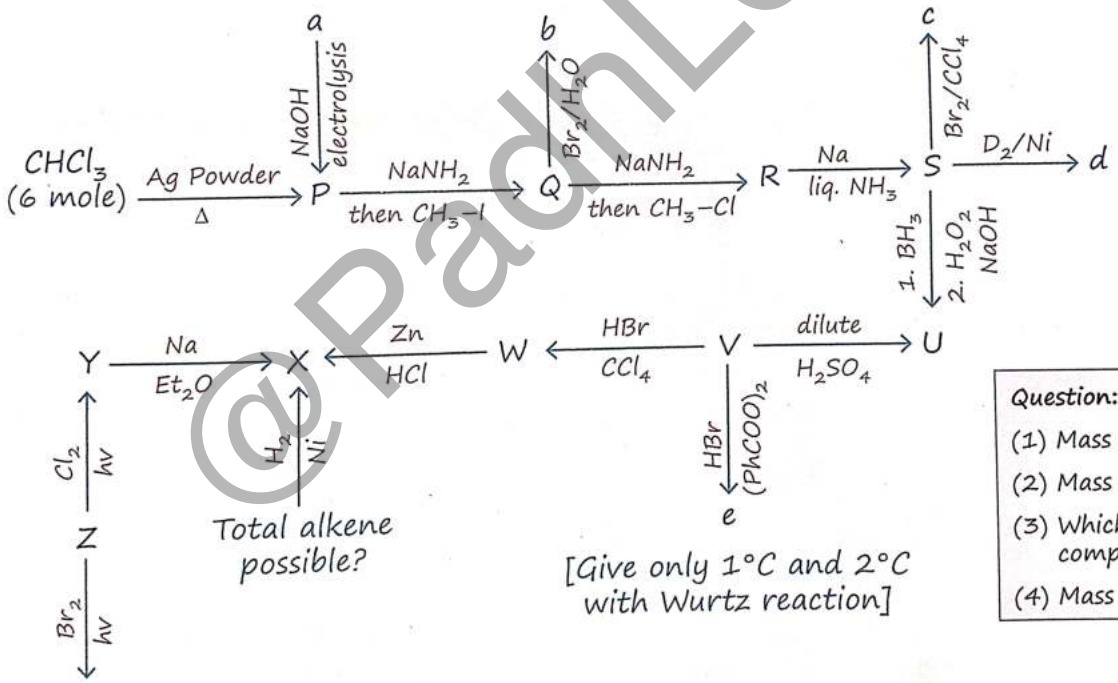
R 20% $\xrightarrow[KOH/\Delta]{KMnO_4}$ S 100% $\xrightarrow[CaO,\ \Delta]{NaOH}$ T x gm $\xleftarrow[770\ K,\ 10\ atm]{Cr_2O_3}$ U

556.

$\xrightarrow[\text{2. NaOH}\\\text{3. Electrolysis}\\\text{4. } Cr_2O_3,\ 770K,\ 10\ atm]{\text{1. } O_3/H_2O}$ P $\xleftarrow[\text{3. } Cl_2/hv\ (minor)\\\text{4. } Na/Et_2O\\\text{5. ??}]{\text{1. } HBr/CCl_4\\\text{2. } Zn/HCl}$ Q $\xrightarrow[\text{3. } HBr/(PhCOO)_2\ [excess]\\\text{4. } Na/Et_2O]{\text{1. } Br_2/hv\\\text{2. Na/dry ether}}$ R

557. 2-butene $\xrightarrow[\text{2. NaOH, CaO/}\Delta\\\text{3. } Cl_2/hv\\\text{4. } HC\equiv CNa]{\text{1. } O_3/H_2O}$ P $\xrightarrow[\text{2. } KMnO_4/KOH,\ \Delta\\\text{3. NaOH/CaO, }\Delta\\\text{4. } O_3/Zn]{\text{1. Red hot Fe tube}}$ Q

558.



Total alkene possible?

[Give only 1°C and 2°C with Wurtz reaction]

Total bromoderivative possible are?

Question:

(1) Mass of Ag required

(2) Mass of (b) produced

(3) Which represent racemic compound

(4) Mass of (Y) present

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Acylation

Aliphatic and aromatic primary and secondary amines react with acid chlorides, anhydrides and esters by nucleophilic substitution reaction. This reaction is known as acylation. You can consider this reaction as the replacement of hydrogen atom of $-NH_2$ or $>N-H$ group by the acyl group. The products obtained by acylation reaction are known as amides. The reaction is carried out in the presence of a base stronger than the amine, like pyridine, which removes HCl so formed and shifts the equilibrium to the right hand side.

Reaction

$$R-NH_2 + R'-C(=O)-Cl \xrightarrow[NEt_3]{\text{Pyridine or}}$$

Mechanism

1. $C_2H_5-\ddot{N}H-H$ + $CH_3-C(=O)-Cl$ $\xrightarrow{\text{Base}}$ $C_2H_5-\overset{\oplus}{N}H-C(CH_3)(O^-)-Cl$ $\longrightarrow$ $C_2H_5-N(H)-C(=O)-CH_3$ + $H-Cl$

   Ethanamine — N-Ethylethanamide

2. $C_2H_5-\ddot{N}H-C_2H_5$ + $CH_3-C(=O)-Cl$ $\xrightarrow{\text{Base}}$ $C_2H_5-\ddot{N}(C_2H_5)-C(=O)-CH_3$ + $H-Cl$

   N-Ethylethanamide — N, N-Diethylethanamide

3. $C_6H_5-\ddot{N}H-H$ + $CH_3-C(=O)-O-C(=O)-CH_3$ $\longrightarrow$ $C_6H_5-\ddot{N}H-C(=O)-CH_3$ + CH_3COOH

   Benzenamine — Ethanoic anhydride — N-Phenylethanamide or Acetanilide

Baba Tea-Stall (BTS)

(i) Reaction is also known as Schotten Baumann reaction.

(ii) Pyridine is added to neutralize the HCl.

(iii) If (a) R = H ⇒ Formylation
R = Ph ⇒ Benzoylation
(b) $R = CH_3$ ⇒ Acetylation.

(iv) In case of acetylation, mass of 42 gm $(-C(=O)-CH_3)$ increases.

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Practice Questions

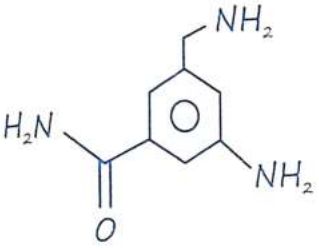
559. $CH_3-NH_2 \xrightarrow[Py.]{CH_3COCl}$

560. Aniline $\xrightarrow[Py.]{Ac_2O}$

561. $\xrightarrow[Py.]{PhCOCl\ (xs)}$

562. $\xrightarrow[(1\ eq.)]{CH_3COCl/py.}$

563. $\xrightarrow[(excess)]{CH_3COCl/Py.}$

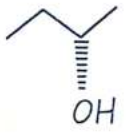
564. 

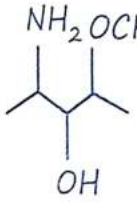
$\xrightarrow{CH_3COCl/Py.}$ (1 eq.)

$\xrightarrow[(2\ eq.)]{CH_3COCl/Py.}$

$\xrightarrow[(excess)]{CH_3COCl/Py.}$

565. Salicylic acid $\xrightarrow[(x_s)]{CH_3COCl/Py.}$

566.  $\xrightarrow{CH_3COCl/Py.}$

567.  $\xrightarrow{CH_3COCl/Py.}$

568. Phenol $\xrightarrow[Py.]{Ac_2O/Py.}$

569. N-methyl aniline $\xrightarrow[Py.]{PhSO_2Cl}$

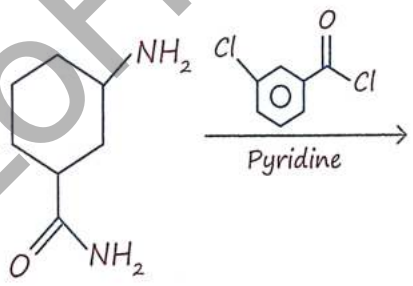
570. $Me_2NH \xrightarrow[Py.]{PhSO_2Cl}$

571. $Me_3N \xrightarrow[Py.]{PhSO_2Cl}$

572. Ethylamine $\xrightarrow[Py.]{TsCl}$

573. Diethyl amine $\xrightarrow{PhSO_2Cl}$

574. $\xrightarrow{PhSO_2Cl}$

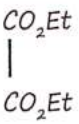
575. 

$\xrightarrow{Pyridine}$

576. Benzyl alcohol $\xrightarrow[H^{\oplus}]{CH_3COOH}$

577. $C_6H_{12}O_6 \xrightarrow[(excess)]{Ac_2O/Py.}$ Vapour density of product is 195.

Find the number of –OH group in $C_6H_{12}O_6$ is:

578. $CH_3NH_2 \xrightarrow[Py.]{}$ 

1 : 1

2 : 1

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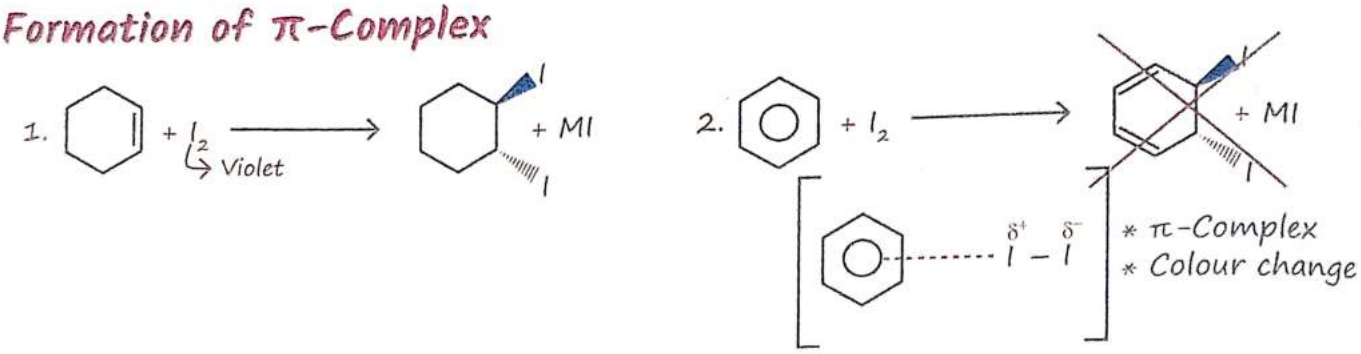
Electrophilic Aromatic Substitution

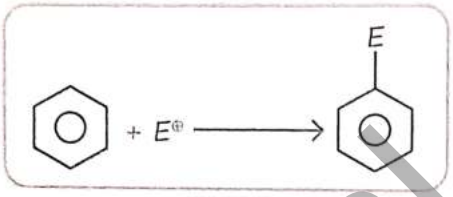
Formation of π-Complex



Electrophilic Aromatic Substitution

An electrophilic substitution reaction is a type of chemical reaction in which an electrophile replaces a hydrogen atom in an aromatic compound.

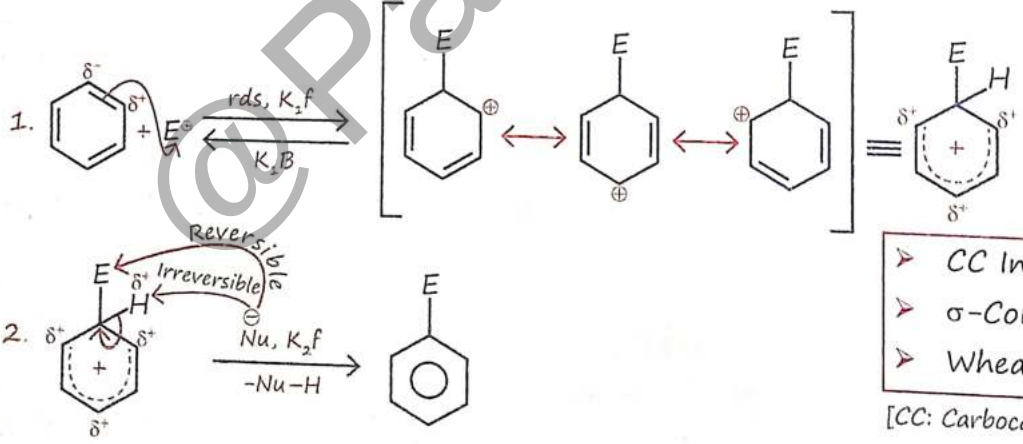
Reaction



When E^+ attacks the benzene ring: "Aromaticity: I'm about to end this man's whole career."

Mechanism

- The electrophile (E^+) attacks the aromatic ring, forming a resonance-stabilized arenium ion (σ-complex) as an intermediate.
- A Nu^- then removes a H^+ from the intermediate, restoring the aromaticity and giving the substituted aromatic product.



- CC Intermediate
- σ-Complex
- Whealand Complex

[CC: Carbocation Intermediate]

Baba Tea-Stall (BTS)

(i) Carbocation intermediate
(ii) Formation of arenium ion (CC) is rate determining step.
(iii) Rate law ⇒ $r = K[Ar-H][E^{\oplus}]$
(iv) Bimolecular reaction
(v) Stability of Carbocation increase ⇒ rate increases
(vi) Electron donating groups will increase the rate
(vii) Reaction can be reversible & irreversible

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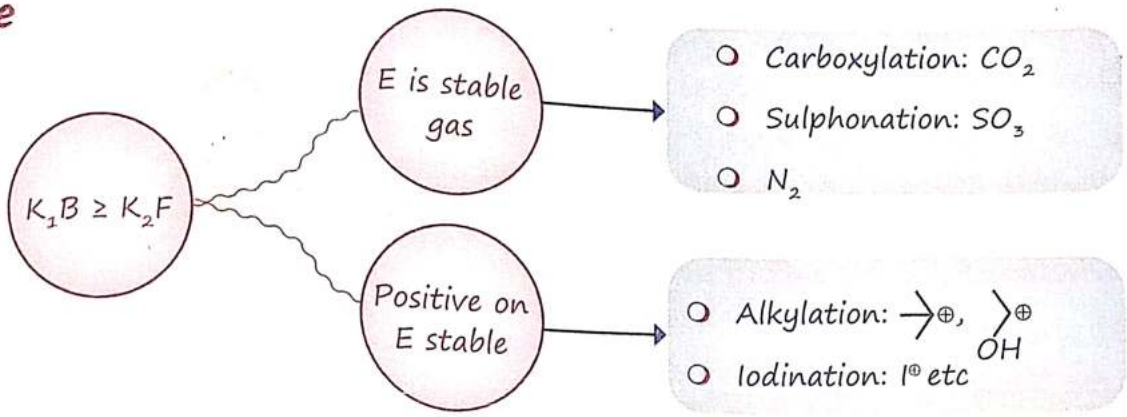
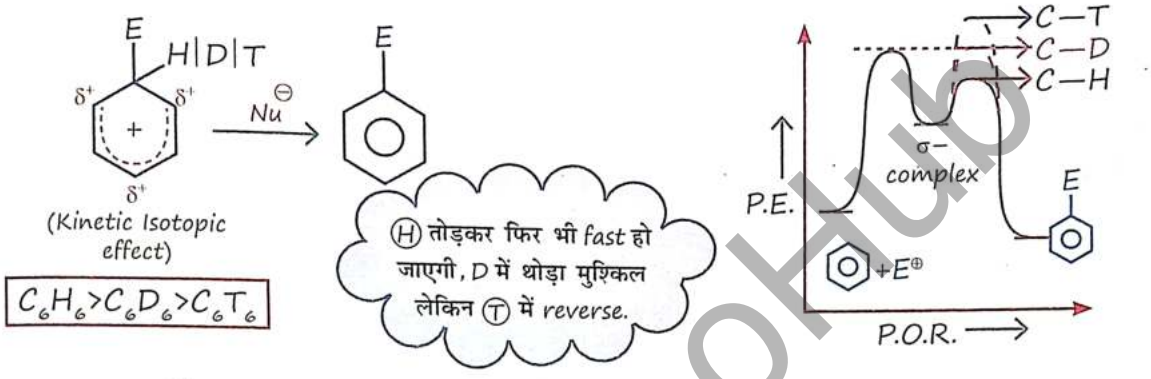
Reversible



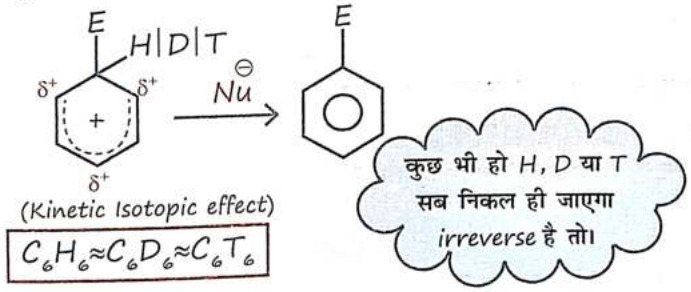


Reversible vs Irreversible EAS

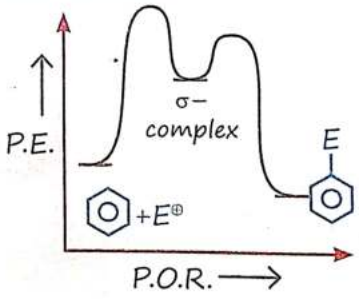
Criteria	Reversible EAS	Irreversible EAS
E^+ Stability	Stable	Reactive
K relation	$K_1B \geq K_2F$	$K_2F >> K_1B$
Nature	Equilibrium	One-way, product-favored
Directionality	Controlled by thermodynamics	Controlled by kinetics

Irreversible

- $K_2F >>> K_1B$



- Bimolecular Reaction



Kinetic Isotope Effect (KIE)

EAS Type	Reactivity Order	Reason
Reversible	$C_6H_6 > C_6D_6 > C_6T_6$	C–H breaking may be involved in RDS
Irreversible	≈ Equal for all	RDS does not involve C–H cleavage

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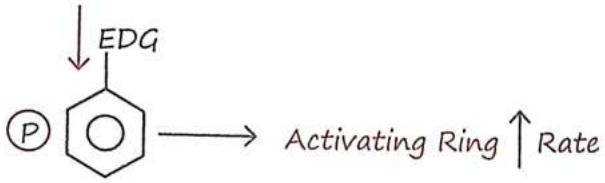
Reactivity

The reactivity of a substituted benzene ring in electrophilic aromatic substitution depends on the nature of the substituent already attached to the ring. These substituents either activate or deactivate the ring toward further substitution.

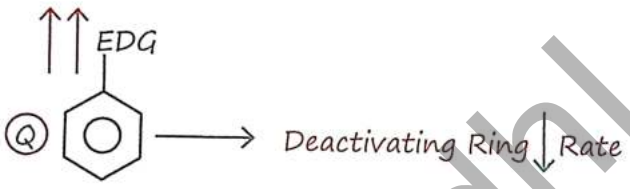
Benzene: "Why' do you deactivate me?"
Robin Hood Group: "I take electron density from the rich and still mess things up for the poor."

Types of Substituents

- EDG (Electron Donating Group):
  Activate the aromatic ring and increase the substitution rate. They donate electrons, stabilizing the intermediate.



- EWG (Electron Withdrawing Group):
  Deactivate the ring and decrease the substitution rate They withdraw electrons, destabilizing the intermediate.



- Reactivity decides by net effect
- Reactivity ⇒ P > R > Q
  - Net Withdrawal (−I) → Net EWG (−R, −H, −I)
  - Net donating (+R, +H, +I)

- Robin Hood (Special case):
  Also deactivates the ring, reducing the reaction rate possibly with unique effects on substitution.

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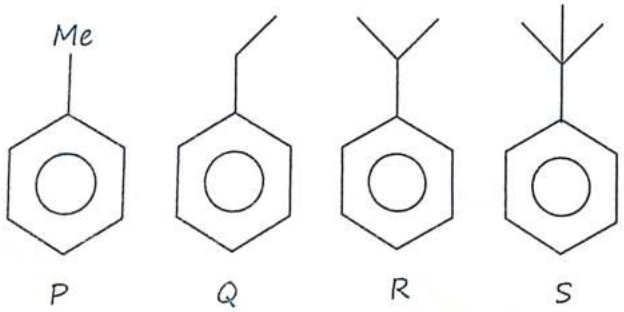
Practice Questions

Q. Compare the rate of EAS?

579.



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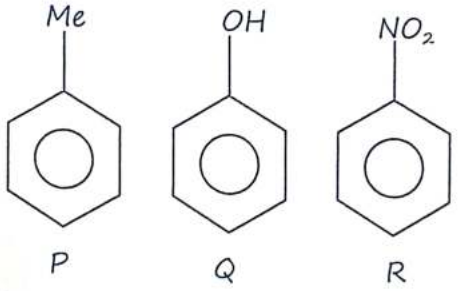

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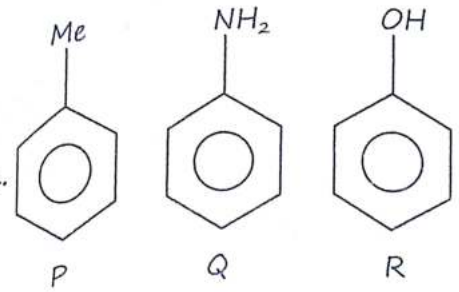

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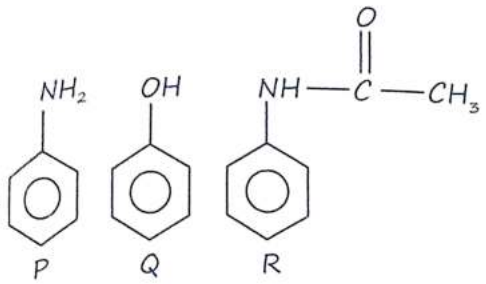

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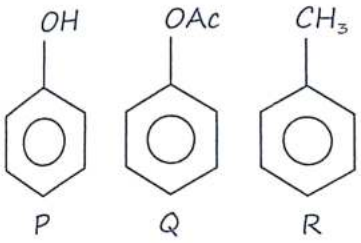

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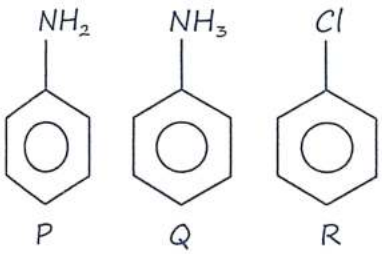

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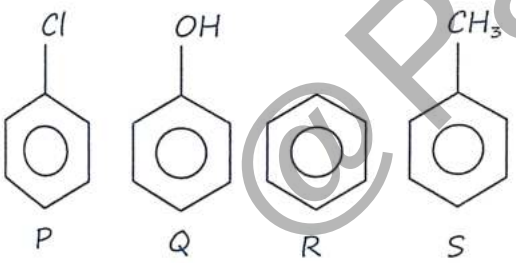

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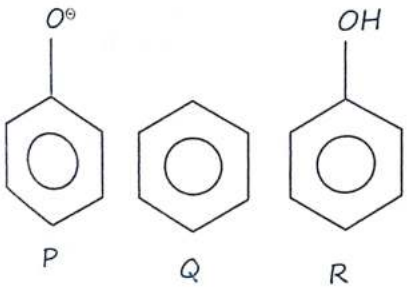

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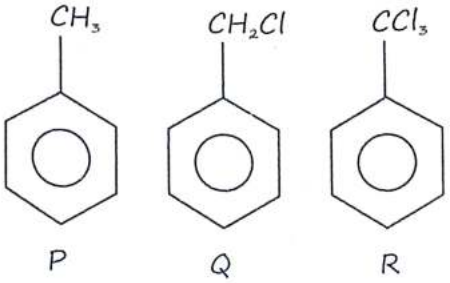

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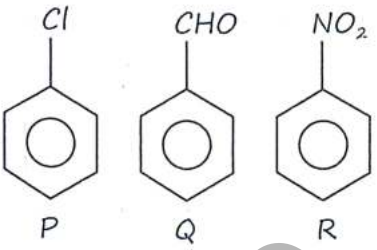

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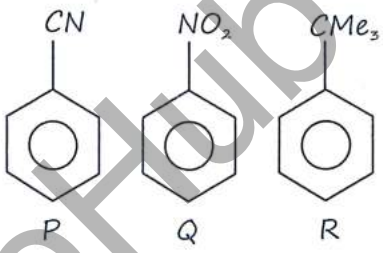

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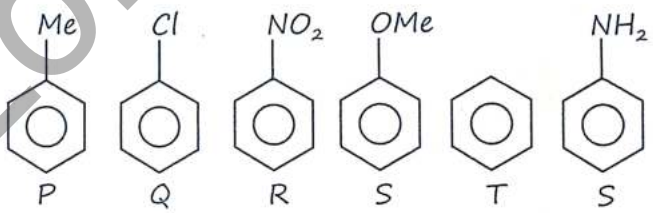

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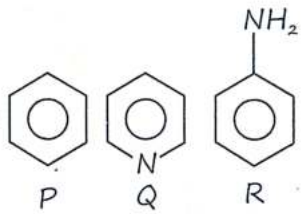

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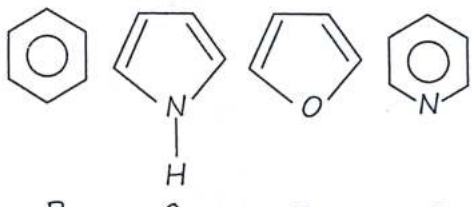

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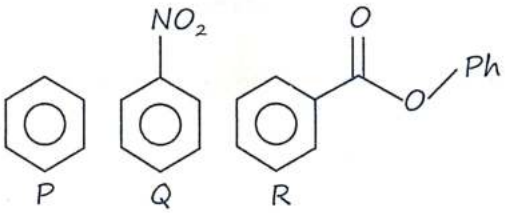

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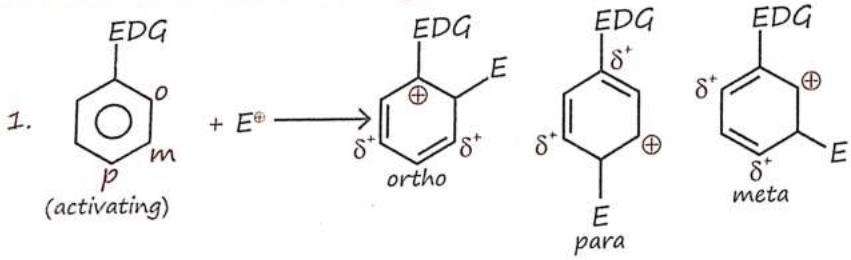
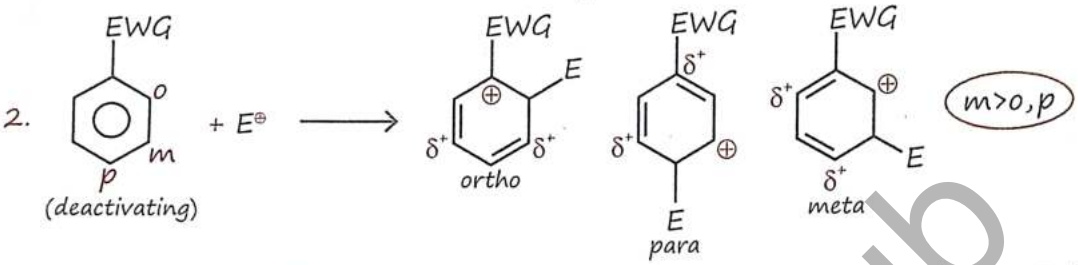
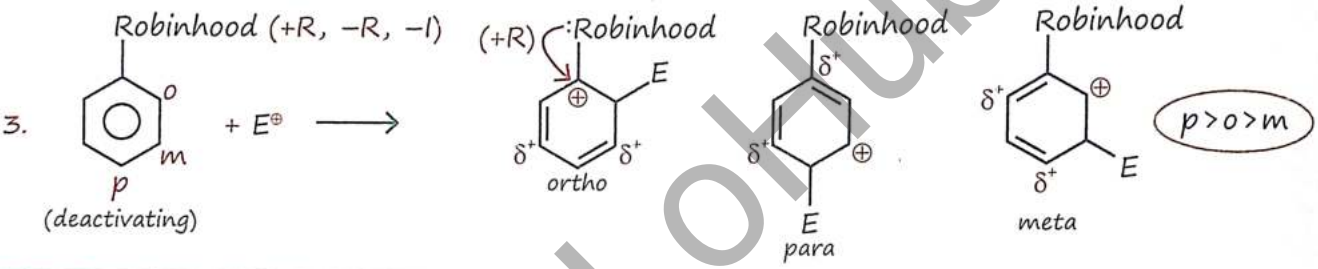
Orientation (decide by stability of positive charge)



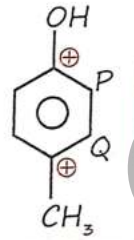
(Hindrance का चक्कर बाबू भैया)





Baba Tea-Stall (BTS)

- G (lp, +H, +I) → o, p > m
- Jaha positive zyada stable ho uske respect mai o/p will be major product.



- Robinhood are net withdrawing w.r.t. Benzene by inductive effect (−I) but o/p directing due to resonance effect (+R, +M).

Major product will be ortho w.r.t. OH and meta w.r.t. $-CH_3$

Recap: Ortho, Meta, Para Directing

Student: "Why are you so moody?"
Benzene: "Because EDG says go Ortho-Para, but EWG pushes Meta!"

Group	Effect on Rate	Directing Position	Examples
Electron Donating Groups	Increases	Ortho & Para	$-OH$, $-NH_2$, $-CH_3$, $-OCH_3$
Electron Withdrawing Groups	Decreases	Meta	$-NO_2$, $-SO_3H$, $-COOH$, $-CN$
Robinhood Groups	Decreases	Ortho & Para	$-Cl$, $-Br$, $-I$, $-\overset{O}{\overset{\|\|}{S}}-OR$, $-NO$

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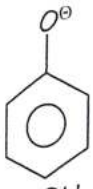
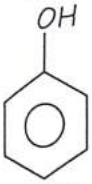
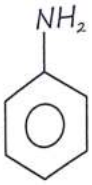
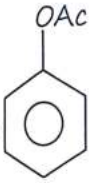
Practice Questions

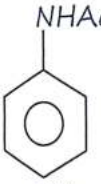
Find activating/deactivating and o/p or m directing?

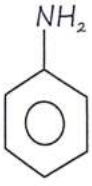
594. 

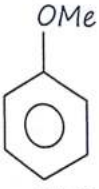
595. 

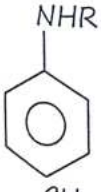
596. 

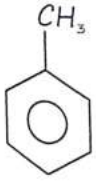
597. 

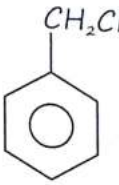
598. 

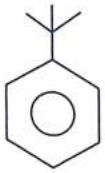
599. 

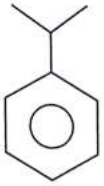
600. 

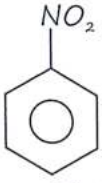
601. 

602. 

603. 

604. 

605. 

606. 

607.

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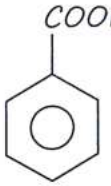
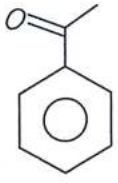
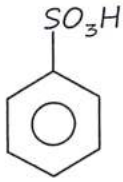
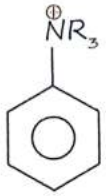
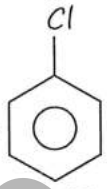
609. 

610. 

611. 

612. 

613. 

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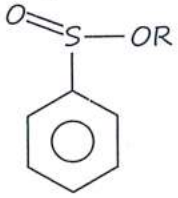
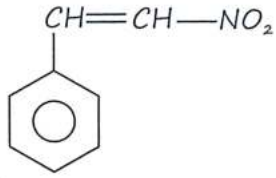
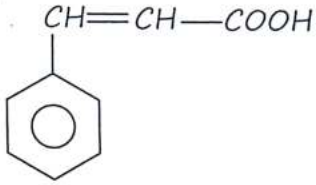
615. 

616. 

617. 

➤ Find E⊕ position?

618.

619.

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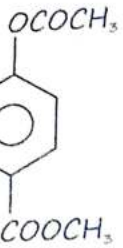
620. 

621.

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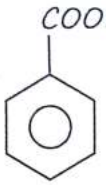
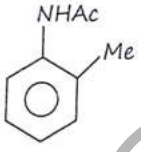
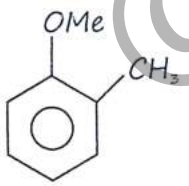
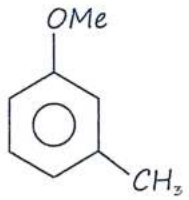
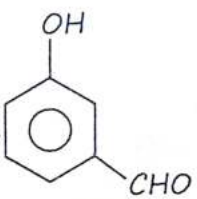
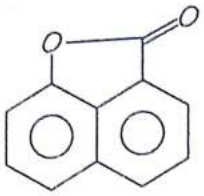
623. 

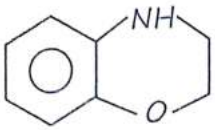
624. 

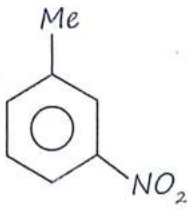
625. 

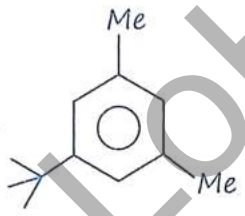
626. 

627. 

628. 

629. 

630. 

631. 

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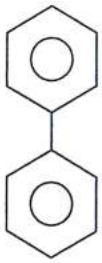
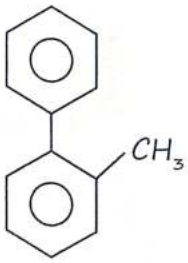
633. 

634. 

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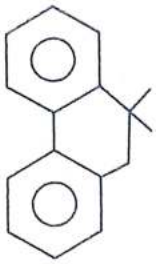
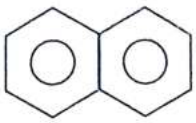
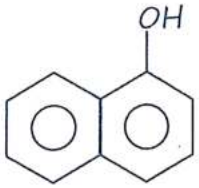
636. 

637. 

638. 

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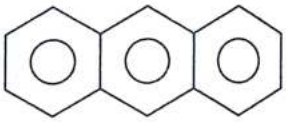
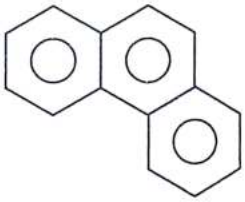
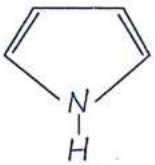
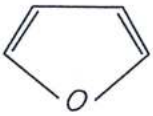
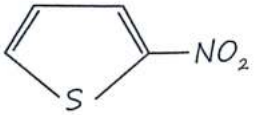
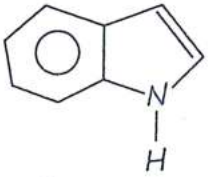
640. 

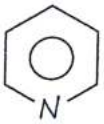
641. 

642. 

643. 

644. 

645. 

646. 

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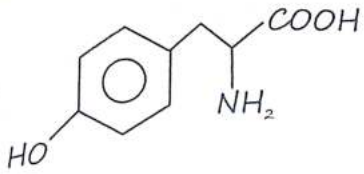
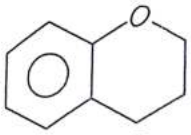
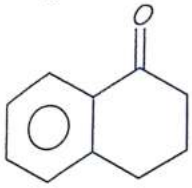
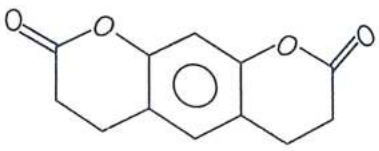
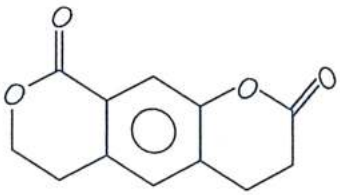
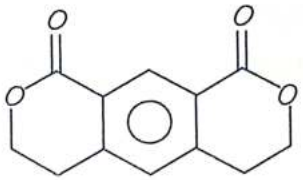
647. 

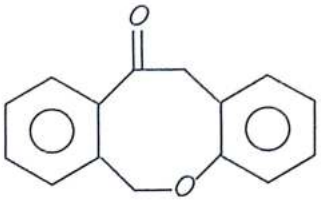
648. 

649. 

650. 

651 

652. 

653. 

Previous Year Questions

33. Identify correct statement/s:

(A) $-OCH_3$ and $-NHCOCH_3$ are activating group.

(B) $-CN$ and $-OH$ are meta directing group.

(C) $-CN$ and $-SO_3H$ are meta directing group.

(D) Activating groups act as ortho - and para directing groups.

(E) Halides are activating groups.

Choose the correct answer from the options given below: [24 Jan, 2025 (Shift-II)]

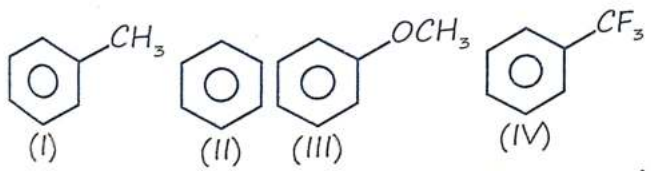
(a) (A), (B) and (E) only

(b) (A), (C) and (D) only

(c) (A) only

(d) (A) and (C) only

Sol. (b) $-CN$ is a meta directing group while $-OH$ is an o/p directing group. Halogens (F, Cl, Br, I) are deactivating groups.

34. 

The correct arrangement for decreasing order of electrophilic substitution for above compounds. [06 April, 2024 (Shift-II)]

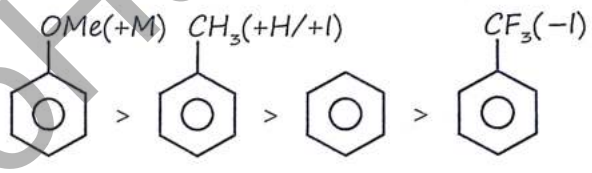
(a) (IV) > (I) > (II) > (III)

(b) (III) > (I) > (II) > (IV)

(c) (II) > (IV) > (III) > (I)

(d) (III) > (IV) > (II) > (I)

Sol. (b) The correct order is:



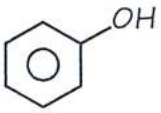
35. The correct order of reactivity towards the electrophilic substitution of the compounds aniline (I), benzene (II) and nitrobenzene (III) is: (2003)

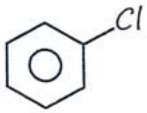
(a) III > II > I (b) II > III > I

(c) I < II > III (d) I > II > III

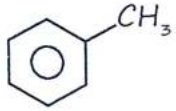
Sol. (d) The order of electrophilic substitution are: I > II > III

36. Which of the following compounds will be most easily attacked by an electrophile? (1999, 1998)

(a) 

(b) 

(c)

(d) 

Sol. (a) Phenol consist of OH that is moderate electron donating group which is also an ortho-para directing group for electrophile in the ring. Hence, will be most easily attacked by the benzene ring as $-OH$ group increases electron density in the ring.

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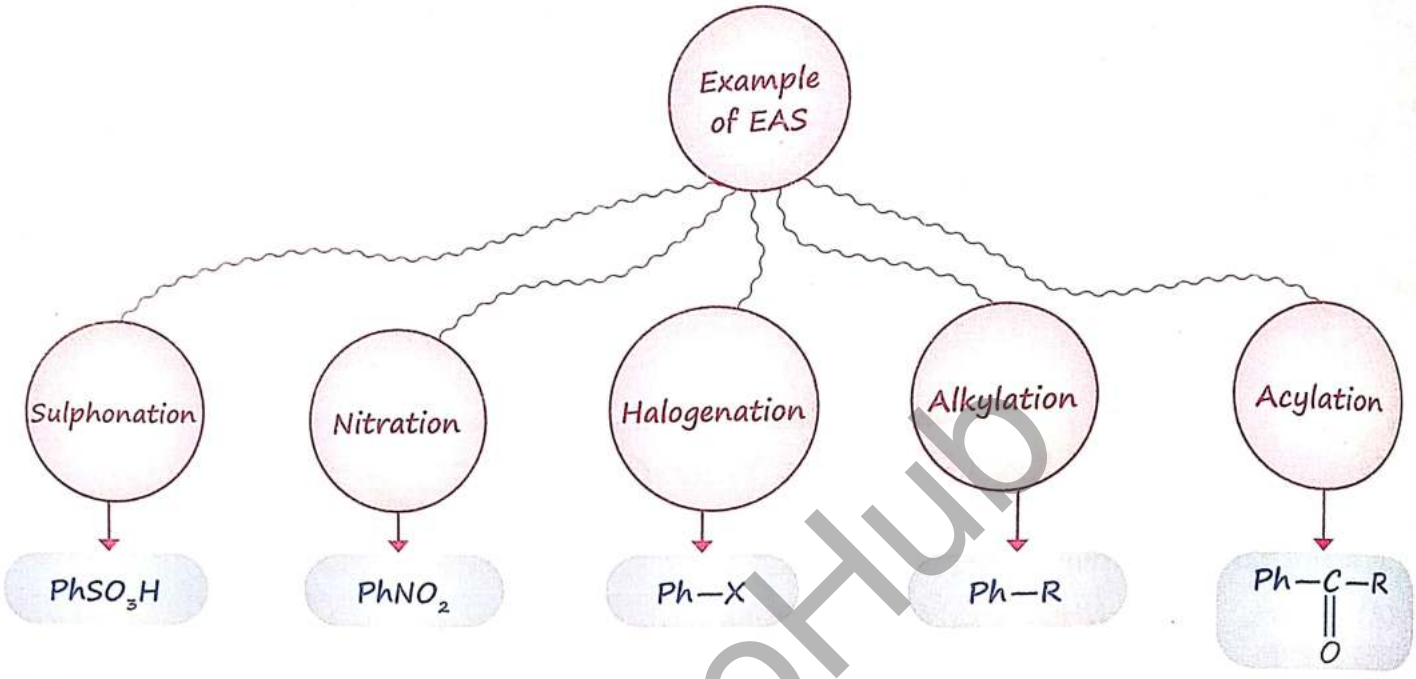
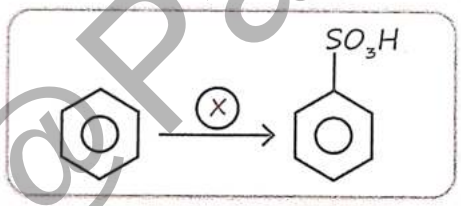
Examples of Electrophilic Aromatic Substitution



I. Sulphonation Reaction

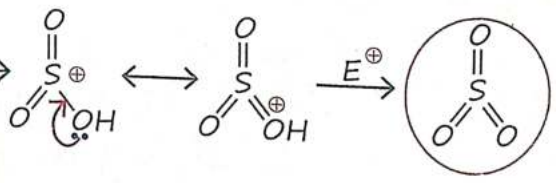
Sulphonation Reaction of benzene, where benzene reacts with sulphonating agents like concentrated H_2SO_4 or oleum ($H_2S_2O_7$) to form benzenesulphonic acid ($C_6H_5SO_3H$). The reaction involves the substitution of a hydrogen atom by a sulphonic acid group ($-SO_3H$).

Reaction



X may be:
1. Conc. H_2SO_4
2. Oleum ($H_2S_2O_7$) → $H_2SO_4 + SO_3$

Generation of Electrophile

1. $HO-\overset{\overset{O}{\|}}{\underset{\underset{O}{\|}}{S}}-OH + H_2SO_4 \longrightarrow HO\ddot{-}\overset{\overset{O}{\|}}{\underset{\underset{O}{\|}}{S}}-\overset{\oplus}{O}H_2 \longrightarrow$  $\xrightarrow{E^{\oplus}}$

2. $H_2S_2O_7 \rightarrow H_2SO_4 + SO_3$

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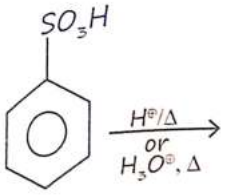
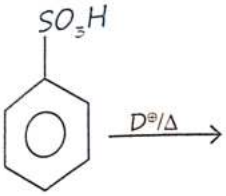
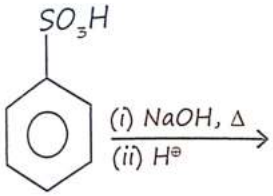
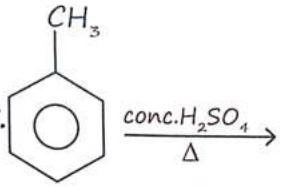
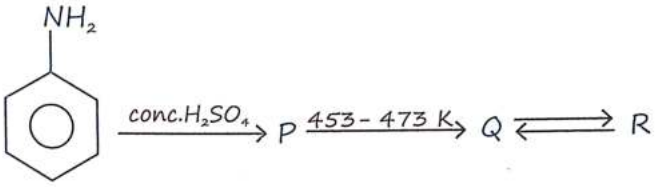
Practice Questions

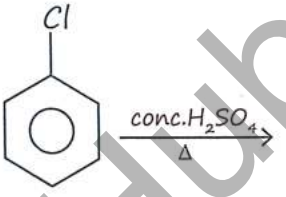
654. 

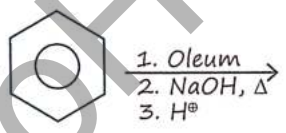
655. 

656. 

657. 

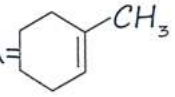
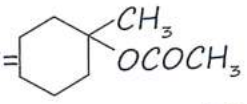
658. 

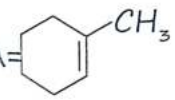
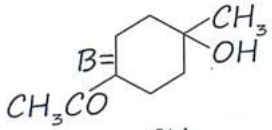
659. 

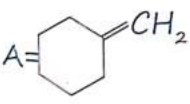
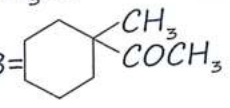
660. 

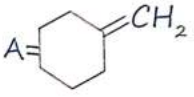
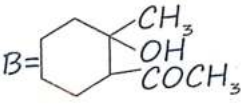
Previous Year Questions

37. Identify the major products A and B respectively in the following set of reactions.
[08 April, 2024 (Shift-I)]

(a) A=  and B= 

(b) A=  and B= 

(c) A=  and B= 

(d) A=  and B= 

Sol. (a)

38. Benzene on nitration gives nitrobenzene in presence of HNO_3 and H_2SO_4 mixture, where:
[20 July, 2021 (Shift-II)]

(a) Both H_2SO_4 and HNO_3 act as a bases

(b) Both H_2SO_4 and HNO_3 act as an acids

(c) HNO_3 acts as a base and H_2SO_4 acts as an acid

(d) HNO_3 acts as an acid and H_2SO_4 acts as a base

Sol. (c)

39. Arrange the following in increasing order of reactivity towards sulphonation with fuming sulphuric acid. Benzene, toluene, methoxy benzene, chlorobenzene (IIT JEE 1988)

Sol. Sulphonation is electrophilic substitution reaction, more strong electron donating group present on ring more is the reactivity due to increase in e– density at o, p-position.

Hence, the correct order is

chlorobenzene < Benzene < toluene < methoxy benzene.

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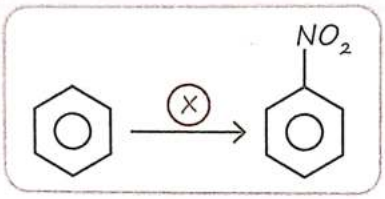
2. Nitration Reaction

An aromatic compound (like benzene) reacts with nitrating agent to form a nitro-substituted aromatic compound.

Reaction



X may be:

1. $HNO_3|H_2SO_4$
2. Conc. HNO_3
3. dilute HNO_3
4. N_2O_5 or CH_3COONO_2
5. NO_2BF_4 or NO_2ClO_4
6. HNO_2 following HNO_3

Generation of Electrophile

1. H_2SO_4 (acid) + HNO_3 (Base) $\longrightarrow$ $\overset{\oplus}{N}O_2$ (Nitronium ion) + H_2O + $SO_4^{\ominus}$

   AB $HO-N(=O)-O$ → Dehydrating agent

   $\overset{\oplus}{N}O_2$ + $H_3O^{\oplus}$ + $2HSO_4^{\ominus}$; $E^{\oplus}$

   *Van't hoff factor (no. of ions) $\longrightarrow$ (4) [3.85 Real]

2. Conc. HNO_3 $\xrightarrow{\text{Self Ionisation}}$ $\overset{\oplus}{N}O_2$; $E^{\oplus}$

3. dil.HNO_3 $\xrightarrow{\text{Self}}$ $\overset{\oplus}{N}O_2$; $E^{\oplus}$; $H_2O-\overset{\oplus}{N}O_2$

4. N_2O_5 $\xrightarrow{\pi^*}$ $Ph-NO_2$

5. HNO_2 (HO−N=O) + $H^{\oplus}$ $\longrightarrow$ $\overset{\oplus}{N}O$ (Nitrosonium ion) $\longrightarrow$ $Ph-N=O$ $\xrightarrow{HNO_3}$ $Ph-NO_2$

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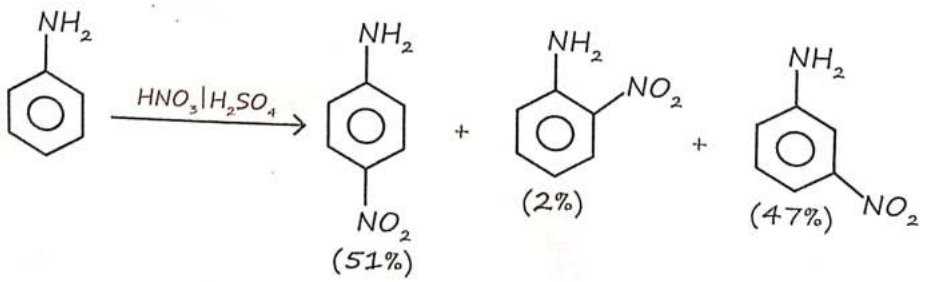
Nitration of Aniline

Nitration of Aniline, where aniline ($C_6H_5NH_2$) reacts with concentrated HNO_3 and H_2SO_4. The reaction is typically challenging due to the protonation of the $-NH_2$ group under acidic conditions, which reduces its activating effect on the benzene ring. That is why significant amount of meta derivative is also formed.



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How to increase Otho, Meta and Para

* % Para ↑↑ ⟶ 1. Block or Protect $-NH_2$ (acylation) 2. Nitration 3. Hydrolysis

* % Meta ↑↑ ⟶ Increase acidic medium (pH↓↓)

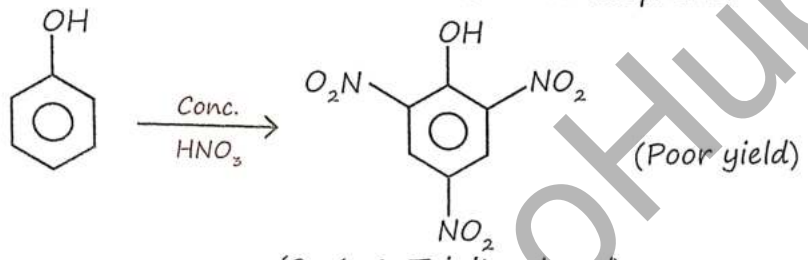
($PhNH_2 \rightarrow Ph\overset{\oplus}{N}H_3$)
(d)

* % Ortho ↑↑ ⟶ 1. Block or Protect $-NH_2$ (acylation) 3. Nitration 4. Hydrolysis

2. Para Block (Sulphonation)

Nitration of Phenol

Nitration of Phenol, where phenol (C_6H_5OH) reacts with concentrated nitric acid (HNO_3). The $-OH$ group activates the benzene ring, leading to formation of 2, 4, 6-Trinitrophenol.

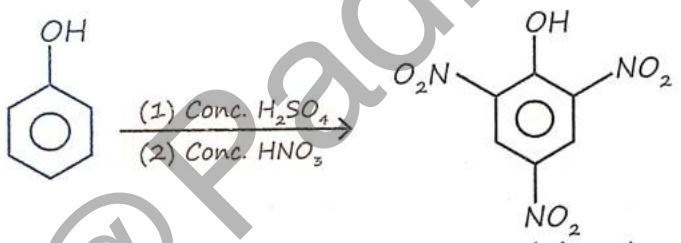


(Poor yield)

(2, 4, 6-Trinitrophenol)

How to Increase Yield

Nowadays picric acid is prepared by treating phenol first with concentrated sulphuric acid which converts it to phenol-2,4-disulphonic acid, and then with concentrated nitric acid to get 2,4,6-trinitrophenol.

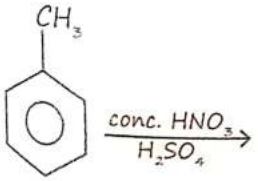


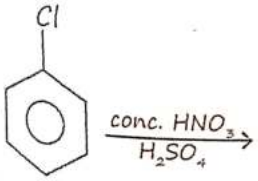
(Good yield)

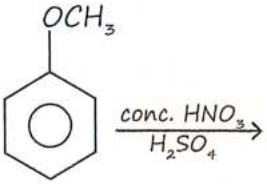
(2, 4, 6-Trinitrophenol)

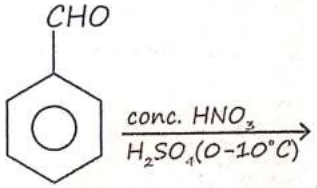
Practice Questions

Q. Draw major and minor product of following reaction.

661. 

662. 

663. 

664. 

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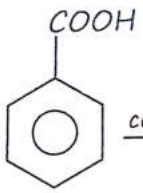
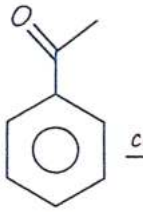
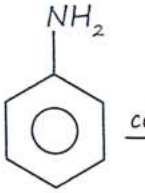
665. 

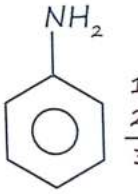
$\xrightarrow[H_2SO_4]{conc.\ HNO_3}$

666. 

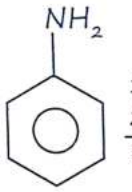
$\xrightarrow[H_2SO_4]{conc.\ HNO_3}$

667. 

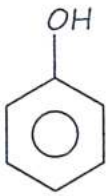
$\xrightarrow[H_2SO_4]{conc.\ HNO_3}$

668. 

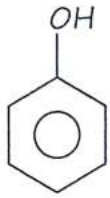
$\xrightarrow[3.\ H_3O^{\oplus}]{1.Ac_2O/Py\ \ 2.\ HNO_3/H_2SO_4}$

669. 

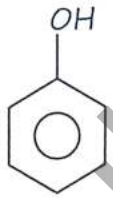
$\xrightarrow[3.\ HNO_3/H_2SO_4]{1.Ac_2O/Py\ \ 2.\ H_2SO_4,\ \Delta}$

670. 

$\xrightarrow[HNO_3]{dilute}$

671. 

$\xrightarrow{conc.\ HNO_3}$

672. 

$\xrightarrow[2.\ Conc.\ HNO_3]{1.\ Conc.\ H_2SO_4}$

673. Partial rate factor for tert butyl benzene at O : M : P is 4.5 : 30 : 75 w.r.t benzene for electrophilic aromatic substitution. Then find % yield of nitration at para position and also find reactivity of butyl benzene than benzene.

674. Nitration of toluene is 25 times faster than benzene. % yield of ortho and para position is 56.5 and 40% respectively then find partial rate factor for para position.

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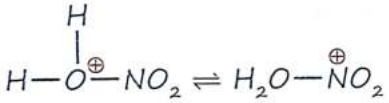
Previous Year Questions

40. Given below are two statements:

Statement-I: Nitration of benzene involves the following step:



Statement-II: Use of Lewis base promotes the electrophilic substitution of benzene.

In the light of the above statements, choose the most appropriate answer from the options given below: [05 April, 2024 (Shift-I)]

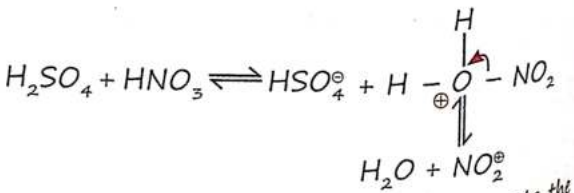
(a) Both Statement-I and Statement-II are incorrect

(b) Statement-I is correct but Statement-II is incorrect

(c) Both Statement-I and Statement-II are correct

(d) Statement-I is incorrect but Statement-II is correct

Sol. (b) Nitration of benzene



Thus, we can see that Lewis acids promote the formation of electrophiles not Lewis base.

41. The compound P, Q and S were separately subjected to nitration using HNO_3/H_2SO_4 mixture. The major product formed in each case respectively, is

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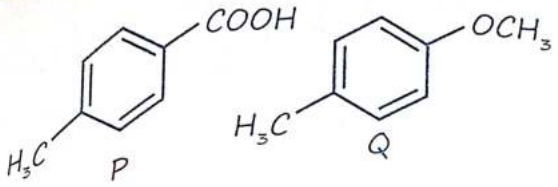
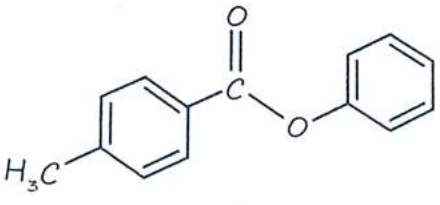


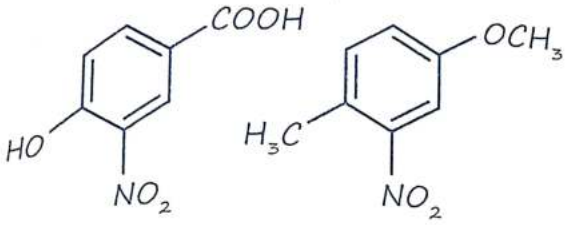
P Q

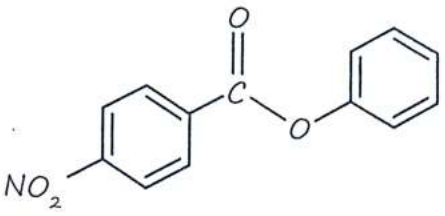


S

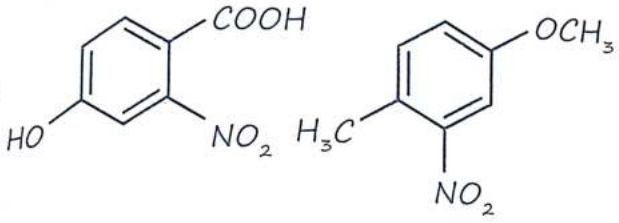
(IIT JEE 2010)

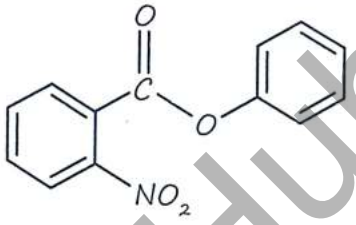
(a)



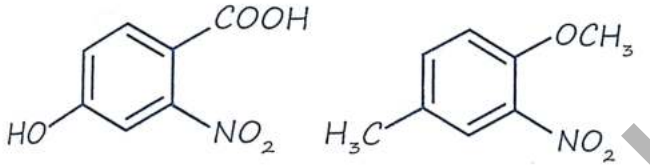


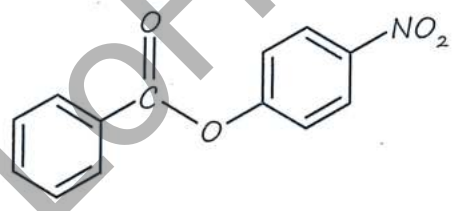
(b)



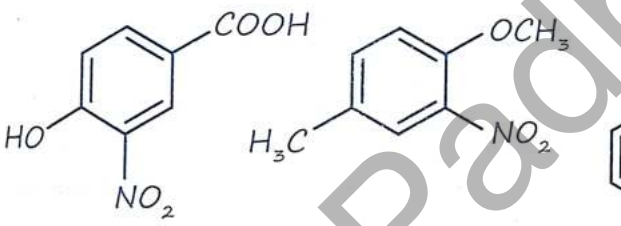


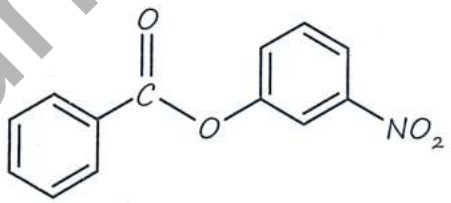
(c)





(d)

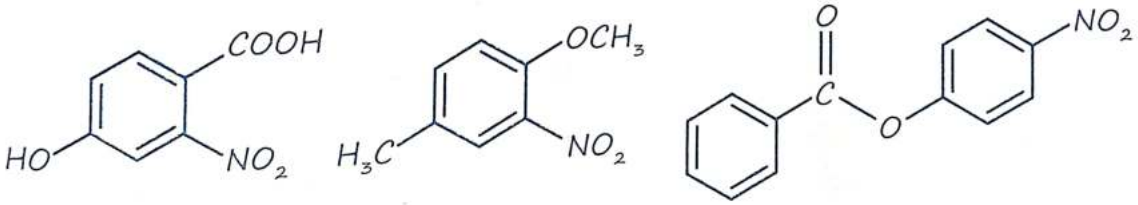




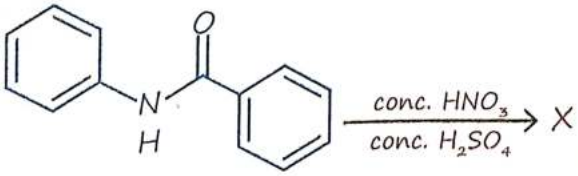
Sol. (c) Ring activating group – (–R), –OH

Ring deactivating group – ($-NO_2$), –CN

HNO_3/H_2SO_4 is nitrating mixture hence there will be attack of $-NO_2$ group (NO_2^+) on reactant. Major product formed in the given reactions.



42. In the following reaction,



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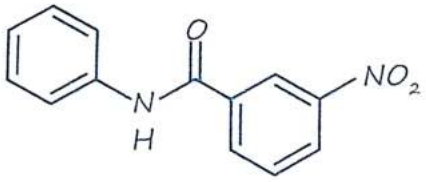
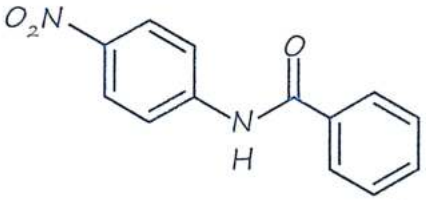
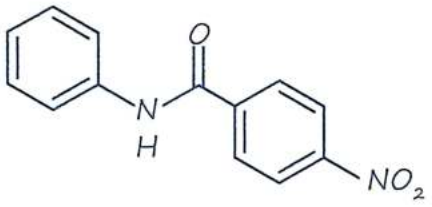
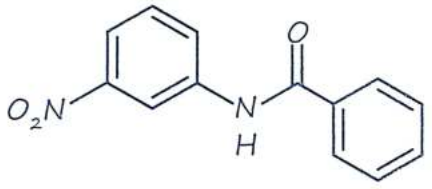
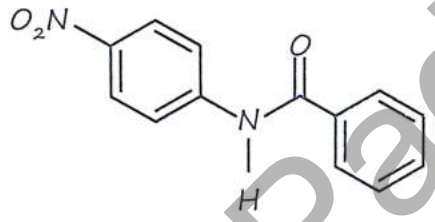
The product X is (IIT JEE 2007)

(a) 

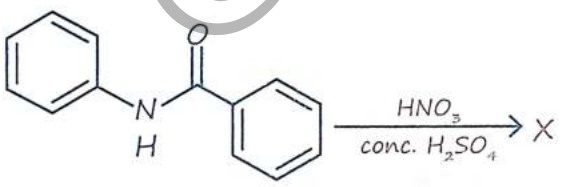
(b) 

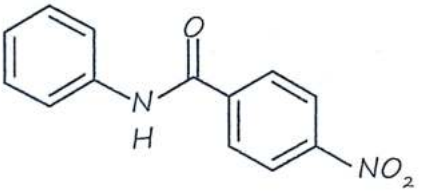
(c) 

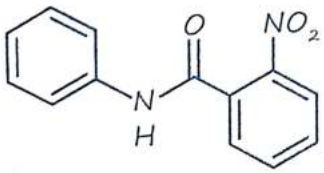
(d) 

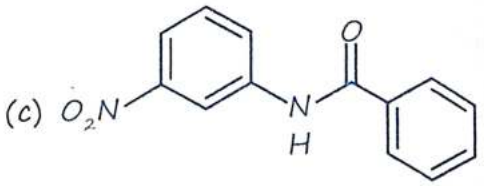
Sol. (b) 

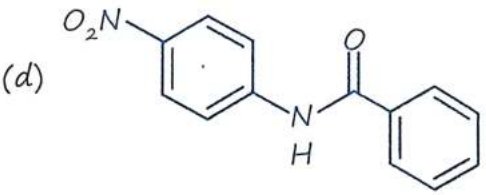
43. What will be the major product in the following mononitration reaction?
[10 Jan, 2019 (Shift-II)]



(a) 

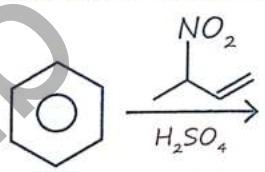
(b) 

(c) 

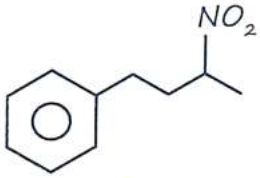
(d) 

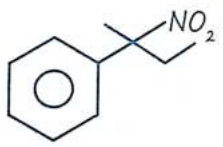
Sol. (d) –NH (activating group) is ortho, para directing.

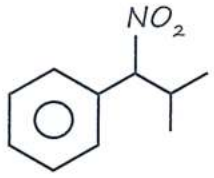
44. The major product of the following reaction is:

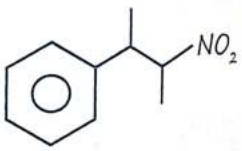


[25 Feb, 2021 (Shift-II)]

(a) 

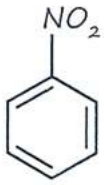
(b) 

(c) 

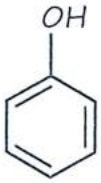
(d) 

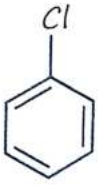
Sol. (d)

45. Which of the following compound is most reactive in electrophilic aromatic substitution? (2020-Covid)

(a) 

(b)

(c) 

(d) 

Sol. (c)

46. The reactive species in the nitration of benzene is: (1994)

(a) NO_3 (b) HNO_3

(c) NO_1^+ (d) NO_1^-

Sol. (c)

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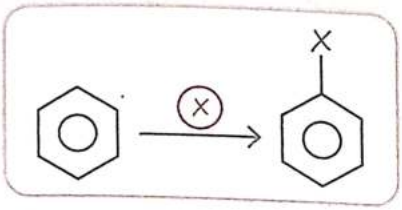
3. Halogenation Reaction

Halogenation Reaction of aromatic compounds, is an electrophilic aromatic substitution reaction where a halogen substitutes a hydrogen atom on the benzene ring.

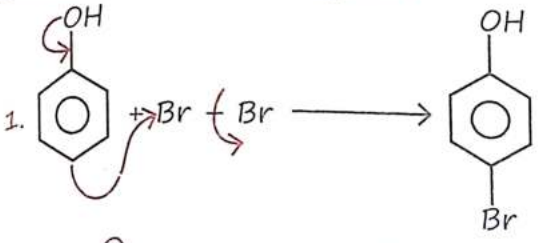
Reaction



X may be:

1. X_2 (in case of activated rings like OMe)
2. X_2/(Lewis acid (Fe, FeX_3, $ZnCl_2$, AlX_3)
3. Interhalogen (I-Cl, I-Br)
4. X_2/PPS (H_2O, CH_3COOH)

Generation of Electrophile

1. 

(without lewis acid)

2. $X-X + LA \longrightarrow X^{\delta+}\cdots X^{\delta-}\cdots LA \longrightarrow (X^{\oplus})(LAX^{\ominus})$

$E^{\oplus}$

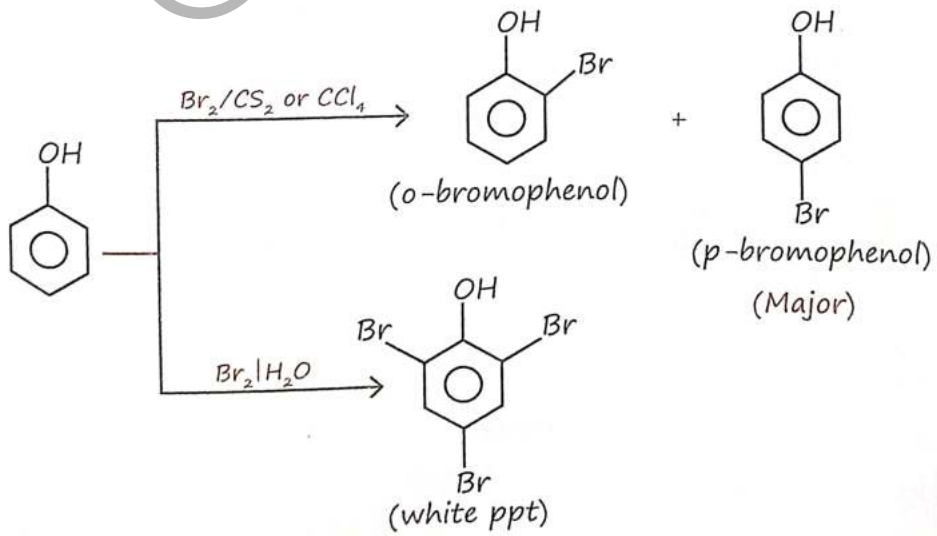
3. X_2 in PPS $\Rightarrow X^{\oplus}$

Halogenation of phenol

Halogenation of Phenol, where phenol reacts with Br_2:

- In CS_2 or CCl_4 (non-polar solvent), mono-substitution occurs.
- In H_2O (polar solvent), phenol undergoes tri-bromination at ortho and para positions, forming 2, 4, 6-tribromophenol as a white precipitate.

The -OH group activates the ring, especially towards electrophilic substitution.



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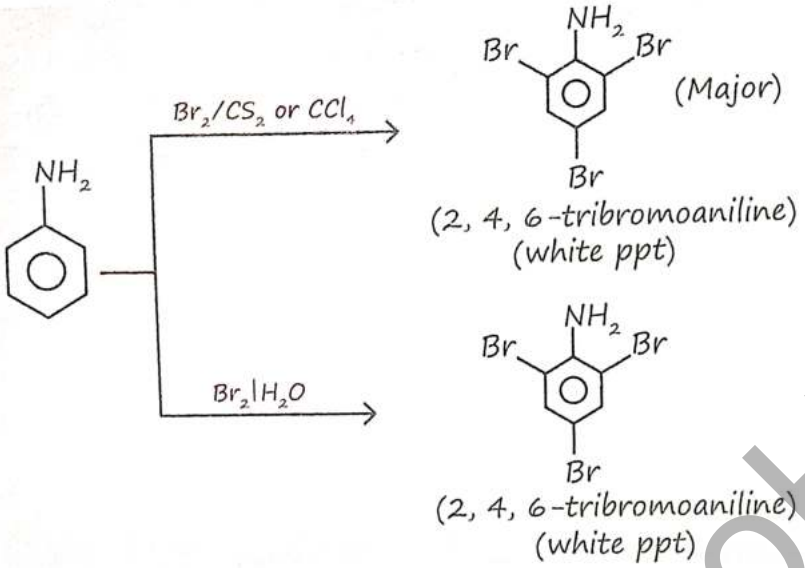
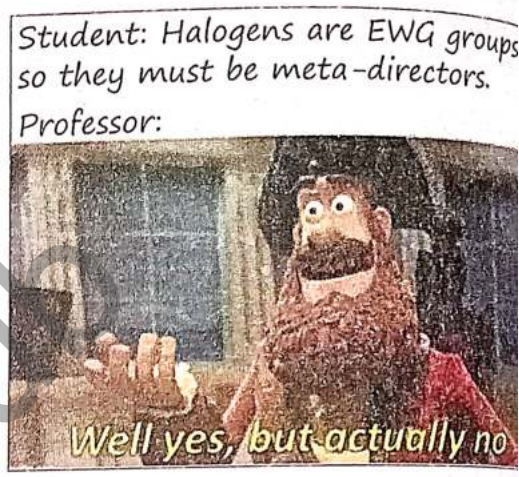
Halogenation of Aniline

Halogenation of Aniline, where aniline reacts with Br_2:

- As aniline is highly activated in both CS_2 or CCl_4 (non-polar solvent) and H_2O (polar solven
  2, 4, 6-tribromoaniline is obtained as a major product.

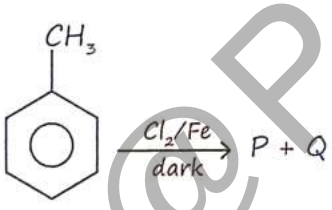
The $-NH_2$ group strongly activates the ring, making it highly reactive toward electrophilic substitutio

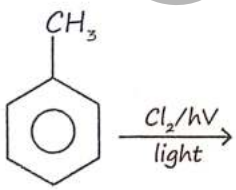


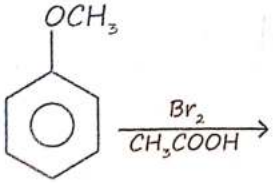


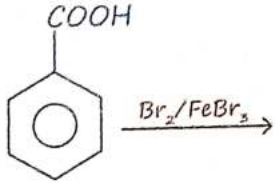
Practice Questions

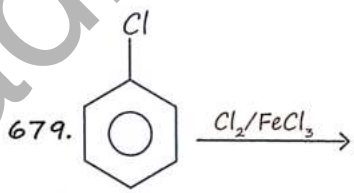
Q. Draw the major products of given reactions.

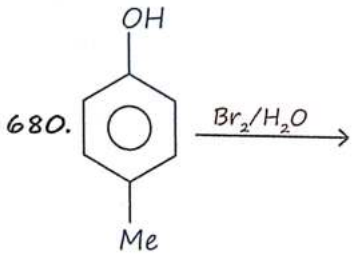
675. 

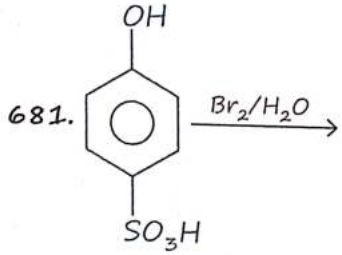
676. 

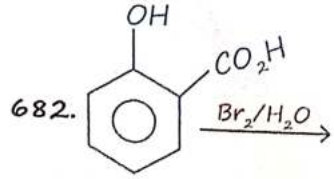
677. 

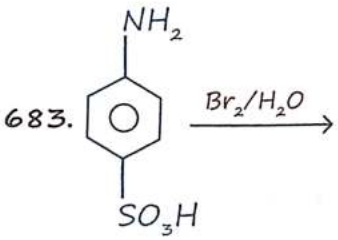
678. 

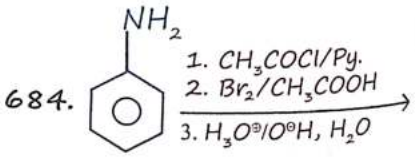
679. 

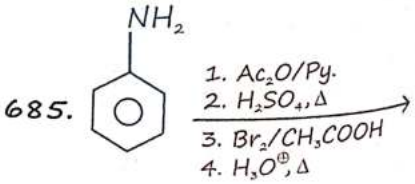
680. 

681. 

682. 

683. 

684. 

685. 

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47. The compound C_7H_8 undergoes the following reactions: (2018)

$$C_7H_8 \xrightarrow{3Cl_2/\Delta} A \xrightarrow{Br_2/Fe} B \xrightarrow{Zn/HCl}$$

The product 'C' is

(a) m-bromotoluene

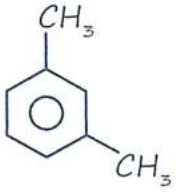
(b) o-bromotoluene

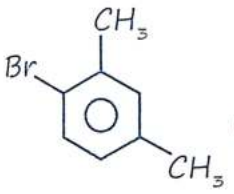
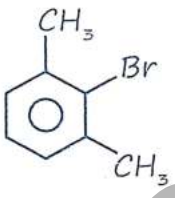
(c) p-bromotoluene

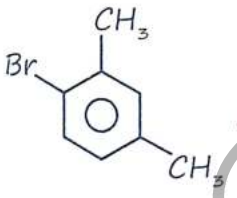
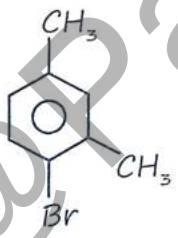
(d) 3-bromo-2, 4, 6-trichlorotoluene

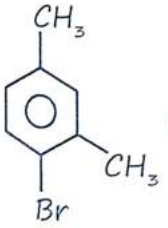
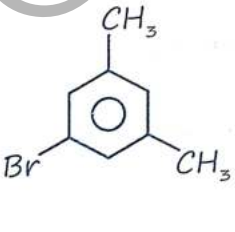
Sol. (a) n-bromotoulene.

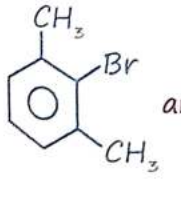
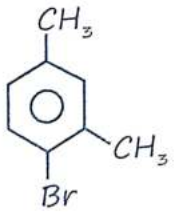
48. What products are formed when the following compound is treated with Br_2 in the presence of $FeBr_3$? (2014)



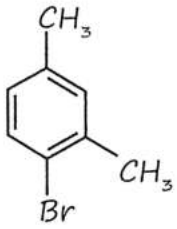
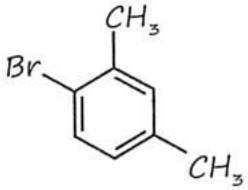
(a)  and 

(b)  and 

(c)  and 

(d)  and 

Sol. (b) CH_3 group is an ortho & para directing group so, electron density will be more on ortho & para site. Hence, the products are:

 and 

49. The reaction of toluene with Cl_2 in presence of $FeCl_3$ gives X and reaction in presence of light gives Y. Thus, X and Y are: (2010 Pre)

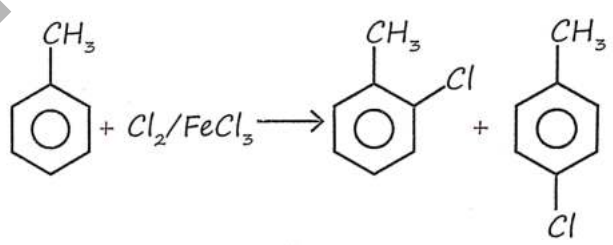
(a) X = Benzal chloride, Y = o-chlorotoluene

(b) X = m-chlorotoluene, Y = p-chlorotoluene

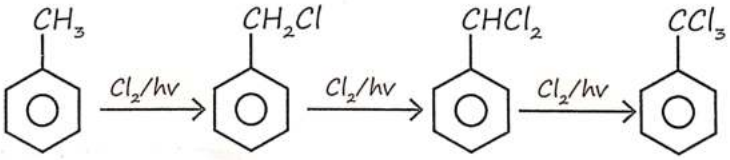
(c) X = o- and p-chlorotoluene, Y = Trichloromethyl benzene

(d) X = Benzyl chloride, Y = m-chlorotoluene

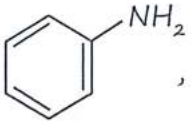
Sol. (c) In presence of Cl_2 in $FeCl_3$, the reaction takes place as:



In presence of sunlight, the reaction takes place as:



50. Which of the following is suitable to synthesize chlorobenzene? (2022)

(a)  , HCl Heating

(b) Benzene, Cl_2, anhydrous $FeCl_3$

(c) Phenol, $NaNO_2$, HCl, CuCl

(d) , HCl

Sol. (b)

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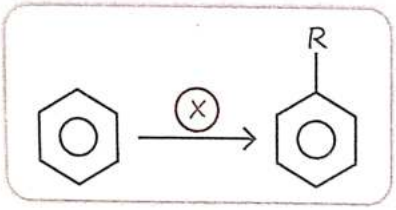
Friedel-Crafts Alkylation

An alkyl halide (R–X) reacts with AlX_3 to generate a carbocation (R^+) that can attack an aromatic ring.

Reaction



X may be:
1. R—X/Lewis acid (Fe, FeX_3, BF_3, AlX_3)
2. R—OH/$H^\oplus$
3. Alkene/$H^\oplus$
4. R—OH or Alkene/Lewis Acid (BF_3)

Generation of Electrophile

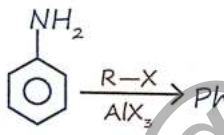
1. $R-X + AlX_3 \longrightarrow \overset{\delta^+}{R}\cdots\overset{\delta^+}{X}\cdots\overset{\delta^-}{AlX_3} \longrightarrow (\overset{\oplus}{R})\ \overset{\ominus}{AlX_4}$
2. $R-OH + \overset{\oplus}{H} \longrightarrow R-\overset{\oplus}{O}H_2 \xrightarrow{-H_2O} (\overset{\oplus}{R})$
3. $R-CH=CH_2 + \overset{\oplus}{H} \longrightarrow (R-\overset{\oplus}{C}H-CH_3)$
4. $R-CH=\overset{\delta^-}{C}H_2 + LA \longrightarrow (R-\overset{\oplus}{C}H-CH_3)$

Baba Tea-Stall (BTS)

(i) Carbocation is Obtained as intermediate

(ii) Rearrangement may be possible

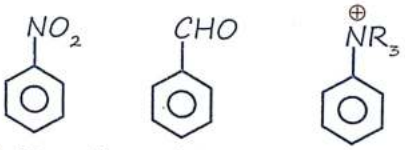
(iii) Highly activated rings doesn't show this reaction



$\xrightarrow[AlX_3]{R-X} Ph-\overset{\oplus}{N}H_2-\overset{\ominus}{AlCl_3}$ ['N' becomes EWG]

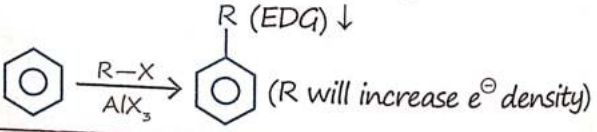
(iv) Highly deactivated rings doesn't show this reaction

For examples:



(i) Deactivated rings can be used as Solvent.

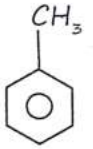
(ii) Polyalkylation occurs during reaction.

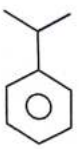


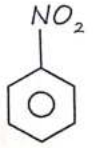
(R will increase $e^\ominus$ density)

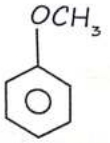
Practice Questions

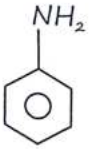
686. Which of the following compound will give f.c. alkylation?

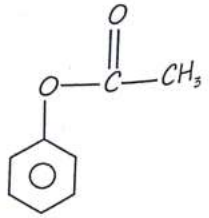
(a) 

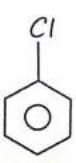
(b) 

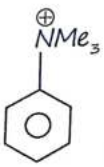
(c) 

(d) 

(e) 

(f) 

(g) 

(h) 

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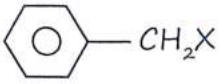
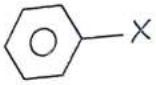
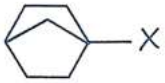
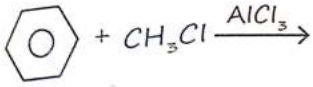
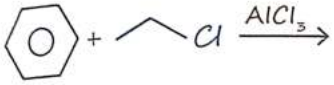
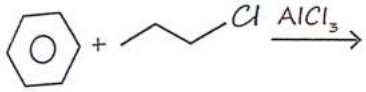
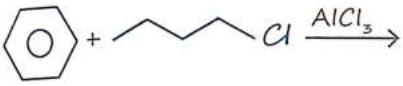
687. Which of the following will be possible for R–X in F.C. alkylation?

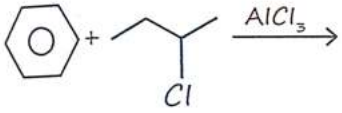
(a)
(b)
(c)
(d) 
(e) 
(f) 

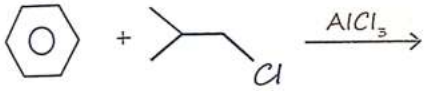
688. 

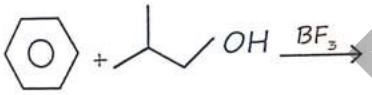
689. 

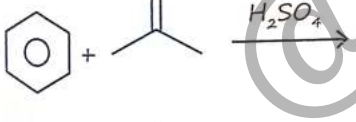
690. 

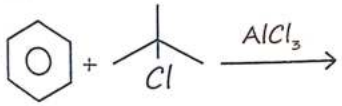
691. 

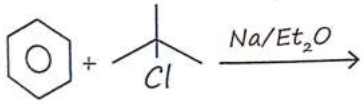
692. 

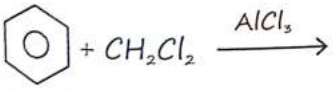
693. 

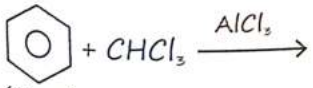
694. 

695. 

696. 

697. 

698. 
(excess)

699. 
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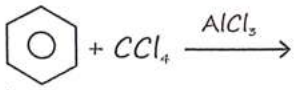
700. 
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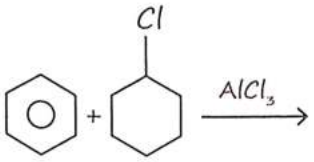
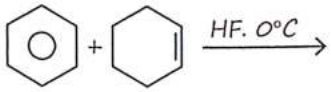
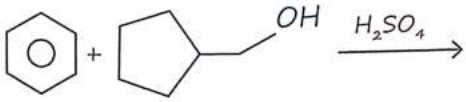
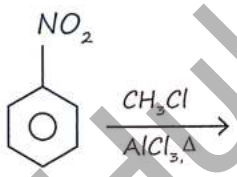
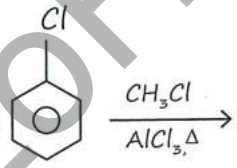
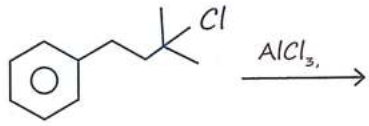
701. 

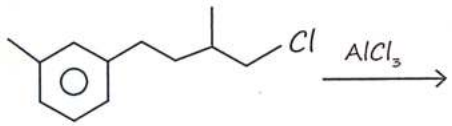
702. 

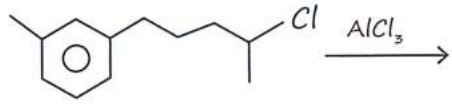
703. 

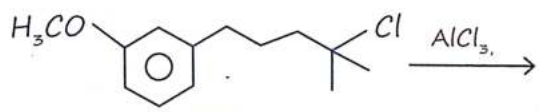
704. 

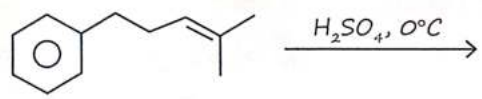
705. 

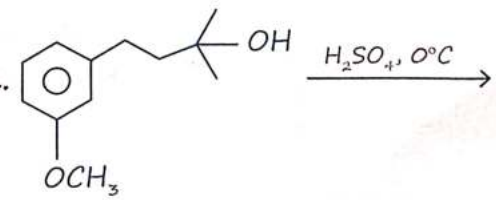
706. 

707. 

708. 

709. 

710. 

711. 

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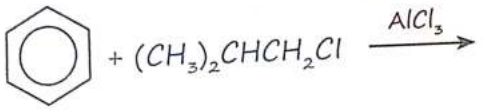
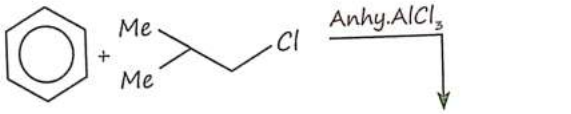
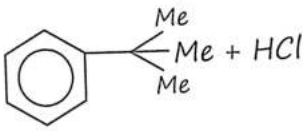
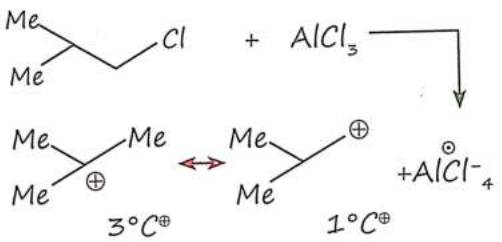
51. Write the structure of the major organic product expected from the following reaction (IIT JEE 1992)



Sol.





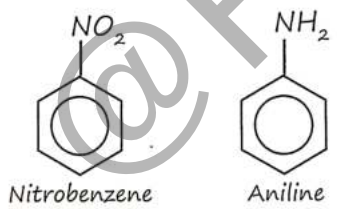


tertiarycarbocation is more stable.

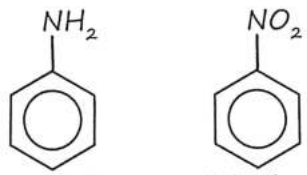
52. Number of compounds from the following which cannot undergo Friedel-Crafts reactions is :___ toluene, nitrobenzene, xylene, cumene, aniline, chlorobenzene, m-nitroaniline, m-dinitrobenzene

[09 April, 2024 (Shift-II)]

Sol. [4] Compounds which do not give Friedel Crafts reaction are:



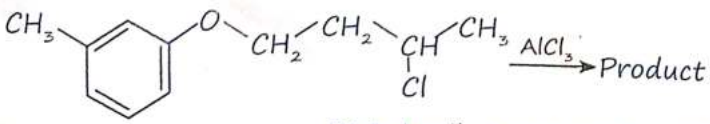
Nitrobenzene Aniline



m-nitroaniline m-dinitrobenzene

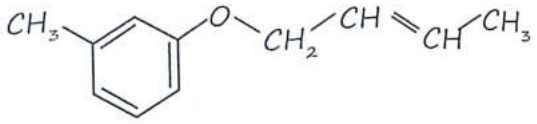
Hence, in total, there are 4 compounds.

53. The major product obtained in the given reaction is

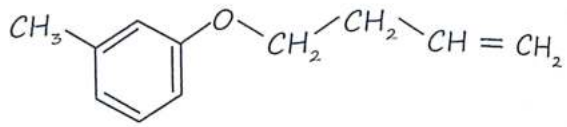


[10 April, 2019 (Shift-II)]

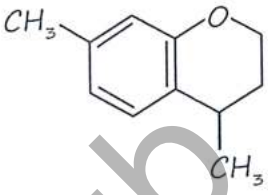
(a)



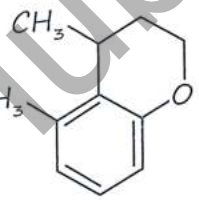
(b)



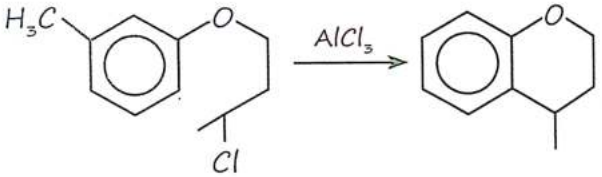
(c)



(d)



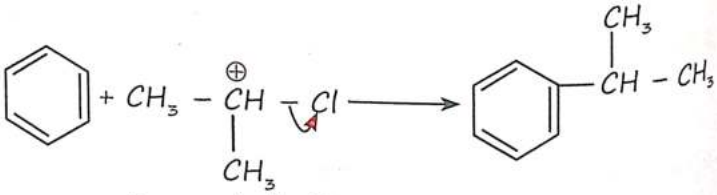
Sol. (c) The given reaction is an example of intramolecular friedel craft alkylation which occurs as:



54. Which of the following can be used as the halide component for Friedel-Crafts reaction? (2016 - II)

(a) Chloroethene (b) Isopropyl chloride
(c) Chlorobenzene (d) Bromobenzene

Sol. (b) In the case of iso-propyl chloride, lonepairs on chlorine reacts with lewis acid and gives 2° carbocation.



Iso-propyl chloride cumene

In chlorobenzene, bromobenzene or chloroethene, lone pair of electron on halogen atom are delocalised and attains the double bond character. So, all of these shows less reactivity with lewis acid.

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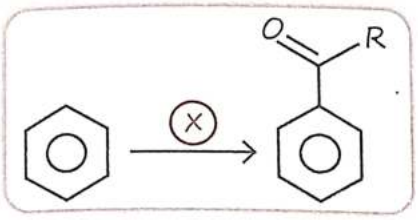
Friedel-Crafts Acylation

It involves the substituting an acyl group into the benzene ring. This reaction occur in the presence of Lewis acid.

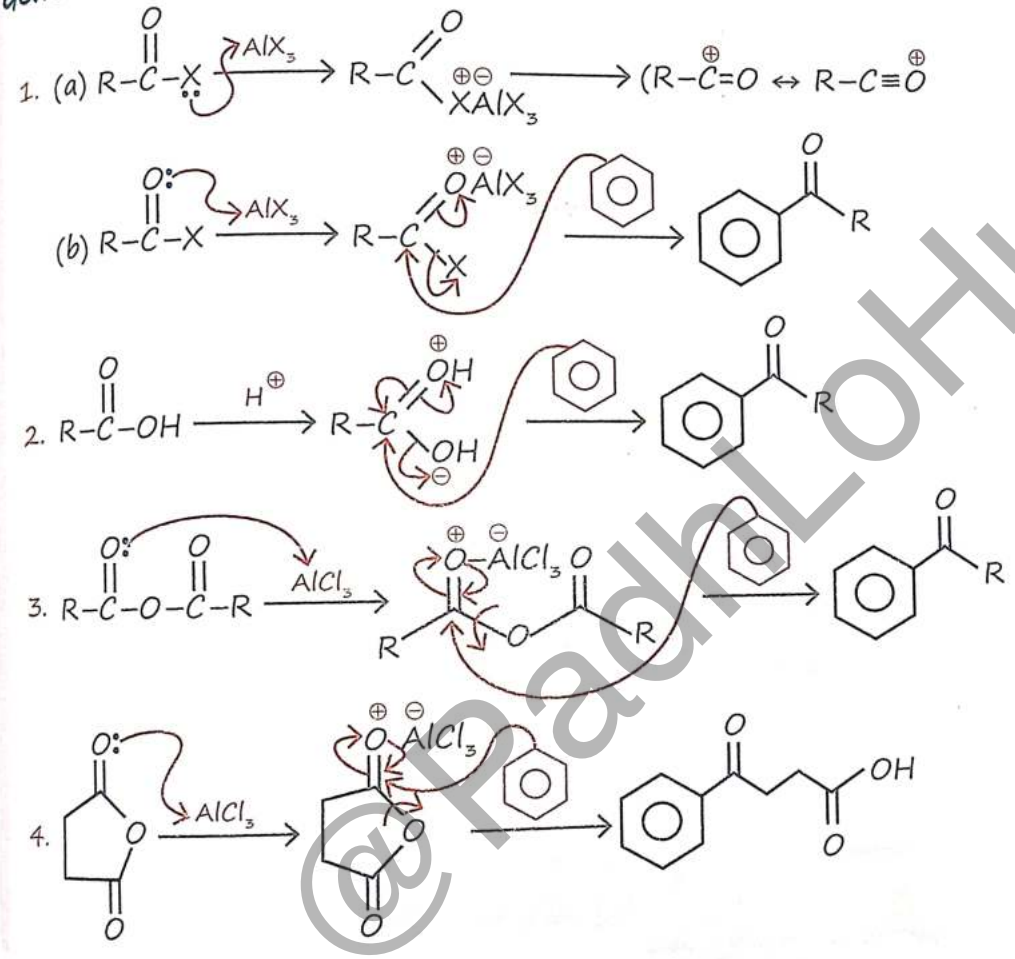
Reaction



X may be:

1. RCOX/Lewis acid (AlX_3)
2. R—COOH/H_2SO_4 or H_3PO_4
3. $(RCO)_2O/AlCl_3$
4. Succinic Anhydride|$AlCl_3$

Generation of Electrophile



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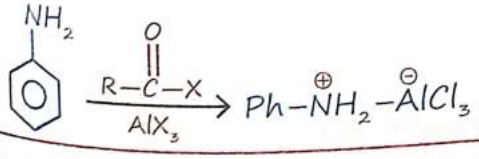
Baba Tea-Stall (BTS)

(i) Acyl cation is obtained as intermediate
(ii) Rearrangement is not possible
(iii) Highly activated rings doesn't show this reaction.

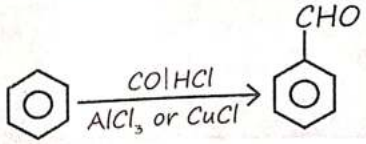


(iv) Excess amount of Lewis acid is required.
(v) Acylation is preferred over alkylation
(vi) Formylation is not possible because CO evaporates.

$$HCOCl \xrightarrow{AlCl_3} CO\uparrow + HCl\uparrow + AlCl_3$$

Gattermann Koch Reaction

Gattermann-Koch Reaction is a specific type of Friedel-Crafts acylation used to introduce a formyl group (-CHO) into an aromatic ring.



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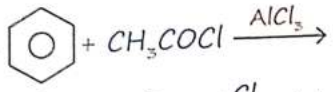
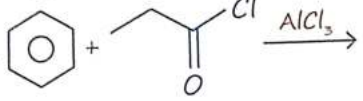
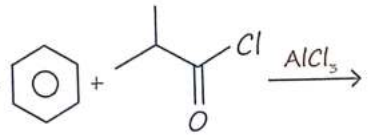
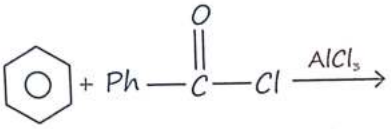
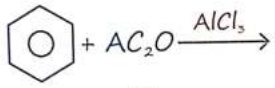
Practice Questions

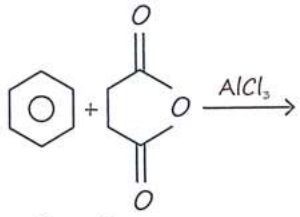
712. 

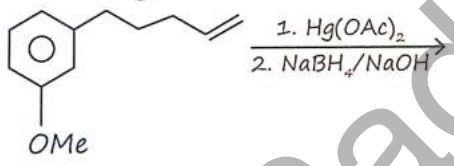
713. 

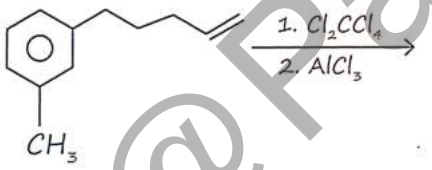
714. 

715. 

716. 

717. 

718. 

719. 

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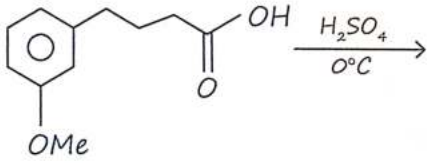
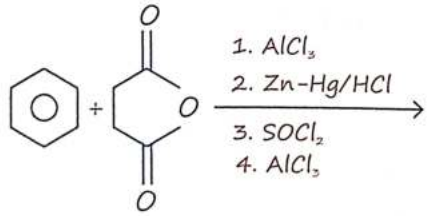
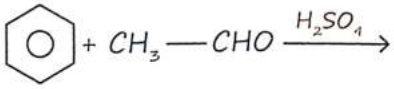
721. 

722. 

723. 
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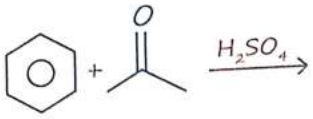
724. 
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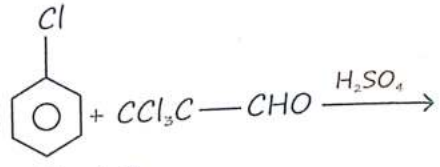
725. 
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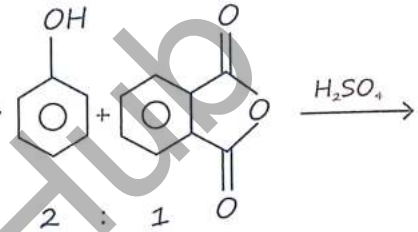
726. 
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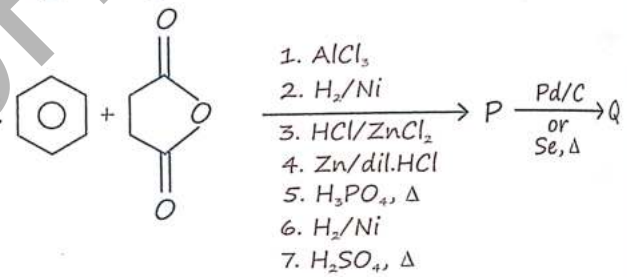
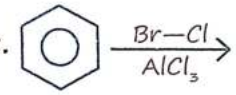
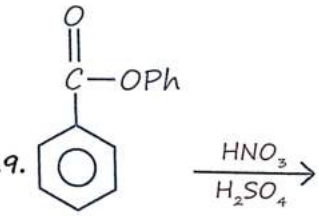
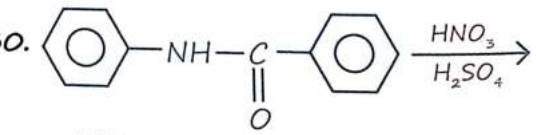
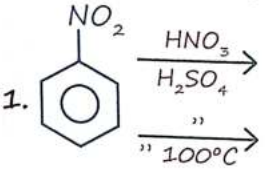
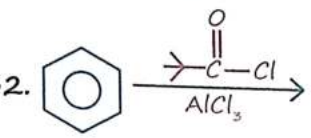
727. 

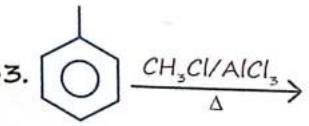
728. 

729. 

730. 

731. 

732. 

733. 

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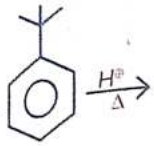
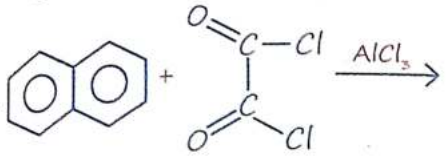
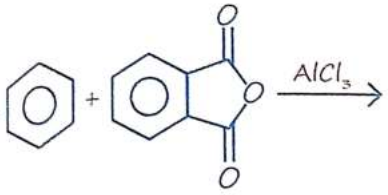
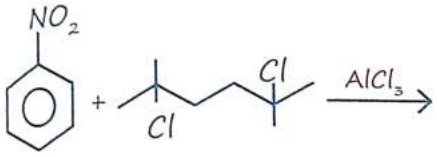
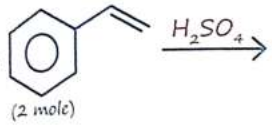
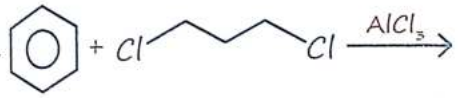
734. 

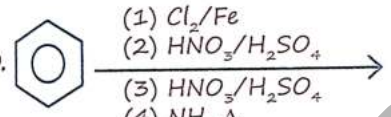
735. 

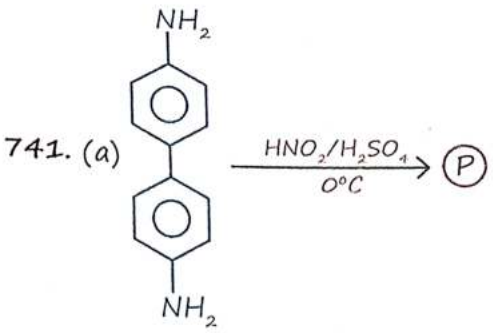
736. 

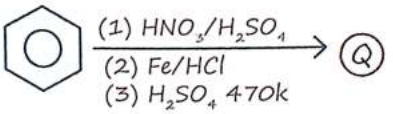
737. 

738. 

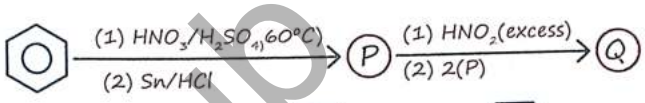
739. 

740. 

741. (a) 

(b) 

(c) $2Q + P \longrightarrow (R)$

742. 

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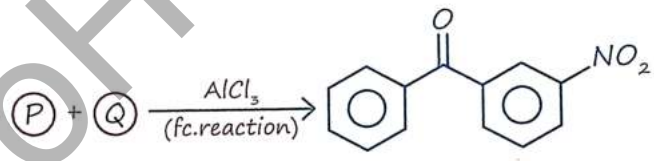
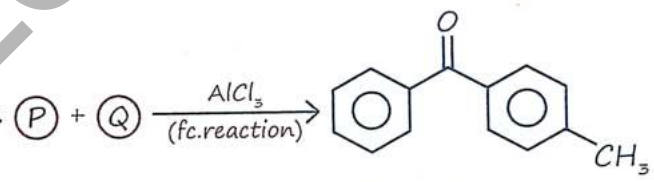
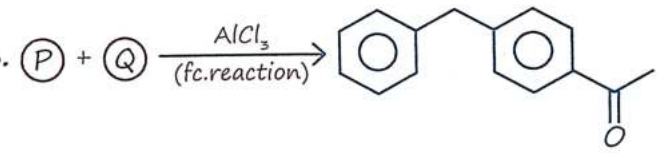
744. 

745. 

746. 

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Previous Year Questions

55. Number of compounds from the following which cannot undergo Friedel-Crafts reactions is :___ toluene, nitrobenzene, xylene, cumene, aniline, chlorobenzene, m-nitroaniline, m-dinitrobenzene
[09 April, 2024 (Shift-II)]

Sol. [4]

56. The correct sequence of reagents used in the preparation of 4-bromo-2-nitroethyl benzene from benzene is: [25 Feb, 2021 (Shift-II)]

(a) HNO_3/H_2SO_4, $Br_2/AlCl_3$, $CH_3COCl/AlCl_3$, Zn – Hg/HCl

(b) $CH_3COCl/AlCl_3$, $Br_2/AlBr_3$, HNO_3/H_2SO_4, Zn/HCl

(c) $Br_2/AlBr_3$, $CH_3COCl/AlCl_3$, HNO_3/H_2SO_4, Zn/HCl

(d) $CH_3COCl/AlCl_3$, Zn – Hg/HCl, $Br_2/AlBr_3$, HNO_3/H_2SO_4

Sol. (d)

57. Benzene reacts with CH_3Cl in the presence of anhydrous $AlCl_3$ to form: (2009)

(a) Chlorobenzene (b) Benzylchloride

(c) Xylene (d) Toluene

Sol. (d)

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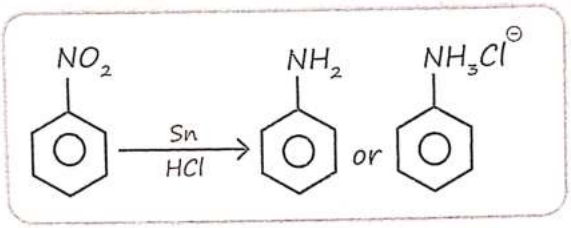
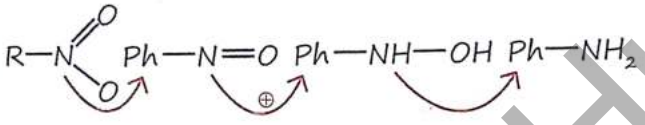
Reduction of Nitro Group to Aniline

Nitrobenzene is reduced to aniline using Sn/HCl. This method gives better yield than catalytic hydrogenation due to less side product formation. Better yield than H_2/Pd because $FeCl_2$ formed get hydrolysed So, HCl will produce.

Reaction



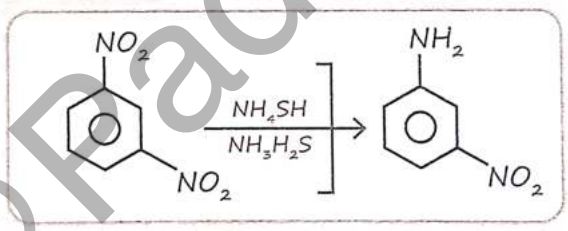
Mechanism



Selective Reduction of Dinitrobenzene

One nitro group in m-dinitrobenzene is selectively reduced to an amine using ammonium sulfide. This reaction gives p-nitroaniline.

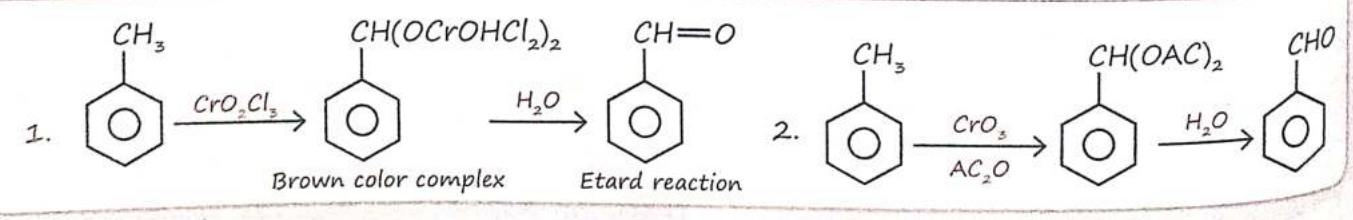
Reaction



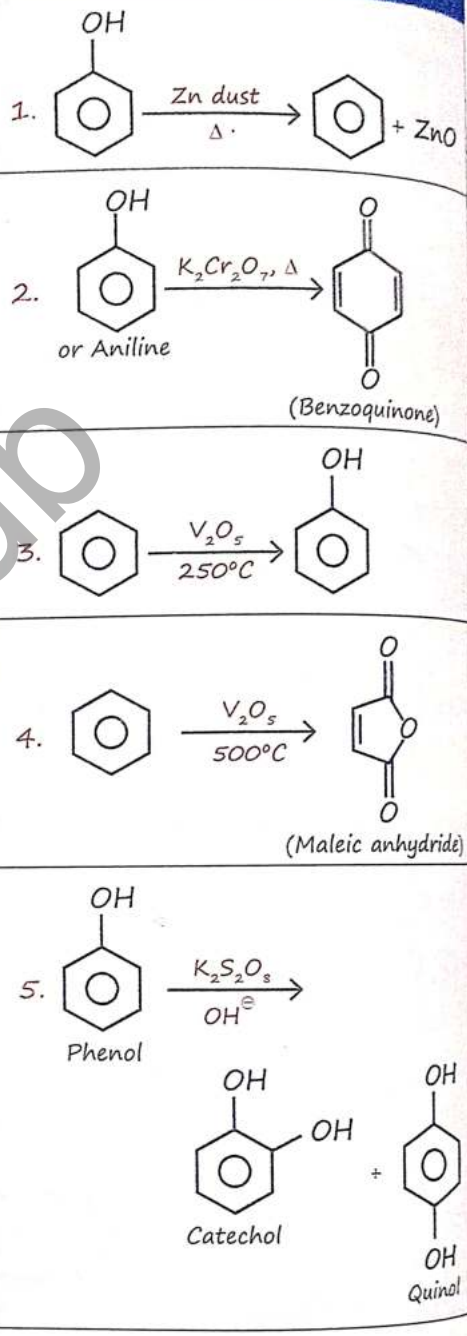
Etard Reaction

Toluene is oxidized using CrO_3 and Ac_2O to form a chromium complex, which upon hydrolysis gives benzaldehyde. This is a selective oxidation at the benzylic position.

Reaction



Some Different Reaction



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Diazotization

Formation of Diazonium Salt

Primary aromatic amines react with $NaNO_2$ and HCl to form diazonium salts through nitrosation steps. The diazonium ion is stabilized by resonance and is key for further substitutions.

1° Amine

Reaction

$$R-NH_2 \xrightarrow{NaNO_2/HCl} R-OH + N_2O\uparrow$$

Mechanism

1. $R-NH_2 \xrightarrow[\oplus NO]{NaNO_2/HCl} R-\overset{\oplus}{N}H_2-N=O$
2. $R-\overset{\oplus}{N}H_2-N=O \longrightarrow R-NH-N=O$
3. $R-NH-N=O \rightleftharpoons R-N=N-OH$
4. $R-N=N-OH \xrightarrow{HCl} R-N=\overset{\oplus}{N} \leftrightarrow R-\overset{\oplus}{N}\equiv N$
5. $R-\overset{\oplus}{N}\equiv N \xrightarrow{rds} R^{\oplus} + N_2\uparrow$
6. $R^{\oplus}$ + $Cl^{\ominus} \rightarrow R-Cl$; $H_2O \rightarrow R-OH$ (major); $O^{\ominus}-N=O \rightarrow R-ONO$; $:N(=O)-O^{\ominus} \rightarrow R-NO_2$; $ROH \rightarrow R-O-R$

2° Amine

Nitrosation of secondary amines forms stable N-nitroso compounds (often carcinogenic).

Reaction

$$R_2NH \xrightarrow{NaNO_2/HCl} R_2N-N=O$$
(Yellow Oily layer)

Mechanism

1. $R_2NH \xrightarrow[\oplus NO]{NaNO_2/HCl} R_2\overset{\oplus}{N}H-N=O$
2. $R_2\overset{\oplus}{N}H-N=O \longrightarrow R_2N-N=O$
(N-Nitroso amine)

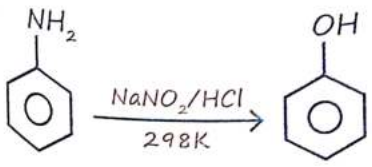
3° Amine

R_3N does not react with $NaNO_2/HCl$ – no diazotization or nitroso compound formation.

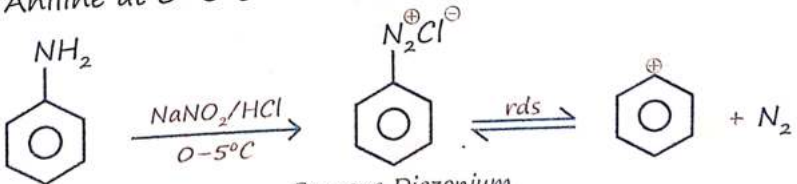
Reaction

$$R_3N \xrightarrow{NaNO_2/HCl} \text{No reaction}$$

- Aniline at high temperature.



- Aniline at 0–5°C



Benzene Diazonium Chloride (BDC)

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Baba Tea-Stall (BTS)

(i) Carbocation is obtained as intermediate

(ii) Rearrangement may be possible

(iii) Nitrosonium ion ($\overset{\oplus}{N}O$) acts as Electrophile

(iv) HNO_2 acts as base while HCl acts as acid

(v) Quantitative analysis of N_2 is used in estimation of amino acid & protein

(vi) Reaction is used to distinct 1°, 2°, 3° aliphatic amines

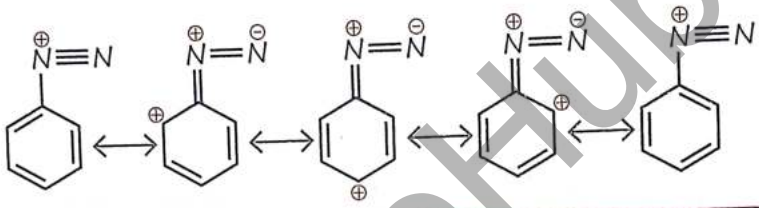
- 1° amine ⟶ N_2 gas evolved
- 2° amine ⟶ Yellow oily layer (N—nitroso amine)

(vii) Sometime Reaction carried out with tilden reagent (NOCl) and R—Cl is obtained as major product.

(viii) Alkyl diazonium salts are unstable at 0–5°C.

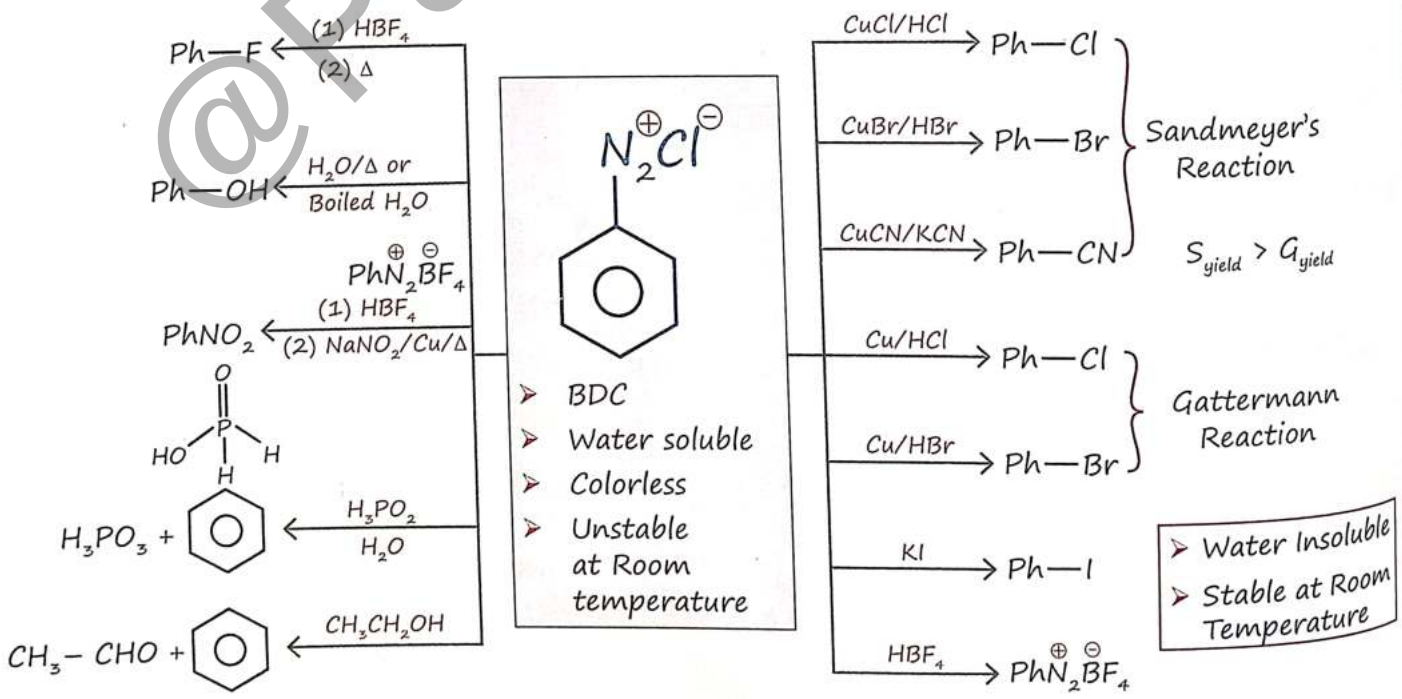
(ix) Primary aromatic amines gives stable salt at 0–5°C so they can be used in azo dye test

(x) Due to high resonance, BDC shows colour in azo dye test.



Unimolecular Aromatic Nucleophile Substitution (S_NIAr)

Diazonium salts react with nucleophiles like Cl^-, OH^-, and Cu^+ to form haloarenes, phenols, and nitroarenes respectively. These are useful for aromatic substitution.



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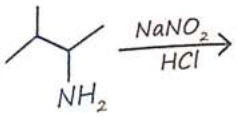
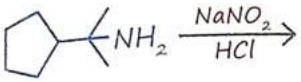
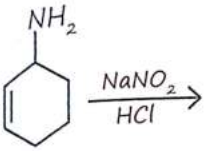
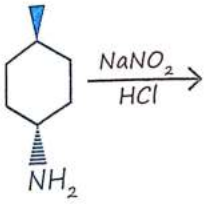
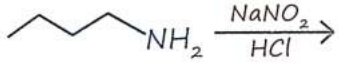
Practice Questions

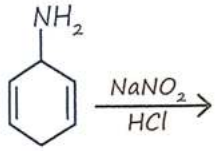
747. 

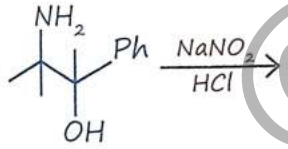
748. 

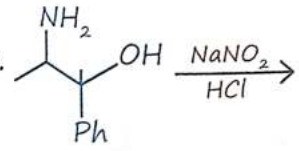
749. 

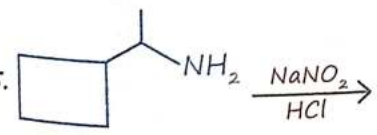
750. 

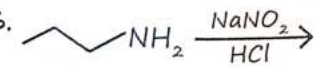
751. 

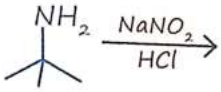
752. 

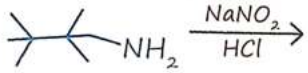
753. 

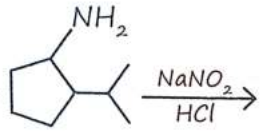
754. 

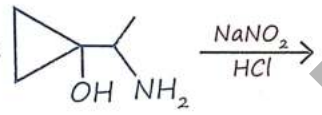
755. 

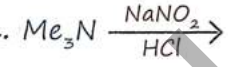
756. 

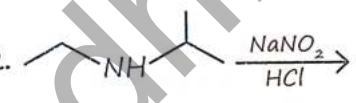
757. 

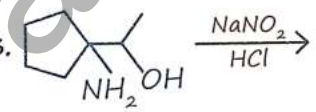
758. 

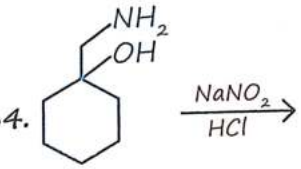
759. 

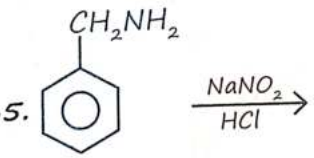
760. 

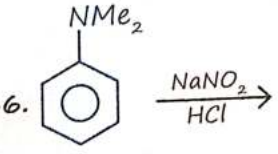
761. 

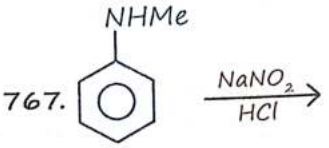
762. 

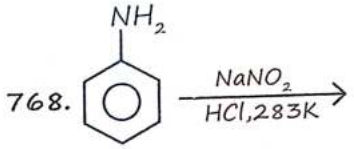
763. 

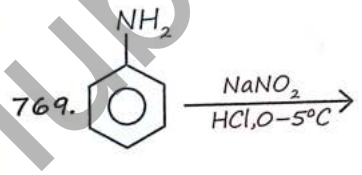
764. 

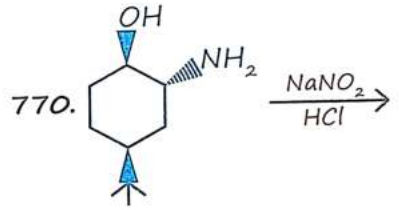
765. 

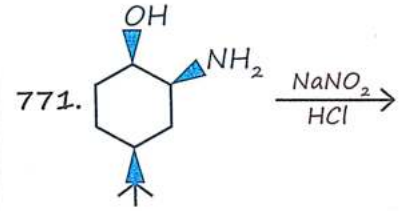
766. 

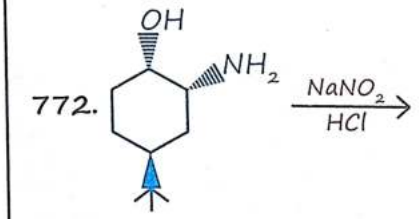
767. 

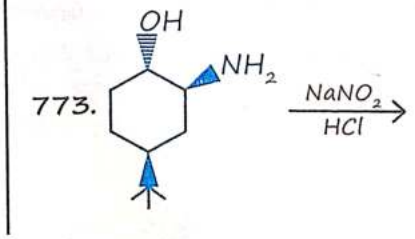
768. 

769. 

770. 

771. 

772. 

773. 

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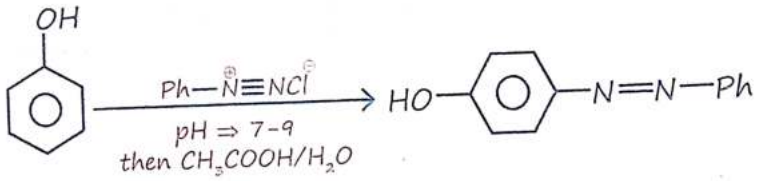
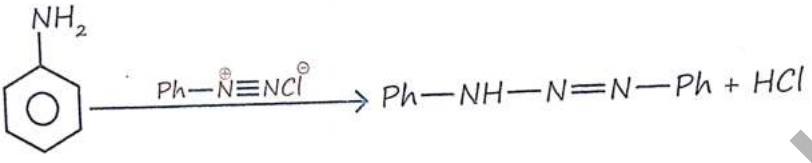
Coupling Reaction

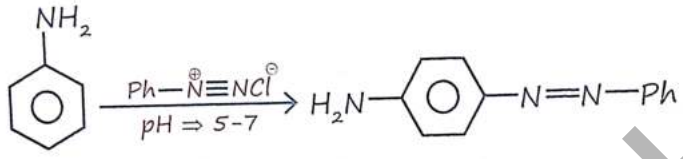
Diazonium salts undergo electrophilic substitution with activated aromatic rings to form azo compounds. This is used in dye chemistry.

(a) 

- Orange colour dye
- Antibiotic
- C—N Coupling

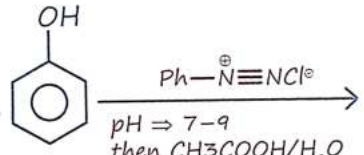
(b) 

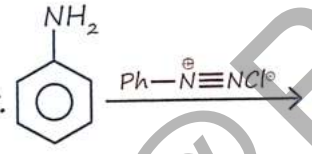
- N—N coupling

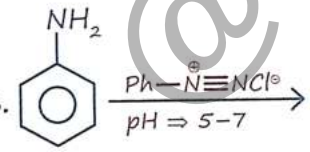
(c) 

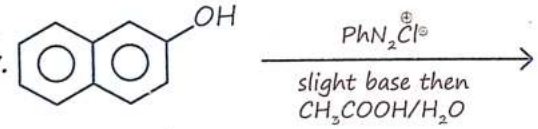
- Yellow colour dye
- C—N coupling

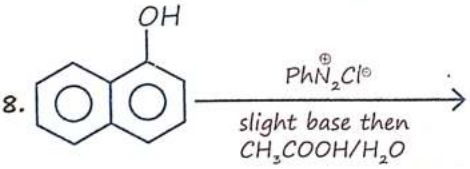
Practice Questions

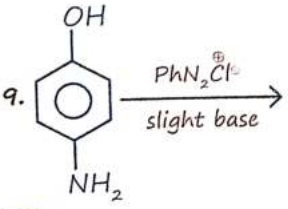
774. 

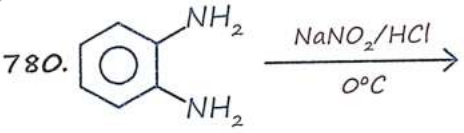
775. 

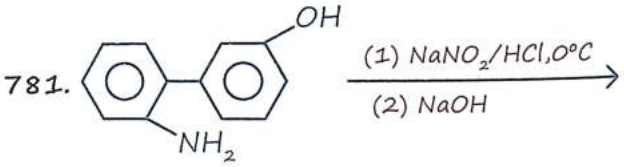
776. 

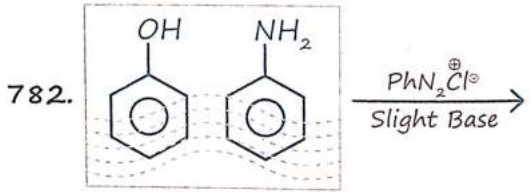
777. 

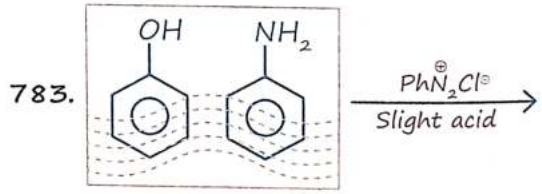
778. 

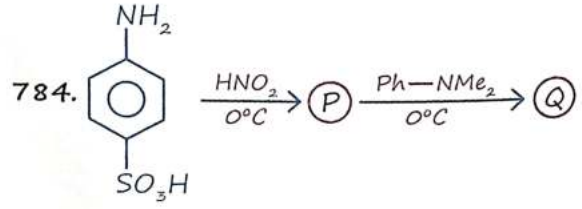
779. 

780. 

781. 

782. 

783. 

784. 

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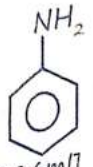
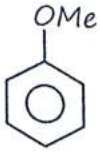
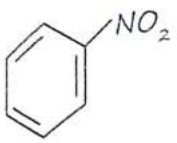
785. 
[18.6ml]
[d = 1 gm/ml]

$\xrightarrow[(2)\ Ph-NMe_2]{(1)\ NaNO_2/HCl, 0°C}$ (P) [x gm]
[20%]

786. 
$\xrightarrow{PhN_2\overset{\oplus}{Cl}^{\ominus}}$

Previous Year Questions

58. Consider the following sequence of reactions:

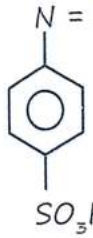


(i) Sn + HCl
(ii) $NaNO_2$, HCl 0°C
(iii) Cu_2Cl_2
(iv) Na, Ether
→ A Product

Molar mass of the product formed (A) is _____ g mol^{-1}. [22 Jan, 2025 (Shift-I)]

Sol. [154]

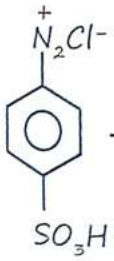
59. Choose the correct colour of the product for the following reaction.
[24 Jan, 2023 (Shift-II)]


+1-Naphthyl amine →

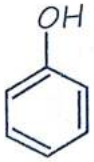
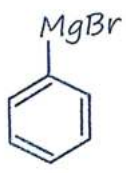
(a) Yellow (b) White
(c) Red (d) Blue

Sol. (c)

60. Identify the product in the following reaction: (2023)

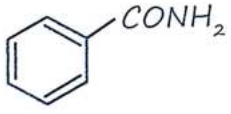
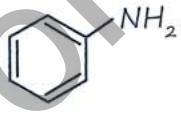
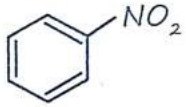


(i) Cu_2Br_2/HBr
(ii) Mg/dry ether
(iii) H_2O
→ Prodcut

(a)
(b) 
(c)
(d) 

Sol. (c)

61. A given nitrogen-containing aromatic compound A reacts with Sn/HCl, followed by HNO_2 to give an unstable compound B. B, on treatment with phenol, forms a beautiful coloured compound C with the molecular formula $C_{12}H_{10}N_2O$. The structure of compound A is: (2016-II)

(a)
(b) 
(c) 
(d) 

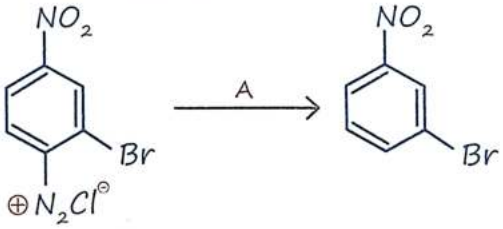
Sol. (d)

62. Which of the following will be most stable diazonium salt $RN^{2+}X^-$? (2014)

(a) $C_6H_5N_2^+X^-$ (b) $CH_3CH_2N_2^+X^-$
(c) $C_6H_5CH_2N_2^+H^-$ (d) $CH_3N_2^+X^-$

Sol. (a)

63. In the reaction: (2013)


; A is-

(a) $HgSO_4/H_2SO_4$ (b) Cu_2Cl_2
(c) H_3PO_2 and H_2O (d) H^+/H_2O

Sol. (c)

64. Choose the correct sequence of reagents in the conversion of 4-nitrotoluene to 2-bromotoluene. (2023-Manipur)

(a) $NaNO_2/HCl$; Sn/HCl; Br_2; H_2O/H_3PO_2
(b) Sn/HCl; $NaNO_2/HCl$; Br_2; H_2O/H_3PO_2
(c) Br_2; Sn/HCl; $NaNO_2/HCl$; H_2O/H_3PO_2
(d) Sn/HCl; Br_2; $NaNO_2/HCl$; H_2O/H_3PO_2

Sol. (c)

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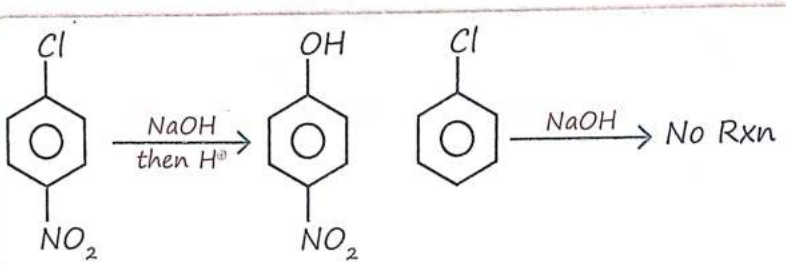
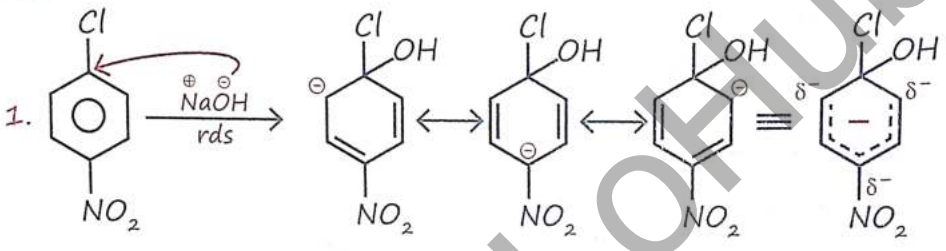
Bimolecular Nucleophilic Aromatic Substitution (S_N2Ar)

In this reaction, an electron-withdrawing group (like NO_2) activates an aromatic ring for substitution by a nucleophile (e.g., OH^- or NH_2^-) at a position ortho or para to it. The mechanism involves the formation of a Meisenheimer complex, followed by loss of a leaving group like halide.

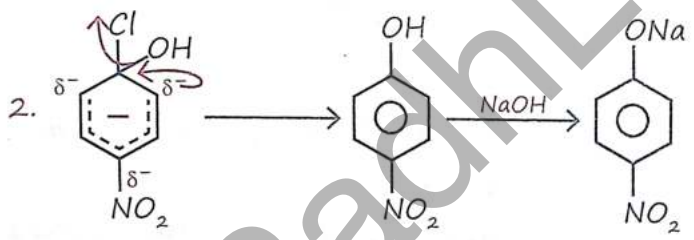
Reaction



Mechanism



m-Complex Carbanion



Baba Tea-Stall (BTS)

(i) Carbonion Intermediate (Meisenheimer-complex)

(ii) Formation of m-complex is RDS

(iii) Rate of reaction ∝ stability of m-complex (anion)

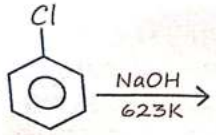
(iv) Stability ∝ EWG at ortho / para position ∝ EWG of leaving group

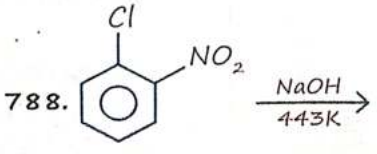
(v) EWG at O/P position will increase the rate of reaction

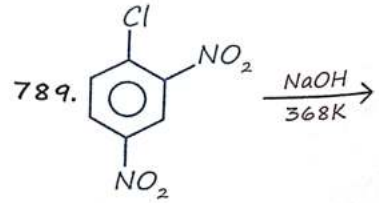
(vi) −I effect of leaving group will increase the rate

(Ar—F > Ar—Cl > Ar—Br > Ar—I)

Practice Questions

787. 

788. 

789. 

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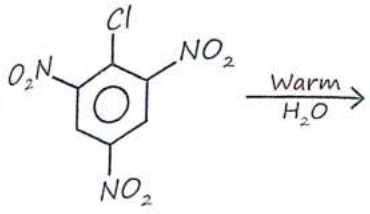
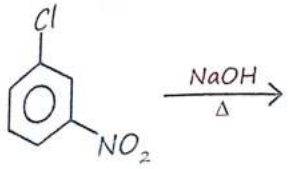
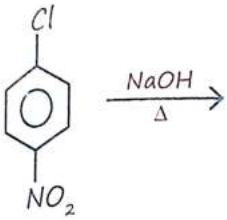
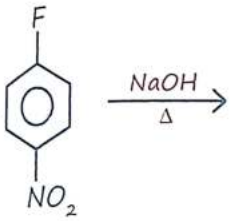
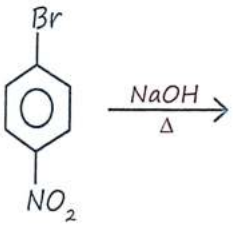
790. 

791. 

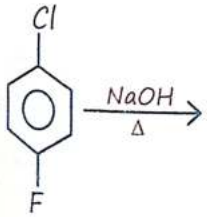
792. Rate of 787–791.

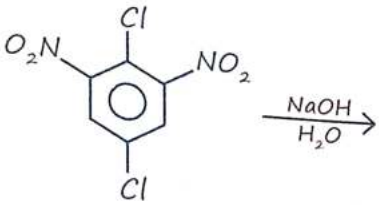
793. 

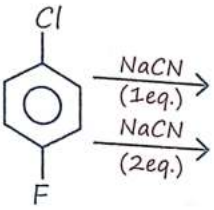
794. 

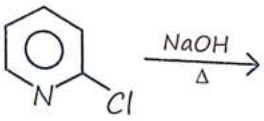
795. 

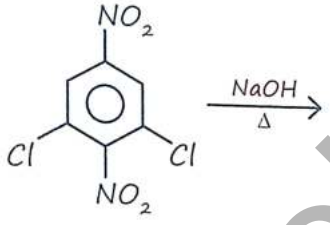
796. Rate of 793–795.

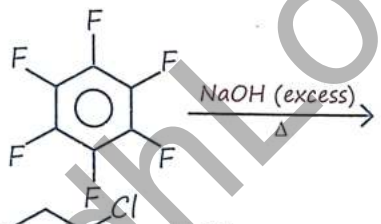
797. 

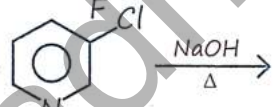
798. 

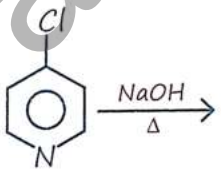
799. 

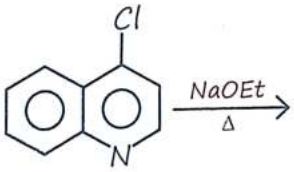
800. 

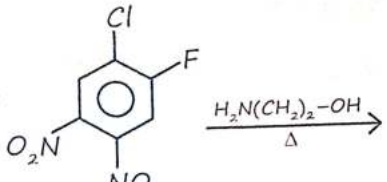
801. 

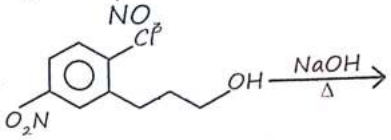
802. 

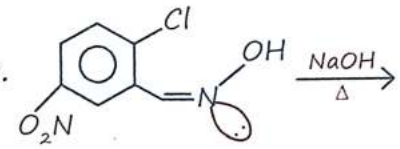
803. 

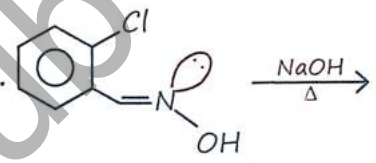
804. 

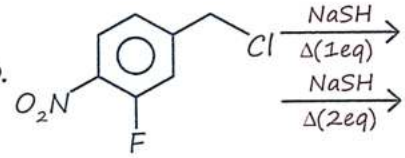
805. 

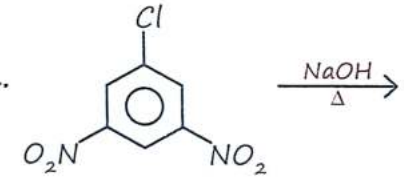
806. 

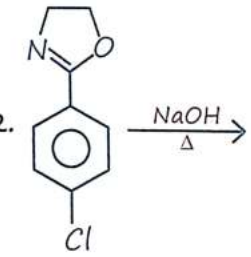
807. 

808. 

809. 

810. 

811. 

812. 

Previous Year Questions

65. Aryl halides are less reactive than alkyl halides towards nucleophilic reagents. Give a reason. (IIT JEE 1994)

Sol. Aryl halides are less reactive towards nucleophilic substitution reaction as compared to alkyl halides because of resonance stabilization in aryl halides. Due to resonance, the C–Cl bond becomes shorter and stronger and cannot be easily replaced by nucleophiles.

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5 Haloalkanes and Haloarenes

Introduction of Haloalkanes and Haloarenes

Haloalkanes: Alkanes that have halogen atoms attached to the sp^3 hybridized carbon of an alkyl halide.

Haloarenes: Aromatic hydrocarbons that have halogen atoms attached to the sp^2 hybridized carbon of an aryl halide.

Classification

On the basis of no. of Halogen Atoms

- Monohalogen compounds : $-CH_3 - X$
- Dihalogen compounds :

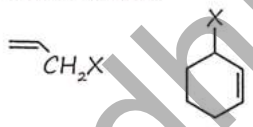
$$\begin{array}{l} CH_2\text{–}X \\ | \\ CH_2\text{–}X \end{array}$$

- Polyhalogen compounds :

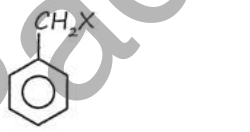
$$\begin{array}{lll} CH_2 - & CH - & CH_2 \\ | & | & | \\ X & X & X \end{array}$$

Compounds containing sp^3 C – X bond

- Alkyl Halides : R – X
- Allylic Halides:

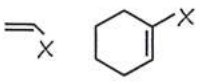


- Benzylic Halides:

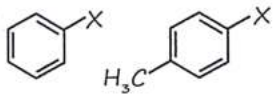


Compounds containing sp^2 C – X bond

- Vinylic Halides :



- Aryl Halides :



Reactions by Haloalkane

Substitution Reaction	When a group /atom replace by another group /atom: $R - X + Nu^{\ominus} \longrightarrow R\text{–}Nu + X^{\ominus}$ (S_N) (i) S_N2 (ii) S_N1 (iii) S_Ni (iv) S_NNGP (v) S_NAE (vi) S_NEA (vii) S_N1Ar (viii) S_N2Ar
Elimination Reaction	(i) E_1 (ii) E_2 (iii) E_1Cb (iv) E_i
Reaction with Metal	(i) Grignard formation: $R\text{–}X + Mg \longrightarrow \overset{\delta-}{R} - \overset{\delta+}{MgX}$ (ii) Wurtz reaction: $2R - X + 2Na \longrightarrow R - R + 2NaX$

One tiny halogen atom can flip an entire molecule's reactivity. That's power in one substitution!

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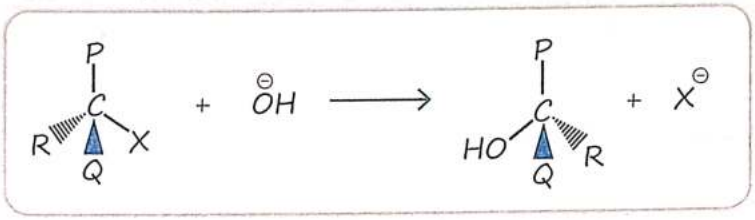
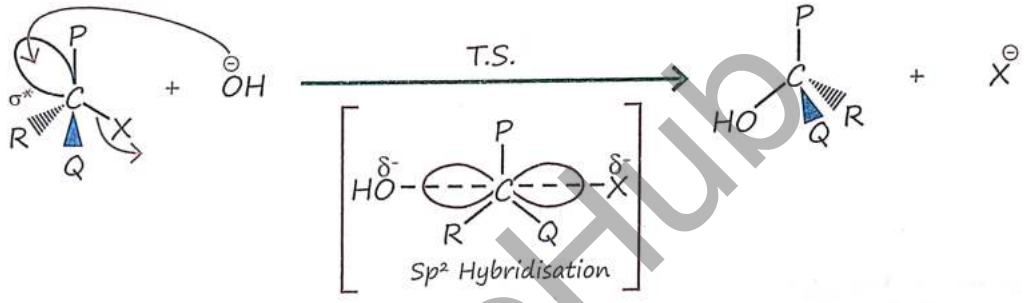
Bimolecular Nucleophilic Substitution (S_N2)

In S_N2, the nucleophile attacks the electrophilic carbon from the backside in a single concerted step, displacing the leaving group. This leads to inversion of configuration (Walden inversion) and is favored in primary alkyl halides.

Reaction



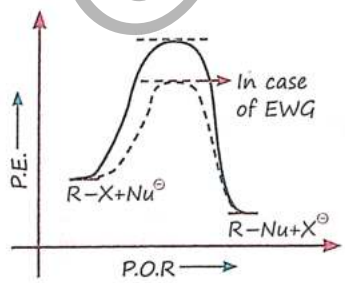
Mechanism

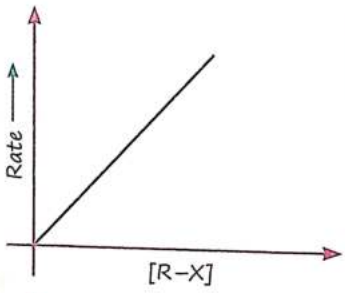


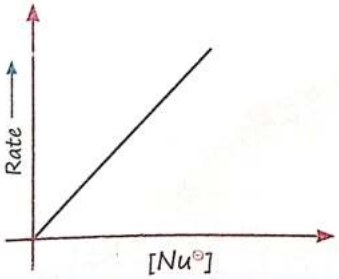
Baba Tea-Stall (BTS)

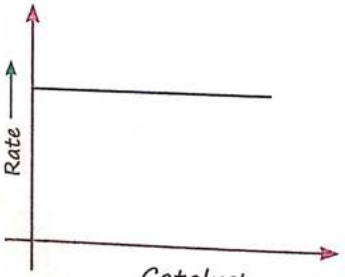
(i) Single step process
(ii) Via T.S. So no carbocation and no rearrangement
(iii) In Pentavalent T.S.
(iv) Highly unstable T.S.
(v) T.S. involves $3C-4e^{\ominus}$ bond
(vi) sp^2 hybrid T.S.
(vii) Anion like character in T.S.
(viii) EWG will stabilise the T.S. So rate will increase
(IX) No kinetic isotopic effect observed.
(x) Elemental effect observed.
$(R-\overset{35}{Cl} > R-\overset{37}{Cl})$
(xi) In Walden inversion or umbrella inversion or bond inversion occurs.
(xii) Bimolecular reaction
(xiii) Rate law $\Rightarrow r = K[R-X]^1 [Nu^{\ominus}]^1$
(xiv) Second order reaction

Graphs:









"Me trying to explain Walden inversion to friends: 'Imagine holding an umbrella in a storm...'"

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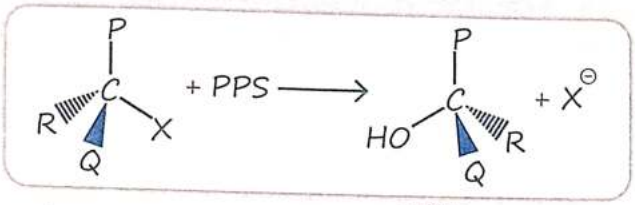
(Unimolecular nucleophilic substitution)

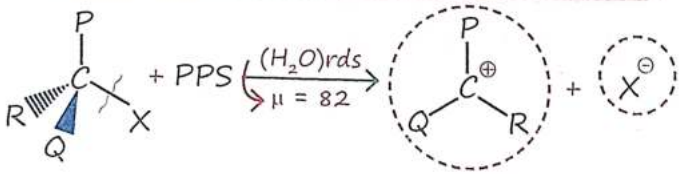
The S_N1 reaction proceeds in two steps: first, the leaving group departs, forming a carbocation intermediate; then the nucleophile attacks. It is common with tertiary alkyl halides and leads to racemization due to the planar nature of the carbocation.

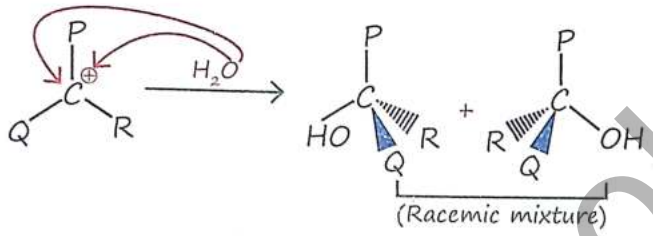
Reaction



"Tertiary alkyl halide: I break up with my LG first, then party with the nucleophile."

Mechanism

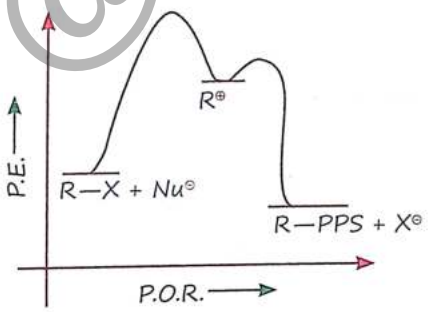
1. 

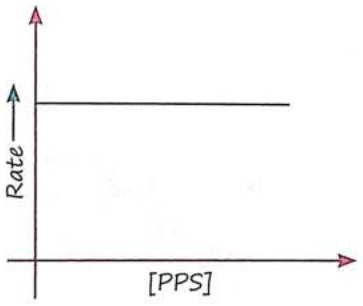
2. 

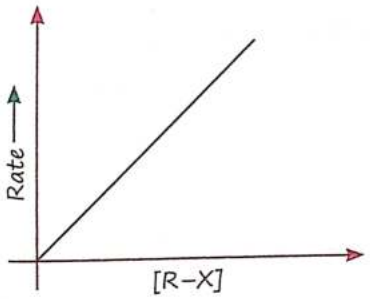
- Solvation energy
- Solvolysis (PPS)
- Hydrolysis (H_2O)

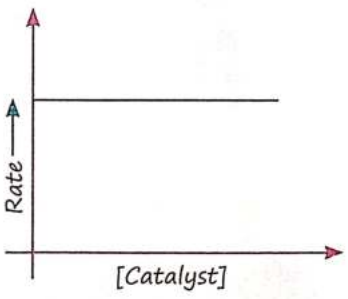
Baba Tea-Stall (BTS)

(i) Two step process
(ii) Carbocation obtained as intermediate
(iii) Rearrangement possible
(iv) Formation of CC is rds.
(v) Unimolecular reaction
(vi) Racemic mixture obtained
(vii) Both retention & Inversion occurs.
(viii) No kinetic Isotopic effect Observed.
(ix) Dielectric constant (Polarity) increases the rate.
(x) Branching will help the dissociation.
(xi) Elemental effect observed.
$(R-\overset{35}{Cl} > R-\overset{37}{Cl})$
(xii) Rate law = $k\,[R-X]^1\,[\cancel{PPS}]^{excess}$
(xiii) Pseudo first order reaction
(xiv) Graph:









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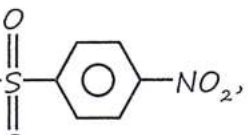
Factors of S_N1 and S_N2

1. Structure of substrate: Draw T.S.

- S_N1 is favored by tertiary (3°) substrates because they form stable carbocations. Secondary (2°) can also react; primary (1°) generally do not.
- S_N2 prefers primary (1°) substrates due to less steric hindrance, while tertiary (3°) substrates are usually unreactive in S_N2.

2. Nature of leaving group

A good leaving group is essential for both S_N1 and S_N2. Groups like I^-, Br^-, Cl^-, and tosylate leave easily because they are stable after departure. Poor leaving groups (like OH^- or NH_2^-) hinder the reaction unless converted into better ones (e.g., water in acidic medium).

Good	Moderate	Poor
$-\overset{\oplus}{O}H_2$, $-\overset{\oplus}{N}_2$, $-I$, $-O-\underset{\underset{O}{\|}}{\overset{\overset{O}{\|}}{S}}-$  , $-NO_2$, $-O-\underset{\underset{O}{\|}}{\overset{\overset{O}{\|}}{S}}-CF_3$, $-OTf$	$-Cl$, $-Br$, $-OAc$, $-OBs$, $-OTS$, $-O-\underset{\underset{O}{\|}}{\overset{\overset{O}{\|}}{S}}-Me$, $-OMs$	$-F$, $-OH$, $-OR$, $-\overset{\ominus}{N}R_3$, $-\overset{\oplus}{S}R_2$

3. Solvent Effect

- S_N1 reactions are favored by polar protic solvents (like water, alcohol) because they stabilize the carbocation and the leaving group.
- S_N2 reactions are favored by polar aprotic solvents (like acetone, DMSO, DMF) as they don't solvate the nucleophile, keeping it highly reactive for backside attack.

"Protic = Promotes S_N1"
"Aprotic = Accelerates S_N2"

(a) **S_N2 reaction:** Generally PAS is preferred

(b) **S_N1 reaction:** PPS is used

4. Nucleophile Strength

- S_N2 requires a strong nucleophile (like OH^-, CN^-, I^-) since it participates in the rate-determining step.
- S_N1 is almost independent of nucleophile strength because the nucleophile attacks after the slow carbocation formation step.

(a) **S_N2 Reaction:** Good $Nu^\ominus$ required

High $[Nu^\ominus]$

(b) **S_N1 reaction:** Poor $Nu^\ominus$

Not depend on $[Nu^\ominus]$

5. Branching in Substrate

- S_N1 is favored by more branching (especially tertiary carbon) because it stabilizes the carbocation intermediate through hyperconjugation and inductive effects.
- S_N2 is hindered by branching as steric bulk around the electrophilic carbon blocks the nucleophile's backside attack.

(a) **S_N2 Reaction:** Branching ↑⇒ rate↓ [...]

(b) **S_N1 reaction:** Branching ↑⇒ rate↑

S_N1: "The more branching, the better I feel."
Carbocation: "It's my support system!"
S_N2 trying a 3° carbon: "Knock knock—"
Branches: "No room. Try the methyl next door."

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Comparison of S_N1 and S_N2

	S_N1	S_N2
Characteristics of mechanism	Two elementary steps: Step 1: $R\!-\!\ddot{X}: \rightleftharpoons R^{\oplus} + :\ddot{X}:^{\ominus}$ Step 2: $R^{\oplus} + :Nu^{\ominus} \rightarrow R - Nu$ Ionisation of alkyl halide (step 1) is rate-determining	Single step: $^{\ominus}Nu \rightarrow R\!-\!\ddot{X}: \rightarrow Nu\!-\!R + :\ddot{X}:^{\ominus}$ Nucleophile displaces leaving group; bonding to the incoming nucleophile accompanies cleavage of the bond to the leaving group.
Rate-determining transition state	$^{\delta+}R \cdots \ddot{X}:^{\delta-}$	$^{\delta-}Nu \cdots R \cdots \ddot{X}:^{\delta-}$
Molecularity	Unimolecular	Bimolecular
Kinetics and rate law	First order: Rate = k [alkyl halide]	Second order: Rate = k [alkyl halide] [nucleophile]
Relative reactivity of halide leaving groups	RI > RBr > RCl >> RF	RI > RBr > RCl >> RF
Effect of structure on rate	$R_3CX > R_2CHX > RCH_2X > CH_3X$ Rate is governed by stability of carbocation that is formed in ionization step. Tertiary alkyl halides can react only by the S_N1 mechanism; they never react by the S_N2 mechanism.	$CH_3X > RCH_2X > R_2CHX > R_3CX$ Rate is governed by steric effects (crowding) in transition state). Methyl and primary alkyl halides can react only by the S_N2 mechanism: they never react by the S_N1 mechanism.
Effect of nucleophile on rate	Rate of substitution is independent of both concentration and nature of nucleophile. Nucleophile does not participate until after rate-determining step.	Rate depends on both nature of nucleophile and its concentration.
Effect of solvent on rate	Rate Increases with increasing polarity of solvent as measured by its dielectric constant e.	Polar aprotic solvents give fastest rates of substitution, solvation of Nu is minimal and nucleophilicity is greatest.
Stereochemistry	Not stereospecific racemization accompanies inversion when leaving group is located at a chirality center.	Stereospecific: 100% inversion of configuration at reaction site. Nucleophile attacks carbon from side opposite bond to leaving group.
Potential for rearrangements	Carbocation intermediate capable of rearrangement.	No carbocation intermediate; no rearrangement

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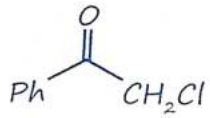
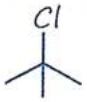
Try Yourself

Q. Draw transition state of the following substrate:

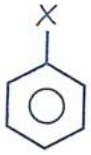
(a)  (b) CH_3-Cl (c) (d)  (e) $Me-O-CH_2Cl$

Q. Compare the leaving group tendency.

(a) $I^{\ominus}$, $Br^{\ominus}$, $Cl^{\ominus}$, $F^{\ominus}$ (b) $CH_3COO^{\ominus}$, $(AcO)^{\ominus}$, $CH_3SO_3^{\ominus}$, $CF_3SO_3^{\ominus}$, $(TfO^{\ominus})$

(c) $TsO^{\ominus}$, $NsO^{\ominus}$, $BsO^{\ominus}$, $PhSO_3^{\ominus}$ (d) O_2, N_2 (e) $Cl^{\ominus}$, $\overset{\ominus}{O}H$,

Q. Compare the rate of reaction w.r.t S_N1 and S_N2?

(a)  (b) (c) (c) Neopentyl halide

Practice Questions

1. Draw transition state and compare the rate on PPS/PAS?

(a) $CH_3 - I + NaOH \longrightarrow$

(b) $CH_3 - \overset{\oplus}{N}Me_3 + NaOH \longrightarrow$

(c) $CH_3 - I + NH_3 \longrightarrow$

(d) $CH_3 - \overset{\oplus}{S}Me_3 + NH_3 \longrightarrow$

2. Compare the Nucleophilicity?

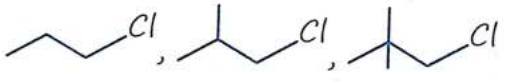
(a) $\overset{\ominus}{O}H$ $\overset{\ominus}{C}N$ $(CH_3)_3C-O^{\ominus}$

(b) $I^{\ominus}$ $Br^{\ominus}$ $Cl^{\ominus}$ $F^{\ominus}$

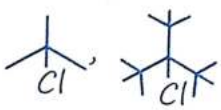
(c) $\overset{\ominus}{C}H_3$ $\overset{\ominus}{N}H_2$ $\overset{\ominus}{O}H$ $F^{\ominus}$

(d) NH_2-NH_2, NH_3

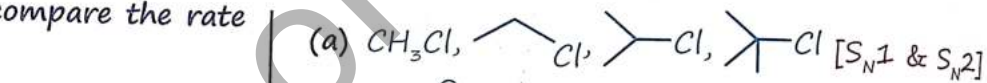
3. Compare the rate of S_N1 and S_N2:

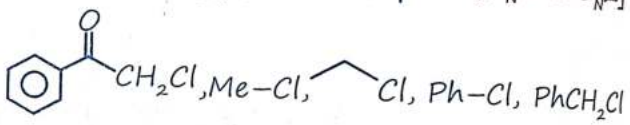
(a) 

(b)

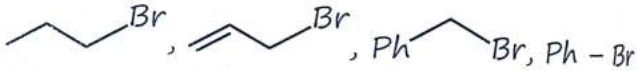
(c) 

4. For following compounds, compare rate of given reaction?

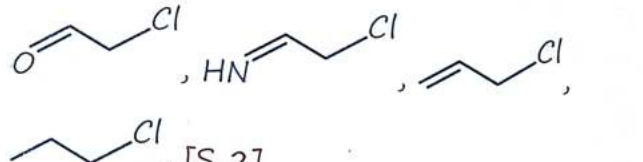
(a) CH_3Cl,  $[S_N1$ & $S_N2]$

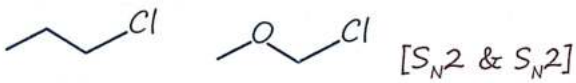
(b)  $[S_N2]$

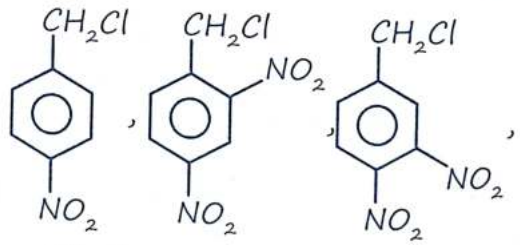
(c) CH_3Br CD_3Br $[S_N2]$

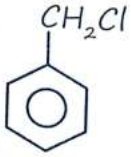
(d)  $[S_N1$ & $S_N2]$

(e) $CH_3-CH_3-\overset{37}{Cl}$ $[S_N2]$

(f)  $[S_N2]$

(g)  $[S_N2$ & $S_N2]$

(h) 

 $[S_N2]$

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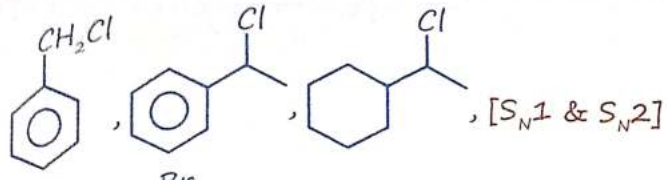
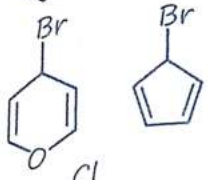
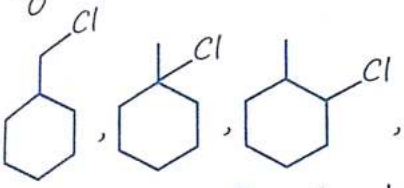
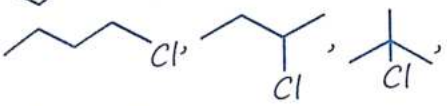
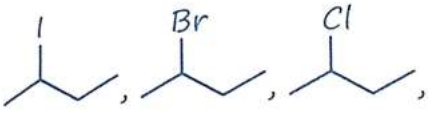
(i)  , [S_N1 & S_N2]

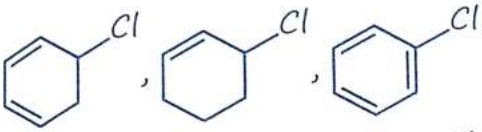
(j)  [S_N2]

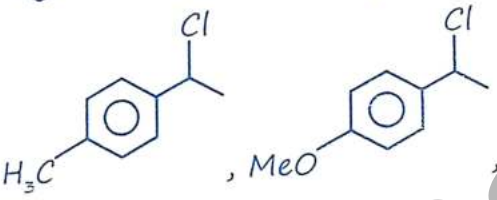
(k)  , [S_N1 & S_N2]

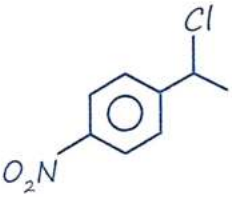
(l) 

[S_N1 & S_N2]

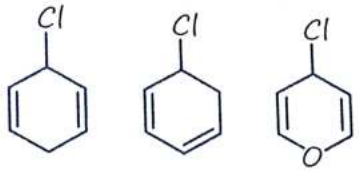
(m)  , [S_N1 & S_N2]

(n)  [S_N1 & S_N2]

(o) 

 [S_N1 & S_N2]

(p) [S_N1]

(q)  [S_N1]

(r) H_2O CH_3CH_2OH $HCONH_2$ [S_N1]

(s) NaOH NaSH [S_N2]

(t) [NaOH], [NaOH], [NaSH] [S_N2]
1M 2M 2M

(u) [H_2O] [H_2O] [S_N1]
1M 2M

(v) [$PhCH_2Cl$] [$PhCH_2Cl$] [S_N1 & S_N2]
1M 2M

(w) [S_N1]

(x) [S_N1 & S_N2]

(y) $PhCH_2Cl$, CH_3OCH_2Cl, $CH_2 = CH-CH_2Cl$
[S_N1 & S_N2]

5. In S_N1 reaction, 100% racemic mixture not obtained?

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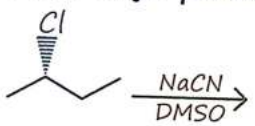
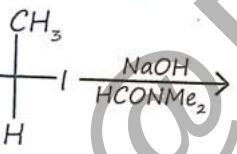
Reagent	Nucleophile ($Nu^{\ominus}$)	Substitution product R—Nu	Class of main product
NaOH (KOH)	$\overset{\ominus}{O}H$	R—OH	Alcohol
H_2O			
NaOR'			
NaI			
NH_3			
$R'NH_2$			
R'R''NH			
KCN			
AgCN			
KNO_2			
$AgNO_2$			
R'COOAg			
$LiAlH_4$			
$R'^{\ominus}M^{\oplus}$			

Practice Questions

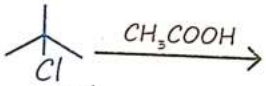
6. Draw major product of the following reactions:

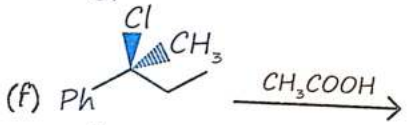
(a) 

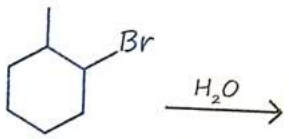
(b) $CH_3-Br \xrightarrow[DMF]{NaC\equiv CH}$

(c) 

(d)

(e) 

(f) 

(g) 

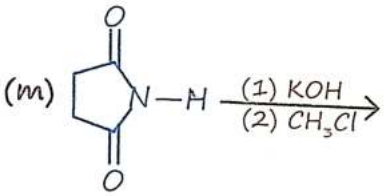
(h)

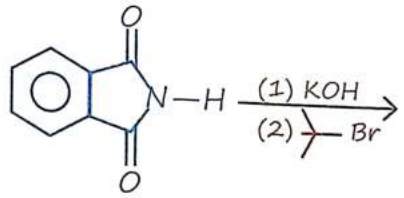
(i) $CH_3Cl + NH_3 \xrightarrow[(excess)]{}$

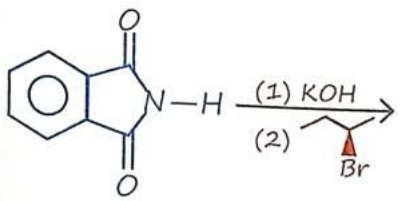
(j) $CH_3Cl + NH_3 \xrightarrow[(excess)]{}$

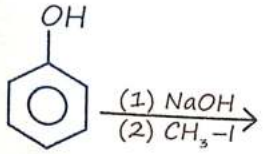
(k) $PhCH_2Cl + NH_3 \longrightarrow (P)$

(l) $(P) + 2CH_3Cl \longrightarrow$

(m) 

(n) 

(o) 

(p) 

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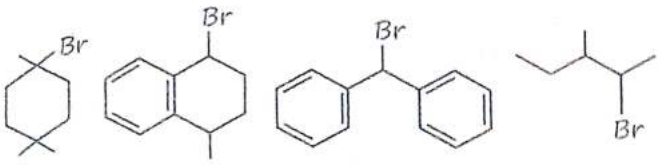
Previous Year Questions

1. The ascending order of relative rate of solvolysis of following compounds is:
[23 Jan, 2025 (Shift-II)]



(A) (B) (C) (D)

(a) (D) < (B) < (A) < (C)
(b) (D) < (A) < (B) < (C)
(c) (C) < (D) < (B) < (A)
(d) (C) < (B) < (A) < (D)

Sol. (b) Relative rate of solvolysis ∝ stability of carbocation. So, stability of carbocation will be as:

So, order of relative rate of solvolysis in ascending order is,

(D) < (A) < (B) < (C).

2. Given below are two statements:

Statement-I: The conversion proceeds well in the less polar medium.

$$CH_3-CH_2-CH_2-CH_2-Cl \xrightarrow{HO^-} CH_3-CH_2-CH_2-CH_2-OH + Cl^{(-)}$$

Statement-II: The conversion proceeds well in the more polar medium.

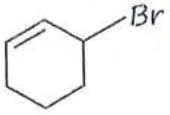
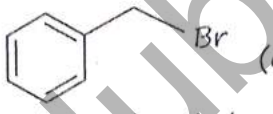
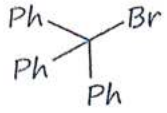
$$CH_3-CH_2-CH_2-CH_2-Cl \xrightarrow{R_3\ddot{N}} CH_3-CH_2-CH_2-CH_2-\overset{R}{\underset{R}{\overset{|}{\underset{|}{N^{(+)}}}}}-R\ \overset{(-)}{Cl}$$

In the light of the above statements, choose the correct answer from the options given below
[24 Jan, 2025 (Shift-I)]

(a) Statement-I is true but Statement-II is false
(b) Statement-I is false but Statement-II is true
(c) Both Statement-I and Statement-II are true
(d) Both Statement-I and Statement-II are false

Sol. (c) This reaction given in statement I will proceed faster in the less polar medium. The reaction given in statement II will proceed faster in the more polar medium.

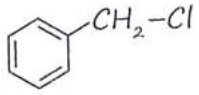
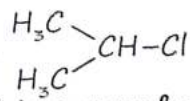
3. Which among the following halides will generate the most stable carbocation in the nucleophilic substitution reaction?
[29 Jan, 2025 (Shift-II)]

(a)
(b) 
(c) 
(d) 

Sol. (d) Triphenylmethyl carbocation ($(C_6H_5)_3C^+$) is the most stable because the positive charge on the central carbon atom is delocalized over the three phenyl rings through resonance.

4. Which among the following halide/s will not show S_N1 reaction: [27 Jan, 2024 (Shift-II)]

A. $H_2C=CH-CH_2Cl$ B. $CH_3-CH=CH-Cl$

C.  D. 

Choose the most appropriate answer from the options given below:

(a) (A), (B) and (D) only
(b) (A) and (B) only
(c) (B) and (C) only
(d) (B) only

Sol. (d) Since $CH_3-CH_3 = \overset{+}{C}H$ is highly unstable therefore, $CH_3-CH = CH-Cl$ cannot give S_N1 reaction.

5. Given below are two statements:

Statement-I: High concentration of strong nucleophilic reagent with secondary alkyl halides which do not have bulky substituents will follow S_N2 mechanism.

Statement-II: A secondary alkyl halide when treated with a large excess of ethanol follows S_N1 mechanism.

In the the light of the above statements, choose the most appropriate from the questions given below:
[30 Jan, 2024 (Shift-II)]

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(a) Statement-I is true but Statement-II is false.
(b) Statement-I is false but Statement-II is true.
(c) Both statement-I and Statement-II are false.
(d) Both statement-I and Statement-II are true.

Sol. (d) Statement-I: Rate of S_N2 depends on the conc. of $Nu^{\ominus}$ as well as conc. of reacting species.
S_N2 reaction is favoured by high concentration of nucleophile (Nu^-) & less crowding in the substrate molecule.
Statement-II: Solvolysis follows S_N1 path.
Both Statements-I & II are correct.

6. Given below are two statements:

Statement-I: S_N2 reactions are 'stereospecific', indicating that they result in the formation only one stereo-isomers as the product.

Statement-II: S_N1 reactions generally result in formation of product as racemic mixtures. In the light of the above statements, choose the correct answer from the options given below:

[08 April, 2024 (Shift-II)]

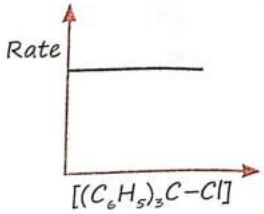
(a) Statement-I is true but Statement-II is false
(b) Statement-I is false but Statement-II is true
(c) Both Statement-I and Statement-II is true
(d) Both Statement-I and Statement-II is false

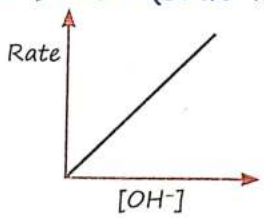
Sol. (c) S_N2 reaction lead to Inversion
S_N1 reaction lead to racemisation.

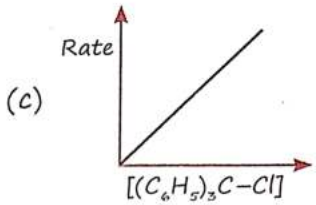
7. The graph which represents the following reaction is:

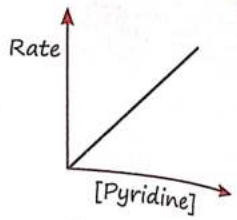
$$(C_6H_5)_3C-Cl \xrightarrow[\text{Pyridine}]{OH^-} (C_6H_5)_3C-OH$$

[1 Feb, 2023 (Shift-II)]

(a)


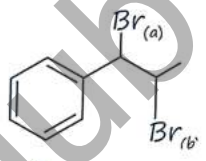
(b)


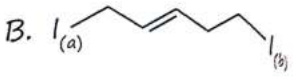
(c)


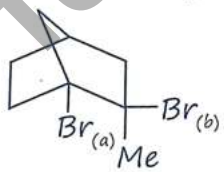
(d)


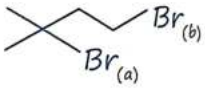
Sol. (c) It is S_N1 reaction so rate of reaction depends on the concentration of alkyl halide only.
$Rate = k[(C_6H_5)_3\ C - Cl]\ [OH^-]$
$Rate \propto [(C_6H_5)_3\ C - Cl]$

8. Choose the halogen which is most reactive towards S_N1 reaction in the given compounds(A, B, C & D) [8 April, 2023 (Shift-I)]

A.


B.


C.


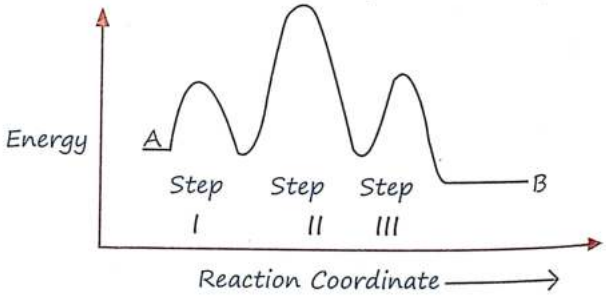
D.


(a) A-$Br_{(b)}$; B-$I_{(b)}$; C-$Br_{(b)}$; D-$Br_{(b)}$
(b) A-$Br_{(a)}$; B-$I_{(a)}$; C-$Br_{(b)}$; D-$Br_{(a)}$
(c) A-$Br_{(b)}$; B-$I_{(a)}$; C-$Br_{(a)}$; D-$Br_{(a)}$
(d) A-$Br_{(b)}$; B-$I_{(a)}$; C-$Br_{(a)}$; D-$Br_{(a)}$

Sol. (b)

9. Consider the following reaction that goes from A to B in three steps as shown below:

[6 April, 2023 (Shift-II)]



Choose the correct

	Number of intermediates	Number of Activated complexes	Rate determining step
(a)	3	2	II
(b)	2	3	II
(c)	2	3	I
(d)	2	3	III

Sol. (b)

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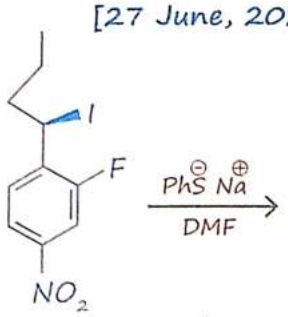
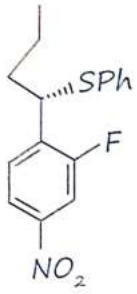
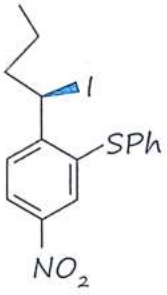
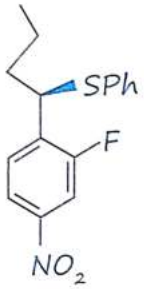
10. The major product of the following reaction is:

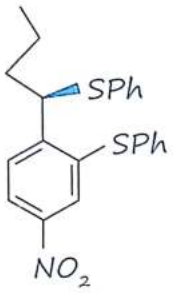
[27 June, 2022 (Shift-I)]



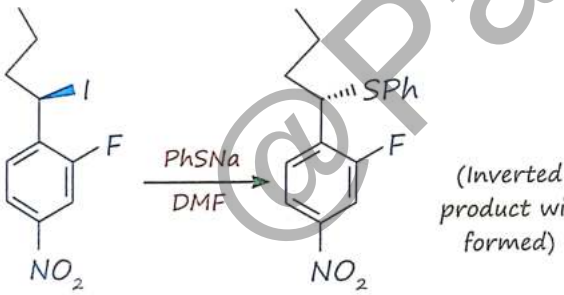
(a) 

(b) 

(c) 

(d) 

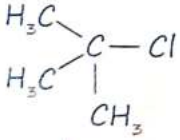
Sol. (a) The reaction will take place via S_N2, and the following inverted product will be formed.

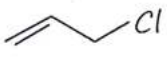


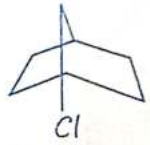
(Inverted product will formed)

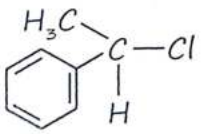
11. Which one of the following compounds is inactive towards S_N1 reaction?

[28 June, 2022 (Shift-I)]

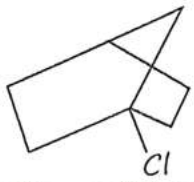
(a) 

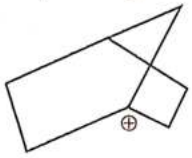
(b) 

(c) 

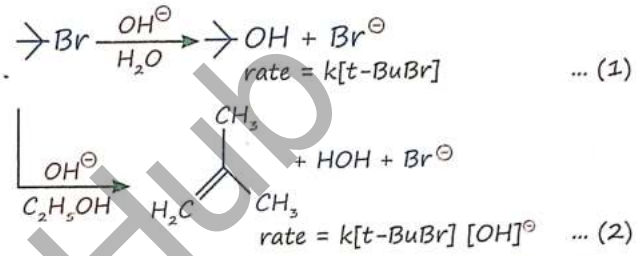
(d) 

Sol. (c) The rate of S_N1 reaction depends on the stability of carbocation.

The compound  cannot react as per S_N1 reaction as its carbocation i.e.

 is unstable and cannot be formed according to Bredt rule.

12. Consider the reaction sequence given below:



Which of the following statements is true?

(a) Changing the concentration of base will have no effect on reaction (2)

(b) Changing the concentration of base will have no effect on reaction (1)

(c) Changing the base from $OH^{\ominus}$ to $^{\ominus}OR$ will have no effect on reaction (2)

(d) Doubling the concentration of base will double the rate of both the reactions.

[2 Sept, 2020 (Shift-II)]

Sol. (b) For reaction 1

rate = k[t − BuBr]

For reaction 2

rate = k[t − BuBr] [OH^-]

Reaction 1 is independent of concentration of OH^- where as reaction 2 is dependent on concentration of OH^-. Hence changing the concentration of base will have no effect on reaction 1.

13. An 'Assertion' and a 'Reason' are given below. Choose the correct answer from the following options:

Assertion (A): Vinyl halides do not undergo nucleophilic substitution easily.

Reason (R): Even though the intermediate carbocation is stabilized by loosely held α-electrons, the cleavage is difficult because of strong bonding. [12 April, 2019 (Shift-II)]

(a) Both (A) and (R) are wrong statements.

(b) Both (A) and (R) are correct statements and (R) is the correct explanation of (A).

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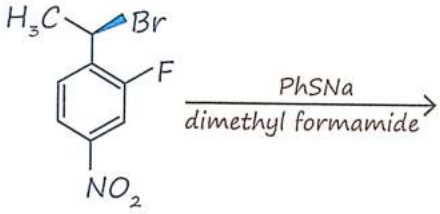
(c) Both (A) and (R) are correct statements but (R) is not the correct explanation of (A).

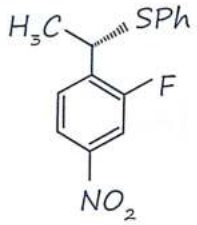
(d) (A) is a correct statement but (R) is a wrong statement.

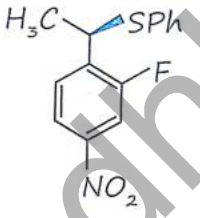
Sol. (d) Due to partial double bond character of C – Cl bond, halogen leaves with great difficulty so vinyl halide do not undergo nucleophilic substitution, assertion is correct.

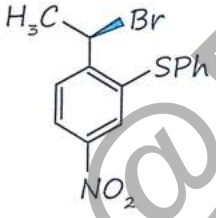
Reason: Intermediate carbocation is not stablised by loosely held π-electrons because empty orbital, being at 90° cannot overlap with p-orbitals of π bond so, reason is wrong.

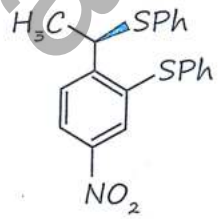
14. **The major product of the following reaction is** (IIT JEE 2008)



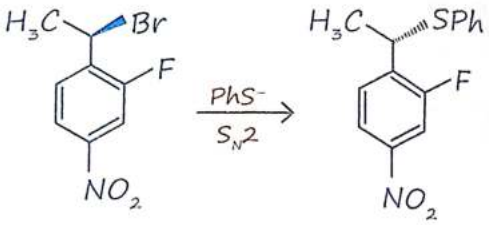
(a)


(b)


(c)


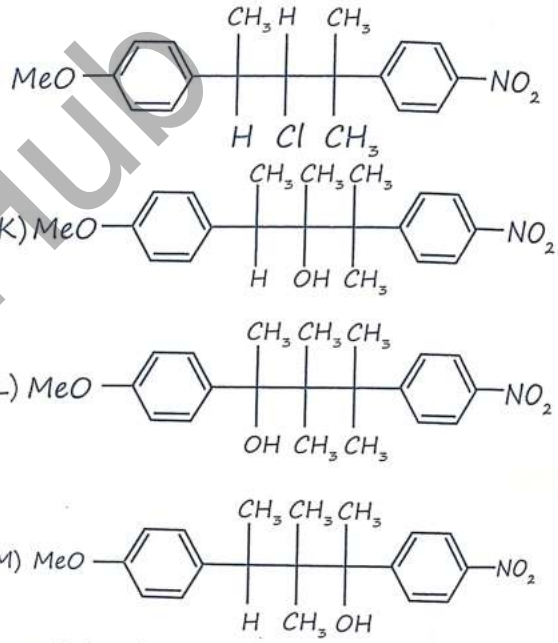
(d)


Sol. (a)



PhS^- is the nucleophile that substitutes with the Br^- through SN^2 mechanism accompained inversion of configuration at α-C.

15. **The following compound on hydrolysis in aqueous acetone will give** (IIT JEE 2005)



It mainly gives

(a) K and L (b) Only K

(c) L and M (d) Only M

Sol. (a)

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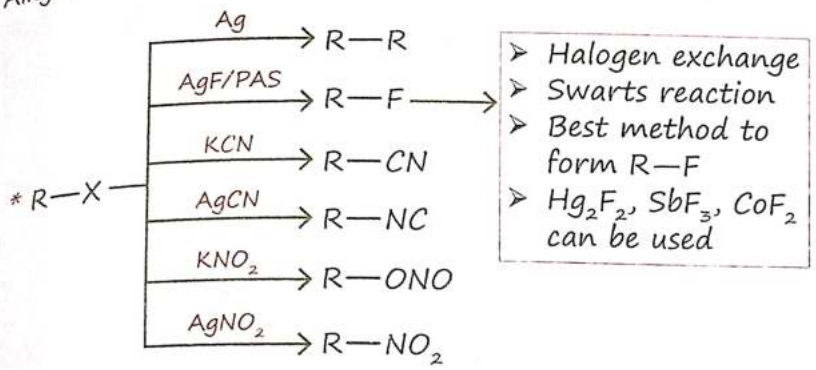
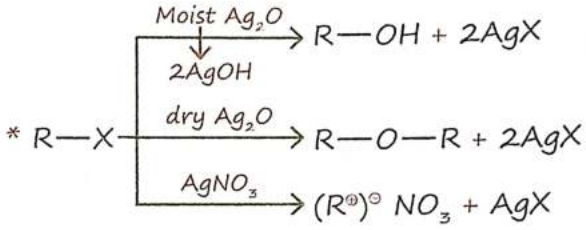
Reaction with Ag

Alkyl halides (R–X) react with silver salts to undergo nucleophilic substitution, forming various products.





Finkelstein reaction

The Finkelstein reaction is a halogen exchange reaction, where an alkyl halide (R–Cl or R–Br) is treated with NaI in acetone to form alkyl iodide (R–I). It works well because NaCl or NaBr precipitates out, driving the reaction forward.

Reaction

➢ R—X + NaI ⟶ No Rxⁿ (Cl, Br)

➢ R—X + NaI $\xrightarrow{\text{acetone}}$ R—I + NaX↓ (PPt)
(NaI: Soluble; R—I + NaX↓: Insoluble)

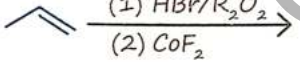
Baba Tea-Stall (BTS)

(i) Driving force of rxⁿ is formation of PPt (ii) Halogen exchange rxⁿ (iii) S_N2 mechᵐ used

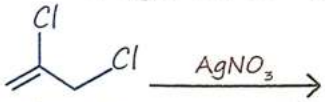
Practice Questions

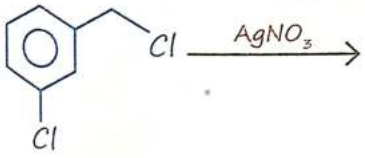
7. $CH_3CH_2Cl \xrightarrow[HCONMe_2]{Hg_2F_2}$

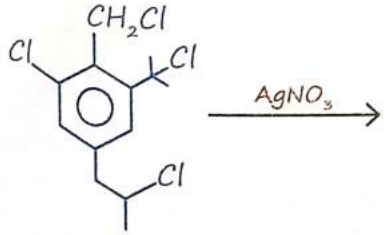
8. $CCl_4 \xrightarrow{2AgF}$

9.  $\xrightarrow[(2)\ CoF_2]{(1)\ HBr/R_2O_2}$

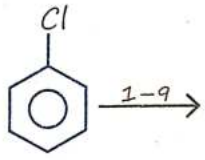
Find Mole of AgCl for (7–9)?

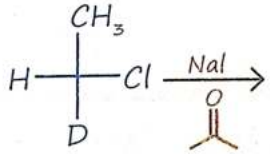
10. 

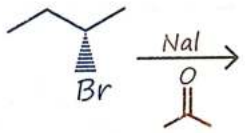
11. 

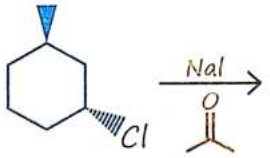
12. 

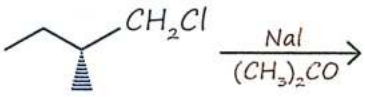
13. $CH_3Br \xrightarrow{1-9}$

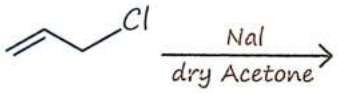
14. 

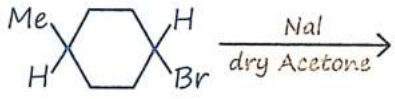
15. 

16. 

17. 

18. 

19. 

20. 

21. $CH_3Cl \xrightarrow{NaOH}$

22. $CH_3Cl \xrightarrow[NaOH]{NaI}$

23. Compare rate of 21 and 22

24. $CH_3COCl \xrightarrow[Pyridine]{NaOH}$

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Reaction of Alcohols with HX

Alcohol (R–OH) reacts with hydrogen halide (HX) in the presence of CCl_4 to form alkyl halide (R–X) and water.

Reaction

$$R-OH \xrightarrow[CCl_4]{HX} R-X + H_2O$$

Mechanism

S_N1:

1. $R-\ddot{O}H \xrightarrow{H-\ddot{X}} R-\overset{\oplus}{O}H_2 + X^{\ominus}$
2. $R-\overset{\oplus}{O}H_2 \longrightarrow R^{\oplus} + H_2O$
3. $R^{\oplus} \xrightarrow{X^{\ominus}} R-X$

S_N2:

1. $R-\ddot{O}H \xrightarrow{H-\ddot{X}} R-\overset{\oplus}{O}H_2 + X^{\ominus}$
2. $R-\overset{\oplus}{O}H_2 \xrightarrow{X^{\ominus}} R-X + H_2O$

Baba Tea-Stall (BTS)

(i) Carbocation obtained so rearrangement possible.

(ii) S_N2 in 1°—OH and S_N1 in 2°—OH, 3°—OH

$$3°-OH > 2°-OH > 1°-OH < CH_3-OH$$

(iii) Rate of reaction for HX: HI > HBr > HCl > HF

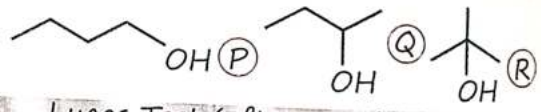
- **Different condition for HX:**
  1. HI ⇒ NaI/H_3PO_4
  2. HBr ⇒ Δ, $NaBr/H_2SO_4$ or H_3PO_4
  3. HCl ⇒ Δ, Lewis acid ($AlCl_3$, $ZnCl_2$, BF_3...), dry HCl

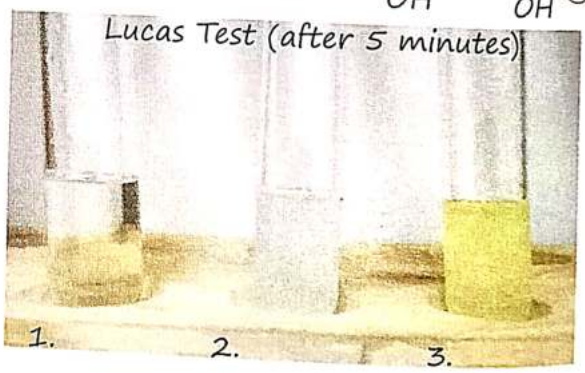
Lucas test or Groove's process

In this test, alcohol reacts with HCl in presence of $ZnCl_2$ (Lucas reagent). Formation of alkyl chloride causes turbidity. The rate of turbidity helps identify the type of alcohol (3° reacts fastest, 1° slowest).

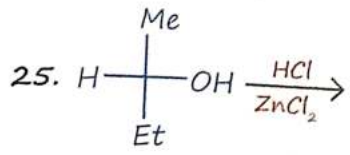
Reaction

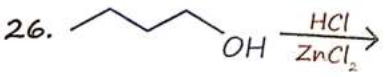
- $R-OH + HCl/ZnCl_2 \longrightarrow R-Cl\downarrow$ (White turbidity)
  - Soluble
  - Insoluble
  - $HCl/ZnCl_2$: Lucas reagent
- 3°—OH ⟶ Immediate white turb
- 2°—OH ⟶ 5 to 7 min.] S_N1
- 1°—OH ⟶ No turbidity at R.T.] S_N2

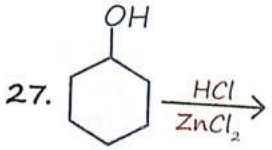
Ex: 

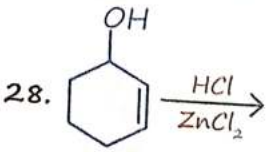


Practice Questions

25. 

26. 

27. 

28. 

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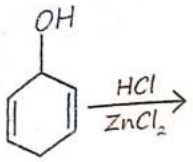
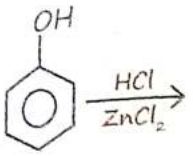
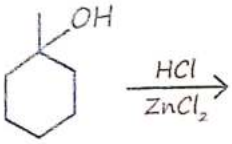
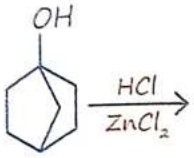
29. 

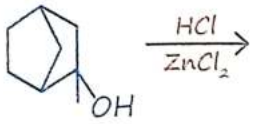
30. 

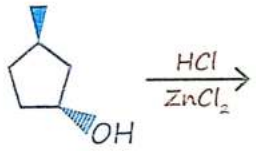
31. (a) Compare rate 27, 28, 29, 30

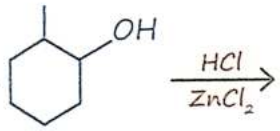
(b) Compare rate 32, 33, 34, 37

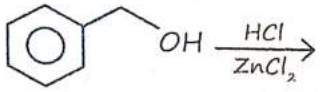
32. 

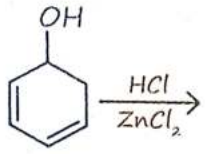
33. 

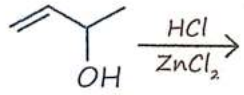
34. 

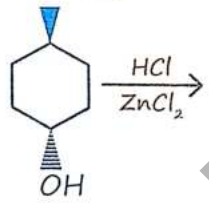
35. 

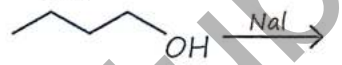
36. 

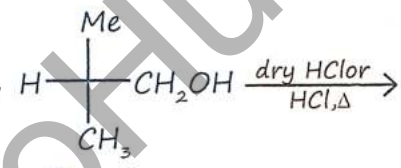
37. 

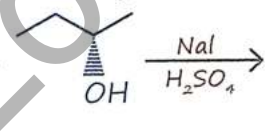
38. 

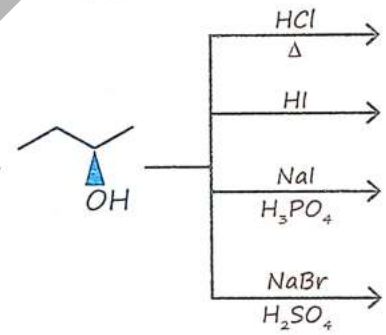
39. 

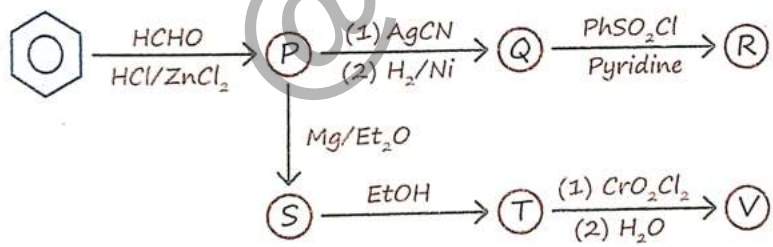
40. 

41. 

42. 

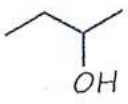
43. 

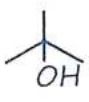
44. 

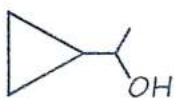
45. 

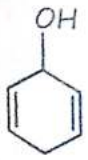
46. How many of the following will produce immediate turbidity with Lucas?

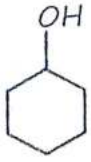
(a)

(b) 

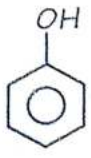
(c) 

(d) 

(e) 

(f) 

(g)

(h) 

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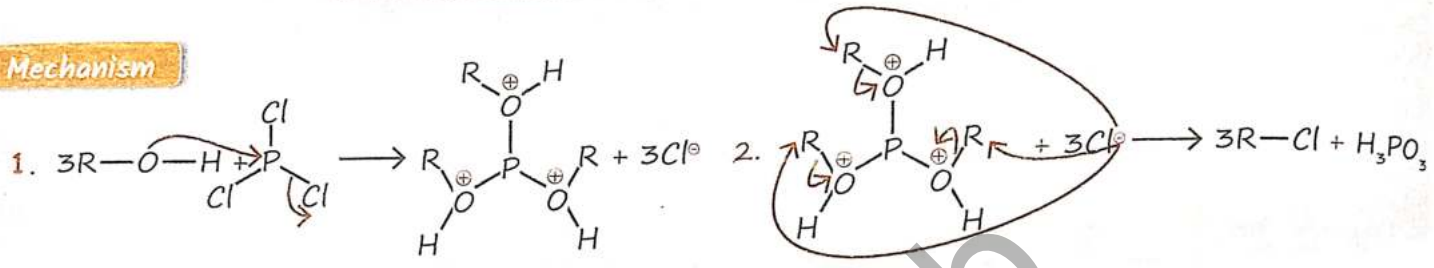
Reaction with PCl_3

Alcohols react with phosphorus trichloride to give alkyl halide and phosphorous acid. The reaction follows S_N2 mechanism with inversion, and is driven by the formation of a strong P = O bond.

Reaction

$$3R{-}O{-}H + PCl_3 \longrightarrow 3R{-}Cl + H_3PO_3$$

Mechanism



Baba Tea-Stall (BTS)

(i) S_N2 Reaction

(ii) Inversion occurs

(iii) Driving force is formating of P—O bond

(iv) 6 times S_N occurs

(v) Red P/X_2 can be used (X = Br, I)

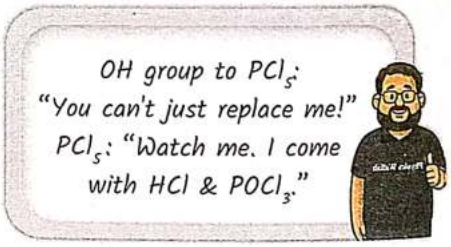
PX_3

Reaction with PCl_5

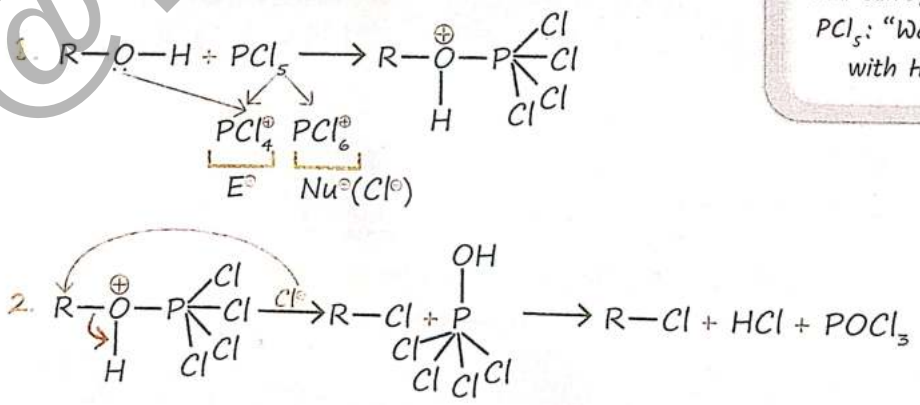
Alcohol reacts with phosphorus pentachloride to form alkyl chloride, hydrogen chloride, and $POCl_3$. This S_N2 reaction also involves inversion and is favored by the formation of a stable P=O bond.

Reaction

$$R{-}O{-}H + PCl_5 \longrightarrow R{-}Cl + H{-}Cl + POCl_3$$



Mechanism



Baba Tea-Stall (BTS)

(i) S_N2 mechanism

(ii) Inversion occurs

(iii) Formation of P = O bond is driving force

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With ether

Ethers react with phosphorus pentachloride via S_N2 followed by S_N1 mechanism to give alkyl chlorides and $POCl_3$. The reaction involves inversion followed by retention, and is driven by P=O bond formation.

Reaction

$$R-O-R' + PCl_5 \longrightarrow R-Cl + R'Cl + POCl_3$$

Mechanism

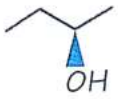
1. $R-\ddot{O}-R' + PCl_5 \longrightarrow \overset{R}{\underset{R'}{\overset{\oplus}{O}}}-PCl_4$

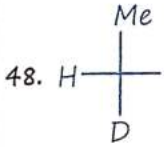
$PCl_4^{\oplus}$ ($E^{\oplus}$) $PCl_6^{\ominus}$ ($Nu^{\ominus}(Cl^{\ominus})$)

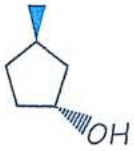
- S_N2 mech^m then S_Ni mech^m
- Inversion occurs then retention
- formation of P=O bond is dividing force.

2. $\overset{R}{\underset{R'}{\overset{\oplus}{O}}}-PCl_4 \xrightarrow[S_N2]{Cl^{\ominus}} R-Cl + \underset{\delta^-}{R'}-\ddot{O}-P(Cl)_2 \xrightarrow[S_Ni]{4MCTS} R'-Cl + POCl_3$

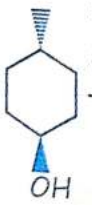
Practice Questions

47.  $\xrightarrow{PCl_3}$

48.  $\xrightarrow{PCl_3}$

49.  $\xrightarrow[Br_2]{Red\ P}$

50. $\xrightarrow[(3)\ NaOH]{(1)\ Red\ P/Cl_2,\ (2)\ NaI/\ \underset{}{\overset{O}{\parallel}}\ (4)\ PCl_3}$

51.  $\xrightarrow[(3)\ NaOH]{(1)\ Red\ P/Cl_2,\ (2)\ NaI/\ \underset{}{\overset{O}{\parallel}}\ (4)\ HCl/ZnCl_2}$

52. $R-O-R' \xrightarrow{PCl_5}$

53. $R-CHO \xrightarrow{PCl_5}$

54. $R-\underset{\underset{O}{\parallel}}{C}-R' \xrightarrow{PCl_5}$

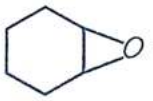
55. $R-\underset{\underset{O}{\parallel}}{C}-O-R \xrightarrow{PCl_5}$

56. $R-COOH \xrightarrow{PCl_5}$

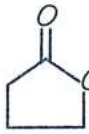
57. $R-\underset{\underset{O}{\parallel}}{C}-O-\underset{\underset{O}{\parallel}}{C}-R$

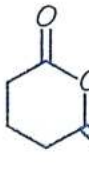
58. $\xrightarrow{PCl_5}$

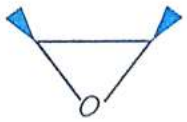
59. $R-\underset{\underset{O}{\parallel}}{C}-NH_2 \xrightarrow{PCl_5}$

60.  $\xrightarrow{PCl_5}$

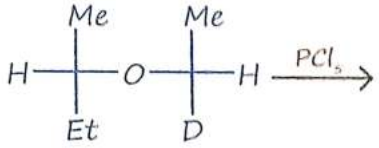
61. $R-COCl \xrightarrow{PCl_5}$

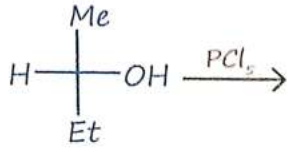
62.  $\xrightarrow{PCl_5}$

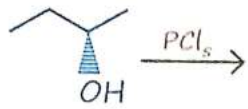
63.  $\xrightarrow{PCl_5}$

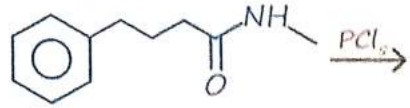
64.  $\xrightarrow{PCl_5}$

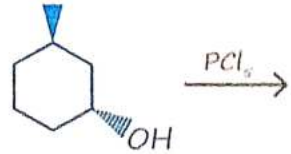
65. $\xrightarrow{PCl_5}$

66. 

67. 

68. 

69. 

70. 

71. $R-CHO \xrightarrow{SF_4}$

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Darzen Reaction

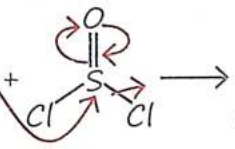
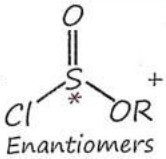
Alcohol reacts with thionyl chloride ($SOCl_2$) to form alkyl chloride, SO_2 and HCl.

The reaction follows S_Ni mechanism and retention of configuration occurs, with best yield observed when R–Cl is formed along with gaseous products.

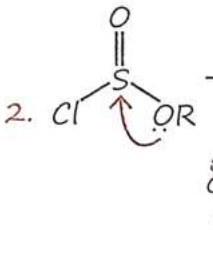
Reaction

$$R-O-H + SOCl_2 \longrightarrow R-Cl + HCl\uparrow + SO_2\uparrow$$

Mechanism

1. $R-O-H$ +  →  + $HCl\uparrow$

Enantiomers

2.  → $R-Cl + SO_2\uparrow$

OH group: "You'll replace me, right?"
$SOCl_2$: "Yes. But I'll do it gently, and you won't even change sides."

Baba Tea-Stall (BTS)

(i) 4 MCTS
(ii) Ion pair
(iii) Best yield of R–Cl (SO_2 and HCl are gases)
(iv) S_Ni mechanism
(v) Retention occurs

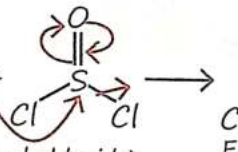
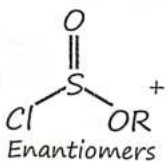
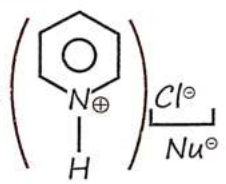
With Pyridine/NEt_3

Alcohol reacts with $SOCl_2$ in presence of pyridine or triethylamine to give alkyl chloride with inversion of configuration (S_N2 mechanism). The base traps HCl, improving yield and avoiding racemisation.

Reaction

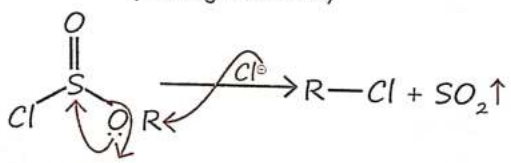
$$R-O-H + SOCl_2 \xrightarrow[NEt_3]{\text{Pyridine}} R-Cl + SO_2\uparrow + \left(\text{pyridinium}\right)Cl^{\ominus}$$

Mechanism

1. $R-O-H$ +  →  + HCl → 

(Thionyl chloride)

Enantiomers

2. 

Baba Tea-Stall (BTS)

(i) S_N2 mechanism
(ii) Inversion

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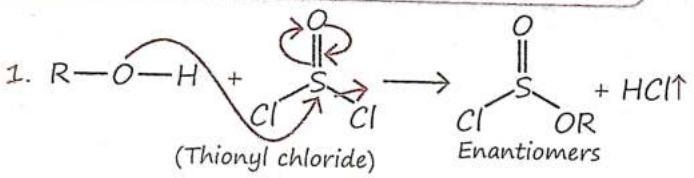
With allylic alcohol

Allylic alcohols react with thionyl chloride ($SOCl_2$) to form allyl halides, along with HCl and SO_2 gases. The reaction proceeds via an S_Ni mechanism with retention of configuration and is favoured by the formation of stable ion pairs and multiple resonance structures.

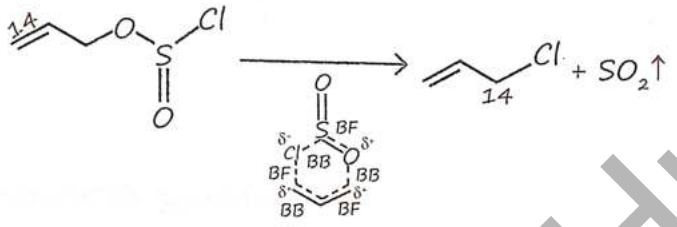
Reaction

$$R-O-H + SOCl_2 \longrightarrow R-Cl + HCl\uparrow + SO_2\uparrow$$

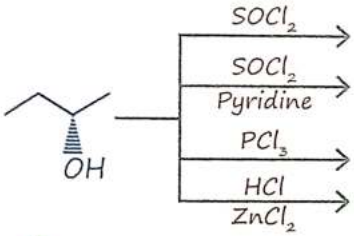
Mechanism

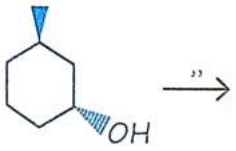
1. 

- 6 MCTS
- Ion pair
- S_Ni mechᵐ

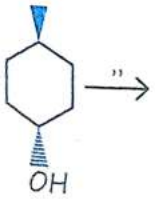
2. 

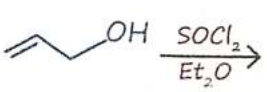
Practice Questions

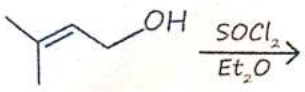
72. 

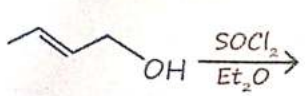
73. 

74. $CH_3CH_2CH_2CH_2OH \xrightarrow{''}$

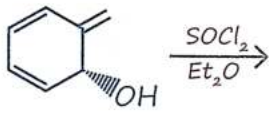
75. 

76. 

77. 

78. 

79. $Me-C\equiv C-\underset{OH}{CH}-CH_3 \xrightarrow[Et_2O]{SOCl_2}$

80. 

81. $R-COOH \xrightarrow[Et_2O]{SOCl_2}$

82. $R-SO_3H \xrightarrow[Et_2O]{SOCl_2}$

83.

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Cleavage of Ether and hydrolysis of ether

Ethers react with HX to form alcohols and alkyl halides; symmetrical ethers give two same alcohols or halides, unsymmetrical ethers may yield different products.

In presence of H_2O and acid, both symmetrical and unsymmetrical ethers hydrolyze to alcohols.

- Symmetrical ether: $R-O-R \xrightarrow{HX} R-\overset{\oplus}{O}(H)-R \xrightarrow{X^\ominus} R-OH + R-X$

  $R-O-R \xrightarrow{H_3O^\oplus} R-\overset{\oplus}{O}(H)-R \xrightarrow{H_2O} 2R-OH$

- Unsymmetrical ether: $R-O-R' \xrightarrow{HX} R-\overset{\oplus}{O}(H)-R' \xrightarrow{X^\ominus} R-OH + R'-X$ or $R'-OH + R-X$

  $R-O-R' \xrightarrow{H_3O^\oplus} R-\overset{\oplus}{O}(H)-R' \xrightarrow{H_2O} R-OH + R'-OH$

- Case 1 (R' is 3° and resonance stable): Under acidic HX, cleavage occurs via S_N1, forming a stable carbocation → gives R–OH + R'–X.

  $R-O-R'$ (3° & stable cc) $\xrightarrow{HX}$ $R-\overset{\oplus}{O}(H)-R' \xrightarrow{X^\ominus} R-OH + R'-X$ (S_N1)

  $R-O-R'$ (3° & stable cc) $\xrightarrow{H_3O^\oplus}$ $R-\overset{\oplus}{O}(H)-R' \xrightarrow{H_2O} R-OH + R'-OH$ (S_N1)

- Case 2 (Both R and R' are 1°): Cleavage follows S_N2, as no stable carbocation is formed → gives R–OH + R'–X directly.

  $R-O-R'$ (1° 1°) $\xrightarrow{HX}$ $R-\overset{\oplus}{O}(H)-R' \xrightarrow{X^\ominus} R-OH + R'-X$ (S_N2)

  $R-O-R'$ (1° 1°) $\xrightarrow{H_3O^\oplus}$ $R-\overset{\oplus}{O}(H)-R' \xrightarrow{H_2O} R-OH + R'-OH$ (S_N2)

- Case 3 (R is 1° and R' is 2°): Reaction may proceed through S_N2 with HX, but hydrolysis (with water) often follows S_N1, depending on the group.

  $R-O-R'$ (1° 2°) $\xrightarrow{HX}$ $R-\overset{\oplus}{O}(H)-R' \xrightarrow{X^\ominus} R-X + R'-X$ (S_N2)

  $R-O-R'$ (1° 2°) $\xrightarrow{H_3O^\oplus}$ $R-\overset{\oplus}{O}(H)-R' \xrightarrow{H_2O} R-OH + R'-OH$ (S_N1)

Baba Tea-Stall (BTS)

(i) Anhydrous HX always show S_N2 only

(ii) Conc. HX or cold HX or HX (1eq.) are same.

(iii) HX/Δ or hot HX or HX (excess) are same.

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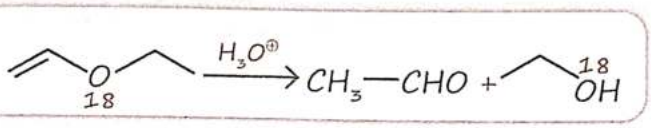
Unsaturated Ether

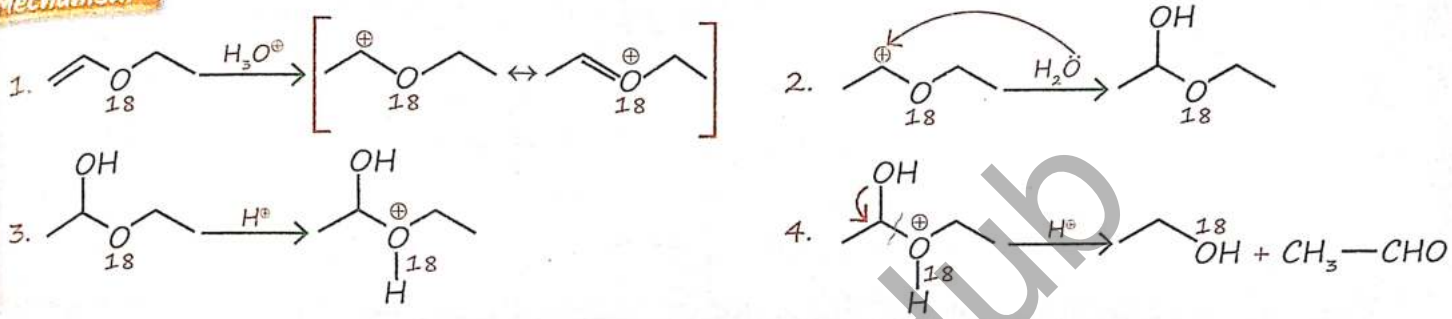
Under acidic hydrolysis, unsymmetrical ethers (like vinyl ethers) undergo protonation followed by rearrangement.

This leads to formation of aldehydes or ketones and alcohols, via resonance-stabilized carbocation or enol intermediates.

Reaction



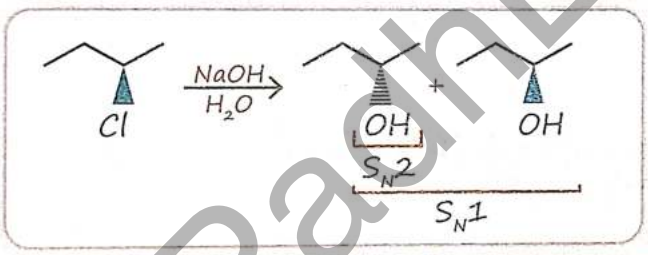
Mechanism



Mixed Case of SN

When both S_N1 and S_N2 pathways are possible, product distribution depends on nucleophile concentration and substrate structure.

Reaction



$$r_{SN} = K_1[R—X]\,[Nu^{\ominus}] + K_2\,[R—X]$$

$$\%S_N2 = \frac{K_1[R—X]\,[Nu^{\ominus}]}{K_1[R—X]\,[Nu^{\ominus}] + K_2\,[R—X]} \times 100$$

Try Yourself

Consider the following compounds:

A. $CH_2=CH-O-CH_2CH_3$

B. $CH_3CH_2-O-CH_3$

C. $CH_3CH(Cl)CH_2CH_3$

Which of the following statements is/are correct?

(a) Compound A, when treated with H_3O^+/heat, gives CH_3CHO and ethanol via a resonance-stabilized intermediate.

(b) Compound B undergoes acid-catalyzed cleavage to give two molecules of ethanol.

(c) Compound A reacts faster in basic medium than acidic medium due to nucleophilic attack on the ether linkage.

(d) In aqueous NaOH, compound C shows both S_N1 and S_N2 products, and $\%S_N2$ increases with higher nucleophile concentration.

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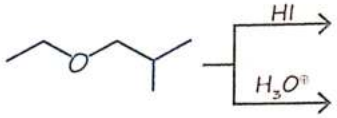
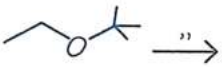
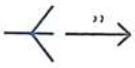
Practice Questions

84. 

85. $CH_3-O-CH_3 \xrightarrow{\text{"}}$

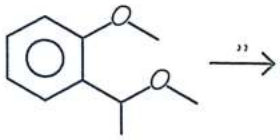
86.

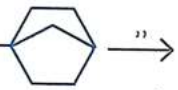
87. 

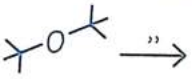
88. $Ph-O-$ 

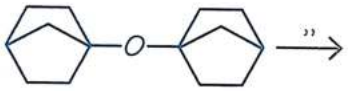
89. $Ph-O-Ph \xrightarrow{\text{"}}$

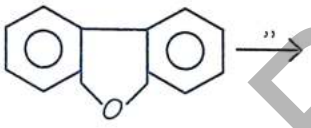
90. $Ph-OCH_3 \xrightarrow{\text{"}}$

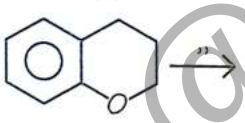
91. 

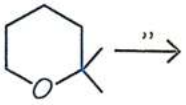
92. $Ph-O-$ 

93. 

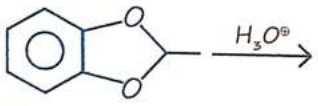
94. 

95. 

96. 

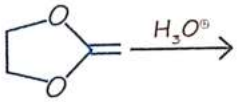
97. 

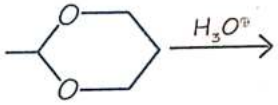
98.

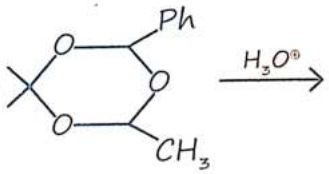
99. 

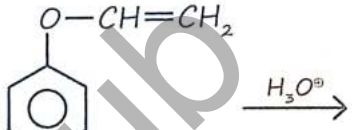
100. $CH_3-CN \xrightarrow{H_3O^{\oplus}}$

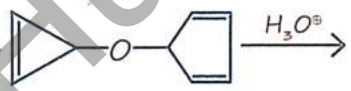
101. $CH_3-NC \xrightarrow{H_3O^{\oplus}}$

102. 

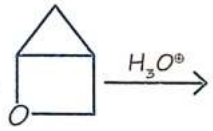
103. 

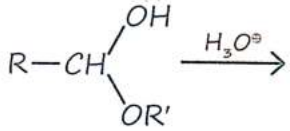
104. 

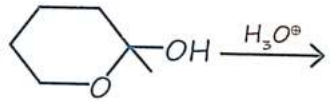
105. 

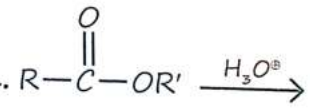
106. 

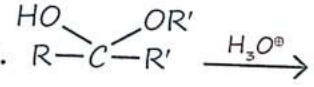
107. $CH_3-CH=NH \xrightarrow{H_3O^{\oplus}}$

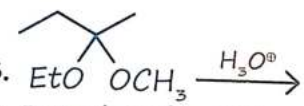
108. 

109. 

110. 

111. 

112. 

113. 

114. Rate law for SN reaction of 2-Bromobutane with $\overset{\ominus}{O}H$ in 75% ethanol and 25% water at 30°C is : $r = 3.2 \times 10^{-5}$.

$[R - X]\,[\overset{\ominus}{O}H] + 1.5 \times 10^{-6}\,[R-X]$. What % S_N2 takes place when $[\overset{\ominus}{O}H] = 1M$ and $[\overset{\ominus}{O}H] = 0.01M$.

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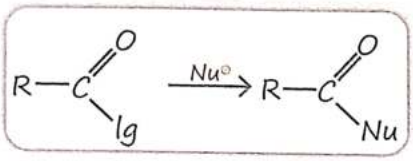
Nucleophilic Substitution Reaction (S_NAE)

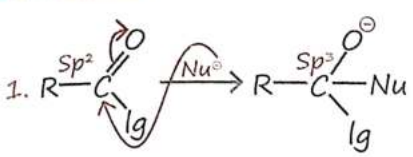
In this reaction, a nucleophile attacks a carbonyl carbon (R–C=O–LG), forming a tetrahedral intermediate. The intermediate is unstable and collapses, eliminating the leaving group (LG) to form the substituted carbonyl compound (R–C=O–Nu).

This involves Addition followed by Elimination.

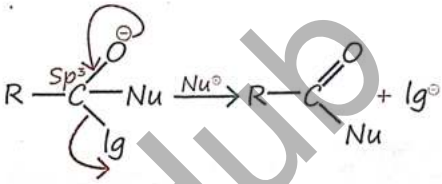
Reaction



Mechanism



- Unstable tetrahedral intermediate
- Addⁿ Rxⁿ



- Elimination Rxⁿ

Practice Questions

115. $CH_3-COCl \xrightarrow{NH_3}$

116. $CH_3-COCl \xrightarrow{RNH_2}$

117. $CH_3-COCl \xrightarrow{ROH}$

118. $CH_3-COCl \xrightarrow{H_3O^{\oplus}}$

119. $CH_3-COCl \xrightarrow{RCOO^{\ominus}Na^{\oplus}}$

120. $CH_3-COCl \xrightarrow{H^{\ominus}(1eq.)}$

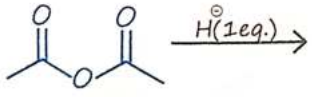
121. $CH_3-COOR' \xrightarrow{NH_3}$

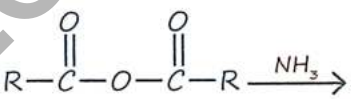
122. $CH_3-COOR' \xrightarrow{RNH_2}$

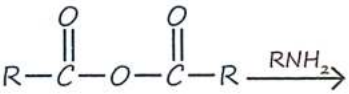
123. $CH_3-COOR' \xrightarrow{OH^{\ominus}/H_2O}$

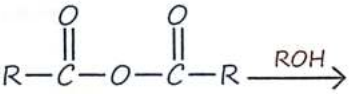
124. $CH_3-COOR' \xrightarrow{H_3O^{\oplus}}$

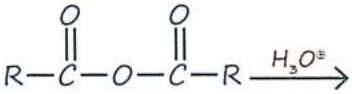
125. $CH_3-COOR' \xrightarrow{H^{\ominus}(1eq.)}$

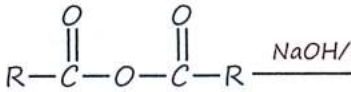
126.  $\xrightarrow{H^{\ominus}(1eq.)}$

127.  $\xrightarrow{NH_3}$

128.  $\xrightarrow{RNH_2}$

129.  $\xrightarrow{ROH}$

130.  $\xrightarrow{H_3O^{\oplus}}$

131.  $\xrightarrow{NaOH/H_2O}$

132. $R-CONH_2 \xrightarrow{H_3O^{\oplus}}$

133. $R-CONH_2 \xrightarrow{OH^{\ominus}/H_2O}$

134. $R-CONH_2 \xrightarrow{Cl^{\ominus}}$

NCERT Gyaan

- Chloramphenicol $\xrightarrow{Use}$ Typhoid fever treat (antibiotic)
- Chloroquine $\xrightarrow{Use}$ Malaria treatment
- Halothane $\xrightarrow{Use}$ Anesthesia in surgery.
- Thyroxine $\xrightarrow[Causes]{Deficiency}$ Goiter disease.

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Polyhalogen Compounds

Polyhalogen compounds watching you skip their properties section: "You'll see me again... in the MCQ."

CH_2Cl_2 (Methylene Chloride)

(i) Use: Paint remover, propellant in a aerosol, drug manufacture. Metal cleaning & finishing
(ii) Less Concentration: Nervous system, hearing, vision
(iii) High Concentration: Dizziness, Nausea, numbness in fingers and toes.
(iv) Direct contract: Intense burning of skin and cornea.

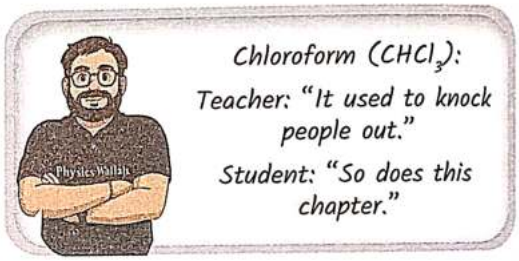
$CHCl_3$ (Chloroform)

(i) Production of freon refrigerator R-22.
(ii) Anaesthetic in Surgery with ether.
(iii) 900ppm inhalation causes headache.
(iv) Exposure to form phosgene gas.

$$2CHCl_3 + O_2 \longrightarrow 2COCl_2 + 2HCl$$

(v) Dark coloured bottle completely filled.
(vi) Sometime EtOH is added.

$$COCl_2 + EtOH \longrightarrow$$



Chloroform ($CHCl_3$):
Teacher: "It used to knock people out."
Student: "So does this chapter."

CHI_3 (Iodoform)

(i) Used as antiseptic due to free iodine not Iodoform.
(ii) Objectionable smell.

CCl_4 (Tetrachloro methane)

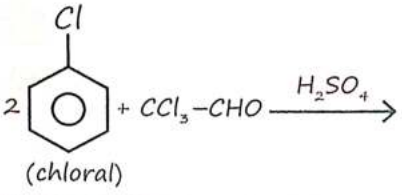
(i) Used in manufacture of refrigerator and aerosol propellant.
(ii) Feedstock in CFC (CF_2Cl_2)
(iii) Degreasing agent, fire extinguisher.
(iv) Irregular heart beat and liver cancer.
(v) Causes Ozone layer depletion.

Freons

(i) Chlorofluoro carbon of methane and Ethane.
(ii) Stable, unreactive, non toxic, easy to liquefiable.
(iii) Produced by $CCl_4 + AgF$
(iv) Aerosol propellant, fridge, AC
(v) Production of 2B pounds till 1974
(vi) Ozone layer depletion

DDT

(i) First Cl containing insecticide
(ii) Paul muller got Noble
(iii) Effective for malaria and Typhus
(iv) Fat soluble so stored in fat tissue of animals

(v) 

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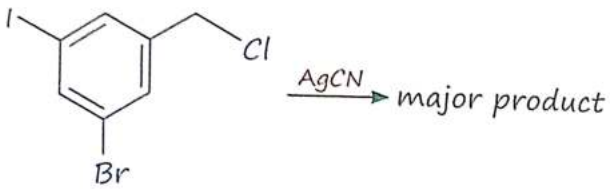
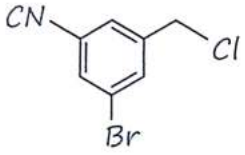
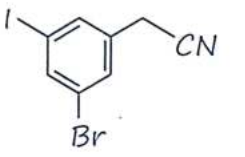
16. The structure of the major product formed in the following reaction is:

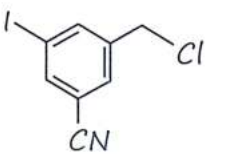
[24 Jan, 2025 (Shift-II)]



(a) 

(b)

(c) 

(d) 

Sol. (b) AgCN is covalent in nature where N is free to donate electron pair forming isocyanide as the main product.

17. The products A and B in the following reactions, respectively are

$$A \xleftarrow{Ag-NO_2} CH_3-CH_2-CH_2-Br \xrightarrow{AgCN} B.$$

[28 Jan, 2025 (Shift-I)]

(a) $CH_3-CH_2-CH_2-NO_2$, $CH_3-CH_2-CH_2-CN$

(b) $CH_3-CH_2-CH_2-ONO$, $CH_3-CH_2-CH_2-NC$

(c) $CH_3-CH_2-CH_2-ONO$, $CH_3-CH_2-CH_2-CN$

(d) $CH_3-CH_2-CH_2-NO_2$, $CH_3-CH_2-CH_2-NC$

Sol. (d) AgCN is covalent compound and $AgNO_3$ is ionic compound.
A is $CH_3-CH_2-CH_2-NO_2$, B is $CH_3-CH_2-CH_2-NC$.

18. $C_2H_5Br \xrightarrow{alc.KOH} A \xrightarrow[CCl_4]{Br_2} B \xrightarrow[Excess]{KCN} C \xrightarrow[Excess]{H_3O^+} D$

[01 Feb, 2024 (Shift-II)]

Acid D formed in above reaction is:

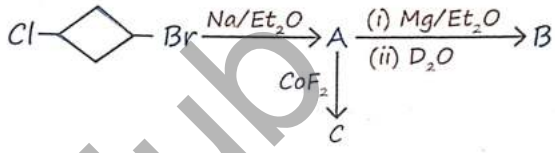
(a) Gluconic acid (b) Succinic acid

(c) Oxalic acid (d) Malonic acid

Sol. (b)

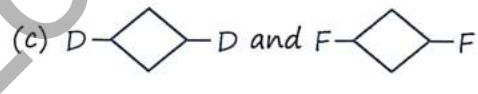
19. In the following sequence of reaction, the major products B and C respectively are:

[09 April, 2024 (Shift-I)]



(a)

(b)

(c) 

(d)

Sol. (a)

20. Assertion (A): Hydrolysis of an alkyl chloride is a slow reaction but in the presence of NaI, the rate of the hydrolysis increases.

Reason (R): I^- is a good nucleophile as well as a good leaving group.

In the light of the above statements, choose the correct answer from the options given below. [24 Jan, 2023 (Shift-I)]

(a) (A) is false but (R) is true.

(b) (A) is true but (R) is false.

(c) Both (A) and (R) are true and (R) is the correct explanation of (A).

(d) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).

Sol. (c) Hydrolysis of alkyl halide is slow in the presence of chlorine because 'Cl' is not a good leaving group or not it is a good Nucleophile as compare to Iodine.

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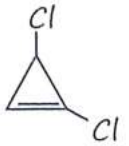
21. The decreasing order of reactivity of the following organic molecules towards $AgNO_3$ solution is [4 Sept, 2020 (Shift-I)]

(A)

(B) 

(C) CH_3CHCH_3 with Cl

(D) $CH_3CHCH_2NO_2$ with Cl

(a) (B) > (A) > (C) > (D)

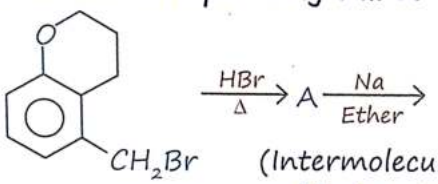
(b) (A) > (B) > (D) > (C)

(c) (A) > (B) > (C) > (D)

(d) (C) > (D) > (A) > (B)

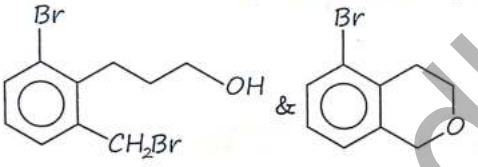
Sol. (b)

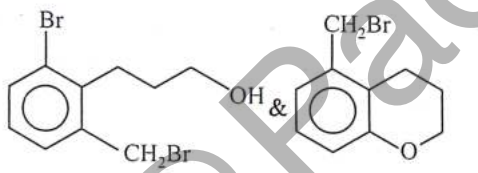
22. In the following reaction sequence, structure of A and B respectively will be

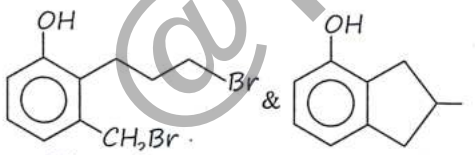


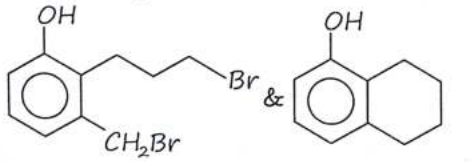
(Intermolecular product) B

[7 Jan, 2020 (Shift-II)]

(a) 

(b) 

(c) 

(d) 

Sol. (d)

23. Match List-I with List-II.

List-I Compound		List-II Use	
(A)	Carbon tetrachloride	(I)	Paint remover
(B)	Methylene chloride	(II)	Refrigerators and air conditioners
(C)	DDT	(III)	Fire extinguisher
(D)	Freons	(IV)	Non Biodegradable insecticide

Choose the correct answer from the options given below: [01 Feb, 2024 (Shift-II)]

(a) (A)-(I), (B)-(II), (C)-(III), (D)-(IV)

(b) (A)-(III), (B)-(I), (C)-(IV), (D)-(II)

(c) (A)-(IV), (B)-(III), (C)-(II), (D)-(I)

(d) (A)-(II), (B)-(III), (C)-(I), (D)-(IV)

Sol. (b) CCl_4 is commonly used in fire extinguishers.

CH_2Cl_2 is frequently used as a paint remover.

Freons (such as CFCs) are used in refrigerators and air conditioners.

DDT is used as a non-biodegradable insecticide.

24. Match List-I with List-II:

List-I (Compound)		List-II (Uses)	
A.	Iodoform	I.	Fire extinguisher
B.	Carbon tetrachloride	II.	Insecticide
C.	CFC	III.	Antiseptic
D.	DDT	IV.	Refrigerants

Choose the correct answer from the options given below: [06 April, 2024 (Shift-I)]

(a) A-I, B-II, C-III, D-IV

(b) A-III, B-II, C-IV, D-I

(c) A-III, B-I, C-IV, D-II

(d) A-II, B-IV, C-I, D-III

Sol. (c)

25. In the following halogenated organic compounds the one with maximum number of chlorine atoms in its structure is: [31 Jan, 2023 (Shift-II)]

(a) Chloral (b) Gammaxene

(c) Chloropicrin (d) Freon -12

Sol. (b)

26. Which of the following compounds is an example of Freon? [11 April, 2023 (Shift-II)]

(a) $C_2Cl_2F_2$ (b) C_2HF_3 (c) $C_2H_2F_2$ (d) C_2F_4

Sol. (a) Freons are chlorofluoro carbon having formula $C_2Cl_2F_2$.

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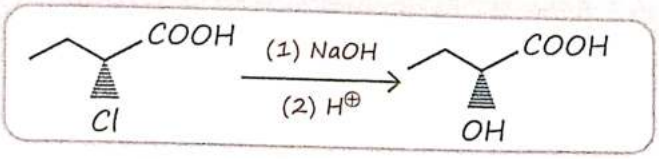
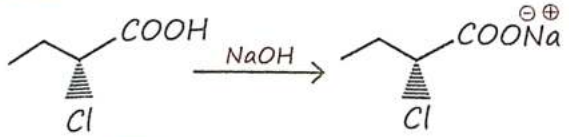
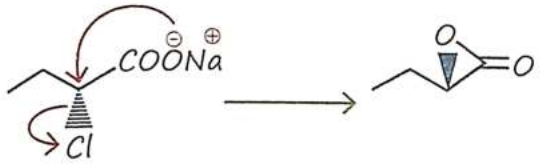
Neighbouring Group Participation (S_NNGP) or Anchimeric Assistance

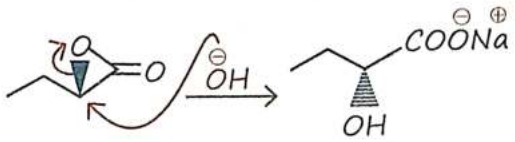
Reaction

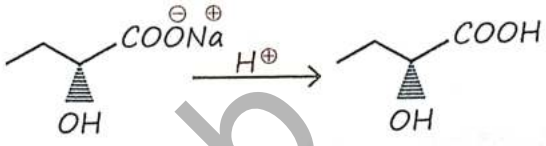


Mechanism

1. 

2. 

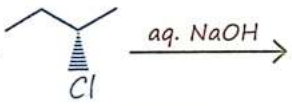
3. 

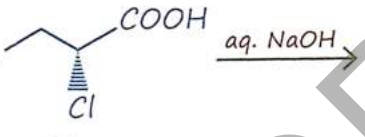
4. 

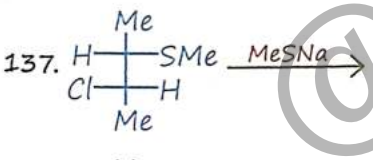
Baba Tea-Stall (BTS)

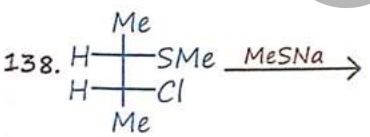
(i) Neighbouring group participation is S_N reaction
(ii) Also known as Anchimeric assistance.
(iii) Generally retained product obtained due to 2 times S_N2
(iv) LG and Internal Nu⊖ must be present and opposite to each other.
(v) Rate of Reaction increases drastically.

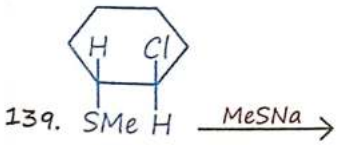
Practice Questions

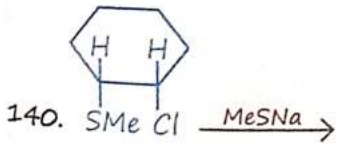
135. 

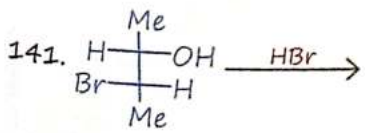
136. 

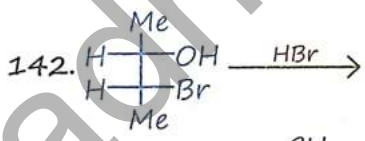
137. 

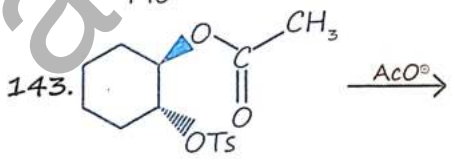
138. 

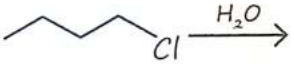
139. 

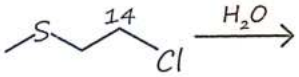
140. 

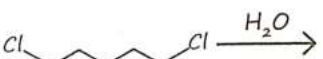
141. 

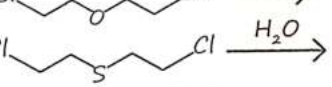
142. 

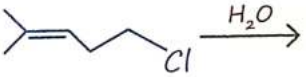
143. 

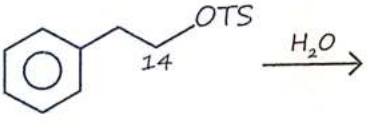
144. (a) 

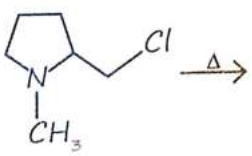
(b) 

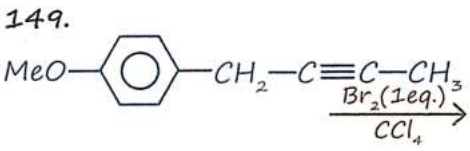
145. (a) 

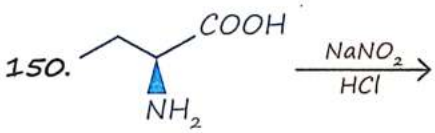
(b) 

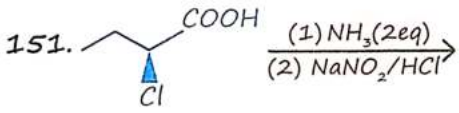
146. 

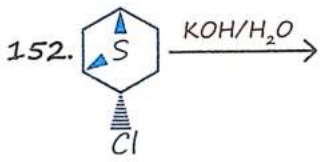
147. 

148. 

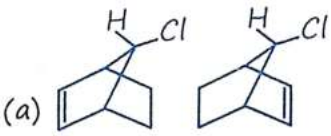
149. 

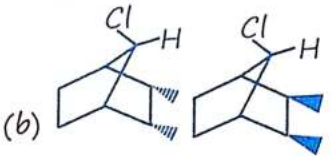
150. 

151. 

152. 

153. Compare rate of S_N2

(a) 

(b) 

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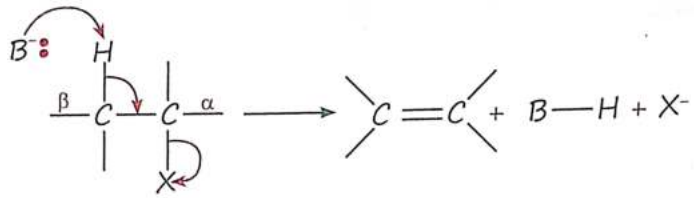
Elimination

When atoms/groups are eliminated from a compound during a reaction, the reaction is known a elimination reaction:

Reaction



B = Base : X = Leaving group

- Reverse of addition reaction.
- During elimination, degree of unsaturation of compound increases.

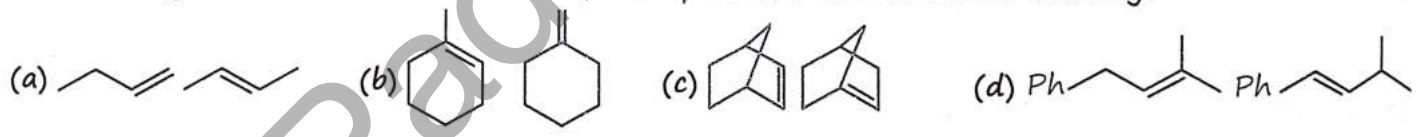
Types of elimination reactions

- 1-1 elimination/α-α elimination/α-elimination.
- 1-2 elimination/α-β elimination/β-elimination.
- 1-3 elimination/α-γ elimination/γ-elimination

Zaitsev and Hoffmann

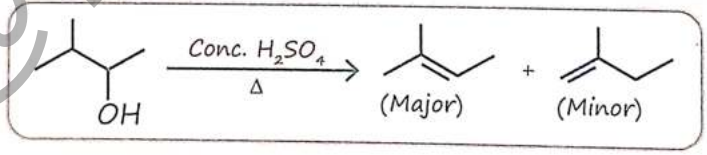
Zaitsev/Saytzeff	Hoffmann
➢ More substituted alkene	➢ Less substituted alkene
➢ Generally more stable	➢ Generally less stable

Q. Find the Saytzeff and Hoffmann alkene in each pair and also find their stability?

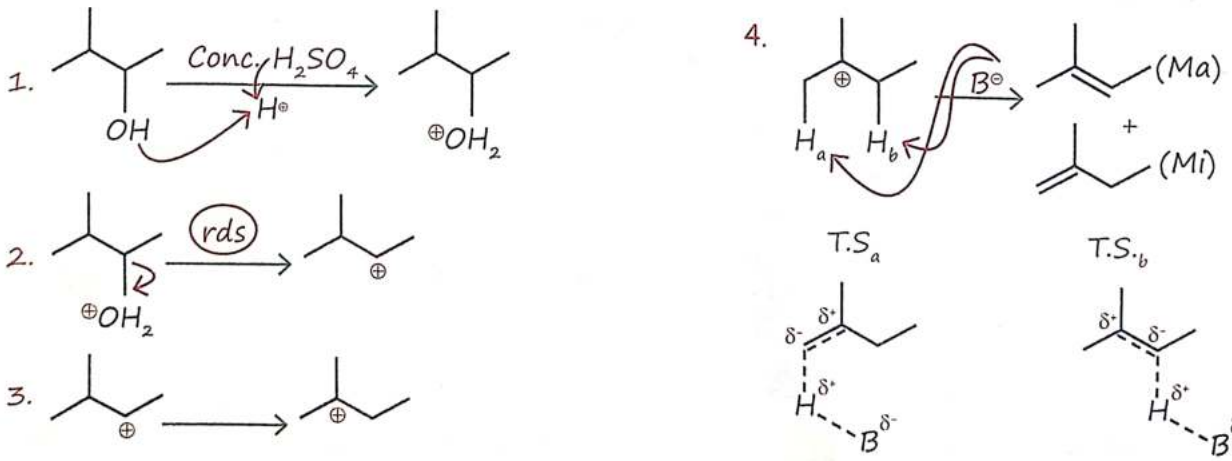


Dehydration of Alcohol

Reaction



Mechanism



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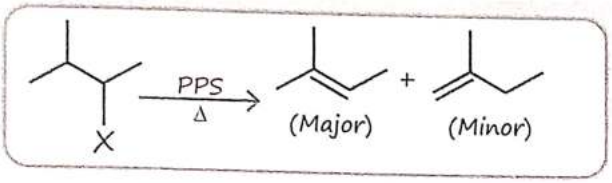
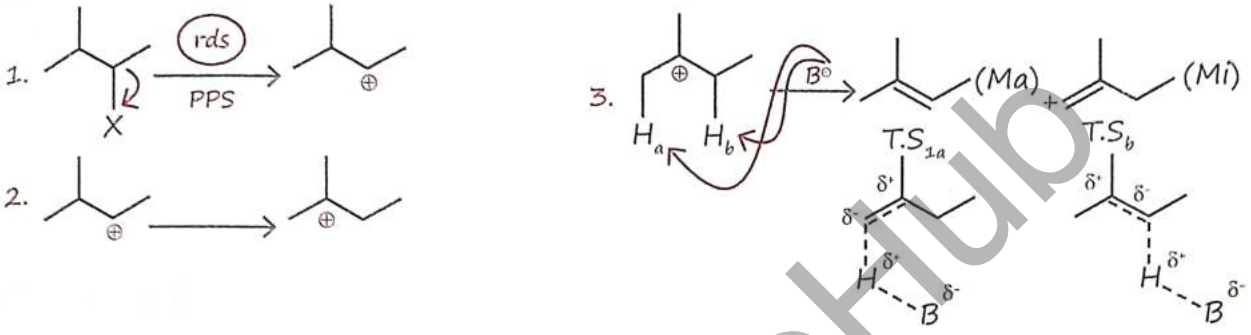
Unimolecular Elimination Reaction

Tertiary alkyl halides eliminate under heat and polar protic solvent forming alkenes via carbocation intermediate. The E1 mechanism shows major product based on Zaitsev's rule, with formation passing through TS_α and TS_β transition states.

Reaction

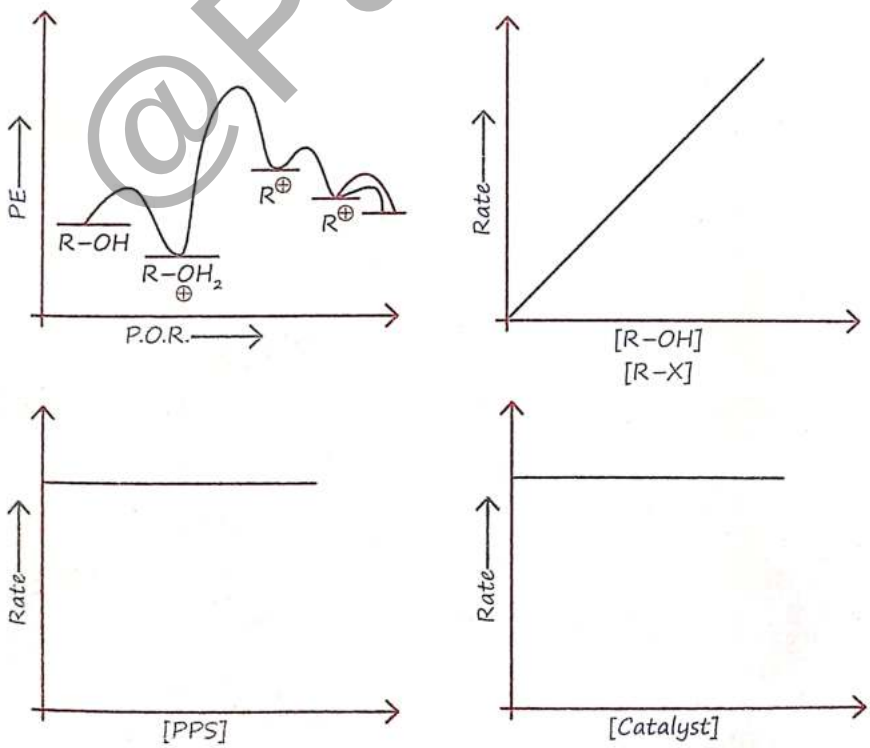


Mechanism



Baba Tea-Stall (BTS)

(i) Carbocation obtained as intermediate.
(ii) Rearrangement may be possible.
(iii) Unimolecular reaction and pseudo 1st order.
$r = k[R-OH][H_2SO_4] = K'[R-OH]$
(iv) Rate of dehydration α stability of carbocation
(v) Reaction involves dehydration, CC and elimination.
(vi) Two step process
(vii) No kinetic isotopic effect present.
(viii) Elemental effect observed
(ix) Generally zaitsev alkene major product
(x) Unimolecular elimination reaction (E1).
(xi) We can use H_2SO_4, H_3PO_4, $KHSO_4$ or Al_2O_3.
(xii) Graph:



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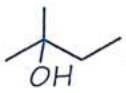
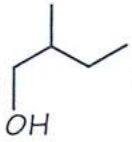
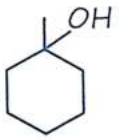
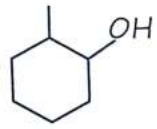
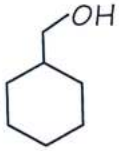
Practice Questions

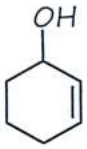
154.  $\xrightarrow[H_2SO_4,\Delta]{Con.}$

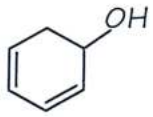
155.  $\xrightarrow[H_2SO_4,\Delta]{Con.}$

156.  $\xrightarrow[H_2SO_4,\Delta]{Con.}$

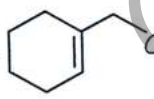
157.  $\xrightarrow[H_2SO_4,\Delta]{Con.}$

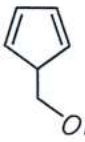
158.  $\xrightarrow[H_2SO_4,\Delta]{Con.}$

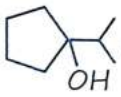
159.  $\xrightarrow[H_2SO_4,\Delta]{Con.}$

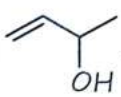
160.  $\xrightarrow[H_2SO_4,\Delta]{Con.}$

161. $\xrightarrow[H_2SO_4,\Delta]{Con.}$

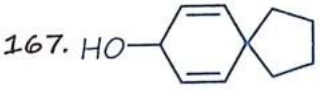
162.  $\xrightarrow[H_2SO_4,\Delta]{Con.}$

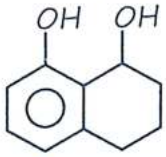
163.  $\xrightarrow[H_2SO_4,\Delta]{Con.}$

164.  $\xrightarrow[H_2SO_4,\Delta]{Con.}$

165.  $\xrightarrow[H_2SO_4,\Delta]{Con.}$

166. $\xrightarrow[H_2SO_4,\Delta]{Con.}$

167.  $\xrightarrow[H_2SO_4,\Delta]{Con.}$

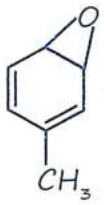
168.  $\xrightarrow[H_2SO_4,\Delta]{Con.}$

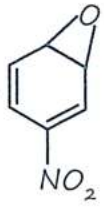
169. $\xrightarrow[H_2SO_4,\Delta]{Con.}$

170. $\xrightarrow[H_2SO_4,\Delta]{Con.}$

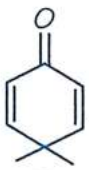
171. $\xrightarrow[H_2SO_4,\Delta]{Con.}$

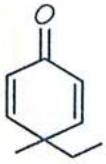
172. $\xrightarrow[H_2SO_4,\Delta]{Con.}$

173.  $\xrightarrow[H_2SO_4,\Delta]{Con.}$

174.  $\xrightarrow[H_2SO_4,\Delta]{Con.}$

175. $\xrightarrow[H_2SO_4,\Delta]{Con.}$

176.  $\xrightarrow[H_2SO_4,\Delta]{Con.}$

177.  $\xrightarrow[H_2SO_4,\Delta]{Con.}$

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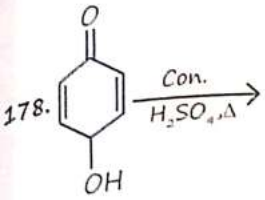
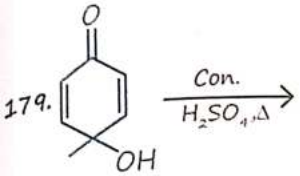
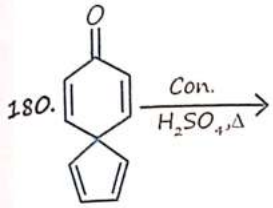
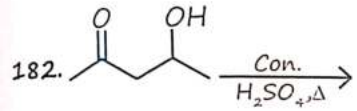
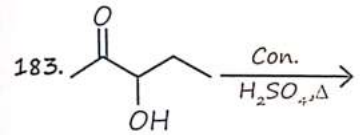
178. 

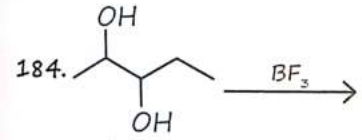
179. 

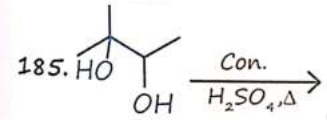
180. 

181.

182. 

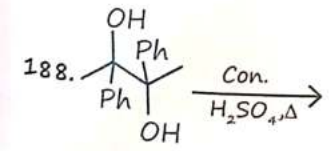
183. 

184. 

185. 

186.

187.

188. 

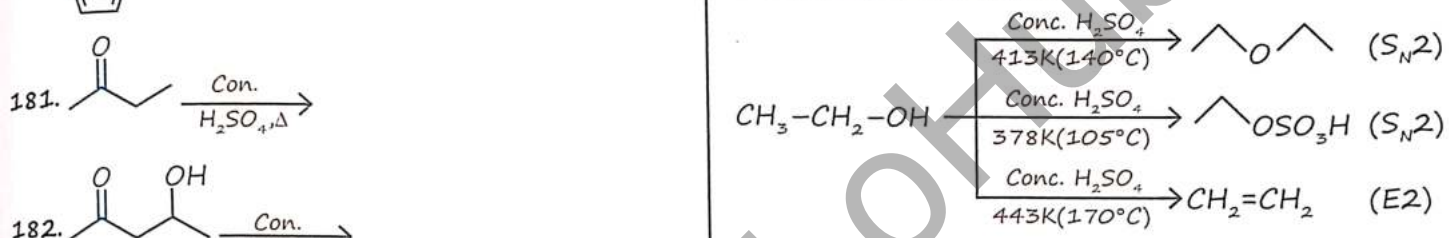
189.

190. Repeat 186-189 of migratory aptitude is:
R > Ar > H

191.

192.

193. Draw mechanism:



194. CH_2-CH_2 $\xrightarrow[H_2SO_4]{cold}$

195.

196.

197.

198.

199.

200.

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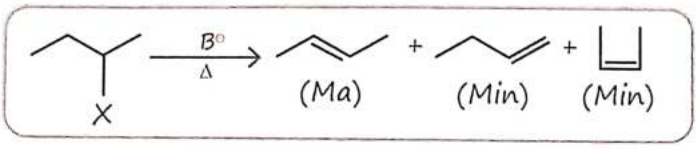
Bimolecular Elimination Reaction

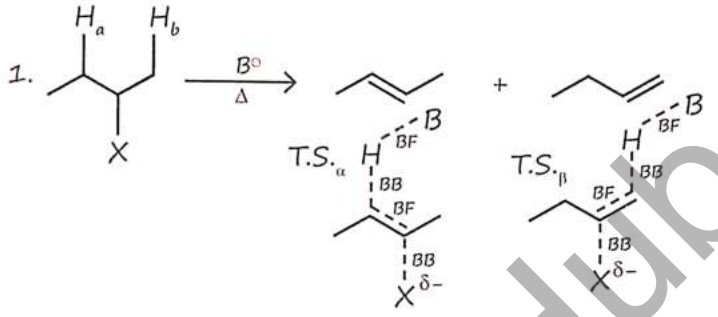
In the E_2 mechanism, a strong base abstracts a β-hydrogen while the leaving group departs in a single step, forming an alkene.

This concerted process requires anti-periplanar geometry, and yields both Zaitsev (major) and Hoffmann (minor) products depending on base and substrate.

Reaction

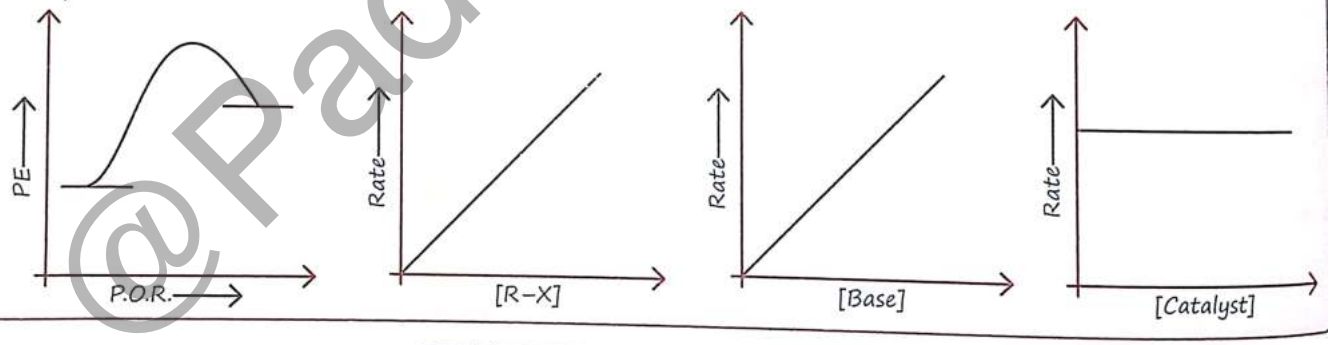


Mechanism



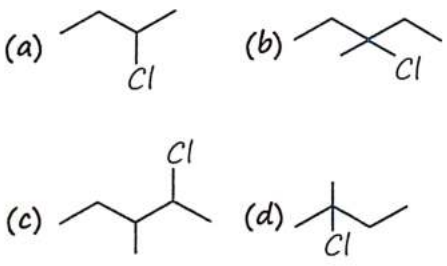
Baba Tea-Stall (BTS)

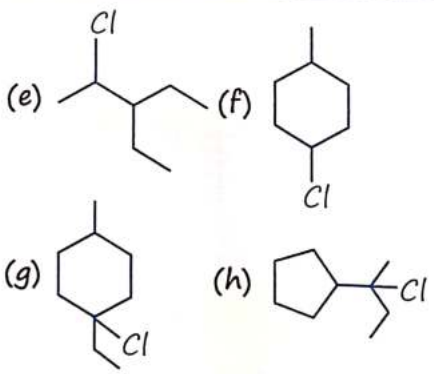
(i) One step process.
(ii) No carbocation so no Rearrangement.
(iii) Transition state is 'Anti Periplanar'.
(iv) Anti elimination is observed.
(v) Bimolecular reaction
(vi) Rate law = $K[R—X]^1\ [B^\ominus]^1$
(vii) Second order reaction
(viii) T must be high as $\boxed{\Delta G = \Delta H - T\Delta S}$
(ix) Kinetic isotopic effect observed.
(x) Elemental effect observed.
(xi) Generally Zaitsev alkene is major
(xii) Graphs:

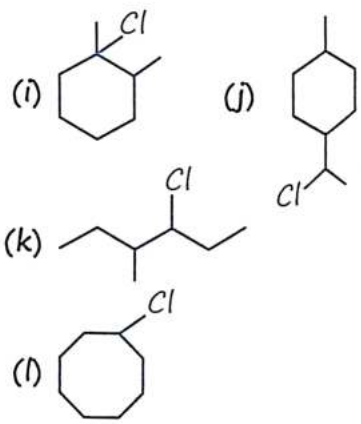


Practice Questions

201. Total number of alkene Obtained on dehydrohalogenation are:







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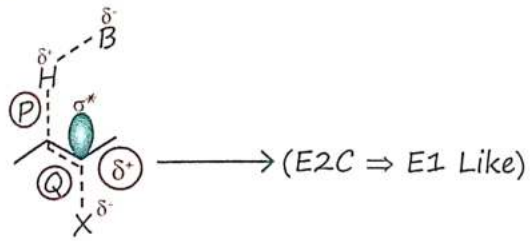
Different Mechanisms of E2

- E2C
- E2(T.S.)
- E2a

➢ P < Q when leaving group is good

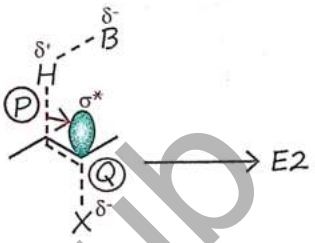
(Moderate base)



→ (E2C ⇒ E1 Like)

Major product is decided by stability of 'Alkene'.

➢ P = Q when leaving group is moderate.

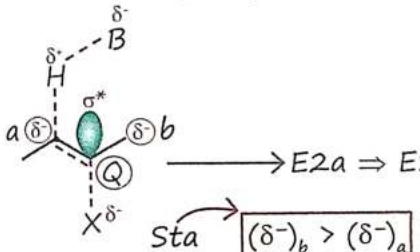


→ E2

Major product is decided by stability of 'Alkene'.

➢ P > Q when leaving group is Poor.

(F, OH, OR, $\overset{\oplus}{N}R_3$, $\overset{\oplus}{S}R_2$)



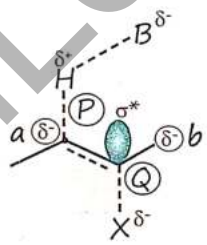
→ E2a ⇒ E1CB like

Sta $(\delta^-)_b > (\delta^-)_a$

Major product is decided by stability of 'δ⁻'.

➢ P > Q when strong base is taken.

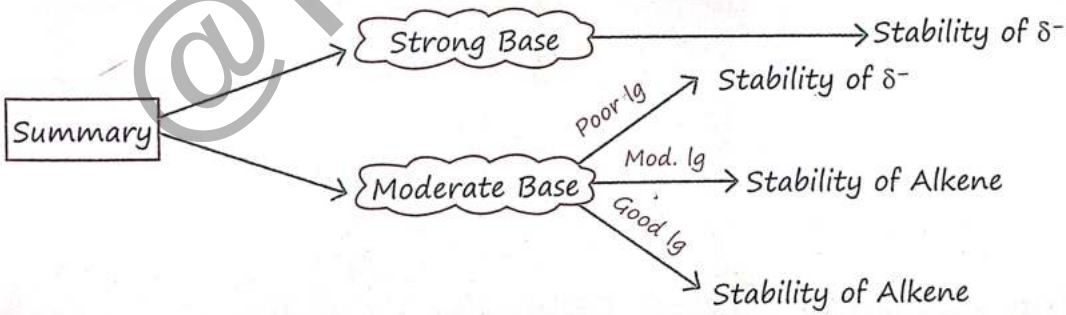
($tBuONa$, $\overset{\oplus}{N}H_2$)



→ E2a ⇒ E1CB like

Sta $(\delta^-)_b > (\delta^-)_a$

Major product is decided by stability of (δ⁻).



Base

- Aqueous KOH promotes S_N2, especially with 1° alkyl halides.
- Alcoholic KOH induces E2 elimination in 2° and 3° alkyl halides, forming alkenes.
- Conditions like alc. KOH at 200°C or RO^-/Δ favor elimination due to stronger basicity and heat.

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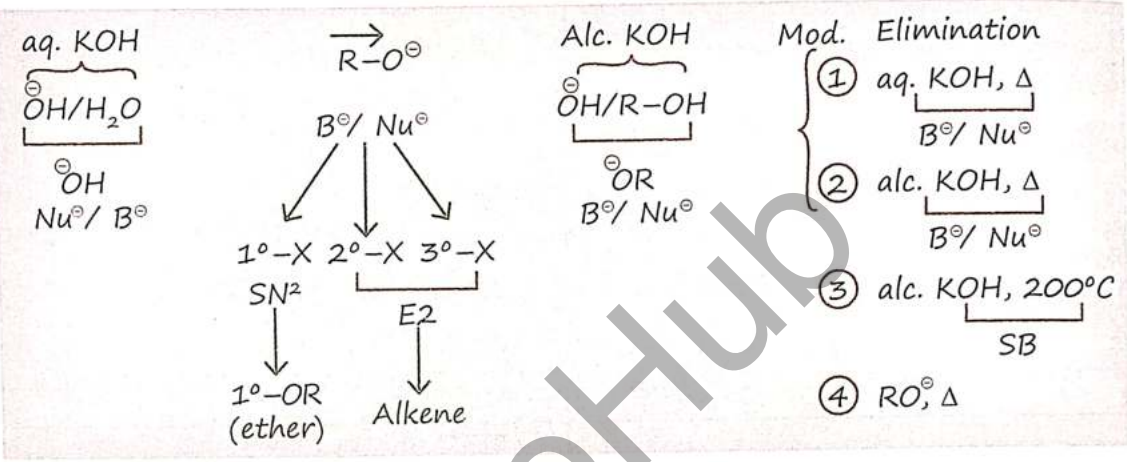
Bases

- Good leaving groups (like Br^-, I^-) promote both SN and E reactions; poor ones (e.g., F^-, OH^-) require stronger conditions.
- Weak bases (e.g., OH^- in H_2O) give S_N2, moderate bases (RO^-/ROH) favor E2, and strong bulky bases (like t-BuO^-) shift major product toward Hoffmann elimination.



Classification of Bases

Poor Base	KOH/H_2O (alc. KOH) ⇒ heat or Δ $Nu^{\oplus}/B^{\ominus} \rightarrow SN^2$ $\quad E^2$
Moderate Base	$RO^{\ominus}$, $RO^{\ominus}/\Delta$, KOH/ROH (alc KOH)/Δ 1°–X $B^{\ominus}/Nu^{\ominus} \rightarrow E2 \rightarrow$ PureE2 $\quad \rightarrow$ PureE2
Strong Base	$O^{\ominus}$, $O^{\ominus}$, $\overset{\ominus}{N}H_2$, LDA, $\ominus Li^{\oplus}$ ($tBuO^{\ominus}$)

Classification of Leaving Group

- Leaving Group
  - Good, Moderate
  - Poor → F, OH, OR, $\overset{\oplus}{N}R_3$, $\overset{\oplus}{S}R_2$

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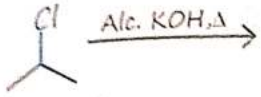
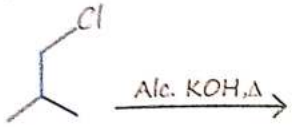
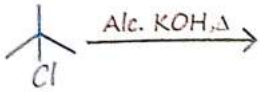
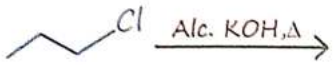
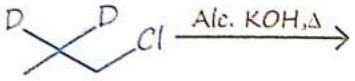
Practice Questions

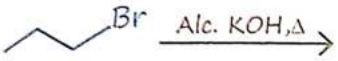
202. 

203. 

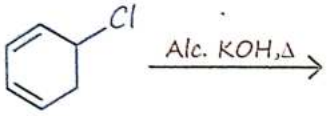
204. 

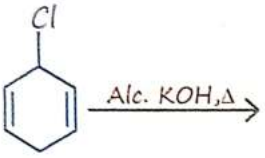
205. 

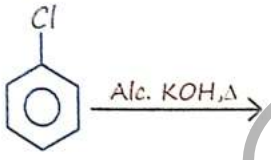
206. 

207. 

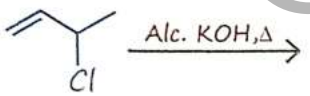
208. Compare rate of 203–208.

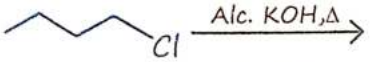
209. 

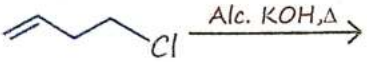
210. 

211. 

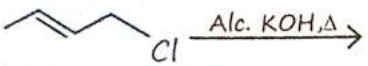
212. Compare rate of 209 and 211.

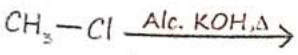
213. 

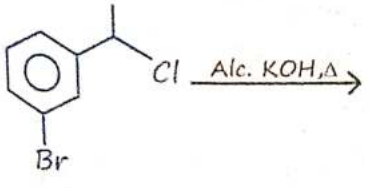
214. 

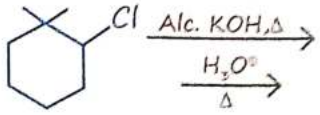
215. 

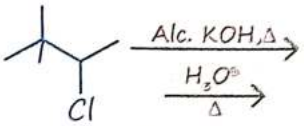
216. Rate of 213–215.

217. 

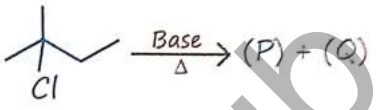
218. 

219. 

220. 

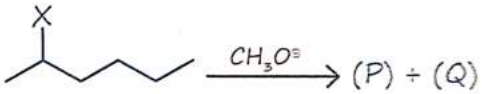
221. 

222. Based on the data, find (P) and (Q).



Base	(P)	(Q)
$\overset{\ominus}{O}H$	85%	15%
$\overset{\ominus}{O}Et$	70%	30%
$t\text{-}BuO^{\ominus}$	22%	78%
$(Et)_3C\text{-}O^{\ominus}$	12%	88%

223. Based on the data, Find (P) and (Q).



Base	(P)	(Q)
I	81%	19%
Br	72%	28%
Cl	67%	33%
F	30%	70%

224. $CH_3-CHCl_2 \xrightarrow[\text{or KOH/EtOH}]{\text{Alc. KOH/}\Delta}$

225. $CH_3-CHCl_2 \xrightarrow[\text{or Moist KOH, 200°C}]{NaNH_2/\Delta}$

226. $CH_3-CHCl_2 \xrightarrow[(2)\ NaNH_2,\Delta]{(1)\ \text{Alc. KOH},\Delta}$

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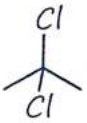
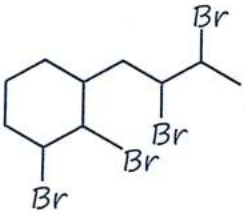
227. $CH_3-CH(Cl)-Cl$ → $NaNH_2$(1eq.) / $NaNH_2$(2eq.) / $NaNH_2$(3eq.) / $NaNH_2$(4eq.)

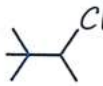
228. $CH_3-CH(Cl)-Cl$ → (1) Alc. KOH, Δ (2) $NaNH_2$, Δ

229.  → (1) $NaNH_2$(3eq.) (2) CH_3-I

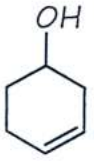
230. $CH_2=CH_2$ → (1) Br_2/CCl_4 (2) Alc.KOH, Δ (3) $NaNH_2$(3eq.) (4) CH_3-I(2eq.) (5) Na/liq. NH_3

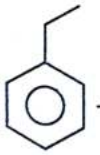
231. → (1) Br_2/CCl_4 (2) Alc. KOH, Δ(xs)

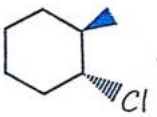
232.  → KOH, EtOH, Δ

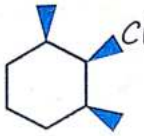
233.  → t BuONa, Δ / t BuOH, Δ

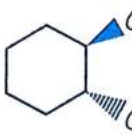
234. → EtONa, Δ

235.  → (1) PCl_3 (2) HBr (3) Alc. KOH/Δ

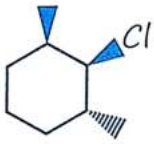
236.  → (1) NBS (2) EtONa (3) $HBr/(PhCO)_2O_2$ (4) CoF_2 (5) tBuONa (6) B_2H_6–THF (7) $H_2O_2/NaOH$

237.  → KOH, EtOH, Δ

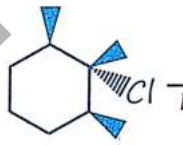
238.  → KOH, EtOH, Δ

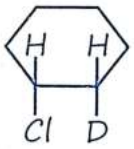
239.  → KOH, EtOH, Δ

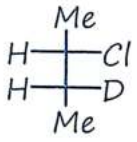
240. → KOH, EtOH, Δ

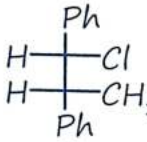
241.  → KOH, EtOH, Δ

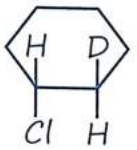
242. → KOH, EtOH, Δ

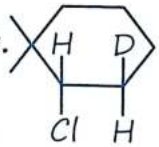
243.  → KOH, EtOH, Δ

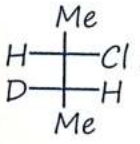
244.  → KOH, EtOH, Δ

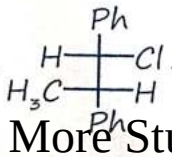
245.  → KOH, EtOH, Δ

246.  → KOH, EtOH, Δ

247.  → KOH, EtOH, Δ

248.  → KOH, EtOH, Δ

249.  → KOH, EtOH, Δ

250.  → KOH, EtOH, Δ

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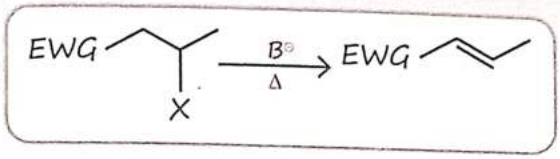
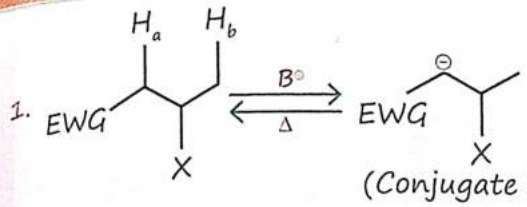
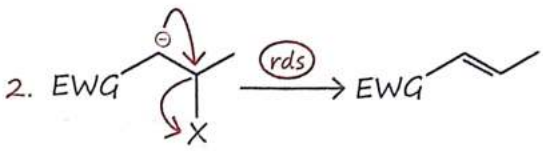
E1CB Reaction

Reaction



Mechanism





Baba Tea-Stall (BTS)

(i) Unimolecular Reaction.

(ii) Two step process.

(iii) Carbanion (CB) obtained as intermediate

(iv) No Rearrangement

(v) Rate law = $K[C-B] = K[R-X][B^{\ominus}]$

(vi) 2nd orde Rxn.

(vii) EWG is must to stabilise the C^-.

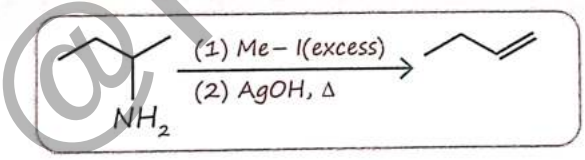
(viii) Kinetic isotopic effect of substrate not present

(ix) Kinetic isotopic effect of solvent observed.

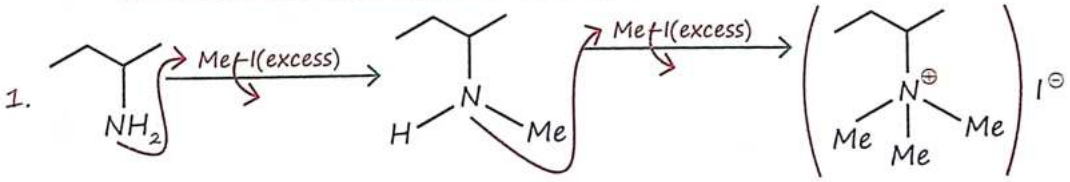
Hoffmann Exhaustive Methylation Elimination (HEME)

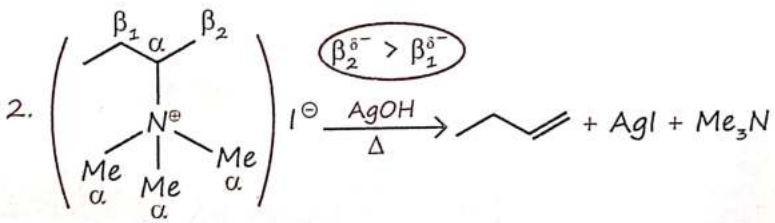
- A primary amine reacts with excess methyl iodide (Me–I) and AgOH/heat, converting into a quaternary ammonium salt.
- Elimination then occurs to give the least substituted alkene (Hoffmann product), along with Me_3N and AgI as byproducts.

Reaction



Mechanism





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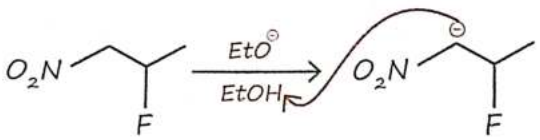
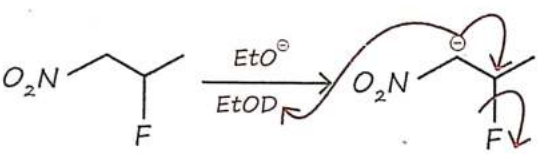
Effect of K.E. of Solvent

(a) 

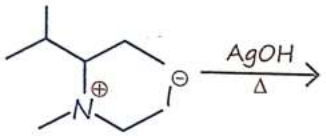
(More Reversible)

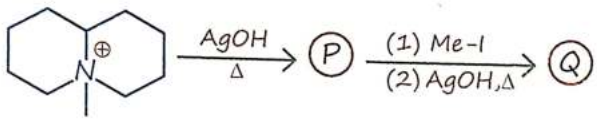
(b) 

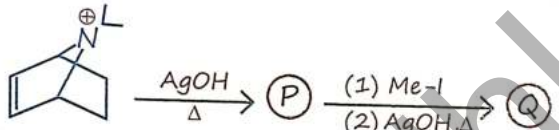
(Less Reversible)
More forward

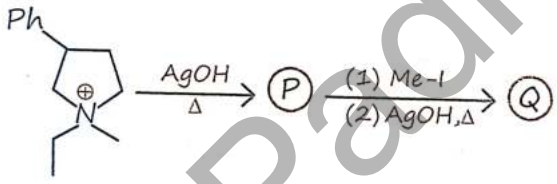
yield of (b) > yield of (a)

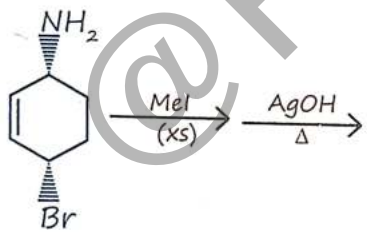
Practice Questions

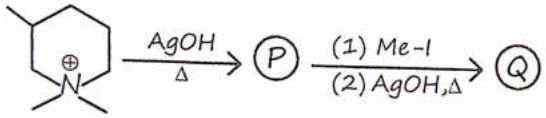
251. 

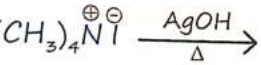
252. 

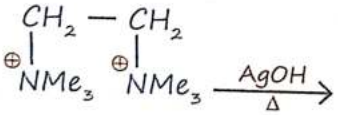
253. 

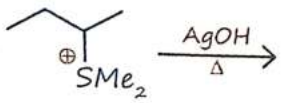
254. 

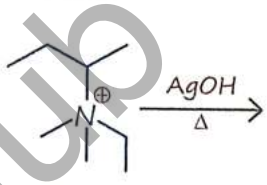
255. 

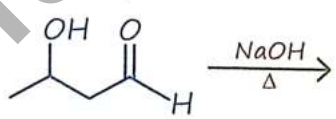
256. 

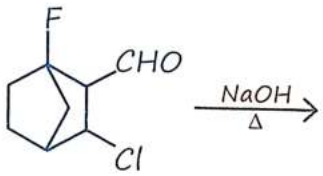
257. 

258. 

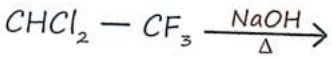
259. 

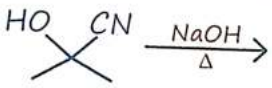
260. 

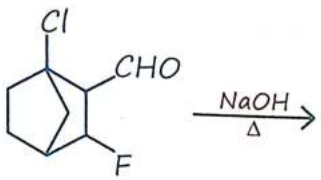
261. 

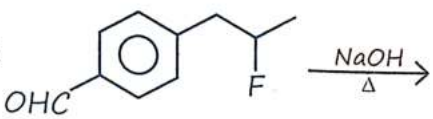
262. 

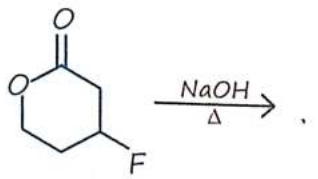
263.

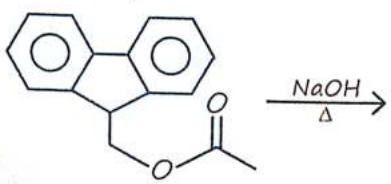
264. 

265. 

266. 

267. 

268. 

269. 

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Williamson's Ether Reaction

It is an important laboratory method for the preparation of symmetrical and unsymmetrical ethers. In this method, an alkyl halide is allowed to react with sodium alkoxide.

$$RO^{\ominus} + R'-X \longrightarrow R-O-R'$$

1°–X (Unbranched)	1°–X (Branched)	2°–X	3°–X	Ph–X X etc.
S_N2	E2 (Ma)	E2 (Ma)	E2	No Rxn

Unbranched 1° Halide: "Smooth S_N2... I'm the perfect fit."

Branched 1° or 2° Halide: "S_N2 gets blocked, so we let E_2 handle it."

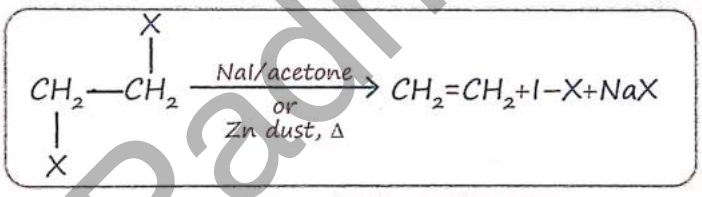
3° Halide: "Forget ether. I'm all about elimination now."

Phenyl/Vinyl Halide: "Nice try. I don't even react."

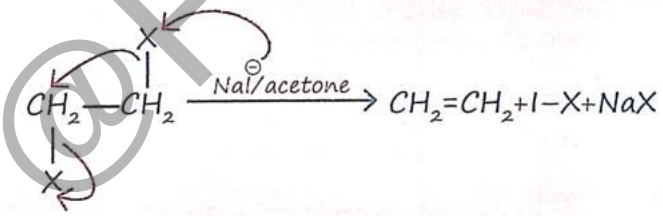
Dehalogenation Reaction

Vicinal dihalides undergo elimination upon treatment with zinc/acetic acid or Na–acetone, forming alkenes. This is an E_2 reaction and follows anti-periplanar geometry for elimination.

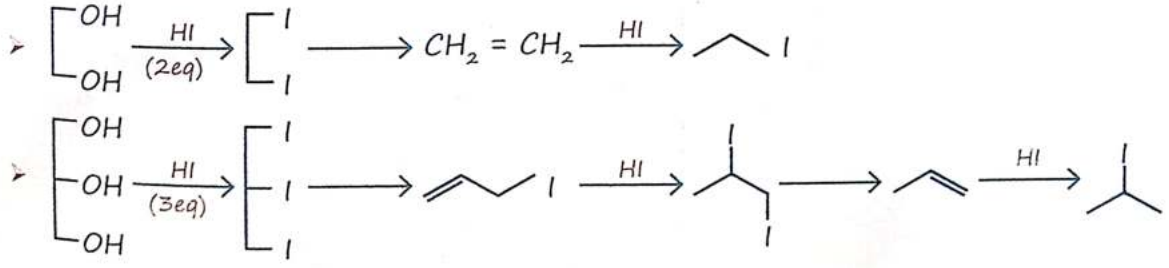
Reaction



Mechanism



Under harsh conditions (heat, excess HX), the reaction may proceed through several intermediates to yield a stable alkene.



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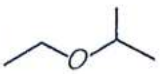
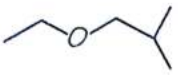
Practice Questions

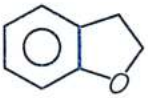
270. $RONa + R'-X \longrightarrow$ 

271. $? \longrightarrow$ 

272. $? \longrightarrow CH_3-O-C(CH_3)_2-CH_3$

273. $? \longrightarrow$

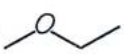
274. $? \longrightarrow$

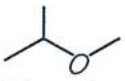
275. $? \longrightarrow$ 

276. $? \longrightarrow$ Anisole

277. $? \longrightarrow Ph-O-Ph$

278. $? \longrightarrow Ph-O-$

279. $? \longrightarrow$ 

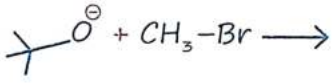
280. $? \longrightarrow$ 

281. $? \longrightarrow Ph-O-Ph-OCH_3$

282. $CH_3ONa + CH_3-X \longrightarrow$

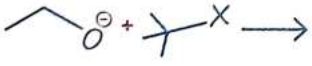
283. $CH_3ONa +$ $\longrightarrow$

284.

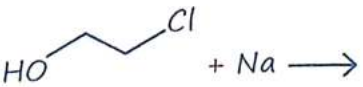
285. 

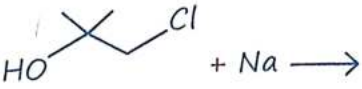
286.

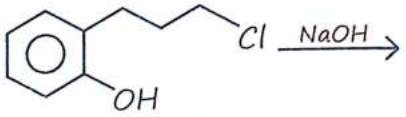
287.

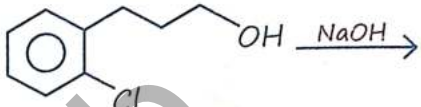
288. 

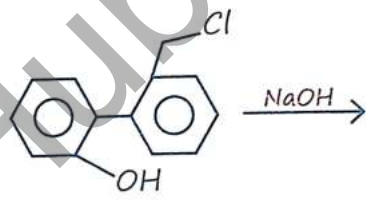
289.

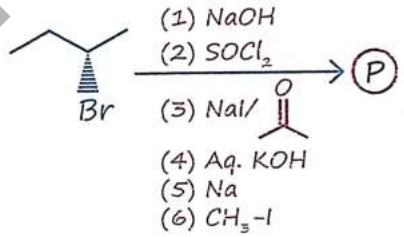
290. 

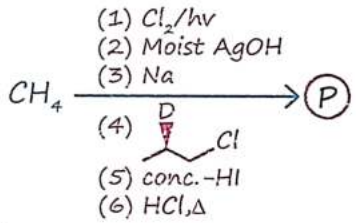
291. 

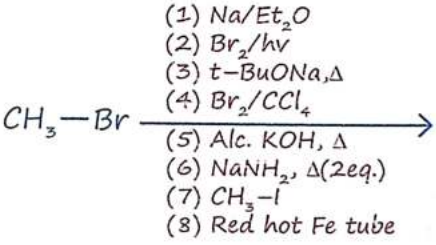
292. 

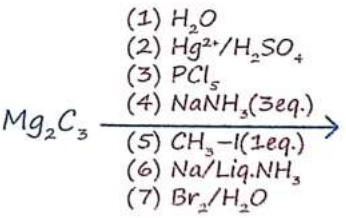
293. 

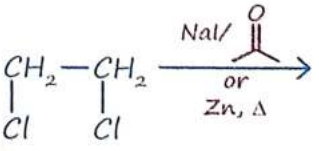
294. 

295. 

296. 

297. 

298. 

299. 

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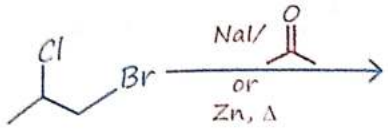
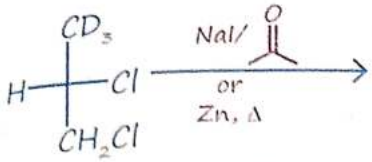
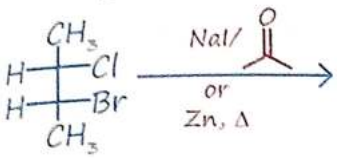
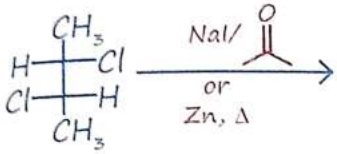
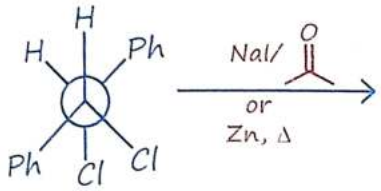
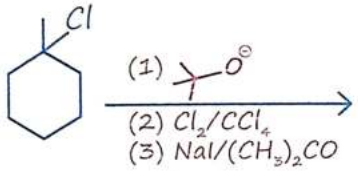
300. 

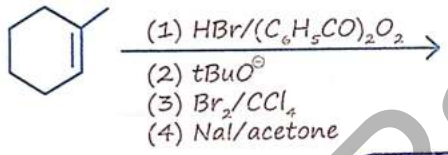
301. 

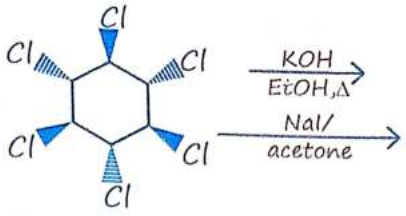
302. 

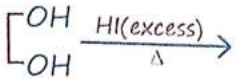
303. 

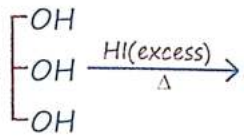
304. 

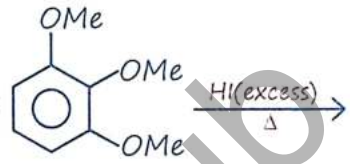
306. 

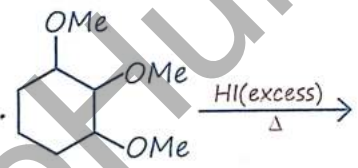
306. 

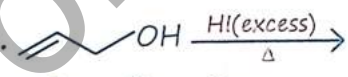
307. 

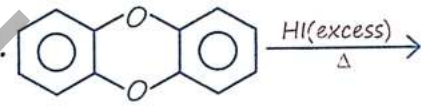
308. 

309. 

310. 

311. 

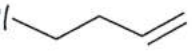
312. 

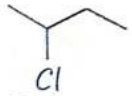
313. 

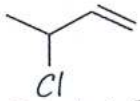
Previous Year Questions

27. The decreasing order of reactivity towards dehydrohalogenation (E_1) reaction of the following compounds is [8 Jan, 2020 (Shift-I)]

(A)

(B) 

(C) 

(D) 

(a) D > B > C > A (b) B > A > D > C
(c) B > D > C > A (d) B > D > A > C

Sol. (a)

28. Given below are two statements: [04 April, 2025 (Shift-II)]

Statement (I): Alcohols are formed when alkyl chlorides are treated with aqueous potassium hydroxide by elimination reaction.

Statement (II): In alcoholic potassium hydroxide, alkyl chlorides form alkenes by abstracting the hydrogen from the b-carbon.

In the light of the above statements, choose the most appropriate answer from the options given below:

(a) Both Statement I and Statement II are incorrect
(b) Statement I is incorrect but Statement II is correct
(c) Statement I is correct but Statement II is incorrect
(d) Both Statement I and Statement II are correct

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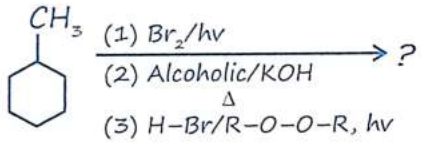
Sol. (b) Statement I is incorrect because aq. KOH causes substitution.

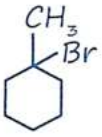
$R-Cl \xrightarrow{(aq.KOH)} R-OH$

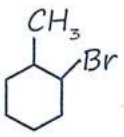
Statement II is correct because alcoholic KOH causes elimination

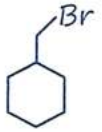
29. Predict the major product of the following reaction sequence:-

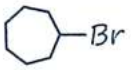


[04 April, 2025 (Shift-I)]

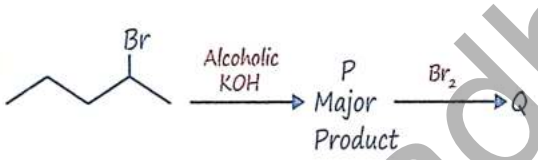
(a) 

(b) 

(c) 

(d) 

Sol. (b)

30. 

Consider the above sequence of reactions. 151 g of 2-bromopentane is made to react. Yield of major product P is 80% whereas Q is 100%.

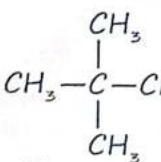
Mass of product Q obtained is _____g.

(Given molar mass in g mol^{-1} H: 1, C : 12, O : 16, Br : 80) [02 April, 2025 (Shift-II)]

Sol. [184]

31. Given below are two statements:

Statement I: CH_3-O-CH_2-Cl will undergo S_N1 reaction though it is a primary halide.

Statement II:  will not undergo SN_2 reaction very easily though it is a primary halide.

In the light of the above statements, choose the most appropriate answer from the options given below: [22 Jan, 2025 (Shift-I)]

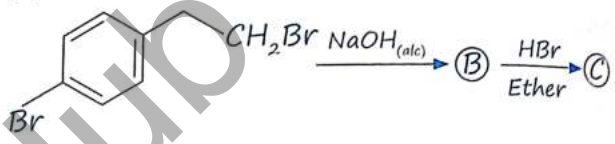
(a) Statement I is incorrect but statement II is correct.

(b) Both statement I and statement II are correct.

(c) Both statement I and statement II are incorrect.

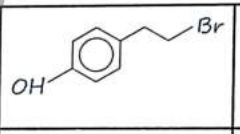
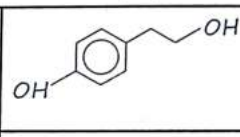
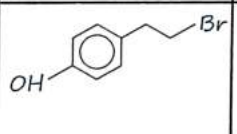
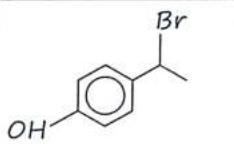
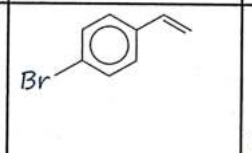
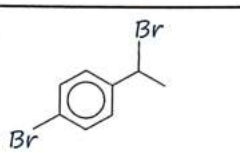
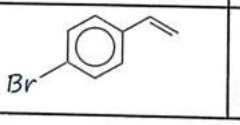
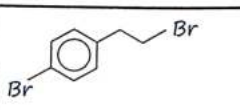
(d) Statement I is correct but statement II is incorrect.

Sol. (b)

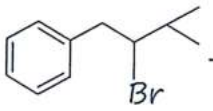
32. 

Identify B and C and how are A and C related?

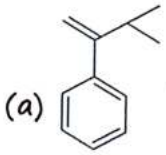
[04 April, 2024 (Shift-I)]

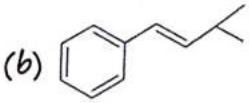
	(A)	(B)	(C)
(a)			functional group isomers
(b)			Derivative
(c)			position isomers
(d)			chain isomers

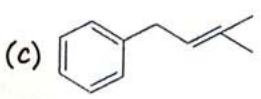
Sol. (c)

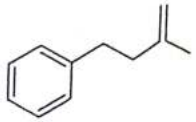
33. 

Product P is [04 April, 2024 (Shift-II)]

(a) 

(b) 

(c) 

(d) 

Sol. (b)

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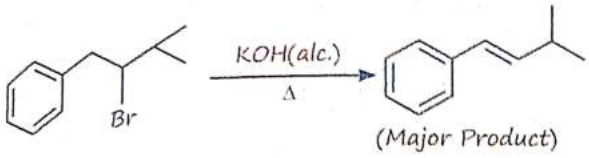


34. The major product of the following reaction is:

$CH_3CH_2CH(Br)-CH_2(Br)$ $\xrightarrow[\text{(ii) } NaNH_2 \text{ in liq. } NH_3]{\text{(i) KOH alc.}}$

[12 Jan, 2019 (Shift-II)]

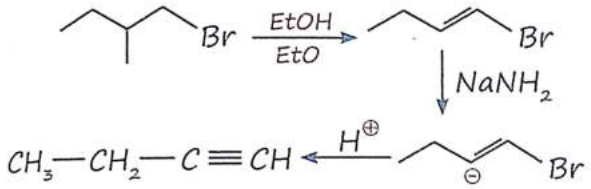
(a) $CH_3CH=C=CH_2$

(b) $CH_3CH_2CH(NH_2)-CH_2(NH_2)$

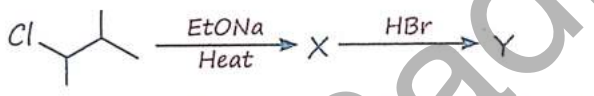
(c) $CH_3CH=CHCH_2NH_2$

(d) $CH_3CH_2C\equiv CH$

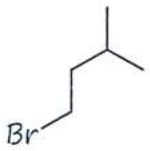
Sol. (b)

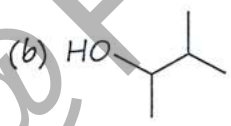


35. The major product 'Y' in the following reaction is:

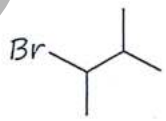


[10 April, 2019 (Shift-II)]

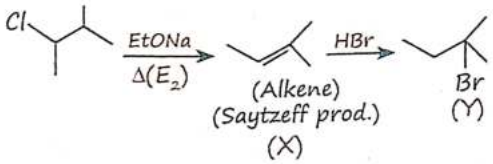
(a) 

(b) 

(c)

(d) 

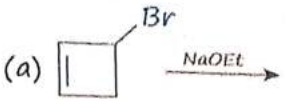
Sol. (c) The reaction occurs as:

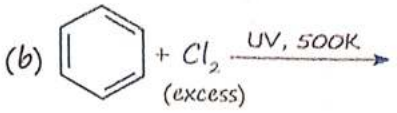


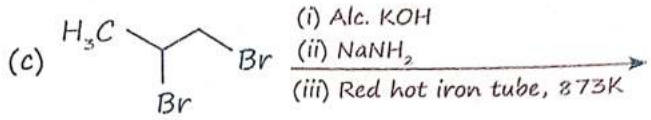
36. Choose the correct option(s) that give(s) an aromatic compound as the major product.

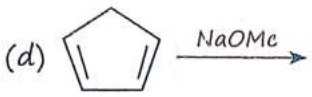
C-16.47 W-54.58 UA-12.34 PC-16.61

(JEE Adv. 2019)

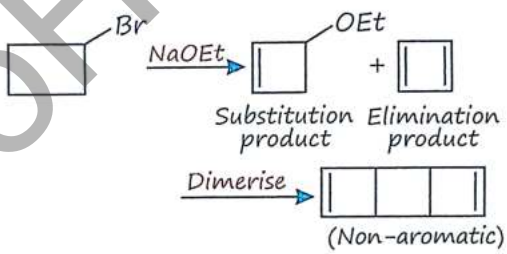
(a) 

(b) 

(c) 

(d) 

Sol. (c,d) For a compound to be aromatic, a cyclic and planar compound is required. It must follow $(4n + 2)$ e^- rule and must have the conjugated system in it.

(a) 

37. The product (C) in the below mentioned reaction is:

$CH_3-CH_2-CH_2-Br \xrightarrow[\Delta]{KOH(alc.)} A \xrightarrow{HBr} B \xrightarrow[KOH_{(aq.)}]{\Delta} C$

[31 Jan, 2024 (Shift-I)]

(a) Propan-1-ol (b) Propene

(c) Propyne (d) Propan-2-ol

Sol. (d) (Propan-2-ol)

38. $CH_3CH_2Br + NaOH \xrightarrow{C_2H_5OH}$ Product A

$\xrightarrow{H_2O}$ Product B

Total number of hydrogen atom in A and B is ____.

[31 Jan, 2024 (Shift-I)]

Sol. [10] $CH_3CH_2Br +$

$NaOH \xrightarrow{C_2H_5OH} CH_2=CH_2$ (A)

$NaOH \xrightarrow{H_2O} CH_3-CH_2-OH$ (B)

Therefore, total number of hydrogen atoms in compounds A and B are 10.

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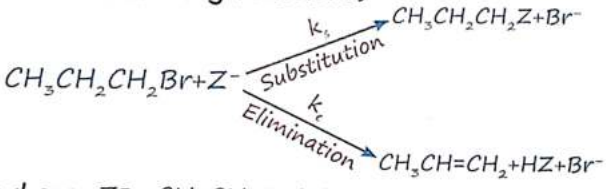
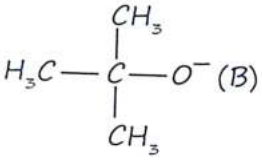
39. For the following reactions,



where, $Z^- = CH_3CH_2O^-$ (A) or



k_s and k_e are respectively, the rate constants for substitution and elimination reactions and $\mu = \frac{k_s}{k_e}$, the correct option is ____.

[7 Jan, 2020 (Shift-II)]

(a) $\mu_A > \mu_B$ and k_e (B) > k_e (A)

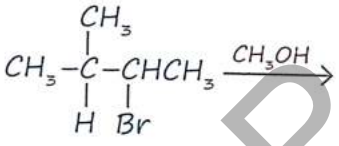
(b) $\mu_A > \mu_B$ and k_e (A) > k_e (B)

(c) $\mu_B > \mu_A$ and k_e (B) > k_e (A)

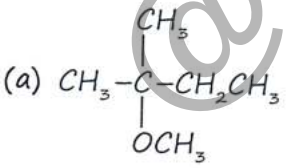
(d) $\mu_B > \mu_A$ and k_e (A) > k_e (B)

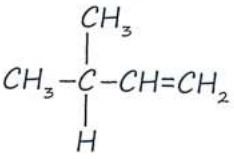
Sol. (a) $Z^{\ominus} = CH_3CH_2 - O^{\ominus}$ favors substitution over elimination where as $Z^{\ominus} = (CH_3)_3CO^{\ominus}$ favors elimination over substitution.

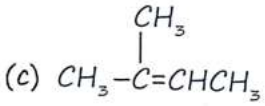
40. The major product of the following reaction is

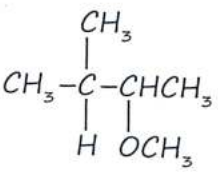


[10 April, 2019 (Shift-I)]

(a) 

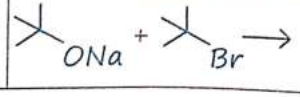
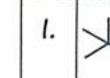
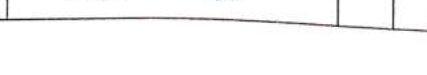
(b) 

(c) 

(d) 

Sol. (a)

41. List-I contains reactions and List-II contains major products.

List-I (Compound)		List-II (Uses)	
P.		I.	
Q.		II.	
R.		III.	
S.		IV.	
		V.	

Match each reaction in List-I with one or more products in List-II and choose the correct option.

C-31.03 W-55.2 UA-13.76 (JEE Adv. 2018)

(a) P → 1, 5; Q → 2; R → 3; S → 4

(b) P → 1, 4; Q → 2; R → 4; S → 3

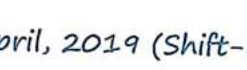
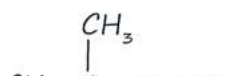
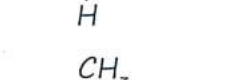
(c) P → 1, 4: Q → 1,2; R → 3, 4; S → 4

(d) P → 4, 5; Q → 4; R → 4; S → 3, 4

Sol. (b)

42. Match the chemical conversion in Column-I with the appropriate reagents in Column-II and select the correct answer using the code given below the lists.

C-49.72 W-41.17 UA-9.12 (JEE Adv. 2013)

Column-I (Compound)		Column-II (Uses)	
P.		I.	(i) $Hg(OAc)_2$; (ii) $NaBH_4$
Q.		II.	NaOEt
R.		III.	Et–Br
S.		IV.	(i) BH_3; (ii) H_2O_2 / NaOH

	P	Q	R	S
(a)	2	3	1	4
(b)	3	2	1	4
(c)	2	3	4	1
(d)	3	2	4	1

Sol. (a)

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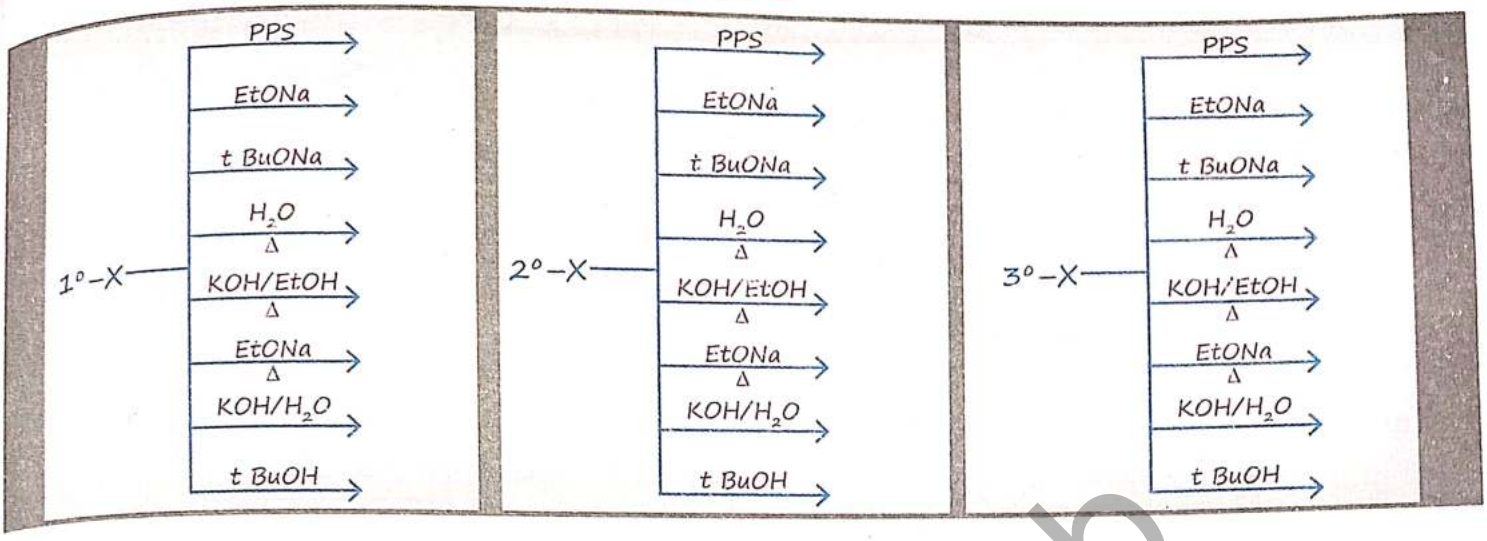
SN Vs E



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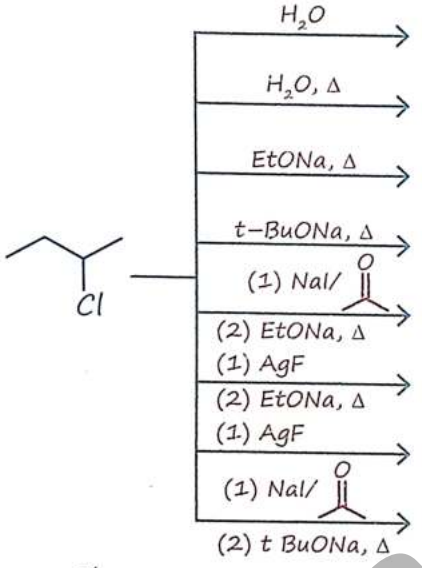
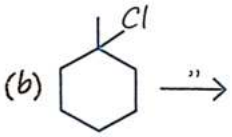
314. The rate law of reaction of $\overset{\ominus}{O}H$ with to form an elimination product in 75% EtOH and 25% H_2O :

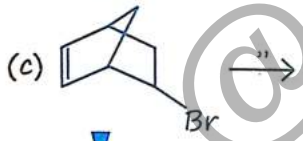
Rate = 7.1×10^{-5} $[(CH_3)_3C - Br]$ $[\overset{\ominus}{O}H]$ + 1.5×10^{-6} $[(CH_3)_3C-Br]$

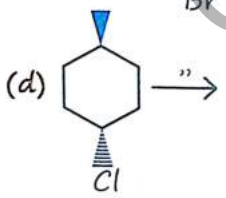
What % of the reaction takes place by E2 pathway when :

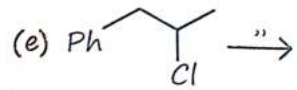
(a) $[\overset{\ominus}{O}H]$ = 5.0 M (b) $[\overset{\ominus}{O}H]$ = 0.0025 M

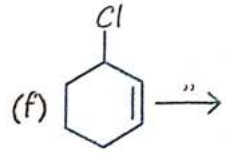
(a) 

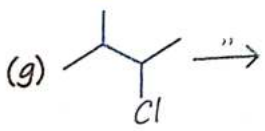
(b) 

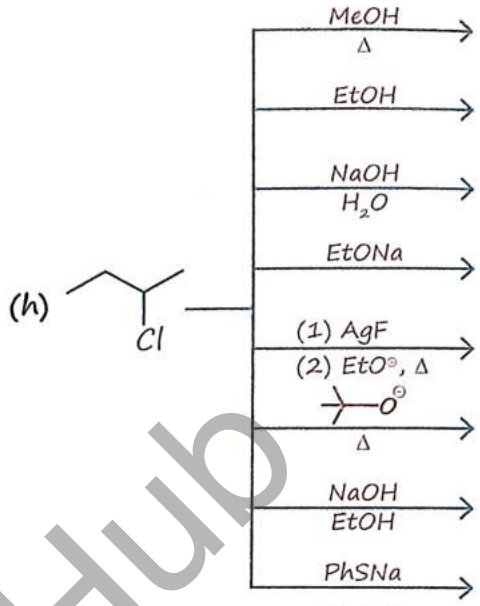
(c) 

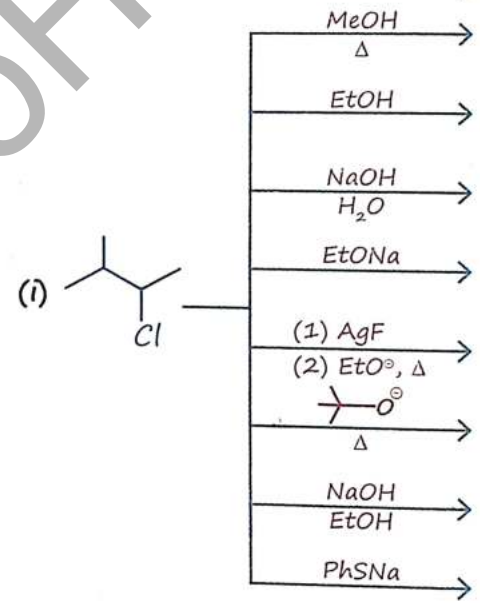
(d) 

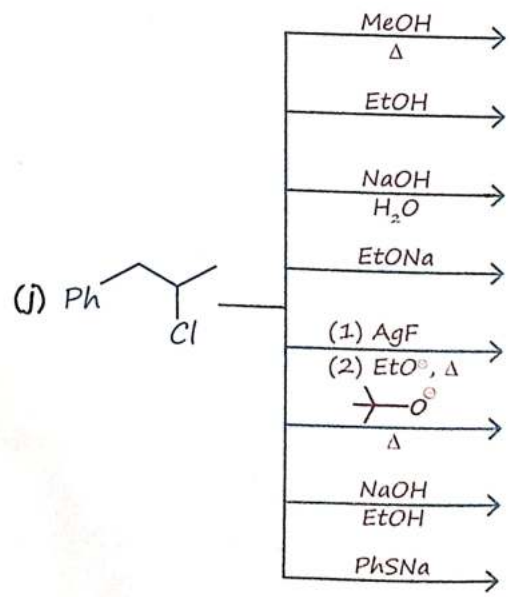
(e) 

(f) 

(g) 

(h) 

(i) 

(j) 

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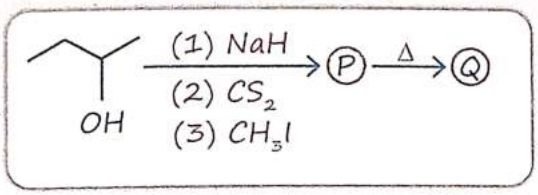
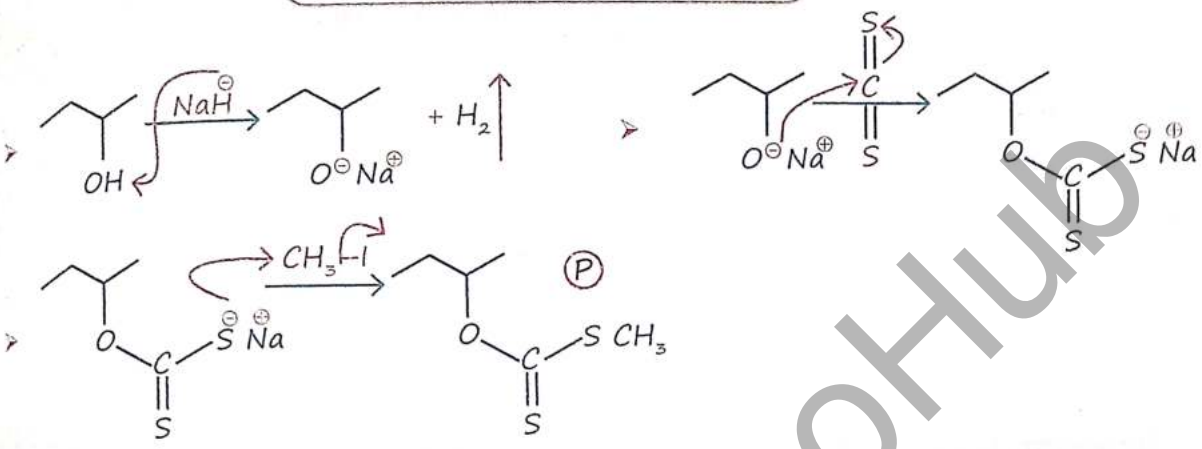
Ei Reaction

An alcohol is converted into a xanthate ester (via CS_2 and CH_3I), which upon heating undergoes Ei elimination to form an alkene.
This reaction proceeds via a six-membered cyclic transition state, releasing CH_3SH and CO_2.

Reaction

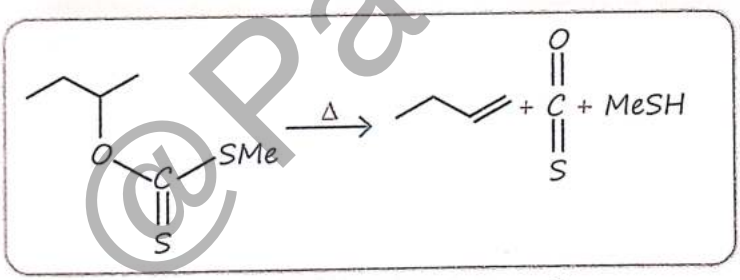




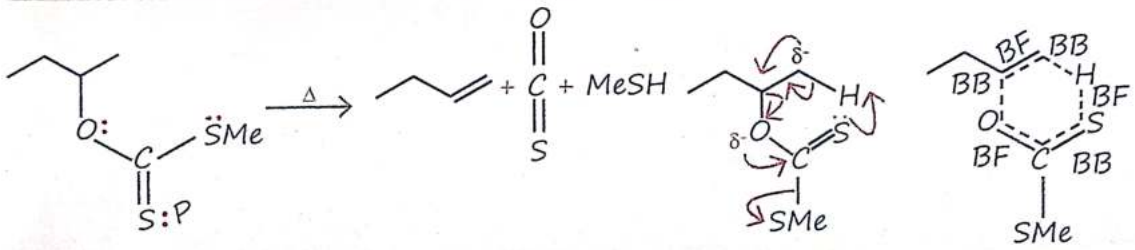
Pyrolysis of Xanthate

Upon heating, xanthate esters undergo syn elimination through a 6-membered cyclic transition state to yield an alkene. It follows the Ei mechanism, releasing volatile by-products like MeSH and CO_2, and is useful for converting alcohols to alkenes.

Reaction



Mechanism –



Baba Tea-Stall (BTS)

(i) 6MCTS involved
(ii) Ei Mechanism
(iii) δ⁻ stability decides
(iv) Syn elimination

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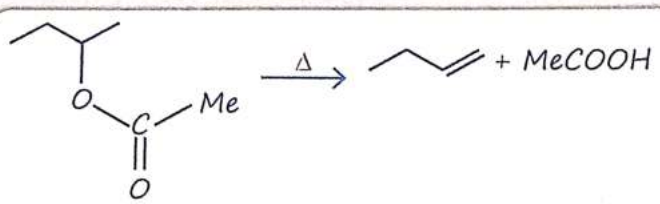
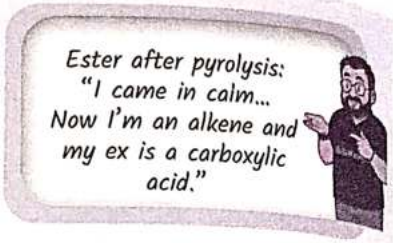
Pyrolysis of Ester

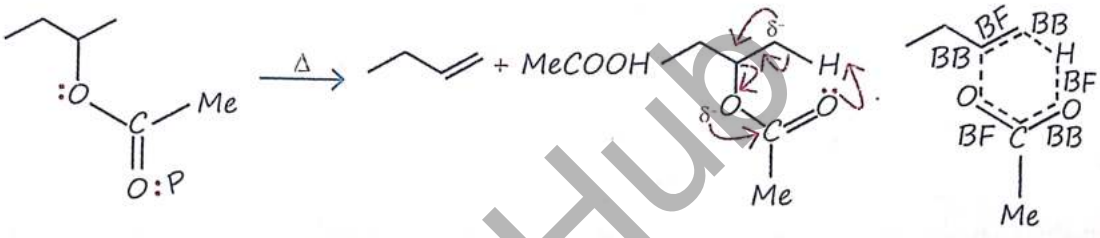
Esters on heating also undergo Ei elimination through a 6-membered cyclic transition state, forming alkenes and acids (like MeCOOH). The elimination is syn, and driven by the formation of a stable π-bond and acidic by-products.

Reaction





Mechanism



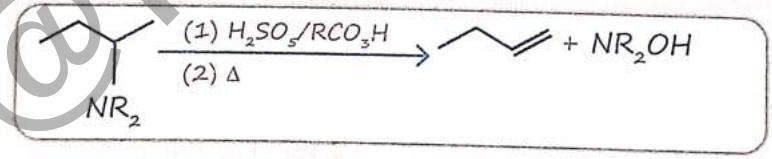
Baba Tea-Stall (BTS)

(i) 6MCTS involved
(ii) Ei Mechanism
(iii) δ⁻ stability decides
(iv) Syn elimination

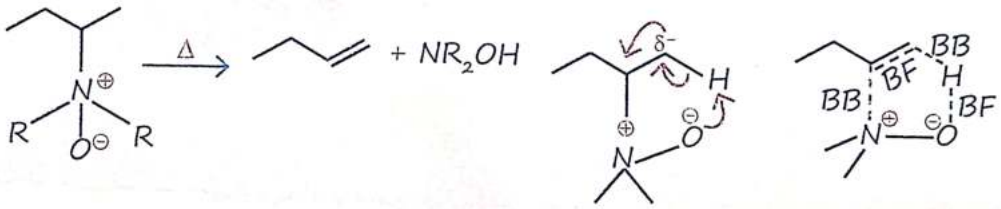
Cope Elimination

Tertiary amines (NR_2) are first oxidized to amine oxides using peracids like H_2O_2. On heating, they eliminate via a 5-membered cyclic transition state to give an alkene and hydroxylamine.
It is a classic example of syn-Ei elimination.

Reaction



Mechanism



Baba Tea-Stall (BTS)

(i) 5 MCTS involved
(ii) Ei Mechanism
(iii) δ⁻ stability decides
(iv) Syn elimination

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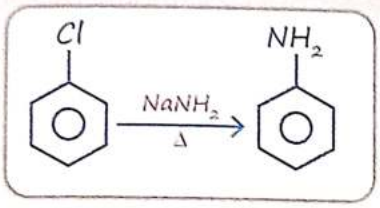
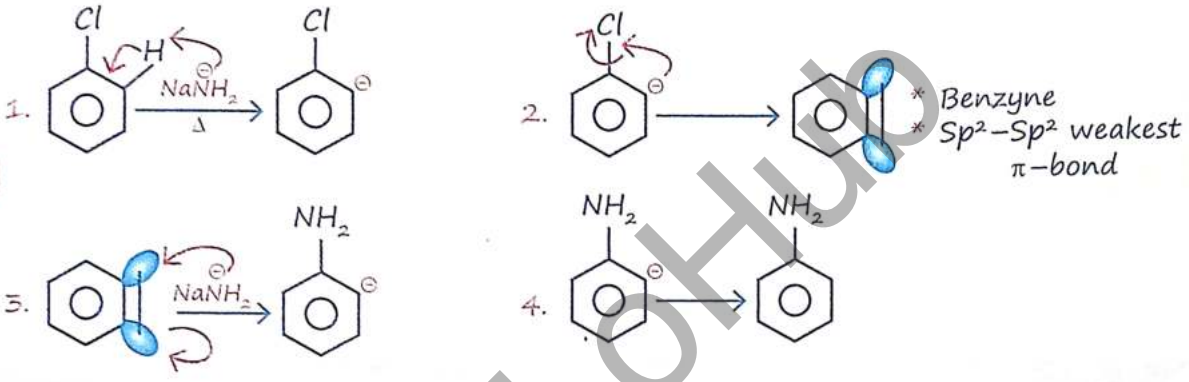
Cine Substitution (S_NEA)

In this reaction, aryl halides (like chlorobenzene) react with strong bases ($NaNH_2$) at high temperatures via the benzyne intermediate.
The nucleophile adds at the position adjacent to the leaving group (cine position), followed by protonation.
It is an example of nucleophilic aromatic substitution via elimination–addition (S_NEA).

Reaction



Mechanism



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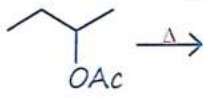
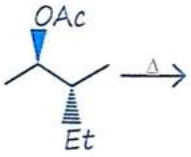
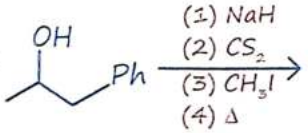
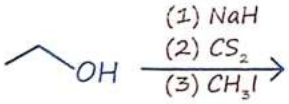
Practice Questions

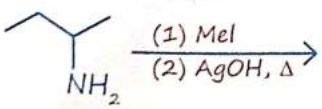
315. $CH_3CH_2OH \xrightarrow[(2)\ \Delta]{(1)\ CH_3COCl/Py.}$

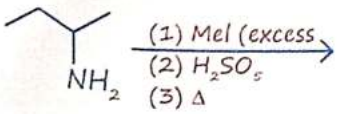
316. 

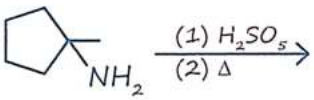
317. 

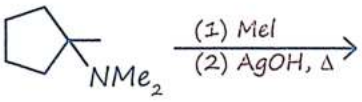
318. 

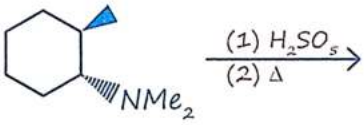
319. 

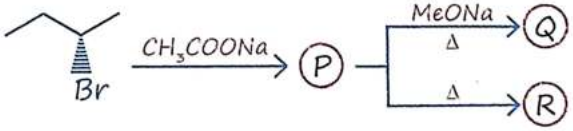
320. 

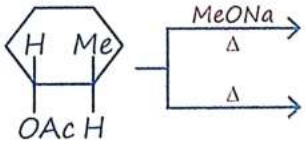
321. 

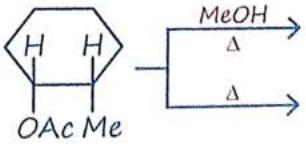
322. 

323. 

324. 

325. 

326. 

327. 

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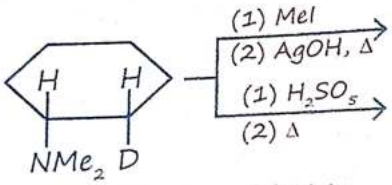
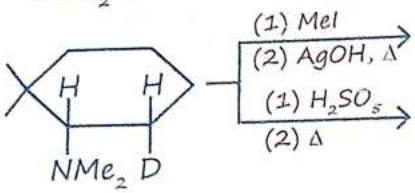
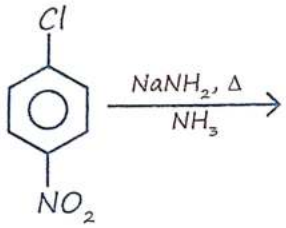
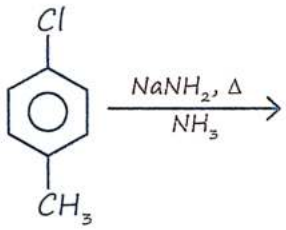
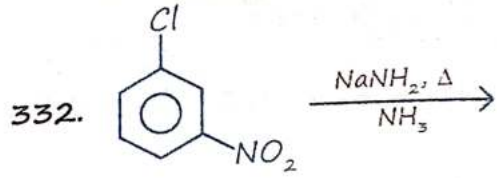
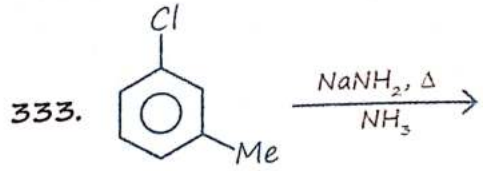
328. 

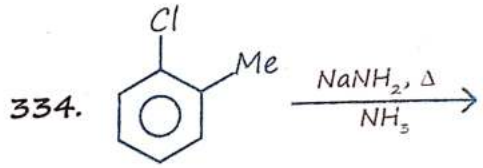
329. 

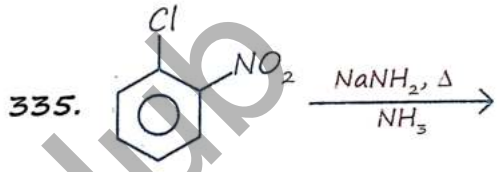
330. 

331. 

332. 

333. 

334. 

335. 

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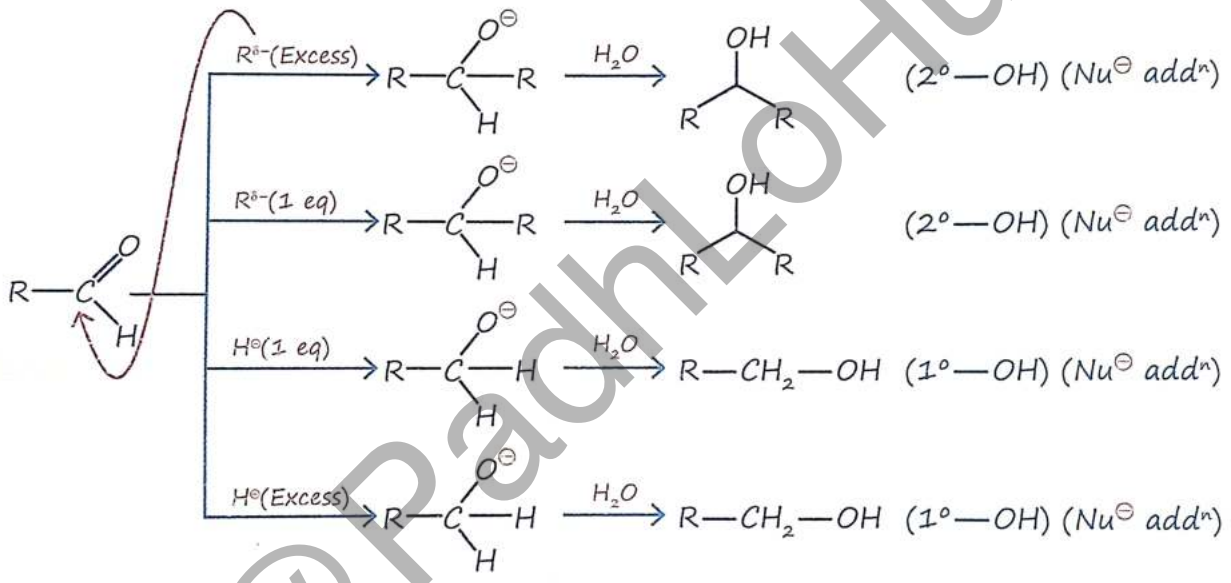
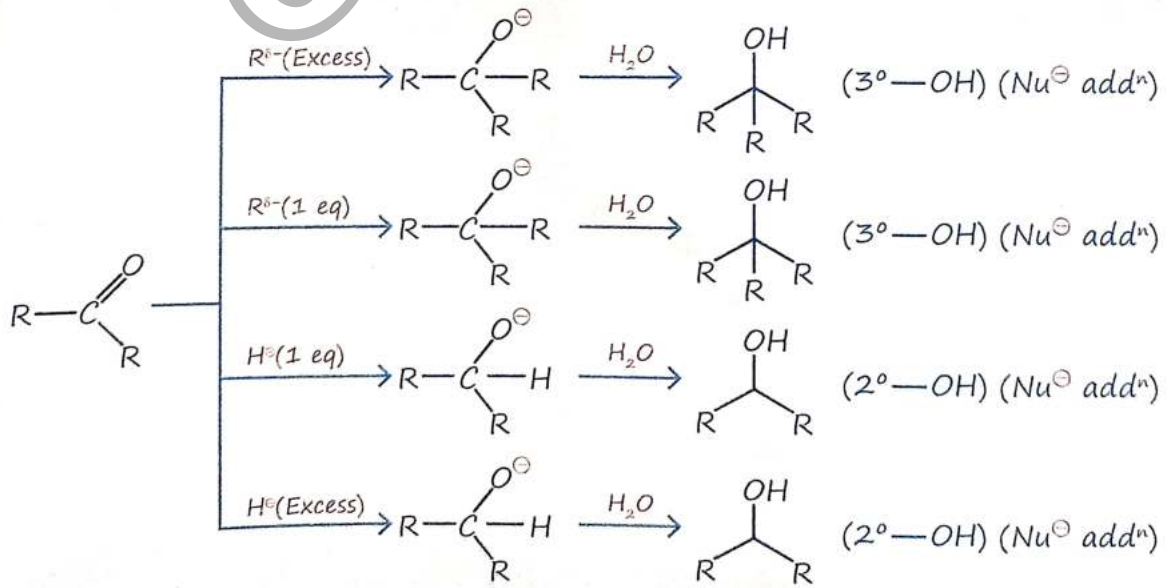
6 Alcohols, Phenols and Ethers

Reduction Basics

Nucleophilic addition reactions of aldehydes and ketones with Grignard (R^-) and hydride (H^-) reagents, showing the formation of different types of alcohols (1°, 2°, 3°) depending on the reagent and substrate used.





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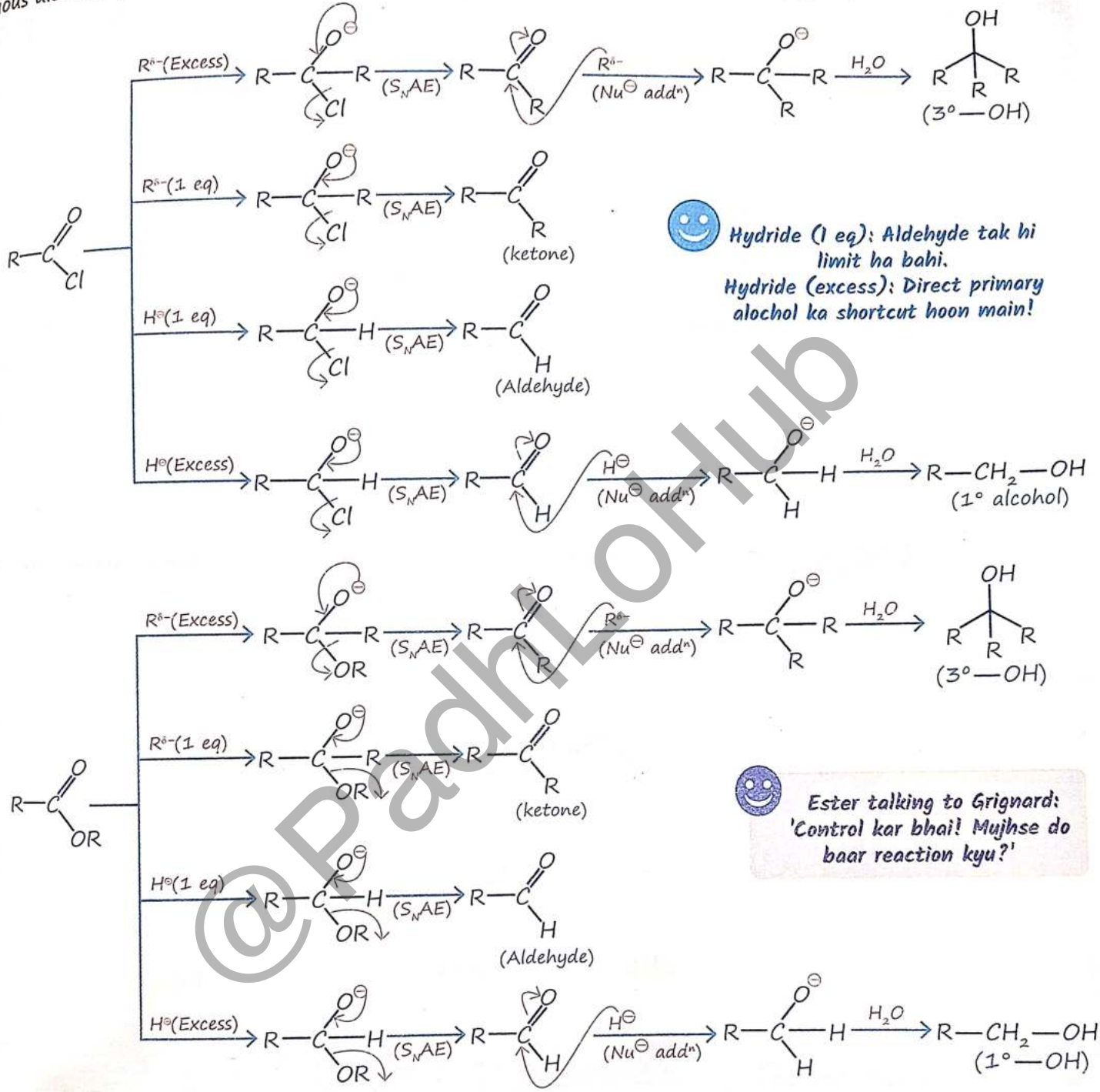
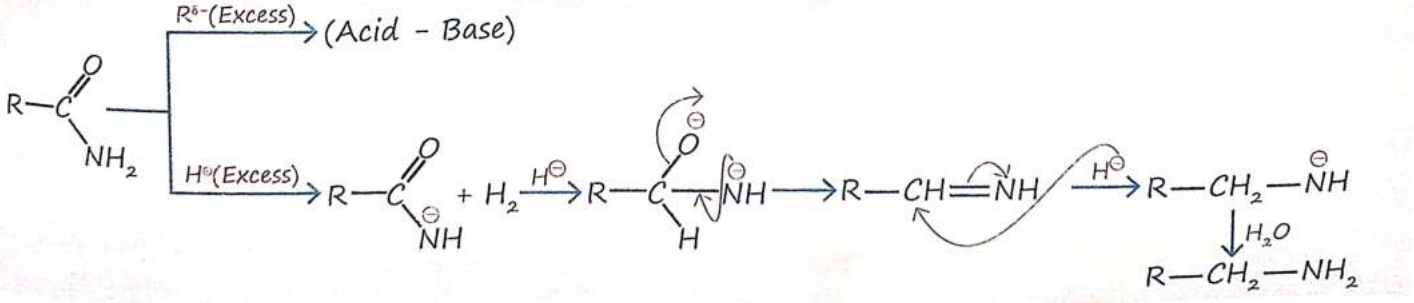
Nucleophilic addition (Nu^- add) to esters and acid chlorides form ketones, aldehydes, and ultimately various alcohols (1°, 2°, 3°) depending on the reagent equivalents and leaving group.



Nucleophilic addition reactions of amides ($R-CONH_2$) with Grignard (R^-) and hydride (H^-) reagents, leading to amines via imine intermediates and ultimately yielding primary amines ($R-CH_2-NH_2$) upon hydrolysis.



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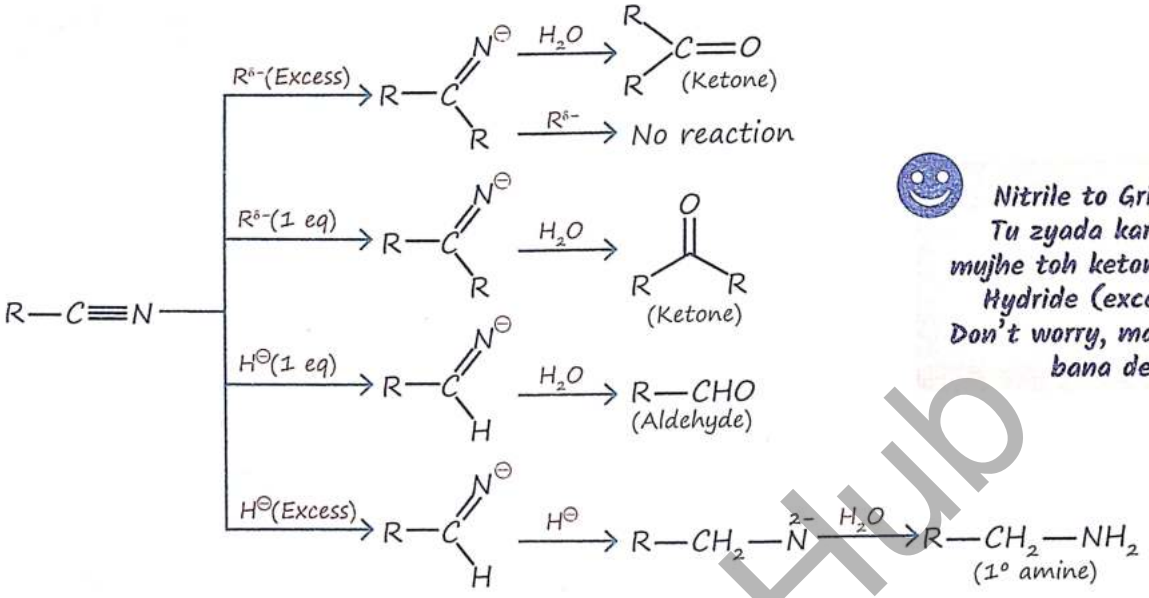
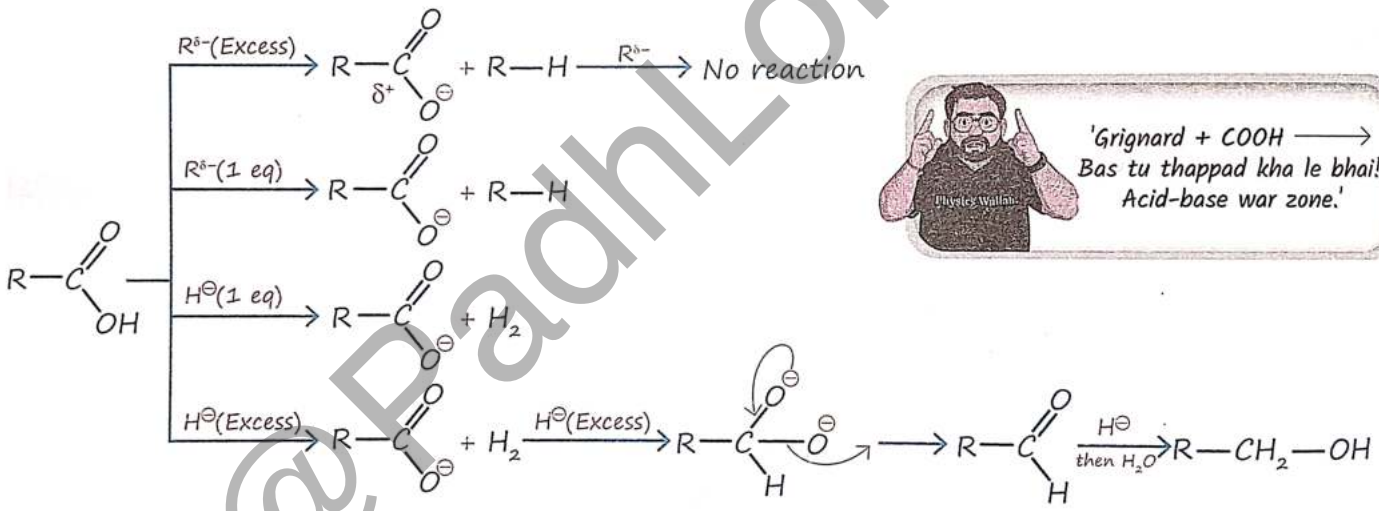
Nucleophilic addition reactions involving nitriles (R–C≡N) and carboxylic acids (R–COOH), showing how Grignard (R^-) and hydride (H^-) reagents lead to ketones, aldehydes, or primary amines depending on the reagent and conditions.



Nitrile to Grignard (excess):
Tu zyada kar raha hai bhai... mujhe toh ketone hi banna tha!
Hydride (excess) to nitrile:
Don't worry, main seedha amine bana deta hoon!



'Grignard + COOH ⟶ Bas tu thappad kha le bhai! Acid-base war zone.'

Recap: Reaction of Grignard Reagent with Carbonyl Compounds

Compound Type	Product	Reaction Type	Notes
Aldehyde / Formaldehyde	2° Alcohol	Nucleophilic Addition	Adds R to C=O → then H_2O gives alcohol
Ketone	3° Alcohol	Nucleophilic Addition	Adds R to two R groups
Ester/Acid Chloride	3° Alcohol/Ketone	S_NAE	1 eq → ketone, excess → alcohol
Nitrile	Ketone	Nu^- Addⁿ	Via imine intermediate
COOH / Amide	No C–C Bond	Acid–Base Reaction	Just salt formation, no addition

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1. Organometallic Compounds

Compound having C–Metal bond are known as organometallic compound.

➢ More ΔE_n, More ionic and more reactive

Example: (i) RMgX → Grignard Reagent

(ii) R_2CuLi → Gillmann Reagent

C	Na	Li	Mg	Cu	Zn	Hg
2.5	0.9	1.0	1.3	2.0	2.2	2.3

RNa RLi RMgX - - - - - - R_2CuLi R_2Zn R_2Cd

Highly reactive (Hard Nucleophile) — Less Reactive (Soft Nucleophile)

Formation of organometallic compounds, including Grignard reagents and Gilmann reagents, through reactions involving halides and metals, with key steps such as transmetallation highlighted.

1. $R-X + Mg \xrightarrow{Et_2O} \overset{\delta-}{R}\underset{(-I)}{-}\overset{\oplus}{Mg}\overset{\ominus}{X}$ (Grignard reagent)

2. $R-X + 2Li \longrightarrow \overset{\ominus\oplus}{RLi} + LiX$

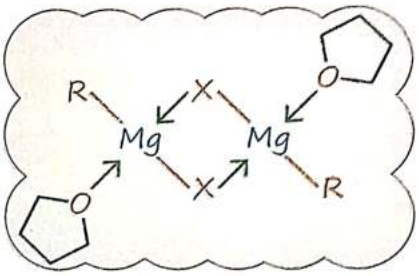
3. $\overset{\delta-}{RLi} + CuX \longrightarrow R_2CuLi + LiX$ (2eq) (Gillmann reagent)

4. $RMgX + CdCl_2 \longrightarrow R_2Cd + 2XMgCl$

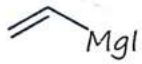
5. $RMgX + CuX \longrightarrow R-Cu + MgX_2$

} Transmetallation

Grignard Reagent

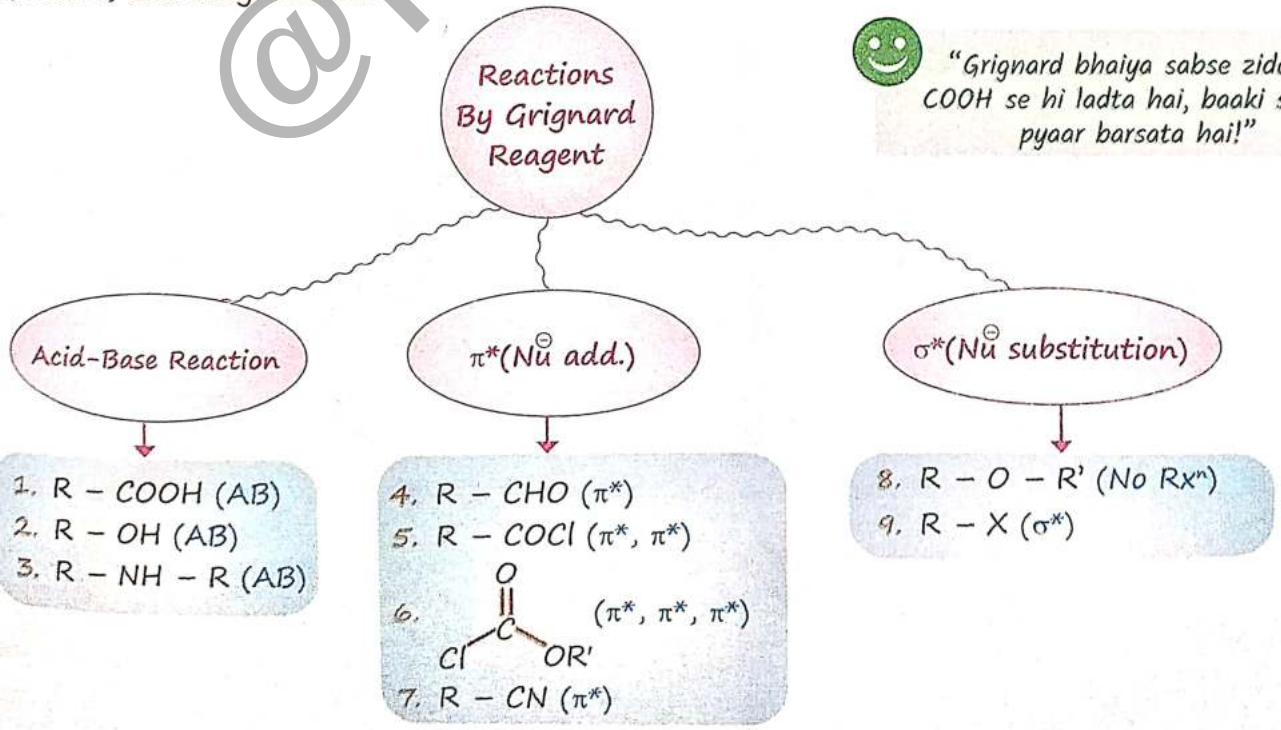


➢ RMgX is an organometallic compound and source of carbanion.

Example: $\overset{\delta-}{R}MgX$  PhMgBr

➢ In inert solvent, it exit as dimer. $2RMgX \rightleftharpoons MgX_2 \rightleftharpoons$ Dimer

Reactions of Grignard reagents into three types: acid-base reactions, nucleophilic addition (to π^* electrophiles), and nucleophilic substitution (to σ^* electrophiles), with examples like carboxylic acids, aldehydes, esters, and alkyl halides.



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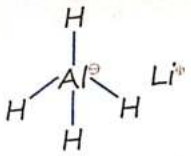
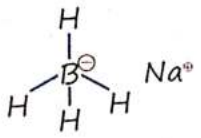
2. Metallic Hydride

1. $LiAlH_4$ (LAH)

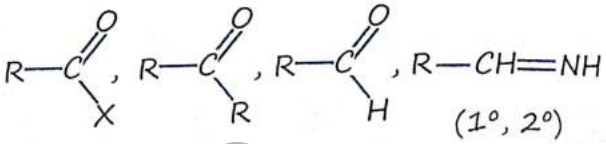


- Solvent ⇒ Polar aprotic solvent
- Nucleophilic Reducing agent
- Highly reactive so not chemoselective
- Reduces all functional group
  - O containing ⇒ alcohol
  - N containing ⇒ amine

2. $NaBH_4$ (SBH)



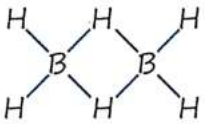
- Solvent ⇒ Polar protic solvent
- Nucleophilic Reducing Agent
- Chemoselective
- Reacts only with:



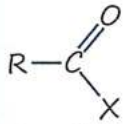
(1°, 2°)

- LBH ⇒ $LiBH_4$ = SBH + ester

3. B_2H_6 (BH_3.THF)



- Solvent ⇒ Polar aprotic solvent
- Electrophilic Reducing Agent
- Chemoselective
- Reacts with all except

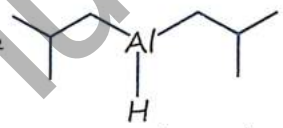


(acid halide)

- Reactivity: R–COOH > Alkene > .....

4. DIBAL–H

$AlH(i\text{-}Bu)_2$



- Solvent ⇒ Non-polar solvent
- Electrophilic Reducing agent
- Chemoselective
- Reacts with all functional group but only once (1 $H^{\ominus}$)
- Reacts at −78°C
- Reactivity: R–COOR' > R–COOH

LAH: Main sab kuch reduce kar dunga!
$NaBH_4$: Bas gentle hoon, sirf aldehyde aur ketone!
DIBAL-H: Thande dimaag se kaam karta hoon – selective hoon!

Recap: Reducing Agents (LAH, $NaBH_4$, DIBAL-H)

Agent	Selectivity	Solvent	Reduces
LAH	Non-selective	Polar aprotic	All C=O and N groups
$NaBH_4$	Mild/ Selective	Polar protic	Aldehyde, Ketone only
DIBAL-H	Very Selective	Non-polar	Esters → Aldehyde

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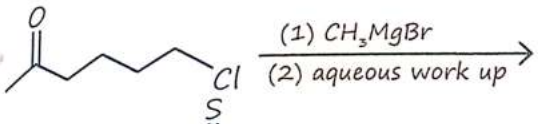
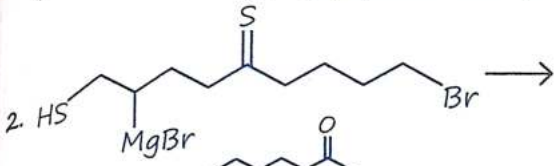
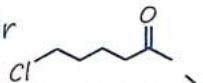
Practice Questions

1. 

2. 

3. CH_3-CHO 

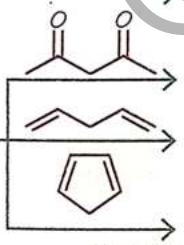
4. CH_3MgBr → H_2O / ROH / ROR

5. CH_3MgBr → RNH_2 / R_2NH / R_3N

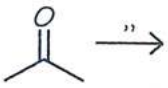
6. CH_3MgBr → RNH_2 / R_2NH / R_3N

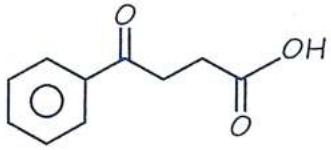
7. CH_3MgBr → HNO_2 / H_2SO_4 / HI

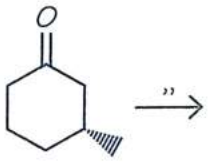
8. CH_3MgBr → CH_3NO_2 / R—COOH / RSH

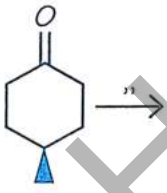
9. CH_3MgBr → 

10. CH_3-CHO → CH_3MgBr then H_2O / LAH then H_2O / SBH EtOH / (1) DIBAL (2) H_2O / (1) B_2H_6 (2) H_2O

11.  → ''

12.  → ''

13.  → ''

14.  → ''

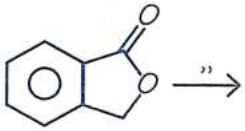
15. CH_3-CHO → CH_3MgBr then H_2O / LAH then H_2O / SBH EtOH / (1) DIBAL (2) H_2O / (1) B_2H_6 (2) H_2O

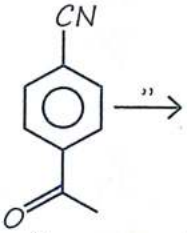
16. CH_3COOCH_3 → ''

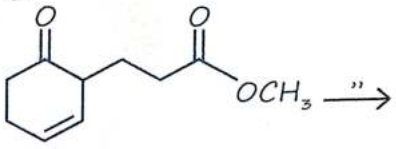
17. $CH_3COOCOCH_3$ → ''

18. $COCl_2$ → ''

19. CH_3-CN → ''

20.  → ''

21.  → ''

22.  → ''

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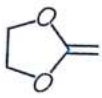
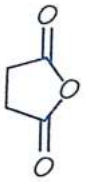
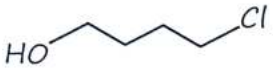
23. $CO_2 \xrightarrow[(2)\ H_2O]{(1)\ CH_3MgBr}$

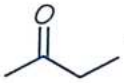
24. $SO_3 \xrightarrow[(2)\ H_2O]{(1)\ CH_3MgBr}$

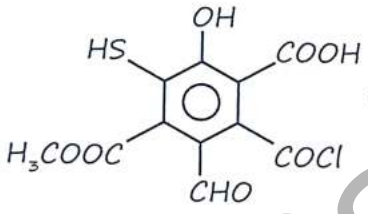
25.  $\xrightarrow[(2)\ H_2O]{(1)\ CH_3MgBr}$

26.  $\xrightarrow[(2)\ H_2O]{(1)\ CH_3MgBr}$

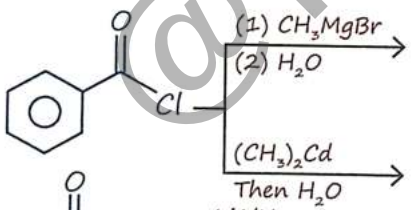
27. $\xrightarrow[(2)\ CO_2 \text{ then } H_2O]{(1)\ Mg/THF}$

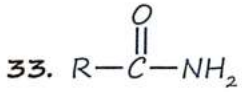
28.  $\xrightarrow[\text{then } H_2O]{CH_3MgBr}$

29.  $\xrightarrow[(2)\ D_2O]{(1)\ Ph\ Mg\ Br}$

30.  $\xrightarrow{\text{'x' } CH_3Mg\ Br}$

31. 'C_4H_8O' $\xrightarrow[\text{then } H_2O]{CH_3MgBr}$ Total products obtained
[Carbonyl compound]

32. 
(1) CH_3MgBr, (2) H_2O
$(CH_3)_2Cd$, Then H_2O

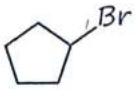
33.  $\xrightarrow[\text{or } B_2H_6]{LiAlH_4}$

34. D-Glucose $\xrightarrow[\text{or } H_2/Ni]{SBH}$

35. D-fructose $\xrightarrow[\text{or } H_2/Ni]{SBH}$

36. $CH_3Br \xrightarrow[(2)\ EtOH]{(1)\ Mg/Et_2O}$

37. $PhCH_2Br \xrightarrow[(2)\ EtOD]{(1)\ Mg/Et_2O}$

38.  $\xrightarrow[(2)\ D_2O]{(1)\ Mg/Et_2O}$

39. $C_5H_{10}O$ [Carbonyl compound] $\xrightarrow[\text{then } H_2O]{CH_3\ MgBr}$ Total products obtained

40. $C_6H_{12}O$ [Carbonyl compound] $\xrightarrow[\text{then } H_2O]{CH_3\ MgBr}$ Total products obtained

41. **Final all possible R —X?**

42. $R-X \xrightarrow[(2)\ H_2O]{(1)\ Mg}$ Ethane

43. $R-X \xrightarrow[(2)\ H_2O]{(1)\ Mg}$ Propane

44. $R-X \xrightarrow[(2)\ H_2O]{(1)\ Mg}$ n-butane

45. $R-X \xrightarrow[(2)\ H_2O]{(1)\ Mg}$ Isopentane

46. $R-X \xrightarrow[(2)\ H_2O]{(1)\ Mg}$ 3-methyl pentane

47. $R-X \xrightarrow[(2)\ H_2O]{(1)\ Mg}$ Methyl Cylohexane

48. $R-X \xrightarrow[(2)\ D_2O]{(1)\ Mg}$

49. $R-X \xrightarrow[(2)\ D_2O]{(1)\ Mg} CH_3-CH_2-C\equiv C-D$

50.
(1) Sn/HCl
(2) HNO_2
(3) CuCN/KCN
(4) PhMgBr
(5) $H_3O^{\oplus}$

51. $CH_3MgCl + CS_2 \longrightarrow$

52. $CH_3MgCl + S \longrightarrow$

53. $CH_3MgCl + O_2 \longrightarrow$
1 : 1

54. $CH_3MgCl + O_2 \longrightarrow$
2 : 1

55. $CH_3MgCl + SO_2 \longrightarrow$

56. $CH_3MgCl + H_2N-Cl \longrightarrow$
2 : 1

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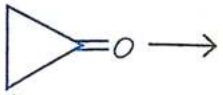
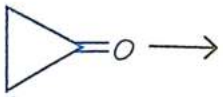
57. $CH_3MgCl + NC-CN \longrightarrow$

58. $CH_3MgCl + NH_2-OEt \longrightarrow$

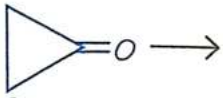
59. $CH_3MgCl + PhO-NH_2 \longrightarrow$

60. $CH_3MgCl +$  $\longrightarrow$

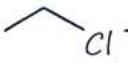
1 : 1

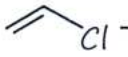
61. $CH_3MgCl +$  $\longrightarrow$

2 : 1

62. $CH_3MgCl +$  $\longrightarrow$

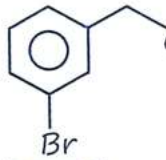
1 : 2

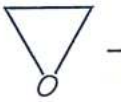
63. $CH_3MgCl +$  $\longrightarrow$

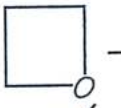
64. $CH_3MgCl +$  $\longrightarrow$

65. $CH_3MgCl +$ $\longrightarrow$

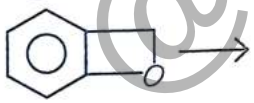
66. $CH_3MgCl + PhCl \longrightarrow$

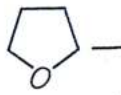
67. $CH_3MgCl +$  $\longrightarrow$

68. $CH_3MgCl +$  $\longrightarrow$

69. $CH_3MgCl +$  $\longrightarrow$

70. $CH_3MgCl +$ $\longrightarrow$

71. $CH_3MgCl +$ 

72. $CH_3MgCl +$  $\longrightarrow$

73. $CH_3MgCl +$ $/AlCl_3 \longrightarrow$

74. $CH_3MgCl + SnCl_4 \longrightarrow$

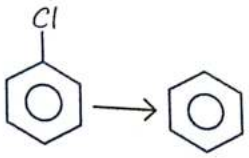
75. $CH_3MgCl + TiCl_3 \longrightarrow$

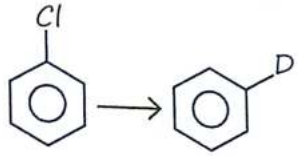
76. $CH_3MgCl + ZnCl_2 \longrightarrow$

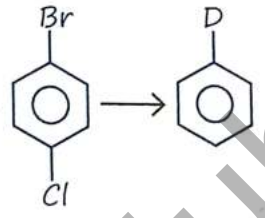
77. $CH_3MgCl + CuCl \longrightarrow$

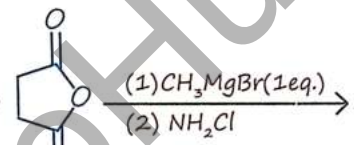
78. $CH_3MgCl + PbCl_4 \longrightarrow$

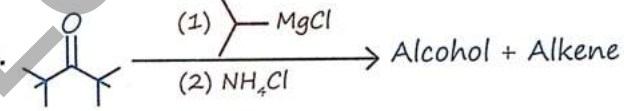
79. Convert the following using Mg/Et_2O and H_2O or D_2O: (75-77)

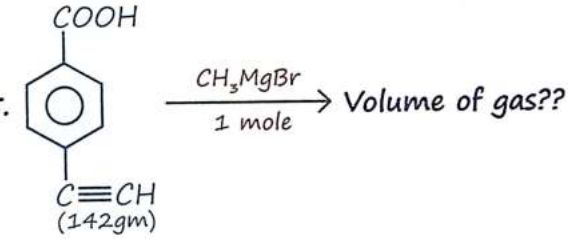
80. 

81. 

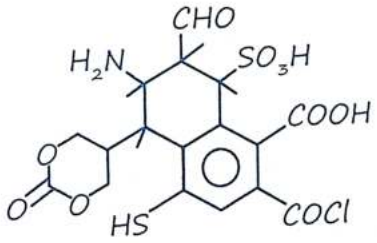
82. 

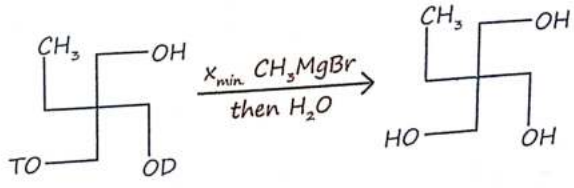
83. 

84. 

85. 

86. Total moles of CH_3MgBr consumed by 1 mole of given compound.

(a) 

(b) 

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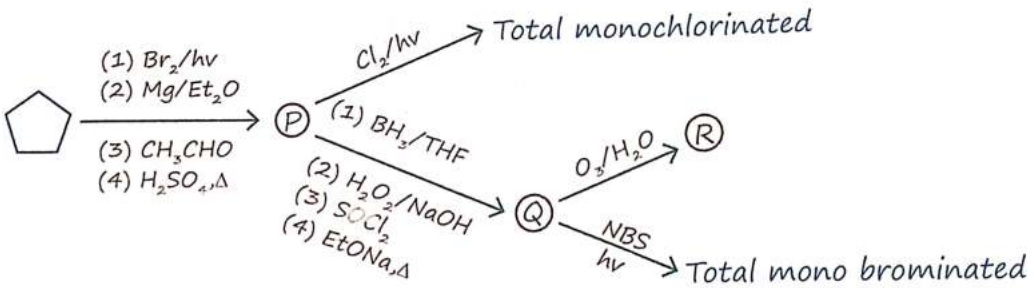
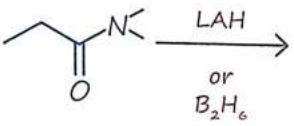
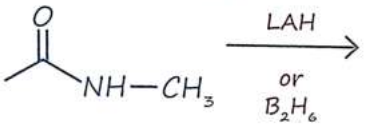
87. 

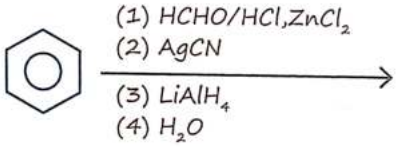
88. $R-NO_2 \xrightarrow[(2)\ H_2O]{(1)\ LAH}$

89. $R-N_3 \xrightarrow[(2)\ H_2O]{(1)\ LAH}$

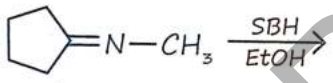
90. $R-NC \xrightarrow[(2)\ H_2O]{(1)\ LAH}$

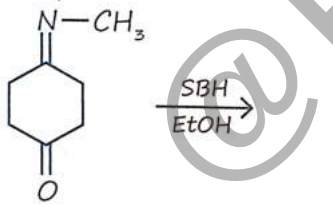
91. 

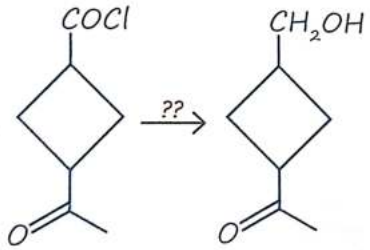
92. 

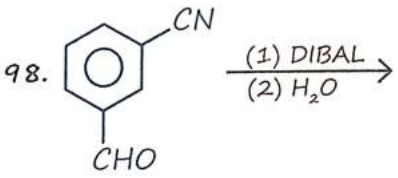
93. 

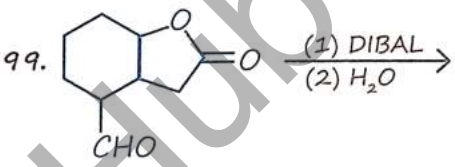
94. $CH_3-CH=NH \xrightarrow[EtOH]{SBH}$

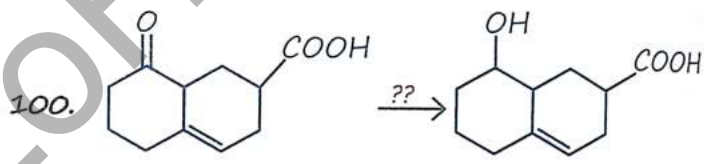
95. 

96. 

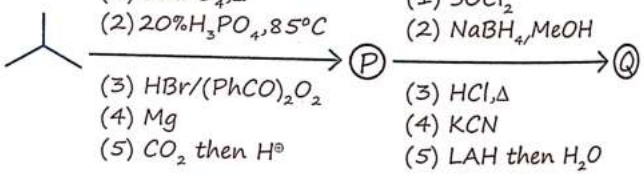
97. 

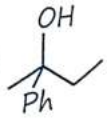
98. 

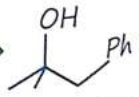
99. 

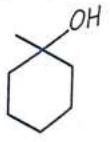
100. 

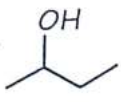
(a) LAH (b) SBH
(c) DIBAL (d) B_2H_6

101. 

102. Carbonyl compound + RMgX ⟶ 

103. Carbonyl compound + RMgX ⟶ 

104. Carbonyl compound + RMgX ⟶ 

105. CH_3-CHO + RMgX ⟶ 

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Previous Year Questions

1. What is the IUPAC name of the organic compound formed in the following chemical reaction? (2021)

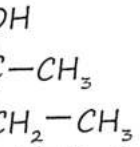
$$\text{Acetone} \xrightarrow[(2)\ H_2O, H^{\oplus}]{(1)\ C_2H_5MgBr, \text{dry ether}} \text{Product}$$

(a) Pentan-2-ol

(b) Pentan-3-ol

(c) 2-methyl butan-2-ol

(c) 2-methyl propan-2-ol

Sol. (c) 

2-Methyl butan-2-ol

2. Reagents which can be used to convert alcohols to carboxylic acids, are (2023-Manipur)

A. $CrO_3 - H_2SO_4$

B. $K_2Cr_2O_7 + H_2SO_4$

C. $KMnO_4 + KOH/H_3O^+$

D. Cu, 573 K

E. $CrO_3, (CH_3CO)_2O$

Choose the most appropriate answer form the options given below:

(a) (B), (C) and (D) only

(b) (B), (D) and (E) only

(c) (A), (B) and (C) only

(d) (A), (B) and (E) only

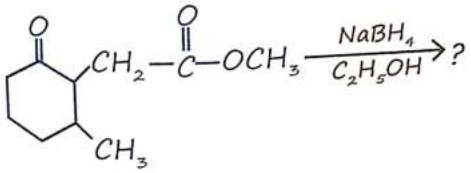
Sol. (c) Reagents given in (A), (B) and (C) convert alcohols to carboxylic acids.

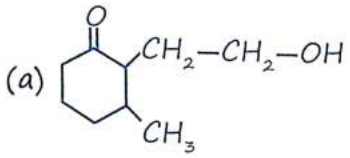
$$R-CH_2-OH \xrightarrow[K_2Cr_2O_7-H_2SO_4 \text{ or}]{CrO_3-H_2SO_4 \text{ or}} R-COOH$$

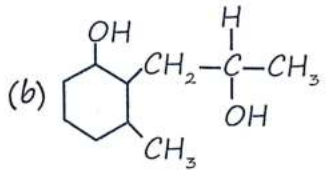
$KMnO_4 + KOH/H_3O^+$

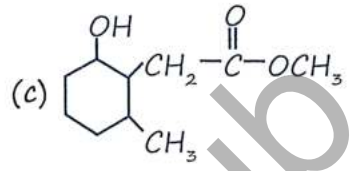
(Strong oxidising agents)

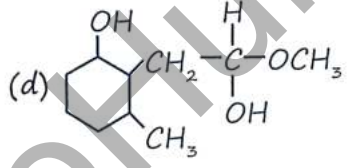
3. The product formed in the following chemical reaction is: (2021)



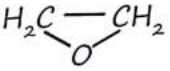
(a) 

(b) 

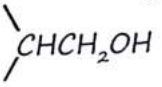
(c) 

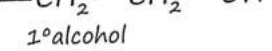
(d) 

Sol. (c) As $NaBH_4$ is a reducing agent, hence it will reduce ketone into secondary alcohol but it will not reduce esters.

4. Reaction  with RMgX leads to formation of: (1998)

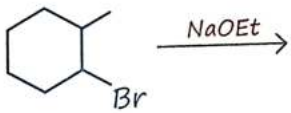
(a) RCHOHR (b) $RCHOHCH_3$

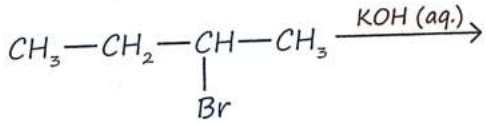
(c) RCH_2CH_2OH (d) 

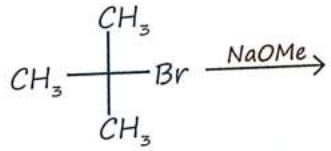
Sol. (c) 

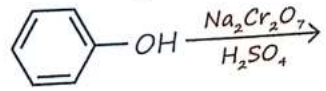
1° alcohol

5. The reactions which cannot be applied to prepare an alkene by elimination, are

A. 

B. 

C. 

D. 

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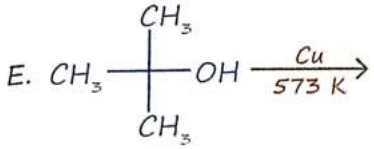
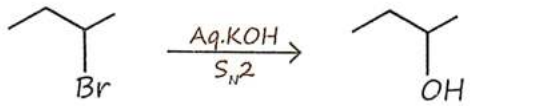
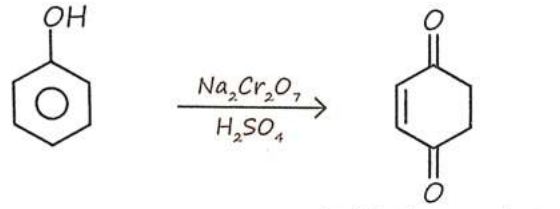
E. 

Choose the correct answer from the options given below: [07 April, 2025 (Shift-I)]

(a) B & E Only (b) B, C & D Only
(c) A, C & D Only (d) B & D Only

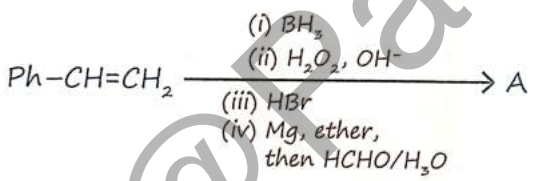
Sol. (d)

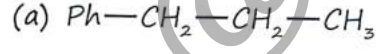
(B) 
(Substitution reaction)

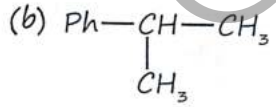
(D) 
(Oxidation reaction)

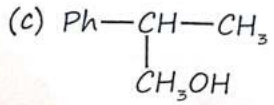
Reactions B and D do not yield alkenes as the products.

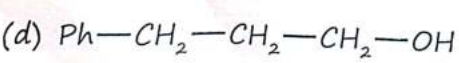
6. The final product A, formed in the following reaction sequence is: [27 Jan, 2024 (Shift-II)]



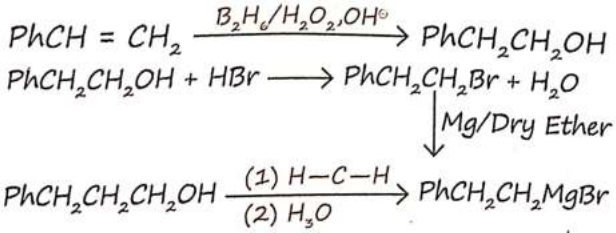
(a) 

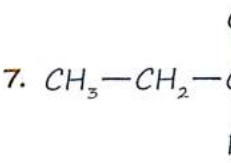
(b) 

(c) 

(d) 

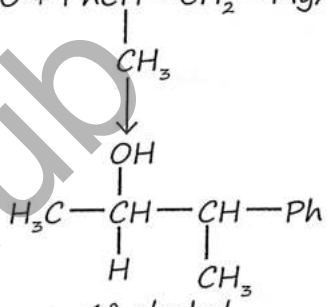
Sol. (d) The reaction is given below



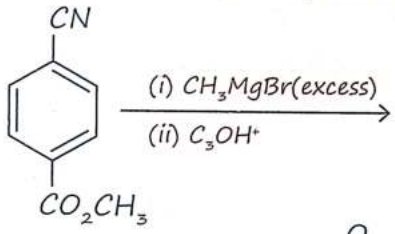
7.  **can not be prepared by:** [12 Jan, 2019 (Shift-I)]

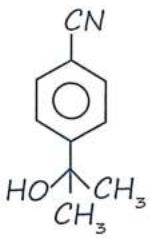
(a) $CH_3CH_2COCH_3 + PhMgX$
(b) $PhCOCH_2CH_3 + CH_3MgX$
(c) $PhCOCH_3 + CH_3CH_2MgX$
(d) $HCHO + PhCH(CH_3)CH_2MgX$

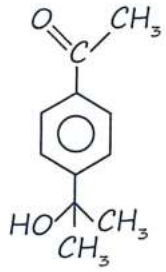
Sol. (d) $HCHO + PhCH(CH_3)CH_2MgX$

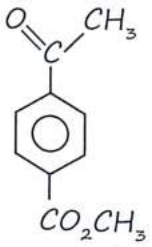

1° alcohol

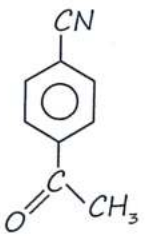
8. Major product of the following reaction is [09 April, 2024 (Shift-II)]

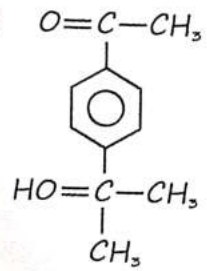


(a) 

(b) 

(c) 

(d) 

Sol. (b) 

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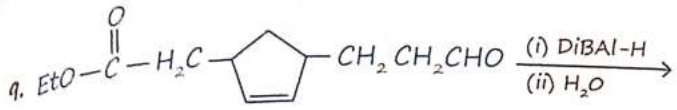
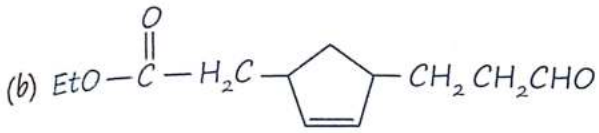
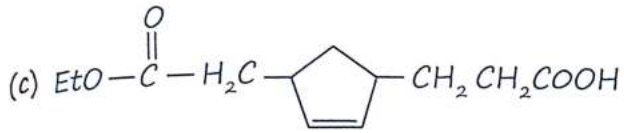
9. 

Consider the above reaction and predict the major product. [26 July, 2022 (Shift-II)]

(a)

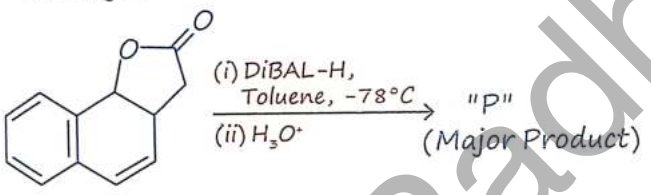
(b) 

(c) 

(d)

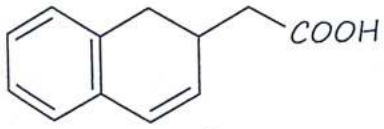
Sol. (a) DiBAL-H reduces cyanide, esters, lactones, amide & carboxylic acid into corresponding aldehyde (partial reduction)

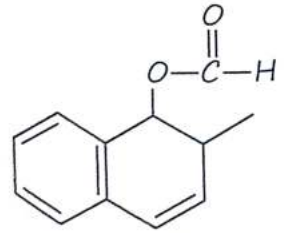
Therefore, ester group is reduced into aldehyde.

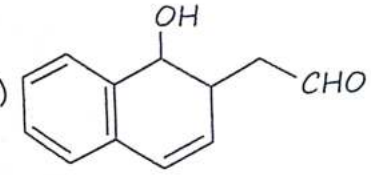
10. 

The product "P" in the above reaction is [16 March, 2021 (Shift-I)]

(a)

(b) 

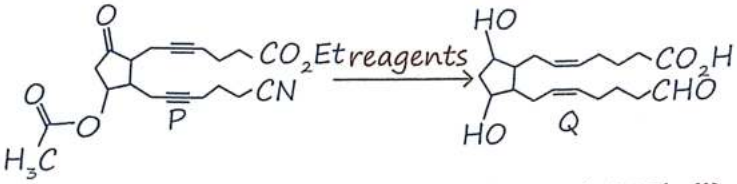
(c) 

(d) 

Sol. (d) DiBAL – H reduces esters to aldehyde and by product is alcohol, when temperature is low and solvent is non-polar.

11. The option(s) with correct sequence of reagents for the conversion of P to Q is(are)

C-7.98 W-35.41 UA-42.64 PC-13.98
(JEE Adv. 2024)



(a) i) Lindlar's catalyst, H_2; ii) $SnCl_2/HCl$; iii) $NaBH_4$; iv) H_3O^+

(b) i) Lindlar's catalyst, H_2; ii) H_3O^+; iii) $SnCl_2/HCl$; iv) $NaBH_4$

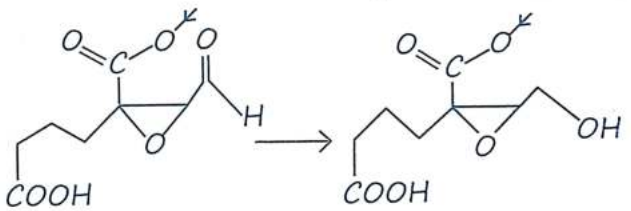
(c) i) $NaBH_4$; ii) $SnCl_2/HCl$; iii) H_3O^+; iv) Lindlar's catalyst, H_2

(d) i) Lindlar's catalyst, H_2; ii) $NaBH_4$; iii) $SnCl_2/HCl$; iv) H_3O^+

Sol. (c, d) All the reagents in (C) and (D) will result the desired product, (Q).

12. Reagent(s) which can be used to bring about the following transformation is (are)

C-5 W-41.12 UA-30.7 PC-23.17
(JEE Adv. 2016)



(a) $LiAlH_4$ in $(C_2H_5)_2O$

(b) BH_3 in THF

(c) $NaBH_4$ in C_2H_5OH

(d) Raney Ni/H_2 in THF

Sol. (c, d) $NaBH_4$ in C_2H_5OH do not reduce acid, ester (–COOR) and epoxide, as it is a mild reducing agent. It reduces only aldehyde to alcohol.

While the $LiAlH_4$ is strong reducing agent that reduces the ester, acid, aldehyde and ketone.

BH_3 in THF and Raney Ni/H_2 in THF reduces acid, ester (–COOR) and epoxide into alcohol.

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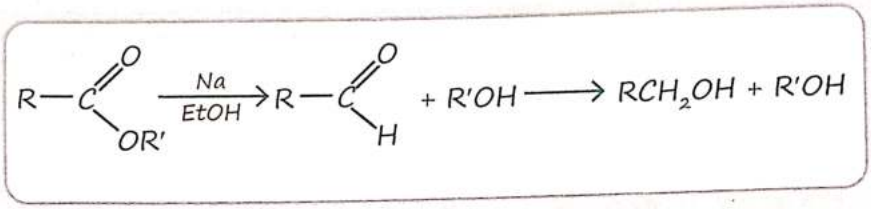
3. Bouveault Blanc Reduction

The Bouveault–Blanc reduction is a classical method used to reduce esters (R–COOR') to primary alcohols ($R-CH_2OH$) using metallic sodium (Na) in the presence of alcohol (typically ethanol, EtOH).

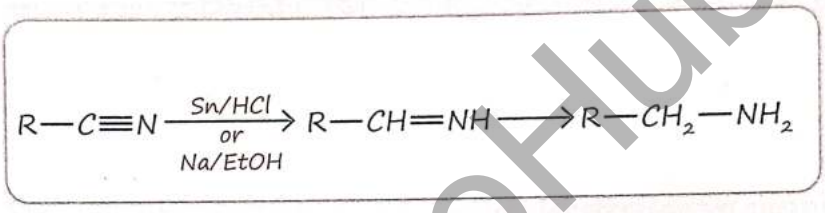
Reaction



4. Mendius Reaction

The Mendius Reaction is a reduction reaction in which nitriles (R–C≡N) are converted to primary amines ($R-CH_2NH_2$) using metal (like Na or Li) in ethanol (EtOH) or other reducing conditions. Reaction with metal acid.

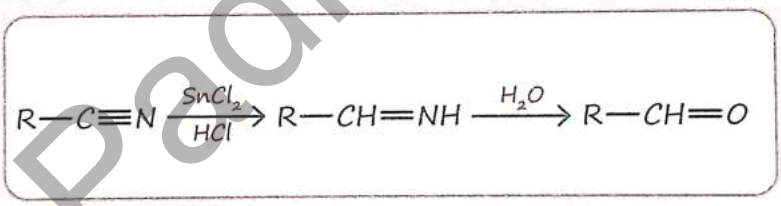
Reaction



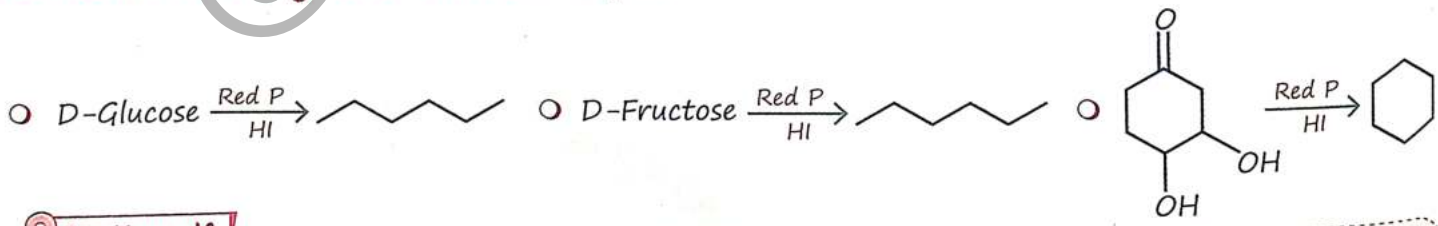
5. Stephen's Reduction

Nitriles are reduced to corresponding imine with stannous chloride in the presence of hydrocholric acid, which on hydrolysis give corresponding aldehyde.

Reaction



6. Reduction by Red P/HI or HI, Δ



Try Yourself

1. $C_6H_5COOC_2H_5 + Na/EtOH \longrightarrow$
2. $C_6H_5CH_2C{\equiv}N + SnCl_2/HCl \longrightarrow$
3. $CH_3CH_2C{\equiv}N + SnCl_2/HCl \longrightarrow$
4. $CH_3CH(OH)CH(OH)CH_2OH + Red\ P/HI \longrightarrow$

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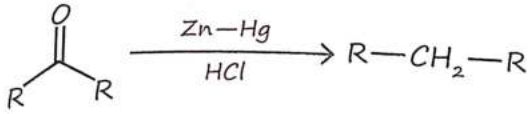
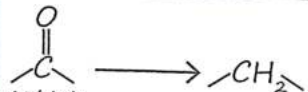
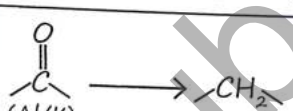
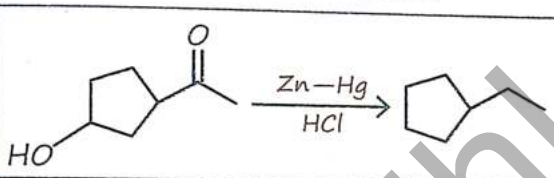
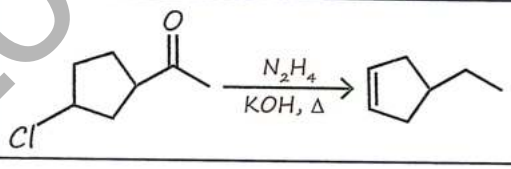
7. Clemmensen Reduction

It reduces ketones or aldehydes to hydrocarbons using zinc amalgam (Zn-Hg) and hydrochloric acid (HCl) under acidic conditions.

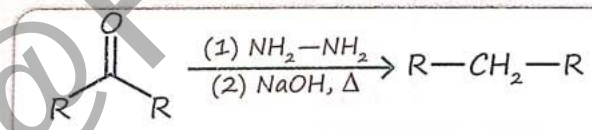


	Clemmensen Reduction	Wolff-Kishner Reduction
Reagent	Zn-Hg/HCl	(i) $NH_2 - NH_2$ (ii) Base, Δ (ethylene glycol)
Reaction		
Intermediate	Free radical and carbanion	Carbanion
Medium	Acidic	Basic
Basic	Acid sensitive must be absent	Base sensitive must be absent
Example		

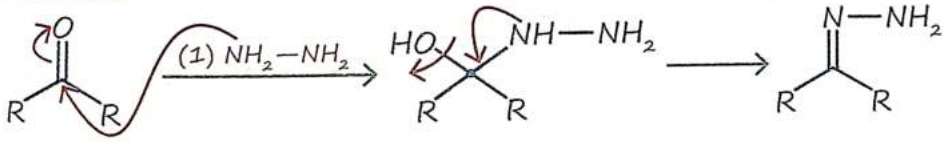
8. Wolff-Kisher Reduction

It reduces ketones or aldehydes to hydrocarbons using hydrazine (NH_2-NH_2) and a strong base (usually KOH) under high heat, in basic conditions.

Reaction



Mechanism

1.


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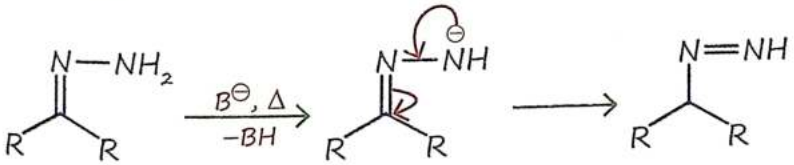
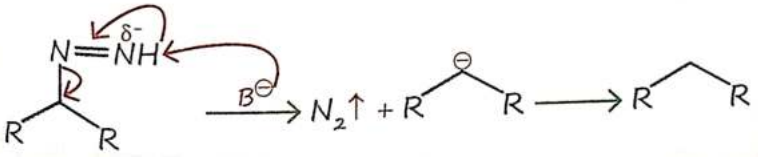


3.


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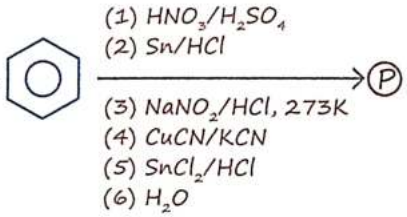
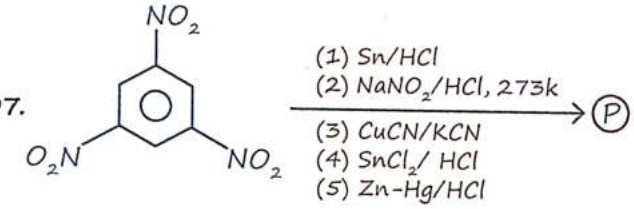
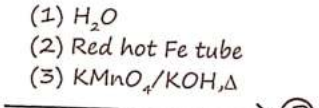
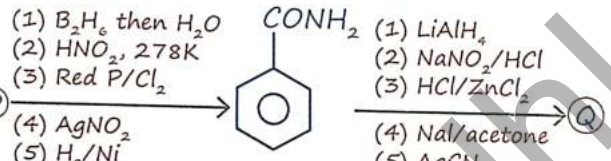
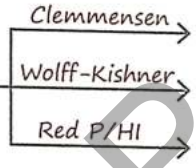
Practice Questions

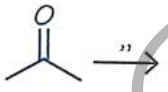
106. 

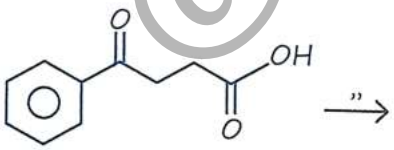
107. 

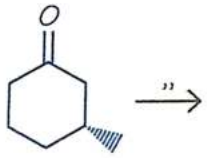
108. Mg_2C_3 

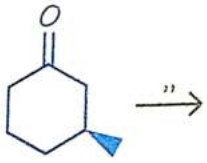
109. (P) 

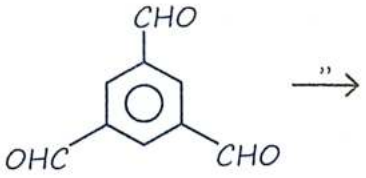
110. CH_3CHO 

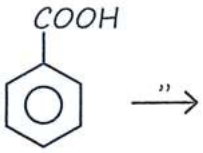
111. 

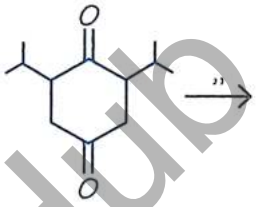
112. 

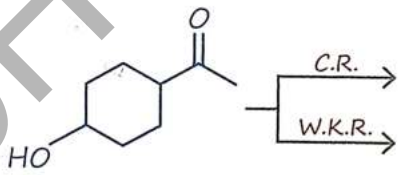
113. 

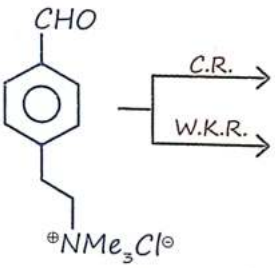
114. 

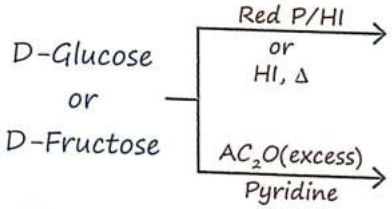
115. 

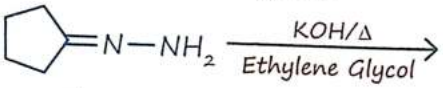
116. 

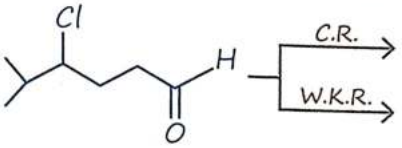
117. 

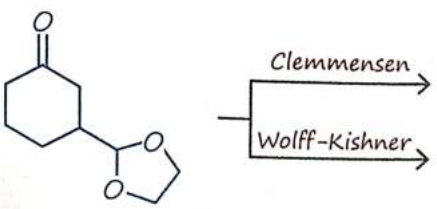
118. 

119. 

120. 

121. 

122. 

123. 

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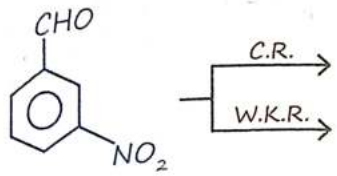
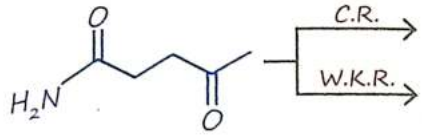
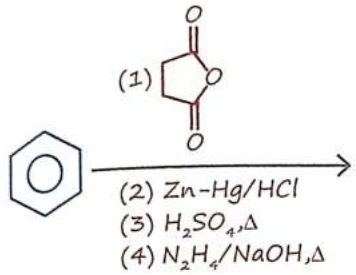
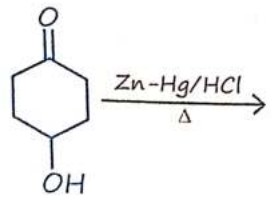
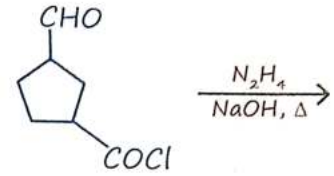
124. 

125. 

126. 

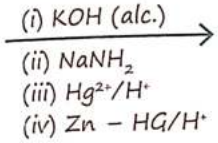
127. 

128. 

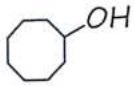
129. Total optically active carbonyl compound of mass 100gm which on reaction with ZnHg/HCl produces optically inactive compounds are.

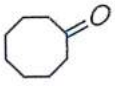
Previous Year Questions

13. 1, 2-dibromocyclooctane



P (Major product) 'P' is [08 April, 2025 (Shift-II)]

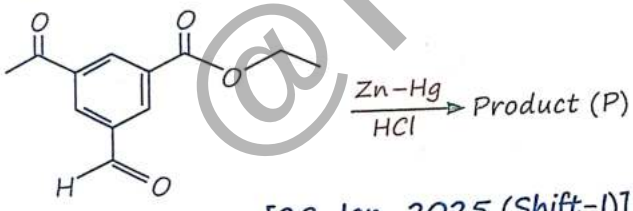
(a) 

(b)
(c) 

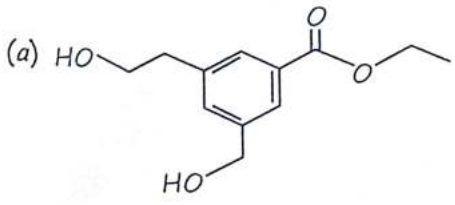
(d)

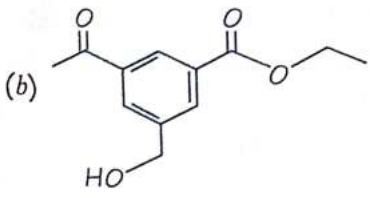
Sol. (b) Cyclooctane

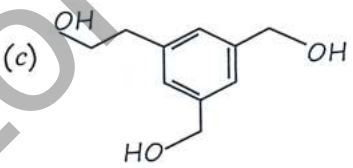
14. The product (P) formed in the following reaction is:

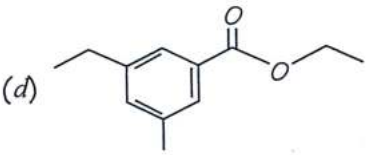


[29 Jan, 2025 (Shift-I)]

(a) 

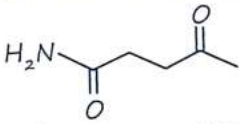
(b) 

(c) 

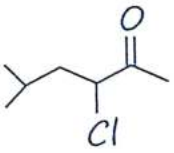
(d) 

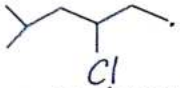
Sol. (d) The given reaction involves a Clemmensen reduction using Zn–Hg/HCl, which selectively reduces ketones or aldehydes to alkanes without affecting other functional groups.

15. Given below are two statements:

Statement-I: 

under Clemmensen reduction conditions will give
.

Statement-II: 

under Wolff-Kishner reduction condition will give 
.

In the light of the above statements, choose the correct answer from the options given below: [24 Jan, 2023 (Shift-II)]

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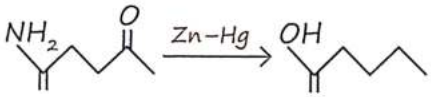
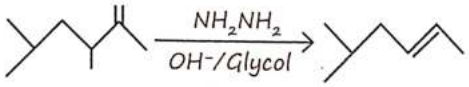
(a) Statement-I is false but Statement-II is true.
(b) Both Statement-I and Statement-II are false.
(c) Statement-I is true but Statement-II is false.
(d) Both Statement-I and Statement-II are true.

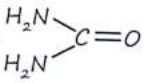
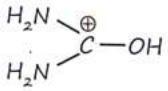
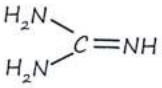
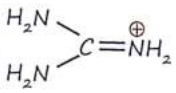
Sol. (c) S-I 

(True)

S-II 

(False)

16. Which amongst the following compound/ species is least basic? (2023-Manipur)

a. 
b. 
c. 
d. 

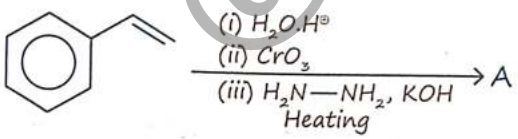
Sol. (b) Compound given in option (b) will be the least basic.

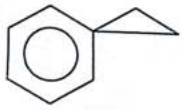
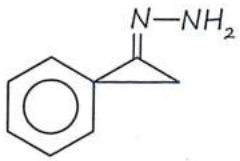
17. When glycerol is treated with excess of HI, it produces: (2010 Mains)

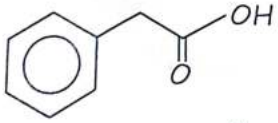
(a) 2-iodopropane (b) Allyl iodide
(c) Propene (d) Glycerol triiodide

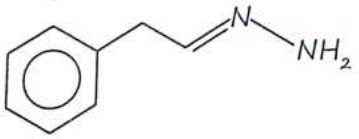
Sol. (a) 2-iodopropane

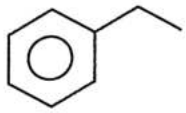
18. The final product A formed in the following multistep reaction sequence is: [29 Jan, 2024 (Shift-I)]



(a) 
(b) 

(c) 

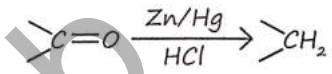
(d) 

Sol. (a) 

19. Clemmensen reduction of a ketone is carried out in the presence of which of the following? (2011 Pre)

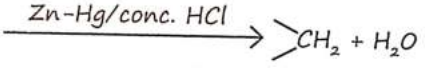
a. H_2 and Pt as catalyst
b. Glycol with KOH
c. Zn-Hg with HCl
d. $LiAlH_4$

Sol. (c) Clemmensen reduction of carboxyl compounds is an acid base reduction.



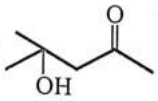
20. Reduction of aldehydes and ketones into hydrocarbons using zinc amalgam and conc. HCl is called: (2007)

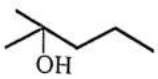
a. Cope reduction
b. Dow reduction
c. Wolf-Kishner reduction
d. Clemmensen reduction

Sol. (d) 

Clemmensen reduction

21. Given below are two statements: One is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): 

can be easily reduced using Zn-Hg/HCl to 

Reason (R): Zn-Hg/HCl is used to reduce carbonyl group to –CH2– group.

In the light of the above statements, choose the correct answer from the options given below: [30 Jan, 2023 (Shift-II)]

(a) (A) is false but (R) is true.
(b) (A) is true but (R) is false.
(c) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(d) Both (A) and (R) are true and (R) is the correct explanation of (A).

Sol. (a)

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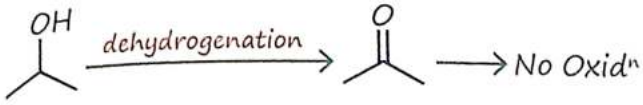
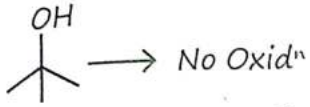
Oxidation

Pattern of Oxidation

1. $CH_3-\overset{-1}{C}H_2-OH \xrightarrow[\text{dehydrogenation}]{2e^{\ominus}} CH_3-\overset{+1}{C}H=O \longrightarrow CH_3-\overset{+3}{C}OOH$

2.  No Oxidⁿ

3.  No Oxidⁿ

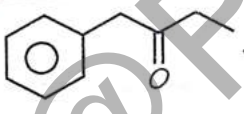
4. $CH_3-\overset{+1}{O}H \longrightarrow H-\overset{O}{\overset{\|}{C}}-H \longrightarrow HCOOH \longrightarrow CO_2$

I. Oxidation of Ketone

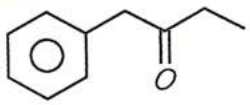
Ketones usually do not get oxidized by potassium permanganate ($KMnO_4$) because they lack the necessary hydrogen on the carbonyl carbon. However, if the ketone has α-hydrogens, $KMnO_4$ can cleave the carbon-carbon bond next to the carbonyl, breaking the molecule into smaller carboxylic acids.

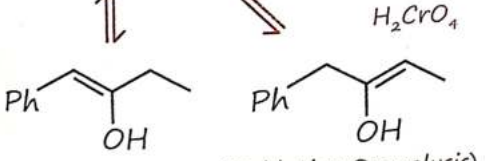
Popoff's Rule states that during the oxidation of an unsymmetrical ketone, the C–CO bond breaks in such a way that the carbonyl group remains attached to the smaller alkyl group.

Reaction

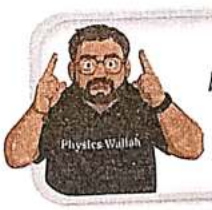
 $\xrightarrow[\text{or Hot } K_2Cr_2O_7 \text{ or } H_2CrO_4]{\text{Hot } KMnO_4}$ $PhCH_2COOH + CH_3COOH + \underbrace{PhCOOH + CH_3CH_2COOH}_{\text{(Major)}}$

Mechanism

1.  $\xrightarrow[\text{or Hot } K_2Cr_2O_7 \text{ or } H_2CrO_4]{\text{Hot } KMnO_4}$ $PhCH_2COOH + CH_3COOH + PhCOOH + CH_3CH_2COOH$



(Oxidative Ozonolysis)



$KMnO_4$ when it sees a ketone without a-hydrogen:

Main kya karu, job chhod du?

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2. Oxidation of alcohol

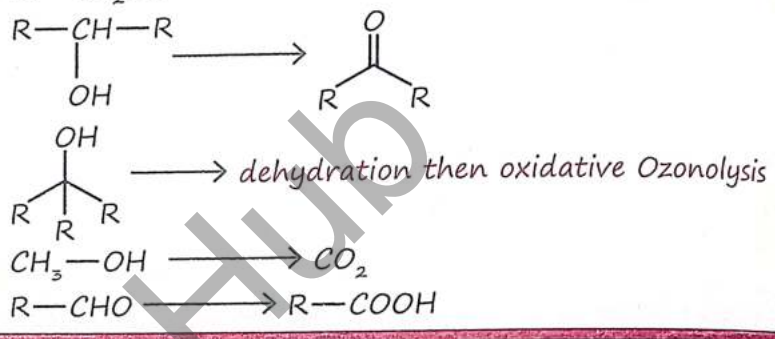
➢ **Oxidation of alcohols:**

- Primary alcohols oxidize to form aldehydes, which can further oxidize to carboxylic acids.
- Secondary alcohols oxidize to form ketones.
- Tertiary alcohols generally do not oxidize easily because they lack hydrogen on the carbon with the –OH group.

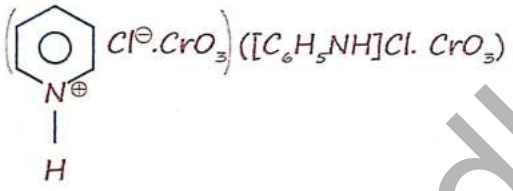
Common oxidizing agents used are potassium permanganate ($KMnO_4$), potassium dichromate ($K_2Cr_2O_7$), and chromium trioxide (CrO_3).

1. $KMnO_4$, Δ
2. $K_2Cr_2O_7$, Δ
3. CrO_3, H_2SO_4 (Jones Reagent)

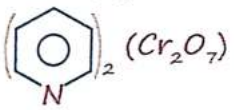
⟹ Strong Oxidising agent



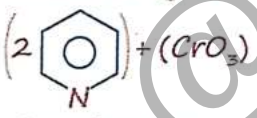
4. PCC (Pyridinium chlorochromate)



5. PDC (Pyridinium dichromate)

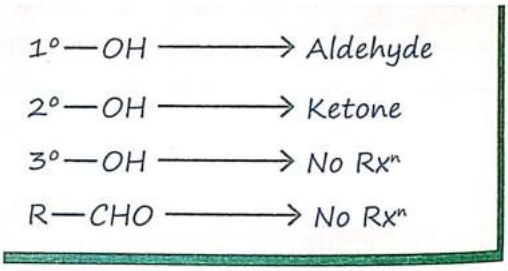


6. Collins Reagent



7. CrO_3/acetone
8. NBS
9. (i) TSCl (ii) DMSO/$NaHCO_3$

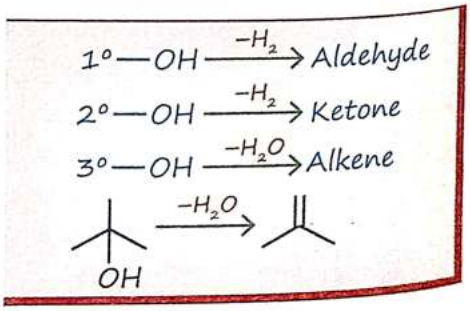
⟹ Mild Oxidising agent



The oxidation of alcohols with copper involves heating a primary or secondary alcohol with copper or copper oxide, which oxidizes the alcohol to an aldehyde (from primary alcohol) or a ketone (from secondary alcohol).

10. Cu, 300°C / Δ

⟹ Mild Oxidising agent



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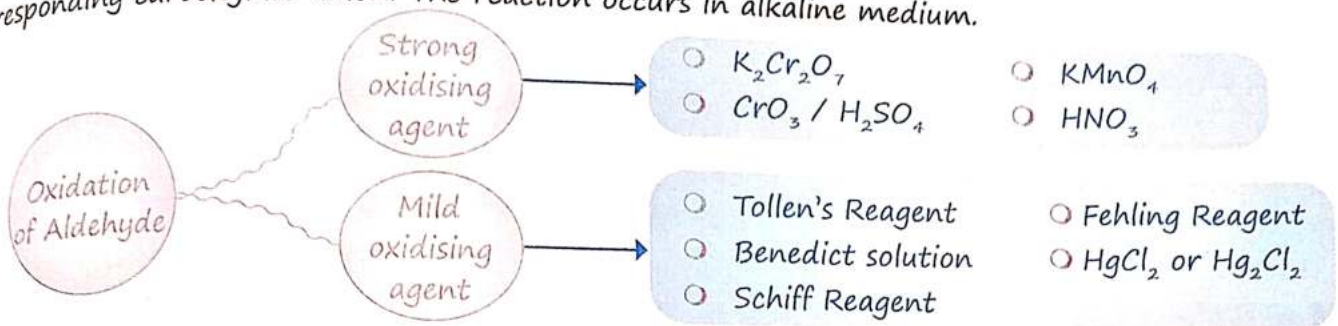
3. Oxidation of aldehyde

One warming an aldehyde with freshly prepared ammoniacal silver nitrate solution (Tollens' reagent), a bright silver mirror is produced due to the formation of silver metal. The aldehydes are oxidised to corresponding carboxylate anion. The reaction occurs in alkaline medium.



(a) Tollen's Reagent

Tollens' test is a chemical test used to detect aldehydes. When an aldehyde is treated with Tollens' reagent (a solution of silver nitrate in ammonia), it gets oxidized to a carboxylic acid, and the silver ions are reduced to metallic silver, forming a shiny "silver mirror" on the test tube.

Tollen's Reagent
- Ammoniacal $AgNO_3$
- $AgNO_3/NH_4OH$
- $Ag_2O/\overset{\ominus}{O}H$
- $[Ag(NH_3)_2]OH$
- $[Ag(NH_3)_2]^{\oplus}$
- Silver Minor test

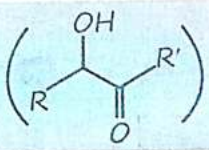
(i) Formation of Tollen's Reagent

1. $2AgNO_3 + 2NaOH \longrightarrow \underset{\text{(Brown)}}{Ag_2O} + 2NaNO_3 + H_2O$
2. $Ag_2O + 2NaNO_3 + H_2O + 4NH_3 \longrightarrow 2[Ag(NH_3)_2]NO_3 + 2NaOH$

(ii) Reaction of Tollen's Reagent

$$R-CHO + 2Ag(NH_3)_2^{\oplus} + 3\overset{\ominus}{O}H \longrightarrow R-\overset{\ominus}{C}OO + 2Ag + 4NH_3 + 2H_2O$$

(iii) Tollen's Test By

- Silver Mirror → —CHO Present (All aldehyde and HCOOH)
- Silver Mirror → —CHO forms ($R-CH(\overset{\ominus}{O}H)(OR') \xrightarrow{\overset{\ominus}{O}H} R-CHO$)
- Silver Mirror → α—hydroxy carboxyl 
- Silver Mirror → Tartaric acid and Ph—NH—OH (Mulliken-Barker)
- White PPT → Terminal Alkyne (Acid Base)

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(b) Fehling Test

➢ **Fehling solution**

(a) aq. $CuSO_4$ + NaOH

(b) Rochelle Salt: $COONa-CH(OH)-CH(OH)-COOK$ $\xrightarrow{Cu^{2+}}$ $COO-CH(O)-CH(O)-COO$

Note: Aromatic –CHO doesn't give Fehling test.

➢ **Fehling Reaction**

$$R-CHO + 2Cu^{+2} + 5\overset{\ominus}{O}H \longrightarrow R-\overset{\ominus}{C}OO + Cu_2O + 3H_2O$$

(Red ppt)

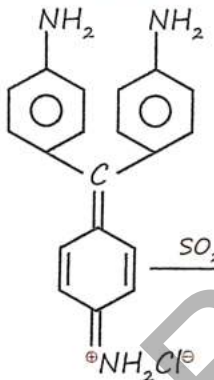
(c) Schiff's Test

➢ Due to slight basic medium, it can't react with hemiacetal.

Ex. CH_3CHO Ph-CHO

- Schiff Base (substituted Imine)
- Schiff Reagent

Reaction Schiff's Reagent is aq. solution of following base decolourised by passing SO_2. Aldehyde restore pink colour of Schiff's reagent.

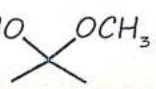
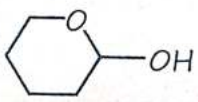


$\xrightarrow{SO_2}$ Colourless solution (Schiff's Reagent) $\xrightarrow[\Delta]{RCHO}$ RCOOH + Pink colour

p-Rosaniline Hydrochloride
Magenta colour (Fuschin)

Try Yourself

Which of the following compounds gives silver mirror with Tollen's reagent and red ppt with Fehling soluiton?

(a) (b) $CH_3-CH(OH)-OCH_3$ (c)  (d) 

Recap: Reducing Agents (LAH, $NaBH_4$, DIBAL-H)

Test	Reagent & Detection	Positive Result	Notes
Tollens' Test	Ammoniacal $AgNO_3$ → detects aldehydes & formic acid	Silver mirror	No reaction with ketones or aromatic ketones
Fehling's Test	Alkaline Cu^{2+} solution → detects aliphatic aldehydes	Red ppt of Cu_2O	Aromatic aldehydes give no reaction
Schiff's Test	Colourless rosaniline dye (SO_2-treated) → detects aldehydes	Pink colour reappears	Doesn't react with hemiacetals or ketones

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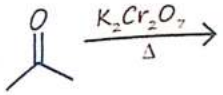
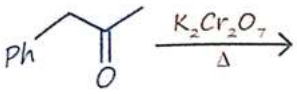
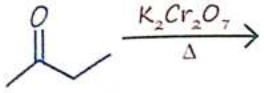
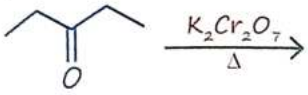
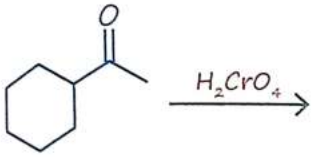
Practice Questions

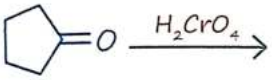
130. 

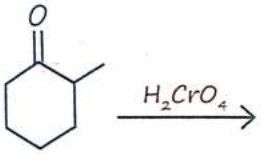
131. 

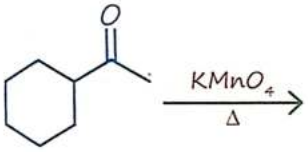
132. 

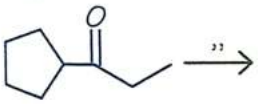
133. 

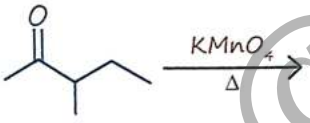
134. 

135. 

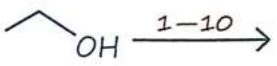
136. 

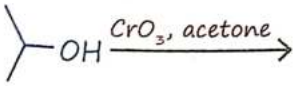
137. 

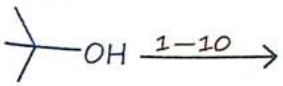
138. 

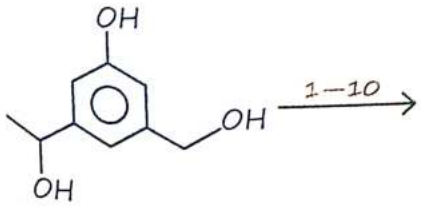
139. 

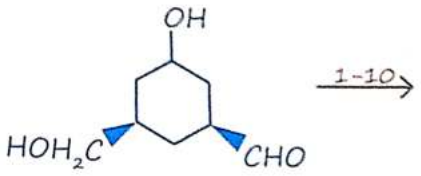
140. $CH_3-OH \xrightarrow{1-10}$

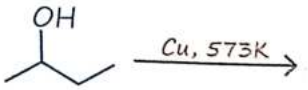
141. 

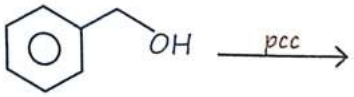
142. 

143. 

144. 

145. 

146. 

147. 

148. Which of the following compound given silver mirror with Tollen's reagent and red ppt with Fehling solution?

T F

149. CH_3-CHO

150. $HCHO$

151.

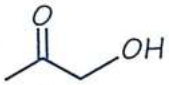
152. CH_3CH_2OH

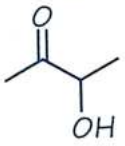
153. $Ph-CHO$

154. $Ph-CH_2CHO$

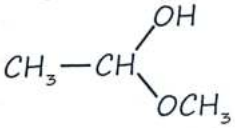
155. $HCOOH$

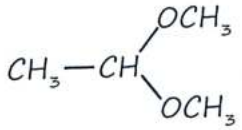
156. CH_3COOH

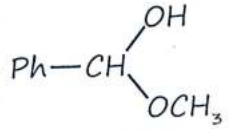
157. 

158. 

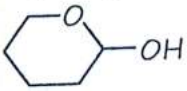
159. $CH_3-C\equiv CH$

160. 

161. 

162. 

163.

164. 

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Carbene

Carbenes are generated by the removal of a leaving group and a proton from a suitable precursor, often by treating a haloform (like chloroform) with a strong base. This creates a highly reactive species with a divalent carbon atom having two nonbonded electrons.

➢ **Generation of Carbene**

(i) $CHCl_3 + KOH \xrightarrow{-H_2O} \overset{\ominus}{C}Cl_3 \xrightarrow[rds\text{-}Cl^{\ominus}]{\Delta} \underset{(DCC)}{\overset{\oplus}{\ominus}CCl_2}$ [α,α- Elimination]

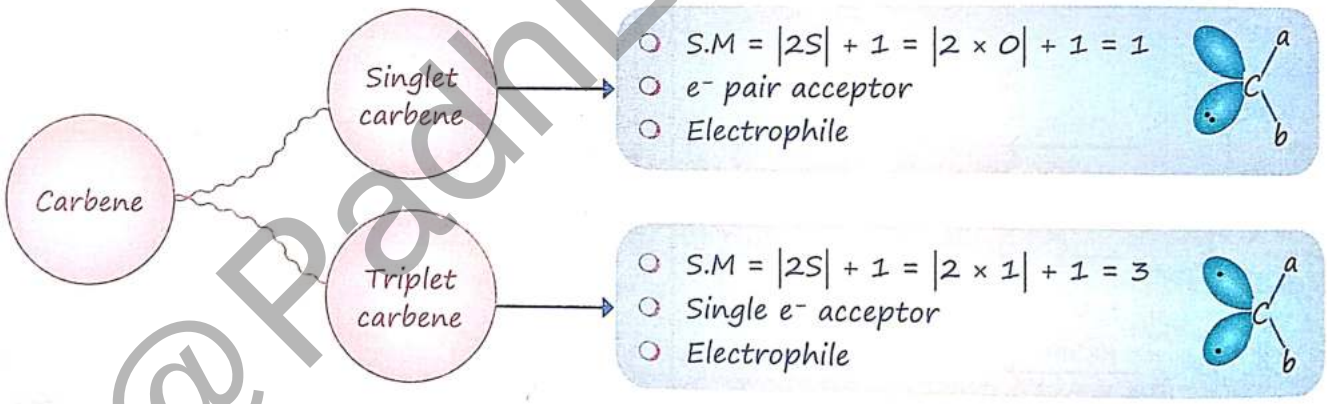
(ii) $Cl_3C - COOH \xrightarrow[\Delta]{Base} Cl_3C\overset{\ominus}{COO} \longrightarrow \overset{\ominus}{C}Cl_3 \xrightarrow[rds\text{-}Cl^{\ominus}]{\Delta} \underset{(DCC)}{\overset{\oplus}{\ominus}CCl_2}$ [Decarboxylation]

(iii) $\overset{\ominus}{C}H_2-\overset{\oplus}{N}_2 \xrightarrow[\Delta-N_2]{h\nu \text{ or}} \overset{\oplus}{\ominus}CH_2$ [Photolysis or Thermolysis]

(iv) $H_2C=C=O \xrightarrow[\Delta CO\uparrow]{h\nu \text{ or}} \overset{\oplus}{\ominus}CH_2$ [Photolysis or Thermolysis]

(ketene)

$H_2C=C=O \leftrightarrow H_2\overset{\ominus}{C}-C\equiv\overset{\oplus}{O}$



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Practice Questions

165. Find the order of stability:

1. $\overset{+}{\underset{-}{C}}H_2$, $\overset{\cdot}{\underset{\cdot}{C}}H_2$
2. $\overset{+}{\underset{-}{C}}H_2$, $\overset{\cdot}{\underset{\cdot}{C}}F_2$
3. $\overset{+}{\underset{-}{C}}H_2$, $\overset{+}{\underset{-}{C}}F_2$
4. $\overset{+}{\underset{-}{C}}F_2$, $\overset{+}{\underset{-}{C}}Cl_2$, $\overset{+}{\underset{-}{C}}Br_2$

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1. Carbylamine Reaction / Isocyanide test

The carbylamine reaction is a test for primary amines where a primary amine reacts with chloroform and a base to produce an isocyanide (carbylamine), characterized by its foul smell. This reaction is used to identify the presence of primary amines in a compound.

Reaction

$$R-NH_2 \xrightarrow[3KOH]{CHCl_3} \overset{+}{R}-\bar{N}\equiv C + 3KCl + 3H_2O$$

(unpleasant smell)

Mechanism

1. $KOH \rightleftharpoons K^{\oplus} + \overset{\ominus}{O}H$
2. $CHCl_3 + \overset{\ominus}{O}H \longrightarrow :CCl_3 + H_2O$ (Trichlorocarbene)
3. $CHCl_3^{\ominus} \xrightarrow{-Cl^{\oplus}} :CCl_2$ (Dichlorocarbene free radical)
4. $R-NH_2 + :CCl_2 \longrightarrow R-\overset{H}{\underset{H}{N^{\oplus}}}-\overset{\ominus}{C}(Cl)-Cl \xrightarrow{2KOH} R-N\equiv C + 2HCl$ (Alkyl isocyanide)

Baba Tea-Stall (BTS)

(i) DCC is obtained as intermediate.
(ii) More $e^{\ominus}$ density on N, more easily attack..
(iii) Only shown by primary amine (1° – NH_2)
(iv) 1 mol of $CHCl_3$ and 3 moles of KOH is required.

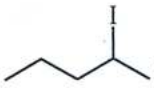
Practice Questions

166. Which of the following amines can produce foul smell with chloroform in potash?

(a) $PhNH_2$ (b) CH_3NH_2
(c) (d) $(CH_3)_2NH$
(e) CH_3CONH_2 (f)
(g) PhNHMe (h) $PhCH_2NH_2$
(i)

167.
(1) Br_2/hv
(2) KOH/EtOH
(3) HBr/R_2O_2
(4) KCN
(5) H_2/Ni
(6) $CHCl_3/KOH$
(7) $LiAlH_4$
(8) $NaNO_2/HCl$

168.
(1) B_2H_6
(2) $H_2O_2/NaOH$
(3) $SOCl_2$
(4) Mg/Et_2O
→ (P)
MeOH → (Q)
(1) HCHO
(2) H_2SO_4, Δ → (R)
(1) CO_2
(2) $SOCl_2$
(3) $(PhCH_2)_2Cd$ → (S)

169. 
(1) HBr/hv
(2) $AgNO_2$
(3) H_2/Ni
NaOH, EtOH, Δ → (P) (19%)
(4) $CHCl_3/KOH$
(5) $H_3O^{\oplus}$ → (Q) + (R)

170. Toluene $\xrightarrow[UV\ light]{Cl_2}$ (P) $\xrightarrow{(X)}$ (Q)
(Q) $\xrightarrow{CHCl_3, KOH}$ (R)
(R) $\xrightarrow[HCl]{NaNO_2}$ (S) $\xrightarrow{PCC}$ (T)

Prepare (T) from following substrate:
(1) Benzene (2) Benzoyl chloride
(3) Benzonitrile (4) Ethyl Benzoate

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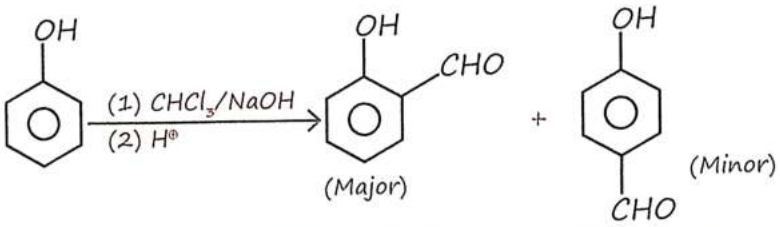
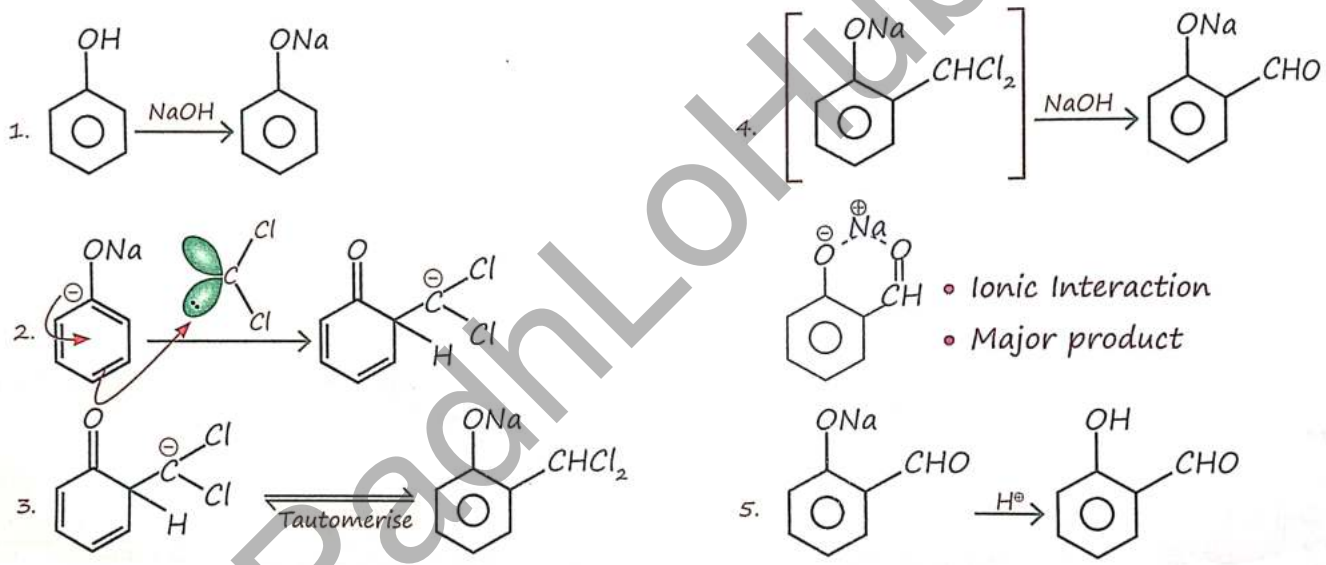
2. Reimer Tiemann reaction

The Reimer-Tiemann reaction involves the formylation of phenols by treating them with chloroform and a strong base, typically producing ortho-hydroxybenzaldehydes. It's commonly used to introduce an aldehyde group adjacent to the hydroxyl group on the aromatic ring.

Reaction



Mechanism



Baba Tea-Stall (BTS)

(i) DCC is obtained as intermediate.
(ii) Ortho is major product due to ionic chelation.
(iii) Another intermediate is O^- substituted benzal chloride
(iv) If any ortho is blocked then para will be major.
(v) If all positions are blocked than attack from para.

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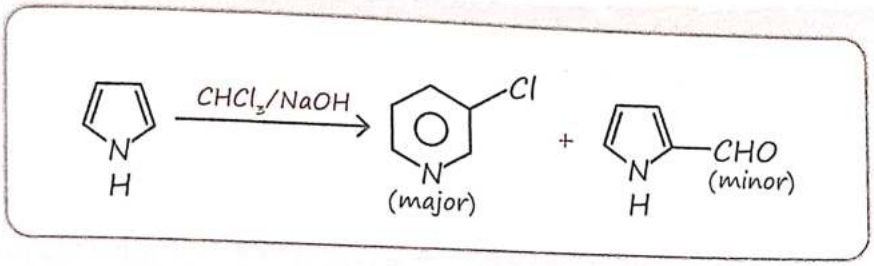
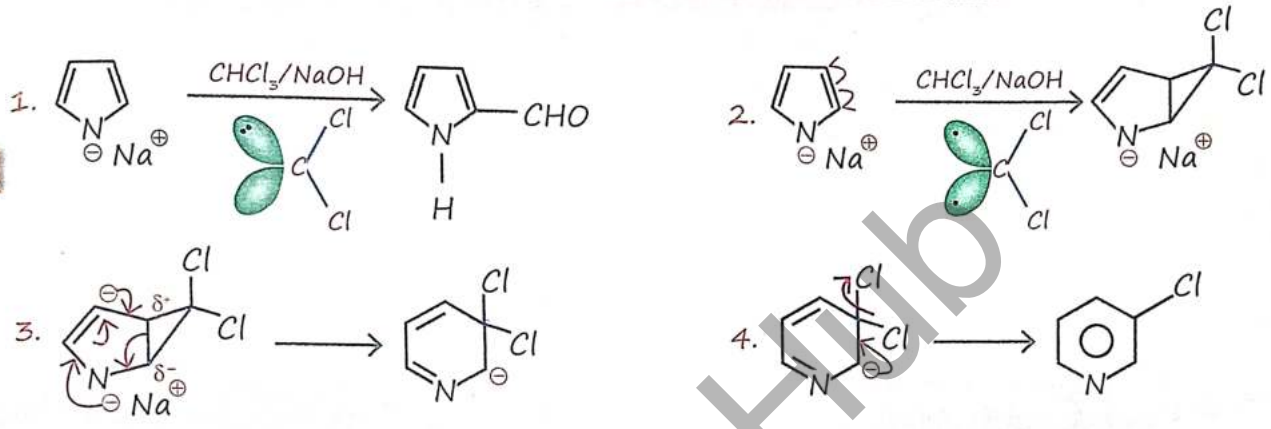
Abnormal Reimer Tiemann Reaction

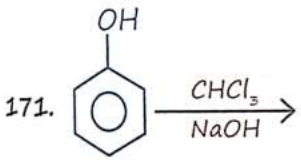
Reaction

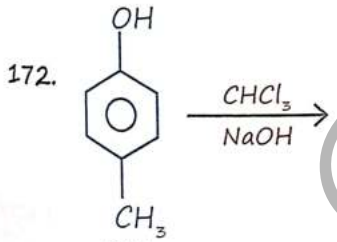


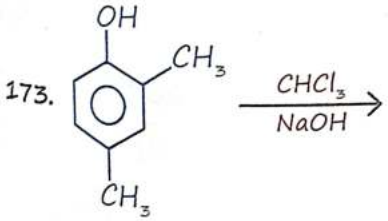
Mechanism

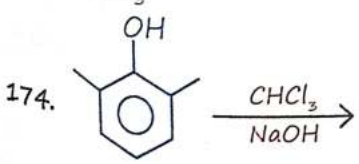


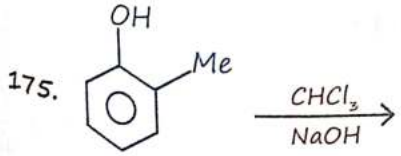
Practice Questions

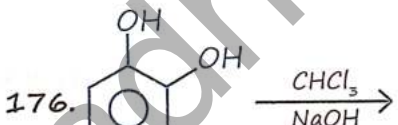
171. 

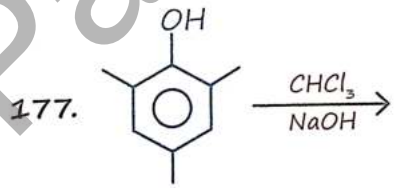
172. 

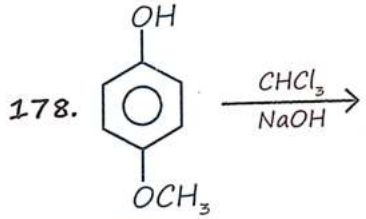
173. 

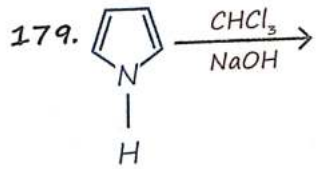
174. 

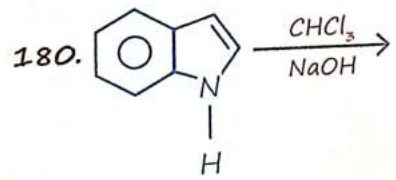
175. 

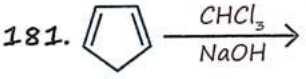
176. 

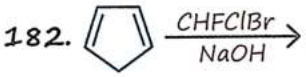
177. 

178. 

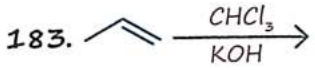
179. 

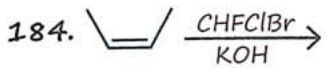
180. 

181. 

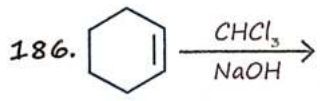
182. 

Reaction with Alkene

183. 

184. 

185.

186. 

187.

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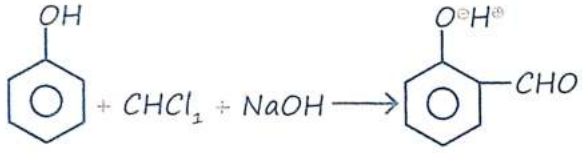
22. In the reaction, the electrophile involved is: (2018)



(a) Dichloromethyl cation ($\overset{\oplus}{C}HCl_2$)

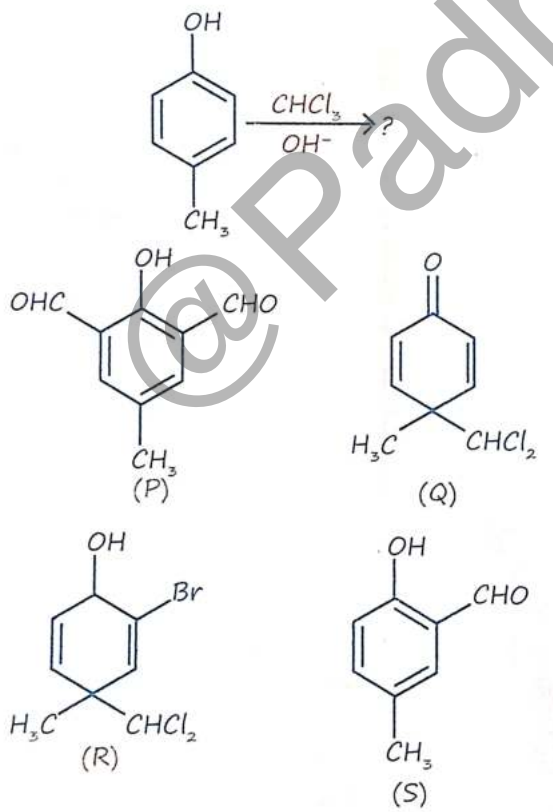
(b) Formyl cation ($\overset{\oplus}{C}HO$)

(c) Dichlorocarbene ($:CCl_2$)

(d) Dichloromethyl anion ($\overset{\ominus}{C}HCl_2$)

Sol. (c) It is Reimer-Tiemann reaction the electrophile formed is $:CCl_2$ (Dichlorocarbene).

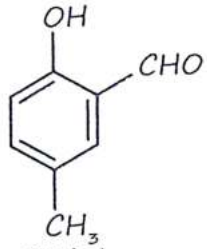
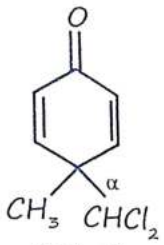
23. In the following reaction, the product (s) formed is/are



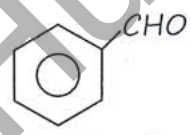
C-27.72 W-44.1 UA-28.18 (JEE Adv. 2013)

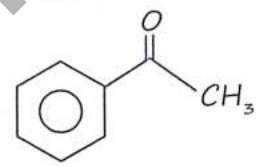
(a) P (major) (b) Q (minor)

(c) R (minor) (d) S (major)

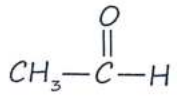
Sol. (b,d) S =  , Q = 

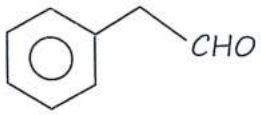
24. The compounds which give positive Fehling's test are:

A. 

B. 

C. $HOCH_2-CO-(CHOH)_3-CH_2-OH$

D. 

E. 

Choose the correct answer from the options given below:

[22 Jan, 2025 (Shift-I)]

(a) A, D and E only

(b) C, D and E only

(c) A, B and C only

(d) A, C and D only

Sol. (b) Aliphatic aldehydes and alpha hydroxy ketones give positive results with Fehling's test. Hence, the compounds (C) which is alpha hydroxy ketone, compounds (D) & (E) which are aliphatic aldehydes give positive Fehling's test.

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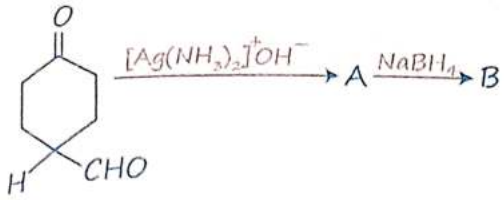
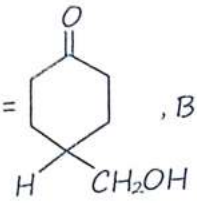
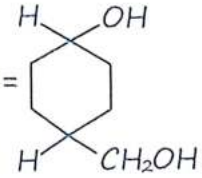
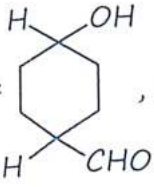
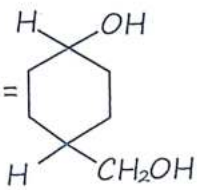
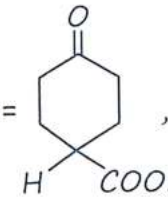
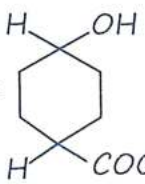
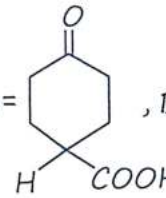
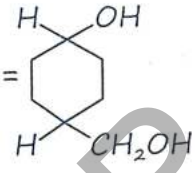
25. The products formed in the following reaction, A and B are [26 July, 2022 (Shift-I)]

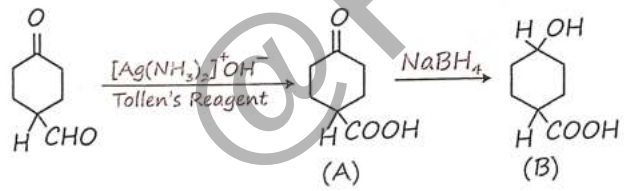


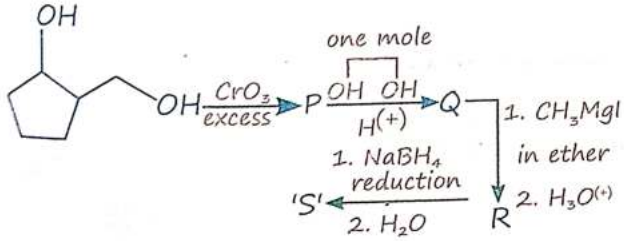
(a) A =  , B = 

(b) A =  , B = 

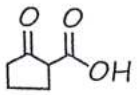
(c) A =  , B = 

(d) A =  , B = 

Sol. (c) 

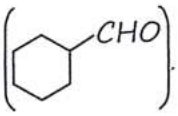
26. 

0.1 mole of compound 'S' will weight _____ g.
(Given molar mass in g mol^{-1} C:12, H:1, O:16)
[29 Jan, 2025 (Shift-I)]

Sol. [13] S = 

0.1 mole of compound (S) weight in gm
= 0.1 × molar mass of compound (S)
= 0.1 × 130 = 13 gm.

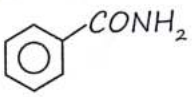
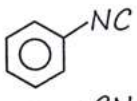
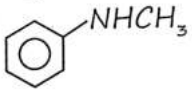
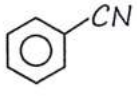
27. From the compounds given below, number of compounds which give positive Fehling's test is___.
Benzaldehyde, Acetaldehyde, Acetone, Acetophenone, Methanal, 4-nitrobenzaldehyde, cyclohexane carbaldehyde.
[29 Jan, 2024 (Shift-I)]

Sol. [3] The compounds that give positive Fehling's test are:
Acetaldehyde (CH_3CHO), Methanal (HCHO), and cyclohexane carbaldehyde  .

28. The compound obtained by heating a mixture of ethyl amine and chloroform with ethanolic potassium hydroxide (KOH) is: (1997)
a. An amide
b. An amide and nitro compound
c. An ethyl isocyanide
d. An alkyl halide

Sol. (c) Isocyanide can be easily characterized by their offensive odour. The reaction is called carbylamine test.

29. Carbylamine test is used to detect the presence of primary amino group in an organic compound. Which of the following compound is formed when this test is performed with aniline?
[25 Feb, 2021 (Shift-II)]

(a) 
(b) 
(c) 
(d) 

Sol. (b) Phenyl isocyanide is formed when carbylamine test is performed with Aniline.

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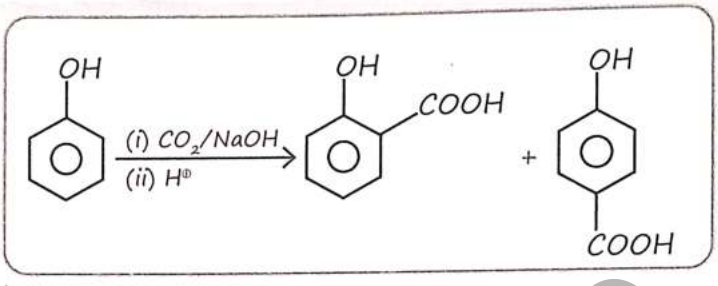
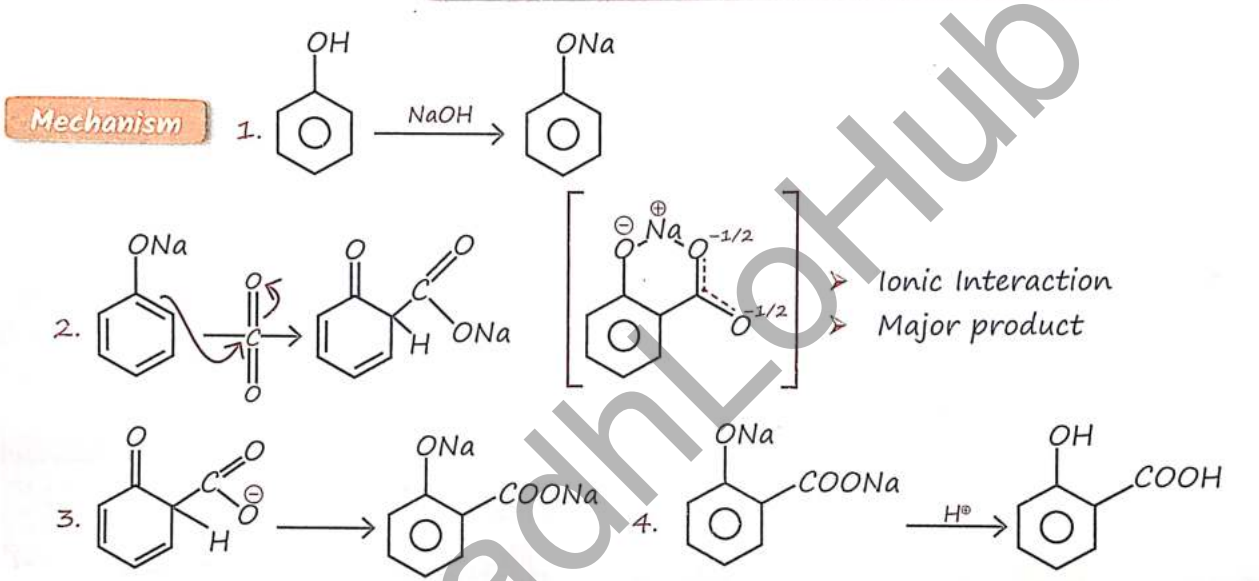
Kolbe's Reaction

In heterocyclic compounds, the Reimer-Tiemann reaction often shows abnormal behavior because the heteroatoms can influence the site of formylation. Instead of the usual ortho position, formylation may occur at other reactive sites on the ring, depending on the electronic effects of the heteroatoms, leading to varied products.

Reaction



Mechanism



Baba Tea-Stall (BTS)

(i) Basic medium is given to increase nucleophilicity of benzene because CO_2 is a weak electrophile.

(ii) Ortho substituted is major product because ionic Chelation.

(iii) We can also use (CCl_4) instead of CO_2.

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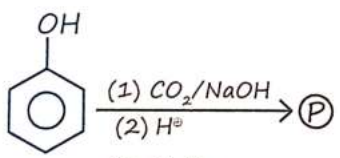
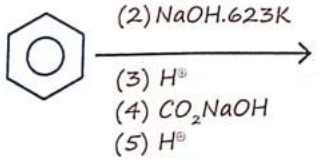
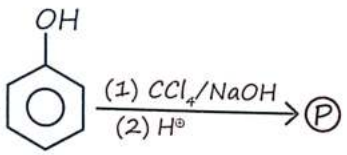
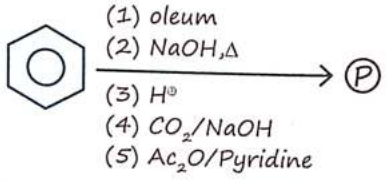
Practice Questions

188. 

189. 

190. 

191. 

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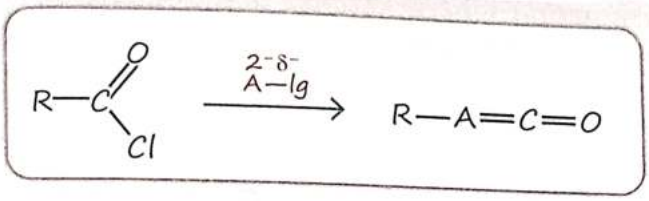
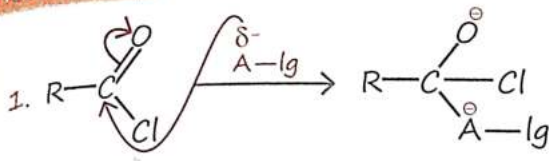
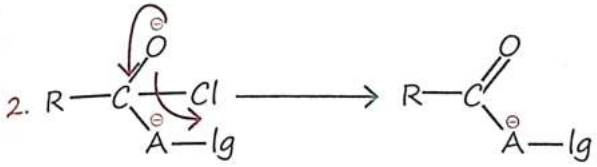
Basic of Rearrangement

Reaction

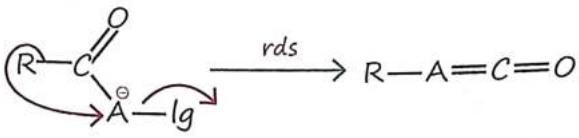


Mechanism

1. 

2. 

- C^+ ⟶ Carbene
- N^+ ⟶ Nitrene
- O^+ ⟶ Nascent oxygen

3. 

- A=C ⟶ Ketene
- A=N ⟶ Isocyanate
- A=O ⟶ $R-\overset{\oplus}{O}=C=O$

4. (a) $R-CH=C=O \xrightarrow{H_3O^{\oplus}} R^{-}CH_2-COOH$

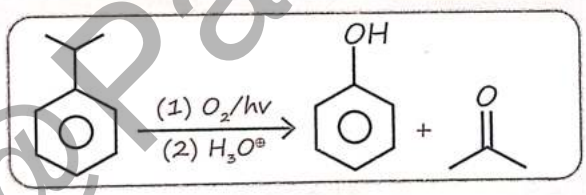
(b) $R-N=C=O \xrightarrow{H_3O^{\oplus}} R-NH_2$

(c) $R-\overset{\oplus}{O}=C=O \xrightarrow{H_3O^{\oplus}} R-OH$

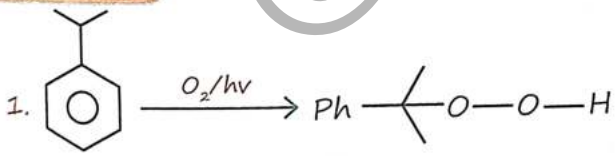
I. Cumene Hydroperoxide Rearrangement

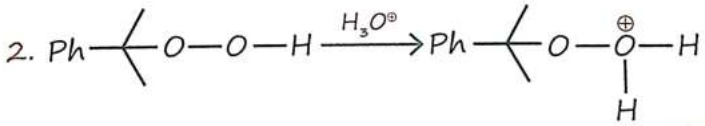
The cumene hydroperoxide rearrangement is an acid-catalyzed reaction where cumene hydroperoxide converts into phenol and acetone. This rearrangement is a key step in the industrial production of phenol and acetone from cumene.

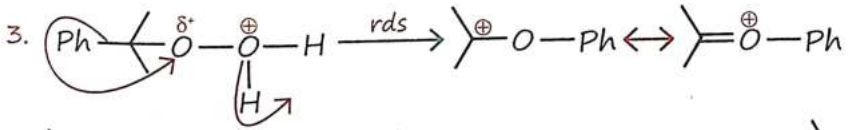
Reaction

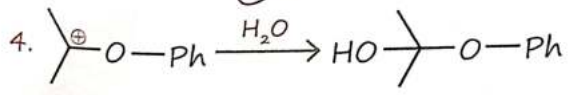


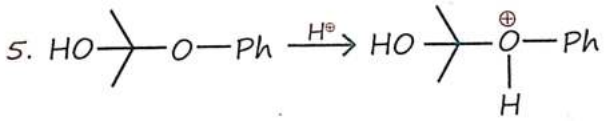
Mechanism

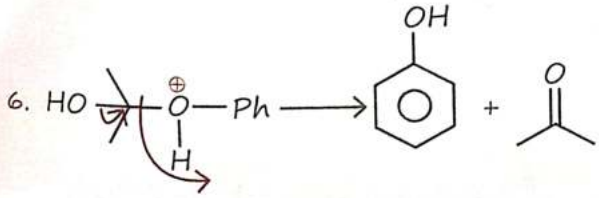
1. 

2. 

3. 

4. 

5. 

6. 

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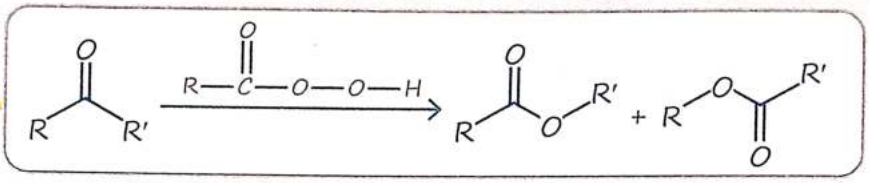
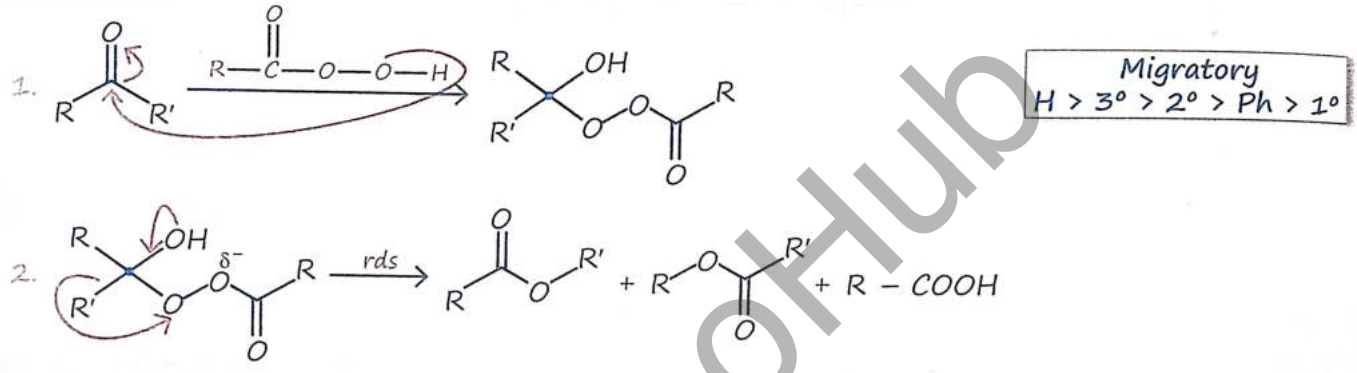
2. Baeyer's Villigers Oxidation

The Baeyer-Villiger oxidation is a reaction where ketones are oxidized by peracids to form esters or lactones. This reaction inserts an oxygen atom next to the carbonyl group, effectively converting ketones into esters.

Reaction



Mechanism



Migratory
H > 3° > 2° > Ph > 1°

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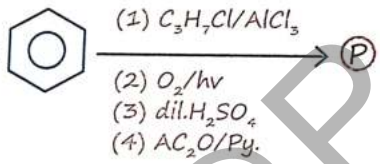
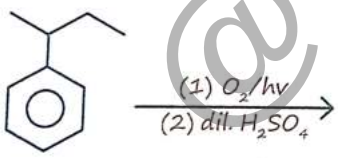
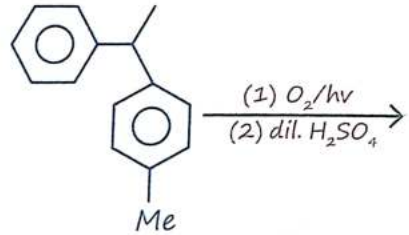
Practice Questions

192. 

193. 

194. 

195. Acetone $\xrightarrow{RCO_3H}$

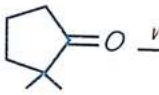
196. Acetophenone $\xrightarrow{RCO_3H}$

197. Acetophenone $\xrightarrow{RCO_3H}$

198. t-Butylphenyl ketone $\xrightarrow{RCO_3H}$

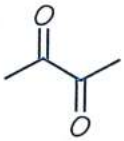
199. Benzaldehyde $\xrightarrow{m\text{-CPBA}}$

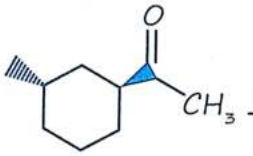
200. Phenyl sec-Butyl ketone $\xrightarrow{m\text{-CPBA}}$

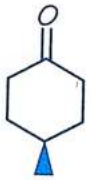
201.  $\xrightarrow{m\text{-CPBA}}$

202. 2-Methyl Cyclohexanone $\xrightarrow{m\text{-CPBA}}$

203. Butanone $\xrightarrow{m\text{-CPBA}}$

204.  $\xrightarrow{m\text{-CPBA}}$

205.  $\xrightarrow{m\text{-CPBA}}$

206.  $\xrightarrow{m\text{-CPBA}}$

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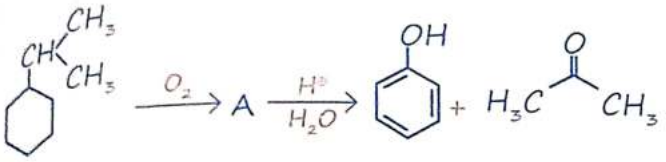
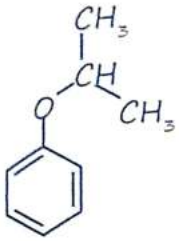
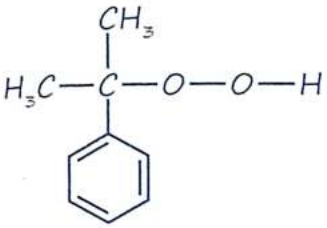
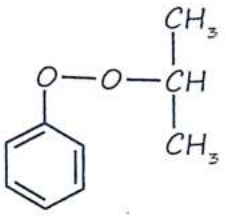
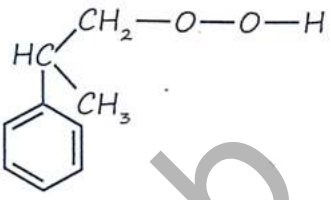
30. The structure of intermediate A in the following reaction, is (2019)



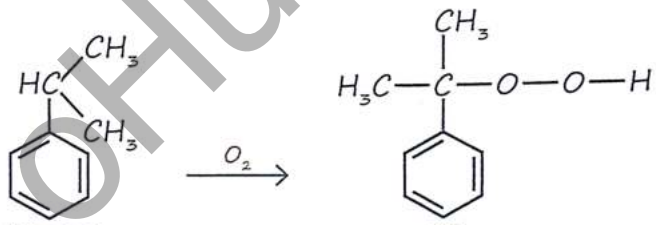
(a) 

(b) 

(c) 

(d) 

Sol. (b)



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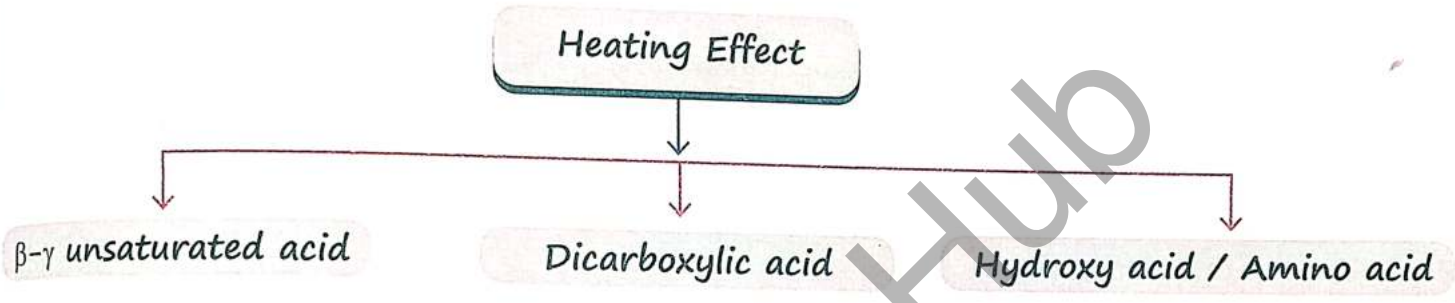
7 Aldehydes, Ketones and Carboxylic acids

Heating Effect

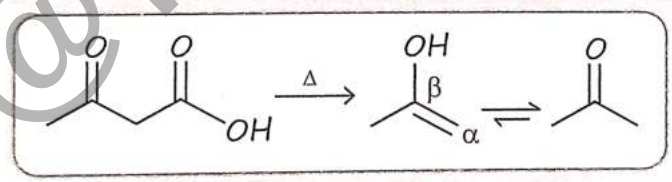


Upon heating, different amines undergo degradation to produce **β-unsaturated acids**, **α, β-unsaturated carboxylic acids**, and **hydroxy acid/amino acid** products, showing the variation in products based on amine types.

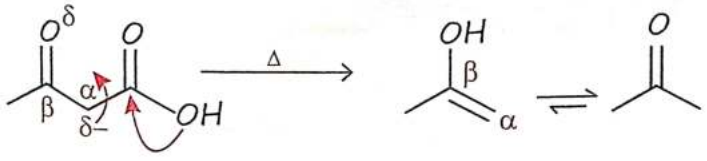
β-γ Unsaturated Acid

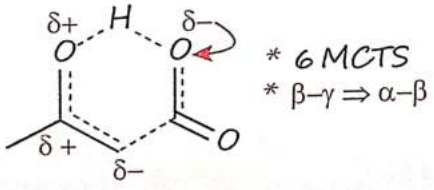
The formation of **β-unsaturated** acids via elimination reactions. It shows the **reaction** (Rxⁿ) and **mechanism** (Mech.) where a hydroxyl group at the β-position undergoes elimination upon heating, resulting in double bond formation. The mechanism includes transition state details involving six-membered cyclic transition states that facilitate β-elimination.

Reaction



Mechanism





* 6 MCTS
* β-γ ⇒ α-β

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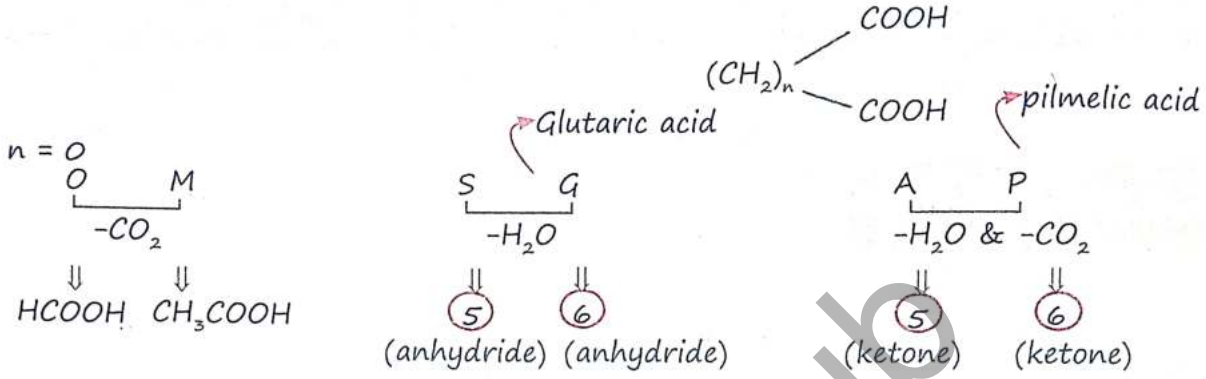
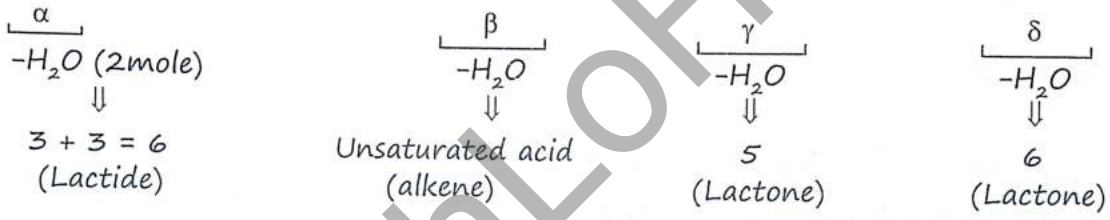
Dicarboxylic Acid

The **reactions of dicarboxylic acids** under heat. It shows different dicarboxylic acids (like oxalic, malonic, succinic, glutaric, adipic, and pimelic acids) undergoing **decarboxylation** or **dehydration** to form products such as acids, anhydrides, and ketones depending on the chain length and functional groups.

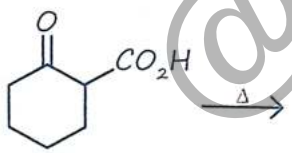


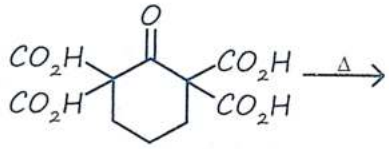
Hydroxy Acid/Amino Acid

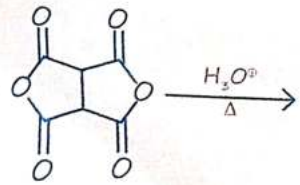


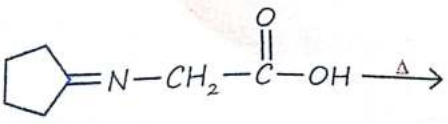
Practice Questions

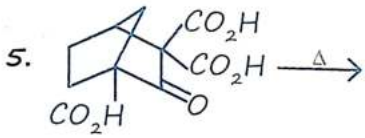
Heating effect:

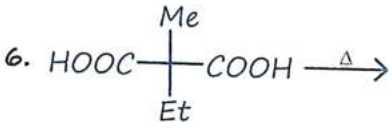
1. 

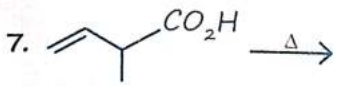
2. 

3. 

4. 

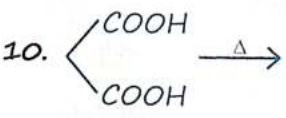
5. 

6. 

7. 

8. $O_2N-CH_2-COOH \xrightarrow{\Delta}$

9. $ON-CH_2-COOH \xrightarrow{\Delta}$

10. 

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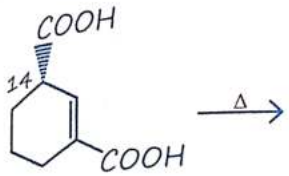
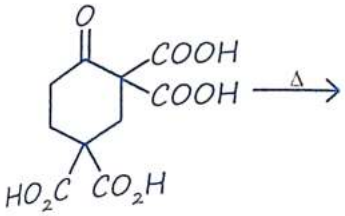
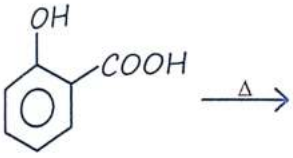
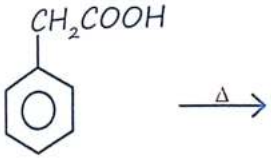
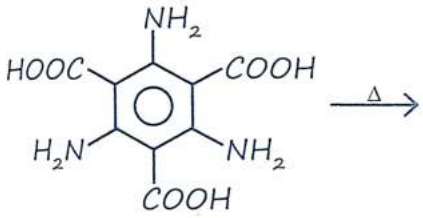
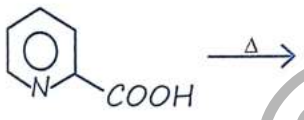
11. 

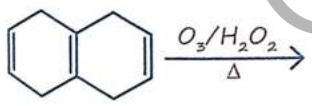
12. 

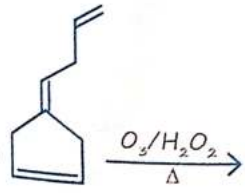
13. 

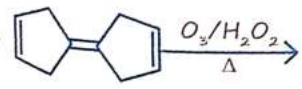
14. 

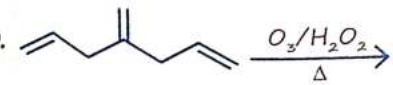
15. 

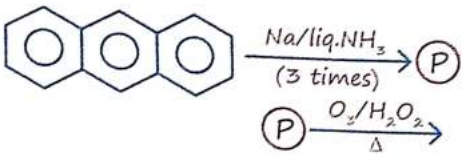
16. 

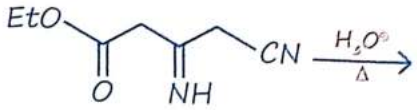
17. 

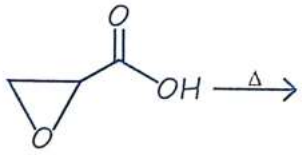
18. 

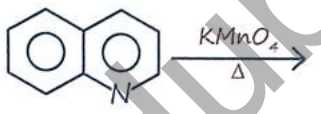
19. 

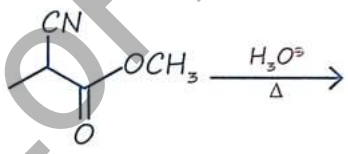
20. 

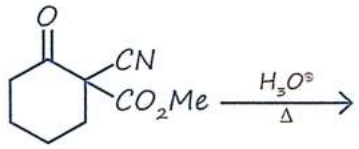
21. 

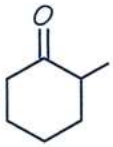
22. 

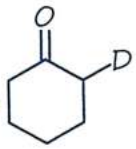
23. 

24. 

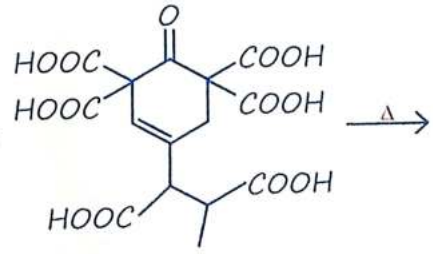
25. 

26. 

27. β-Ketomono acid 

28. β-Ketomono acid 

29. β-γ unsaturated mono acid

30. 

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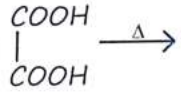
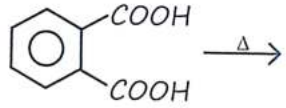
Heating effect of dicarboxylic acids

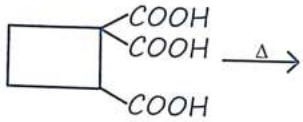
31. 

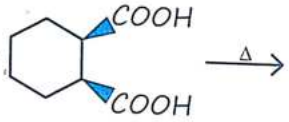
32. Malonic acid $\xrightarrow{\Delta}$

33. Succinic acid $\xrightarrow{\Delta}$

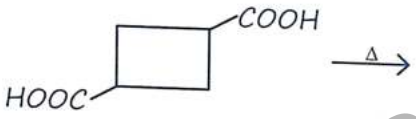
34. Adipic acid $\xrightarrow{\Delta}$

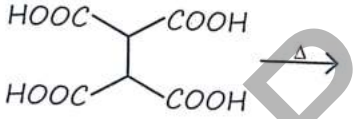
35. 

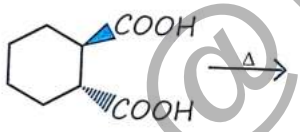
36. 

37. 

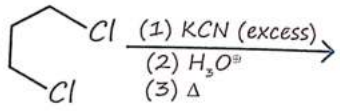
38. $C_4H_4O_4$ $\xrightarrow{\Delta}$ / $\xrightarrow{\Delta}$

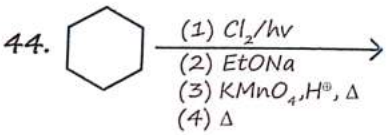
39. 

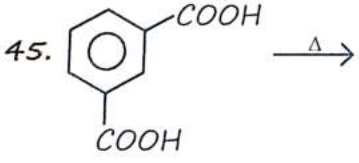
40. 

41. 

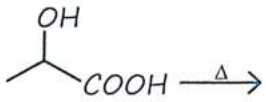
42. $HOOC(CH_2)_5-COOH \xrightarrow{\Delta}$

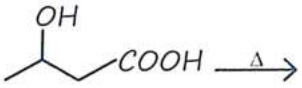
43. 

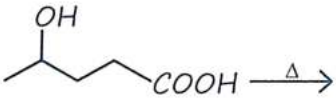
44. 

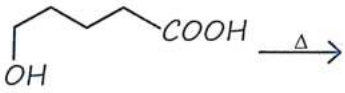
45. 

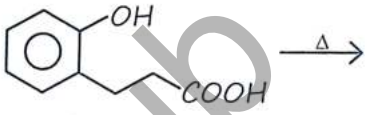
Heating effect of Hydroxy/Amino acids

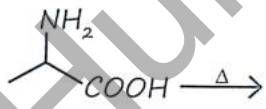
46. 

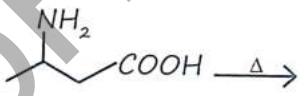
47. 

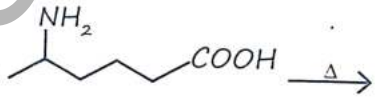
48. 

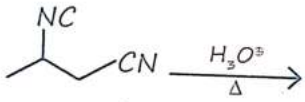
49. 

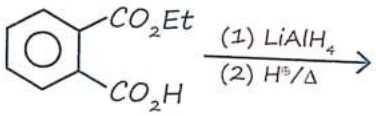
50. 

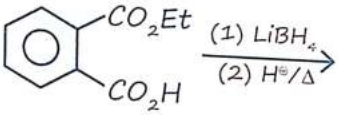
51. 

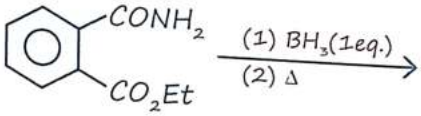
52. 

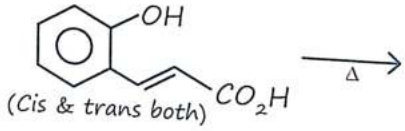
53. 

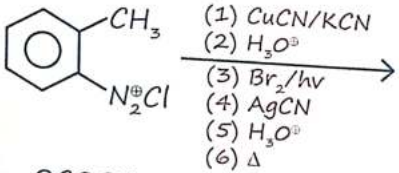
54. 

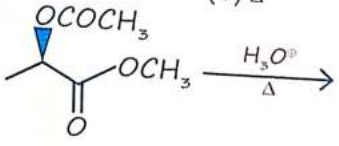
55. 

56. 

57. 

58. 
(Cis & trans both)

59. 

60. 

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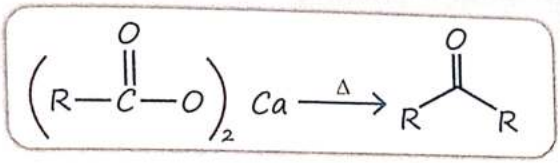
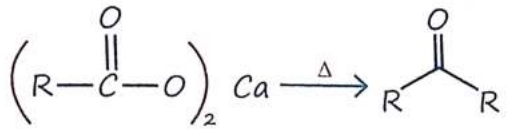
Heating or Distillation of Calcium Salt

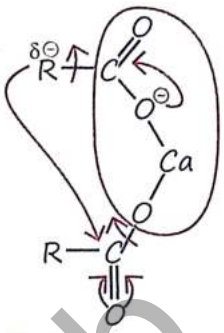
When a calcium salt of a carboxylic acid is dry-distilled (usually with soda lime), it undergoes decarboxylation to produce an alkane. For example, calcium acetate gives methane upon heating.

Reaction



Mechanism



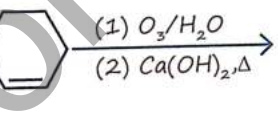


Practice Questions

61. $(CH_3COO)_2Ca \xrightarrow{\Delta}$

62. $(PhCOO)_2Ca \xrightarrow{\text{Dry distillation}}$

63. $EtCOOH \xrightarrow[\Delta]{Ca(OH)_2}$

64. 

65. $(PhCOO)_2Ca + (CH_3COO)_2Ca \xrightarrow{\Delta}$

66. $PhCH_2COONa \xrightarrow[\Delta]{CaCl_2}$

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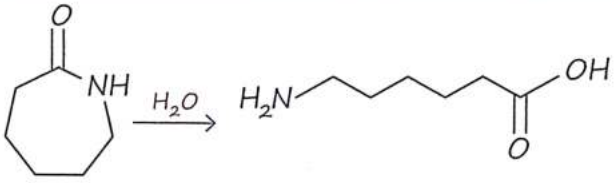
Polymerisation of Amide

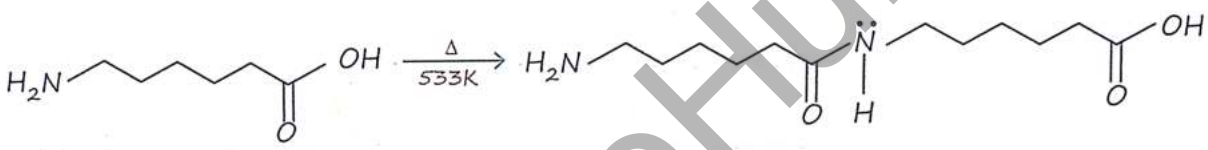
The effect of heat on α-, β-, γ-, and δ-hydroxy or amino acids, leading to different outcomes. For example, α-hydroxy acids form lactides, β-acids give unsaturated acids, and γ/δ-acids cyclise to form lactones or cyclic ketones.

Reaction

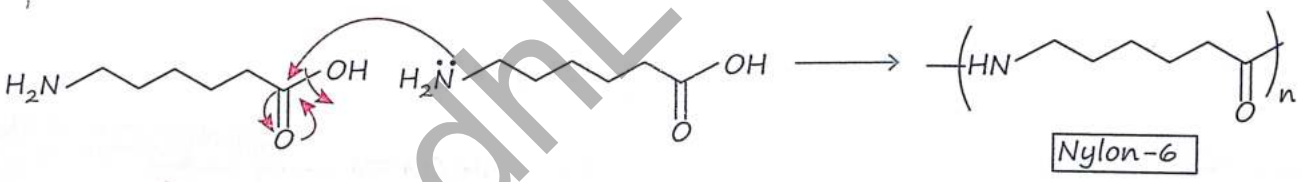


(Caprolactum) (Amino caproic acid)

Mechanism

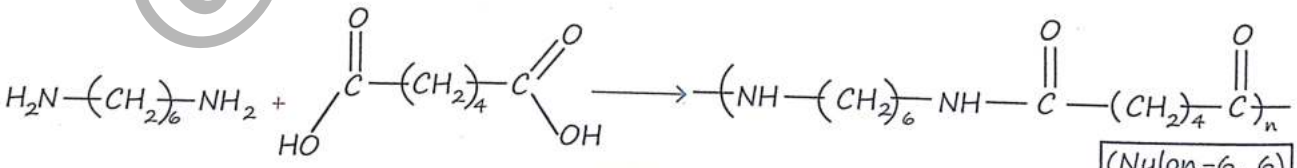


(Amino caproic acid)

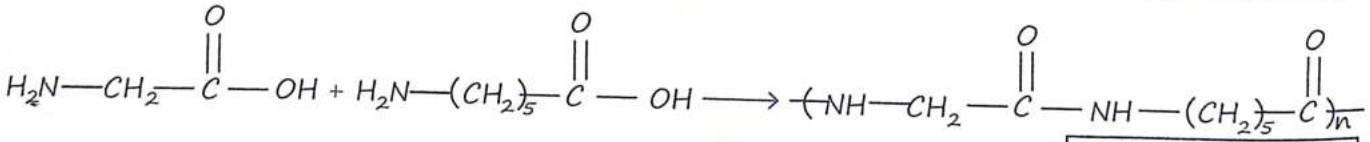


Nylon-6

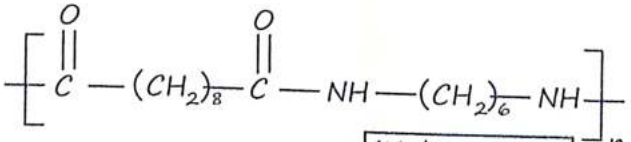
Caprolactam undergoes **ring-opening polymerization** upon heating to form **Nylon-6**, a synthetic polyamide. The process involves the formation of **amide bonds** between monomer units, resulting in a long-chain polymer widely used in textiles, plastics, and engineering materials.



(Nylon-6, 6)



(Nylon-2, Nylon-6)



(Nylon-6, 10)

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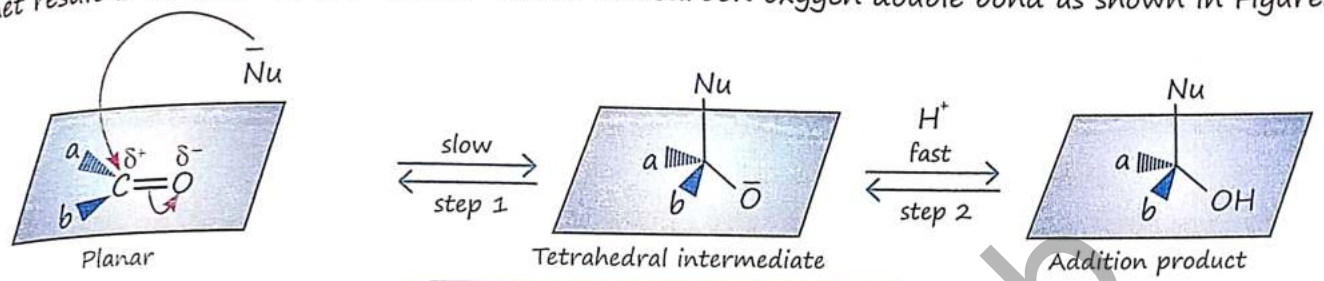
Nucleophilic Addition Reaction

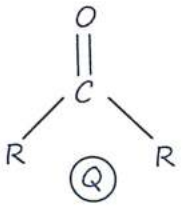
A nucleophile attacks the electrophilic carbon atom of the polar carbonyl group from a direction approximately perpendicular to the plane of sp^2 hybridised orbitals of carbonyl carbon. The hybridisation of carbon changes from sp^2 to sp^3 in this process, and a tetrahedral alkoxide intermediate is produced. This intermediate captures a proton from the reaction medium to give the electrically neutral product. The net result is addition of Nu^- and H^+ across the carbon oxygen double bond as shown in Figure.



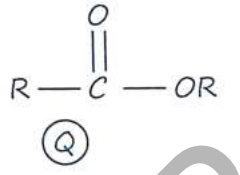
Practice Questions

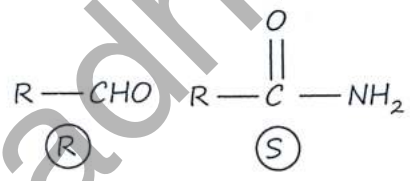
67. Compare the rate of nucleophilic addition Reaction ?

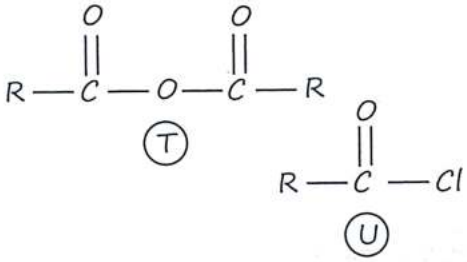
(a) $R-CHO$ (P)



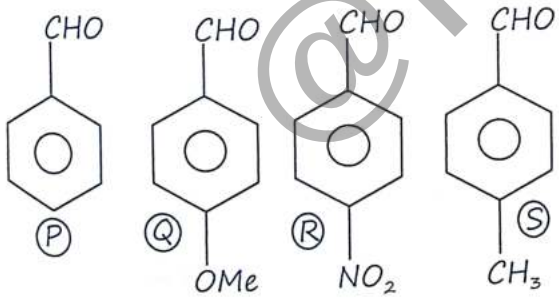
(b) $R-\overset{O}{\overset{\|}{C}}-R$ (P)



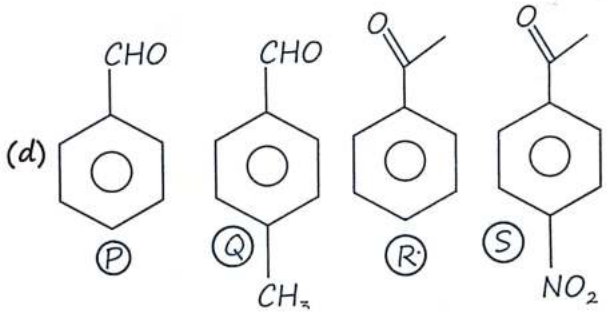




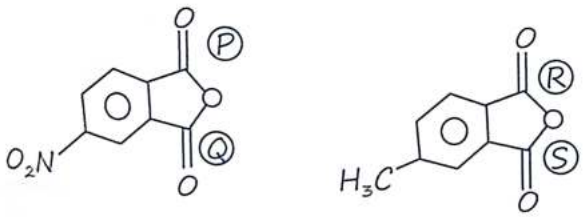
(c)



(d)



(e)



68. Arrange the following compounds in increasing order of their reactivity in nucleophilic addition reactions.

(a) Ethanal, Propanal, Propanone, Butanone.

(b) Benzaldehyde, p-Tolualdehyde, p-Nitrobenzaldehyde, Acetophenone.

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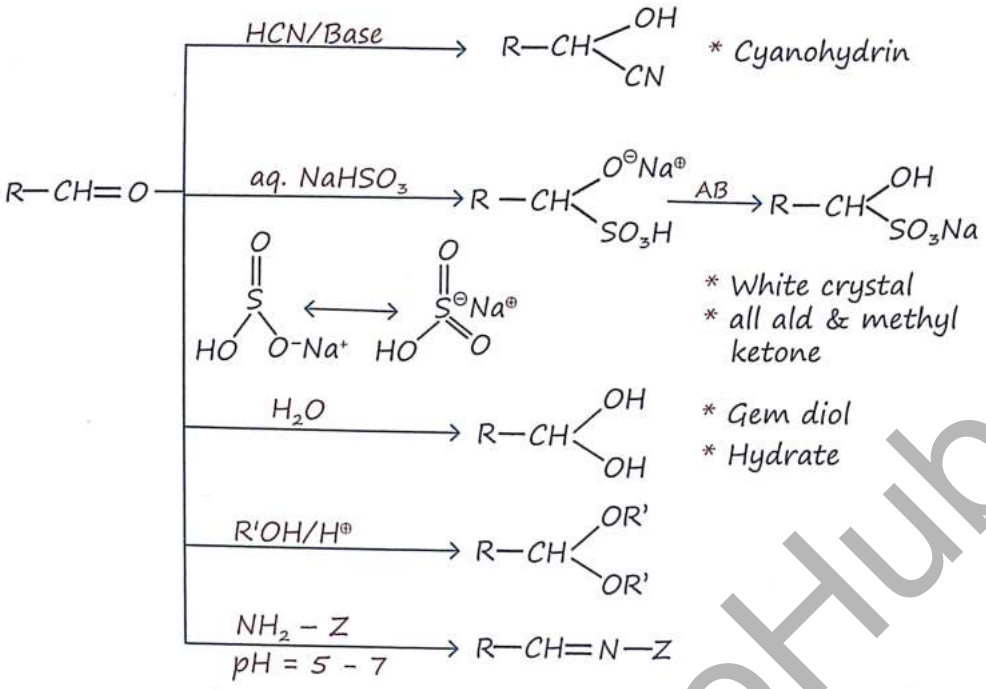
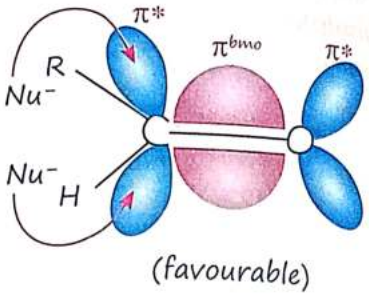
Example of Nucleophilic addition Reaction



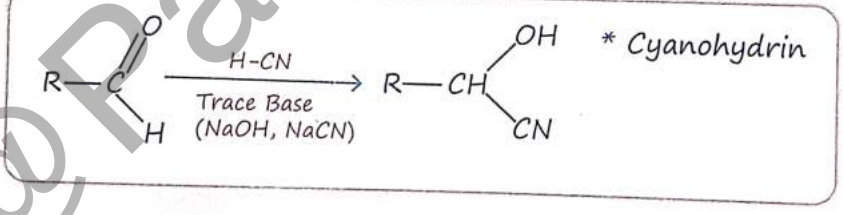


(favourable)

Reaction with H-CN/Base

Aldehydes and ketones react with **hydrogen cyanide (HCN)** in the presence of a base to form **cyanohydrins**. In this nucleophilic addition, the CN^- ion attacks the carbonyl carbon, creating a new C–C bond and introducing both –OH and –CN functional groups.

Reaction



Mechanism

Baba Tea-Stall (BTS)

(i) This reaction occurs very slowly with pure HCN.
(ii) Reaction is catalysed by Base to generate $\overset{\ominus}{CN}$.
(iii) Example of Nucleophilic addition Reaction.

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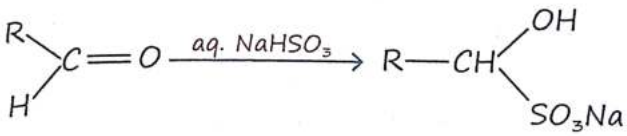
Reaction with Sodium Hydrogen Sulphite:

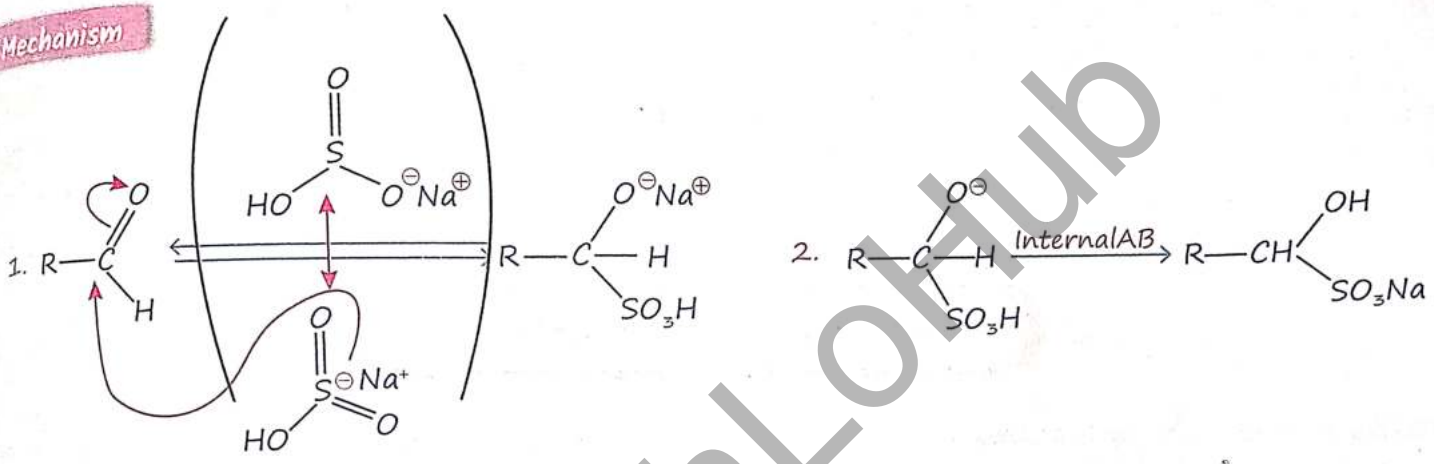
Aldehydes react with **aqueous sodium bisulphite ($NaHSO_3$)** to form **white crystalline, water-soluble bisulphite addition compounds**. This reaction is commonly used to **purify aldehydes**, as the adduct can be easily separated and decomposed to regenerate the original aldehyde.

Reaction



- White crystalline water soluble adduct
- Used to purify aldehydes

Mechanism



Baba Tea-Stall (BTS)

(i) Example of Nucleophilic addition Reaction.
(ii) White crystalline product obtained, so it can be used for separation and purification of aldehyde.
(iii) Crystalline compound is water soluble.
(iv) Reaction can be backward if acidic or basic medium added.

This test is given by
1. All aldehyde (–CHO)
2. All methyl ketone (—C(=O)—CH_3)

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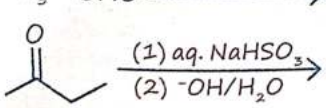
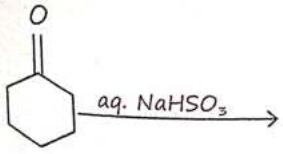
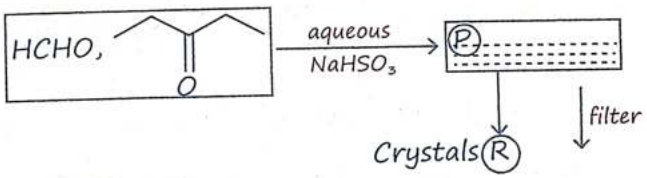
Practice Questions

Reaction with Sodium Hydrogen Sulphite:

69. $CH_3—CHO \xrightarrow{aq.NaHSO_3}$

70. 

71. 

72. 

73. D—Glucose $\xrightarrow{aq.NaHSO_3}$

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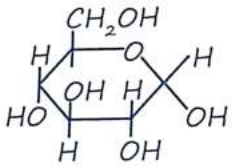
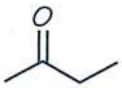
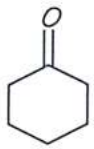
74.  $\xrightarrow{aq.NaHSO_3}$

75. $\xrightarrow[HCN/NaCN]{aq.\ NaHSO_3}$

Reaction with H-CN/Base:

76. $CH_3CHO \xrightarrow{HCN/NaCN}$

77.  $\xrightarrow[(2)\ H_3O^{\oplus}]{(1)\ HCN/NaCN}$

78.  $\xrightarrow[3.\ NaNO_2IHCl]{1.\ HCN/NaCN \\ 2.\ LiAlH_4}$

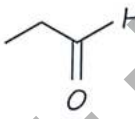
79. (P) $\xrightarrow[2.H_3O^{\oplus}]{1.HCN/NaCN}$ Lactic acid

80. D—Glucose $\xrightarrow{HCN/NaCN}$

81. D-fructose $\xrightarrow{HCN/NaCN}$

82. D-Ribose $\xrightarrow{HCN/NaCN}$ (P)

(P) $\xrightarrow[then\ H_2O]{DIBAl-H}$ (Q) + (R)

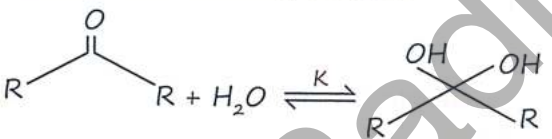
83.  $\xrightarrow[3.\ \Delta]{1.\ HCN/NaCN \\ 2.\ H_3O^{\oplus}}$

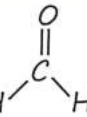
Previous Year Questions

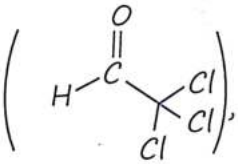
1. Given below are two statements:

[23 Jan, 2025 (Shift-II)]

Consider the following reaction



Statement (I): In the case of formaldehyde  , K is about 2280, due to small substituents, hydration is faster.

Statement (II): In the case of trichloro acetaldehyde  , K is about 2000 due to −I effect of −Cl.

In the light of the above statements, choose the correct answer from the options given below:

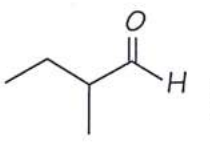
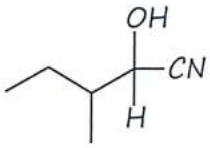
(a) Both Statement I and Statement II are false

(b) Both Statement I and Statement II are true

(c) Statement I is true but Statement II is false

(d) Statement I is false but Statement II is true

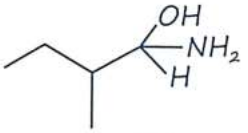
Sol. (b) Both statement-I and statement-II are true.

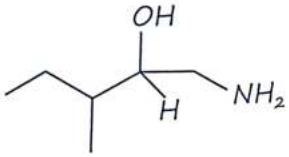
2.  $\xrightarrow[HCN.H_2O]{\text{"X"}}$ 

$\xrightarrow[H_3O^{\oplus}]{LiAlH_4}$ "Y" (Major Product)

Consider the given reaction, Identify 'X' and 'Y'

[26 Aug, 2021 (Shift-II)]

(a) X − HNO_3 Y − 

(b) X − HNO_3 Y − 

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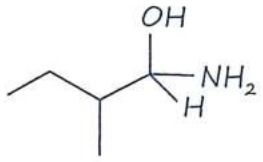
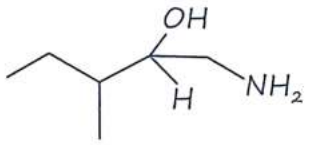
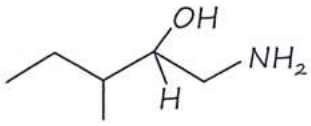
(c) X – NaOH Y – 

(d) X – NaOH Y – 

Sol. (d) X = NaOH and

Y = 

3. In this reaction:

$CH_3CHO + HCN \longrightarrow CH_3CH(OH)CN \xrightarrow{HOH} CH_3CH(OH)COOH$

an asymmetric centre is generated. The acid obtained would be: (2003)

(a) D-isomer

(b) L-isomer

(c) 50% D + 50% L-isomer

(d) 20% D + 80% L-isomer

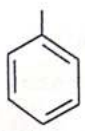
Sol. (c) Lactic acid $[CH_3CH(OH)COOH]$ is an optically active compound due to the presence of asymmetric carbon atom. It exist in D and L-form the ratio of which is found to be (1 : 1), i.e., a racemic mixture.

4. In the reaction $CH_3CHO + HCN \rightarrow CH_3CH(OH)CN$, a chiral centre is produced. This product is: (1995)

(a) Dextrorotatory (b) Racemic mixture

(c) Meso compound (d) Laevorotatory.

Sol. (b)

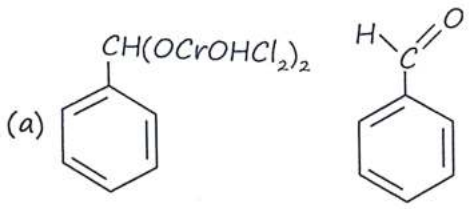
5.  $\xrightarrow[(ii)\ H_3O^+]{(i)\ CrO_2Cl_2,\ CS_2}$ Filter → Residue (A)

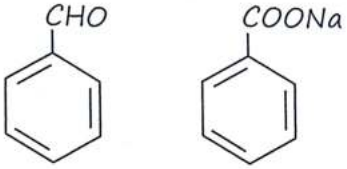
Toluene (excess) (iii) $NaHSO_3$

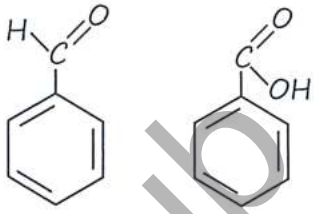
Residue (A) + HCl(dil) → Compound (B)

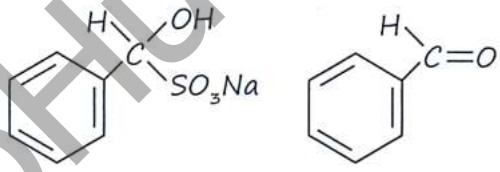
Structure of residue (A) and compound (B) formed respectively is

[22 Jan, 2025 (Shift-II)]

(a) 

(b) 

(c) 

(d) 

Sol. (d)

6. D- (+)-Glyceraldehyde $\xrightarrow[(ii)\ H_2O/H^+]{(i)\ HCN}$ $\xrightarrow{(iii)\ HNO_3}$

The products formed in the above reaction are

[13 April, 2023 (Shift-I)]

(a) Two optically active products.

(b) One optically active and one meso product.

(c) One optically inactive and one meso product.

(d) Two optically inactive products.

Sol. (b)

7. The number of stereoisomers formed in a reaction of (±)Ph(C=O)C(OH)(CN) Ph with HCN is_____. [where Ph is $-C_6H_5$]

[29 July, 2022 (Shift-II)]

Sol. [3]

8. Arrange the following in order of their increasing reactivity towards HCN:

CH_3CHO, CH_3COCH_3, HCHO, $C_2H_5COCH_3$

(IIT JEE 1995)

Sol. $C_2H_5COCH_3 < CH_3COCH_3 < CH_3CHO, < HCHO$

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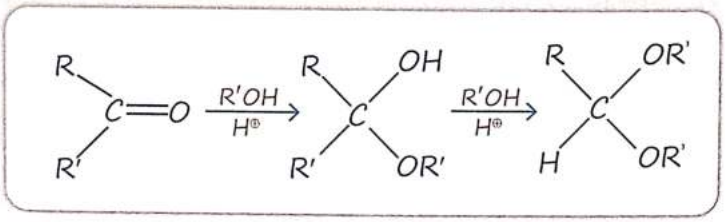
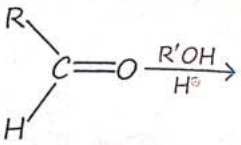
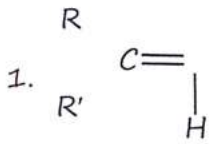
Reaction



Try Yourself



Mechanism

1. 

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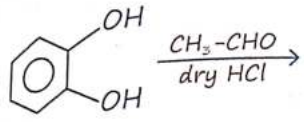
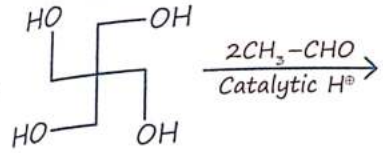
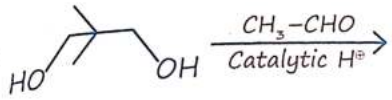
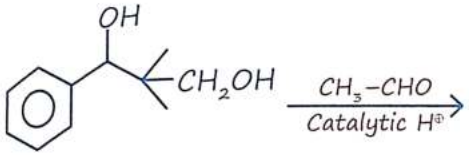
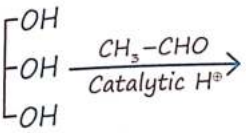
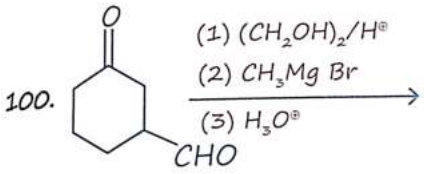
95. 

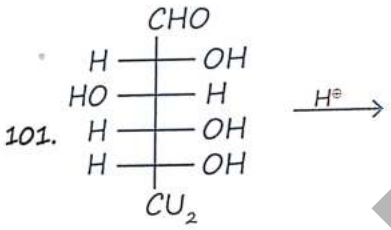
96. 

97. 

98. 

99. 

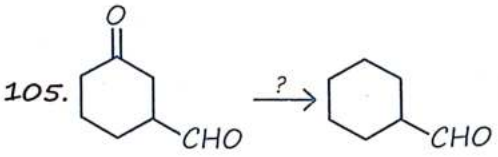
100. 

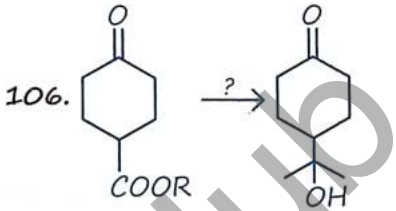
101. 

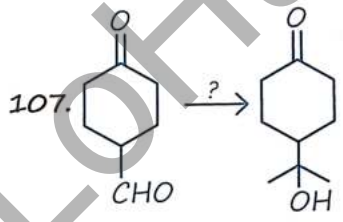
102. D-fructose $\xrightarrow{H^{\oplus}}$

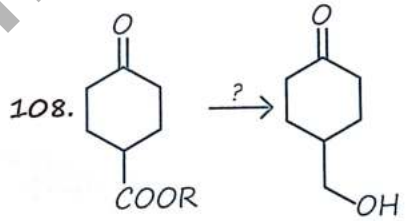
103. D-Galactose $\xrightarrow{H^{\oplus}}$

104. D-Ribose $\xrightarrow{H^{\oplus}}$

105. 

106. 

107. 

108. 

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Previous Year Questions

9. Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R):
Assertion (A): Acetal/Ketal is stable in basic medium.
Reason (R): The high leaving tendency of alkoxide ion gives the stability to acetal/ketal in basic medium.
In the light of the above statements, choose the correct answer from the options given below: [25 Jan, 2023 (Shift-I)]

(a) (A) is true but (R) is false.

(b) (A) is false but (R) is true.

(c) Both (A) and (R) are true and (R) is the correct explanation of A.

(d) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).

Sol. (a)

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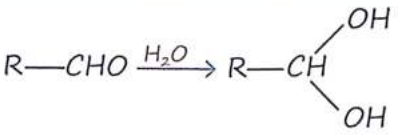
Addition of H_2O

The addition of water to a carbonyl compound (aldehyde or ketone) forms a **gem-diol** (a compound with two hydroxyl groups on the same carbon). This reaction is reversible and generally favored in aldehydes due to their higher reactivity compared to ketones.

Reaction

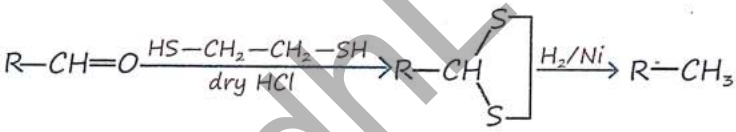


- Gemdiol
- Hydrate

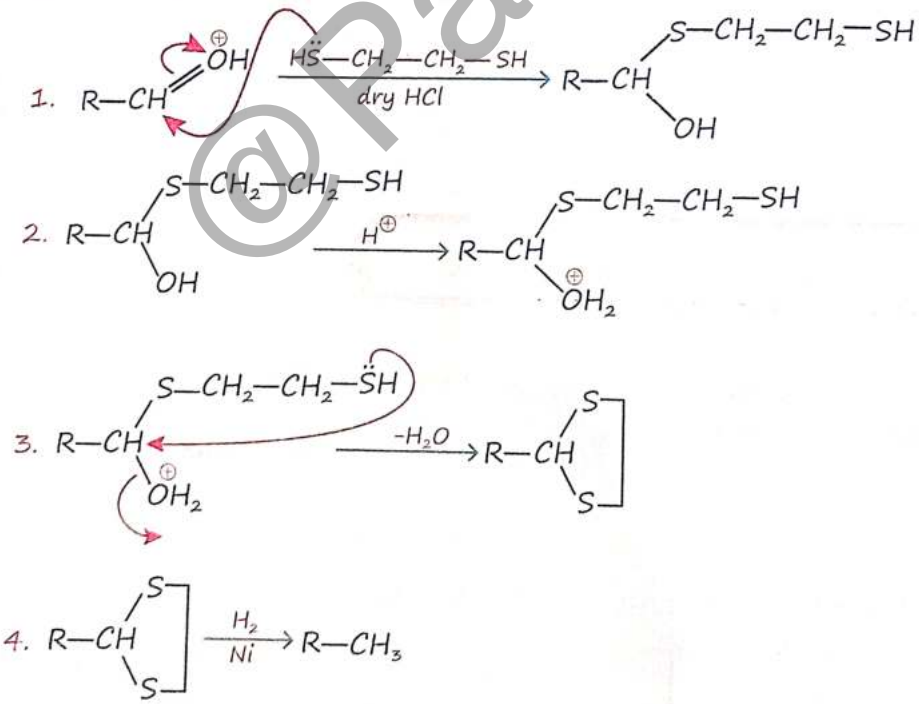
Mozingo Reduction

The Mozingo Reduction involves converting **aldehydes to alkanes** through a sequence of reactions with **dithiols**, forming a **cyclic thioacetal intermediate**, which is then reduced using **Raney nickel** to yield the corresponding **alkane**. This method is used to remove carbonyl oxygen completely (deoxygenation).

Reaction



Mechanism



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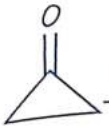
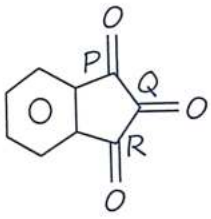
Practice Questions

109. $HCHO \xrightarrow{H_2O/H^\oplus}$

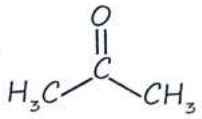
110. $CH_3-CHO \xrightarrow{H_2O/H^\oplus}$

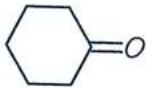
111.  $\xrightarrow{H_2O/H^\oplus}$

112. $Cl_3C-CHO \xrightarrow{H_2O/H^\oplus}$ (Chloral)

113.  $\xrightarrow{H_2O/H^\oplus}$

114. $CH_3-CHO \longrightarrow$

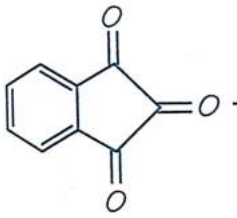
115. $HCHO \longrightarrow$ 

116.  $\xrightarrow[(2)\ H_2|Raney\ Ni]{(1)\ (CH_2SH)_2|dry\ HCl}$

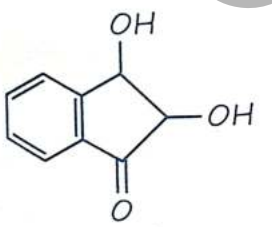
117. Acetophenone $\xrightarrow[(2)\ H_2|Raney\ Ni]{(1)\ (CH_2SH)_2|dry\ HCl}$

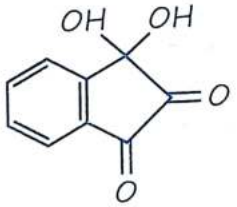
118. Benzophenone $\xrightarrow[(2)\ H_2|Raney\ Ni]{(1)\ (CH_2SH)_2|dry\ HCl}$

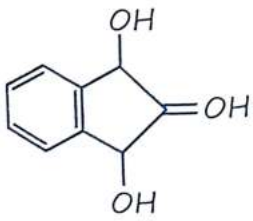
Previous Year Questions

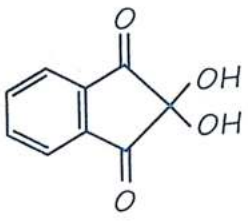
10.  $\xrightarrow{H_2O}$ 'A'

major product 'A' formed in the above reaction is [15 April, 2023 (Shift-I)]

(a) 

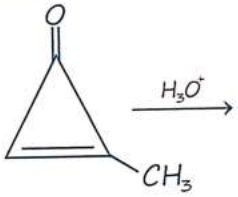
(b) 

(c) 

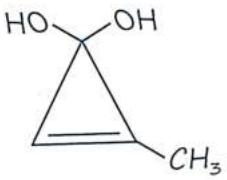
(d) 

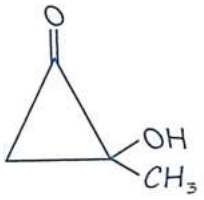
Sol. (d)

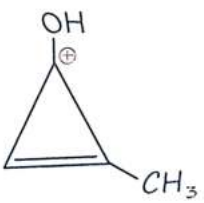
11. The major product in the following reaction is

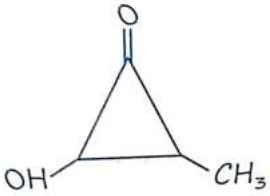


[8 Jan, 2020 (Shift-II)]

(a) 

(b) 

(c) 

(d) 

Sol. (c)

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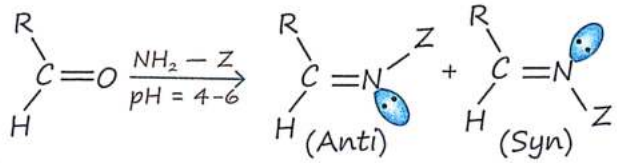
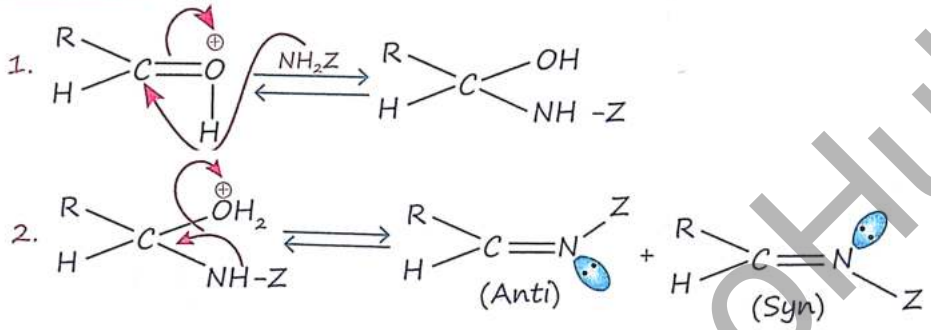
Reaction with NH_2-Z

Reaction of **aldehydes or ketones with substituted hydrazines** (NH_2–Z) to form **hydrazones** via nucleophilic addition followed by elimination of water. The products can exist as **syn and anti isomers**, and this method is commonly used for identification or derivatization of carbonyl compounds

Reaction



Mechanism



Baba Tea-Stall (BTS)

(i) Reaction involves nucleophilic addition followed by elimination.
(ii) Reaction is catalysed by acid to increase electrophilicity.
(iii) 2, 4-DNP test is used for distinction of carbonyl group from others.
(iv) Formation of $>C=N.Z$ favours the equilibrium forward

1.	$-H$ NH_3	Ammonia	$>C=NH$	imine
2.	$-R$ $R-NH_2$	imine	$>C=N-R$	Substituted imine (Schiff's Base)
3.	$-OH$ H_2N-OH	Hydroxylamine	$>C=N-OH$	Oxime
4.	$-NH_2$ H_2N-NH_2	Hydrazine	$>C=N-NH_2$	Hydrazone
5.	$H_2N-HN-C_6H_5$	Phenyl Hydrazine	$>C=N-NH-Ph$	Phenylhydrazone
6.	$H_2N-HN-C_6H_3(NO_2)(NO_2)$	* 2, 4 dinitro phenylhydrazine * 2, 4-DNP * Braddy's Reagent * Borsch reagent	$>C=N-NH-C_6H_3(NO_2)(NO_2)$ (PPt)	Phenylhydrazone
7.	$H_2N-NH-C(=O)-NH_2$	Semicarbazide	$>C=N-NH-C(=O)-NH_2$	Semicarbazone

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Reaction with Secondary Amine

Reaction of **aldehydes with secondary amines**, resulting in the formation of **enamines**. The mechanism involves nucleophilic addition of the amine to the carbonyl group, followed by dehydration under acidic conditions (pH 4–6) to give a stable enamine, which is an important intermediate in organic synthesis.

Reaction

$$CH_3-CHO \xrightarrow[pH=4-6]{R_2NH} CH_2=CH-NR_2 \quad \text{(enamine)}$$

Mechanism

1. $CH_3-CHO \xrightarrow[pH=4-6]{R_2NH} CH_3-CH=\overset{\oplus}{O}H$

2. $CH_3-CH=\overset{\oplus}{O}H \xrightarrow{R_2\ddot{N}H} CH_3-CH(OH)(NR_2)$

3. $CH_3-CH(OH)(NR_2) \xrightarrow{pH=4-6} CH_2=CH-NR_2$

Osazone Formation

Reaction of **aldehydes** with **phenylhydrazine** ($PhNHNH_2$) to form **osazones**. The aldehyde group first forms a **phenylhydrazone**, which then reacts with another molecule of phenylhydrazine at the adjacent carbon to give a **highly conjugated osazone**—a characteristic derivative useful for identification.

Reaction

$$\begin{matrix}CHO\\|\\CH_2OH\end{matrix} \xrightarrow[(3\ eq.)]{PhNHNH_2} \begin{matrix}CH=N-NH-Ph\\|\\CH=N-NH-Ph\end{matrix} \quad \text{(Osazone)}$$

Mechanism

1. $\begin{matrix}CHO\\|\\CH_2OH\end{matrix} \xrightarrow{Ph\ NHNH_2} \begin{matrix}CH=N-NH-Ph\\|\\CH_2OH\end{matrix}$ (Phenylhydrazone) $\rightleftharpoons$ $\begin{matrix}CH-NH-NH-Ph\\\|\\CH-OH\end{matrix} \longrightarrow \begin{matrix}CH-NH\\|\\CH=O\end{matrix} + PhNH_2$

2. $\begin{matrix}CH=NH\\|\\CH=O\end{matrix} \xrightarrow[(2eq.)]{PhNHNH_2} \begin{matrix}CH=N-NH-Ph\\\|\\CH=N-NH-Ph\end{matrix}$ (Osazone)

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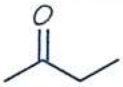
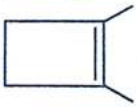
Practice Questions

119. $CH_3-CHO \xrightarrow{1-7}$

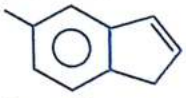
120. $\xrightarrow[(2)\ 1-7]{(1)\ O_3|H_2O_2,\ \Delta}$

121.  $\xrightarrow[H^{\oplus}]{NH_2-OH}$

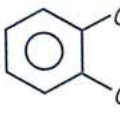
122.  $\xrightarrow{O_2/ZnH_2O}$ (P) ; (P) $\xrightarrow[\Delta]{(NH_4)_2CO_3}$

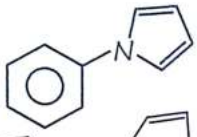
123. (P) $\xrightarrow[or\ P_2O_5,\ \Delta]{Conc.\ H_2SO_4}$

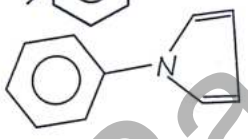
124. (P) $\xrightarrow[\Delta]{H_2S}$

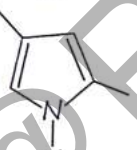
125.  $\xrightarrow[(2)\ NH_3]{(1)\ O_3/Zn}$

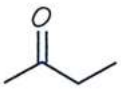
126. $\xrightarrow[(2)\ NH_2-NH_2]{(1)\ O_3/Zn}$

127.  $\xrightarrow{NH_2-NH_2}$

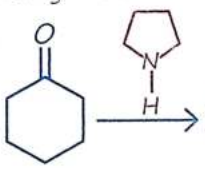
128. (P) + (Q) $\longrightarrow$ 

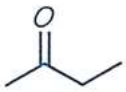
129. P + Q $\xrightarrow{NH_3}$ 

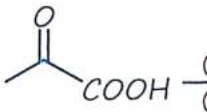
130. P + Q $\xrightarrow{NH_3}$ 

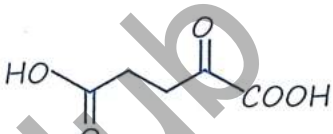
131.  $\xrightarrow{Me_2NH}$

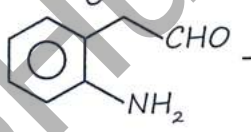
132. $CH_3-CHO \xrightarrow{Me-NH-NH_2}$

133.  (P) ; (P) $\xrightarrow[(2)\ H_2O/H^{\oplus}]{(1)\ CH_3-I}$ (Q)

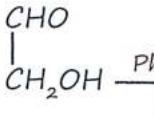
134.  $\xrightarrow[KCN]{NH_4Cl}$

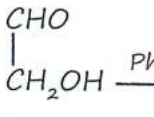
135.  $\xrightarrow[(2)\ H_2/Ni]{(1)\ NH_3}$

136.  $\xrightarrow[(2)\ H_2/Ni]{(1)\ NH_3}$

137.  $\xrightarrow{H^{\oplus}}$

138. $CH_3=CHO \xrightarrow[KCN]{NH_4Cl}$ (P) $\xrightarrow{H_3O^{\oplus}}$ (Q)

139.  $\xrightarrow[(1eq.)]{PhNHNH_2}$

140.  $\xrightarrow[(3eq.)]{PhNHNH_2}$

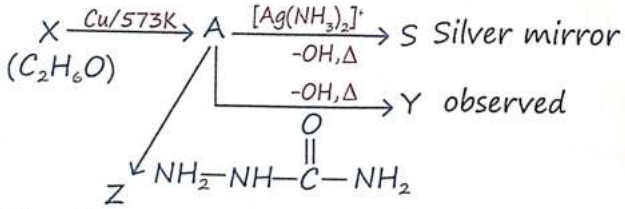
141. D-Glucose $\xrightarrow[(3eq.)]{PhNHNH_2}$

142. D-Fructose $\xrightarrow[(3eq.)]{PhNHNH_2}$

143. D-Mannose $\xrightarrow[(3eq.)]{PhNHNH_2}$

Previous Year Questions

12. Consider the reactions: (2017-Delhi)



Identify A, X, Y and Z

a. A-Ethanol, X-Acetaldehyde, Y-Butanone, Z-Hydrazone

b. A-Methoxymethane, X-Ethanoic acid, Y-Acetate ion, Z-hydrazine

c. A-Methoxymethane, X-Ethanol, Y-Ethanoic acid, Z-Semicarbazide

d. A-Ethanal, X-Ethanol, Y-But-2-enal, Z-Semicarbazone

Sol. (d)

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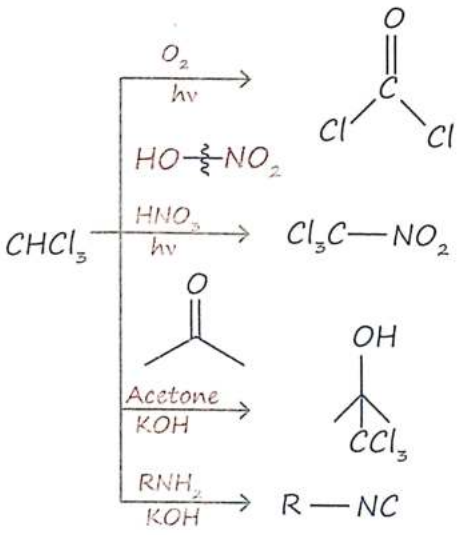
Reactions with Chloroform

Various reactions of chloroform under different conditions:

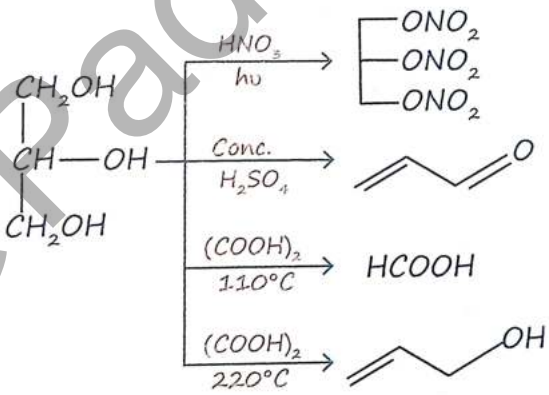
- With O_2 and UV light, it forms phosgene gas, which is highly poisonous.
- With nitric acid or nitrous acid, it forms chloropicrin, a teargas compound.
- In the presence of acetone and base, it yields chlorobutanol (chloral hydrate).
- With primary amines and KOH, it undergoes the carbylamine reaction, producing foul-smelling isocyanides, a test for primary amines.



- Phosgene gas
- Poisonous
- Chloropicrin
- Teargas
- Chloretone
- Carbylamine Reaction
- Foul smell

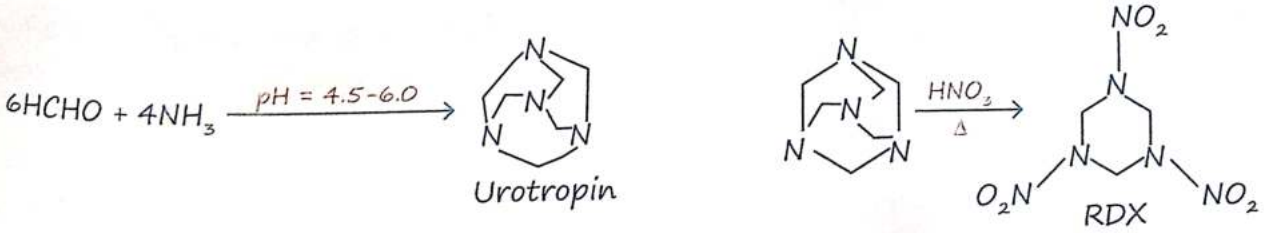
Reactions with Glycerol

Glycerol undergoes nitration with HNO_3 to form **trinitroglycerin**, a powerful explosive. When heated with conc. H_2SO_4, it forms **acrolein**, and oxidation at different temperatures yields **formic acid** or **allyl alcohol**.



Formation of Urotropin

Urotropin (hexamethylenetetramine) is formed by reacting **formaldehyde (HCHO)** with **ammonia (NH_3)** under mildly acidic conditions (pH 4.5–6.0). It results in a **cyclic amine compound**, which upon nitration with nitric acid forms **RDX**, a powerful military explosive.



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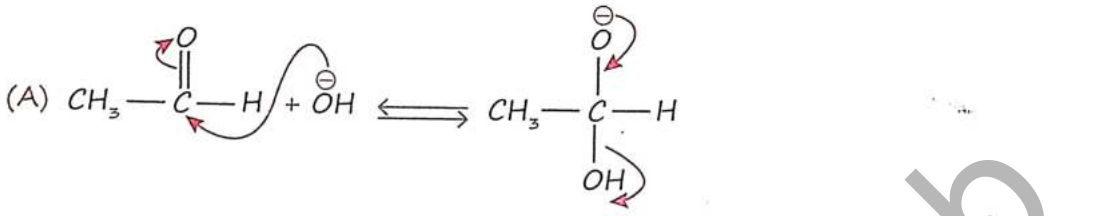
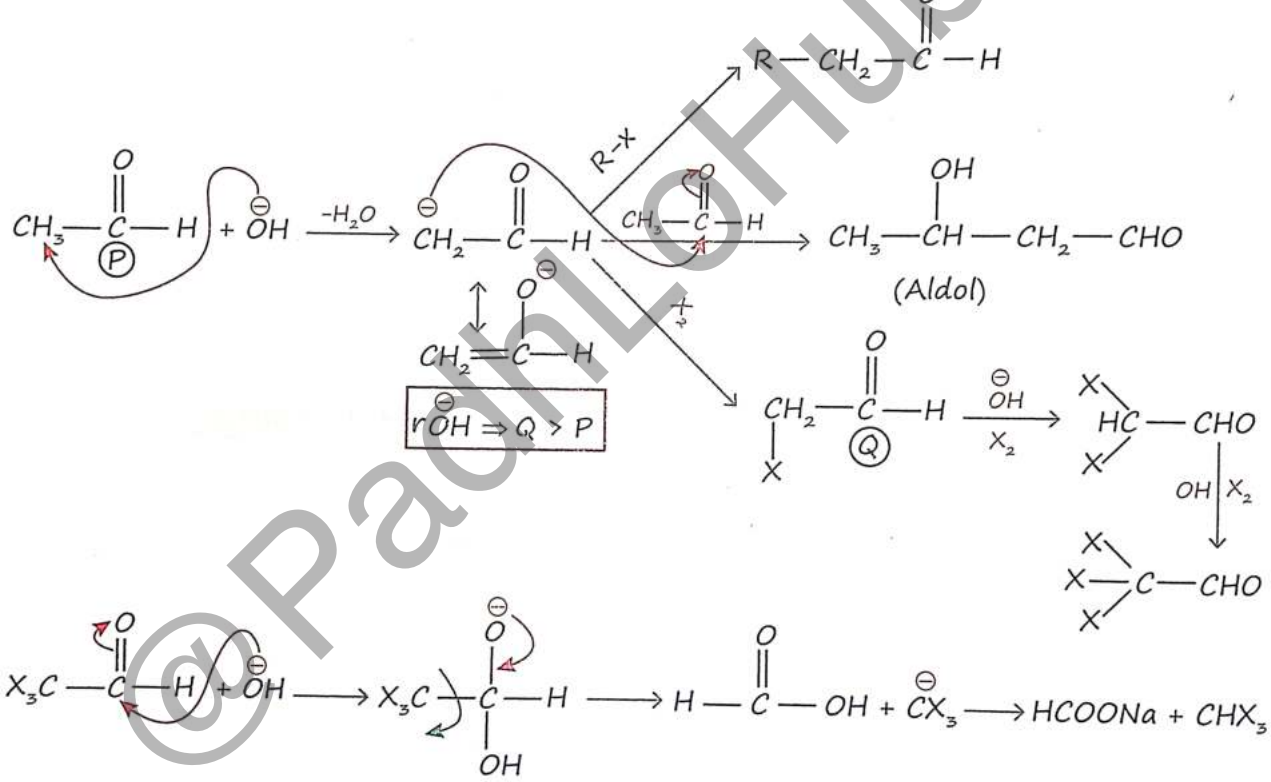
Basics of Name Reactions

When aldehydes or ketones react with a base, they first form an **enolate ion** by removal of an α-hydrogen. The enolate attacks another carbonyl compound to form a **β-hydroxy aldehyde or ketone** (aldol reaction). **Similarly, if the enolate reacts with halogen (X_2), it undergoes multiple α-halogenations**, and in the case of methyl ketones, this leads to the **haloform reaction**, producing a **carboxylate salt** and a haloform (CHX_3) as the final products.



(B)



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Previous Year Questions

13. Which of the following has the most acidic hydrogen? (IIT JEE 2000)

(a) 3-hexanone

(b) 2,4 -hexanedione

(c) 2,5 -hexanedione

(d) 2,3 -hexanedione

Sol. (b) 2,4 Hexanedione has most acidic hydrogen this is because the Carbanion left after the removal of hydrogen ion that is resonance stabilized.

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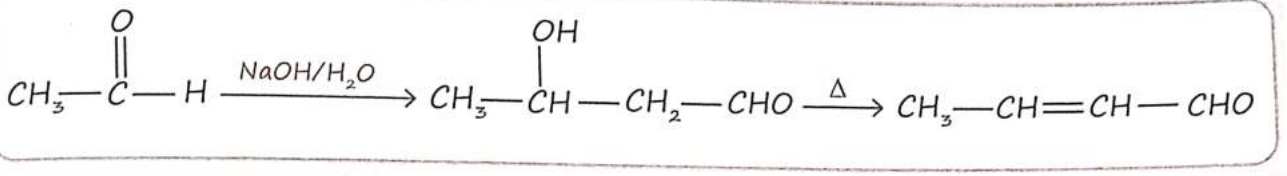
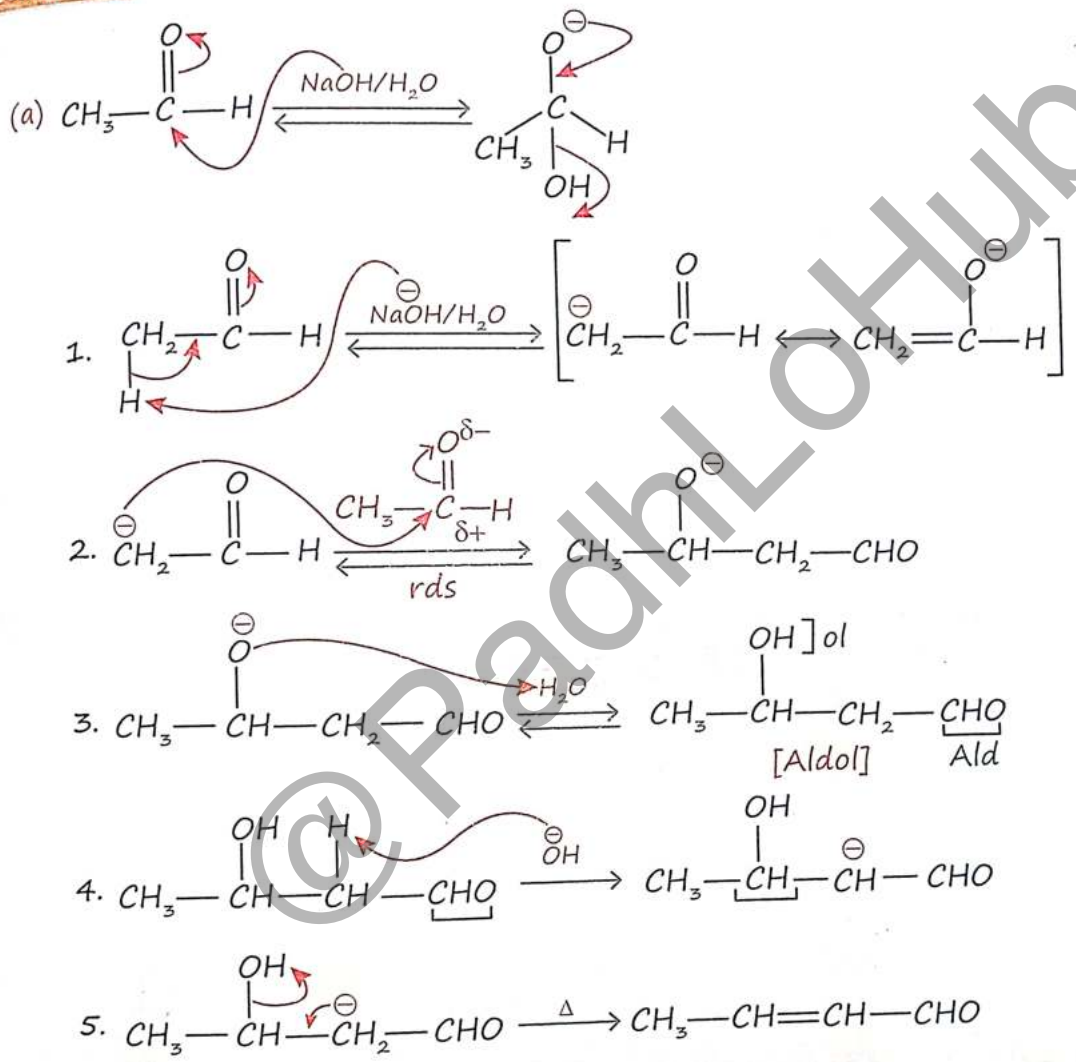
Aldol Reaction

Aldol condensation involves the reaction of **aldehydes or ketones containing α-hydrogens** in the presence of a base to form **β-hydroxy aldehydes or ketones** (aldols).

Reaction



Mechanism



Baba Tea-Stall (BTS)

(i) Reaction involves :

1 Carbanion/Enolate ion
2 Nu⊖addⁿ (rds)
3. Acid Base Reaction (aldol formation)
4. Carbanion
5. Elimination /dehydration } E1CB

(ii) Nucleophilic addition is RDS.

(iii) Complete Reaction is Reversible.

(iv) Instead of NaOH, Δ, we can also use:

1. KOH, Δ
2. $NaHCO_3$, Δ
3. $Ca(OH)_2$, Δ
4. $Ba(OH)_2$, Δ (Commonly used)
5. K_2CO_3, Δ

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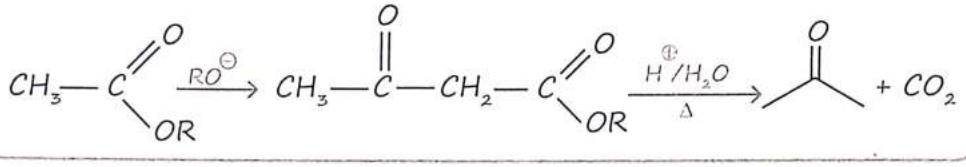
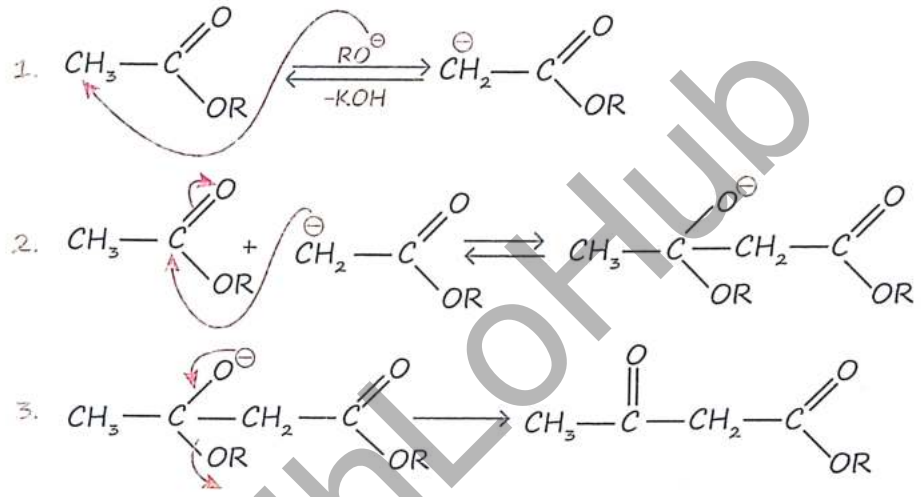
Claisen Condensation

The Claisen condensation is a carbon–carbon bond-forming reaction between two ester molecules (or an ester and a ketone) in the presence of a strong base, forming a β-keto ester. It proceeds via enolate formation from one ester, which then attacks the carbonyl carbon of another, followed by elimination of the alkoxy group. This reaction is useful in synthesizing 1,3-dicarbonyl compounds.

Reaction



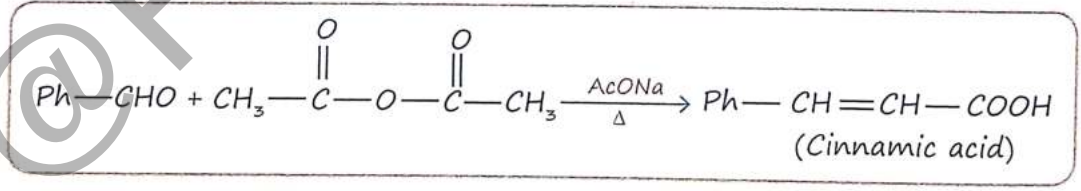
Mechanism



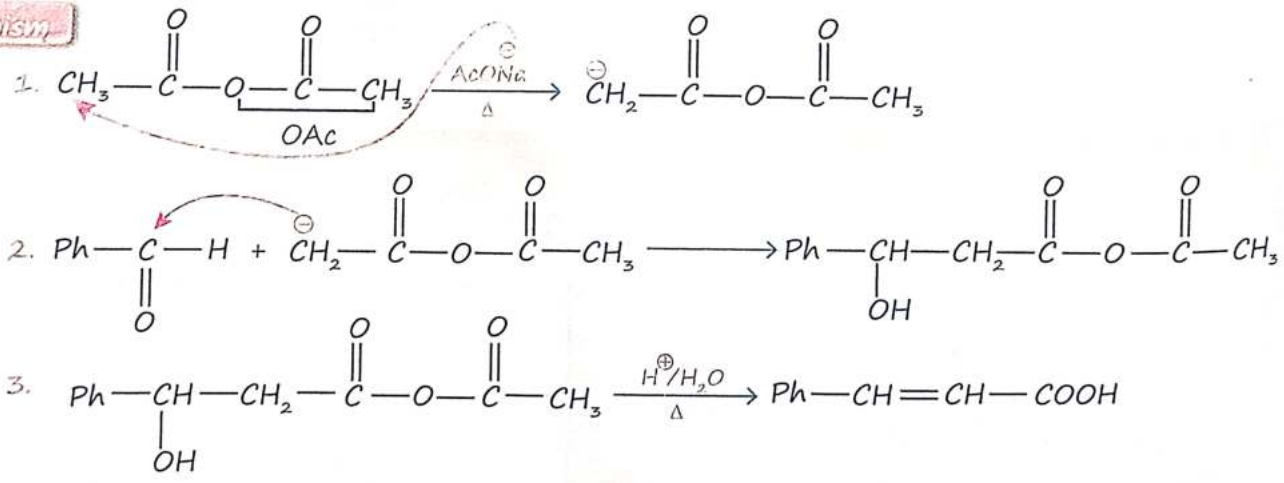
Perkin Condensation

The Perkin reaction involves the condensation of an aromatic aldehyde with an acid anhydride in the presence of a base (like sodium acetate) to form α,β-unsaturated carboxylic acids. It is commonly used in the synthesis of cinnamic acid derivatives

Reaction



Mechanism



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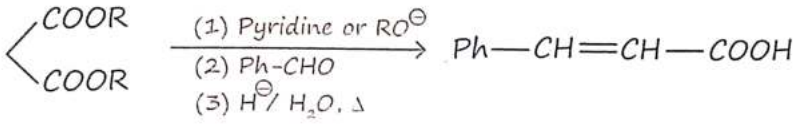
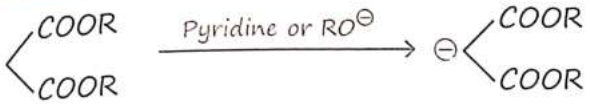
Knoevenagel Reaction

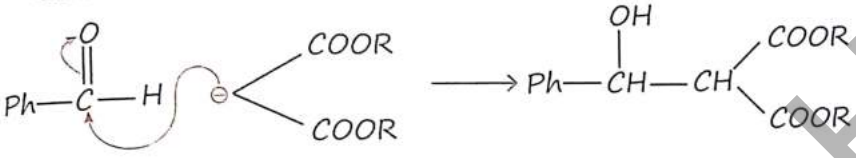
The Knoevenagel reaction is a **base-catalyzed condensation** of an **aldehyde or ketone** with a compound containing active methylene (like malonic ester), forming **α,β-unsaturated compounds**. It proceeds via enolate formation and is used in fine chemical synthesis

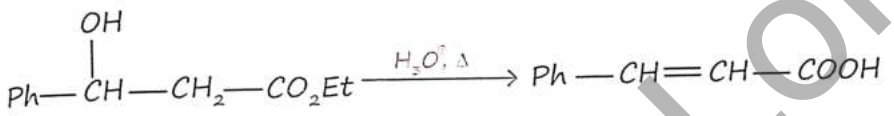
Reaction



Mechanism

1. 

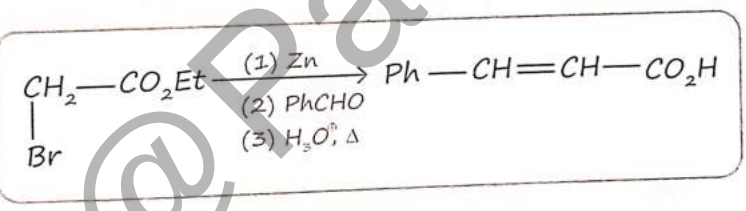
2. 

3. 

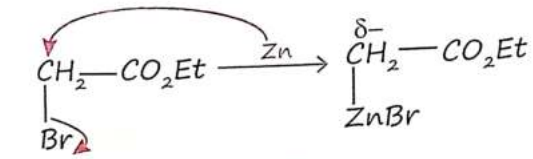
Reformatsky Reaction

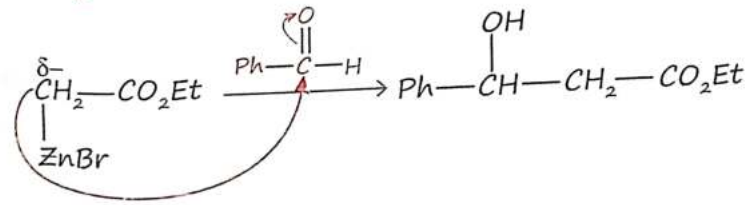
In the Reformatsky reaction, **α-halo esters** react with **aldehydes or ketones** in the presence of **zinc metal** to form **β-hydroxy esters**. It is a C–C bond-forming reaction involving a zinc enolate intermediate

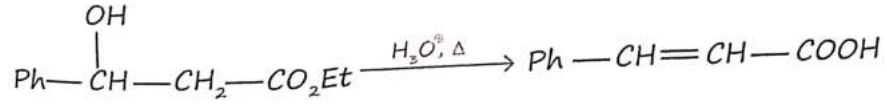
Reaction



Mechanism

1. 

2. 

3. 

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Practice Questions

144. $\xrightarrow{Ba(OH)_2,\ \Delta}$

145. Propanal $\xrightarrow{Ba(OH)_2,\ \Delta}$

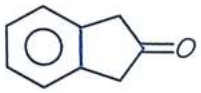
146. $PhCH_2CHO$ $\xrightarrow{Ba(OH)_2,\ \Delta}$

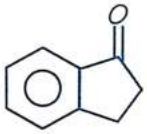
147. Acetophenone $\xrightarrow{Ba(OH)_2,\ \Delta}$

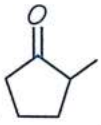
148. Butanone $\xrightarrow{Ba(OH)_2,\ \Delta}$

149. Butanal $\xrightarrow{Ba(OH)_2,\ \Delta}$

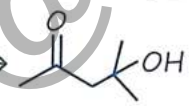
150. $\xrightarrow{Ba(OH)_2,\ \Delta}$

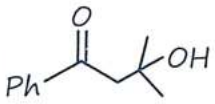
151.  $\xrightarrow{Ba(OH)_2,\ \Delta}$

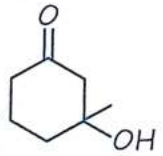
152.  $\xrightarrow{Ba(OH)_2,\ \Delta}$

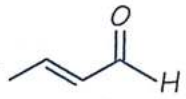
153.  $\xrightarrow{Ba(OH)_2,\ \Delta}$

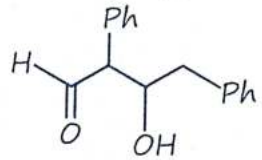
154. $CH_3-CH=CH-CHO$ $\xrightarrow{Ba(OH)_2,\ \Delta}$

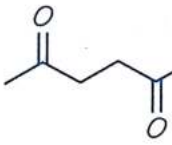
155. (CC) $\xrightarrow{NaOH}$ 

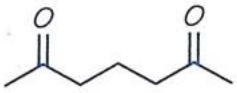
156. (CC) $\xrightarrow{NaOH}$ 

157. (CC) $\xrightarrow{NaOH}$ 

158. (CC) $\xrightarrow[\Delta]{NaOH}$ 

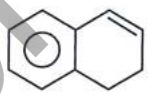
159. (CC) $\xrightarrow{NaOH}$ 

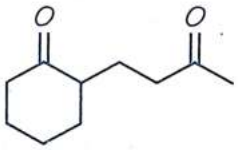
160.  $\xrightarrow{NaOH,\ \Delta}$

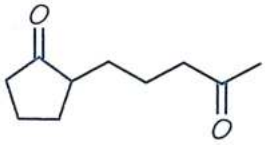
161.  $\xrightarrow{NaOH,\ \Delta}$

162. $\xrightarrow{NaOH,\ \Delta}$

163. $\xrightarrow[(2)\ NaOH,\ \Delta]{(1)\ O_3/Zn,\ H_2O}$

164.  $\xrightarrow[(2)\ NaOH,\ \Delta]{(1)\ O_3/Zn,\ H_2O}$

165.  $\xrightarrow{NaOH,\ \Delta}$

166.  $\xrightarrow{NaOH,\ \Delta}$

167. $CH_3-CHO + HCHO$ (1 : 1) $\xrightarrow{NaOH,\ \Delta}$

168. $CH_3-CHO + HCHO$ (1 : 2) $\xrightarrow{NaOH,\ \Delta}$

169. $CH_3-CHO + HCHO$ (1 : 3) $\xrightarrow{NaOH,\ \Delta}$

170. PhCHO + Acetone $\xrightarrow{NaOH,\ \Delta}$

171. $Ph-CHO + CH_3CH_2CHO$ $\xrightarrow{NaOH,\ \Delta}$

172. Acetophenone + HCHO $\xrightarrow{NaOH,\ \Delta}$

173. Acetophenone + CH_3-CHO $\xrightarrow{NaOH,\ \Delta}$

174. $CH_3-NO_2 + HCHO$ $\xrightarrow{NaOH,\ \Delta}$

175. $CH_3-NO_2 + HCHO$ (1 : 3) $\xrightarrow{NaOH,\ \Delta}$

176. $CH_3-CN + HCHO$ $\xrightarrow{NaOH,\ \Delta}$

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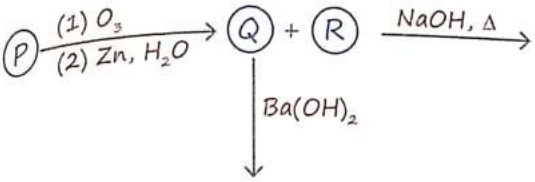
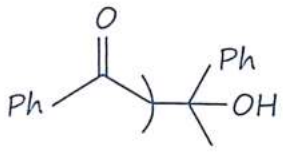
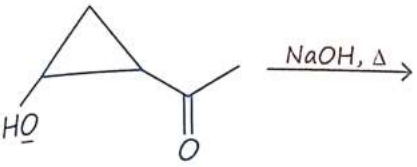
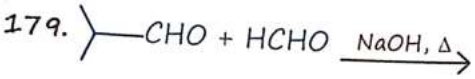
177. 

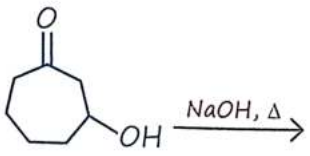


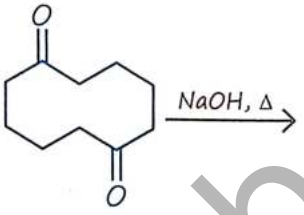
178. 

179. 

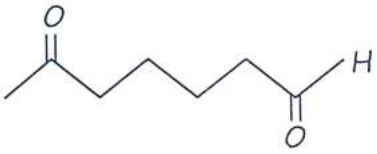
180. $CH_3-CHO + HCHO + Ph-CHO \xrightarrow{NaOH, \Delta}$

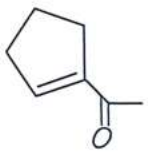
181. $CH_3-CHO + HCHO + Acetone \xrightarrow{NaOH, \Delta}$

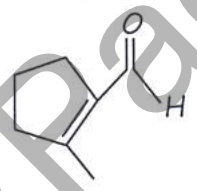
182. 

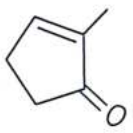
183. 

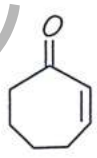
Previous Year Questions

14. When  undergoes intramolecular aldol condensation, the major product formed is [08 April, 2025 (Shift-II)]

(a) 

(b) 

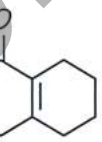
(c) 

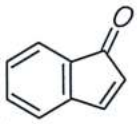
(d) 

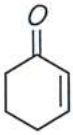
Sol. (a)

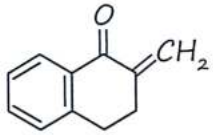
15. Aldol condensation is a popular and classical method to prepare α,β-unsaturated carbonyl compounds. This reaction can be both intermolecular and intramolecular. Predict which one of the following is not a product of intramolecular aldol condensation?

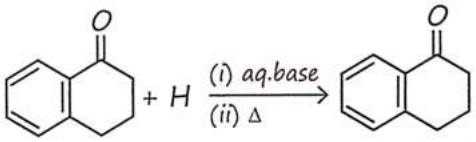
[04 April, 2025 (Shift-I)]

(a) 

(b) 

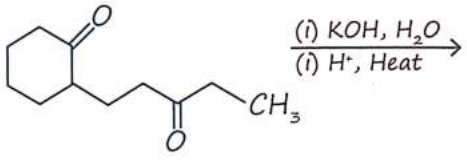
(c) 

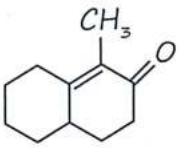
(d) 

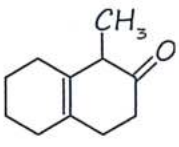
Sol. (d) 

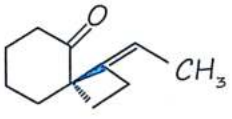
The above reaction is intermolecular aldol.

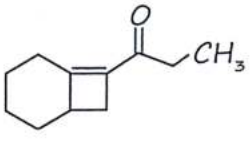
16. The major product of the following reaction is (JEE Adv. 2015)



(a) 

(b) 

(c) 

(d) 

Sol. (a)

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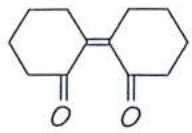
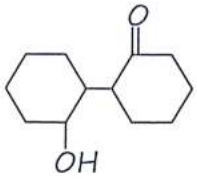
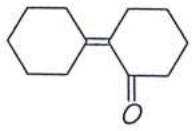
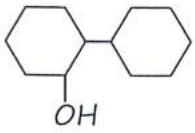
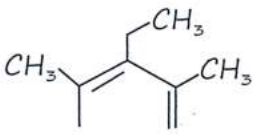
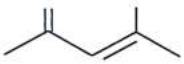
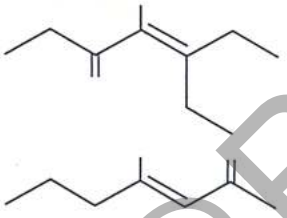
17. Which of the following product is formed when cyclohexanone undergoes aldol condensation followed by heating? (2017-Delhi)

(a) 
(b) 
(c) 
(d) 

Sol. (c)

18. Which one of the following is not formed when acetone reacts with 2-pentanone in the presence of dilute NaOH followed by heating? (2022)

(a) 
(b) 
(c) 
(d)

Sol. (c)

19. Which one of the following esters cannot undergo Claisen self-condensation? (1998)

(a) $CH_3CH_2CH_2COOC_2H_5$

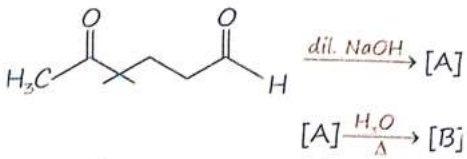
(b) $C_6H_5COOC_2H_5$

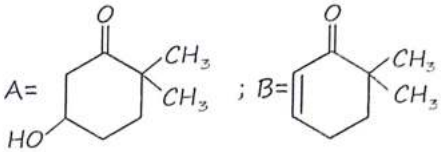
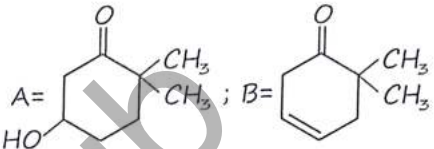
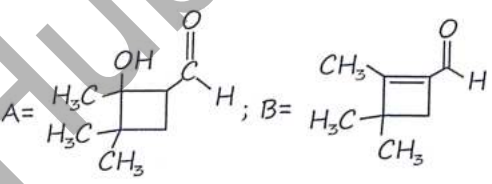
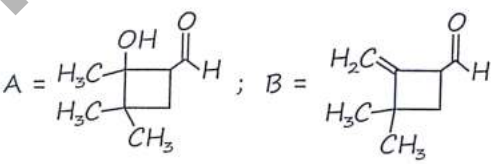
(c) $C_6H_5CH_2COOC_2H_5$

(d) $C_6H_{11}CH_2COOC_2H_5$

Sol. (b) The esters that have active methylene group ($-CH_2-$), show Claisen condensation reaction. As $C_6H_5 - COOC_2H_5$ has no α-hydrogen atom or active methylene group so, it cannot undergo Claisen-Condensation reaction.

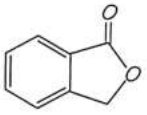
20. In the following reactions, products A and B are [12 Jan, 2019 (Shift-I)]



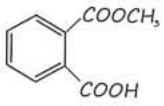
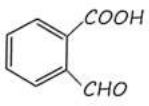
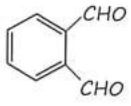
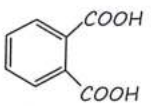
(a) 
(b) 
(c) 
(d) 

Sol. (a)

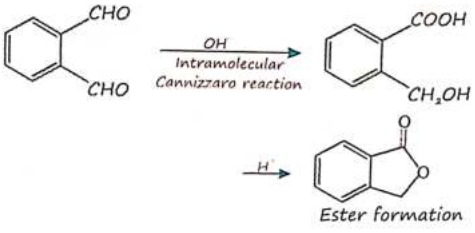
21. Which of the following reactants on reaction with conc. NaOH followed by acidification gives following lactone as the main product?


Lactone

(IIT JEE 2006)

(a) 
(b) 
(c) 
(d) 

Sol. (c) ∵ Conc. NaOH is used, there's a chance of cannizzaro reaction.



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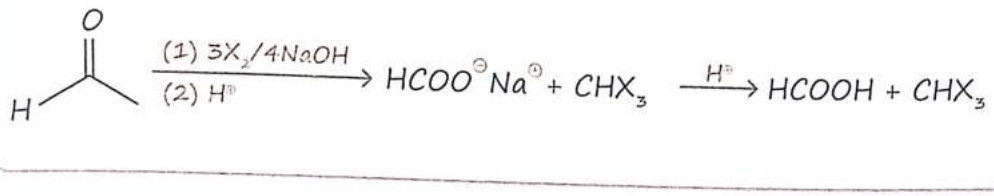
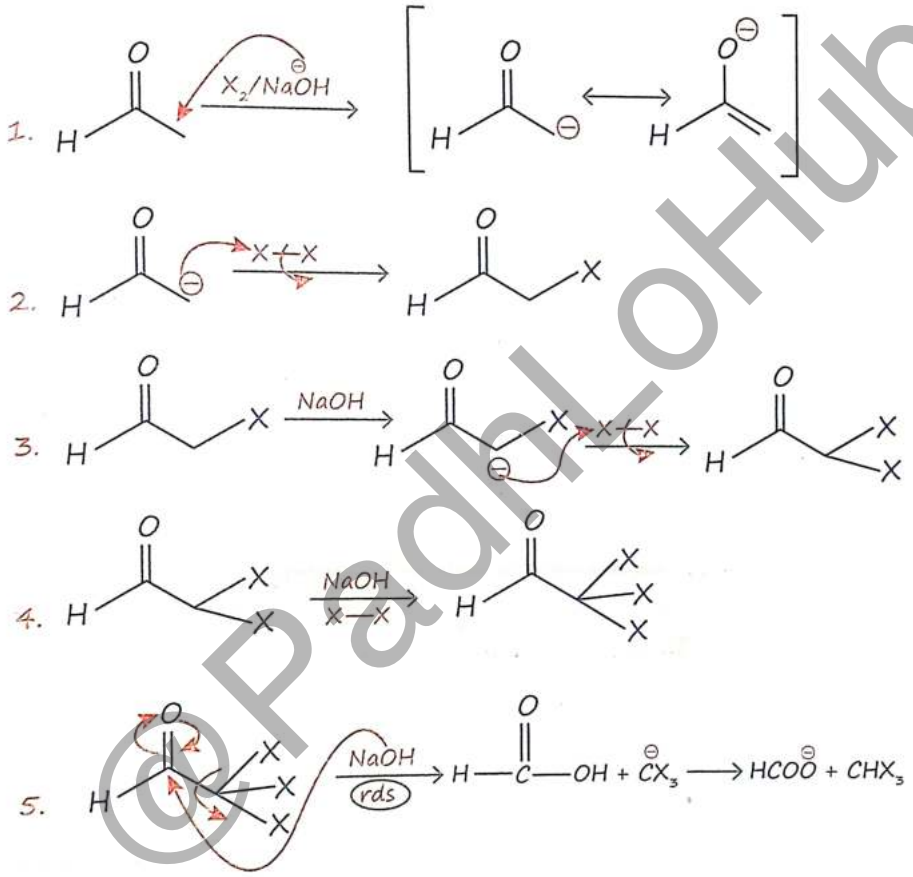
Haloform Reaction

The haloform reaction occurs when a **methyl ketone** reacts with **halogen (X2)** and **base**, leading to the formation of a **carboxylate salt** and a **haloform (CHX3)**. It is used to test for the presence of methyl ketone groups.

Reaction

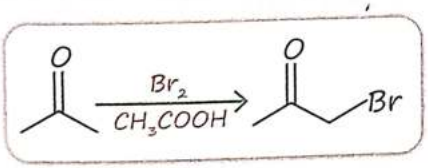


Mechanism



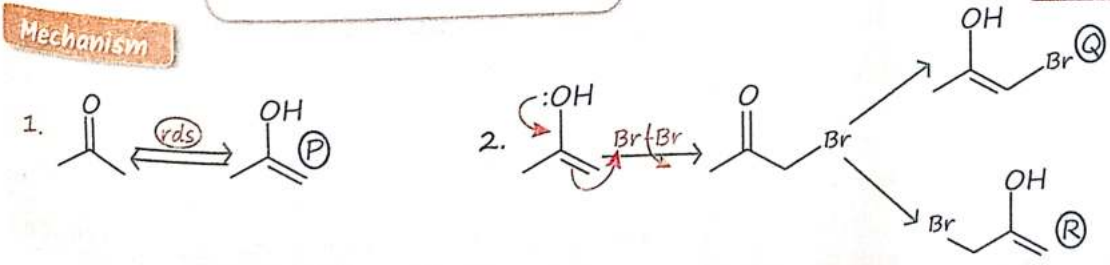
Acidic Haloform

Reaction



- ✓ rate is independent on $[Br_2]$
- ✓ $Nu^{\ominus} \Rightarrow P > R > Q$

Mechanism



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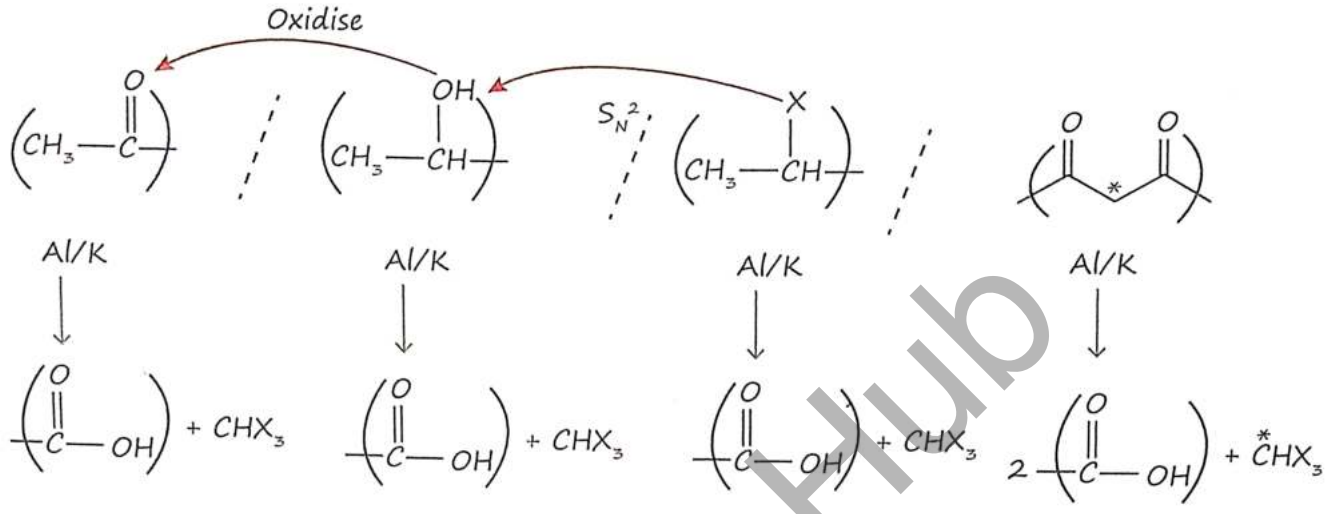
Summary

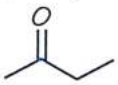
- Reaction Given by:



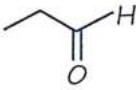
- Reagent:
  1. $X_2/NaOH$
  2. NaOX
  3. $\overset{\ominus}{O}X$
  4. Bleaching Powder ($CaOCl_2 \rightarrow \overset{\ominus}{O}Cl$)
  5. Electrolysis of aq. NaCl
  6. Sodium hypohalite

Practice Questions

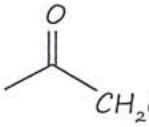
184. $Br_2/NaOH$

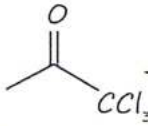
185.  NaOI

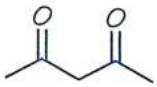
186. CH_3CHO NaOI

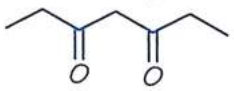
187.  NaOI

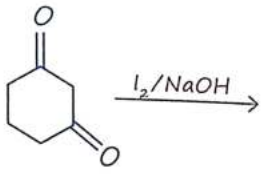
188. Acetophenone NaOI

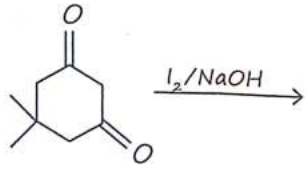
189.  NaOCl

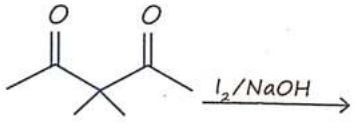
190.  Bleaching Powder

191.  $I_2/NaOH$

192.  $I_2/NaOH$

193.  $I_2/NaOH$

194.  $I_2/NaOH$

195.  $I_2/NaOH$

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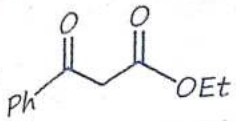
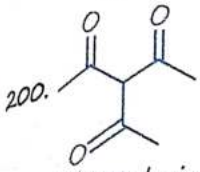
196.  $\xrightarrow{NaOCl}$

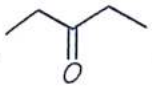
197. $CH_3CO_2Et \xrightarrow{NaOCl}$

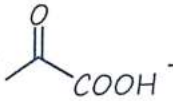
198. $CH_3COCl \xrightarrow{NaOCl}$

199. $CH_3C(=O)OH \xrightarrow{NaOCl}$

200.  $\xrightarrow{NaOCl}$

201. Electrolysis of Ethanolic solution of aq. NaCl $\longrightarrow$

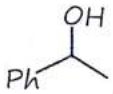
202.  $\xrightarrow{NaOCl}$

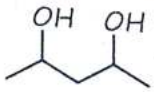
203.  $\xrightarrow{NaOCl}$

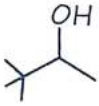
204. $\xrightarrow{NaOI}$

205. $\xrightarrow{NaOI}$

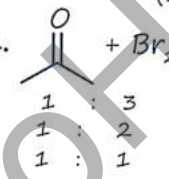
206. 2-Butanol $\xrightarrow{NaOI}$

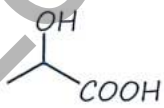
207.  $\xrightarrow{NaOI}$

208.  $\xrightarrow{NaOI}$

209.  $\xrightarrow{NaOI}$

210. CaC_2 (12.8gm) $\xrightarrow[\text{(1) } H_2O \\ \text{(2) Red hot Fe tube} \\ \text{(3) } CH_3COCl/AlCl_3 \\ \text{(4) NaOI} \\ \text{(5) } NH_3, \Delta \\ \text{(6) NaOBr} \\ \text{(7) } HNO_2 \text{ (70\%)} \\ \text{(8) CuCN/KCN} \\ \text{(9) } CH_3MgBr \text{ then } H_2O \\ \text{(10) } Br_2/NaOH \text{ (80\%)} \\ \text{(11) NaOH, CaO/}\Delta]{}$ (P) (xgm)

211. 

212. 

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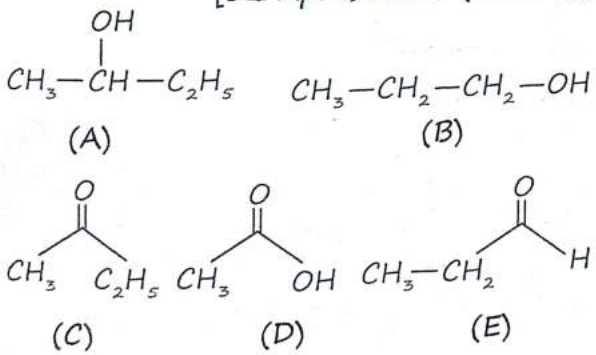
Previous Year Questions

22. Which among the following compounds give yellow solid when reacted with NaOI/NaOH? [04 April, 2025 (Shift-II)]



(A) (B) (C) (D) (E)

Choose the correct answer from the options given below:

(a) (B), (C) and (E) Only
(b) (A) and (C) Only
(c) (C) and (D) Only
(d) (A), (C) and (D) Only

Sol. (b) The compounds give iodoform test having (CH_3CO-) and ($CH_3CH(OH)-$) group.

23. Among the following, the number of compounds which will give positive iodoform reaction is [6 April, 2023 (Shift-II)]

(a) 1-Phenylbutan-2-one
(b) 2-Methylbutan-2-ol
(c) 3-Methylbutan-2-ol
(d) 1-Phenylethanol
(e) 3,3-dimethylbutan-2-one
(f) 1-Phenylpropan-2-ol

Sol. (d)

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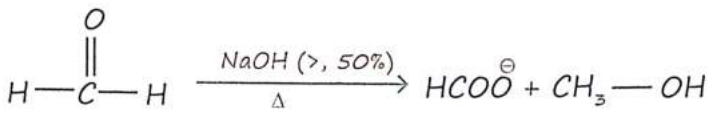
Cannizzaro Reaction

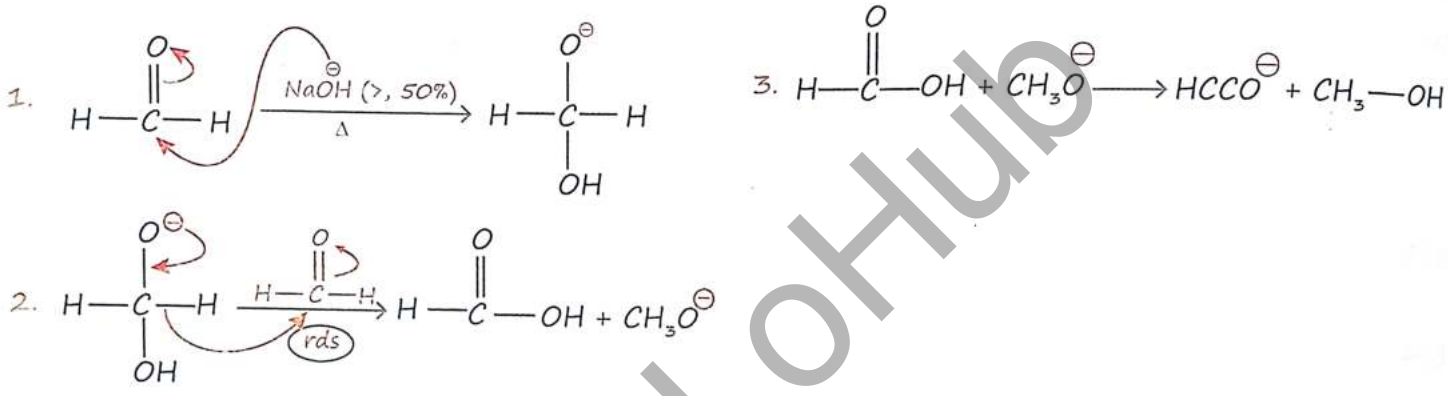
The Cannizzaro reaction occurs when **aldehydes without α-hydrogens** (like formaldehyde or benzaldehyde) react with **concentrated base** (e.g., NaOH), undergoing **disproportionation**: one molecule gets **oxidized to a carboxylic acid** and the other gets **reduced to an alcohol**.

Reaction



$$H-CHO \xrightarrow[\Delta]{NaOH\ (>,\ 50\%)} HCOO^{\ominus} + CH_3-OH$$

Mechanism



Baba Tea-Stall (BTS)

(i) Self cannizzaro is Redox as well as Disproportionation.
(ii) More than 50% Base & Heat is required.
(iii) Hydride migration is RDS.
(iv) Kinetic isotopic effect observed.
(v) In case cross and Intramolecular, easy attack on C = O ⇒ Oxidation and other will be reduced.
(vi) EWG will increase the rate and strong donating group presence will inhibit the reaction
(vii) It occurs when no α-H present.

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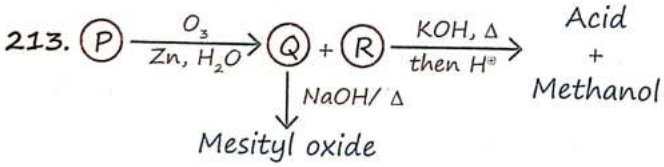
Practice Questions

213. 

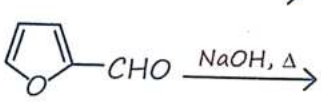
214. $HCHO \xrightarrow{NaOH,\ \Delta}$

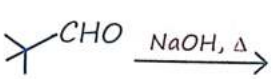
215. $H-CO-D \xrightarrow{NaOH,\ \Delta}$

216. $H-CO-D \xrightarrow{NaOD,\ \Delta}$

217. $PhCHO \xrightarrow{NaOH,\ \Delta}$

218. $Ph-CO-D \xrightarrow{NaOH,\ \Delta}$

219. 

220. 

221. $HCHO + PhCHO \xrightarrow[\Delta]{NaOH}$

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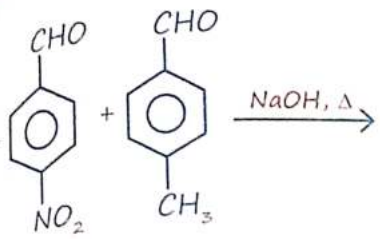
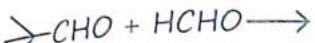
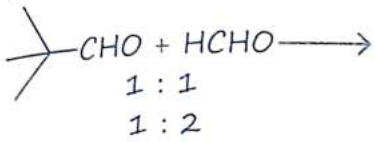
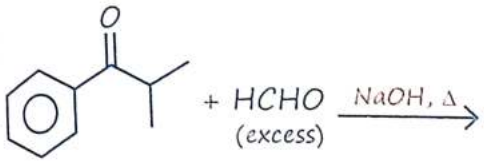
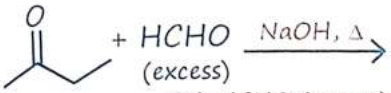
222. 

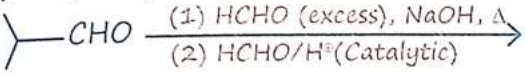
223. 

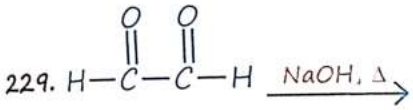
224. 

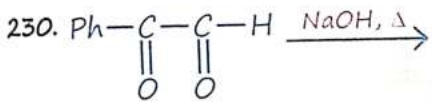
225. 

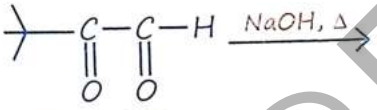
226. $CH_3 - CHO + HCHO \xrightarrow[\Delta]{NaOH}$ (excess)

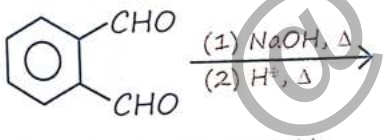
227. 

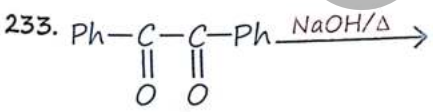
228. 

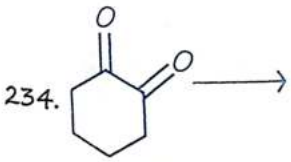
229. 

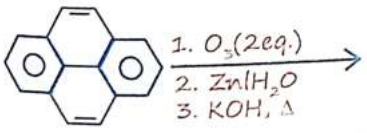
230. 

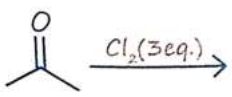
231. 

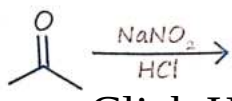
232. 

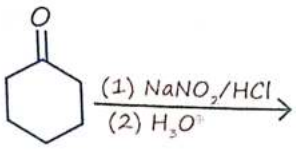
233. 

234. 

235. 

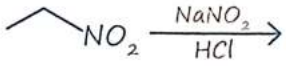
236. 

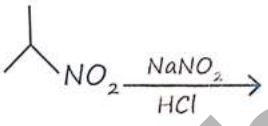
237. 

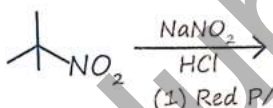
238. 

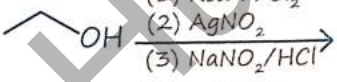
239. $CH_3CHO \xrightarrow[(3)\ NaOH,\ \Delta]{(1)\ NaNO_2/HCl,\ (2)\ H_3O^+}$

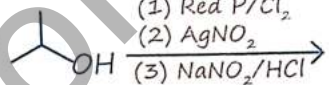
240. $CH_3COOH \xrightarrow{Cl_2}$

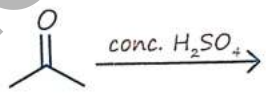
241. 

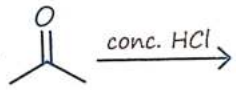
242. 

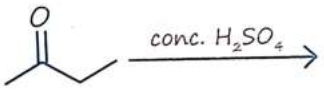
243. 

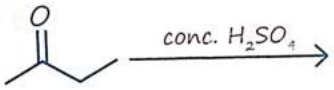
244. 

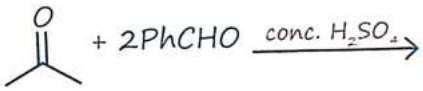
245. 

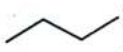
246. 

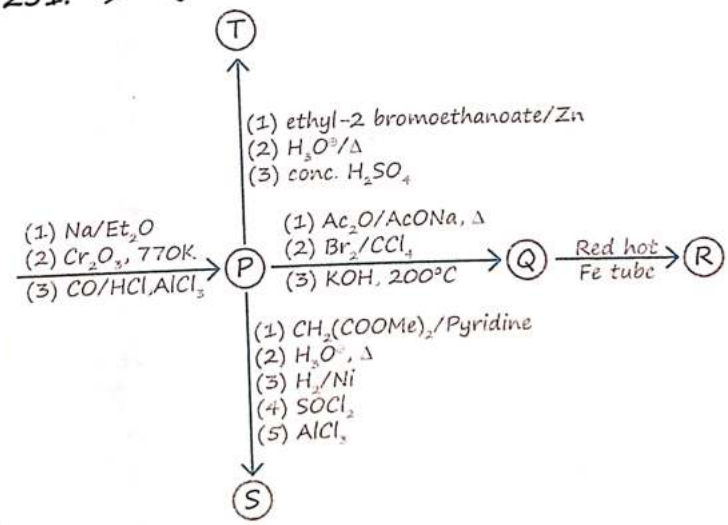
247. 

248. 

249. 

250. 

251. 



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252. If x = Total compounds which can show aldol condensation.
y = Total compounds which can show haloform reaction.
z = Total compounds which can show cannizzaro reaction.

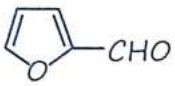
253. CH_3-CHO

254.

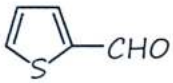
255. $Ph-CHO$

256. $Ph-\overset{\overset{O}{\|}}{C}-CH_3$

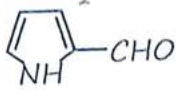
257. $HCHO$

258. 

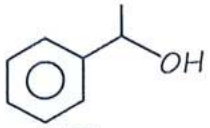
259. Cl_3C-CHO

260. 

261.

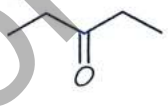
262. 

263.

264. 

265.

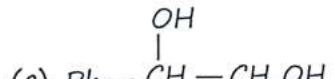
266.

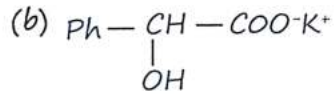
267. 

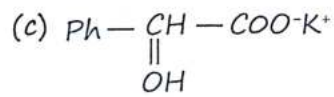
Previous Year Questions

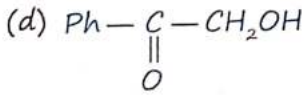
24. The major product (P) in the following reaction is:

[03 April, 2025 (Shift-II)]

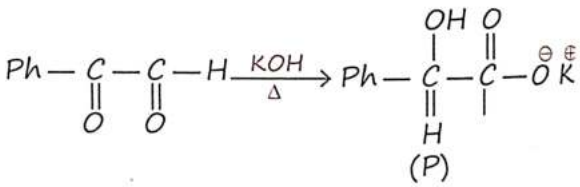
(a) 

(b) 

(c) 

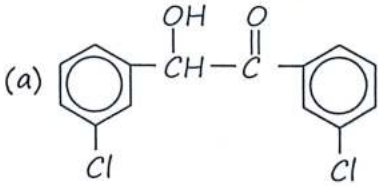
(d) 

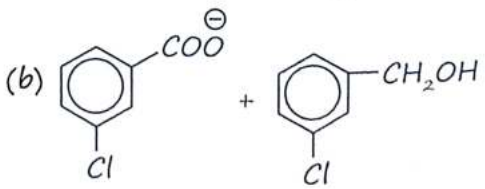
Sol. (b) (Intramolecular cannnizzaro reaction)



25. m-chlorobenzaldehyde on treatment with 50% KOH solution yields.

[30 Jan, 2024 (Shift-II)]

(a) 

(b) 

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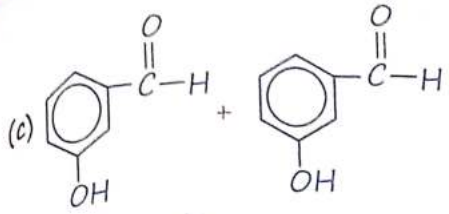
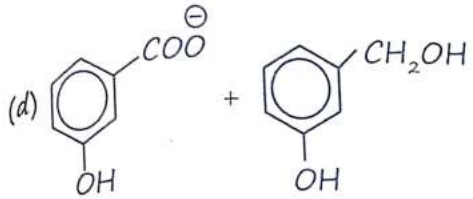
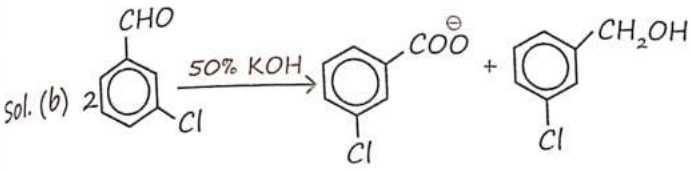
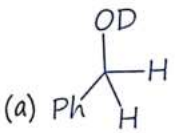
(c) 

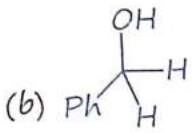
(d) 

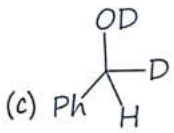
Sol. (b) 

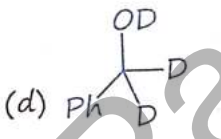
26. The correct structure of product 'A' formed in the following reaction is,

[28 June, 2022 (Shift-I)]

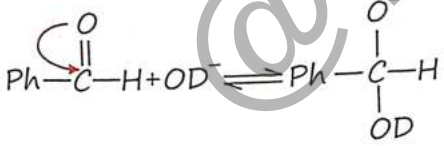
(a) 

(b) 

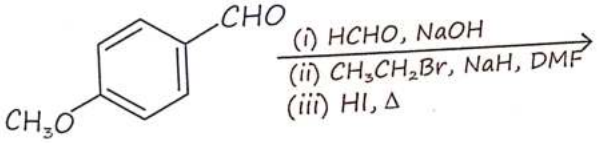
(c) 

(d) 

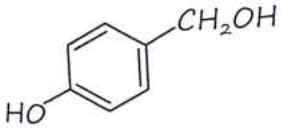
Sol. (a) The given reaction is Canizzaro reaction whose mechanism is given below:

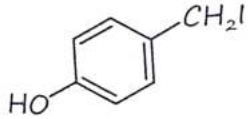


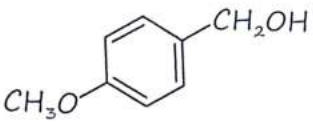
27. Identify A in the following chemical reaction.

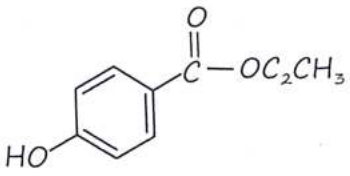


[26 Feb, 2021 (Shift-II)]

(a) 

(b) 

(c) 

(d) 

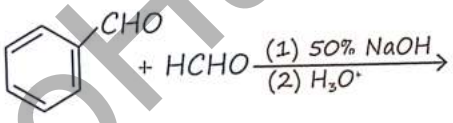
Sol. (b) Step I:- Cannizaro reaction will take place.

Step II: Base will remove hydrogen of benzylic alcohol.

Step III: Formation of ether will take place by S_N^2 reaction.

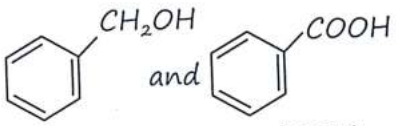
Step IV: S_N^1 reaction will take place.

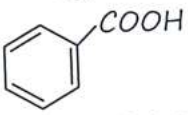
28. Major products of the following reaction are

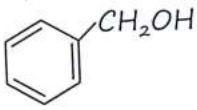


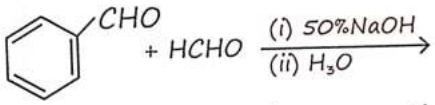
[10 April, 2019 (Shift-I)]

(a) CH_3OH and HCO_2H

(b) 

(c) CH_3OH and 

(d) HCOOH and 

Sol. (d) 

This is cross cannizaro reaction so more reactive carbonyl compound is oxidized and less reactive is reduced so answer

29. The Cannizzaro's reaction is not given by

(IIT JEE 1983)

(a) trimethyl acetaldehyde

(b) acetaldehyde

(c) benzaldehyde

(d) formaldehyde

Sol. (b) Carbonyl compounds lacking alpha hydrogen atom give Cannizzaro reaction. Acetaldehyde has alpha hydrogen atom so it doesn't give Canizarro reaction.

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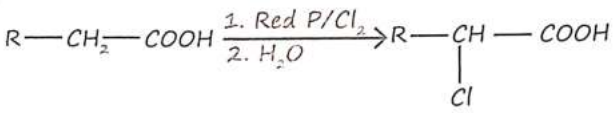
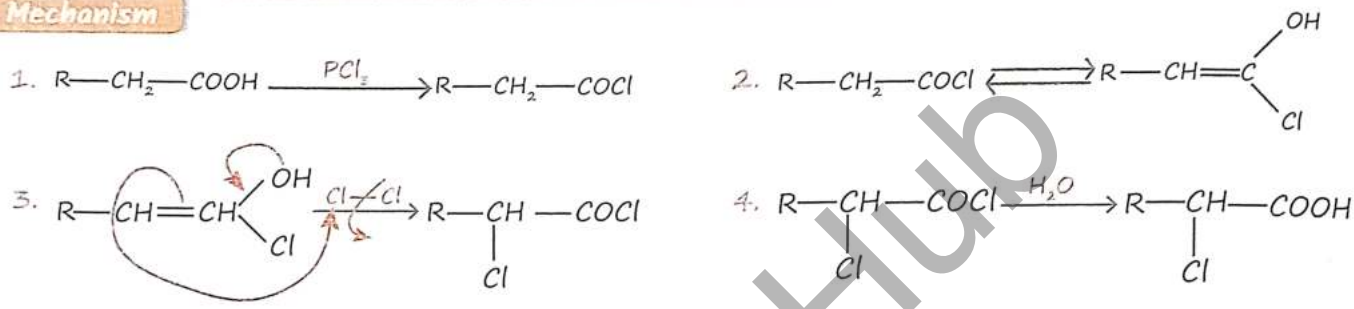
Hell Volhard Zeliensky Reaction

The HVZ reaction involves the α-halogenation of carboxylic acids. In the presence of halogen (X_2) and red phosphorus, the α-hydrogen is substituted by a halogen, forming α-halo acids—an important step in amino acid and intermediate synthesis.

Reaction



Mechanism



Baba Tea-Stall (BTS) :

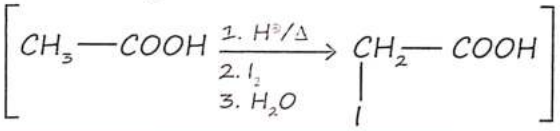
(i) Example of electrophilic Substitution Reaction.

(ii) Reaction is used for halogenation of α-carbon of carboxylic acid.

(iii) Catalytic amount of Red P & excess of Cl_2 is used.

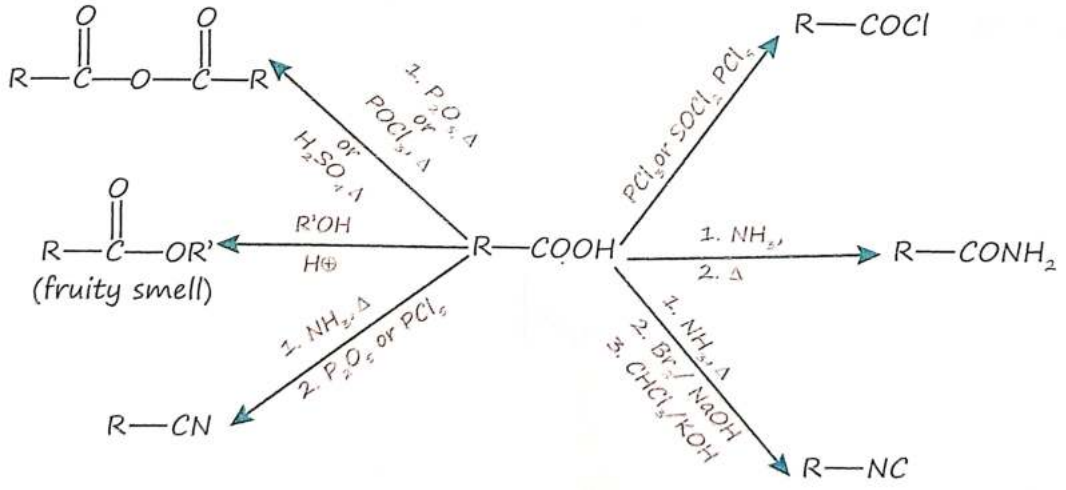
(iv) NCERT uses H_2O but it may or may not be given.

(v) In case of I_2, Simonini Reaction is used.



Preparation of Acid Derivative

Conversion of carboxylic acids (R–COOH) into various acid derivatives like acid chlorides (R–COCl), amides (R–$CONH_2$), nitriles (R–CN), esters (R–COOR'), and anhydrides (R—CO—O—CO—R) in the presence of PCl_5, $SOCl_2$, NH_3, ROH, and P_2O_5,



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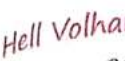
Practice Questions

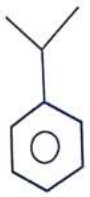
➢ Hell Volhard Zeliensky reaction

268. $CH_3-COOH \xrightarrow{Red\ P/Cl_2}$

269. 4-chlorophenyl acetic acid $\xrightarrow{Red\ P/Cl_2}$

270.  $\xrightarrow[(3)\ NaNO_2/HCl]{(1)\ Red\ P/Cl_2,\ (2)\ NH_3(2eq.)}$

271. $CH_3-COOH \xrightarrow[2.\ KOH/H_2O]{1.\ Red\ P/Cl_2} \xrightarrow{3.\ H^\oplus}$

272.  $\xrightarrow[2.\ H^\oplus/H_2O]{1.\ O_2/hv}$ (P) + (Q)

(P) $\xrightarrow[2.\ Red\ P/Cl_2]{1.\ KMnO_4/\Delta}$ (R)

(Q) $\xrightarrow[2.\ O_3/Zn,\ H_2O]{1.\ Zn,\ \Delta}$ (S)

➢ Preparation of acid derivative

273. $R-\underset{\underset{O}{\|}}{C}-X \xrightarrow[{}^{\ominus}OH/H_2O]{H^\oplus/H_2O\ or}$

274. $R-\underset{\underset{O}{\|}}{C}-OR \xrightarrow[{}^{\ominus}OH/H_2O]{H^\oplus/H_2O\ or}$

275. $R-\underset{\underset{O}{\|}}{C}-O-\underset{\underset{O}{\|}}{C}-R \xrightarrow[{}^{\ominus}OH/H_2O]{H^\oplus/H_2O\ or}$

276. $R-NC \xrightarrow{H^\oplus/H_2O}$

277. $R-\underset{\underset{NH_2}{|}}{\overset{\overset{O}{\|}}{C}} \xrightarrow[\Delta]{H^\oplus/H_2O}$

278. $R-\underset{\underset{NH_2}{|}}{\overset{\overset{O}{\|}}{C}} \xrightarrow[\Delta]{\overset{\oplus}{O}H/H_2O}$

279. $R-C\equiv N \xrightarrow[H_2O_4]{95\%}$

280. $R-C\equiv N \xrightarrow[H_2O_4]{98\%}$

➢ Hydrolysis

281. $R-N=C=O \xrightarrow{\overset{\ominus}{O}H/H_2O}$

282. $R-N=C=O \xrightarrow{H^\oplus/H_2O}$ "

283. $R-CH=C=O \xrightarrow{\overset{\ominus}{O}H/H_2O}$

284. $R-CH=C=O \xrightarrow{\overset{\oplus}{H}/H_2O}$

285. $R-\overset{\oplus}{O}=C=O \xrightarrow{\overset{\oplus}{H}/H_2O}$

286. $R-CH=C=O \xrightarrow{\overset{\oplus}{H}/ROH}$

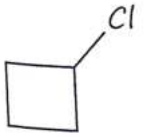
287. $R-CH=C=O \xrightarrow{\overset{\oplus}{H}/R-NH_2}$

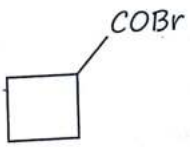
288. $R-CH=C=O \xrightarrow{HCl}$

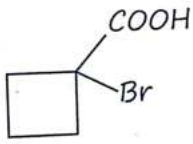
289. $Ph-C\equiv N \xrightarrow{\overset{\oplus}{H}/EtoH}$

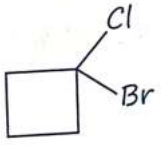
Previous Year Questions

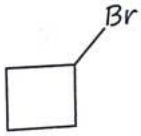
30. Identify the product in the following reaction (2023-Manipur)

 $\xrightarrow[(iv)\ H_2O]{(i)\ KCN,\ (ii)\ H_2O/HCl,\ \Delta,\ (iii)\ Br_2/red\ phosphorus}$ Product

a. 

b. 

c. 

d. 

Sol. (b)

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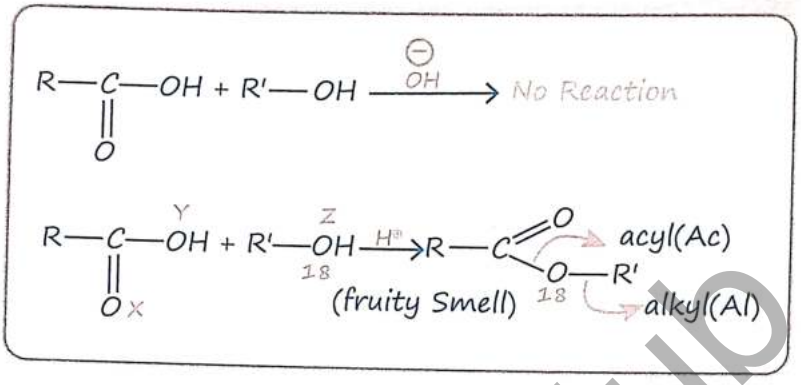
Esterification

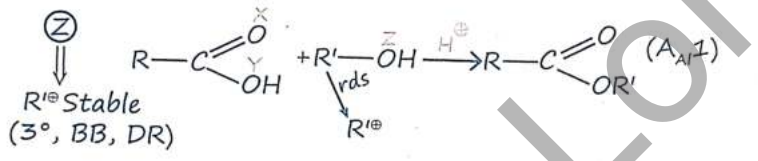
Esterification is the reaction between a **carboxylic acid** and an **alcohol** in the presence of a **strong acid catalyst** (like concentrated H_2SO_4) to form an **ester** and **water**. It is a reversible reaction known as **Fischer–Speier esterification**, commonly used to produce esters with **pleasant fruity smells**.

Reaction

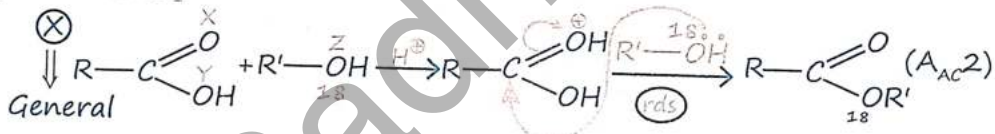


Mechanism

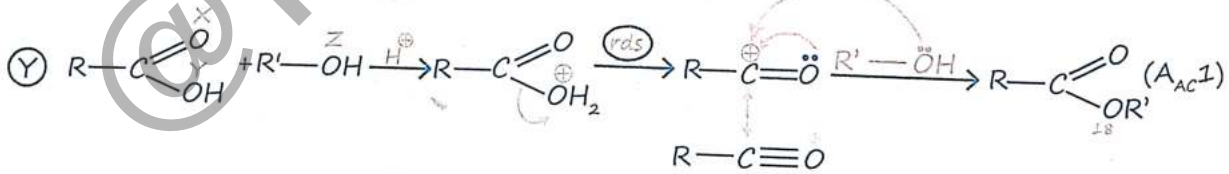
[When $R^{\oplus}$ is highly stable]

1. 

[General case]

2. 

[When R is highly bulky]

3. 

Baba Tea-Stall (BTS)

(i) Esterification is not possible in basic medium.
(ii) Various mechanism like $A_{AC}1$, $A_{AC}2$, $A_{Al}1$, $A_{Al}2$ are possible.
(iii) Excess of alcohol is required.
(iv) With Alcohol, we can use Ac_2O/Py or RCOCl/Py or R-COOH/$H^{\oplus}$.

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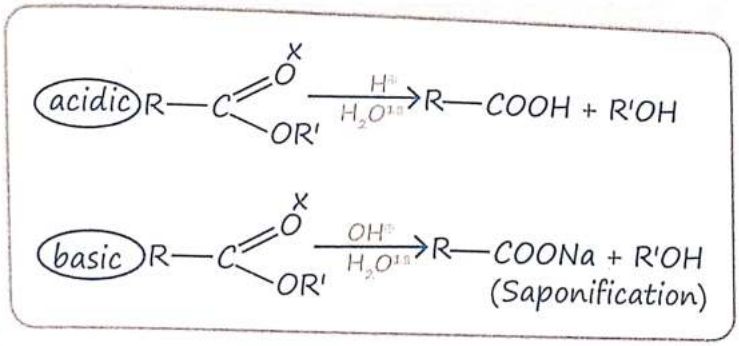
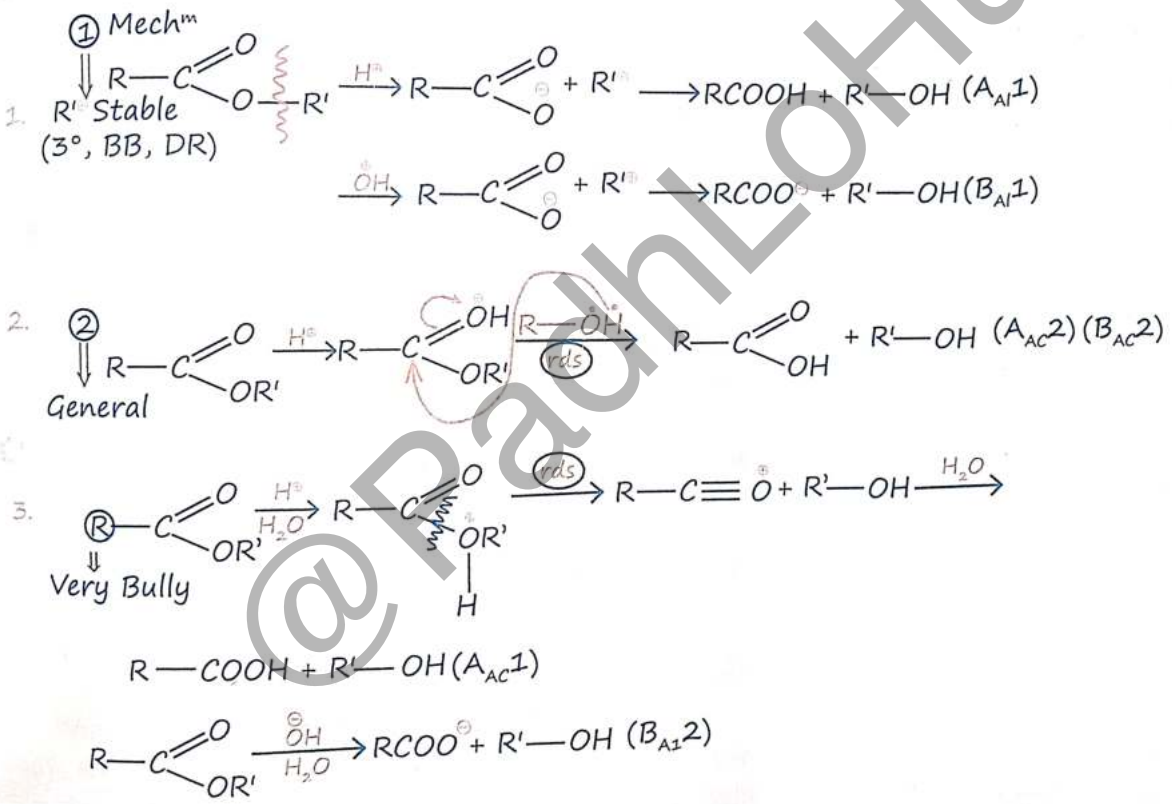
Hydrolysis of Ester

Hydrolysis of esters is the process where an ester reacts with water to produce a carboxylic acid and an alcohol. Under acidic conditions, it's a reversible reaction, while in basic conditions (saponification), it proceeds irreversibly, yielding a carboxylate salt and an alcohol.

Reaction

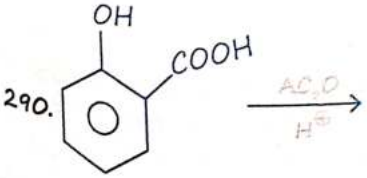


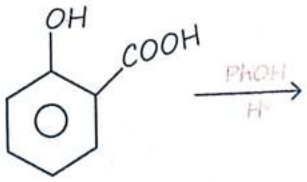
Mechanism



Practice Questions

Esterification:

290. 

291. 

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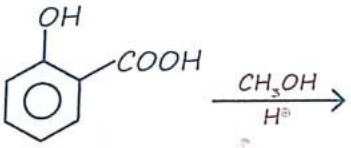
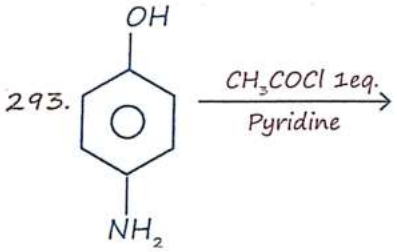
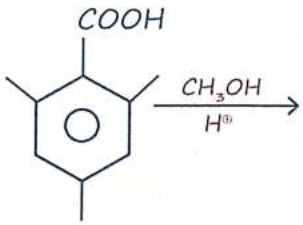
292. 

293. 

294. 

295. $CH_3-COOH \xrightarrow[H^{\oplus}]{} OH$

➢ Hydrolysis of ester

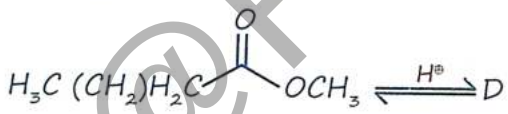
296. Many sweet-smelling compounds contain an ester functional group, R – CO – O – R', where R/R' = alkyl, aryl groups. Hydrolysis of an ester is an exhaustively studied reaction:

$RCOOR' + H_2O \rightleftharpoons RCOOH + R'OH$

The hydrolysis can be catalyzed by an acid (A) or a base (B). The reaction may be unimolecular (1) or bimolecular (2), and may involve breaking of acyl–O bond (AC) or alkyl–O bond (AL). Thus, there are eight different mechanisms possible as listed in the table below.

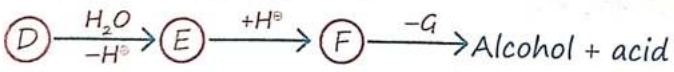
Acid catalyzed		Base catalyzed	
Unimolecular	Biomolecular	Unimolecular	Bimolecular
$A_{AC}1$	$A_{AC}2$	$B_{AC}1$	$B_{AC}2$
$A_{AL}1$	$A_{AL}2$	$B_{AL}1$	$B_{AL}2$

297. Consider the acidic hydrolysis of methylpentanote in aqueous H_2SO_4.

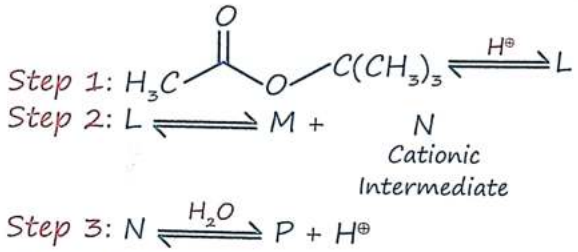


(i) Draw the structure of the most favorable conjugate acid (D) of methyl petanoate.

(ii) D is attacked by a water molecule (nucleophile) and, through a series of reactions, is broken into a carboxylic acid and an alcohol. Identify the species E (a neutral molecule), F and G (both charged species).



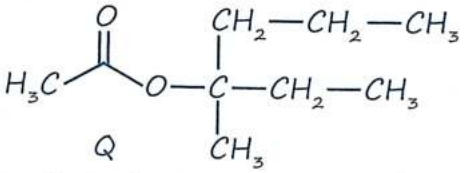
298. In the hydrolysis of K in moderately concentrated aq. H_2SO_4, the preferred mechanism involved conjugated acid (L) of K. L undergoes decomposition to form a stable product M and a cation N. N on reaction with water forms P.



(i) Draw the structures of L, M, M, and P.

(ii) Identify the slowest step in the reaction mechanism.

299. An optically active sample of ester Q is completely hydrolyzed under aqueous acidic conditions.



In the hydrolysis using moderately concentrated H_2SO_4, two additional compounds R and S (other than the expected alcohol and carboxylic

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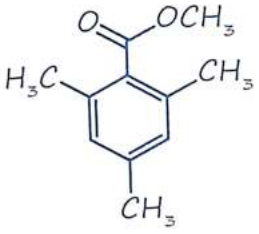
acid) having identical molecular formulae are formed in small amounts.

(i) Draw the structures of R and S.

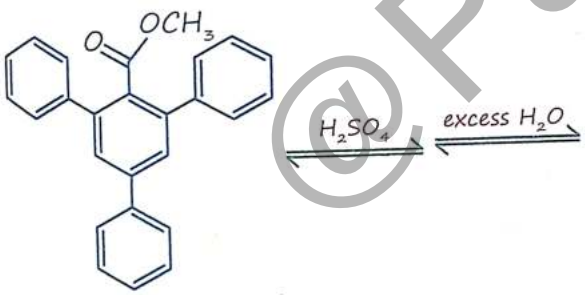
(ii) When the hydrolysis is carried out in aqueous ethanol containing H_2SO_4, another compound T which is not an acid/alcohol/ester is also formed. Identify T.

300. Hydrolysis of U is difficult under reaction conditions, most likely due to bulky methyl groups near the carbonyl group. The ester is hydrolysed by dissolving in concentrated H_2SO_4, and then pouring the reaction mixture into a large excess of cold water. Draw the structure of cationic intermediate V and neutral compound W formed in concentrated acid before pouring into water.



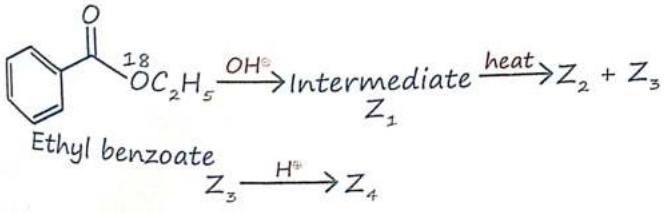
U

301. When ester X is dissolved in concentrated H_2SO_4, and the mixture is poured in water, an unusual product Y is obtained which is neither an acid nor an alcohol. Draw the structure of Y.



X

302. An ester can also be hydrolyzed by heating with aqueous NaOH followed by acidification to yield the carboxylic acid. For alkaline hydrolysis of ethyl benzoate labelled with ^{18}O, the rate of hydrolysis depends on the concentrations of both ester and alkali.

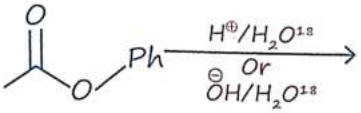


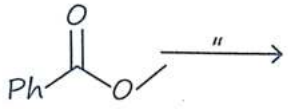
Ethyl benzoate

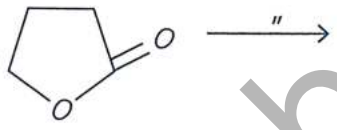
$Z_3 \xrightarrow{H^\oplus} Z_4$

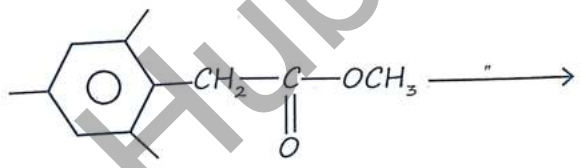
(i) Draw the structures of the species/product Z1 to Z4 along with the label of 18O.

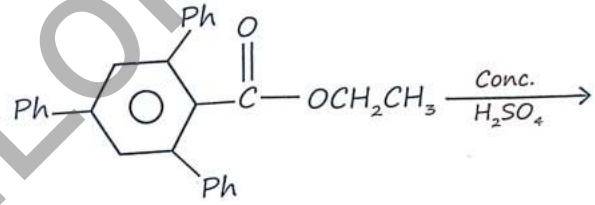
(ii) Using the Table in page 4, label the above reaction mechanism appropriately.

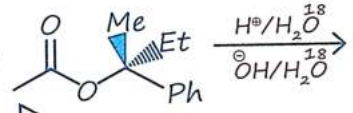
303. 

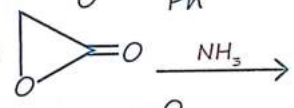
304. 

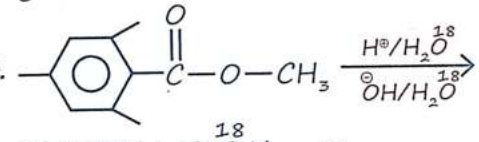
305. 

306. 

307. 

308. 

309. 

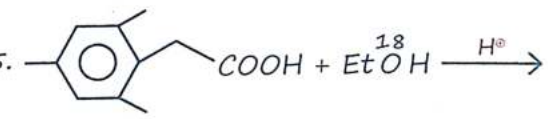
310. 

311. $PhCOOH + Et\overset{18}{O}H \xrightarrow{H^\oplus}$

312. $EtCOOH + Ph\overset{18}{O}H \xrightarrow{H^\oplus}$

313. $PhCOOH + Me_3C-OH \xrightarrow{H^\oplus}$

314.

315. 

316.

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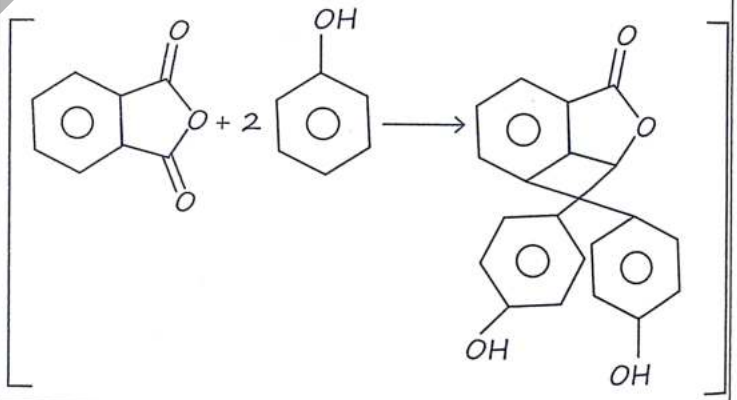
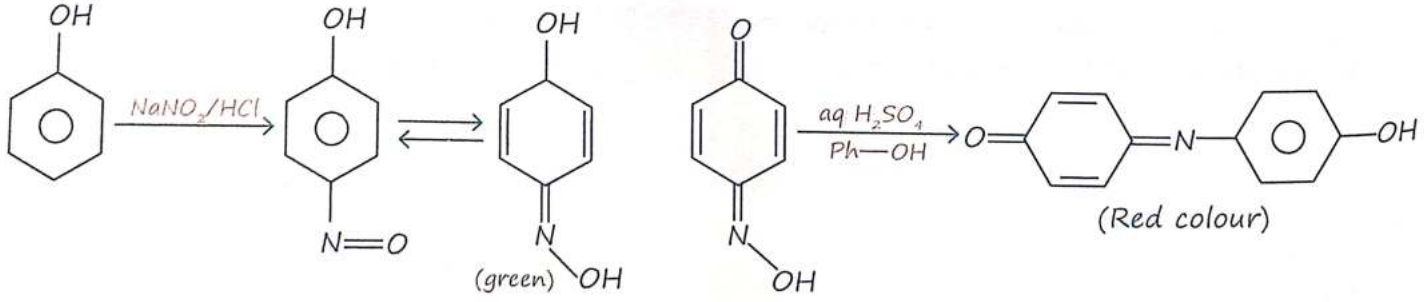
Distinguish Test for Functional Group

Test of Alcohol and Phenol

Alcohol	Phenol
➢ Active Metal test [R–OH + Na ⟶ $RO^{\ominus}Na^{\oplus}$ + 1/2 H_2]	➢ Active metal test [ROH + Na ⟶ $RO^{\ominus}Na^{\oplus}$ + 1/2 H_2]
➢ Lucastest 3°–oH ⟶ Immediate white turbidity 2°–oH ⟶ 5 to 7 min. 1°–oH ⟶ No turbidity at R.T.	➢ Solubility test [PhOH + NaOH ⟶ PhONa + H_2O]
➢ Victor meyer test (RBC) 1°–oH ⟶ Red (Nitrolic) 2°–oH ⟶ Blue (pseudo nitro) 3°–oH ⟶ Colour less	➢ Litmus test [B ⟶ R]
➢ Acylation [ROH + CH_3COCl ⟶ ROAc + HCl↑]	➢ Bromine water test [PhOH + Br_2/H_2o ⟶ White PPt]
➢ Ceric Ammonium Nitrate test (CAN) $[(NH_4)_2[Ce(NO_3)_6]$ + 3ROH ⟶ $2NH_4NO_3$ + $[Ce(NO_3)_4(ROH)_3]$ (Red colour)	➢ Phenophtalein Indicator 
	➢ Neutral $FeCl_3$ [3PhOH + $FeCl_3$ ⟶ $Fe(OPh)_3$ + 3HCl] (Violet)
	➢ Libermann Nitroso test [GRB test]

Liebermann nitroso test



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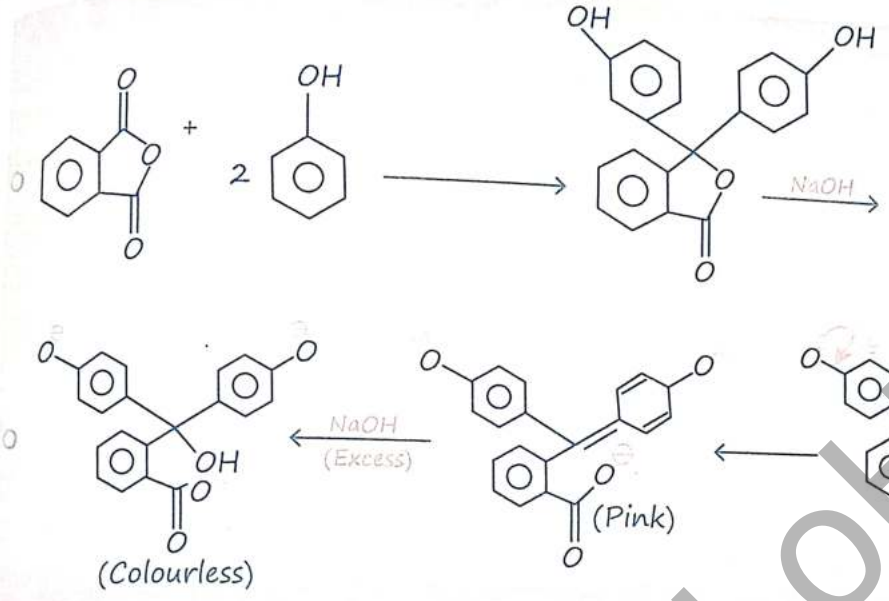
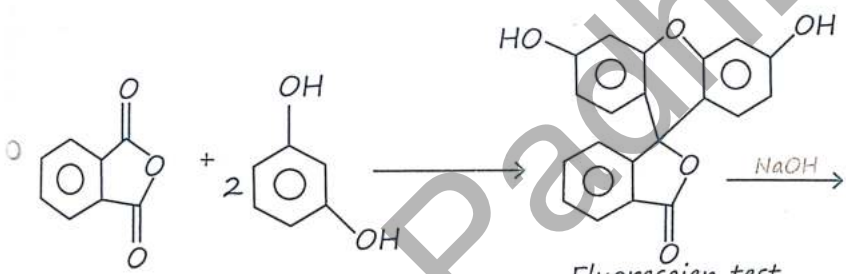
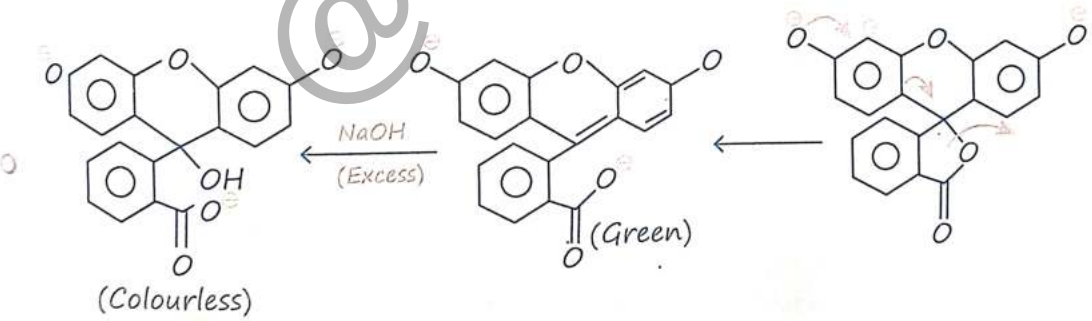
Phenolphthalein dye test

Phenolphthalein dye test, used to detect **phenol groups**. Phenol reacts with **phthalic anhydride** in the presence of **concentrated sulphuric acid**, forming a **colourless intermediate**. On treatment with excess NaOH, it develops a **pink-coloured dye**, indicating the presence of phenol by forming **phenolphthalein**, a pH indicator.





Fluorescien test



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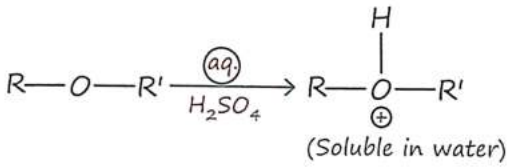
Test of Ether

Ethers react with cold H_2SO_4, especially at higher temperatures, forming alkyl hydrogen sulphates, which are soluble in water—indicating ether cleavage.

- Cold H_2SO_4



(Soluble in water)

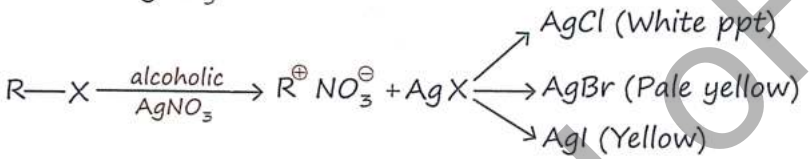
Test of Haloalkane

Haloalkanes are tested using alcoholic $AgNO_3$, producing precipitates of silver halides:

- AgCl (white),
- AgBr (pale yellow),
- AgI (yellow),

helping distinguish between halide types based on colour.

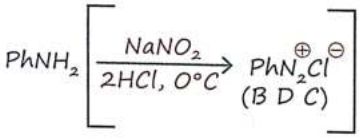
- Alcoholic $AgNO_3$

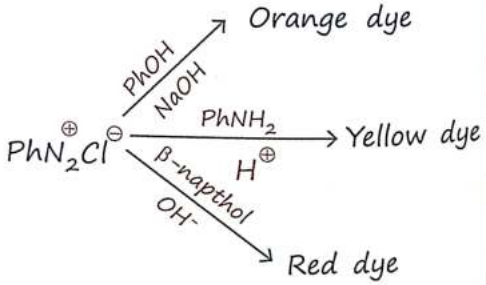


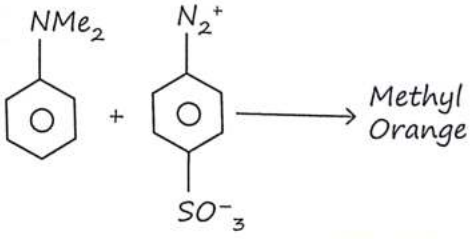
Test of Amines

- Libermann Nitroso test
  - $1^\circ NH_2 \rightarrow N_2\uparrow$
  - $2^\circ NH \rightarrow$ Yellow layers
  - $3^\circ N \rightarrow$ No Rx^n
- Solublity in aq HCl
- Fractional distillation (separation + distillation)
- Hinsberg test
  - $1^\circ NH_2 \rightarrow$ Base soluble sulphonamide
  - $2^\circ NH \rightarrow$ Base insoluble sulphonamide
  - $3^\circ N \rightarrow$ No Rx^n
- Diethyl Oxalate
  - $1^\circ NH_2 \rightarrow$ Solid
  - $2^\circ NH \rightarrow$ liquid
  - $3^\circ N \rightarrow$ gas
- Mustard oil test
  - $1^\circ NH_2 \rightarrow$ Mustard oil smell (R– NCS)
  - $2^\circ NH \rightarrow \times$
  - $3^\circ N \rightarrow \times$
- Carbylamine test
  - $1^\circ NH_2 \rightarrow$ RNC, foul smell
  - $2^\circ NH \rightarrow \times$
  - $3^\circ N \rightarrow X$

- Litmus test (R → B)
- Azo dyo test (Only for aniline & derivative)







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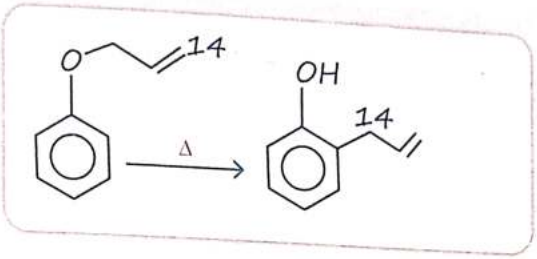
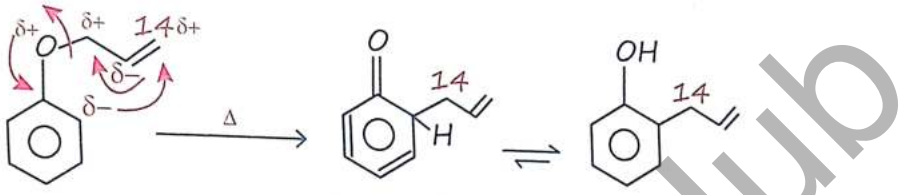
Claisen Rearrangement

The Claisen rearrangement is a reaction where allyl aryl ethers are heated, causing the allyl group to shift to the ortho-position of the aromatic ring. It results in the formation of ortho-allyl phenols and is widely used for creating new carbon–carbon bonds in organic synthesis.

Reaction

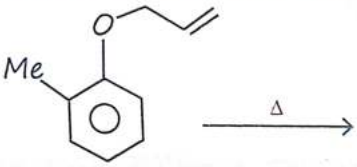


Mechanism



Try Yourself

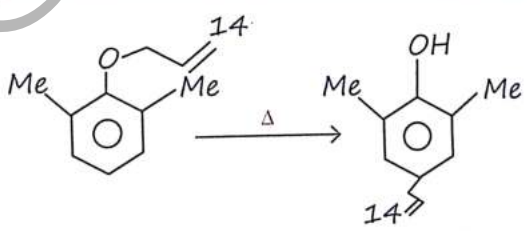
➢ When Ortho is Blocked (For JEE Advanced)



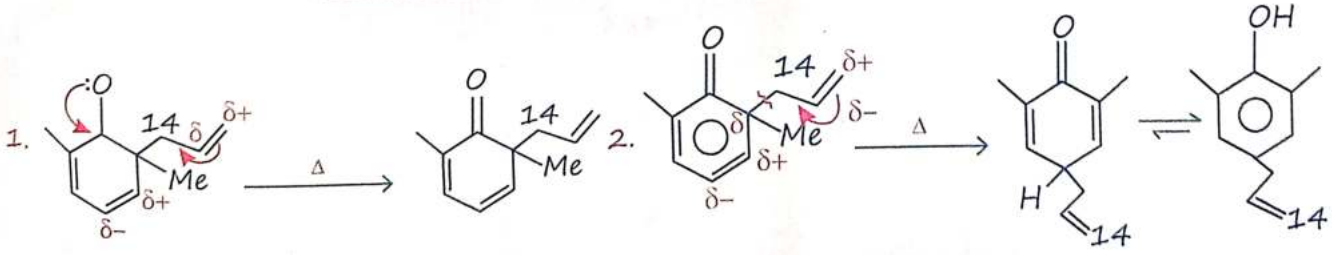
Both Ortho Blocked

When both ortho positions of the aromatic ring are blocked in a Claisen rearrangement, the allyl group migrates to the para position instead. This alternative shift still allows the reaction to proceed, yielding a para-allyl phenol as the major product.

Reaction



Mechanism



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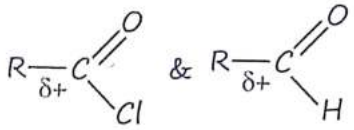
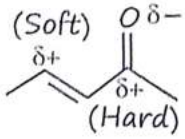
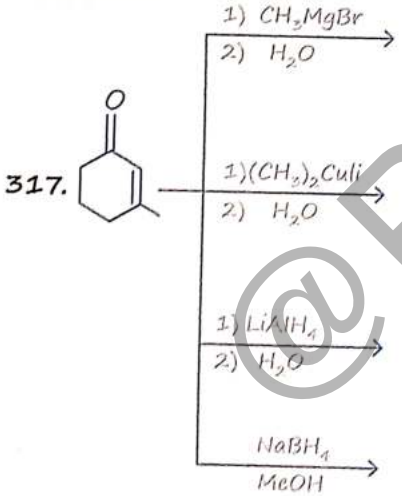
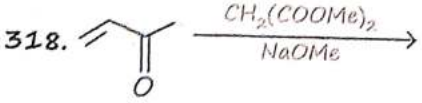
Soft-Hard

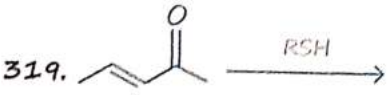
HSAB concept, which states that **hard acids/bases prefer to react with other hard species**, and **soft acids/bases prefer soft partners**. It distinguishes between **absolute electrophiles** (like carbonyl compounds) and **absolute nucleophiles** (like RMgX, $LiAlH_4$), and notes how reagents like $NaBH_4$ selectively reduce carbonyl groups. It also highlights that in **conjugated systems** (e.g., cinnamic acid derivatives), **soft nucleophiles** can reduce double bonds as well.

- HSAB: Hard acid/$E^{\oplus}$ prefers Hard $Nu^{\ominus}$ /Base & Soft acid $E^{\oplus}$ prefer Soft $Nu^{\ominus}$/Base
- Absolute $E^{\oplus}$ (Hard) ⟶  &
- Absolute Nu (Hard) ⟶ RNa, RLi, RMgX, $LiAlH_4$, $NaBH_4$ → absolute Hard; → Soft
- 
- In case of cinnamic bond (Ph – CH = CH – CH –), $LiAlH_4$ – C = C – also to (Ph – CH_2 – CH_2 – C–)

Practice Questions

317. 

318. 

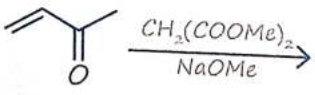
319. 

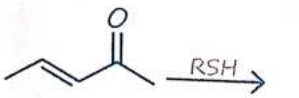
320. $CH_3-CH=CH-CHO$ → $LiAlH_4$ + then H_2O; $NaBH_4$ / MeOH

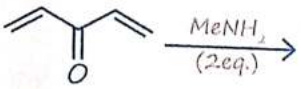
321. $CH_3-CH=CH-COCl$ → $LiAlH_4$ + then H_2O; $NaBH_4$ / MeOH

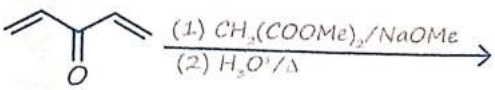
322. $Ph-CH=CH-CHO$ → $LiAlH_4$ + then H_2O; $NaBH_4$ / MeOH

323. $Ph-CH=CH-COOMe$ → $LiAlH_4$ + then H_2O; $NaBH_4$ / MeOH

324. 

325. 

326. 

327. 

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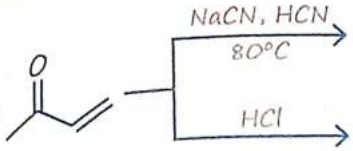
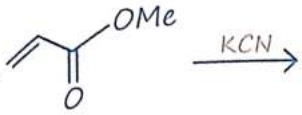
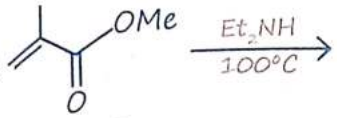
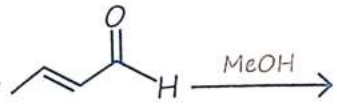
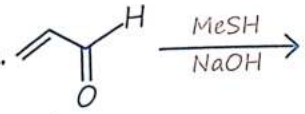
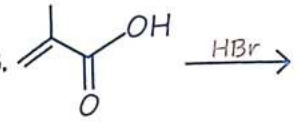
328. 

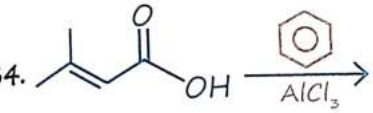
329. 

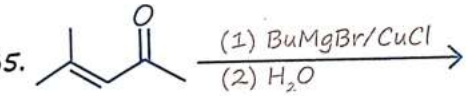
330. 

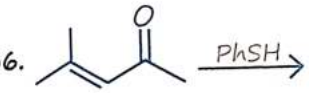
331. 

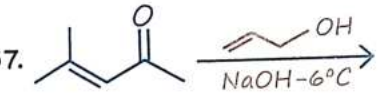
332. 

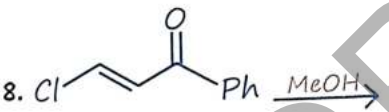
333. 

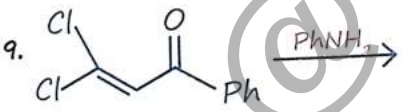
334. 

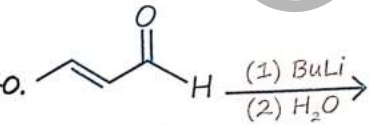
335. 

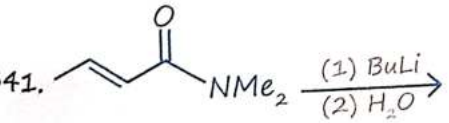
336. 

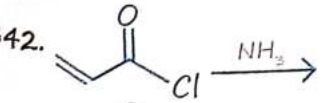
337. 

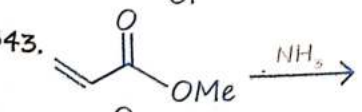
338. 

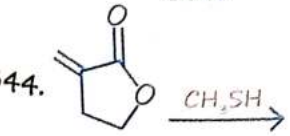
339. 

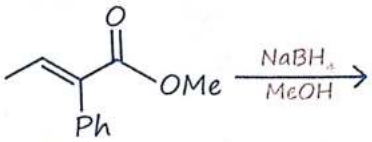
340. 

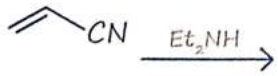
341. 

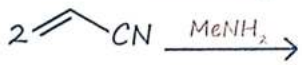
342. 

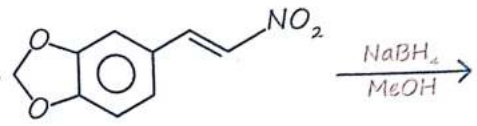
343. 

344. 

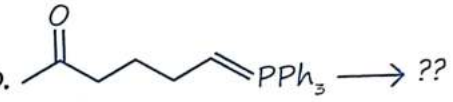
345. 

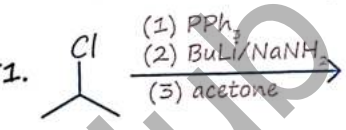
346. 

347. 

348. 

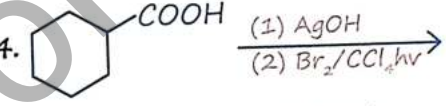
349. $CH_3 - CHO + Ph_3P = CH_2 \longrightarrow$??

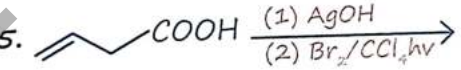
350. 

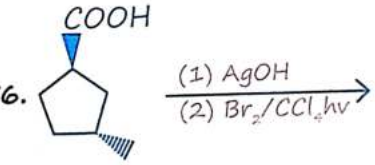
351. 

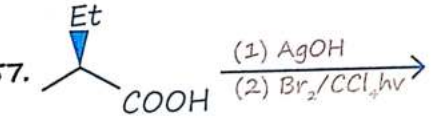
352. CC + ylide $\longrightarrow$

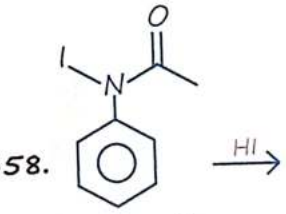
353. CC + ylide $\longrightarrow$

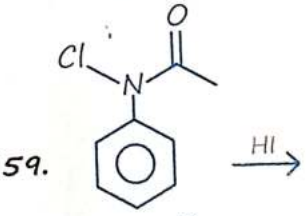
354. 

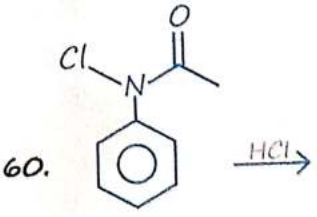
355. 

356. 

357. 

358. 

359. 

360. 

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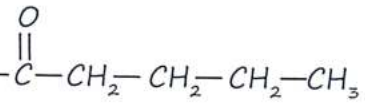
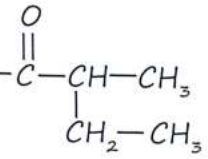
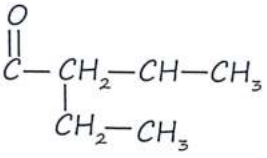
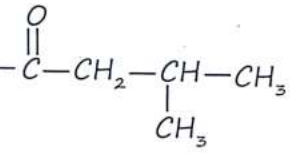
31. "P" is an optically active compound with molecular formula $C_6H_{12}O$. When "P" is treated with 2,4-dinitrophenylhydrazine, it gives a positive test. However, in presence of Tollens reagent, "P" gives a negative test. Predict the structure of "P":

[07 April, 2025 (Shift-II)]

(a) 

(b) 

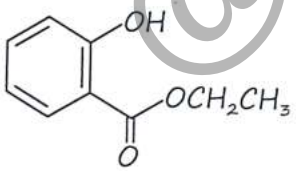
(c) 

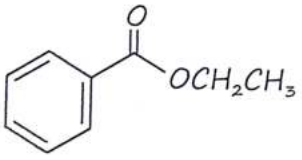
(d) 

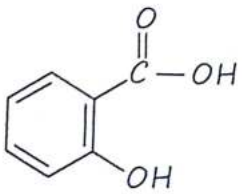
Sol. (b)

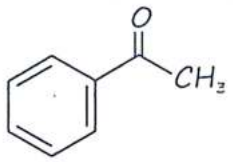
32. Which one of the following compounds will give orange precipitate when treated with 2,4-Dinitrophenyl hydrazine?

[27 July, 2021 (Shift-I)]

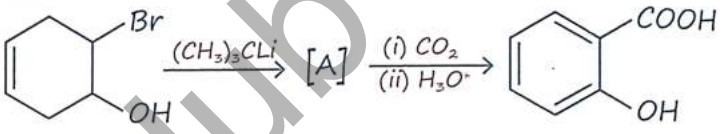
(a) 

(b) 

(c) 

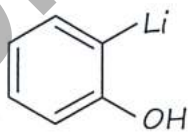
(d) 

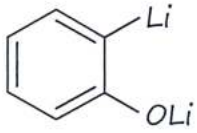
Sol. (d) All carbonyl compounds whether aldehyde or ketone give orange precipitate with 2,4-dinitrophenyl hydrazine (Brady's reagent)

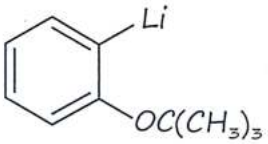
33. 

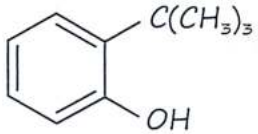
In the given conversion, the compound A is:

[29 June, 2022 (Shift-I)]

(a) 

(b) 

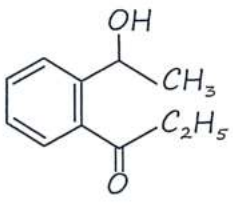
(c) 

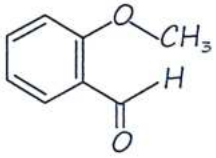
(d) 

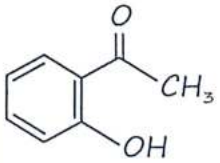
Sol. (b) The given reaction sequence occurs are:

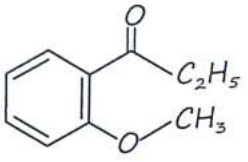
34. An organic compound neither reacts with neutral ferric chloride solution nor with Fehling solution, It however, reacts with Grignard reagent and gives positive iodoform test. The compound is

[8 April, 2019 (Shift-I)]

(a) 

(b) 

(c) 

(d) 

Sol. (a)

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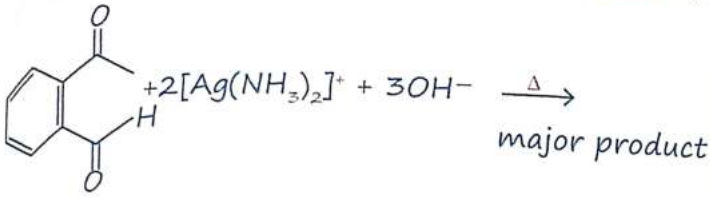
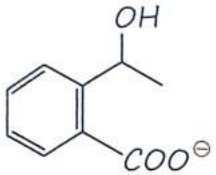
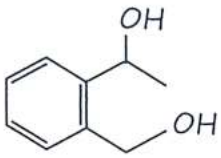
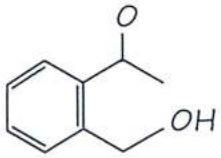
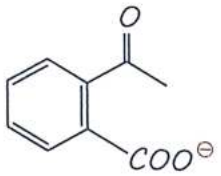
35. Identify the major product obtained in the following reaction: (2023)



(a) 

(b) 

(c) 

(d) 

Sol. (d) Tollen's reagent oxidises aldehydes into carboxylate ion whereas ketone is not oxidised by tollen's reagent.

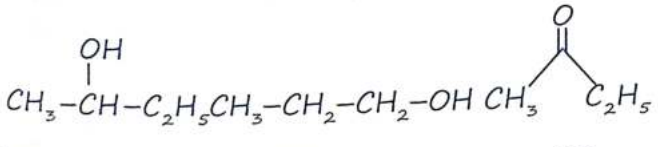
36. CH_3CHO and $C_6H_5CH_2CHO$ can be distinguished chemically by: (2012 Pre)

(a) Fehling solution test
(b) Benedict test
(c) Iodoform test
(d) Tollen's reagent test

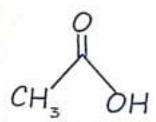
Sol. (c) Aldehydes & ketones have $CH_3-\overset{O}{\overset{\|}{C}}-$ group. These groups give a yellow colour ppt. & when boiled with iodine & NaOH (alkali) this is called iodoform reaction.

$$\underset{\text{(Acetaldehyde)}}{CH_3-\overset{O}{\overset{\|}{C}}-H} + 4NaOH + 3I_2 \rightarrow \underset{\text{(Iodoform)}}{CHI_3} + HCOONa + 3NaI + 3H_2O$$

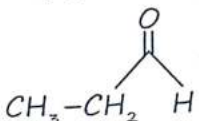
37. Which among the following compounds give yellow solid when reacted with NaOI/NaOH? [04 April, 2025 (Shift-II)]



(A) (B) (C)



(D)



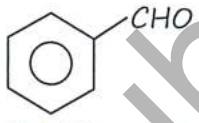
(E)

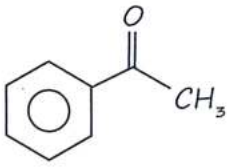
Choose the correct answer from the options given below:

(a) (B), (C) and (E) Only
(b) (A) and (C) Only
(c) (C) and (D) Only
(d) (A), (C) and (D) Only

Sol. (b)

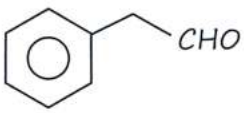
38. The compounds which give positive Fehling's test are:

A. 

B. 

C. $HOCH_2-CO-(CHOH)_3-CH_2-OH$

D. $CH_3-\overset{O}{\overset{\|}{C}}-H$

E. 

Choose the correct answer from the options given below: [22 Jan, 2025 (Shift-I)]

(a) A, D and E only (b) C, D and E only
(c) A, B and C only (d) A, C and D only

Sol. (b)

39. From the compounds given below, number of compounds which give positive Fehling's test is___.

Benzaldehyde, Acetaldehyde, Acetone, Acetophenone, Methanal, 4-nitrobenzaldehyde, cyclohexane carbaldehyde. [29 Jan, 2024 (Shift-I)]

Sol. [3]

40. Which of the following compounds will give silver mirror with ammoniacal silver nitrate? [09 April, 2024 (Shift-II)]

(A) Formic acid (B) Formaldehyde
(C) Benzaldehyde (D) Acetone

Choose the correct answer from the options given below:

(a) C and D only (b) A, B and C only
(c) A only (d) B and C only

Sol. (b)

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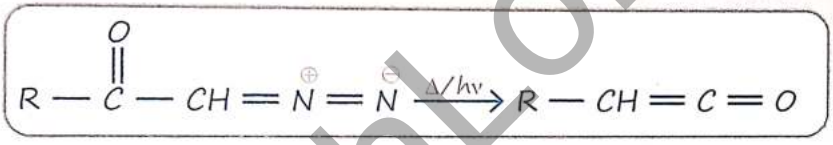
Chemistry of Nitrenes

Nitrenes are highly reactive species with a neutral nitrogen atom having only six valence electrons (like a nitrogen radical). They are electron-deficient and behave like reactive intermediates in many organic reactions. Nitrenes can be generated by the loss of nitrogen gas (N_2) from azides or by photolysis of certain compounds.

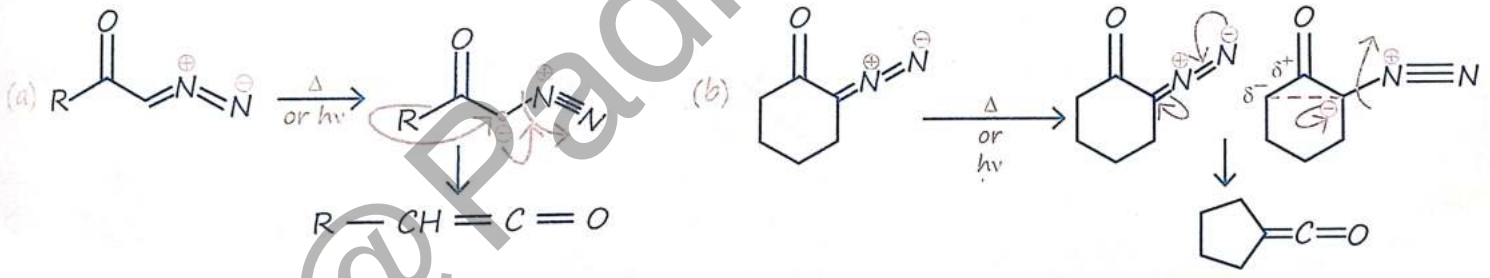
Wolff Rearrangement

Wolff rearrangement involves the transformation of α-diazoketone into a ketene, typically initiated by heat or light (photolysis).

Reaction



Mechanism

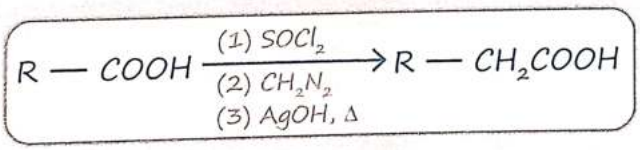


Wolff flashes, Eistert extends

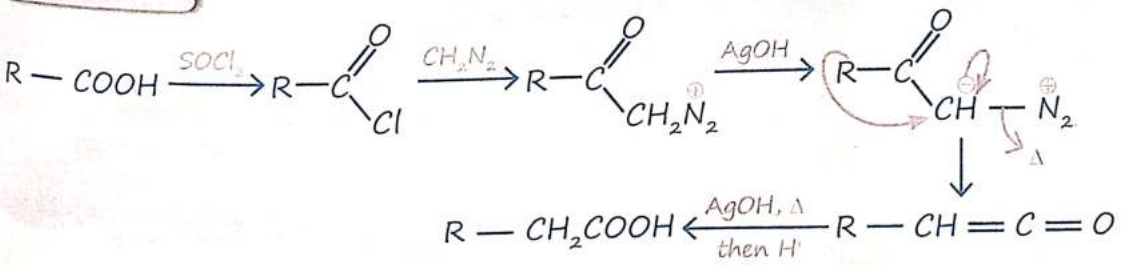
Arndt Eistert rearrangement

Conversion of carboxylic acids to higher homologous acids via diazoketone formation and Wolff rearrangement.

Reaction



Mechanism



Arndt-Eistert rearrangement opens doors for creating bigger acids.

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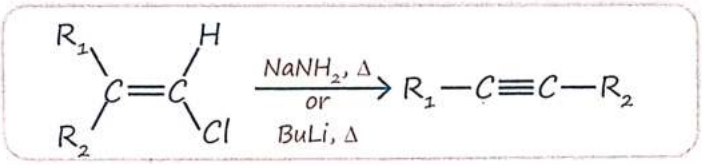
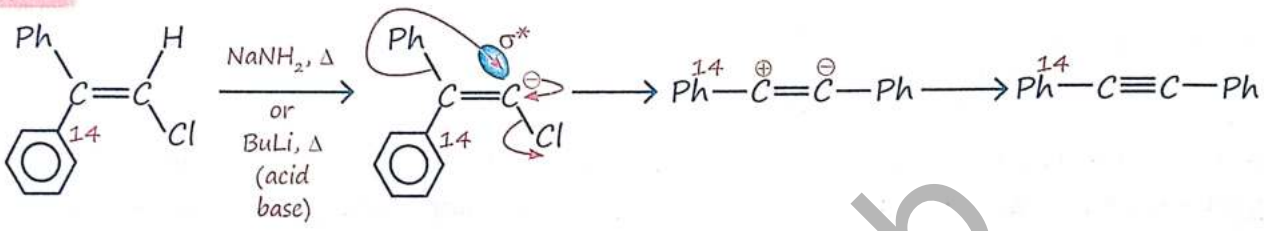
Frist Reaction (1, 1-elimination)

In Frist Reaction both leaving groups depart from the same carbon, and a rearrangement occurs to stabilize the carbocation or carbanion intermediate, ultimately forming an alkyne.

Reaction



Mechanism

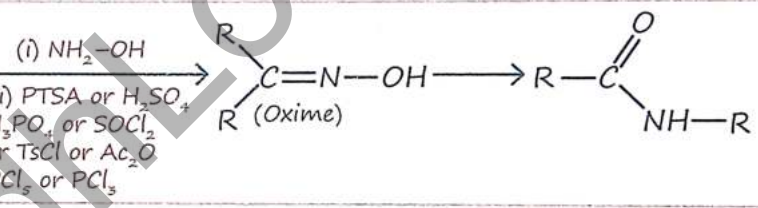


Beckmann Rearrangement

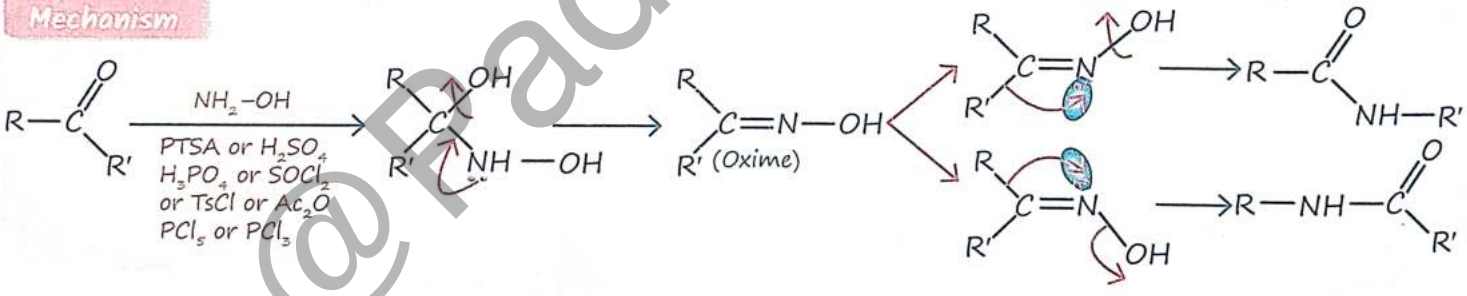
Oximes undergo acid-catalyzed rearrangement to form amides; in aldo-oximes, nitriles are favoured if a stable carbocation is formed during rearrangement.

Reaction

➢ Ketone:



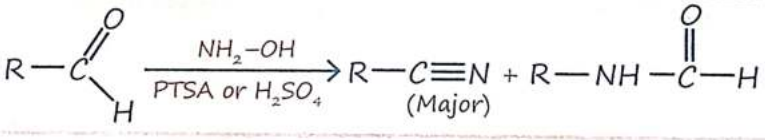
Mechanism



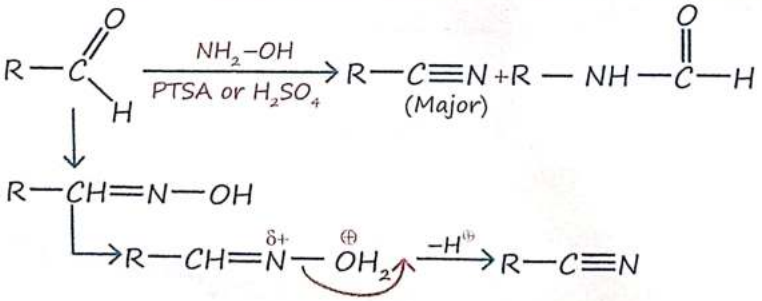
Abnormal Beckmann

Reaction

➢ Aldehyde:



Mechanism



Beckmann Rearrangement:
Oxime mein acid ka tadka, amide ya nitrile ka jackpot pakka!

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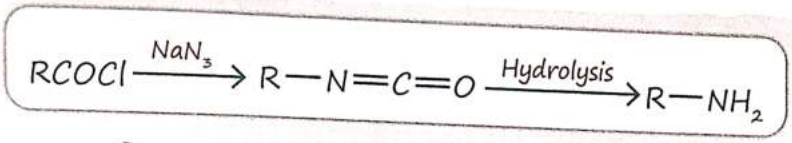
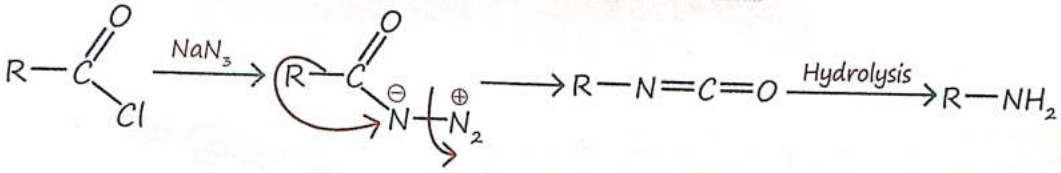
Curtius Rearrangement

An acyl azide thermally decomposes to form an isocyanate, which on hydrolysis gives a primary amine with the loss of one carbon atom.

Reaction



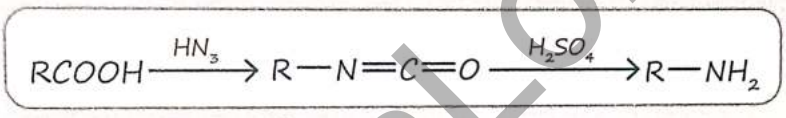
Mechanism



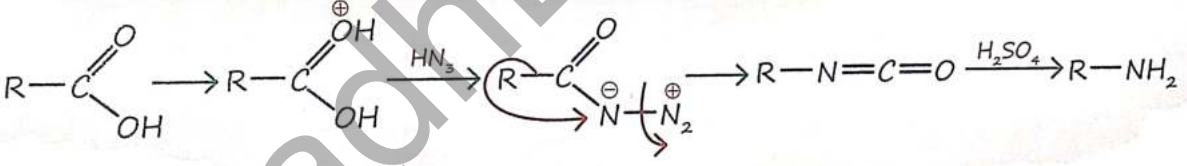
Schmidt Rearrangement

Carboxylic acids react with hydrazoic acid (HN_3) under acidic conditions to give primary amines via isocyanate intermediates.

Reaction

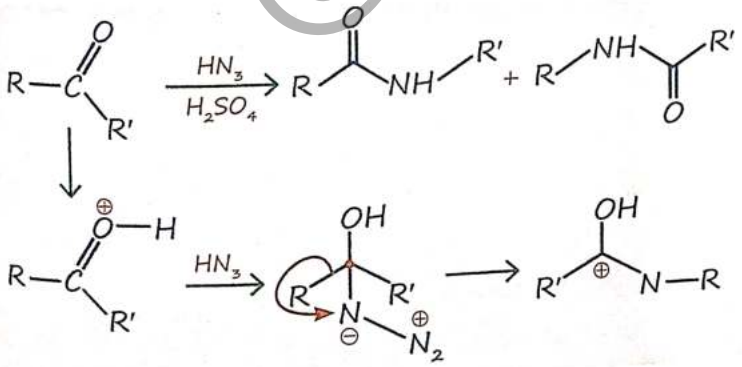


Mechanism

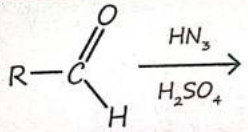


➢ Schmidt Reaction on Ketones and Aldehydes

Ketones give amides and aldehydes give nitriles when treated with HN_3 and acid, via rearrangement through isocyanate intermediates.



Try Yourself



Note: Abnormal Beckmann and Schmidt give cyanide if $\begin{matrix} R \\ \diagdown \\ C=O \\ \diagup \\ R' \end{matrix}$ contains R = H | Resonance | HOH_2C- | 3°

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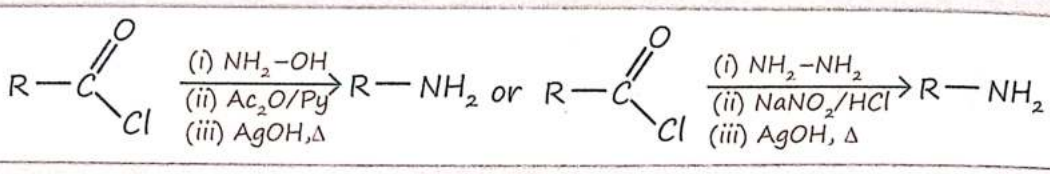
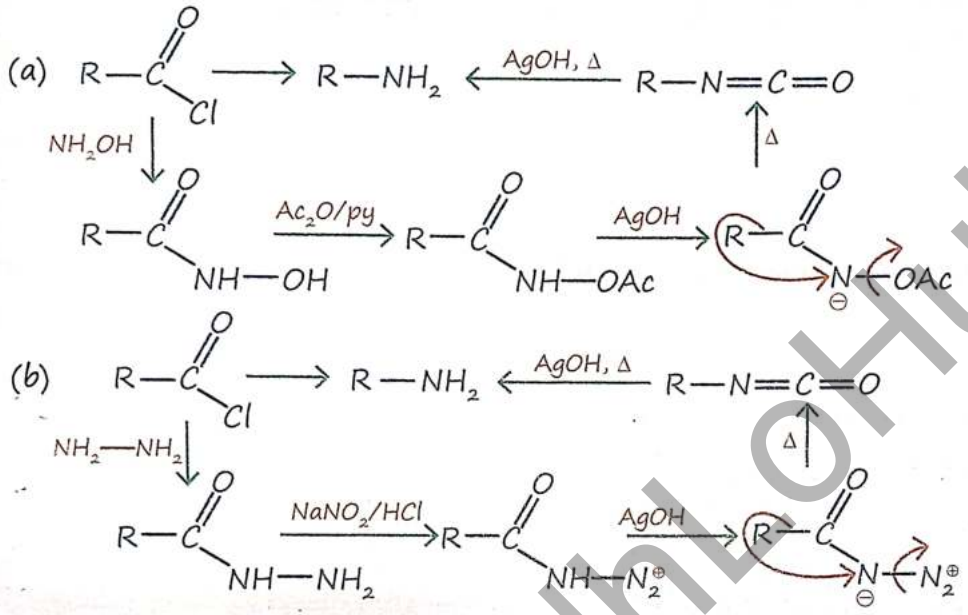
Lossen Rearranagement

Hydroxamic acids are converted into isocyanates via acylation and thermal rearrangement, which then hydrolyze to form amines.

Reaction



Mechanism



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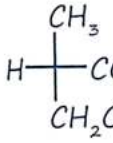
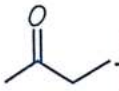
Practice Questions

1. $R-COOH$ → (1) $HN_3/H^{\oplus}$ | (1) $SOCl_2$, (2) NaN_3, (3) $H^{\oplus}/H_2O$ | (1) PCl_5, (2) CH_2N_2, (3) Ag_2O/Δ | (1) PCl_3, (2) $(CH_3)_2Cd$, (3) NH_2OH, (4) $H^{\oplus}$

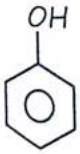
2. $\xrightarrow[H_2SO_4]{N_3H}$

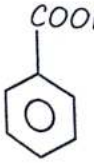
3.  $\xrightarrow[H_2SO_4]{N_3H}$

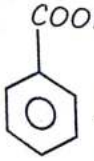
4.  $\xrightarrow[(ii)\ PCl_5]{(i)\ NH_2OH}$

5. $\xrightarrow[(2)\ H_2SO_4]{(1)\ NH_2OH}$ (3) H_2O, 533K

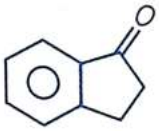
6. $CH_3COOH \xrightarrow{CH_2N_2}$

7.  $\xrightarrow{CH_2N_2}$

8.  $\xrightarrow{CH_2N_2}$

9.  $\xrightarrow[(2)\ CH_2N_2]{(1)\ SOCl_2}$ (3) Ag_2O, Δ

10. $\xrightarrow{CH_2N_2}$

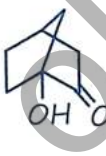
11.  $\xrightarrow[H^{\oplus}]{NH_2OH}$ P + Q

$P \xrightarrow{H_2SO_4} R \xrightarrow[\text{then } H_2O]{LiAlH_4} S$

$Q \xrightarrow{PCl_5} T \xrightarrow[\text{then } H_2O]{B_2H_6} U$ (more basic than S)

12. $Ph-CHO \xrightarrow[(2)\ H_2SO_4]{(1)\ NH_2OH}$

13. $\xrightarrow[H_2SO_4]{NH_2OH}$

14.  $\xrightarrow[H_2SO_4]{NH_2OH}$

15. Q Can be obtained by reaction of Benzonitrile and methyl magnesium bromide followed by Hydrolysis.

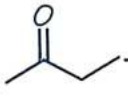
$R-OH \xrightarrow[\Delta]{H_3PO_4} P \xrightarrow[Zn/H_2O]{O_3} Q \xrightarrow{NH_2-OH} R + S$

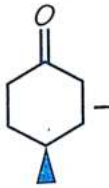
$R \xrightarrow{P_2O_5} T \xrightarrow[\Delta]{H^{\oplus}/H_2O} U + V$

+

$S \xrightarrow{H_2SO_4} W \xrightarrow[\Delta]{H^{\oplus}/H_2O} X + Y$

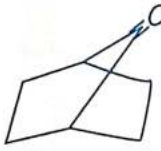
X is more basic than V.

16.  $\xrightarrow[H_2SO_4]{HN_3}$

17.  $\xrightarrow[H_2SO_4]{HN_3}$

18. $CH_3-CHO \xrightarrow[H_2SO_4]{HN_3}$

19. $\xrightarrow[H_2SO_4]{HN_3}$

20.  $\xrightarrow[H_2SO_4]{HN_3}$

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Preparation of Amines

Hoffmann Bromamide Rearrangement

Primary amides react with Br_2 and NaOH to give primary amines with one carbon less via formation of an isocyanate intermediate.

Reaction

$$R-C(=O)-NH_2 \xrightarrow[Br_2]{4\ NaOH} R-NH_2 + Na_2CO_3 + 2NaBr$$

Mechanism

1. $R-C(=O)-NH_2 \xrightarrow[(AB)]{NaOH} R-C(=O)-\overset{\ominus}{N}H$

2. $R-C(=O)-\overset{\ominus}{N}H \xrightarrow{Br+Br} R-C(=O)-NH-Br$

3. $R-C(=O)-NH-Br \xrightarrow[(AB)]{NaOH} R-C(=O)-\overset{\ominus}{N}-Br$

4. $R-C(=O)-\overset{\ominus}{N}-Br \longrightarrow R-N=C=O$ (Intermediate)

5. $R-N=C=O \xrightarrow{2NaOH} R-NH_2 + Na_2CO_3$

Baba Tea-Stall (BTS)

(i) 1 mole of Br_2 and 4 mol of NaOH is taken.

(ii) Migration at $e^{\ominus}$ deficient 'N' (Nitrene like) takes place. by alkyl or aryl group.

(iii) Intermediate Obtained is $R-N=C=O$

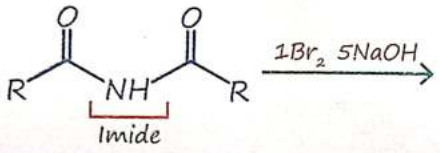
(iv) Rearrangement is completely intramolecular process.

(v) During rearrangement, configuration never changes.

(vi) In the presence of NaOH, Br_2 reacts, to form (NaOBr).

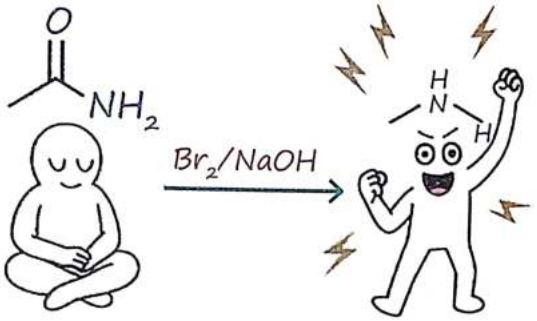
Try Yourself

- For Imide (For JEE Advanced)



Hoffmann Shortcut:
"Hoffmann doesn't do extra — he deletes carbon like unnecessary drama."

Jugaad



Amide before Br_2/NaOH:
"I'm stable, I'm chill."

After reaction:
"I lost a carbon and gained a new identity- AMINE MODE ACTIVATED!"

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Recap: Reaction Comparison

Feature	Curtius	Schmidt	Lossen	Hoffmann
Starts From	Acyl Azide	Carboxylic Acid	Hydroxamic Acid	Amide
Intermediate	Isocyanate	Isocyanate	Isocyanate	Isocyanate
Final Product	1° Amine	1° Amine	1° Amine	1° Amine
Carbon Loss?	Yes	Yes	Yes	Yes

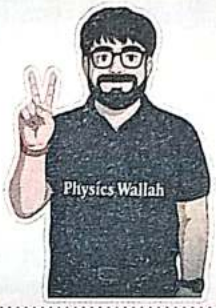
One Intermediate to Rule Them All

Acyl Azide → Curtius → Isocyanate → Amine

Carboxylic Acid → Schmidt → Isocyanate → Amine

Hydroxamic Acid → Lossen → Isocyanate → Amine

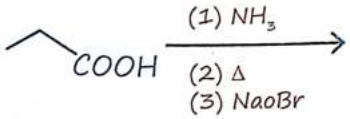
Amide → Hoffmann → Isocyanate → Amine

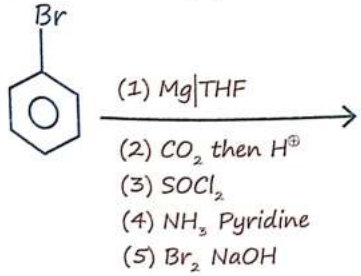


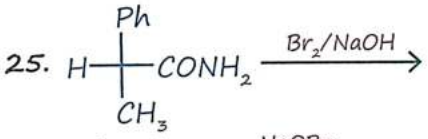
Practice Questions

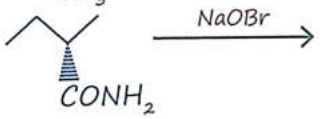
21. $CH_3-\underset{\underset{O}{\|}}{C}-NH_2 \xrightarrow{Br_2/NaOH}$

22. $Ph-\underset{\underset{O}{\|}}{C}-NH_2 \xrightarrow{NaOBr}$

23. 

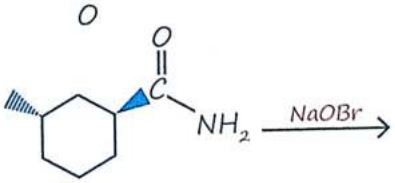
24. 

25. 

26. 

27. $Ph-\underset{\underset{O}{\|}}{C}-NH_2 + CH_3-\overset{15}{\underset{\underset{O}{\|}}{C}}-NH_2 \xrightarrow{Br_2\ NaOH}$

28. $Ph-\underset{\underset{O}{\|}}{C}-NHBr \xrightarrow{\ominus OBr}$

29. 

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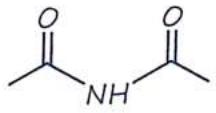
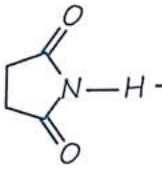
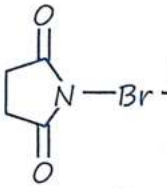
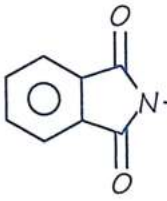
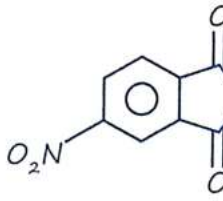
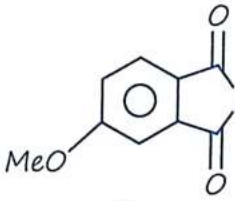
30.  $\xrightarrow{Br_2/NaOH}$

31.  $\xrightarrow{Br_2/NaOH}$

32.  $\xrightarrow{Br_2\ NaOH}$

33.  $\xrightarrow{Br_2/NaOH}$

34.  $\xrightarrow{Br_2/NaOH}$

35.  $\xrightarrow{Br_2/NaOH}$

36. $CH_3-\overset{O}{\overset{\|}{C}}-NH_2 \xrightarrow{Br_2/NaOH}$ / $\xrightarrow{Br_2/NaOD}$

37. $CH_3-\overset{O}{\overset{\|}{C}}-ND_2 \xrightarrow{Br_2/NaOH}$ / $\xrightarrow{Br_2/NaOD}$

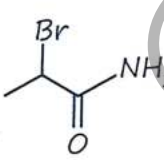
38. $\xrightarrow[AlCl_3]{CH_3Cl}$ P $\xrightarrow[\Delta]{H_2SO_4}$ Q $\xrightarrow[AlCl_3]{CH_3COCl}$ R $\xrightarrow[\Delta]{H^\oplus}$ S $\xrightarrow[KOH,\Delta]{KMnO_4}$ T $\xrightarrow[\Delta]{NH_3}$ U $\xrightarrow{\Delta}$ V $\xrightarrow[(2)\ PhCH_2Cl]{(1)\ KOH}$ W

W $\xrightarrow{KOH\,|\,H_2O,\ H^\oplus}$ Y $\xleftarrow[NaOH]{Br_2}$ X

39. n–hexane $\xrightarrow[770K]{Cr_2O_3}$ P $\xrightarrow[AlCl_3]{CH_2Cl_2}$ Q $\xrightarrow{KCN}$ R $\xrightarrow{H_2/Ni}$ S $\xleftarrow{NaOBr}$ T $\xrightarrow{H_3O^\oplus}$ U $\xrightarrow[\Delta]{H_2SO_4}$ V

40. Find mass of 'U'

$CaC_2 \xrightarrow{H_2O}$ P $\xrightarrow[\text{Then } CH_3-I]{NaNH_2}$ Q (4gm) $\xrightarrow[\text{Fe tube}]{\text{Red hot}}$ R (60%) $\xrightarrow[KOH,\Delta]{KMnO_4}$ S (40%) $\xrightarrow[\Delta]{NH_3}$ T (80%) $\xrightarrow[NaOH]{Br_2}$ U (75%)

41.  $\xrightarrow{Br_2/NaOH}$

42. $Ph-CH=CH-CONH_2 \xrightarrow{Br_2/NaOH}$

43. $Ph-C\equiv C-CONH_2 \xrightarrow{Br_2/NaOH}$

44. $\xrightarrow{Br_2/NaOH}$

45. Ethyne $\xrightarrow[813k]{Fe\ tube}$ P $\xrightarrow[AlCl_3]{CH_3COCl}$ Q $\xrightarrow[KOH,\Delta]{KMnO_4}$ R $\xrightarrow[\Delta]{NH_3}$ S $\xrightarrow{NaOBr}$ T $\xrightarrow[0°C]{HNO_2}$ U $\xrightarrow{CuCN/KCN}$ V $\xrightarrow[H_2O]{DIBAl}$ W

W $\xrightarrow[\text{then } H_2O]{CH_3MgBr}$ X $\xrightarrow[358K]{H_3PO_4}$ Y $\xleftarrow[(PhCO)_2O_2]{HBr}$ Z $\xleftarrow[Et_2O]{Mg}$ A $\xleftarrow[\text{then } H^\oplus]{CO_2}$ B $\xleftarrow{SOCl_2}$ C $\xleftarrow{AlCl_3}$ D $\xleftarrow{H_2/Ni}$ E $\xleftarrow[\Delta]{Al_2O_3}$ F $\xleftarrow{Br_2/hv}$ G

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Previous Year Questions

1. Identify the major product C formed in the following reaction sequence: (2024)

$$CH_3-CH_2-CH_2-I \xrightarrow{NaCN} A \xrightarrow[\text{Partial hydrolysis}]{OH^-} B \xrightarrow[Br_2]{NaOH} C \text{ major}$$

(a) butanamide
(b) α-bromobutanoic acid
(c) propylamine
(d) butylamine

Sol.(c) Propylamine

2. Reaction of propanamide with ethanolic sodium hydroxide and bromine will give (2020-Covid)
(a) Methylamine (b) Propylamine
(c) Aniline (d) Ethylamine

Sol. (d) The reaction is Hoffmann bromide reaction

$$\underset{\text{(Propanamide)}}{CH_3CH_2CONH_2} \xrightarrow[Br_2]{NaOH/Ethanol} \underset{\text{(Ethylamine)}}{CH_3CH_2NH_2}$$

3. Acetamide is treated with the following reagents separately. Which one of these would yield methyl amine? (2010 Pre)
(a) PCl_5 (b) $NaOH-Br_2$
(c) Soda lime (d) Hot conc. H_2SO_4

Sol. (b) $NaOH-Br_2$

4. Intermediates formed during reaction of $R-\underset{\underset{O}{\|}}{C}-NH_2$ with Br_2 and KOH are: (2001)
(a) RCONHBr and RNCO
(b) RNHCOBr and RNCO
(c) RNH - Br and RCONHBr
(d) $RCONBr_2$

Sol. (a) RCONHBr and RNCO

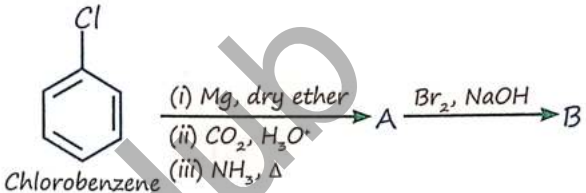
5. Amides can be converted into amines by reaction named after: (1999)
(a) Perkin (b) Claisen
(c) Hoffmann (d) Kekule

Sol. (c) Amide can be converted into amine by Hoffmann - Bromamide reaction.

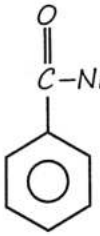
6. Indicate which nitrogen compound amongst the following would undergo Hoffmann reaction (i.e., reaction with Br_2 and strong KOH) to furnish the primary amine ($R-NH_2$)? (1989)
(a) $RCONHCH_3$ (b) $RCOONH_4$
(c) $RCONH_2$ (d) $R-CO-NHOH$

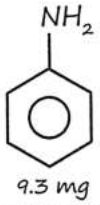
Sol. (c) The amide ($-CONH_2$) group is converted into primary amino group ($-NH_2$) by Hoffmann's bromamide reaction.

7. Consider the following sequence of reactions:



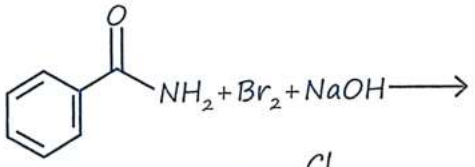
11.25 mg of chlorobenzene will produce .....x 10^{-1} mg of product B.
(Consider the reactions result in complete conversion.)
[Given molar mass of C,H,O,N and Cl as 12,1,16,14 and 35.5 g mol^{-1} respectively]
[28 Jan, 2025 (Shift-I)]

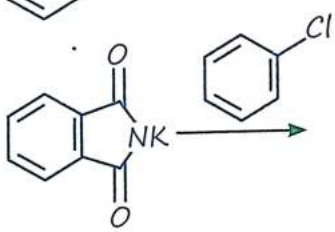
Sol. [93] A = 

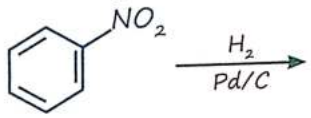
, B = 

9.3 mg
≈ 93 × 10^{-1} mg

8. How many of the transformation given below would result in aromatic amines?
[31 Jan, 2023 (Shift-I)]

(a) 

(b) 

(c) 

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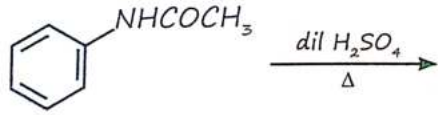
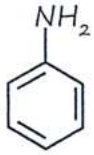
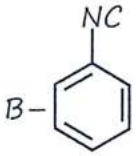
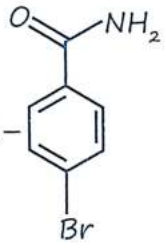
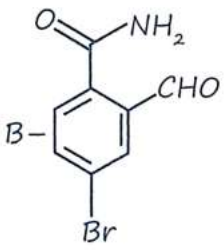
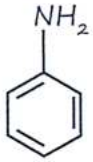
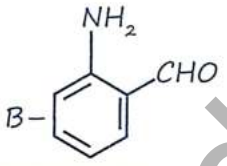
(d) 

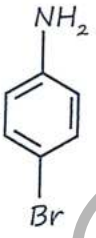
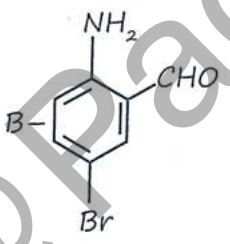
Sol. [3] Aromatic amines will produced in the reactions (a), (c) and (d).

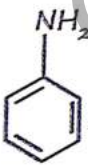
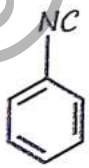
9. **Hoffmann bromamide degradation of benzamide gives product A, which upon heating with $CHCl_3$ and NaOH gives product B. The structures of A and are B:** [17 March, 2021 (Shift-I)]

(a) A–  B– 

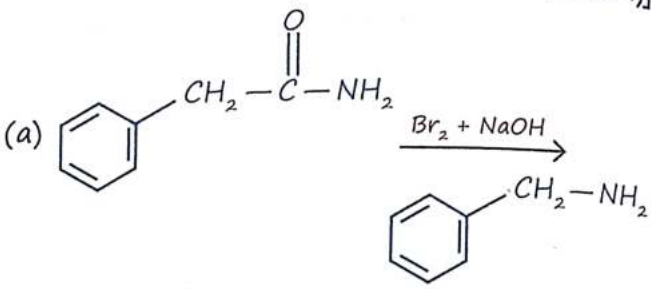
(b) A–  B– 

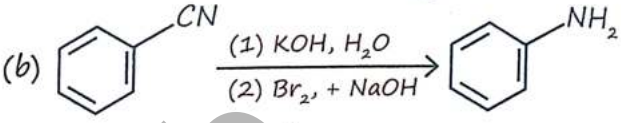
(c) A–  B– 

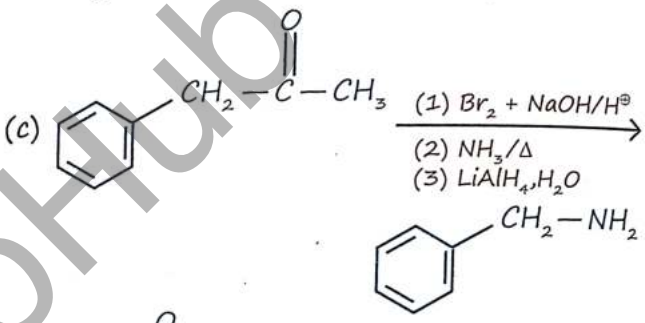
(d) A–  B– 

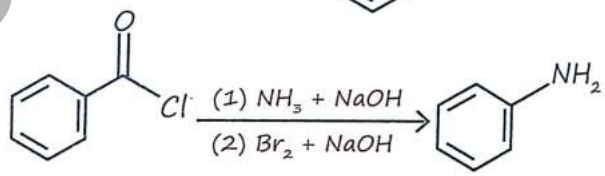
Sol. (a) A =  , B = 

10. **Which of the following reaction DOES NOT involve Hoffmann Bromamide degradation?** [16 March, 2021 (Shift-I)]

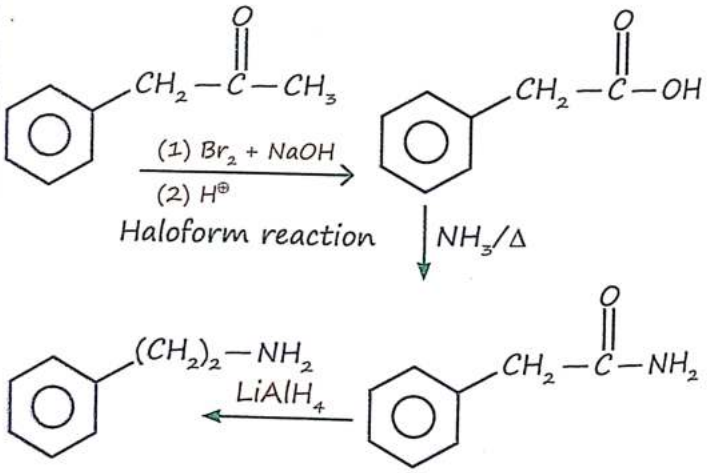
(a) 

(b) 

(c) 

(d) 

Sol. (c)



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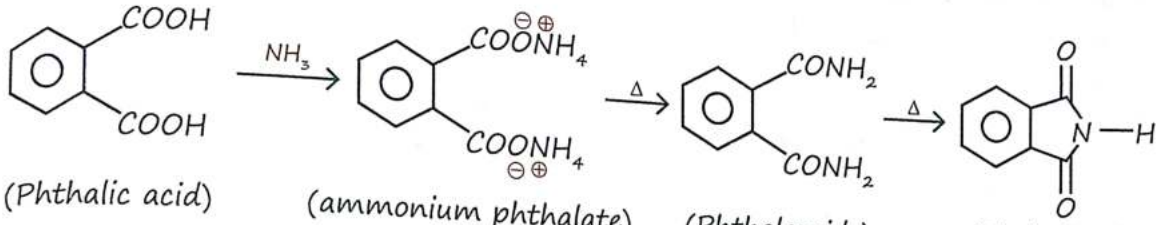
Gabriel Phthalimide Synthesis

- Formation of phthalimide from phthalic acid.

Phthalic acid reacts with ammonia to form ammonium phthalate, which on heating gives phthalamide, and further heating produces phthalimide.



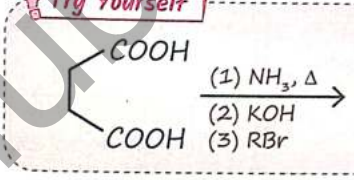
(Phthalic acid) (ammonium phthalate) (Phthalamide) (Phthalimide)

- Gabriel Phthalimide Amine Synthesis:

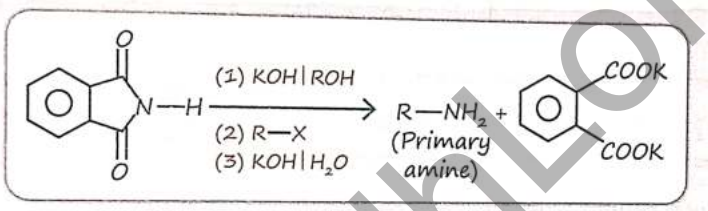
Gabriel Phthalimide Amine Synthesis is used to prepare primary amines ($R-NH_2$).

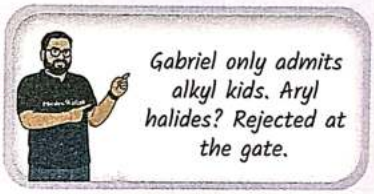
Phthalimide reacts with KOH and an alkyl halide (R–X) through nucleophilic substitution, followed by hydrolysis, to give the primary amine and potassium phthalate.

Try Yourself

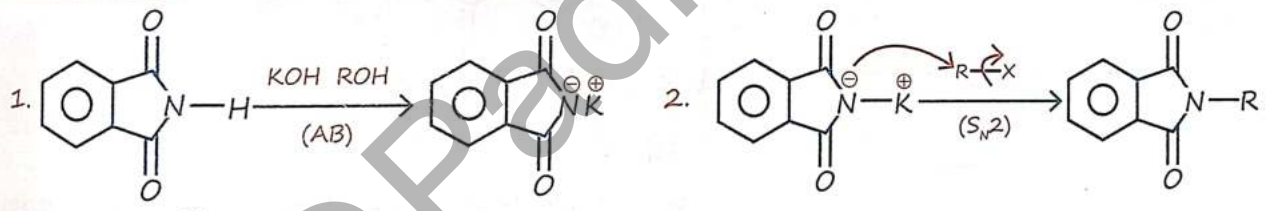


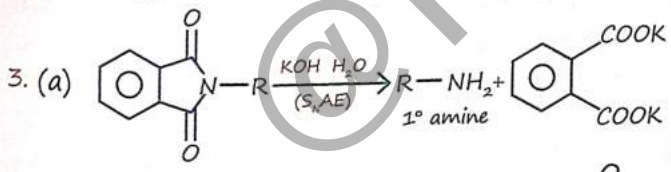
Reaction

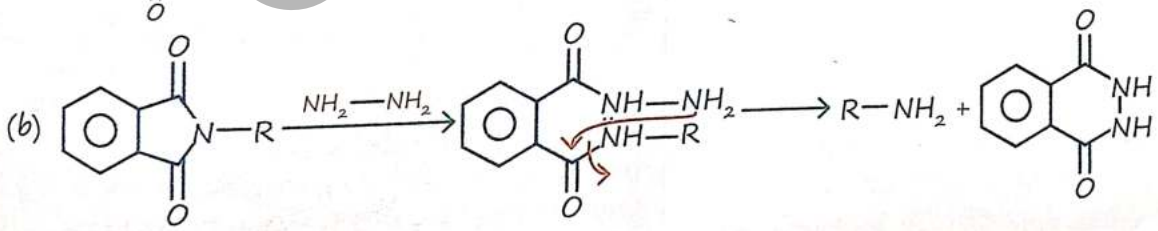




Mechanism







Baba Tea-Stall (BTS)

(i) Best method to synthesize primary amine.
(ii) For acid base, alcoholic KOH is taken.
(iii) R—X must be 1° or 2° halide.
(iv) S_N2 mechanism.
(v) Inversion takes place.
(vi) Only H_2O must not be taken, either acidic or basic medium is taken for hydrolysis.
(vii) Reaction is used prepare amino acid.

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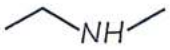
Practice Questions

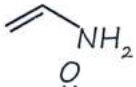
46. Which of the following amines can be obtained good yield by gabriel Phthalimide Synthesis?

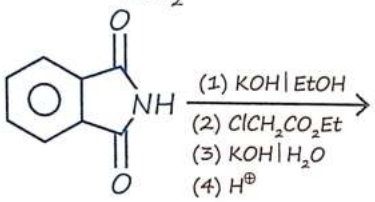
(a) $PhNH_2$ (b) CH_3NH_2

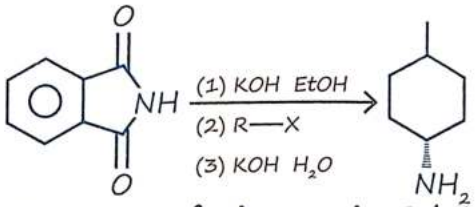
(c) (d) $(CH_3)_2NH$

(e) CH_3CONH_2 (f) 

(g) PhNHMe (h) $PhCH_2NH_2$

(i) 

47. 

48. 

49. The source of nitrogen in Gabriel synthesis of amines is _____. [NCERT Exemplar]

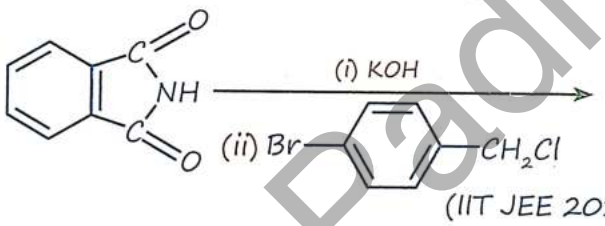
(a) Sodium azide, NaN_3
(b) Sodium nitrite, $NaNO_2$
(c) Potassium cyanide, KCN
(d) Potassium phthalimide, $C_6H_4(CO)_2NK^+$

50. Which of the following amines can be prepared by Gabriel synthesis. [NCERT Exemplar]

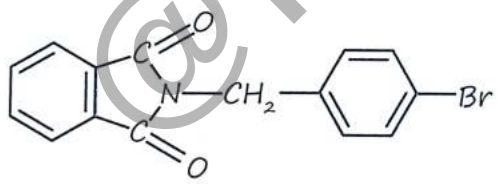
(a) Isobutyl amine
(b) 2-Phenylethylamine
(c) N-methylbenzylamine
(d) Aniline

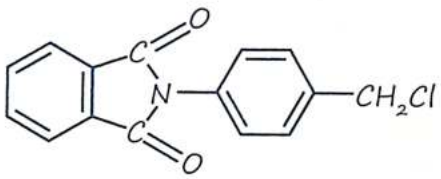
Previous Year Questions

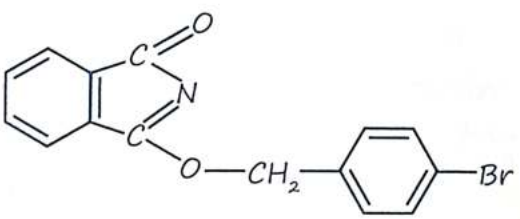
11. The major product of the following reaction is

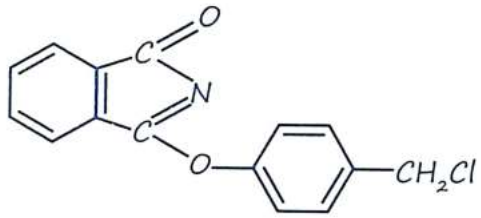


(IIT JEE 2011)

(a) 

(b) 

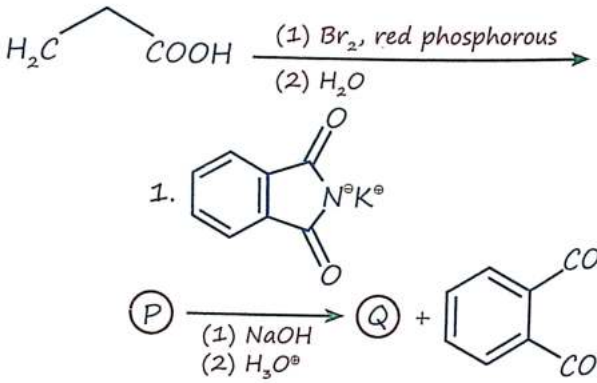
(c) 

(d) 

Sol. (a)

12. Considering the reaction sequence given below, the correct statement(s) is/are:

C2.05 W20.77 UA47.80 (JEE Adv. 2022)



(a) (P) can be reduced to a primary alcohol using $NaBH_4$

(b) Treating (P) with conc. NH_4OH solution followed by acidification gives (Q).

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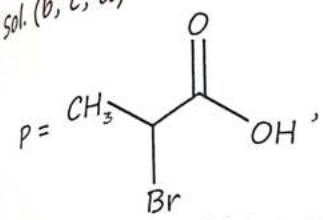
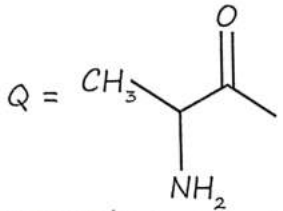
(c) Treating (Q) with a solution of $NaNO_2$ in aq. HCl liberates N_2

(d) (P) is more acidic than CH_3CH_2COOH.

Sol. (b, c, d)

P = 

Q = 

13. Method by which Aniline cannot be prepared is: (2015 Re)

(a) Potassium salt of phthalimide treated with chlorobenzene followed by hydrolysis with aqueous NaOH solution

(b) Hydrolysis of phenylisocyanide with acidic solution

(c) Degradation of benzamide with bromine in alkaline solution

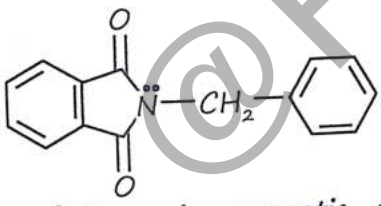
(d) Reduction of nitrobenzene with H_2/Pd in ethanol

Sol. (a) Aryl halides are less reactive towards Gabriel phthalimide reaction.

14. Phthalimide is made to undergo following sequence of reactions. [04 April, 2024 (Shift-II)]

Phthalimide $\xrightarrow[\text{(2) Benzylchloride}]{\text{(1) KOH}}$ 'P'

Total number of π-bonds present in product 'P' is/are

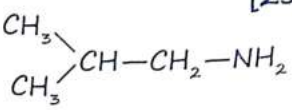
Sol. [8] P = 

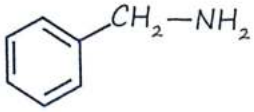
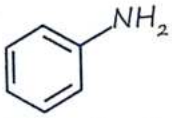
15. Number of isomeric aromatic amines with molecular formula $C_8H_{11}N$, which can be synthesized by Gabriel Phthalimide synthesis is [6 April, 2023 (Shift-II)]

Sol. [*] Gabriel synthesis is used for the preparation of primary amine. Primary Aromatic amine can't be synthesised by this method because aryl halide do not undergo nucleophilic substitution reaction.

NTA Answer Key [5], PW Answer Key [6]

16. The total number of amines among the following which can be synthesized by Gabriel synthesis is: [25 Feb, 2021 (Shift-II)]

(1)  (2) $CH_3CH_2NH_2$

(3)  (4) 

Sol.[3] Primary amines can be synthesized by Gabriel synthesis. 3 Amines can be synthesized.

17. Given below are two statements, one is labelled as Assertion (A) and other is labelled as Reason (R).

Assertion (A): Gabriel phthalimide synthesis cannot be used to prepare aromatic primary amines.

Reason (R): Aryl halides do not undergo nucleophilic substitution reaction.

In the light of the above statements, choose the correct answer from the options given below: [25 July, 2021 (Shift-I)]

(a) Both (A) and (R) true but (R) is not the correct explanation of (A).

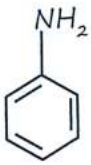
(b) (A) is false but (R) is true.

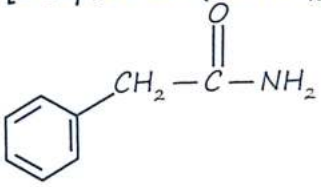
(c) Both (A) and (R) true and (R) is correct explanation of (A).

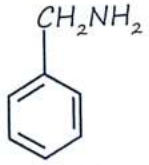
(d) (A) is true but (R) is false.

Sol. (c) Both (A) and (R) true and (R) is correct explanation of (A).

18. Which of the following compounds can be prepared in good yield by Gabriel phthalimide synthesis? [6 Sep, 2020 (Shift-II)]

(a) 

(b) 

(c) 

(d) $CH_3 - CH_2 - NHCH_3$

Sol. (c) Gabriel phthalimide synthesis gives 1° amine in good yield.

19. Which of the following amines can be prepared by Gabriel phthalimide reaction? [8 Apr, 2019 (Shift-I)]

(a) Neo-pentylamine (b) n-butylamine

(c) Triethylamine (d) t-butylamine

Sol. (b) Gabriel phthalimide reaction is used to prepare primary aliphatic amine. n-butyl amine is primary amine among the following option.

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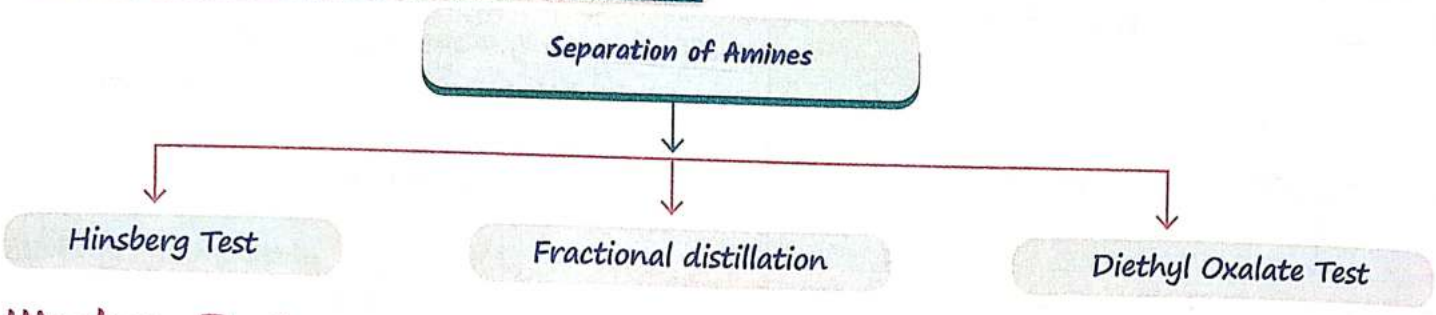
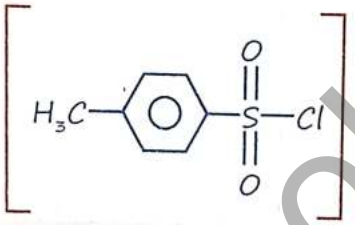
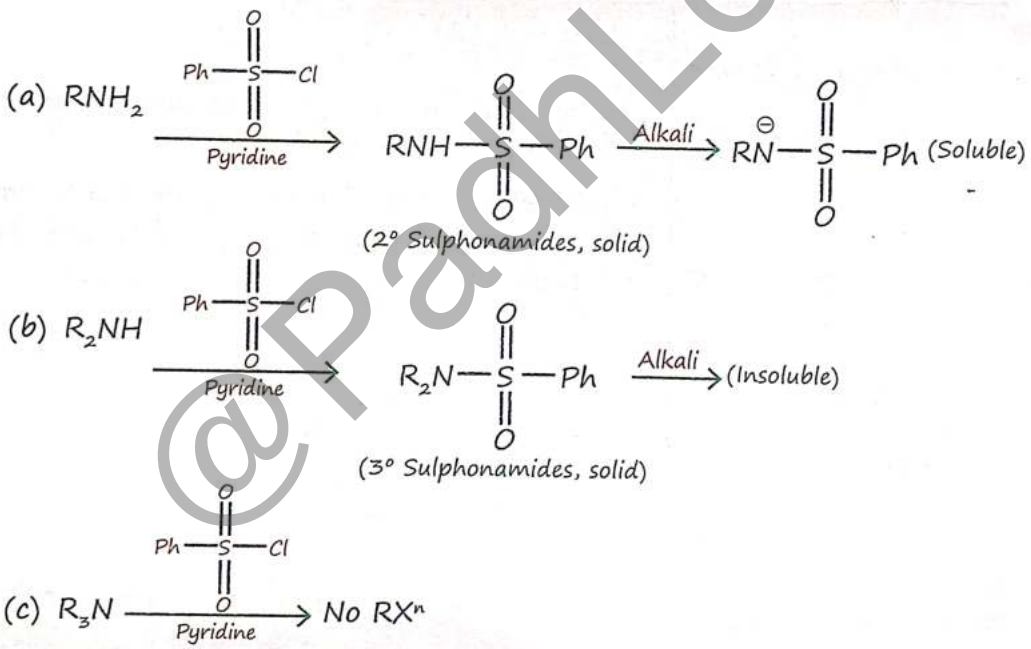
Separation of Amines



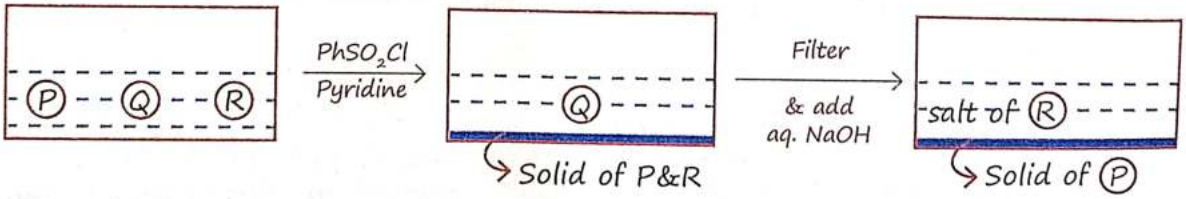
Hinsberg Test

- The Hinsberg test helps to distinguish primary, secondary, and tertiary amines.
- Amines react with benzenesulphonyl chloride (Hinsberg reagent) in the presence of a base.
- Primary amines form soluble sulphonamides, secondary amines form insoluble sulphonamides, and tertiary amines do not react.
- We can also use tosyl chloride 

जुगाड



- Used to separate 1°, 2° & 3° amine



Find P, Q and R ⇒

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Practice Questions

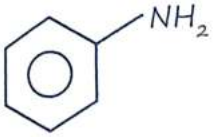
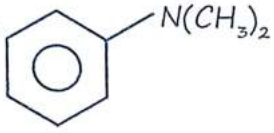
51. Which of the following compounds on reaction with hinsberg reagent produces base insoluble sulphonamide

(a) $PhNH_2$ (b) CH_3NH_2 (c) $PhCONMe_2$ (d) $(CH_3)_2NH$ (e) CH_3CONH_2

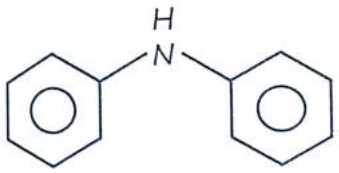
(f) (g) $PhNHMe$ (h) $PhCH_2NH_2$ (i) $EtCONHMe$

Previous Year Questions

20. Which among the following react with Hinsberg's reagent? [23 Jan, 2025 (Shift-I)]

A.  B. 

C. $CH_3 - NH_2$ D. $N(CH_3)_3$

E. 

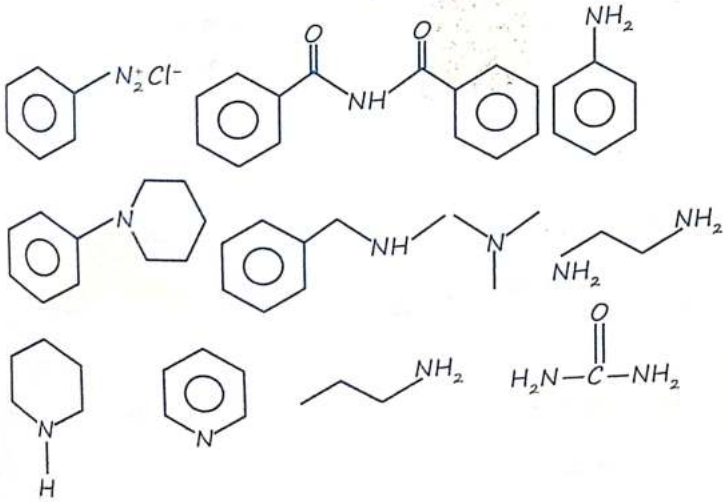
Choose the correct answer from the option given below:

(a) A, B and E only (b) C and D only

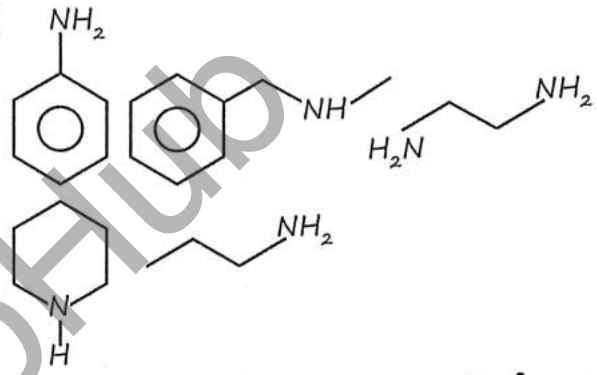
(c) A, C and E only (d) B and D only

Sol. (c) Hinsberg reagent reacts with primary and secondary amines. Compounds given in B and D are tertiary amines, hence, do not react with Hinsberg reagent.

21. Number of compounds which give reaction with Hinsberg's reagent is_____. [01 Feb, 2024 (Shift-II)]



Sol. [5]



22. Which statement is NOT correct for p-toluenesulphonyl chloride? [26 Jun, 2022 (Shift-II)]

(a) It is known as Hinsberg's reagent

(b) It is used to distinguish primary and secondary amines.

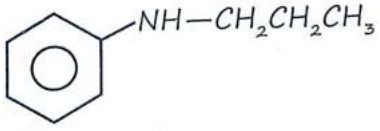
(c) On treatment with secondary amine, it leads to a product, that is soluble in alkali.

(d) It doesn't react with tertiary amines.

Sol. (c) Product formed from secondary amine is insoluble in alkali.

23. An amine on reaction with benzenesulphonyl chloride produces a compound insoluble in alkaline solution. This amine can be prepared by ammonolysis of ethyl chloride. The correct structure of amine is: [26 Feb, 2021 (Shift-I)]

(a) $CH_3CH_2CH_2NH-CH_3$

(b) 

(c) $CH_3CH_2NH_2$

(d) $CH_3CH_2CH_2NHCH_2-CH_3$

Sol. (d) Since, given amine on reaction with $PhSO_2Cl$ produces a compound insoluble in Alkaline solution, it is a 2° Amine.

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9 Biomolecules

Living systems, though made of non-living atoms, grow, sustain, and reproduce through complex chemical interactions. This branch of study, called biochemistry, explores key biomolecules like carbohydrates, proteins, nucleic acids, and lipids, which play vital roles in life processes along with vitamins and minerals.

Carbohydrates

Carbohydrates are organic compounds made up of carbon, hydrogen, and oxygen, typically following the formula $C_x(H_2O)_y$. They serve as a primary source of energy in living organisms and include sugars, starches, and cellulose. Chemically, they are defined as polyhydroxy aldehydes or ketones, or compounds that yield them on hydrolysis.

New Definition

- Optically active polyhydroxy carbonyl (Al/K)

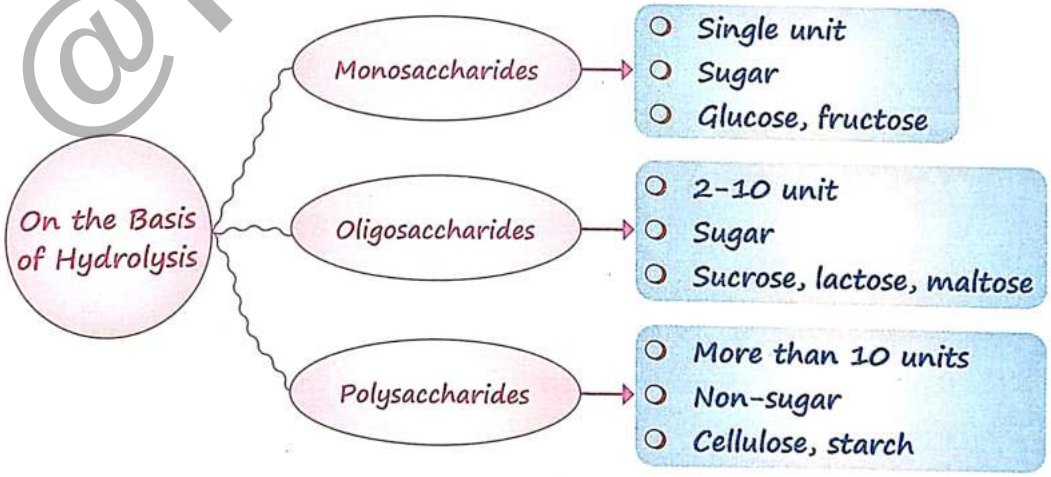
or

Compounds which gives optically active Polyhydroxy carbonyl on hydrolysis.

Classification of Carbohydrate

Carbohydrates are classified based on their structure and hydrolysis behavior into the following categories:

(a) On the basis of hydrolysis



(b) On the basis of reducing and Non-reducing sugar

Sugar which reduces Tollen's and Fehling solution is known as reducing sugar

1. —CHO present or —CHO forms (Hemiacetal) 2. α-hydroxy carbonyl

Ex 1. Glucose: Reducing Ex 2. Fructose: Reducing Ex 3. Sucrose: Non-Reducing

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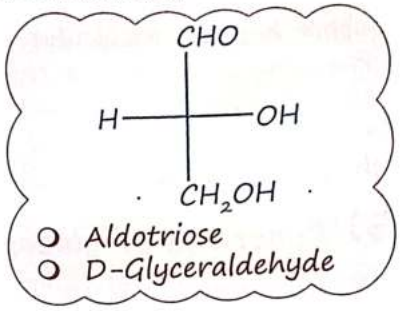
(c) **On the basis of numbers of carbon monosaccharides classification**

3C	Triose	Aldose ketose	Aldotriose ketotriose
4C	Tetrose	Aldose ketose	Aldotriose ketotriose
5C	Pentose	Aldose ketose	Aldotriose ketotriose
6C	Hexose	Aldose ketose	Aldotriose ketotriose



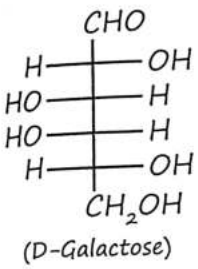
- Aldotriose
- D-Glyceraldehyde

D/L Configuration

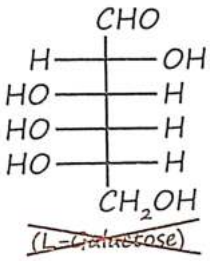
The D- and L- configuration refers to the position of the –OH group or $-NH_2$ group on the chiral carbon farthest from the carbonyl group in a Fischer projection.

- D-form: –OH group is on the right
- L-form: –OH group is on the left

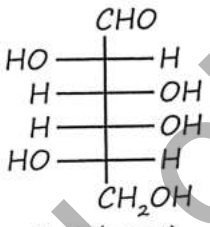
These configurations are based on glyceraldehyde and do not indicate optical rotation.



(D-Galactose)



~~(L-Galactose)~~

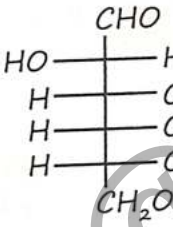


(L-Galactose)

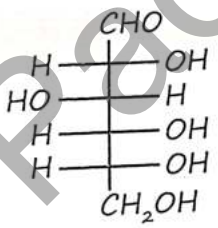
Galactose का configuration fixed हैं इसलिए केवल *l carbon* को *ulta* करके नही चलेगा।

D-family

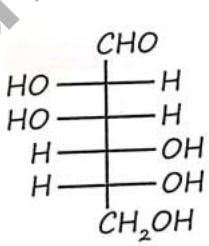
D-allose



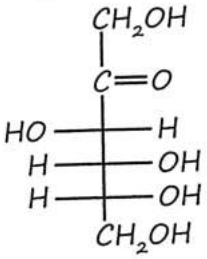
D-altrose



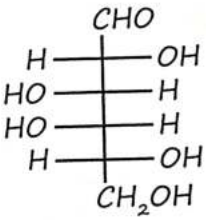
D-Glucose



D-Mannose



D-Fructose



D-Galactose

Epimers

Epimers are a specific type of diastereomers, which are stereoisomers that are not mirror images of each other. They differ in configuration at only one specific chiral (asymmetric) carbon atom in a molecule.

Example:

- D-Glucose and D-mannose ⟶ C-2 epimer
- D-Allose and D-Altrose ⟶ C-2 epimer
- D-Glucose and D-Galactose ⟶ C-4 epimer
- D-Allose and D-Glucose ⟶ C-3 epimer

Monosaccharides

A carbohydrate that cannot be hydrolysed further to give simpler unit of polyhydroxy aldehyde or ketone is called a monosaccharide. About 20 monosaccharides are known to occur in nature. Some common examples are glucose, fructose, ribose, etc

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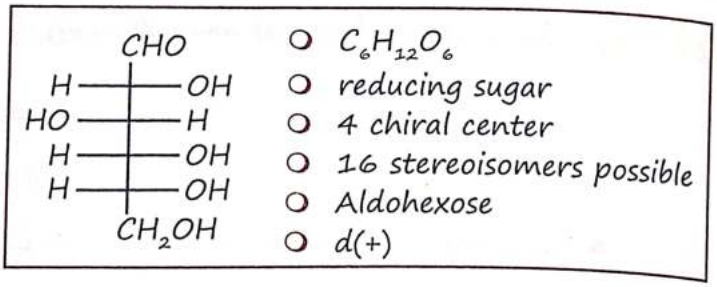
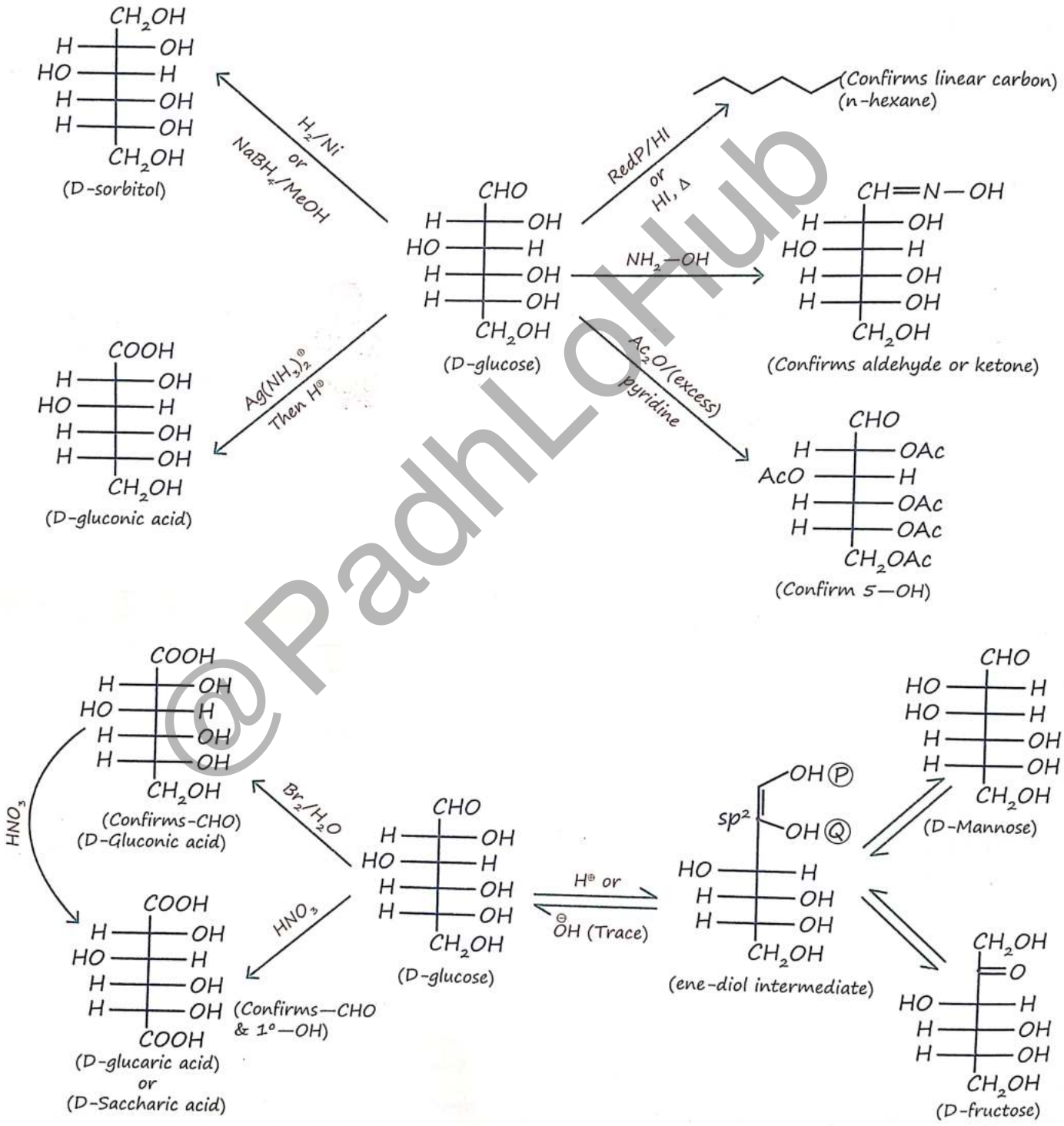
1. Glucose

Glucose has the molecular formula $C_6H_{12}O_6$ and is a reducing sugar with 4 chiral centers, allowing 16 stereoisomers. It is an aldohexose and exists in the d- (+)- form, meaning it rotates plane-polarized light to the right.



(a) Reactions of glucose



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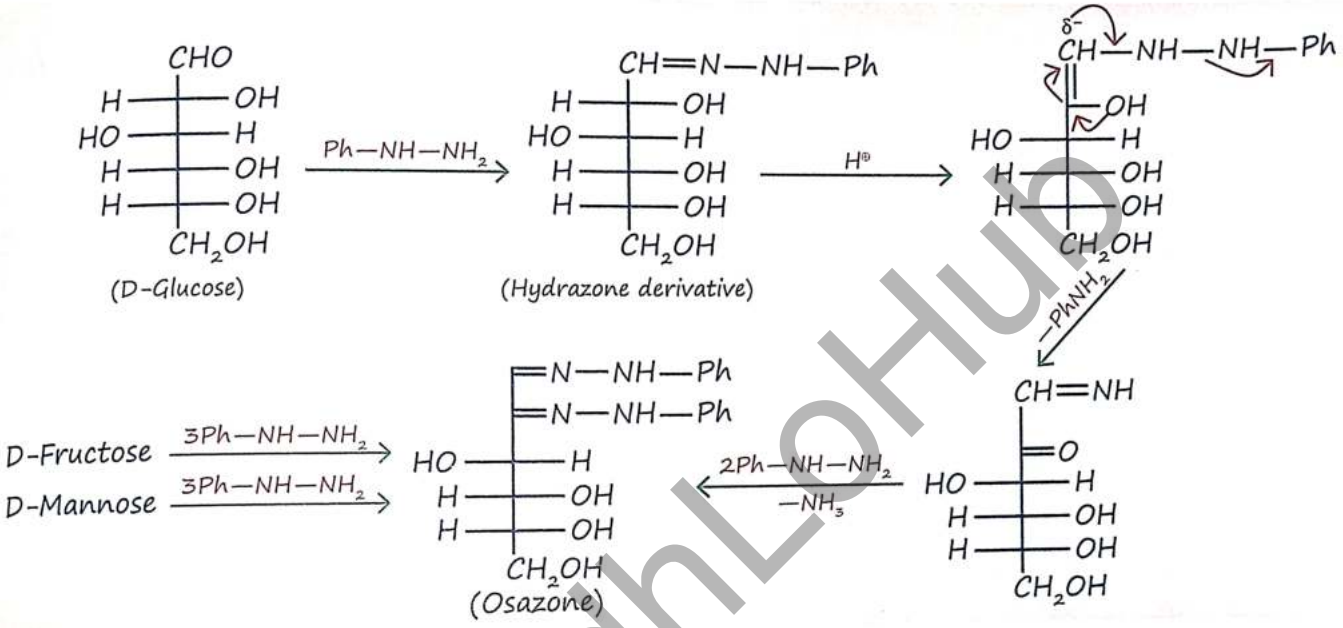
Osazone

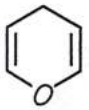
Osazone formation is a chemical test used to identify reducing sugars like glucose and fructose.
When glucose is heated with phenylhydrazine, it forms a characteristic crystalline compound called an osazone.
This reaction involves the carbonyl group (C=O) and the –OH group on the next carbon.
Interestingly, glucose and fructose form the same osazone, as the reaction only affects C-1 and C-2 carbons, which are similar in both.



Note

- D-Glucose, D-Fructose can be differentiate by Br_2/H_2O.
- D-Glucose, D-Mannose & D-Fructose remains in equilibrium due to ene-diol intermediate.

D-Glucose ⇌ D-Mannose ⇌ D-Fructose

- All C-2 epimers can form same osazone.
- All C-2 epimers form same ene-diol intermediate.
- 1.  (Pyran) 2. (furan)

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Cyclic structures of glucose

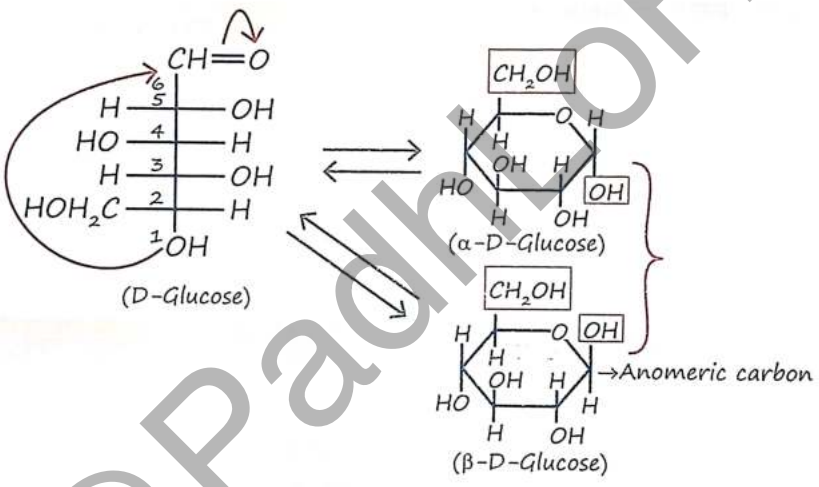
Cyclic structure of glucose, supported by the failure of tests like Schiff's and $NaHSO_3$ to detect the –CHO group, indicating it's not free. Glucose forms cyclic hemiacetal structures—mainly pyranose and less commonly furanose—confirming it exists in multiple forms in solution.

1. Schiff's test X ⟶ —CHO X
2. $NaHSO_3$ test X ⟶ —CHO X
3. Glucopentaacetate does not reacts with NH_2—OH.

Above three observation suggest that glucose exist in some other forms also. Glucose exist in three forms.

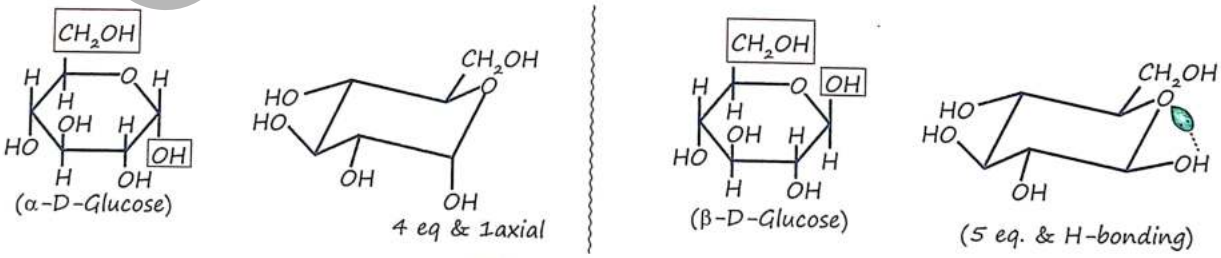
➢ Haworth projections

The α- and β- forms of glucose, which differ in the configuration around the anomeric carbon (C-1), are known as anomers. Glucose forms a six-membered cyclic structure called pyranose, resembling pyran, a ring with five carbon atoms and one oxygen. This cyclic form is best represented using the Haworth projection.



- Ring-chain tautomers
- Epimers
- Anomers
- Pyranose form

➢ Stability of α & β forms



β is more stable then α

α ⇌ Open ⇌ β

36% 0.2% 63.8%

Net rotation of glucose is found to be dextro (d) 52.5°.

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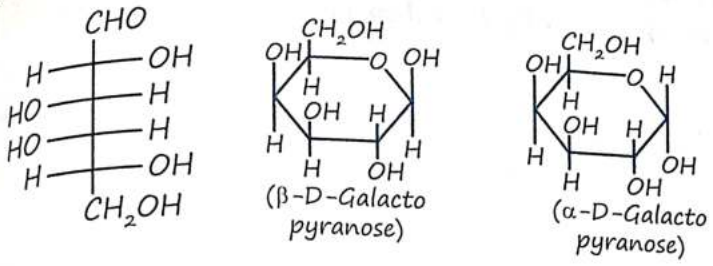
α, β Anomers of given sugar

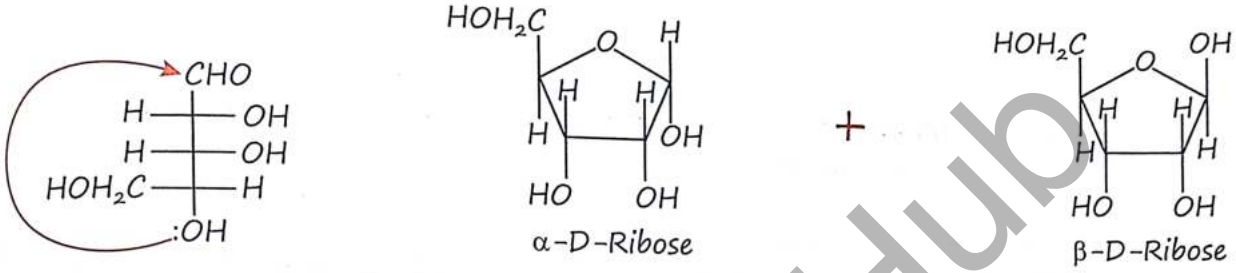
1. (a) D-Galactose (Pyranose)



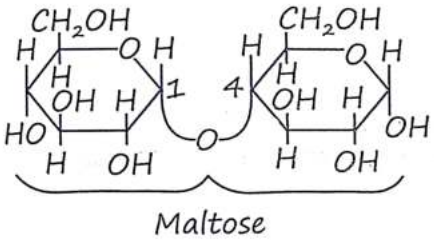
Try Yourself

Q. Drow α, β anomers of given sugar
(i) D-Fructose (Furanose)
(ii) D-Ribose (Furanose)

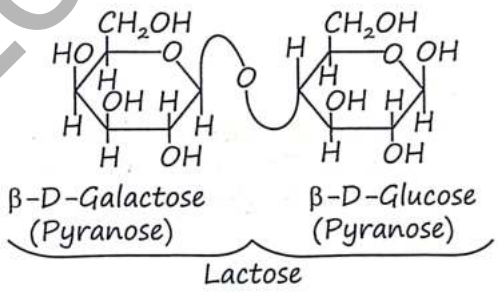
(b) D-Ribose (Furanose)



2. α-D-Glucose & α D-Glucose:

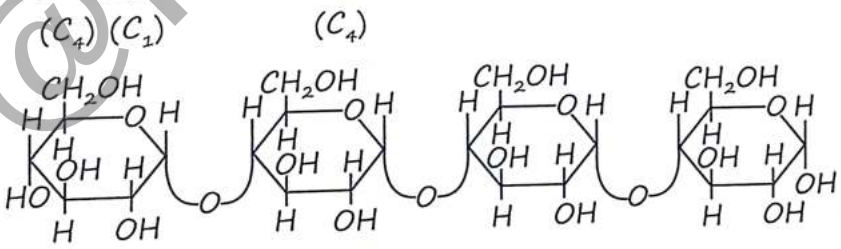


3. β-D-Galactose & β D-Glucose:



Combine the given Sugar:

4. α-D-Glucose & α D-Glucose & α D-Glucose:



Try Yourself

Combine the given Sugar:

(a) α-D-Glucose & β D-fructose:
(C1) (C2)

(b) α-D-Glucose & α D-Glucose & α D-Glucose:
(C1) (C4) (C1) (C6)

(c) β-D-Glucose & β D-Glucose & β D-Glucose:
(C1) (C4) (C1) (C4)

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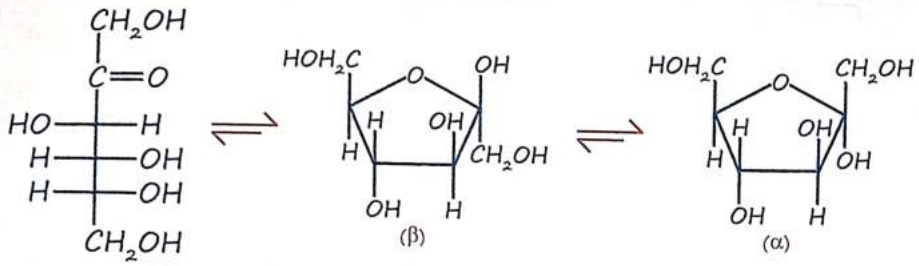
2. Fructose

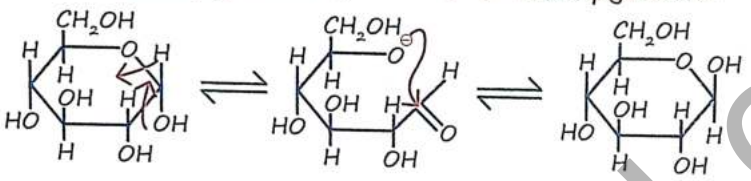
Structure of fructose, a ketohexose with the formula $C_6H_{12}O_6$. It is laevorotatory (−92°) and a reducing sugar with 3 chiral centers, giving rise to 8 stereoisomers. Fructose exists in cyclic forms—α and β furanose—formed by reaction between the ketone group and a hydroxyl group.



- Laevorotatory (−92°)
- $C_6H_{12}O_6$
- 3 Chiral center
- 8 Stereoisomers
- Reducing sugar

Mutarotation

Definition: Change in specific rotation with respect to time and acquired an equilibrium value of rotation in mild basic solution is known as mutarotation.



Examples

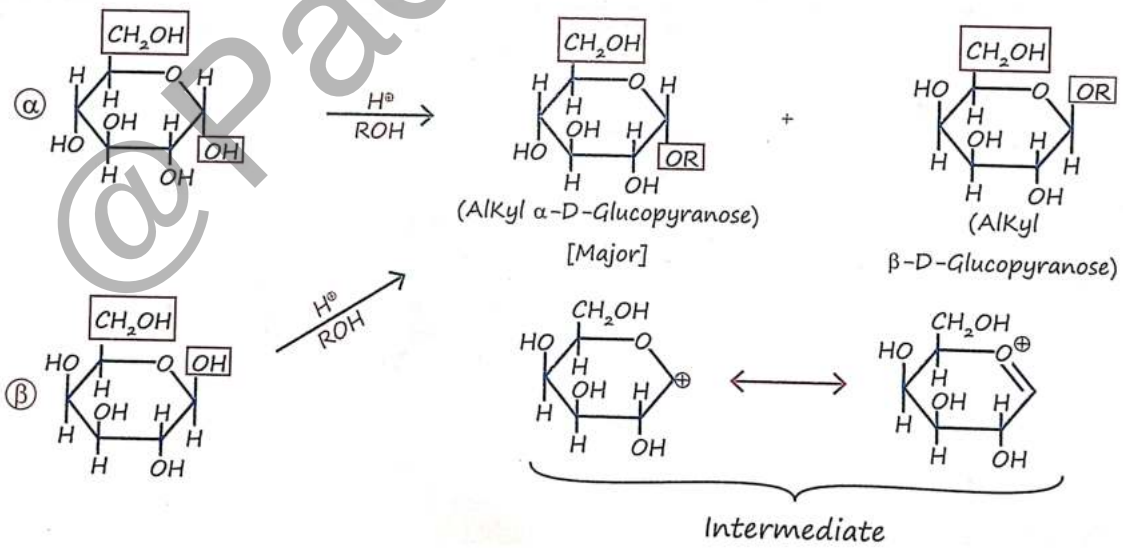
Ex: α-Glucose (x°) ⇌ Open chain (y°) ⇌ α-Glucose (z°)

t = 0	100% (x°)		
t = eq	36% (x°)	0.2%	63.8%

Net rotation: $\frac{36 \times x}{100} + \frac{0.2 \times y}{100} + \frac{63.8 \times z}{100} = \underbrace{+52.5°}_{\text{D-Glucose}}$

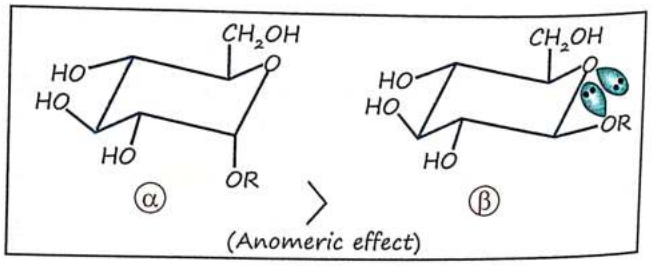
Glycoside

When glucose reacts with alcohol (ROH) in the presence of an acid catalyst (H^+), it forms alkyl α- and β-D-glucopyranosides. This reaction occurs at the anomeric carbon and prevents mutarotation, leading to the formation of stable glycosidic bonds.



Anomeric effect

The anomeric effect is a stereoelectronic phenomenon observed in cyclic sugars (especially in pyranose forms), where the axial position of a substituent on the anomeric carbon is favored over the equatorial one, despite being less sterically favorable.



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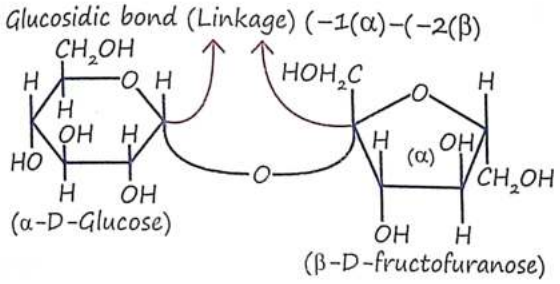
Disaccharides

Disaccharides are carbohydrates made by joining two monosaccharide (simple sugar) units through a glycosidic bond.

1. Sucrose

Sucrose is a disaccharide composed of glucose and fructose linked via a glycosidic bond between the α-anomeric carbon of glucose (C_1) and the β-anomeric carbon of fructose (C_2).

It is non-reducing because both anomeric carbons are involved in the bond. Sucrose is commonly known as table sugar and is naturally found in fruits and plants.

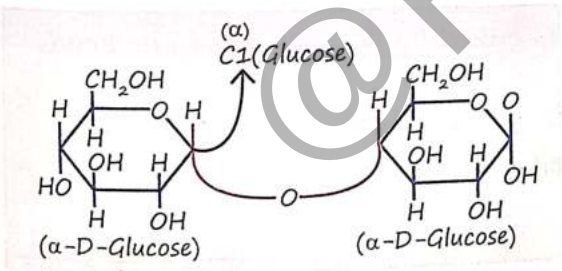


- Sucrose
- Dextrorotatory
- 1 Acetal and 1 ketal present
- Non-Reducing sugar
- Cane sugar
- 9 chiral center
- Both anomeric used (C_1–C_2)
- Sucrose $\xrightarrow{H_2O}$ D-(+)-Glucose (+52°) + D-(–)-Fructose (–92°) (Invert sugar)
- Net rotation = –20°

Invert sugar is a mixture of glucose and fructose made by breaking down sucrose. It's sweeter than regular sugar and often used in syrups, candies, and beverages.

2. Maltose

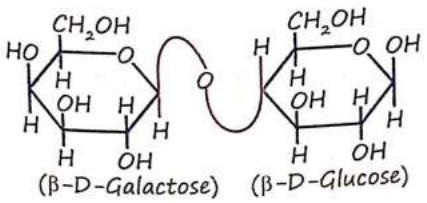
- Disaccharide of two glucose units.
- Linked by an α(1 → 4) glycosidic bond.
- It is a reducing sugar because the second glucose has a free anomeric carbon.



- Maltose
- Anomeric of Glucose is free
- 1 acetal and 1 hemiacetal
- Reducing sugar
- 10 chiral center

3. Lactose

- Disaccharide of glucose and galactose.
- Linked by a β(1→4) glycosidic bond.
- Also a reducing sugar, as glucose has a free anomeric carbon.



- Lactose
- Milk sugar
- 10 chiral centre
- Reducing sugar
- 1 hemiacetal and 1 Acetal

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◆ Polysaccharides

Carbohydrates which yield a large number of monosaccharide units on hydrolysis are called polysaccharides.
Some common examples are starch, cellulose, glycogen, gums, etc.

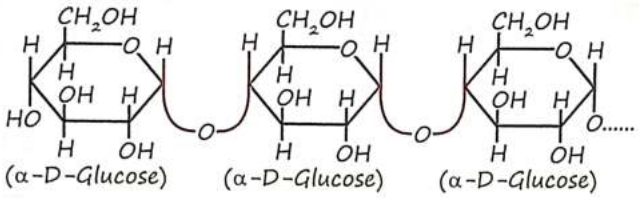
1. Starch

Starch is a polysaccharide made up of glucose units and serves as the main storage carbohydrate in plants.
It consists of two components:

1. Amylose – unbranched, with α(1→4) linkages.
2. Amylopectin – branched, with α(1→4) and α(1→6) linkages at branch points.

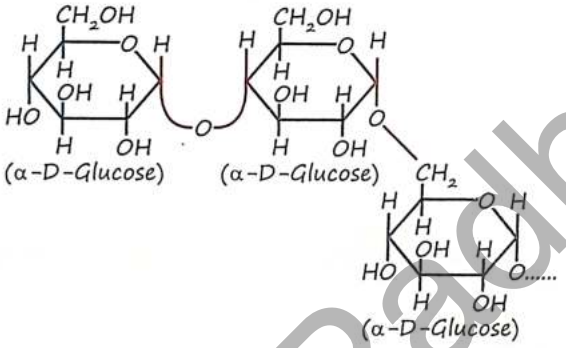
Starch is insoluble in cold water and gives a blue color with iodine (due to amylose).

(a) Amylose



- Starch
- Amylose
- Linear chain
- 15-20% present
- Water soluble

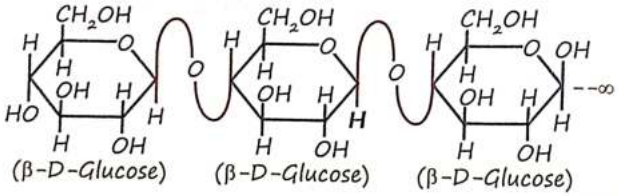
(b) Amylopectin



- Starch
- Amylopectin
- Branched
- 80-85%
- Water insoluble

2. Cellulose

Cellulose is a structural polysaccharide made of β-D-glucose units linked by β(1→4) glycosidic bonds.
It is the main component of plant cell walls.
Forms long, unbranched chains held together by hydrogen bonds, giving it high tensile strength.
Humans cannot digest cellulose due to the lack of the enzyme cellulase.



- Cellulose
- Plants

3. Glycogen

Glycogen is the storage polysaccharide in animals, especially found in the liver and muscles.
It is made of α-D-glucose units.
Has a structure similar to amylopectin but is more highly branched, with α(1→4) and α(1→6) glycosidic bonds.
It is quickly broken down to release glucose when the body needs energy.
Hence, glycogen is often called animal starch.

- Animal starch
- Found in yeast, fungi
- Muscles, Brain
- Cross linked of amylopectin

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Summery

Disaccharides

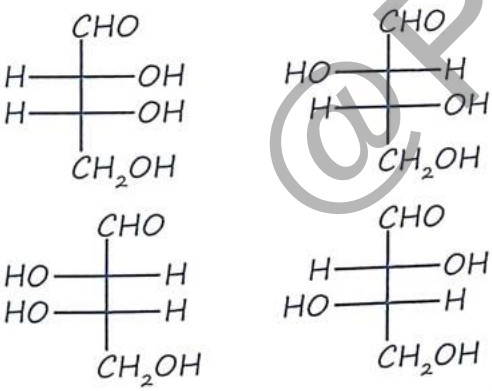
1. $\alpha(C_1) + \beta(C_2) \longrightarrow$ Sucrose (Non-Reducing)
   (Glu) (Fru)
2. $\alpha(C_1) + \alpha(C_4) \longrightarrow$ Maltose (Reducing)
   (Glu) (Glu)
3. $\beta(C_1) + \beta(C_4) \longrightarrow$ Lactose (Reducing)
   (Gal) (Glu)

Polysaccharides

1. α-Glu + α-Glu ⟶ Amylose
   (C1) (C4)
2. α-Glu + α-Glu (C1) (C4) + α-Glu + α-Glu (C1) (C6) ⟶ Amylopectin } Starch (Human)
3. β-Glu + β-Glu ⟶ Cellulose (Plants)
   (C1) (C4)
4. More branched amylopectin ⟶ Glycogen (Animals) (Found in Liver Brain, found in Yeast and Fungi)

Previous Year Questions

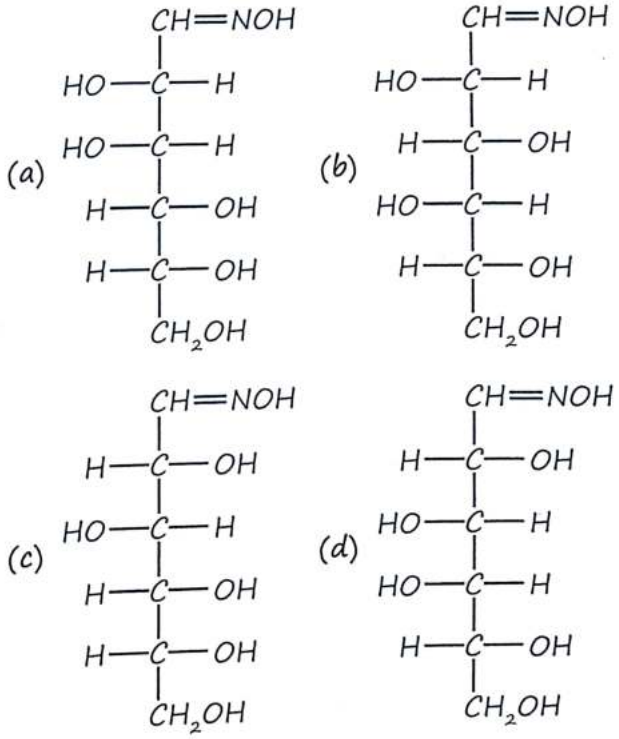
1. The correct corresponding order of names of four aldoses with configuration given below: (2016 - II)



(a) L-erythrose, L-threose, L-erythrose, D-threose
(b) D-threose, D-erythrose, L-threose, L-erythrose
(c) L-erythrose, L-threose, D-erythrose, D-threose
(d) D-erythrose, D-threose, L-erythrose, L-threose

Sol. (d)

2. D(+) glucose reacts with hydroxyl amine and yields an oxime. The structure of the oxime would be: (2014)



Sol. (c)

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3. Which one of the following statements is not true regarding (+) lactose? (2011 Pre)

(a) On hydrolysis (+) lactose gives equal amount of D (+) glucose and D (+) galactose

(b) (+) lactose is a β-glucoside formed by the union of a molecule of D (+) glucose and a molecule of D (+) galactose

(c) (+) lactose is a reducing sugar and does not exhibit mutarotation

(d) (+) lactose, $C_{12}H_{22}O_{11}$ contains 8 -OH groups

Sol. (c)

4. Fructose reduces Tollen's reagent due to: (2010 Mains)

(a) Asymmetric carbons

(b) Primary alcoholic group

(c) Secondary alcoholic group

(d) Enolisation of fructose followed by conversion to aldehyde by base

Sol. (d)

5. Glucose molecule reacts with 'X' number of molecules of phenylhydrazine to yield osazone. The value of 'X' is: (1998)

(a) Four (b) One

(c) Two (d) Three

Sol. (d)

6. The α-D-glucose and β-D-glucose differ from each other due to difference in carbon atom with respect to its: (1995)

(a) Number of OH groups

(b) Size of hemiacetal ring

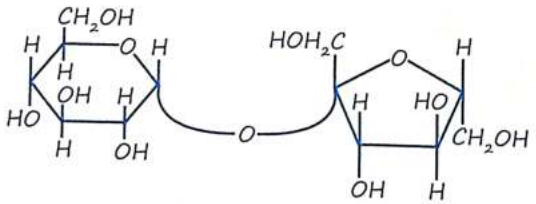
(c) Conformation

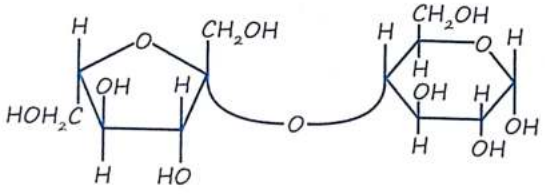
(d) Configuration

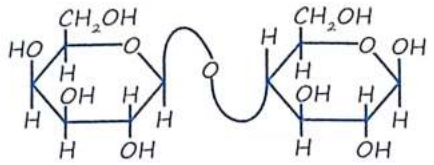
Sol. (d)

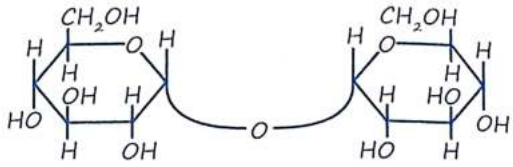
7. A disaccharide X cannot be oxidised by bromine water. The acid hydrolysis of X leads to a laevorotatory solution. The disaccharide X is

C-20.13 W-28.08 UA-51.79 (JEE Adv. 2023)

(a) 

(b) 

(c) 

(d) 

Sol. (*)

8. For 'invert sugar', the correct statement(s) is (are) (Given: specific rotations of (+) - sucrose, (+) - maltose, L - (-) - glucose and L-(+)-fructose in aqueous solution are +66°, +140°, -52° and +92°, respectively)

C-6.19 W-37.26 UA-37.24 PC-19.31 (JEE Adv. 2016)

(a) Invert sugar is prepared by acid catalyzed hydrolysis of maltose.

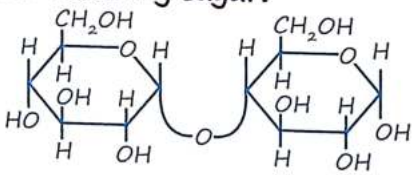
(b) Invert sugar is an equimolar mixture of D-(+) -glucose and D-(-)- fructose.

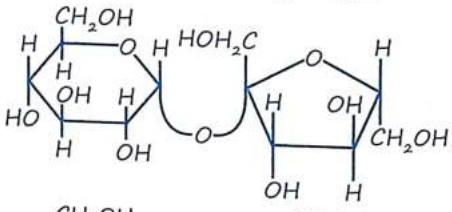
(c) Specific rotation of invert sugar is -20.

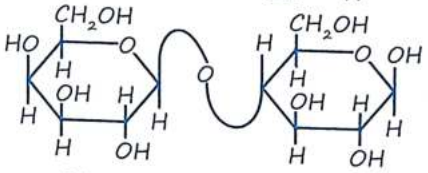
(d) On reaction with Br_2 water, invert sugar forms saccharic acid as one of the products.

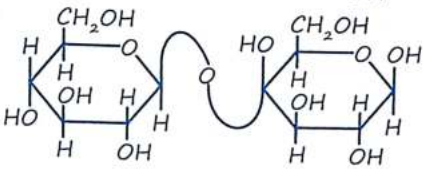
Sol. (*)

9. In disaccharides. if the reducing groups of monosaccharides i.e. aldehydic or ketonic groups are bonded, these are non-reducing sugars. Which of the following disaccharide is a non-reducing sugar?

(a) 

(b) 

(c) 

(d) 

Sol. (*)

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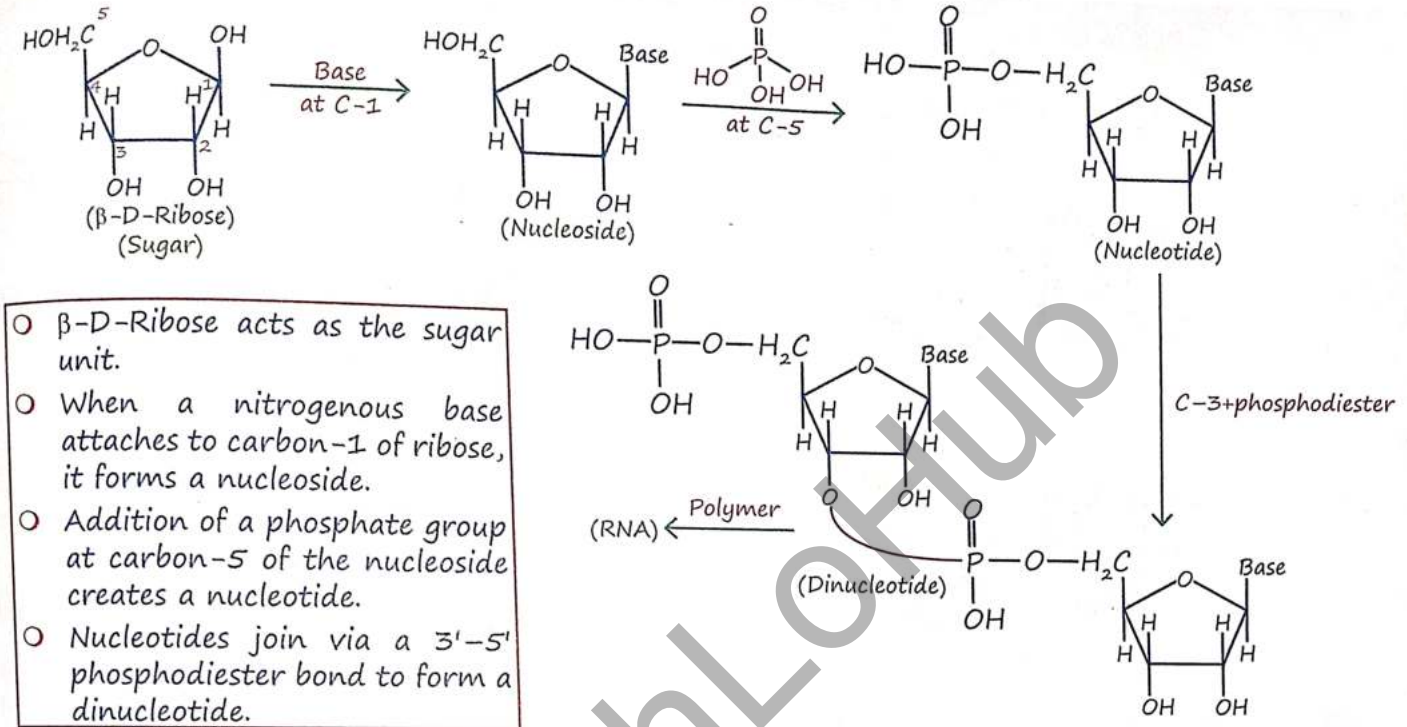
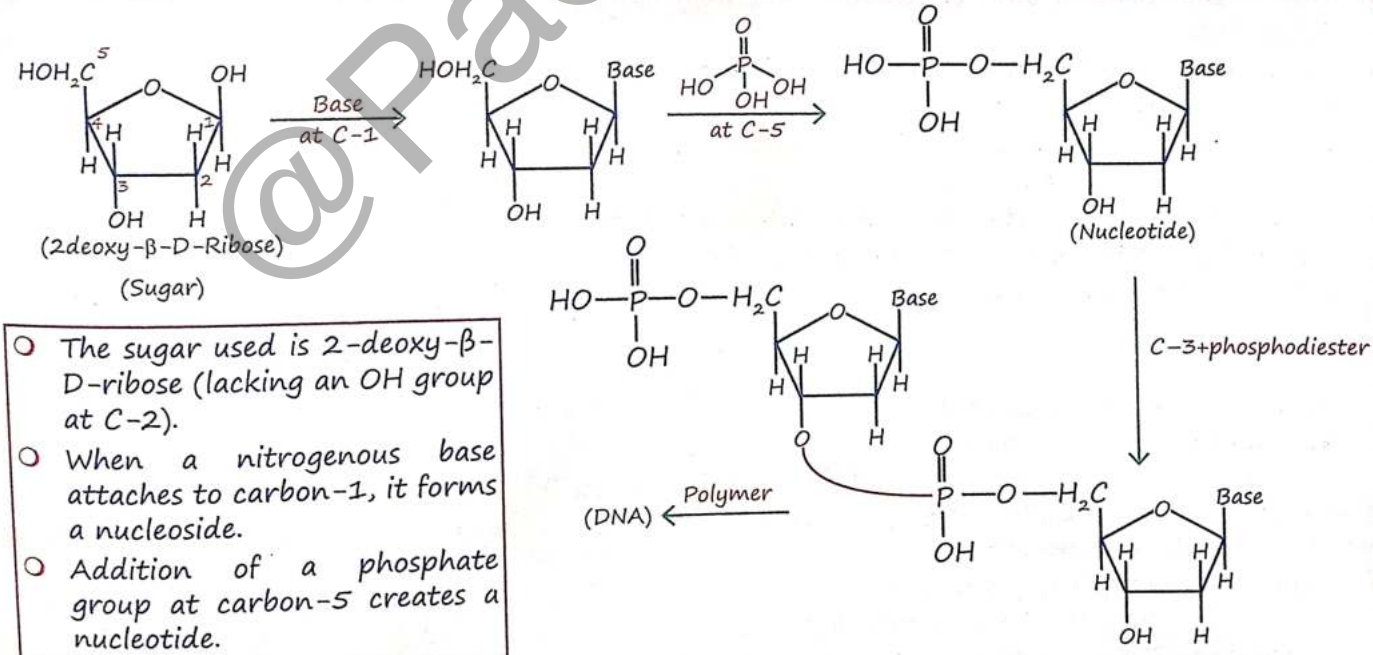
Nucleic Acid

RNA (Ribonucleic Acid)



- β-D-Ribose acts as the sugar unit.
- When a nitrogenous base attaches to carbon-1 of ribose, it forms a nucleoside.
- Addition of a phosphate group at carbon-5 of the nucleoside creates a nucleotide.
- Nucleotides join via a 3'–5' phosphodiester bond to form a dinucleotide.

DNA (Deoxyribonucleic Acid)



- The sugar used is 2-deoxy-β-D-ribose (lacking an OH group at C-2).
- When a nitrogenous base attaches to carbon-1, it forms a nucleoside.
- Addition of a phosphate group at carbon-5 creates a nucleotide.
- Nucleotides link via a 3'–5' phosphodiester bond to form DNA polymer.

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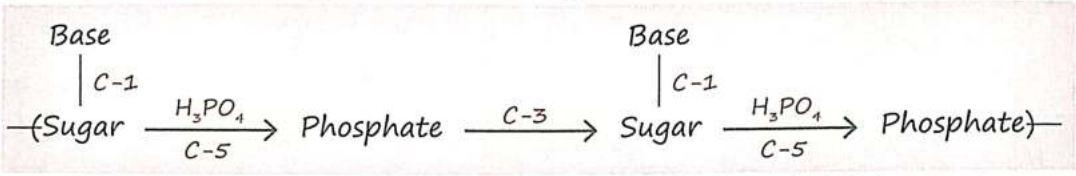
Dinucleotide

Structure of a nucleic acid (DNA or RNA), showing repeating units of sugar and phosphate linked by phosphodiester bonds. Each sugar is attached to a nitrogenous base at carbon-1 (C-1) and a phosphate group at carbon-5 (C-5), continuing with the next sugar at carbon-3 (C-3).



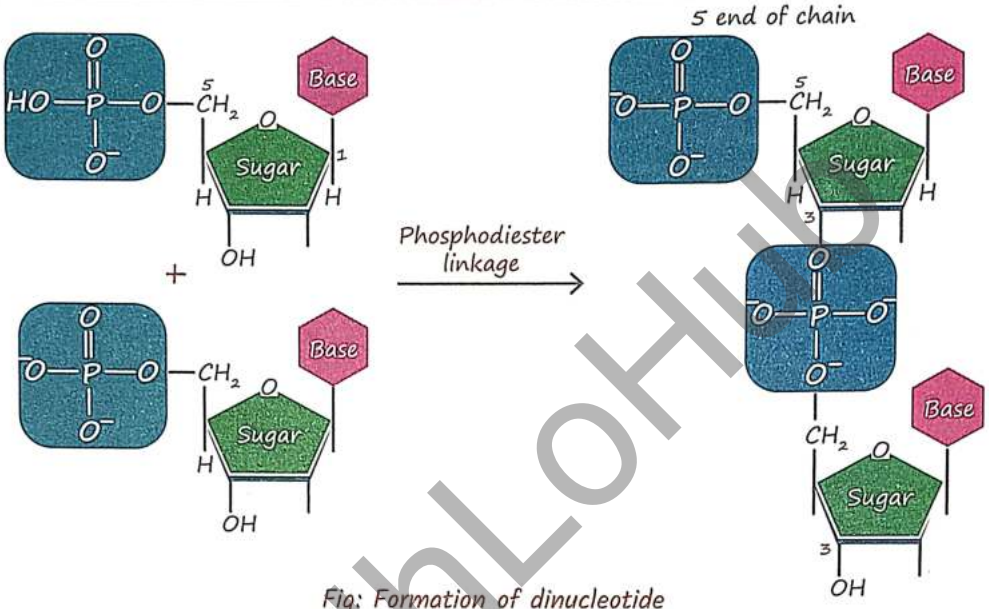


Fig: Formation of dinucleotide

Biological Function of Nucleic Acid

- DNA is the chemical basis of heredity and may be regarded as the reserve Functions of genetic information.
- DNA is exclusively responsible for maintaining of Nucleic the identity of different species of organisms over millions of years.
- DNA molecule is capable of self duplication during cell division and identical DNA strands are transferred to daughter cells.
- Another important function of nucleic acids is the protein synthesis in the cell. Actually, the proteins are synthesized by various RNA molecules in the cell but the message for the synthesis of a particular protein is present in DNA.

RNA

- Structure Single stranded helix.
- RNA molecule are of 3 types
  1. Messenger RNA [m-RNA] 2. Ribosomal RNA [r-RNA] 3. Transfer RNA [t-RNA]

Biological Functions of Nucleic acids: DNA is the chemical basis of heredity and may be regarded as the reserve of genetic information.

- **Duplication during Cell Division:**
  - DNA can replicate itself during cell division.
  - Identical DNA strands are passed on to daughter cells.
- **Protein Synthesis:**
  - RNA molecules synthesize proteins in the cell.
  - DNA carries the instructions (message) for synthesizing specific proteins.

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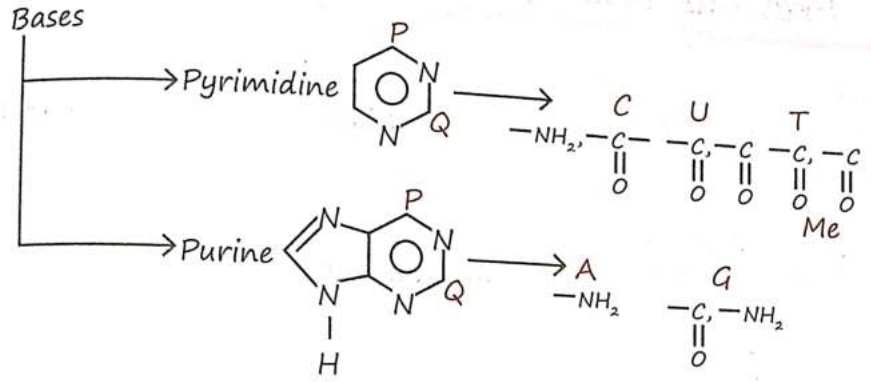
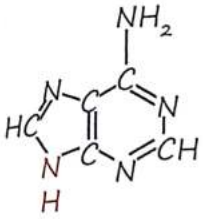
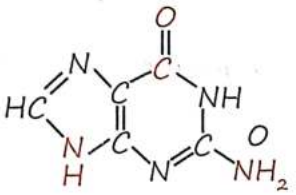
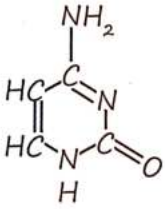
Bases

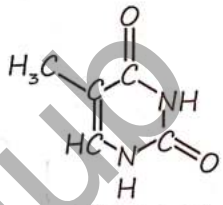


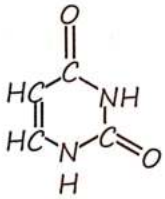
- DNA – AGCT
- RNA – AGCU


Adenine (A)


Guanine (G)


Cytosine (C)


Thymine (T)


Uracil (U)

Bonding between Bases

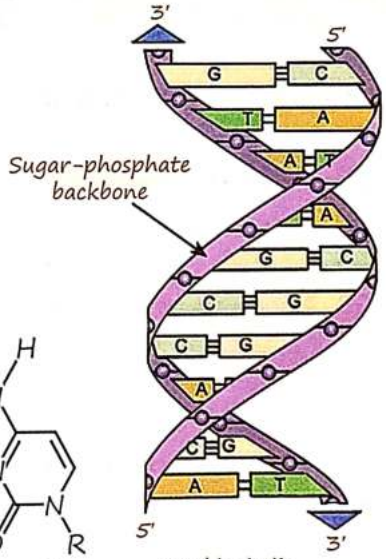
complementary

- A = T → 2-H bonding
- G ≡ C → 3-H bonding

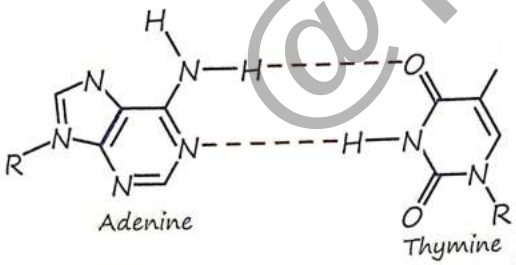
} DNA

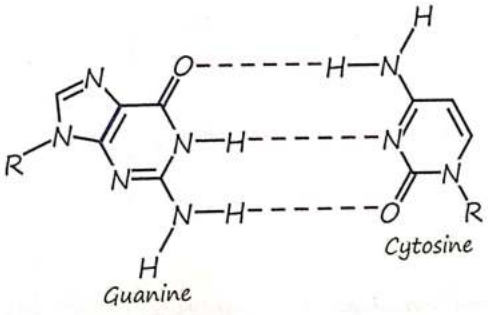
- H-bonding
- Conversion of code

Conversion of code
DNA ⇒ AGC TAC TGC
RNA ⇒ UCG AUG ACG



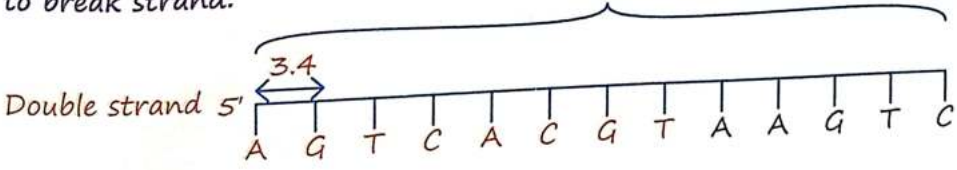
Double-helix represenation


Adenine
Thymine


Guanine
Cytosine

Practice Questions

1. Length of one trand = YÅ if distance between continuous pair is 3.4Å and find total energy required to break strand.



[Given: Average energy per H-bond for A-T base pair = 1.0 kcal mol^{-1}, G-C base pair = 1.5 kcal mol^{-1}, and A-U base pair = 1.25 kcal mol^{-1}. Ignore electrostatic repulsion between the phosphate groups.]

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10. Given below are two statements: (2023)

Statement-I: A unit formed by the attachment of a base to 1' position of sugar is known as nucleoside

Statement-II: When nucleoside is linked to phosphorous acid at 5'-position of sugar moiety, we get nucleotide.

In the light of the above statements, choose the correct answer from the options given below:

(a) Statement-I is false but Statement-II is true.

(b) Both Statement-I and Statement-II are true.

(c) Both Statement-I and Statement-II are false.

(d) Statement-I is true but Statement-II is false.

Sol. (d) When nucleoside is linked to phosphoric acid at 5'-position of sugar moiety, we get nucleotide.

11. In DNA, the complimentary bases are: (2008)

(a) Uracil and adenine; cytosine and guanine

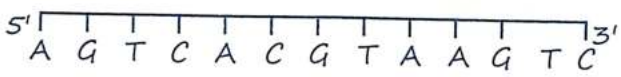
(b) Adenine and thymine; guanine and cytosine

(c) Adenine and thymine; guanine and uracil

(d) Adenine and guanine; thymine and cytosine

Sol. (b) Complimentary base pairing in DNA takes place between C ≡ G with 3 H bonds & A = T with 2 H bonds.

12. For a double strand DNA, one strand is given below:



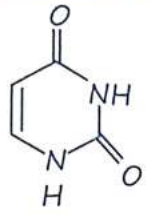
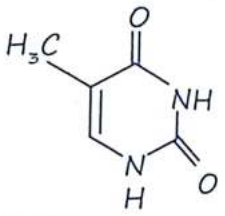
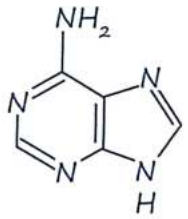
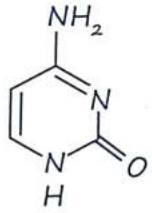
The amount of energy required to split the double strand DNA into two single strands is ____ kcal mol^{-1}.

[Given: Average energy per H-bond for A-T base pair = 1.0 kcal mol^{-1}, G-C base pair = 1.5 kcal mol^{-1}, and A-U base pair = 1.25 kcal mol^{-1}. Ignore electrostatic repulsion between the phosphate groups.]

C-4.83 W-71.7 UA-23.48 (JEE Adv. 2024)

Sol. [4]

13. Match List-I with List-II.

List-I		List-II	
A.	Adenine	I.	
B.	Cytosine	II.	
C.	Thymine	III.	
D.	Uracil	IV.	

Choose the correct answer from the options given below: [24 Jan, 2025 (Shift-II)]

(a) (A)-(III), (B)-(IV), (C)-(I), (D)-(II)

(b) (A)-(IV), (B)-(III), (C)-(II), (D)-(I)

(c) (A)-(III), (B)-(IV), (C)-(II), (D)-(I)

(d) (A)-(III), (B)-(I), (C)-(IV), (D)-(II)

Sol. (c)

14. Sugar moiety in DNA and RNA molecules respectively are [29 June, 2022 (Shift-I)]

(a) β-D-2-deoxyribose, β-D-deoxyribose

(b) β-D-2-deoxyribose, β-D-ribose

(c) β-D-ribose, β-D-2-deoxyribose

(d) β-D-deoxyribose, β-D-2-deoxyribose

Sol. (b) RNA contains b-D- ribose, while DNA possess b-D-2- deoxyribose sugar.

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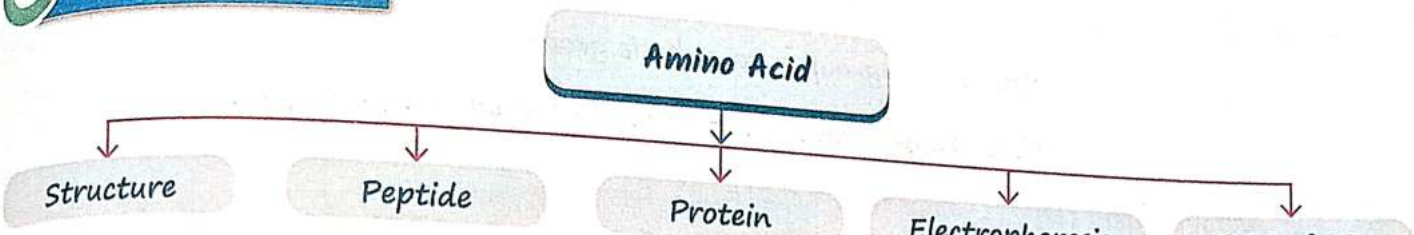
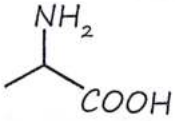
Amino Acid

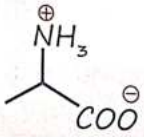


Amino Acid

Compound in which carboxylic acid as well as amine functional group is present are known as amino acid.



Crystalline solid
M.P. Very high



H_2O soluble
Amphoteric in nature

I. Classification

α-Amino Acid

- **Essential Amino Acid** → Which does not synthesize in human body should be taken from outside through diet.
- **Non-Essential Amino Acid** → Which are synthesize in human body does not required form outside.

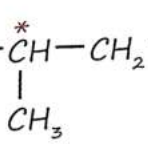
- Most abundant biomolecule of living system.
- All proteins are polymers of α-amino acid
- Amino acid ≥ 100 unit ⇒ Polypeptide
- Amino acid < 100 ⇒ Proteins
- 51 amino acid ⇒ Insulin

G - Glycine —H (G)

A - A' amine $—CH_3$ (A)

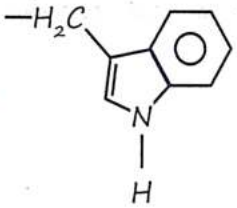
V - Valine (V)

L - Leucine $—CH_2$ (L)

I - Isoleucine  (I)

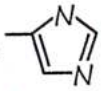
P - Phenylalanine $—CH_2Ph$ (F)

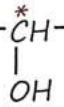
M - Methionine $—CH_2CH_2SMe$ (M)

T - Tryptophan  (W)

Ly - Lysine $—(CH_2)_4—NH_2$ (K)

Ar - Arginine $—(CH_2)_3$ — Guanidine (R)

His - Histidine  (Imidazole) (H)

Thero - Theronine  (T)

Baba Tea-Stall (BTS)

(i) Obtained by hydrolysis of proteins (α-amino acid)
(ii) Glycine ⇒ Sweet taste, Tyrosine ⇒ cheese
(iii) Usually colorless and crystalline
(iv) High melting point and water soluble
(v) Most naturally occuring amino acid exist in L-form
(vi) All amino acid are optically active except Glycine
(vii) All are Essential amino acid except G and A

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2. Classification based on no. of Acidic and Basic Groups

(a) **Neutral amino acid:** Total acidic group = total basic group

(b) **Acidic amino acid:** Acidic group > basic group ⇒ toward anode on electrophoresis

Ex: Aspartic acid (D) Glutamic acid (E)

($-CH_2COOH$) ($-CH_2CH_2COOH$)

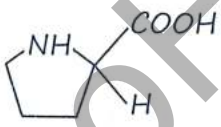
(c) **Basic amino acid:** Acidic group < basic group ⇒ towards cathode on electrophoresis

Ex: Lysine Arginine Histidine

$-(CH_2)_4NH_2$ $-(CH_2)_3$Guanidine $-CH_2$-Imidazole

Baba Tea-Stall (BTS)

(viii) pH values are: Arg > Lys > Hist

(ix) All are primary amino acid except proline 

(x) Amides of aspartic and glutamic acid are:

Asparagine ($-CH_2CONH_2$) (N) and Glutamine ($-CH_2CH_2CONH_2$) (Q)

(xi) Only given amino acid contains two chiral centre:

1. Isoleucine: $-\overset{*}{C}H(CH_3)-CH_2-CH_3$

2. Threonine: $-\overset{*}{C}H(OH)-CH_3$

(xii) Amino acid and functional group

Threonine (T) → 2° —OH ($-\overset{*}{C}H(OH)-CH_3$)

Serine (S) → 1° —OH ($-CH_2OH$)

Tyrosine (Y) → Phenol ($-CH_2PhOH$)

Methionine (M) → Thioether ($-(CH_2)_2SCH_3$)

Cysteine (C) → Thiol ($-CH_2SH$)

Cystine → Disulphide ($-CH_2-(S)_2-CH_2-$)

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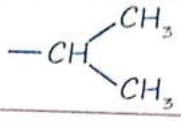
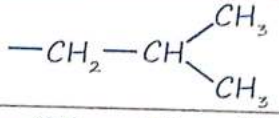
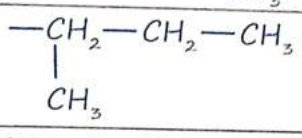
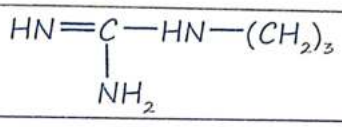
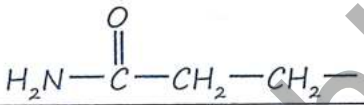
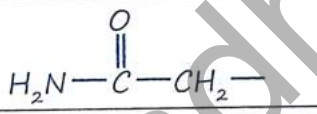
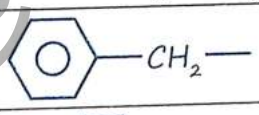
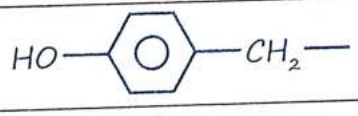
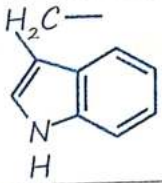
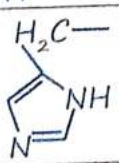
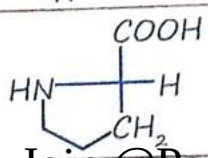
S. No.	Amino acids	Structures	3 Letter abbreviation	1 Letter abbreviation
1.	Glycine	—H	Gly	G
2.	Alanine	$-CH_3$	Ala	A
3.	Valine*		Val	V
4.	Leucine*		Leu	L
5.	Isoleucine*		ILe	I
6.	Arginine*		Arg	R
7.	lysine*	$H_2N-(CH_2)_4-$	Lys	K
8.	Glutamic acid	$HOOC-CH_2-CH_2-$	Glu	E
9.	Aspartic acid	$HOOC-CH_2-$	Asp	D
10.	Glutamine		Gln	Q
11.	Asparagine		Asn	N
12.	Threonine*	$H_3C-CHOH-$	Thr	T
13.	Serine	$HO-CH_2-$	Ser	S
14.	Cysteine	$HS-CH_2-$	Cys	C
15.	Methionine*	$H_3C-S-CH_2-CH_2-$	Met	M
16.	Phenylalanine		Phe	F
17.	Tyrosine		Tyr	Y
18.	Tryptophan*		Trp	W
19.	Histidine*		His	H
20.	Proline		Pro	P

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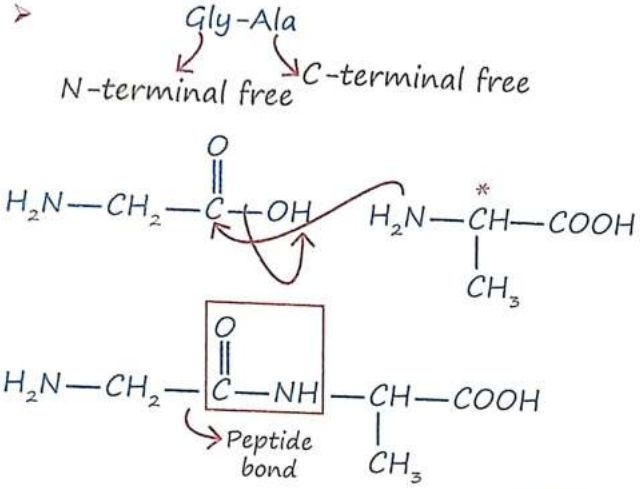
Protein

Definition: Polymer of α-amino acid

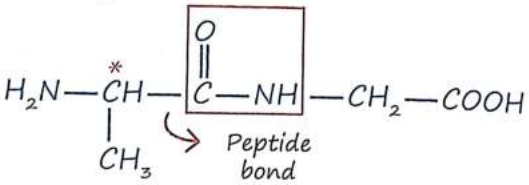
Peptide: When two or more similar or different α-amino acids combines by removal of water molecule then formation of amide linkage takes place and this amide linkage in known as peptide linkage or peptide bond.

Example:

➢ Gly-Ala



➢ Ala-Gly



Practice Questions

Calculation of Peptide

2. Glycine, alanine
   (a) Dipeptide: _____
   (b) Tripeptide: _____
   (c) Decapeptide: _____
   (d) Cyclic tripeptide: _____

3. G, A, V, F
   (a) Tetrapeptide: _____
   $\downarrow H_2O$
   G, A, V, F
   (b) Same as above: NH_2 is chiral centre _____

* Same as (b) with on partial hydrolysis GA never obtained

_____

* Same as (a) with on partial hydrolysis GA never obtained

4. Find total chiral centre in:
   (a) Gly-Ala-Val-Thr-His Ile-Leu
   (b) V L D T Y I F A Q E
   (c) Alacyl Phenylalacyl Leucyl thereocyl Glycyl Histidine

5. Find total Sp^2 Carbon in
   (a) Gly-Ala-Val-Thr-His Ile-Leu
   (b) V L D T Y I F A Q E
   (c) Alacyl Phenylalacyl Leucyl thereocyl Glycyl Histidine

6. Find total $>C=O$ group
   (a) Gly-Ala-Val-Thr-His Ile-Leu
   (b) V L D T Y I F A Q E
   (c) Alacyl Phenylalacyl Leucyl thereocyl Glycyl Histidine

7. Find total p bond $e^{\ominus}$ in:
   (a) Gly-Ala-Val-Thr-His Ile-Leu
   (b) V L D T Y I F A Q E
   (c) Alacyl Phenylalacyl Leucyl thereocyl Glycyl Histidine

8. Find the major product
   (a) Asn-Tyr $\xrightarrow[(excess)]{Ac_2O/Py}$
   (b) Ser-Lys $\xrightarrow[(excess)]{Ac_2O/Py}$
   (c) D-E $\xrightarrow[(excess)]{NaHCO_3}$?? x mole $CO_2\uparrow$

9. No. of peptide linkage in a octapeptide which on reaction $NaNO_2/HCl$ doesn't produce N_2 gas.

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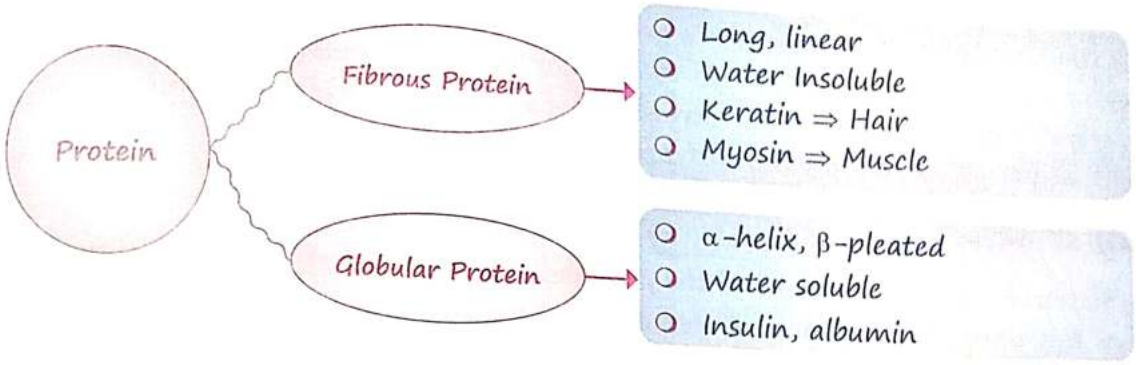
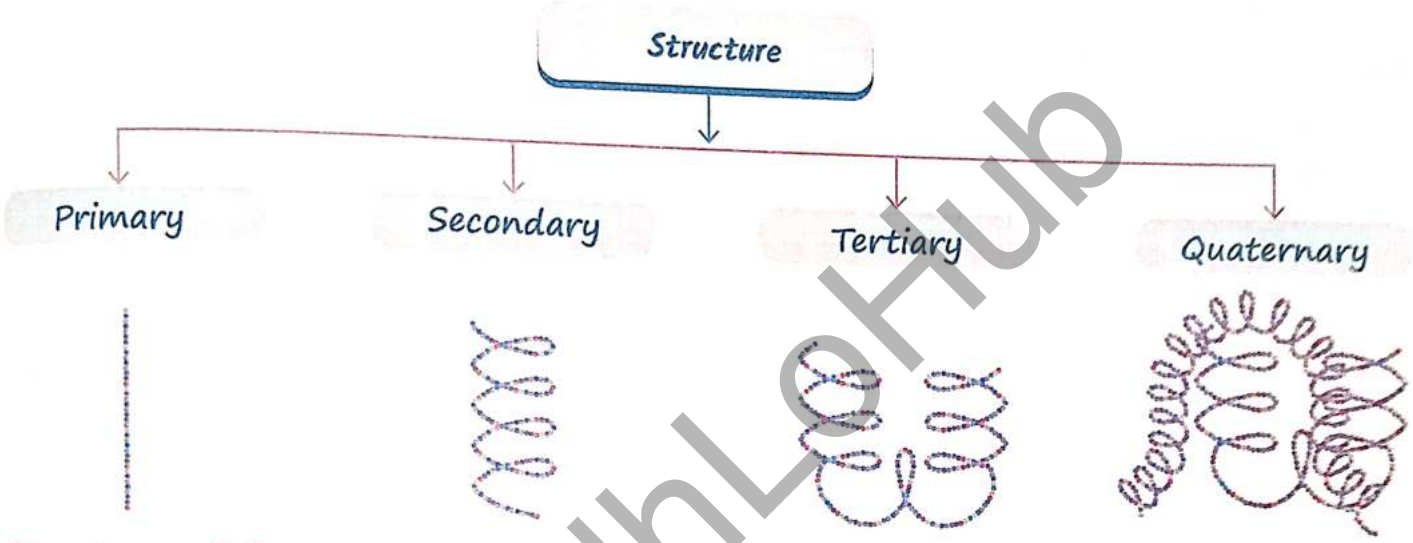
Classification of Protein





Structure of Proteins

Depending on the complexity, the structure of proteins may be distinctly studied as primary structure, secondary structure and tertiary structures.

Primary Structure

Primary Structure of Protein Each polypeptide chain in a protein has amino acids linked with each other in a specific sequence and it is this sequence of amino acids that is said to be the primary structure of that protein. Any change in this primary structure i.e., the sequence of amino acids creates a different protein.

Secondary Structure

The secondary structure of a protein refers to the shape in which a long polypeptide chain can exist. There are two different conformations of the peptide linkage present in proteins viz. α-helix and β-pleated sheet structure. These structures arise due to the regular folding of the backbone of the polypeptide chain due to hydrogen bonding between C=O and –NH– groups of the peptide bond.

- α-Helix

α-Helix is one of the most common ways in which a polypeptide chain forms all possible hydrogen bonds by twisting into a right handed screw (helix) with the –NH group of each amino acid residue hydrogen bonded to the C=O of an adjacent turn of the helix.

- β - pleated

In β - pleated sheet structure all peptide chains are stretched out to nearly maximum extension and laid side by side, which are held together by inter molecular hydrogen bonds. These structures resembles the pleated folds of drapery.

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Tertiary Structure

The tertiary structure of proteins represents overall folding of the polypeptide chains i.e., further folding of the secondary structure. It gives rise to two major molecular shapes viz. fibrous and globular. The main forces which stabilise the 2° and 3° structures of proteins are hydrogen bonds, disulphide linkages, Vander Waals and electrostatic forces of attraction.

Quaternary structure

Quaternary structure of proteins Some of the proteins are composed of two or more polypeptide chains referred to as sub-units. The spatial arrangement of these subunits with respect to each other is known as quaternary structure.

Denaturation of Protein

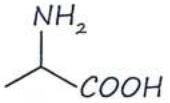
1. Every protein present in its native form.
2. Every protein has its own biological function.
3. If any physical or chemical change like change in pH or temperature denatured the structure of protein as H-bonding disturbs changes its functionality is known as denaturation of protein.
4. Secondary and tertiary structures breaks upto primary structure but primary structure remains same during denaturation of protein.
5. The coagulation of egg white on boiling is a common example of denaturation. Another example is curdling of milk which is caused due to the formation of lactic acid by the bacteria present in milk.

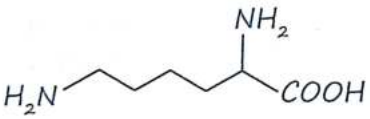
Electrophoresis

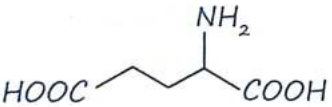
- Point at which no movement towards cathode or anode then that point is known as Isoelectric point (no effect of electric field).
- pI $\Rightarrow [\text{cations}]_{max} = [\text{anion}]_{max}$

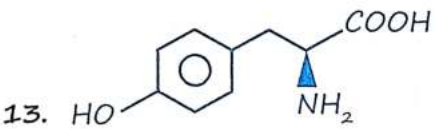
Practice Questions

Find PI for the given compounds in aqueous solution:

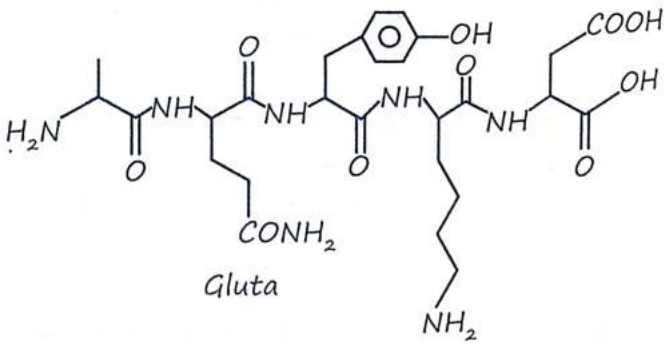
10. 
actual pKa = 2.34, 9.69

11. 
pKa = 2.18, 8.95, 10.79

12. 
pKa = 2.19, 9.67, 4.25

13. 
pKa = 2.20, 9.11, 10.07

14. **Find net charge in the given compound**


Gluta

pH = 2, 7, 12

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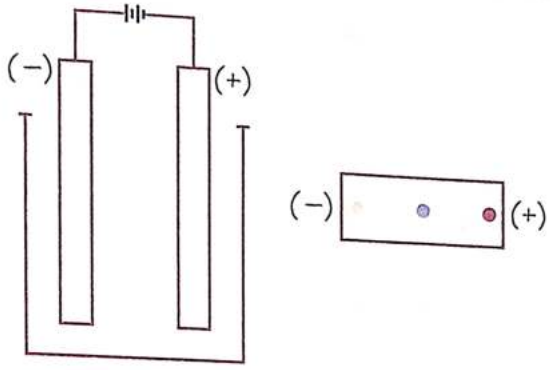
15. On electrophoresis of solution of (G), (H) and (I) at pH = 5, separated compound represented by (·). Find (G), (G) and (I).



16. Isobutane $\xrightarrow[(3)\ HBr/(PhCOO)_2\ (4)\ Mg/tHF\ (5)\ CO_2\ then\ H^{\oplus}]{(1)\ KMnO_4/KOH\ (2)\ H_2SO_4/\Delta}$ (P)

(P) $\xrightarrow[(2)\ H_2/Pd,\ BaSO_4]{(1)\ SOCl_2}$ (Q) $\xrightarrow[(2)\ H_3O^{\oplus}]{(1)\ NH_4Cl/KCN}$ (S) $\xrightarrow{CH_3OH/H^{\oplus}}$ (U)

(P) $\xrightarrow[(2)\ NH_3\ (2eq)]{(1)\ Red\ P/Cl_2}$ (R) $\xrightarrow{AC_2P/Py}$ (T)

Structure synthesis

[Natural form]

17. (T) + (U) $\xrightarrow[(2)\ Partial\ hydrolysis]{(1)\ H^{\oplus}}$ (V) + CH_3COOH + CH_3OH

18. (R) $\xrightarrow{HCl}$ (W) $\xrightarrow[Br_2]{Red\ P}$ (X)

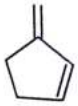
19. (R) $\xrightarrow{aq.\ NaHCO_3}$ (A) $\xrightarrow[CaO,\ \Delta]{NaOH}$ (B)

(R) $\xrightarrow[aq.\ HCl]{NaNO_2}$ (C) $\xrightarrow{\Delta}$ (D)

(R) $\xrightarrow{NOCl}$ (Y) $\xrightarrow[(2)\ x^{\oplus}]{(1)\ aq.\ KOH}$ (Z)

20. (P) $\xrightarrow[(2)\ Cr_2O_3,\ 770\ K\ (3)\ KHnO_4,|Koss,\ \Delta\ (4)\ x^{\oplus}]{(1)\ NaOH,\ Electrolysis}$ (E) (Shows heating effet) $\xrightarrow[(2)\ Na/liq.NH_3\ (3)\ O_3/H_2O\ (4)\ EtoH\ (excess)/TSOH\ (5)\ Br_2]{(1)\ NaOH,\ CaO,\ \Delta}$ (F) $\xrightarrow[(2)\ \Delta\ (3)\ DMSO/NaHCO_3\ (4)\ NH_3\ (5)\ NaBH_4/NCOH]{(1)\ H_3O^{\oplus}(-EtoH)}$ (G) (Koop synthesis)

21. (E) $\xrightarrow[(2)\ KOH/EtoH\ (3)\ (F)]{(1)\ NH_3,\ \Delta}$ $\xrightarrow[(2)\ 4\text{-Bromo-1-butaramine}\ (3)\ HCl.H_2O/\Delta]{(1)\ KOH/EtoH}$ (E) + (H)

22.  $\xrightarrow[(2)\ NH_3\ (excess)/PTSA\ (3)\ H_2,\ pd/C]{(1)\ O_3/H_2O}$ (I) (Reductive amination)

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15. Which of the statements about "Denaturation" given below are correct? (2011 Mains)

Statements:

(A) Denaturation of proteins causes loss of secondary and tertiary structures of the protein

(B) Denaturation leads to the conversion of double strand of DNA into single strand

(C) Denaturation affects primary structure which gets distorted

(a) (A) and (B) (b) (A), (B) and (C)

(c) (B) and (C) (d) (A) and (C)

Sol. (a) Denaturation means distortion in protein structure of enzyme thus inhibiting enzymatic activity. Denaturation of protiens at high temperature causes the loss of secondary & tertiary structure of proteins.

Also in case of DNA, it leads to distortion from double stranded to single – stranded structure.

16. The correct statement in respect of protein haemoglobin is that it: (2004)

(a) Maintains blood sugar level

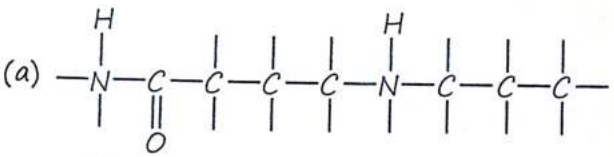
(b) Acts as an oxygen carrier in the blood

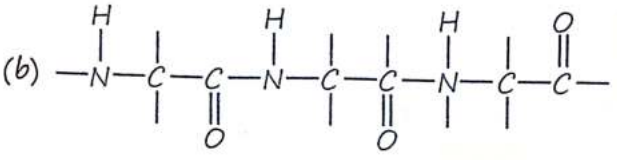
(c) Forms antibodies and offers resistance to diseases

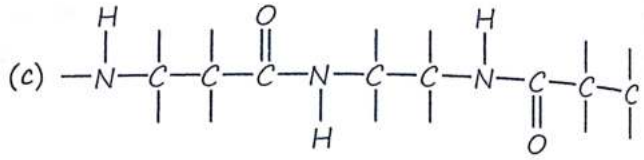
(d) Functions as a catalyst for biological reactions

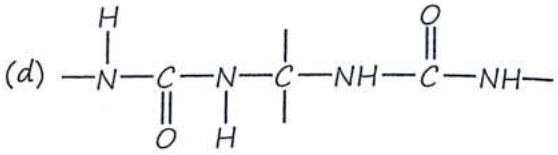
Sol. (b) Haemoglobin acts as an oxygen carrier in the blood. In this protein, 4 Fe^{2+} ions of each haemoglobin can bind with 4 molecules of O_2 & form oxyhaemoglobin.

17. Which one of the following structures represents the peptide chain? (2004)

(a) 

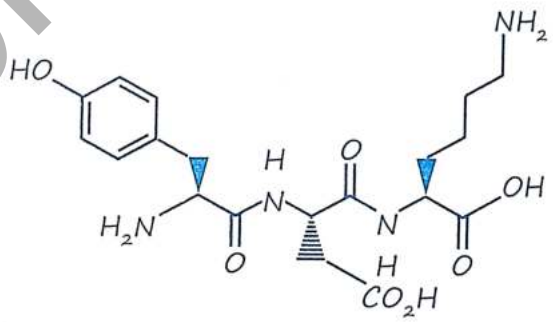
(b) 

(c) 

(d) 

Sol. (b) A peptide bond is formed by joining the carboxyl group of one amino acid to the amino group of another. The structure given in option (b) represents the peptide chain.

18. The structure of a peptide is given below:

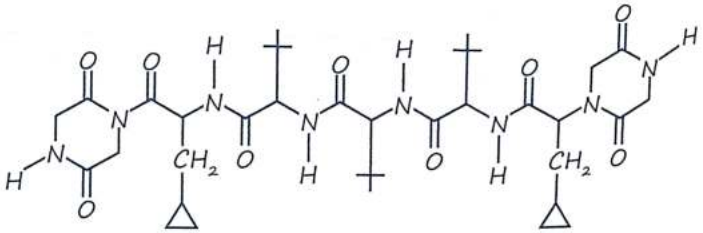


If the absolute values of the net charge of the peptide at pH = 2, pH = 6, and pH = 11 are $|z_1|$, $|z_2|$, and $|z_3|$, respectively, then what is $|z_1| + |z_2| + |z_3|$?

C-2.8 W-29.3 UA-67.9 (JEE Adv. 2020)

Sol. [5]

19. The total number of distinct naturally occurring amino acids obtained by complete acidic hydrolysis of the peptide shown below is



C-1.82 W-89.24 UA-8.94 (JEE Adv. 2014)

Sol. [1]

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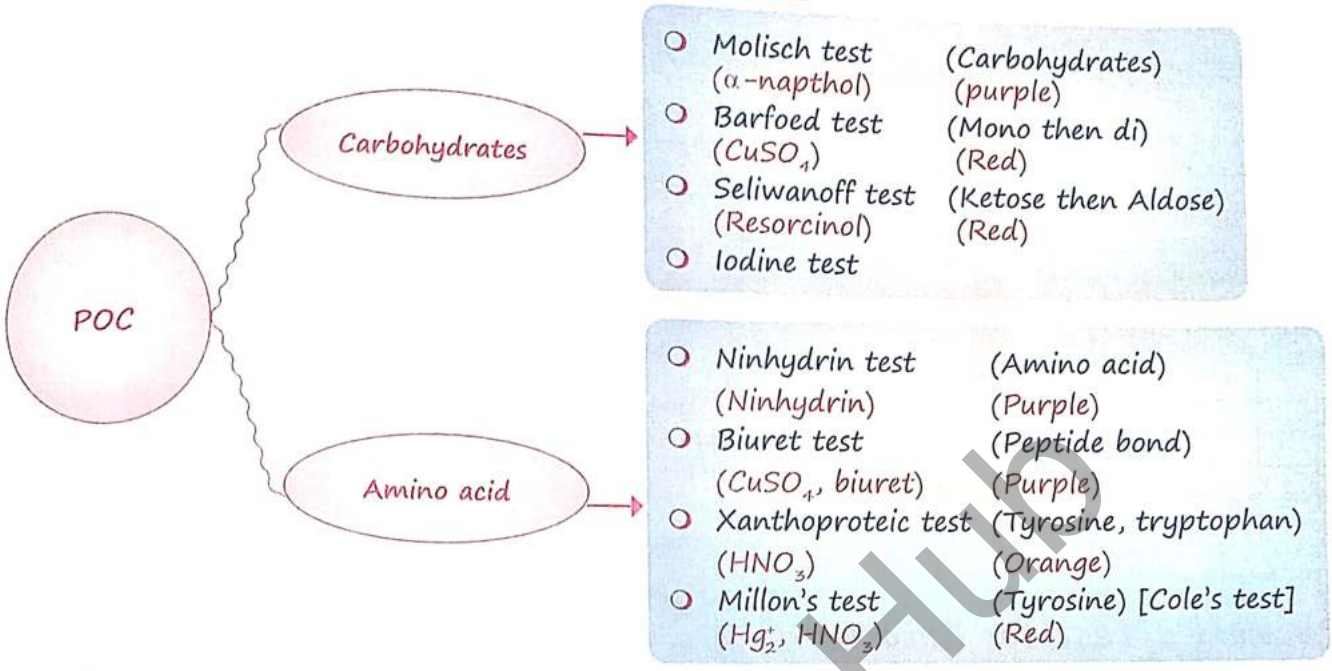
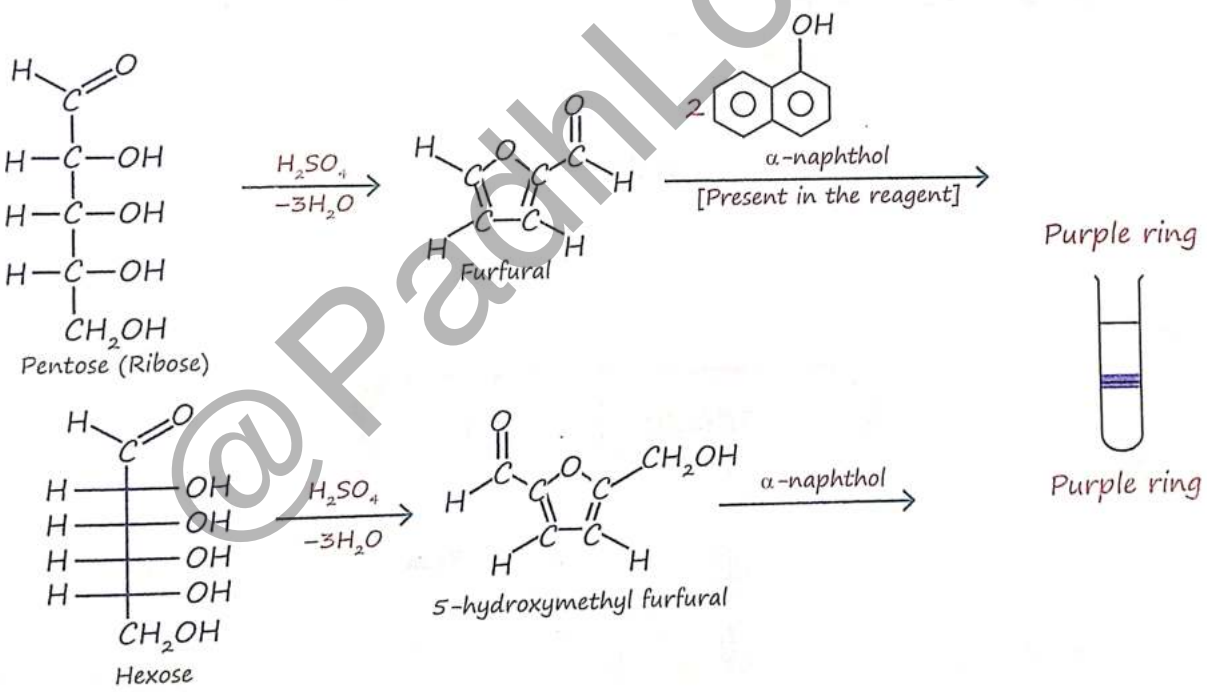
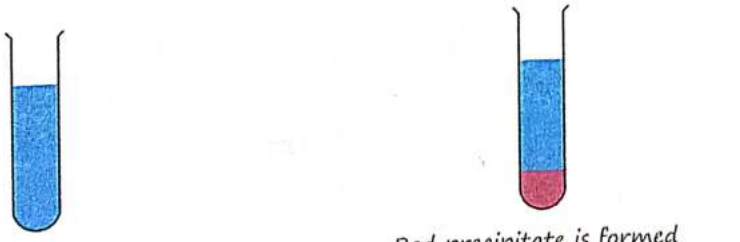
Distinguish Test of Biomolecules



Molisch's Test



Barfoed's Test



Solution remains blue
Carbohydrate is absent

Red precipitate is formed
Faster ⇒ Monosaccharide is present
Slower ⇒ Disaccharide is present

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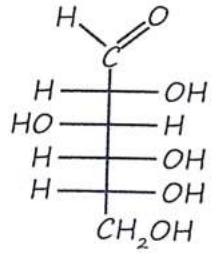
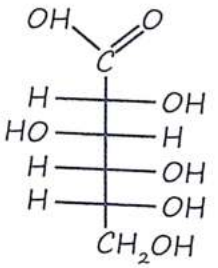
Barfoed's test-Definition, Principle, Procedure, Result, Uses



D-Glucose

+ $2Cu^{++}$ Cupric ion (copper (II)) + $2H_2O$ Water



D-Gluconic acid

+ $Cu_2O\downarrow$ Cuprous oxides (Copper (I)) (Red precipitate) + $4H^+$

Blue Solution
Carbohydrates absent

Red Precipitation
Within few minutes - monosaccharides after
3 minutes - disaccharides

Seliwanoff's Test for Ketoses

Ketose monosacch. (fructose) Deep cherry red +ve test

Ketose containing. Disacch. (Sucrose) Cherry red +ve test

Aldose faint pink/red –ve test

Control (DW) Colour Orange –ve test

Practice Questions

23. D-Ribose $\xrightarrow[(3)\ SOCl_2\ (4)\ Cu_2N_2\ (5)\ ACOH\ (6)\ Ba(OH)_2]{(1)\ Br_2/H_2O\ (2)\ AC_2O\ (excess)}$ (P)

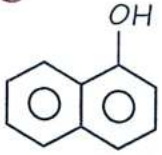
24. D-Glucose $\xrightarrow[(3)\ \Delta\ (4)\ \overset{\ominus}{O}H/H_2O]{(1)\ NH_2-OH/H^{\oplus}\ (2)\ AC_2O\ (excess)}$ (P) [Wohl method]

25. D-Galcatose $\xrightarrow[(3)\ Fe^{+2}/\ H_2O_2]{(1)\ Br_2/H_2O\ (2)\ Ca/(OH)_2}$ (R) [Ruff's method]

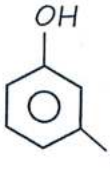
26. L-Glucose $\xrightarrow[(3)\ Zn/AcOH]{(1)\ 3\ PhNHNH_2\ (2)\ H^{\oplus}/H_2O}$ (S)

27. L-Ribose $\xrightarrow[(3)\ H_3O^{\oplus}]{(1)\ HCN/NaCN\ (2)\ H_2,pd,BaSO_4}$ (T) [Kiliani fischer]

28. D-fructose (x) $\xrightarrow[\Delta]{Conc.\ H_2SO_4}$ (P)

29. (P) + 2  $\xrightarrow{H^{\oplus}}$ (Q) $\xrightarrow{-H^{\ominus}}$ [... colour]

30. (X) + $\overset{+2}{CuSO_4}$ $\xrightarrow{H_2O}$ (R) [...colour of...]

31. (P) + 2  $\xrightarrow{H^{\oplus}}$ (S) $\xrightarrow{-H^{\ominus}}$ [... colour]

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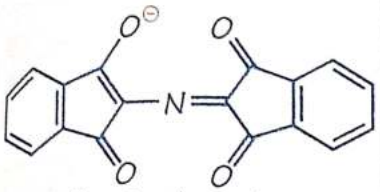
Ninhydrin Test

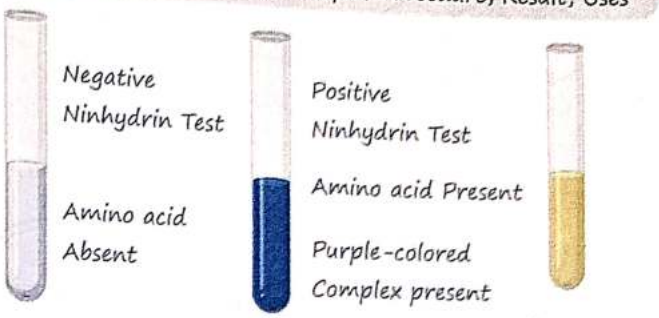
Detection of all amino acids, peptides and proteins

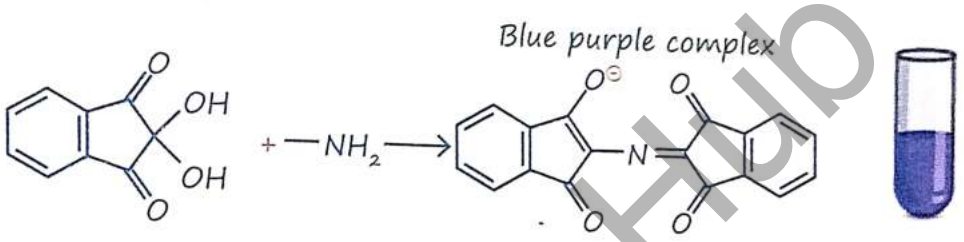
Ninhydrin Test- Definition, Principle, Procedure, Result, Uses

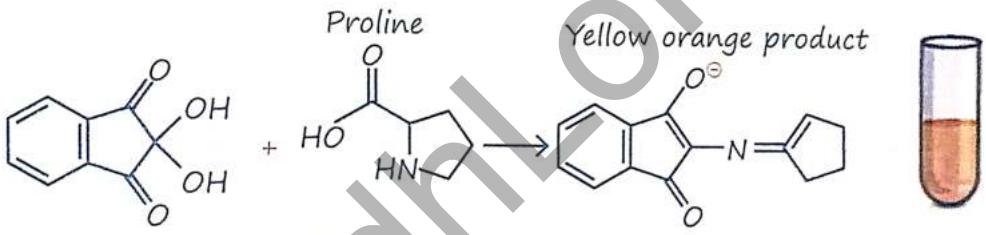


Blue-Purple product
Ruhemann's Purple

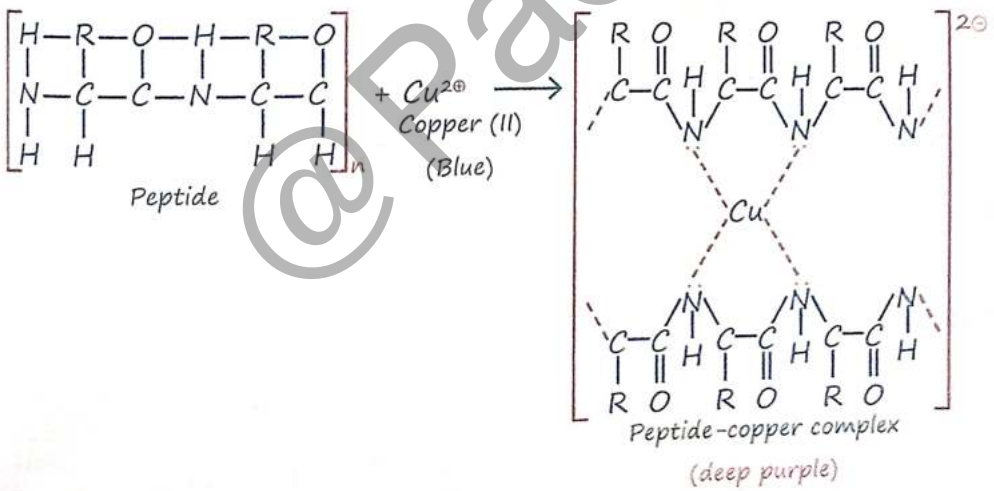
Positive Nagative Positive

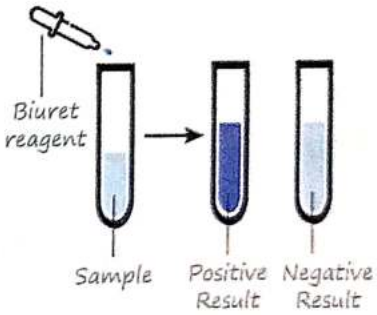




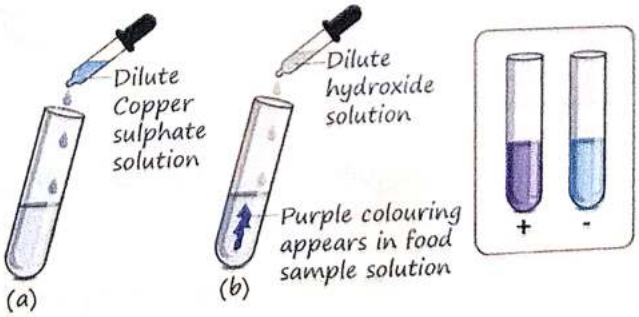


Biuret Test For Protein





Test for Protein



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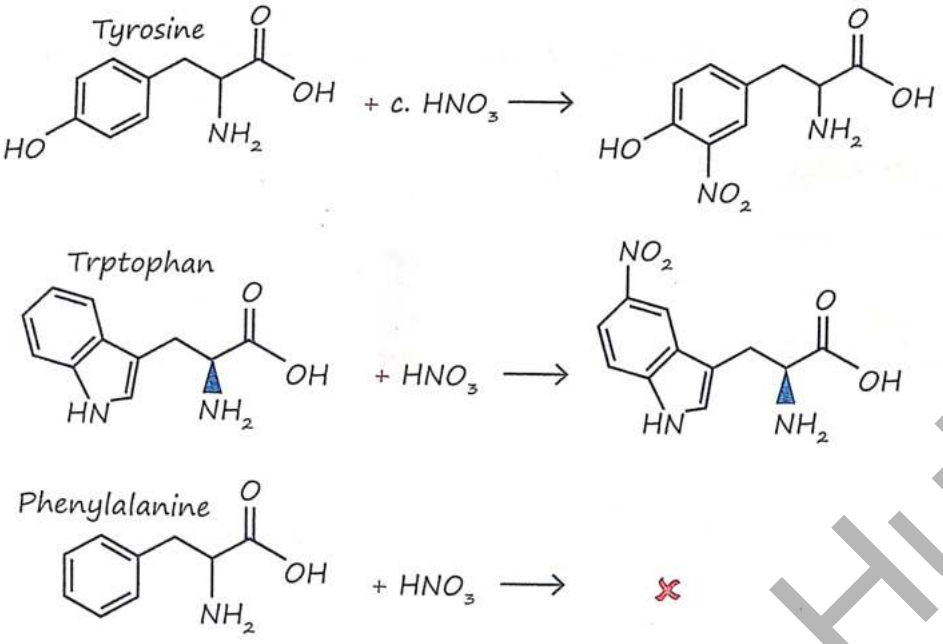
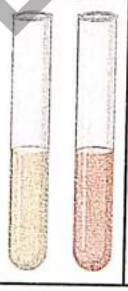
Xanthoproteic Test

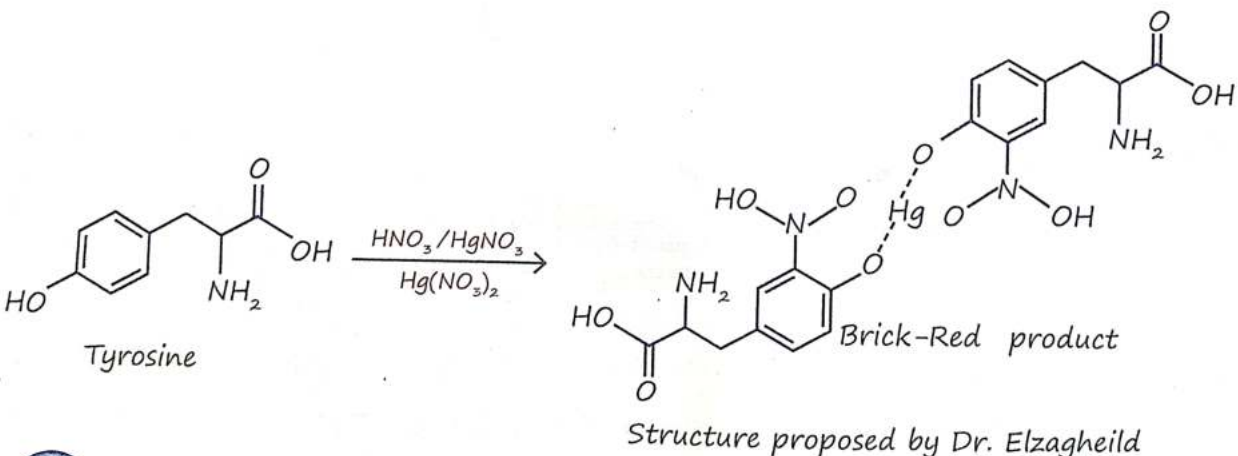


Xanthoproteic Test definition, Principle, Procedure, Result, Uses

	Xanthoproteic test Negative Absence of aromatic amino acids (tyrosine and tryptophan) Absence of the dark yellow or orange color		Xanthoproteic test positive Presence of aromatic amino acids (tyrosine and tryptophan) Presence of the dark yellow or orange color

Million's Test (Cole's Test)

(Test for Detection of Tyrosine)



Structure proposed by Dr. Elzagheild

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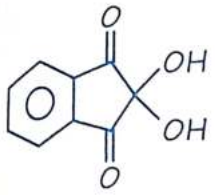
Practice Questions

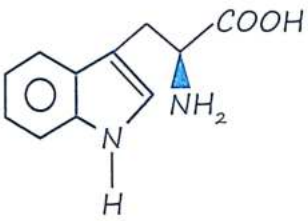
32. 

$\xrightarrow[(2)\ \Delta]{(1)\ \text{Valine [O]}}$ (P) $\xrightarrow{H_3O^{\oplus}}$ (R) + (Q) (Foul smell with $CHCl_3$/KOH)

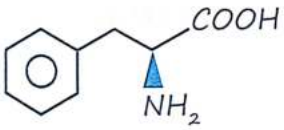
(S) + (X) $\xrightarrow{Ph = 4.5\text{-}6.8}$ (T) $\xrightarrow{-H^{\oplus}}$ [... colour]

(R) $\xrightarrow[2.\ QS/H_2SO_4]{1.\ HCN/NaCN}$ (K) $\xrightarrow[2.\ SOCl_2\ 3.\ H_2/Pd,\ BaSO_4]{1.\ H_2/Ni}$ (L) $\xrightarrow[2.\ KCN/HCN\ 3.\ H_3O^{\oplus}]{1.\ NH_3}$ (N)

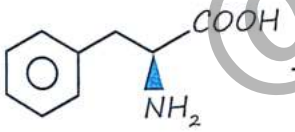
33. Dipeptide [M–O] (x) $\xrightarrow{Cu^{+2}}$ [Complex]$^{+2}$ $\xrightarrow{H_2N-\overset{O}{\overset{\|}{C}}-NH-\overset{O}{\overset{\|}{C}}-NH_2}$ [.... colour]



$\xrightarrow{HNO_3}$ (U) [... Colour]



$\xrightarrow{HNO_3}$ (V) [... colour]



$\xrightarrow[Hg(NO_3)_2]{HNO_3/Hgno_3}$ (V) [... colour]

Previous Year Questions

20. Biuret test is not given by- [AIEEE-2010]

(a) proteins (b) carbohydrates

(c) polypeptides (d) urea

Sol. (b)

21. Which of the following compounds can be detected by Molisch's test? [JEE Main 2012]

(a) Sugars (b) Amines

(c) Primary alcohols (d) Nitro compounds

Sol. (a)

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22. Fructose and glucose can be distinguished by [JEE Main 2019]

(a) Fehling's test (b) Barfoed's test

(c) Benedict's test (d) Seliwanoff's test

Sol. (d)

23. Which of the following tests cannot be used for identifying amino acids? [JEE Main 2019]

(a) Barfoed test (b) Ninhydrin test

(c) Xanthoproteic test (d) Biuret test

Sol. (a)

24. The correct observation in the following reaction is: [JEE Main 2020]

$$\text{Sucrose} \xrightarrow[\text{Cleavage (Hydrolysis)}]{\text{Glycosidic bond}} \text{A+B} \xrightarrow[\text{Reagent}]{\text{Seliwanoff}}$$

(a) Formation of blue colour

(b) Formation of violet colour

(c) Formation of red colour

(d) Gives no colour

Sol. (c)

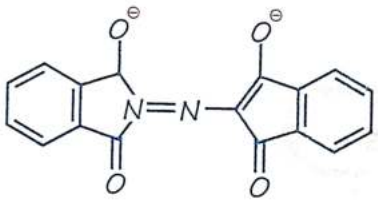
25. Seliwanoff test and Xanthoproteic test are used for the identification of ___ and ___ respectively. [JEE-Mains (26-Feb_Shift-2)2021]

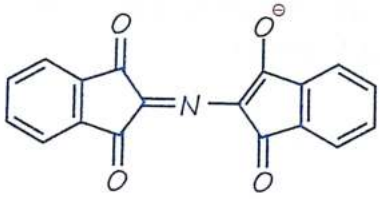
(a) ketoses, proteins (b) proteins, ketoses

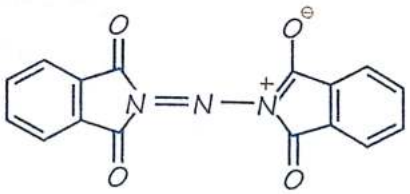
(c) aldoses, ketoses (d) ketoses, aldoses

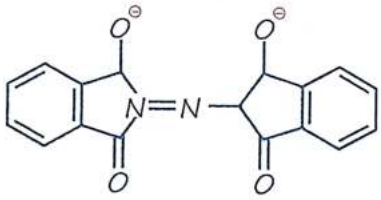
Sol. (a)

26. The correct structure of Rhumann's Purple, the compound formed in the reaction of ninhydrin with proteins is: [JEE-Mains (20-July_Shift-1)2021]

(a) 

(b) 

(c) 

(d) 

Sol. (b)

27. Which one among the following chemical tests is used to distinguish monosaccharide from disaccharide? [JEE-Mains (27-July_Shift-1)2021]

(a) Barfoed test (b) Seliwanoff's test

(c) Iodine test (d) Tollen's test

Sol. (a)

28. Which one of the following tests used for the identification of functional groups in organic compounds does not use copper reagent? [JEE-Mains (27-August_Shift-2)2021]

(a) Seliwanoff's test

(b) Barfoed's test

(c) Biuret test for peptide bond

(d) Benedict test

Sol. (a)

29. A sugar 'X' dehydrates very slowly under acidic condition to give furfural which on further reaction with resorcinol gives the coloured product after sometimes. [JEE-Mains (26-July_Shift-2)2022]

(a) Aldopentose (b) Aldotetrose

(c) Oxalic acid (d) Ketotetrose

Sol. (a)

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Enzyme

Life is possible due to the coordination of various chemical reactions in living organisms. An example is the digestion of food, absorption of appropriate molecules and ultimately production of energy. This process involves a sequence of reactions and all these reactions occur in the body under very mild conditions. This occurs with the help of certain biocatalysts called enzymes. Almost all the enzymes are globular proteins. Enzymes are very specific for a particular reaction and for a particular substrate. They are generally named after the compound or class of compounds upon which they work. For example, the enzyme that catalyses hydrolysis of maltose into glucose is named as maltase.

$$\underset{\text{Maltose}}{C_{12}H_{22}O_{11}} \xrightarrow{\text{Maltase}} \underset{\text{Glucose}}{2\,C_6H_{12}O_6}$$

Sometimes enzymes are also named after the reaction, where they are used. For example, the enzymes which catalyse the oxidation of one substrate with simultaneous reduction of another substrate are named as oxidoreductase enzymes. The ending of the name of an enzyme is -ase.

Enzymes are needed only in small quantities for the progress of a reaction. Similar to the action 10activation energy for acid hydrolysis of sucrose is 6.22 $kJmol^{-1}$, while the activation energy is only 2.15 $kJmol^{-1}$ when hydrolysed by the enzyme, sucrase. Mechanism for the enzyme action has been discussed.

Enzyme	Source	Enzymatic reaction
Invertase	Yeast	Sucrose → Glucose and fructose
Zymase	Yeast	Glucose → Ethyl alcohol and carbon dioxide
Diastase	Malt	Starch → Maltose
Maltase	Yeast	Maltose → Glucose
Urease	Soyabean	Urea → Ammonia and carbon dioxide Proteins
Pepsin	Stomach	Proteins → Amino acids

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Previous Year Questions

30. The incorrect statements regarding enzymes are: [06 April, 2024 (Shift-II)]

A. Enzymes are biocatalysts.

B. Enzymes are non-specific and can catalyse different kinds of reactions.

C. Most Enzymes are globular proteins.

D. Enzyme - oxidase catalyses the hydrolysis of maltose into glucose.

Choose the correct answer from the option given below:

(a) (B) and (C) (b) (B), (C) and (D)

(c) (B) and (D) (d) (A), (B) and (C)

Sol. (c) Enzymes are reactions specific and one catalyst cannot catalyse more than one reaction.

Enzyme maltase causes hydrolyses of maltose into glucose.

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Vitamines

1. Certain organic compounds are required in small amounts in our diet but their deficiency causes specific diseases. These compounds are called vitamins.
2. Most of the vitamins cannot be synthesised in our body but plants can synthesise almost all of them, so they are considered as essential food factors.
3. Organic compounds required in the diet in small amounts to perform specific biological functions for normal maintenance of optimum growth and health of the organism.

fat soluble: K,E,D,A

Water soluble: B_1, B_2, B_6, B_{12}, C

S. No.	Name of Vitamins	Sources	Deficiency Diseases
1.	Vitamin A	Fish liver oil, carrots, butter & milk	Xerophthalmia (hardening of cornea of eye) Night blindness
2.	Vitamin B1 (Thiamine)	Yeast, milk, green vegetables and cereals	Beri beri (loss of appetite, retarded growth)
3.	Vitamin B2 (Riboflavin)	Milk, eggwhite, liver, kidney	Cheilosis (fissuring at corners of mouth and lips), digestive disorders and burning sensation of the skin.
4.	Vitamin B6 (Pyridoxine)	Yeast, milk, egg yolk, cereals and grams	Convulsion
5.	Vitamin B12	Meat, fish, egg and curd	Pernicious anemia (RBC deficient in hemoglobin
6.	Vitamin C (Ascorbic acid)	Citrus fruits, amla and green leafy vegetables	Scurvy (bleeding gums)
7.	Vitamin D	Exposure to sunlight, fish and egg yolk	Rickets (bone deformities in children) and osteomalacia (soft bones and joint pain in adults)
8.	Vitamin E	Vegetable oils like wheat germ oil, sunflower oil, etc.	Increased fragility of RBCs and muscular weakness
9.	Vitamin K	Green leafy vegetables	Increased blood clotting time

Previous Year Questions

31. Which of the following is not a fat soluble vitamin? (2011 Mains)

(a) Vitamin E (b) Vitamin A

(c) Vitamin B complex (d) Vitamin D

Sol. (c) Vitamin B complex (B3) is water soluble. This vitamin cannot be stored in the body and hence, must be consumed each day. As it is dissolve in water, it gets easily absorbed into the tissue of the body.

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Hormones

Hormones are molecules that act as intercellular messengers.

These are produced by endocrine glands in the body and are poured directly in the blood stream which transports them to the site of action.

In terms of chemical nature, some of these are steroids, e.g., estrogens and androgens; some are poly peptides for example insulin and endorphins and some others are amino acid derivatives such as epinephrine and norepinephrine.

Hormones have several functions in the body. They help to maintain the balance of biological activities in the body.

The role of insulin in keeping the blood glucose level within the narrow limit is an example of this function.

Insulin is released in response to the rapid rise in blood glucose level.

On the other hand hormone glucagon tends to increase the glucose level in the blood.

The two hormones together regulate the glucose level in the blood.

Epinephrine and norepinephrine mediate responses to external stimuli. Growth hormones and sex hormones play role in growth and development.

Growth hormones and sex hormones play role in growth and development. Thyroxine produced in the thyroid gland is an iodinated derivative of amino acid tyrosine.

Abnormally low level of thyroxine leads to hypothyroidism which is characterised by lethargyness and obesity.

Increased level of thyroxine causes hyperthyroidism.

Low level of iodine in the diet may lead to hypothyroidism and enlargement of the thyroid gland.

This condition is largely being controlled by adding sodium iodide to commercial table salt ("Iodised" salt).

Steroid hormones are produced by adrenal cortex and gonads (testes in males and ovaries in females).

Hormones released by the adrenal cortex play very important role in the functions of the body.

For example, glucocorticoids control the carbohydrate metabolism, modulate inflammatory reactions, and are involved in reactions to stress.

The mineralocorticoids control the level of excretion of water and salt by the kidney.

If adrenal cortex does not function properly then one of the results may be Addison's disease characterised by hypoglycemia, weakness and increased susceptibility to stress.

The disease is fatal unless it is treated by glucocorticoids and mineralocorticoids.

Hormones released by gonads are responsible for development of secondary sex characters.

Testosterone is the major sex hormone produced in males.

It is responsible for development of secondary male characteristics (deep voice, facial hair, general physical constitution) and estradiol is the main female sex hormone.

It is responsible for development of secondary female characteristics and participates in the control of menstrual cycle. Progesterone is responsible for preparing the uterus for implantation of fertilised egg.

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Hormones Overview

Inter-cellular messengers produced by endocrine glands, secreted directly into the bloodstream and transported to target tissues

Chemical Nature of Hormones:

Steroids (estrogens, androgens)

Polypeptides (insulin, endorphin).

Amino acid derivatives (epinephrine, norepinephrine).

Functions of Hormones

- **Maintaining Biological Balance:** insulin regulates blood glucose levels.
- **Response to External Stimuli:** epinephrine and norepinephrine mediate responses.
- **Growth and Development:** Growth hormones and sex hormones play roles.
- **Thyroxine and Thyroid Function:**
  - Thyroxine regulates metabolic rate, abnormal levels lead to hypothyroidism or hyperthyroidism.
  - Iodine deficiency can cause hypothyroidism, prevented by iodized salt.
- **Steroid Hormones:**
  - Produced by adrenal cortex and gonads.
  - Essential for various bodily functions including carbohydrate metabolism and control of water balance.
- **Role of Sex Hormones:**
  - Testosterone in males and estradiol in females control development of secondary sex characteristics.
  - Progesterone prepares uterus for implantation of fertilized egg.

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Previous Year Questions

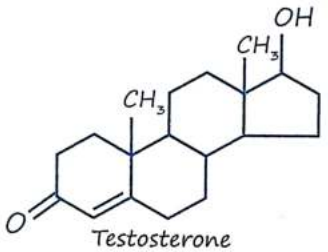
32. **Which of the following hormones is produced under the conditions of stress which stimulate glycogenolysis in the liver of human beings?** (2014)

(a) Thyroxin (b) Insulin

(c) Adrenaline (d) Estradiol

Sol. (c) Adrenaline hormone helps to release fatty acids from fat and glucose from liver glycogen under the condition of stress. Adrenaline hormone is produced by adrenal glands when a brain sends a message about a stressful situation. It is also known as 'flight of fight hormone'.

33. Testosterone, which is a steroidal hormone, has the following structure.

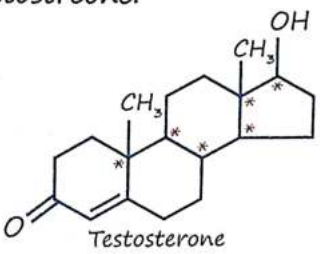


Testosterone

The total number of asymmetric carbon atom/s in testosterone is _____.

[1 Feb, 2023 (Shift-II)]

Sol. [6] There are six assymmetric centres present in testostreone.



Testosterone

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10 Polymers

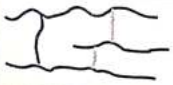
Polymers are large molecules, also known as macromolecules, composed of repeating structural units called monomers. These units are chemically linked together in long chains or networks, forming materials with diverse properties and applications.

Classification of Polymer

Based on Sources

Natural	Semi-Synthetic	Synthetic
Animal and Plants	1/2 Man made + 1/2 Natural	Man Made
Natural Rubber, Starch, Cellulose, Protein	Cellulose-acetate (Rayon)	Plastic, Fibre, Synthetic Rubber

Based on Structure

Linear	Branched	Cross-Linked
Long and Straight Chain	Having Branches	Bi or Trifunctional groups
HDPE, PVC	LDPE	Strong Covalent bonds, Bakelite
		

Based on Forces

Thermosetting > Fibre > Thermoplastic > Elastomer

Elastomer	Fibre	Thermoplastic	Thermoplastic
Having Elastic Properties	High tensile strength & Modulous	Linear & some branched	Cross linked
Weakest forces	Strong forces like H-bonding	Force b/w fibre & Elastomer	Strongest forces
Stretchable & Some Cross links to regain shape	Close packing so Crystalline in nature	Soft on heating & Hard on cooling	Not Reused
Rubber, Buna-N, Buna-S, Neoprene	Polyester, Polyamide	PVC, Polythene	(Heat + x) Bakelite, Urea, Melamine

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1.	Ethene			
2.	Propene			
3.	Vinylchloride			
4.	Styrene			
5.	Acrylonitrile			
6.	Tetrafluoroethene			
7.	Buta-1, 3-diene			
8.	Isoprene			
9.	Chloroprene			
10.	Buta-1, 3-diene + stryene			
11.	Buta-1, 3-diene + Acrylonitrile			
12.	Ethyne			

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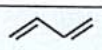
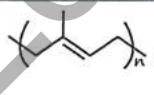
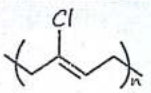
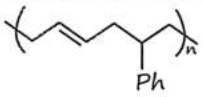
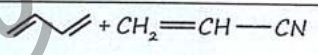
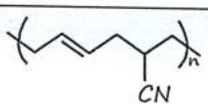
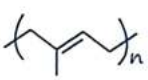
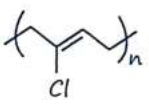
1.	Ethene	$CH_2 = CH_2$	$+CH - CH_2+_n$	Polythene
2.	Propene	$CH_3 - CH = CH_2$	$+CH(CH_3) - CH_2+_n$	Polypropene
3.	Vinylchloride	$CH_2 = CH - Cl$	$+CH_2 - CH(Cl)+_n$	PVC
4.	Styrene	$CH_2 = CH - Ph$	$+CH_2 - CH(Ph)+_n$	Polystyrene
5.	Acrylonitrile	$CH_2 = CH - CN$	$+CH_2 - CH(CN)+_n$	PAN
6.	Tetrafluoroethene	$F_2C = CF_2$	$+CF_2 - CF_2+_n$	Teflon
7.	Buta-1, 3-diene			Rubber
8.	Isoprene			All cis : Polyisoprene (Natural Rubber) All trans : Gattapurcha
9.	Chloroprene			Neoprene
10.	Buta-1, 3-diene + stryene			Buna-S
11.	Buta-1, 3-diene + Acrylonitrile			Buna-N
12.	Ethyne	$HC \equiv CH$	$+CH = CH+_n$	Polythyne

Practice Questions

1.  $\xrightarrow{H_2|Ni}$ P Polymer of ??

2.  $\xrightarrow{H_2|Ni}$ P Polymer of ??

3. Natural Rubber $\xrightarrow[Zn|H_2O]{O_3}$

4. 100 $CH_3 - CH = CH_2 \xrightarrow[h\nu]{(PhCOO)_2}$ mass of product

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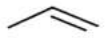
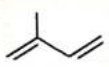
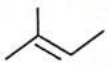
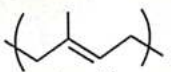
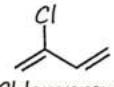
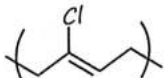
Addition Polymer

- Chain growth
- Mostly ene/yne
- Radical/Anion

Low density Polythene	High density Polythene
LDPE	HDPE
Flexible	Toughly & hard
Poor conductor of electricity	Closely packed
Squeeze bottles	Toys, bucket, dustbins
High Temp. (333k-570k)	Low Temp. (33k)
High Pressure (100 atm)	Zeigler Natta ($TiCl_4 + AlEt_3$)
Branched	Linear

Monomer	Reaction/Intermediate	Polymer of Name	Uses
$CH_2 = CH_2$ (ethene)	$CH_2 = CH_2 \xrightarrow[\text{High P}]{\text{High T}}$ Radical	$\left(CH_2-CH_2\right)_n$ (LDPE)	Squeeze bottles, flexible pipe, toys
$CH_2 = CH_2$ (ethene)	$CH_2 = CH_2 \xrightarrow[\text{333K, Zeigler Natta}]{\text{Low P}}$	$\left(CH_2-CH_2\right)_n$ (HDPE)	Buckets, Dustbins
$F_2C = CF_2$ (Tetra fluoroethyene)	$F_2C = CF_2 \xrightarrow[\text{High T, High P}]{\text{Per Sulphate}}$	$\left(F_2C-CF_2\right)_n$ (Teflon)	Nonstick surface coating, Oil seal and Gasket
$CH_2=CH-CN$ (Acrylo nitrile)	$\xrightarrow[\text{Radical}]{\Delta}$	$\left(CH_2-\underset{\displaystyle CN}{\underset{\vert}{CH}}\right)_n$ (PAN)	Substitute for wool in orlan/ acrilan
$CH_2=CH-Cl$ (Vinyl chloride	$\xrightarrow[\text{Radical}]{\Delta}$	$\left(CH_2-\underset{\displaystyle Cl}{\underset{\vert}{CH}}\right)_n$ (PVC)	Flooring, Pipes, Raincoat
	$\xrightarrow[\text{Radical}]{\Delta}$	$\left(CH_2-\underset{\displaystyle CH_3}{\underset{\vert}{CH}}\right)_n$	Toys, Pipe, Popes, fibre
Styrene ($Ph-CH=CH_2$)		$\left(\underset{\displaystyle Ph}{\underset{\vert}{CH}}-CH_2\right)_n$ (Polystryene)	Television of Radio cabinets, Wrapping Material
 (Isoprene)		 Natural Rubber	Rubber
 (Chloroprene)		 (Neoprene)	Synthetic Rubber Convey or Belt Gasket, hoses

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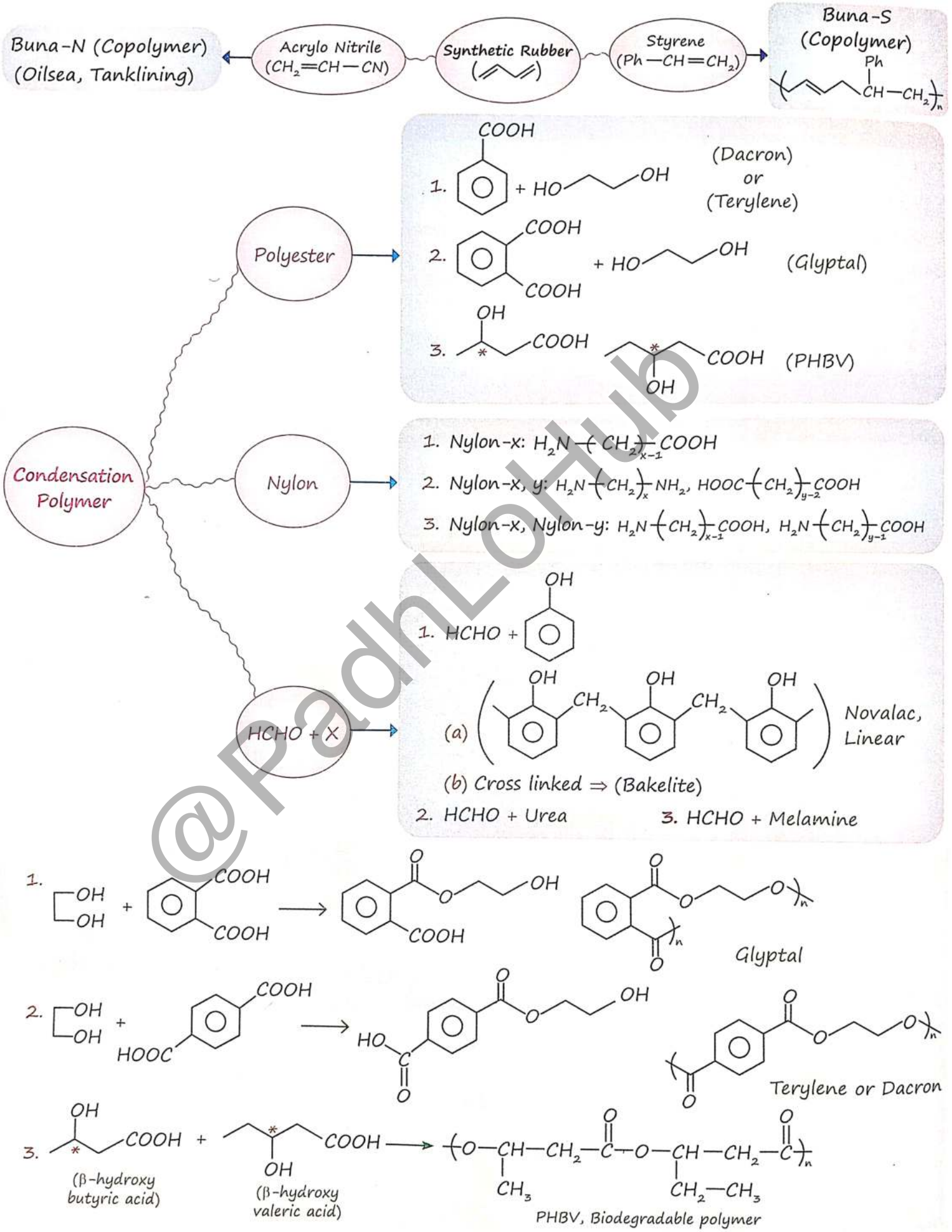

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Nylon-x(no of c)

$H_2N-(CH_2)_{x-1}-COOH$

Homo: $H_2N-(CH_2)_5-COOH \xrightarrow{\text{Polymer}} -(NH-(CH_2)_5-C(=O))_n-$

(amino caproic acid) → (Nylon-6)

CO: $H_2N-CH_2-COOH + H_2N-(CH_2)_5-COOH \xrightarrow{\text{Polymer}} -(NH-CH_2-C(=O)-NH-(CH_2)_5-C(=O))_n-$

(Glycine) + (amino caproic acid) → (Nylon-2, Nylon-6)

Biodegradable

Nylon-x, y

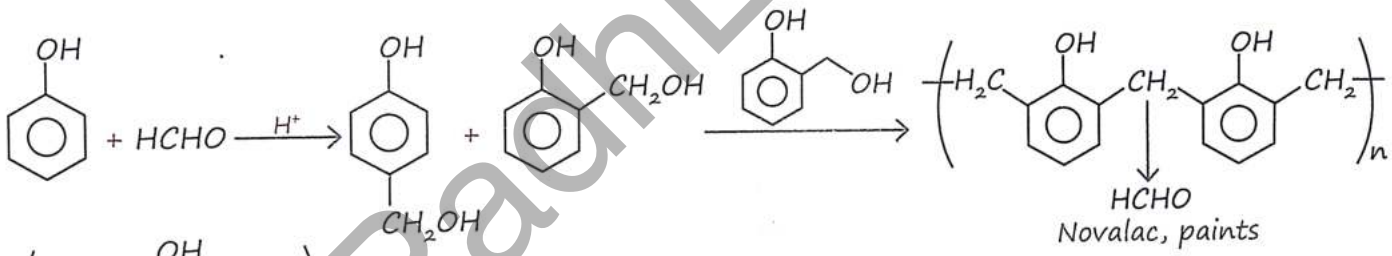
$H_2N-(CH_2)_x-NH_2 + HOOC-(CH_2)_{y-2}-COOH$

1. $H_2N-(CH_2)_6-NH_2 + HOOC-(CH_2)_4-COOH \longrightarrow -(NH-(CH_2)_6-NH-C(=O)-(CH_2)_4-C(=O))_n-$

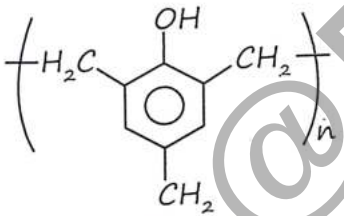
(Adipic acid) → (Nylon-6, 6)

2. $-(C(=O)-(CH_2)_8-C(=O)-NH-(CH_2)_6-NH)_n-$

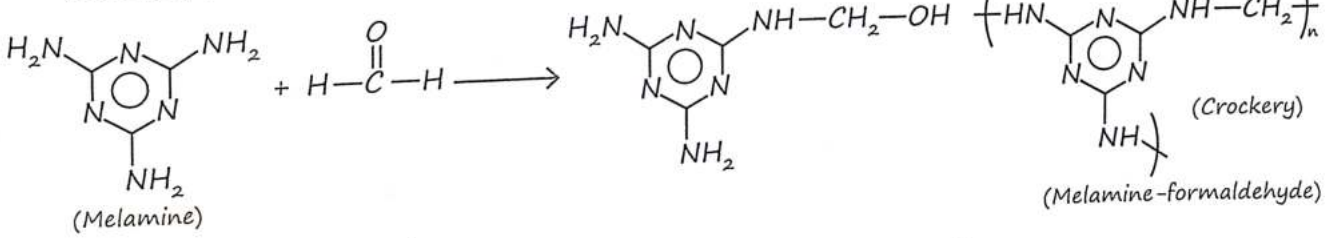
(Nylon-6, 10)

3. 

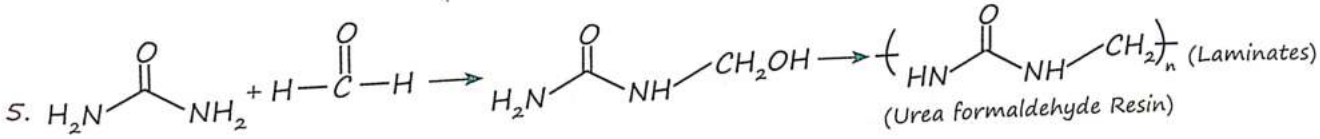
Novalac, paints



Cross link, Bakelite

4. 

(Melamine) → (Crockery), (Melamine-formaldehyde)

5. 

(Urea formaldehyde Resin) → (Laminates)

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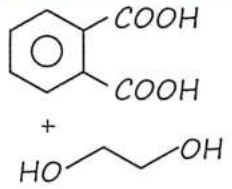
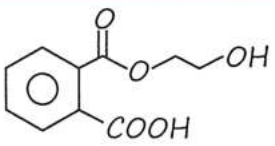
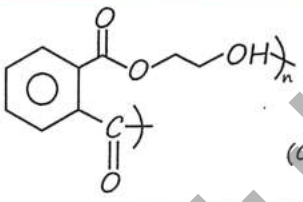
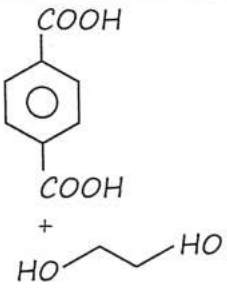
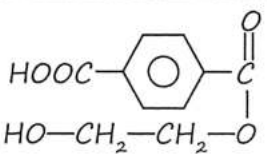
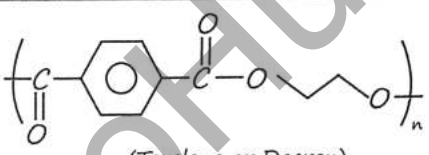
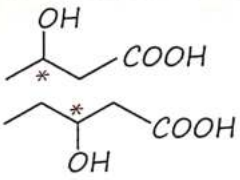
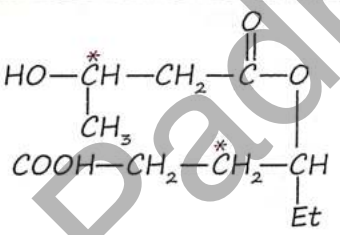
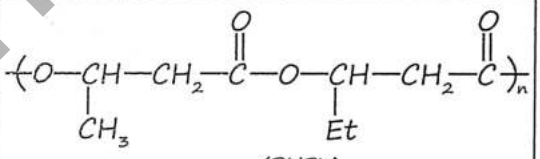
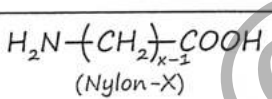
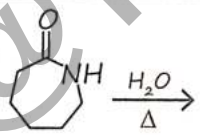
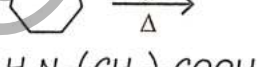
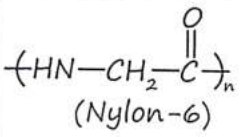
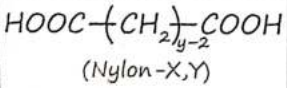
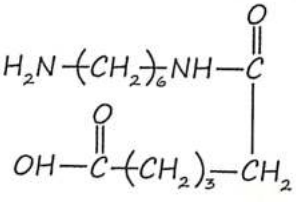
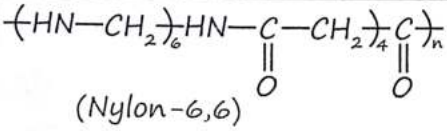
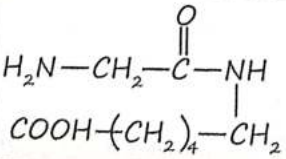
Condensation Polymer: ($-H_2O$, $-ROH$, $-HCl$ removal)

1.	Polyester: Dacron, Glyptal, PHBV
2.	Polyamide: Nylon-x, Nylon-x, y; Nylon-x, Nylon-y
3.	HCHO + x : Phenol, Melamine, Urea

Monomer	Reaction/Intermediatic	Polymer of name	Uses
		 (Glyptal)	Paints
		 (Terelene or Dacron)	Textile
		 (PHBV)	Biodegradable Orthopaedic
 (Nylon-X)	 	 (Nylon-6)	Tyre Cords Fibre
 (Nylon-X,Y)		 (Nylon-6,6)	Bristles for Busters
Nylon X, Nylon Y		(Nylon 2, Nylon6)	Biodegradable

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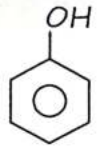
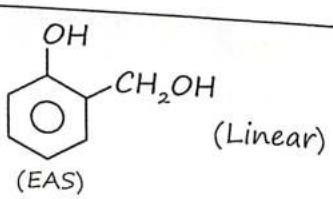
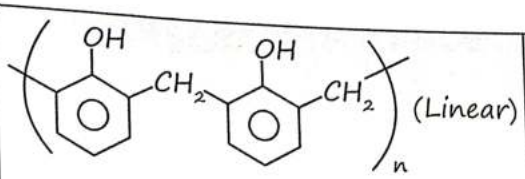
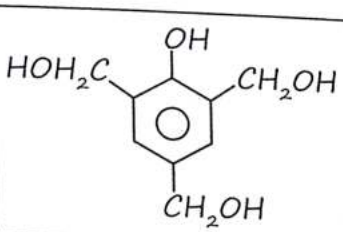
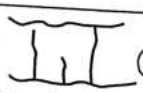
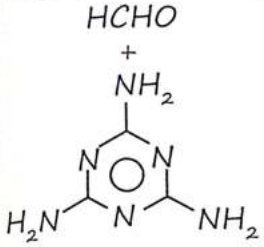
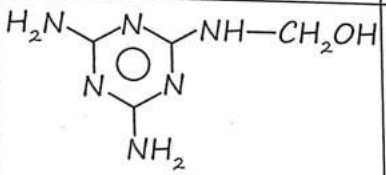
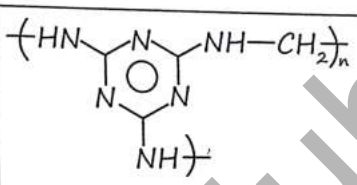
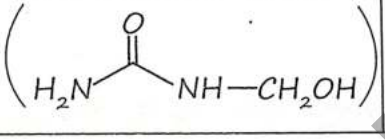
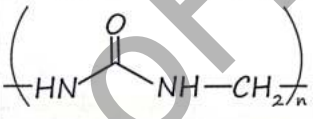
HCHO + 	 (EAS) (Linear)	 (Novalac)	Paints
Ph—OH + HCHO		 (Cross link) (Bakelite	Comb electric Switches
HCHO + 			Unbreakable Crockery
HCHO + Orea			Wood Laminates

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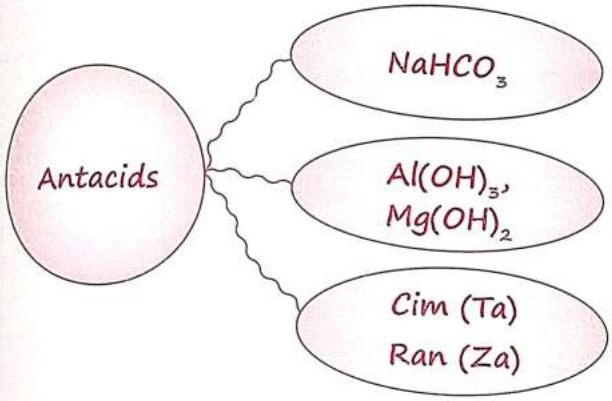
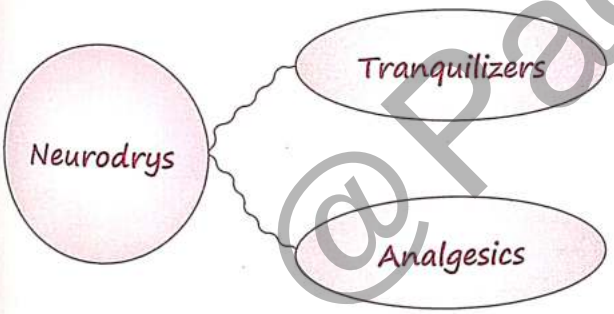
11 Chemistry in Everyday Life

Chemistry in Everyday life



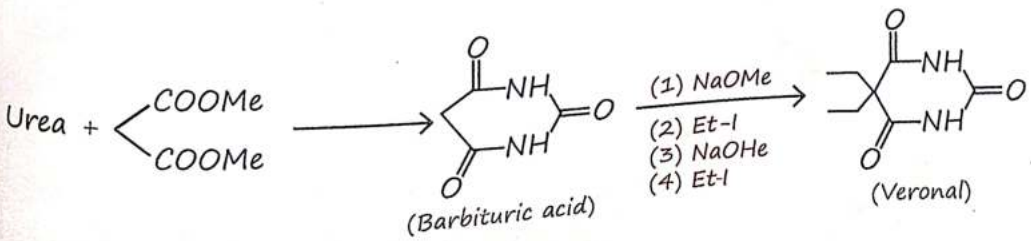
Antihistamines ⟶ Bro (Dil, Di) Ter (Sel) (Antiallergic)



Tranquilizers ⟶ V.V.S. Nora↑↑ (excitement ⇒ sleeping pills)

M.E.C. Nora↓ (Hypertension)

I.P. Nora↓↓ (Depression)



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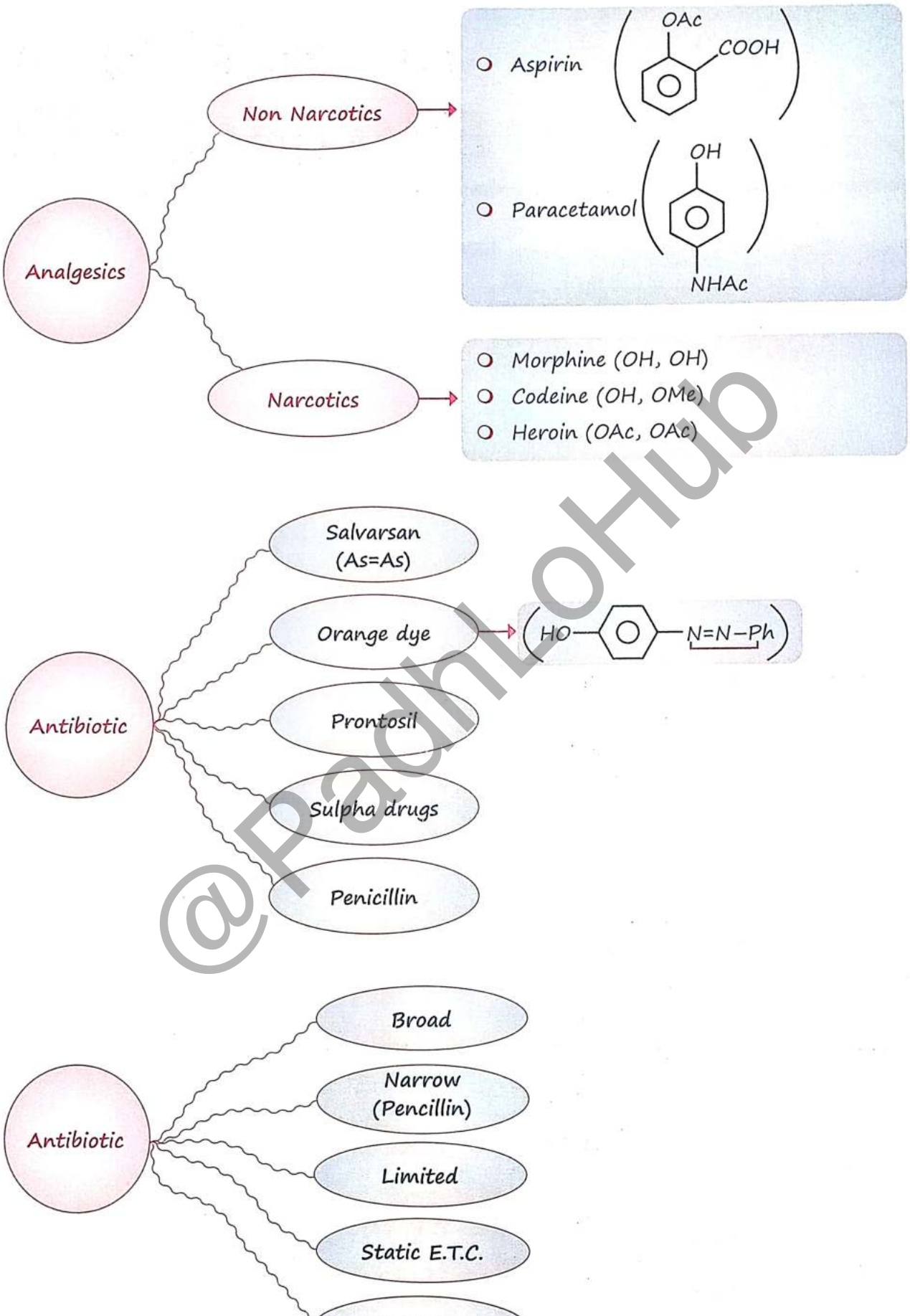

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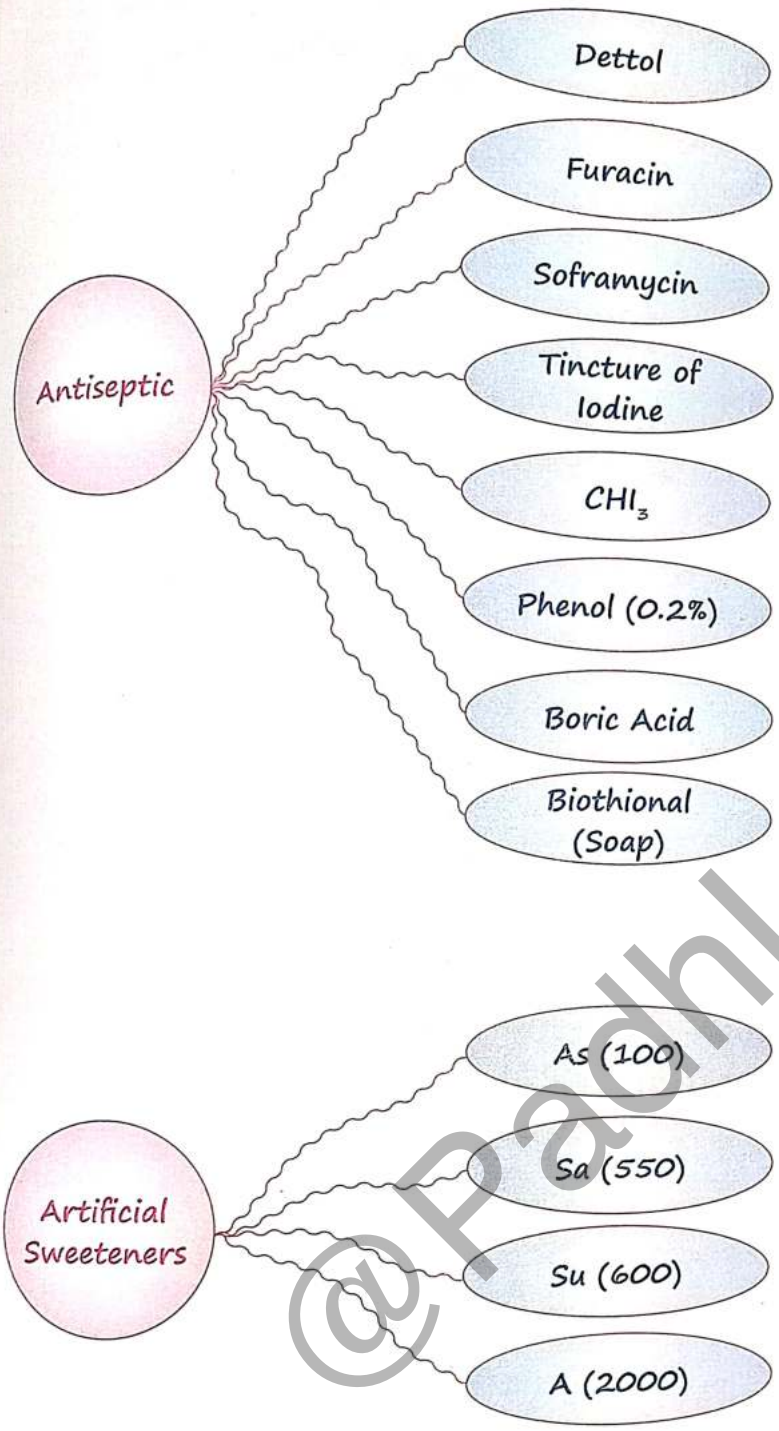

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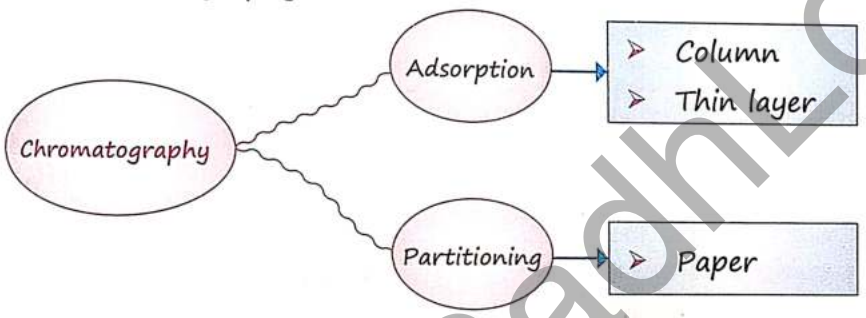
12 Practical Organic Chemistry

Methods of Purification of Organic Compounds

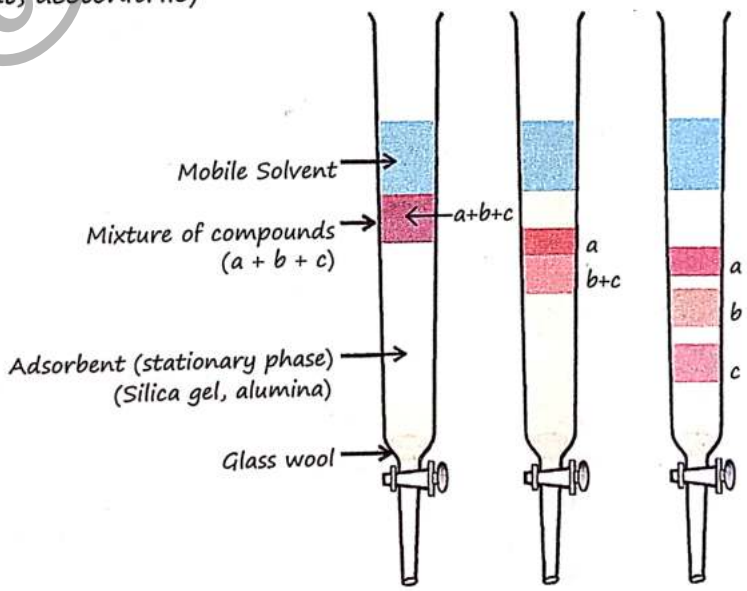
The common techniques used for purification are as follows:

(i) Sublimation
(ii) Crystallisation
(iii) Distillation
(iv) Differential extraction and
(v) Chromatography



Main:

(i) Stationary phase (Silica gel, Alumina)
(ii) Component (Mixture)
(iii) Mobile phase (Solvent, acetonitrile)



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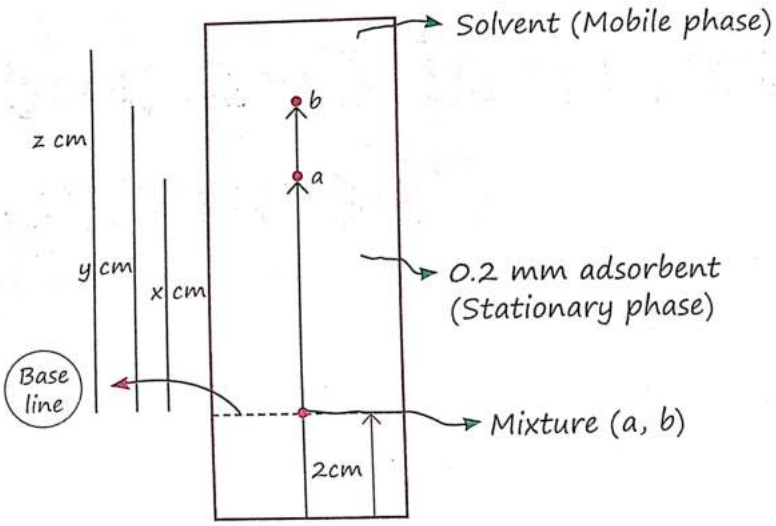

$$(Rf)_a = \frac{\text{Distance by (a) from base line}}{\text{Distance by solvent form base line}}$$

$$(Rf)_b = \frac{\text{Distance by (b) from base line}}{\text{Distance by solvent form base line}}$$

If colourless component present:

(i) UV chamber

(ii) Iodine solution

(iii) Spray (amino acid ninhydrine) (Purple)

Type	Conditions	Examples
(A) Simple distillation	(i) When liquid sample has non volatile impurities (ii) When boiling point difference is 80°K or more.	(i) Mixture of chloroform (BB = 334K) and Aniline (BP = 457K) (ii) Mixture of Ether (BP = 308K) and Toluene (BP = 384 K) (iii) Hexane (342 K) and Toluene
(B) Fractional distillation	When BP difference is 10°	(i) Crude oil in petroleum industry (ii) Acetone (329) and Methyl alcohol (338K)
(C) Distillation under reduced pressure (Vacuum distillation)	When liquid boils at higher temperature and it may decompose before BP is attained.	(i) Concentration of sugar juice (ii) Recovery of glycerol from spent lye. (iii) Glycerol
(D) Steam distillation	When the substance is immiscible with water and steam volatile. $P = P_1 + P_2$ Vapour Pressure = Vapour pressure of organic liquid + Vapour pressure of water	(i) Aniline is separated from water (ii) Turpentine oil (iii) Nitro Benzene (iv) Bromo Benzene (v) Naphthalene (vi) o-Nitrophenol

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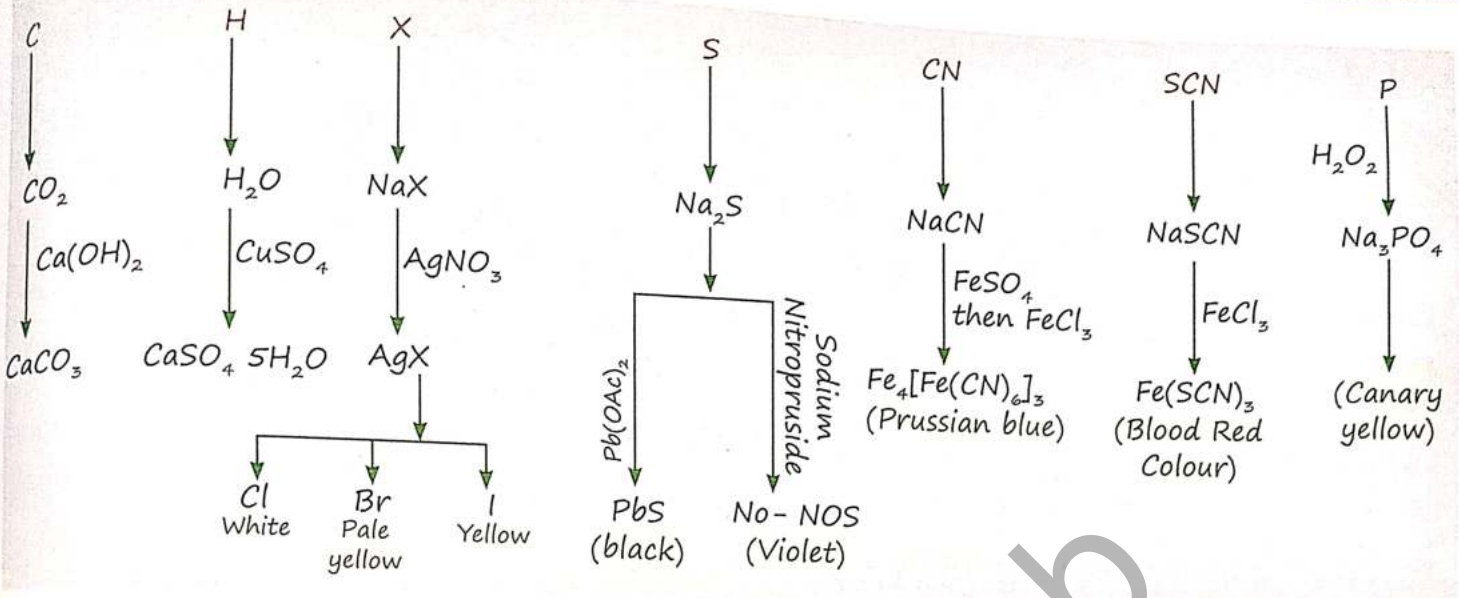

Element	Extract	Reaction with	Product	Colour
C				
H				
N				
S				
N,S				
X				
P				

Element	Extract	Reaction with	Product	Colour
C	CO_2	$Ca(OH)_2$	$CaCO_3$	Milky
H	H_2O	$CuSO_4$	$CuSO_4.5H_2O$	Blue
N	NaCN	$FeSO_4$ then $FeCl_3$	$Fe_4[Fe(CN)_6]_3$	Prussian Blue
S	Na_2S	$Pb(OCOCH_3)_3$ and $Na_2[Fe(CN)_5)NO]$	PbS and $Na_2[Fe(CN)_5 NOS]$	Black and Violet
N,S	NaSCN	$FeCl_3$	$Fe(SCN)_3$	Blood Red
X	NaX	Conc. HNO_3 then $AgNO_3$	AgCl, AgBr, AgI	White, Pale Yellow, Yellow
P	Na_3PO_4	$(NH_4)_2 MoO_4 + HNO_3$	$(NH_4)_3PO_4.12MoO_3$	Canary Yellow

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Duma's Method

$C_xH_yN_z + (2x + y/2)\ CuO \longrightarrow xCO_2 + y/2H_2O + z/2N_2 + (2x + y/2)\ Cu$

Let the mass of organic compound = m g

Volume of nitrogen collected = V_1 mL

Room temperature = T_1K

Volume of nitrogen at STP = $\frac{P_1V_1 \times 273}{760 \times T_1}$ (Let it be V mL)

Where p_1 and V_1 are the pressure and volume of nitrogen, p_1 is different from the atmospheric pressure at which nitrogen gas is collected

The value of p_1 is obtained by the relation;

p_1= Atmospheric pressure – Aqueous tension 22400 mL N_2 at STP weighs 28 g.

V mL N_2 at STP weighs = $\frac{28 \times V}{22400}$ g

Percentage of nitrogen = $\frac{28 \times V \times 100}{22400 \times m}$

Kjeldahl's Method

Organic compound + $H_2SO_4 \longrightarrow (NH_4)_2SO_4 \xrightarrow{2NaOH} Na_2SO_4 + 2NH_3 + 2H_2O$

$2NH_3 + H_2SO_4 \longrightarrow (NH_4)_2SO_4$

Let the mass of organic compound taken = m g

Volume of H_2SO_4 of molarity, M, taken = V mL

Volume of NaOH of molarity, M, used for titration of excess of H_2SO_4 = V_1 mL

V_1mL of NaOH of molarity M = $V_1/2$ mL of H_2SO_4 of molarity M

Volume of H_2SO_4 of molarity M unused = $(V - V_1/2)$ mL

$(V- V_1/2)$ mL of H_2SO_4 of molarity M = $2(V - V_1/2)$ mL of NH_3 solution of molarity M.

1000 mL of 1 M NH_3 solution contains 17g NH_3 or 14 g of N

$2(V - V_1/2)$ mL of NH_3 solution of molarity M contains:

$\frac{14 \times M \times 2(V - V_1/2)}{1000}$

Percentage of N = $\frac{14 \times M \times 2(V - V_1/2)}{1000} \times \frac{100}{m} = \frac{1.4 \times M \times 2(V - V/2)}{m}$

Kjeldahl method is not applicable to compounds containing nitrogen in nitro and azo groups and nitrogen present in the ring (e.g. pyridine) as nitrogen of these compounds does not change to ammonium sulphate under these conditions.

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Element	Method	Compound	Formulae
C	Liebig	CO_2	$\frac{12 \times m_2 \times 100}{44 \times m}$
H	Liebig	H_2O	$\frac{2 \times m_1 \times 100}{18 \times m}$
N	Duma's	$C_xH_yN_z + (2x + y/2)\ CuO \longrightarrow xCO_2 + y/2H_2O + z/2N_2 + (2x + y/2)\ Cu$	$\frac{28 \times V \times 100}{22400 \times m}$
N	Kjeldahl's	$2NH_3 + H_2SO_4 \longrightarrow (NH_4)_2SO_4$ [$-NO_2$, $-NO$, $-N{=}N-$ Ring N, Pyridine] ×	$\frac{1.4 \times M \times 2\left(V - \frac{V}{2}\right)}{m}$
X	Carius	AgX	$\frac{\text{atomic mass of X} \times m_1}{\text{molecular mass of AgX}}$
S	Carius	$BaSO_4$	$\frac{32 \times m_1 \times 100}{233 \times m}$
P	Carius	$(NH_4)_3PO_4.12MoO_3$ or $Mg_2P_1O_7$	$\frac{31 \times m_1 \times 100}{1877 \times m}$ % $\frac{62 \times m_1 \times 100}{222 \times m}$ %
O	–	CO	$\frac{32 \times m_1 \times 100}{88 \times m}$ %

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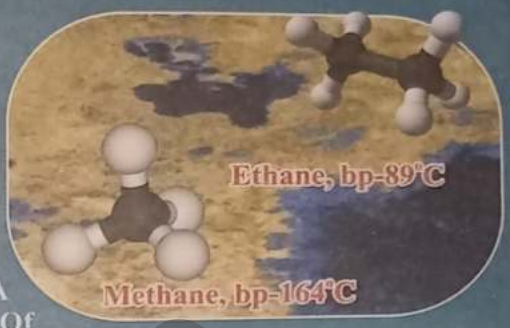
Summary

NOMENCLATURE OF ORGANIC COMPOUNDS



A colorized Radar Image Of The Surface Of Titan, Saturn's Largest Moon, Taken By The Cassini Spacecraft. The Light Tan Areas Correspond To A Solid Surface With High Radar Reflectivity. The Dark Areas Of Low Reflectivity Are Thought To Be Lakes Of Methane And Ethane, Which Rain Down On The Moon And Flow In Rivers Along Its Surface.

Format For IUPAC Name

s–prefix	+	p–prefix	+	word root	+	p–suffix	+	s–suffix
Substituents with locants		-cyclo -Bicyclo		Alk word Carbon in parentchain		Type of Multiple bond		According to main functional group given in priority table

(a) Locant :- Locants are separated by (,) comma.
- Locants and alphabets are separated by hyphen (–). [2,3-dimethyl pentane]
- di, tri, iso, neo and cyclo are neither separated by comma nor by hyphen

(b) Prefix :-Prefix are written in alphabetical order before root word.
- prefix → 1° or p prefix; 2° or p prefix
  Cyclo is 1° prefix and used for cyclic compound.

(c) Word root :- According to number of carbons in parent C-chain.
Ex. 1-Meth 2-Eth 3-Prop 4-But 5-Pent

(d) Primary suffix :- Accordingto saturation and unsaturation written in alphabetical order
C–C → ane C=C → ene C≡C → yne

(e) Secondary Suffix :- According to highest priority functional group

S. NO.	Functional group	Prefix	Suffix/Suffix*
1.	(carboxylic acid)COOH	carboxy	oic acid/carboxylic acid
2.	$-SO_3H$ (sulphonic acid)	sulpho	sulphonic acid
3.	$-(RCO)_2O$(Anhydride)	×	oic anhydride
4.	(ester)–COOR	Alkoxycarbonyl or Alkanoyloxy	oate
5.	–COX (acid halide)	halo formyl	oyl halide
6.	$-CONH_2$ (amide)	Carbonyl	carboxamide
7.	–CN (cyanide)	cyano	Nitrile or carbonitrile
8.	–CHO (aldehyde)	Oxo or formyl	Al or carbaldehyde
9.	–CO–(Ketone)	keto/oxo	one
10.	–OH (alcohol)	hydroxyl	ol
11.	–SH (thio alcohol)	mercapto	thiol
12.	$-NH_2$ (amine)	amino	amine

- For selection of chain and Naming of Compound given order can be used :
  [Functional Group > Multiple bond > Substituents >Alphabatical order]

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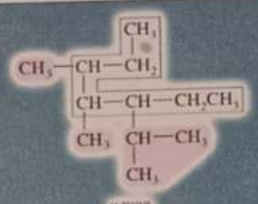
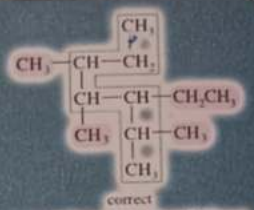
Naming Of Hydrocarbon & Derivatives

Rule-1

1. Find the longest continuous chain of carbon atoms, and use the name of this chain as the base name of the compound.
2. When there are two longest chains of equal length, use the chain with the greater number of substituents as the main chain.



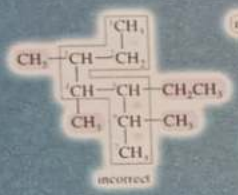
wrong
seven-carbon chain, but only three substituents



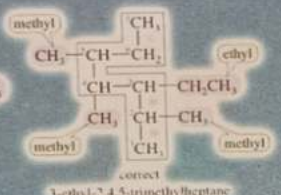
correct
seven-carbon chain, four substituents

Rule-2

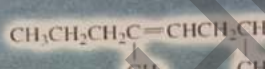
1. Number the longest chain, beginning with the end of the chain nearest a substituents.



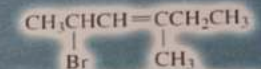
incorrect



correct
3-ethyl-2,4,5-trimethylheptane



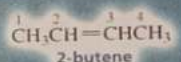
2,5-dimethyl-4-octene
not
4,7-dimethyl-4-octene
because 2 < 4



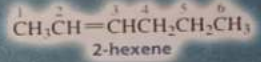
2-bromo-4-methyl-3-hexene
not
5-bromo-3-methyl-3-hexene
because 2 < 3

2. Number the longest continuous chain containing the functional group in the direction that gives the functional group suffix the lowest possible number.

1-butene



2-butene

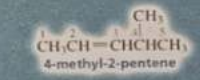


2-hexene

2-propyl-1-hexene

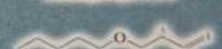
the longest continuous chain has eight carbons but the longest continuous chain containing the functional group has six carbons, so the parent name of the compound is hexene

3. When there are both a functional group suffix and a substituents, the functional group suffix gets the lowest possible number.
Substituents are stated in alphabetical order.



4-methyl-2-pentene

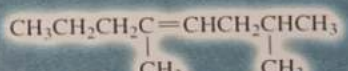
3-methyl-3-heptene



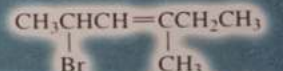
4-pentoxy-1-butene

4-methyl-1,3-pentadiene

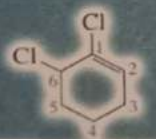
4. A substituents receives the lowest possible number only if there is no functional group suffix or if the same number for the functional groupsuffix is obtained in both directions.



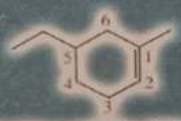
2,5-dimethyl-4-octene
not
4,7-dimethyl-4-octene
because 2 < 4



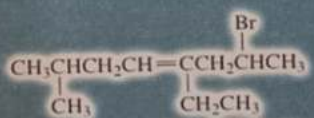
2-bromo-4-methyl-3-hexene
not
5-bromo-3-methyl-3-hexene
because 2 < 3



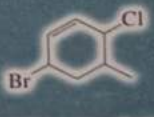
1,6-dichlorocyclohexene
not
2,3-dichlorocyclohexene
because 1 < 2



5-ethyl-1-methylcyclohexene
not
4-ethyl-2-methylcyclohexene
because 1 < 2



2-bromo-4-ethyl-7-methyl-4-octene
not
7-bromo-5-ethyl-2-methyl-4-octene
because 4 < 5



6-bromo-3-chloro-4-methylcyclohexene
not
3-bromo-6-chloro-5-methylcyclohexene
because 4 < 5

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GENERAL ORGANIC CHEMISTRY

Citric acid and ascorbic acid (vitamin C) are two water-soluble organic acids found in citrus fruits. Citric acid plays a central role in metabolism, and ascorbic acid is needed for growth and repair of collagen, a structural protein in connective tissue. Unlike most animals, humans cannot synthesize their own ascorbic acid. Ascorbic acid deficiency causes scurvy, with debilitating and often deadly symptoms resulting from damage to connective tissues.

ELECTRONIC DISPLACEMENT EFFECT

1. Order of +I series : $-\overset{\ominus}{C}H_2 > -\overset{\ominus}{N}H > -O^{\ominus} > -COO^{\ominus} > -C(CH_3)_3 > -CH(CH_3)_2 > CH_2CH_3 > CT_3 > -CD_3 > CH_3 > -T > -D > -H$

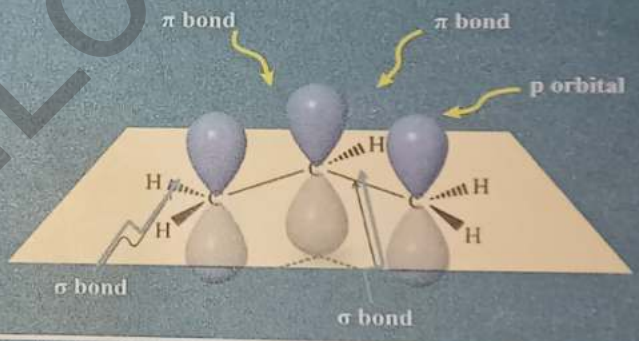
2. Order of –I series : $-\overset{\oplus}{N}F_3 > -\overset{\oplus}{N}R_3 > -\overset{\oplus}{N}H_3 > -NO_2 > -CN > -COOH > -F > -Cl > -Br > -I > -OR > -OH > -NH_2 > -C\equiv CH > -Ph > -CH=CH_2$

3. Stability of Intermediate (= Resonance effect, I = Inductive effect)

(i) Carbocation (+) $\propto$ +R, +H, +I $\propto \frac{1}{-R, -I}$

(ii) Free radical (·) $\propto$ +R, –R, +I $\propto \frac{1}{-I}$

(iii) Carbanion (–) $\propto$ –R, –I $\propto \frac{1}{+R, +H, +I}$



4. Acidic Strength : $HA \underset{}{\overset{K_a}{\rightleftharpoons}} H^{\oplus} + A^{\ominus}$

$K_a = \frac{[H^{\oplus}][A^{\ominus}]}{[HA]}$ $\quad pK_a = -\log K_a$

$\propto$ Stability of conjugate base

$\propto$ Electron withdrawing group (–R, –I)

5. Basic Strength :

$B^{\ominus} + H^{\oplus} \underset{}{\overset{K_b}{\rightleftharpoons}} H\text{–}B$

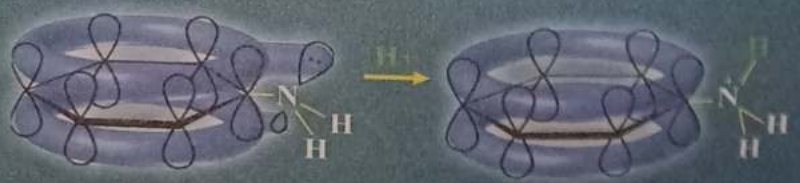
$K_b = \frac{[HB]}{[B^{\ominus}][H^{\oplus}]}$

$\propto$ Stability of conjugate acid

$\propto$ Electron donating tendency

$\propto$ Localised lone pair

$\propto$ Solvation effect (SIP)



aniline
stabilized by overlap with the ring

anilinium ion
no overlap is possible

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$$HA + B^- \rightleftharpoons H\text{–}B + A^\ominus$$

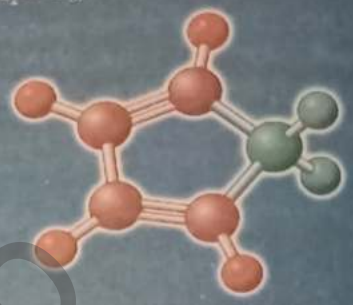
- Reaction always moves from strong acid to weak acid.
- If reaction goes in forward direction, HA is stronger acid than HB and HA is soluble in aq. BQ.
- In case of aqueous NaOH if acid is stronger than water then HA is soluble in aqueous NaOH.

$$HA + NaOH \rightleftharpoons H_2O + Na^{\oplus}A^{\ominus}$$

- In case of aqueous $NaHCO_3$, If HA is stronger than conjugate acid (H_2CO_3) then HA will be soluble is aqueous $NaHCO_3$ & gives brisk effervescence of CO_2 gas.

$$HA + NaHCO_3 \rightleftharpoons H_2CO_3 + Na^{\oplus}A^{\ominus}$$

$H_2CO_3 \rightarrow H_2O + CO_2$

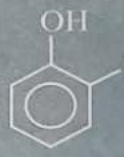


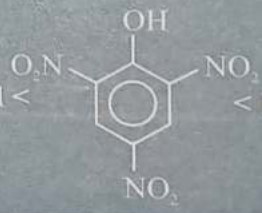
Cyclopentadiene (pKa=15)

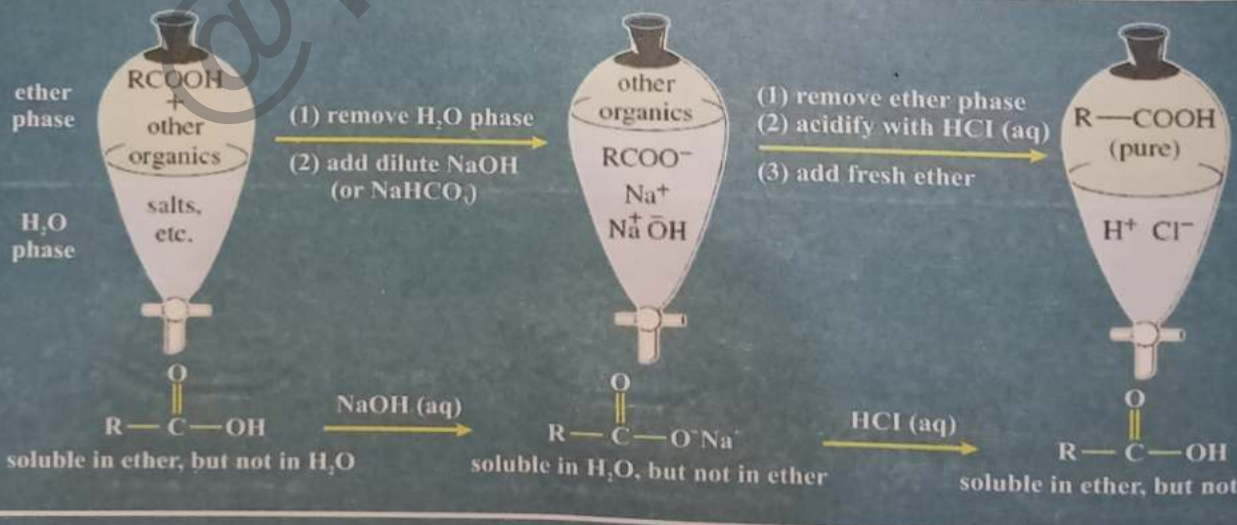
- **NOTE** : Solubility is decided by given factors : -

Solubility
- Salt formation
- Already soluble (due to H-bonding) (CH_3OH, Et–OH, , ,)

- **SOLUBILITY TABLE** :

Alkane < Alkene < H_2 < R–NH_2 < R–C≡CH < R–OH < H_2O < Ph–OH <  < H_2CO_3 <

< R–COOH < Picric acid < R–SO_3H < Squaric acid <  < Inorganic acid (like HBr, HCl)



soluble in ether, but not in H_2O — soluble in H_2O, but not in ether — soluble in ether, but not in H_2O

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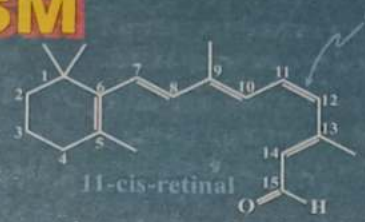
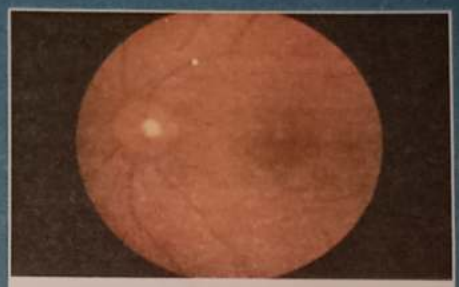
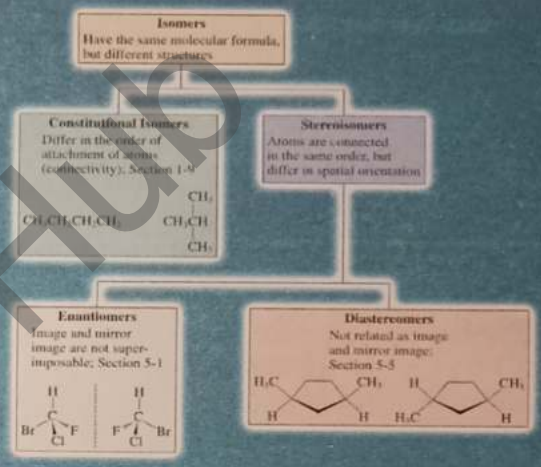
ISOMERISM




view inside the human eye

When visible light with the appropriate energy is absorbed by rhodopsin, the complexedcis-retinal is isomerized to the transisomer. This process is fantastically fast. This change in geometry triggers a response in the rod nerve cells, which is transmitted to the brain and perceived as vision.

1. Isomers have the same molecular formula but are different compounds. Constitutional (structural) isomers differ in the order in which the individual atoms are connected. Stereo isomers have the same connectivity but differ in the three-dimensional arrangement of the atoms. Mirror- image stereo isomers are related to each other as object and mirror image.

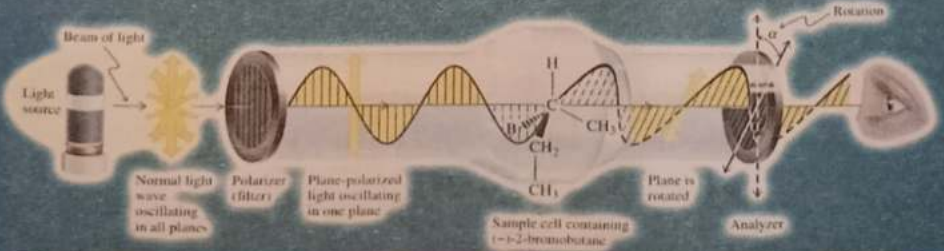


2. A carbon atom bearing four different substituents (asymmetric carbon) is an example of a chiral atom.

an asymmetric center

3. An object that is not super impossible on its mirror image is chiral.
4. Chiral molecules can't have plane of symmetry or centre of symmetryor alternating axis of] symmetry.

5. Most of the physical properties of enantiomers are the same. A major exception is their interaction with plane-polarized light: One enantiomer will rotate the polarization plane clockwise (dextrorotatory), the other counterclockwise (levorotatory). This phenomenon is called optical activity. The extent of the rotation is measured in degrees and is expressed by the specific rotation.



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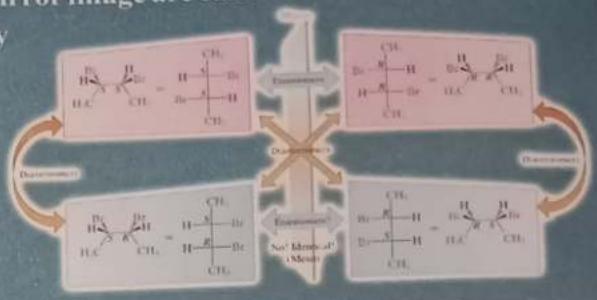
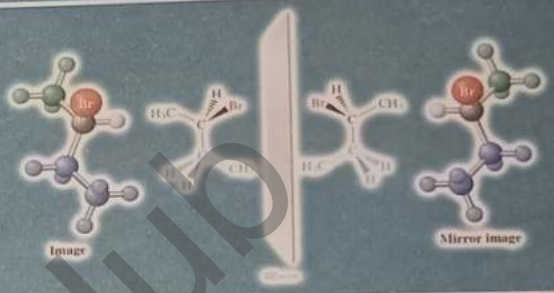
6. Two stereo isomers that are non-superimposable mirror image are called as enantiomers.
7. Molecule containing chiral centers and a symmetry except axis of symmetryis called mesocompound.
8. Diastereomers are stereoisomers that are not related to each other as object to mirror image. Cis and trans isomers of cyclic compounds are examples of diastereomers.



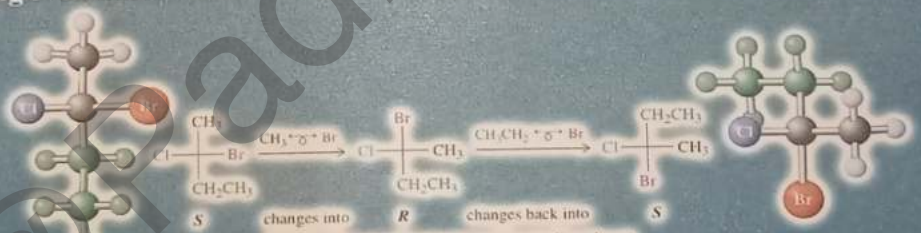
9. A compound containing one stereo center is chiral and exists as a pair of enantiomers. A 1:1 mixture of enantiomers is a race mate (racemic mixture).



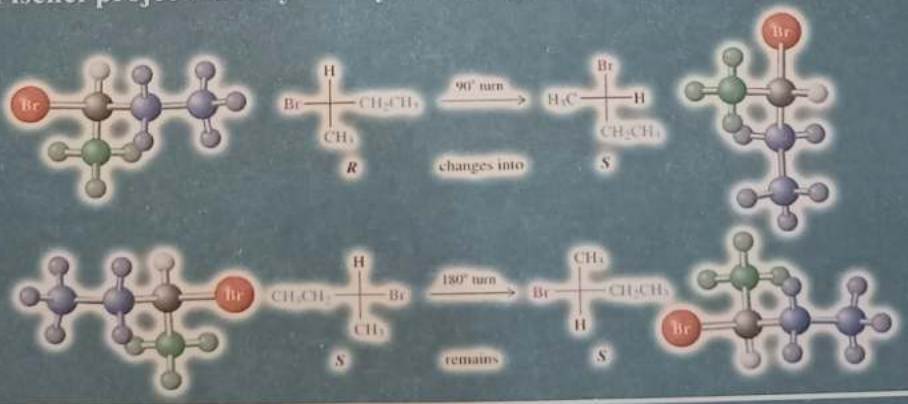
10. Clockwise specifies R if the lowest priority substituents is on a hatched wedge. Counterclockwise specifies S if the lowest priority substituents is on a hatched wedge.

← if you forgot which direction corresponds to which configuration imagine driving a car and turning the steering wheel clockwise to make a right turn or turning it counterclockwise to make a left turn.

11. By changing even times at one chiral centre, configuration doesn't changes



12. Rotating a Fischer projection may or may not change the absolute configuration



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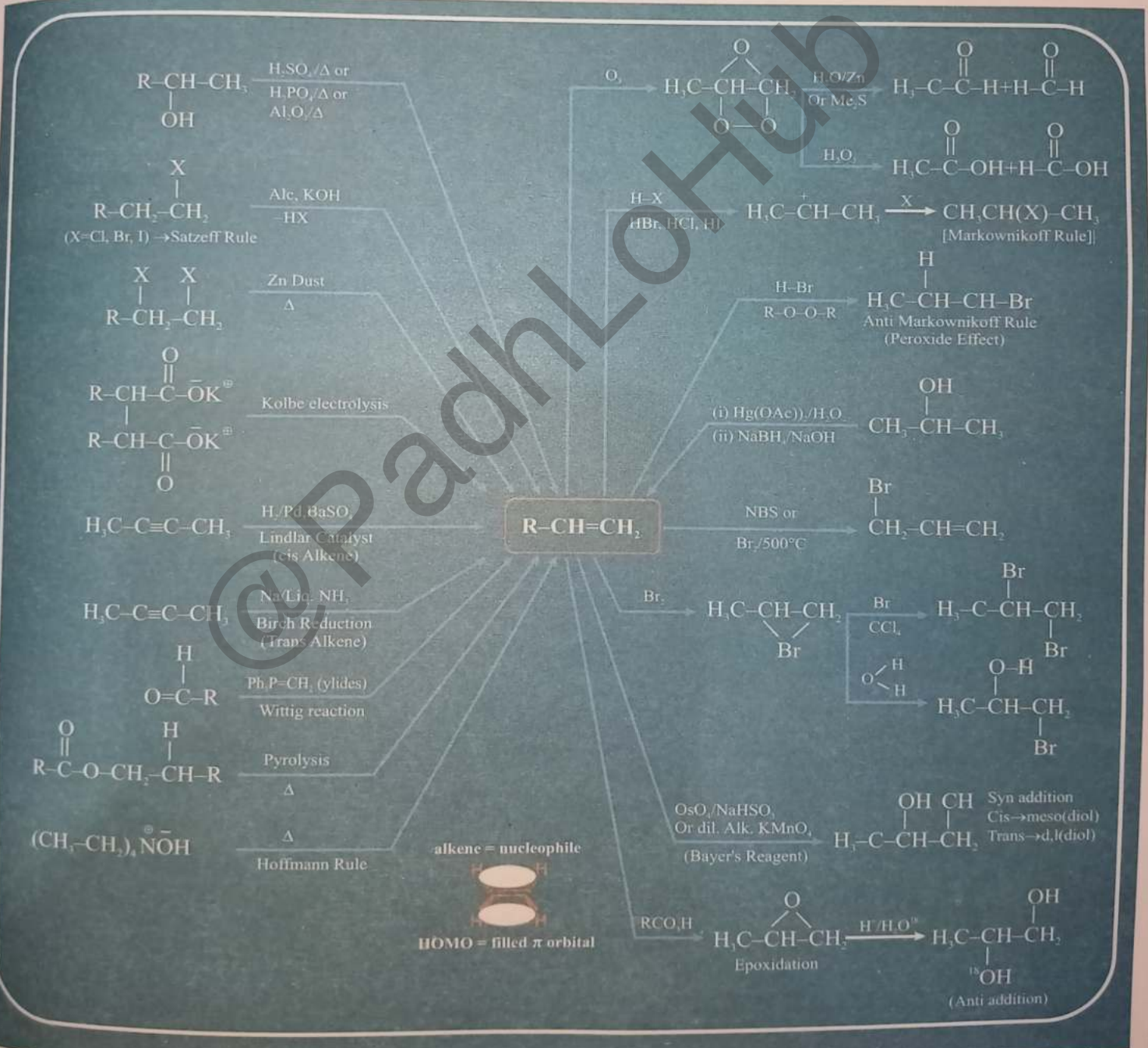
HYDROCARBON

The branched alkane 2,2,4-trimethylpentane is an important component of gasoline and the standard on which the "octane rating" system for fuel effi ciency is based. The car engine shown above requires high-octane fuel to achieve the performance for which it is famous.

REACTIONS OF ALKENE



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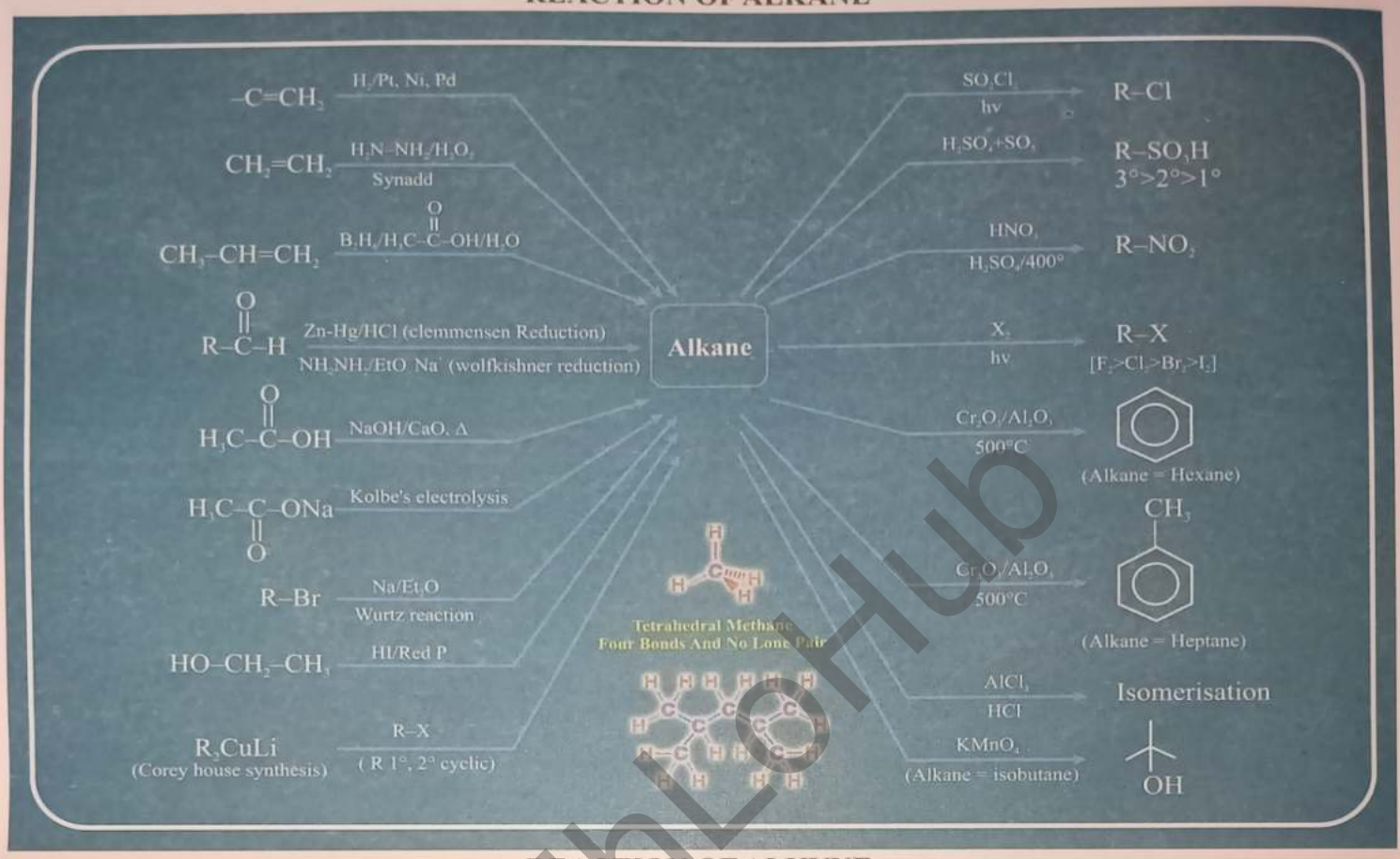
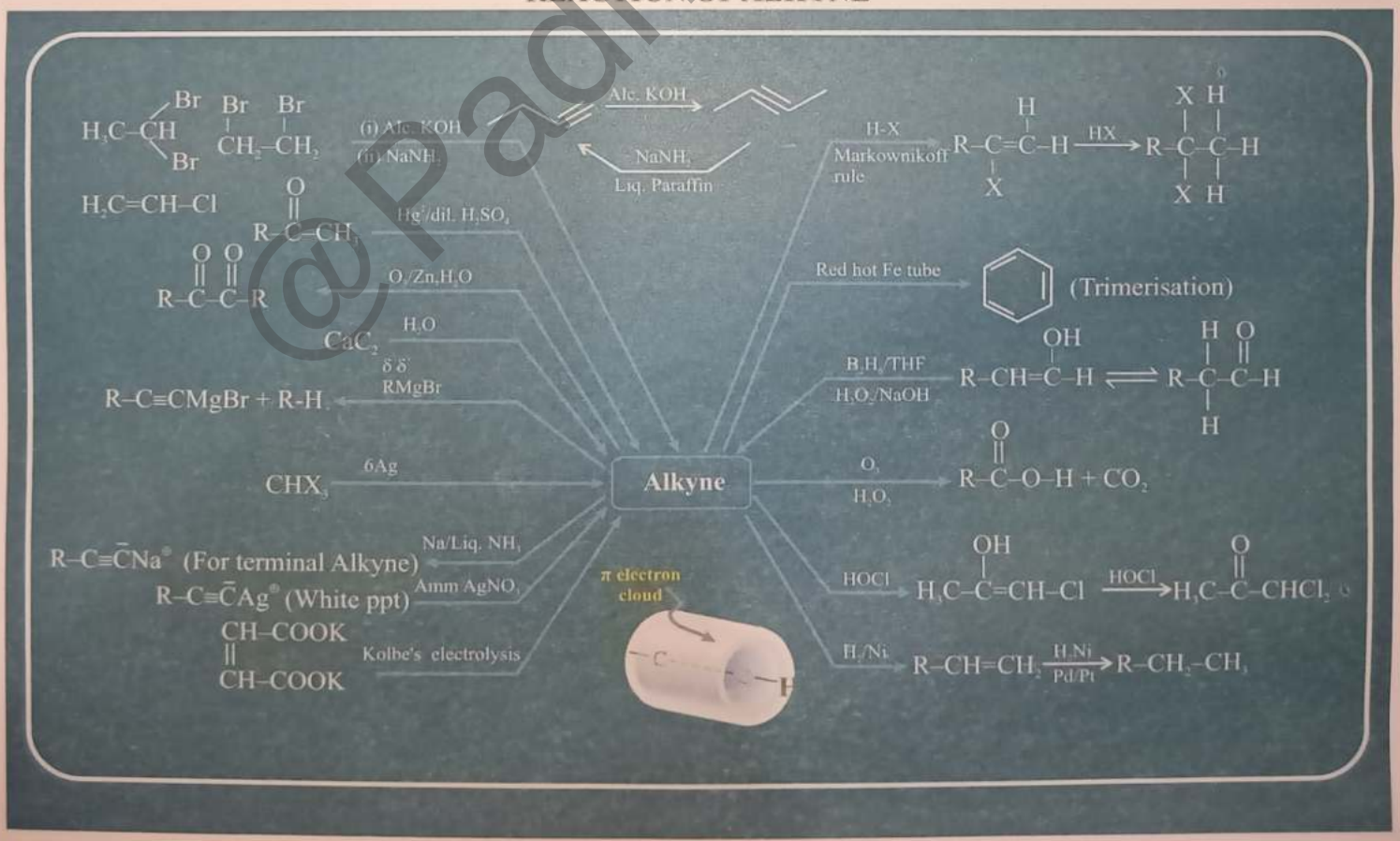
REACTION OF ALKANE



REACTION OF ALKYNE



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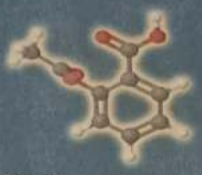
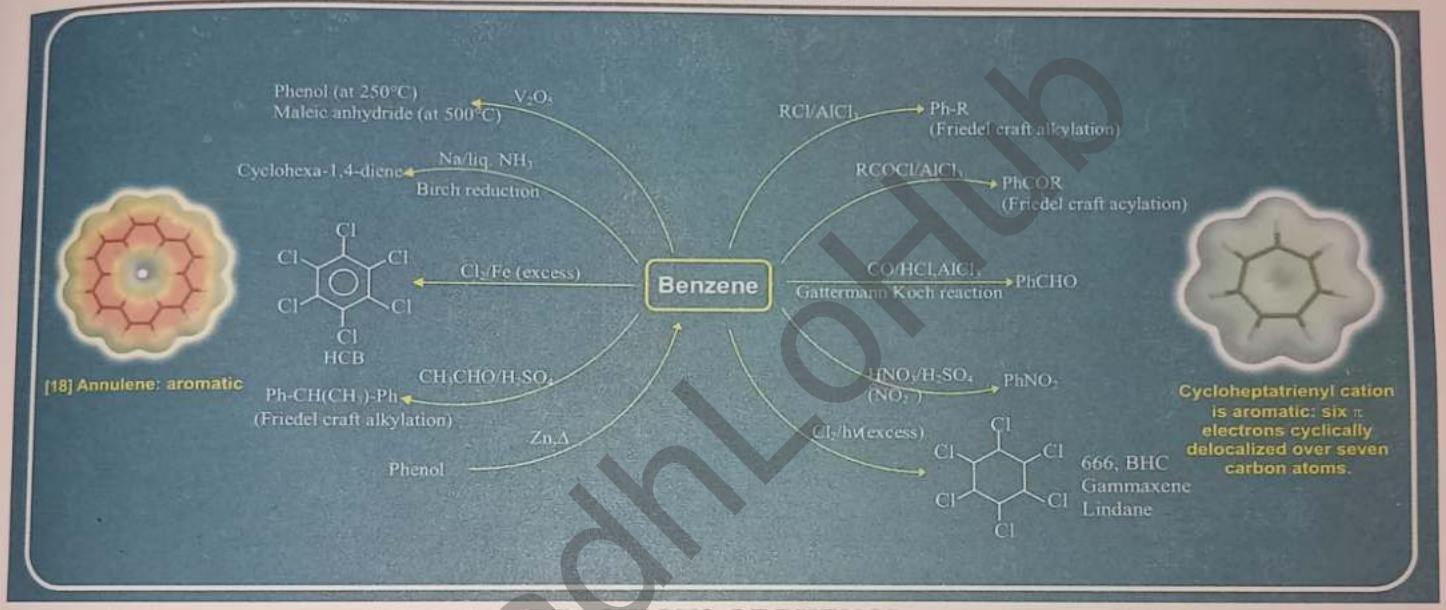
AROMATIC COMPOUND

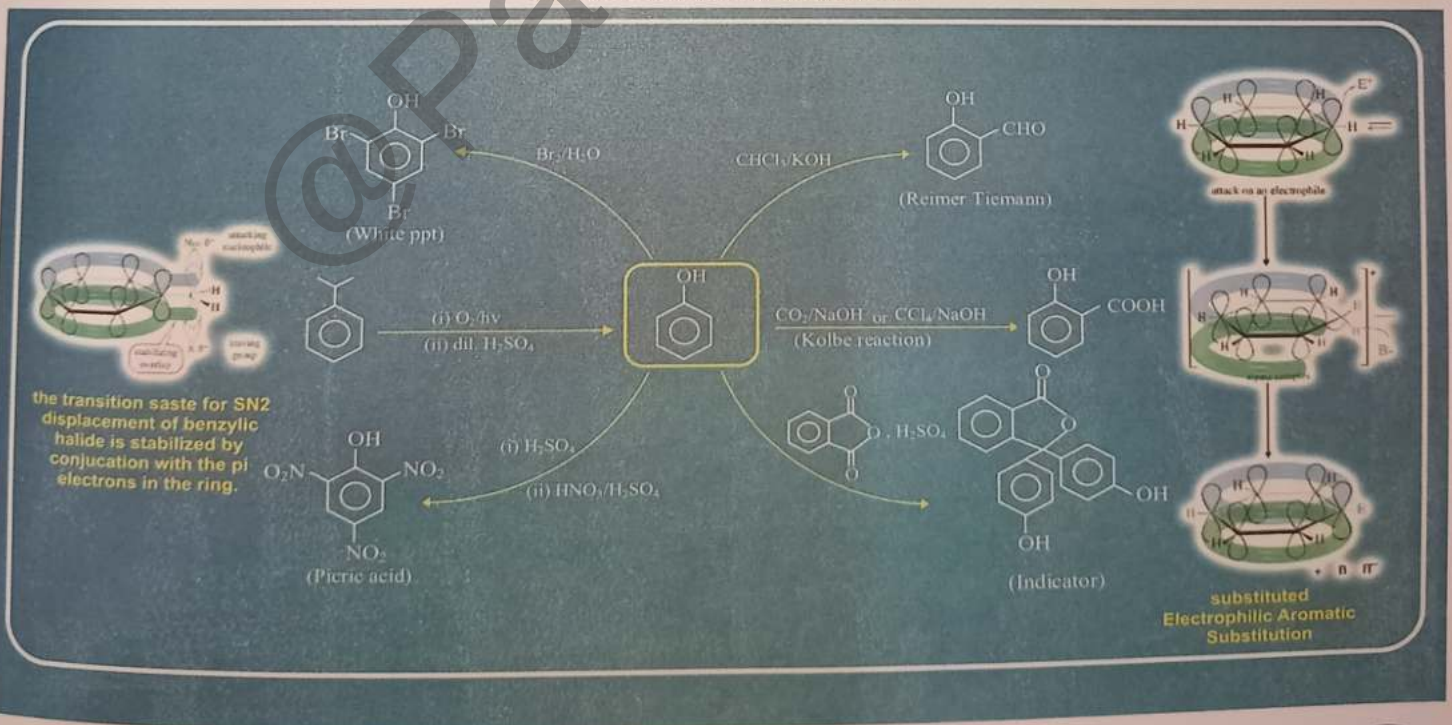


Aspirin, prepared industrially by selective electrophilic aromatic substitution of phenol, is arguably the blockbuster drug of all times. Its active metabolite, 2-hydroxybenzoic acid (salicylic acid), obtained from the bark of the white willow tree, has been used for four millennia for the treatment of inflammation and to relieve pain or discomfort caused by arthritis, soft-tissue injuries, and fever. Aspirin was ironically marketed together with another drug, heroin, whose addictive side effects were not

REACTIONS OF BENZENE



REACTIONS OF PHENOL



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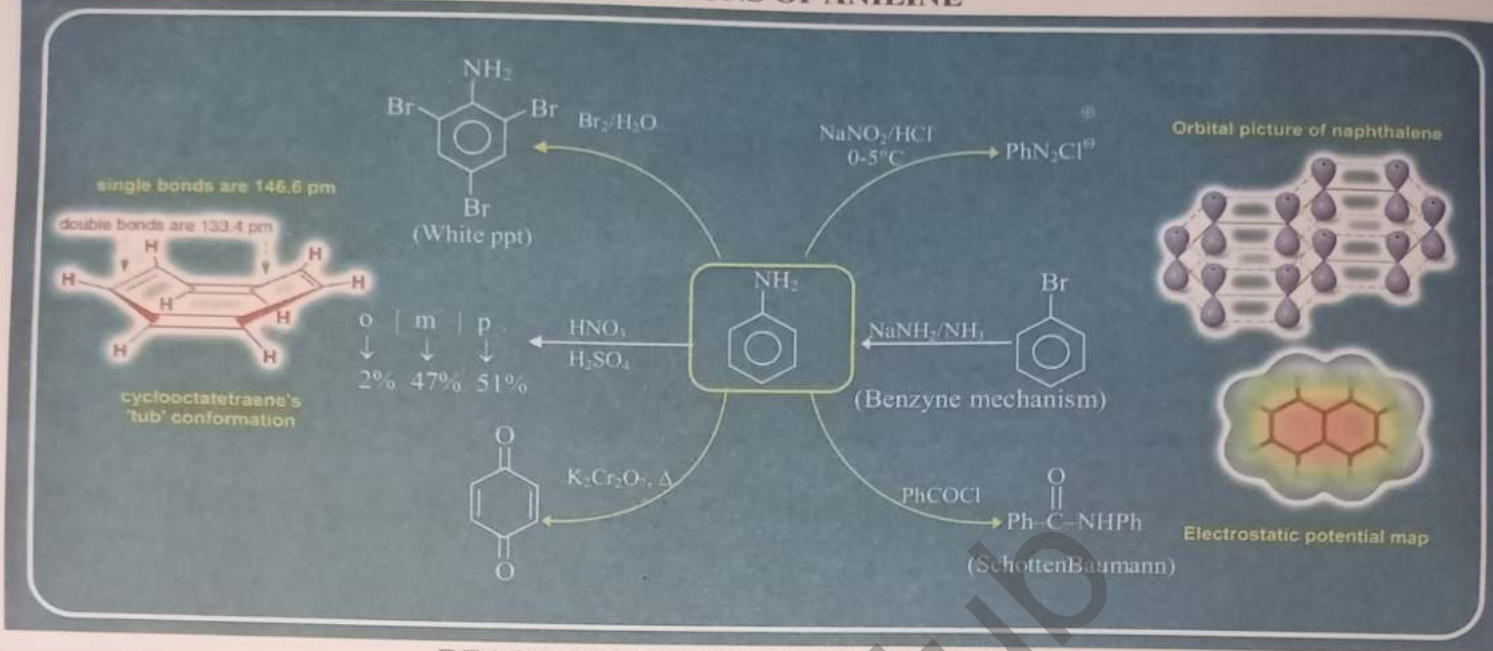
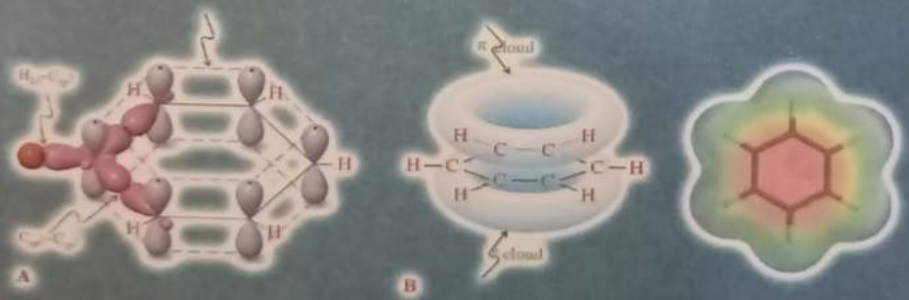
REACTIONS OF ANILINE



REACTIONS OF DIAZONIUM CHLORIDE



Orbital picture of the bonding in benzene
(A) The framework is depicted as straight lines except for the bonding to one carbon, in which the p orbital and the sp2 hybrids are shown explicitly.
(B) The six overlapping p orbitals in benzene form a pi-electron cloud located above and below the molecular plane.
(C) The electrostatic potential map of benzene shows the relative electron richness of the ring and the even distribution of electron density over the six carbon atoms.

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ALKYL HALIDE

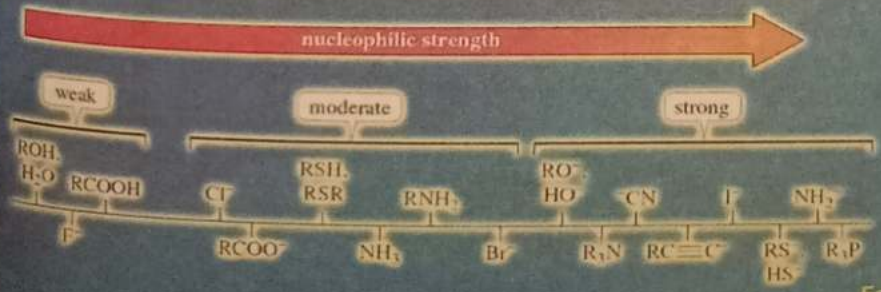
In The Body, The Nitrogen Of The Amino Group In Noradrenaline Attacks The Methyl Group In Sadenoylmethionine By Nucleophilic Substituion To Give Adrenaline. Adrenaline Is A "fight Or Flight' Hormone, Released Into The Blood Stream During Strees And Emergencies, And Is Responsible For The "rush" Felt During Thrilling Experiences.

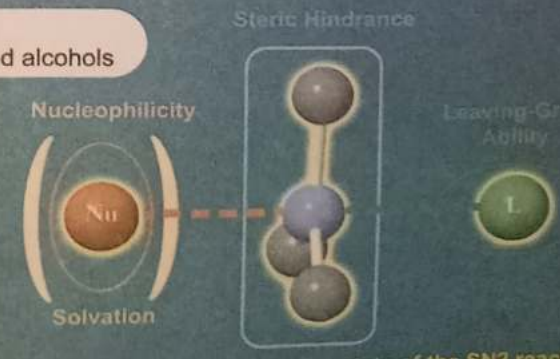
REACTIONS OF ALKYL HALIDE

Comparison S_N1 and S_N2 / E1 and E2

Properties	S_N1	S_N2	E1	E2
Kinetics	Pseudo 1st order	2nd order	Pseudo 1st order	2nd order
Rate	K [RX]	K [RX][Nu:⁻]	K [RX]	K [RX][B:⁻]
Stereochemistry	Racemisation	Inversion	No Special geometry	Anti-periplanar
Substrate	3°>2°>1°>MeX	MeX>1°>2°>3°	3°>2°>>>1°	3°>2°>1°
Nucleophile	Week	Strong	Not important	Needs StrongBase
Solvent	Good ionizing	Faster in aprotic	Good ionizing	Polaritynot important
Leaving Group	Needs good LG	Needs Good LG	Needs good LG	Needs good LG
Rearrangement	Possible	Not Possible	Possible	Not Possible

Some Common Nucleophiles
Listed in increasing order of nucleophilicity in polar solvents such as water and alcohols





Factors that influence the transition state of the SN2 reaction: nucleophilicity, solvation, steric hindrance, and leaving-group.

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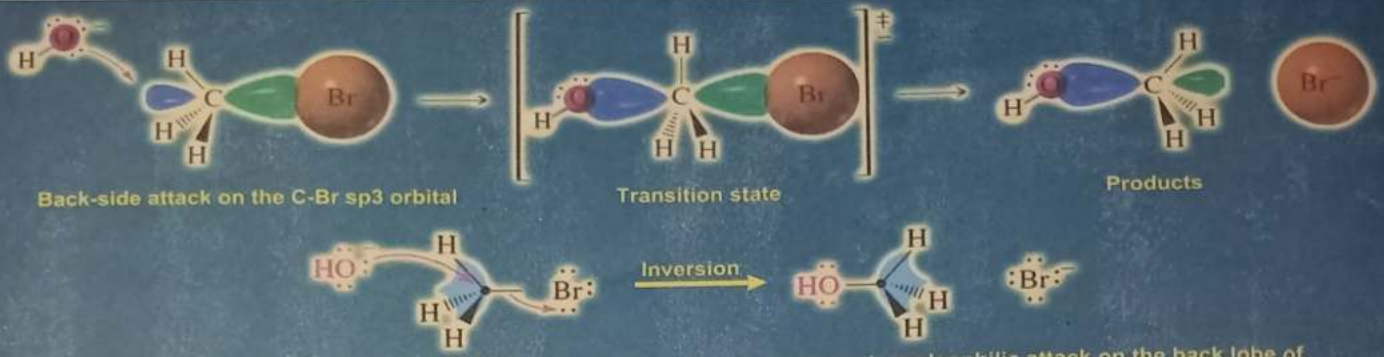
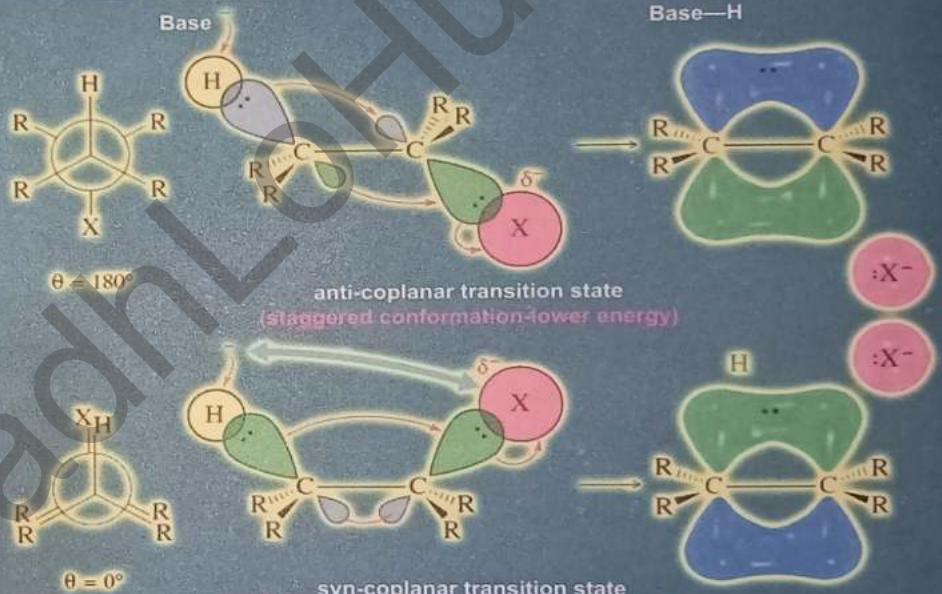
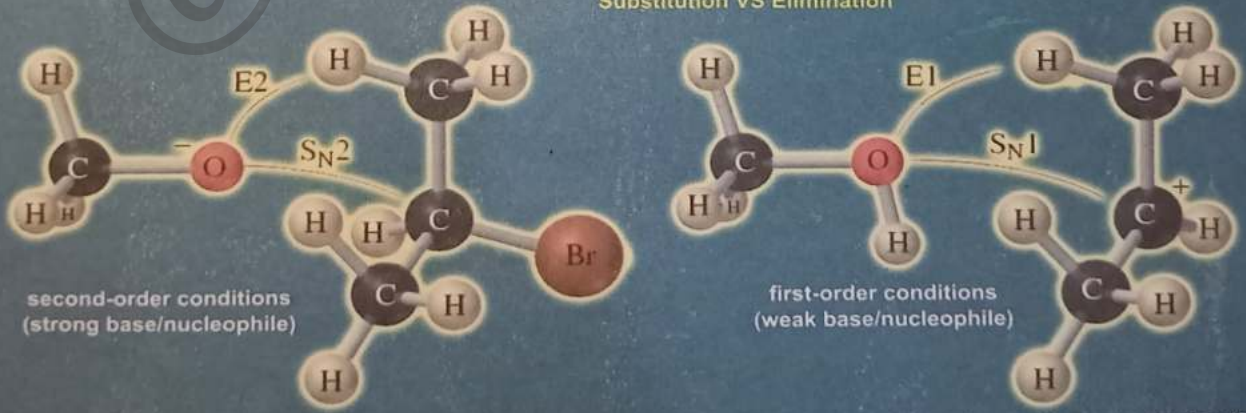


Back-side attack in the S_N2 reaction. The S_N2 reaction takes place through nucleophilic attack on the back lobe of carbon's sp^3 hybrid orbital. This back-side attack inverts the carbon atom's tetrahedron, like a strong wind inverts an umbrella.

In the transition state of the S_N1 ionization, the leaving group is taking on a negative charge. The C—X bond is breaking, and a polarizable leaving group can still maintain substantial overlap.

Concerted transition states of the E2 reaction. The orbitals of the hydrogen atom and the halide must be aligned so that they can begin to form a pi bond in the transition state





Under second-order conditions (strong base/nucleophile), a secondary alkyl halide might undergo either substitution (SN2) or elimination (E2). Under first-order conditions (weak base/nucleophile), SN1 and E1 are possible.

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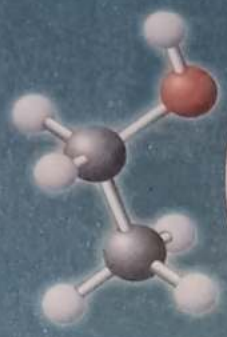
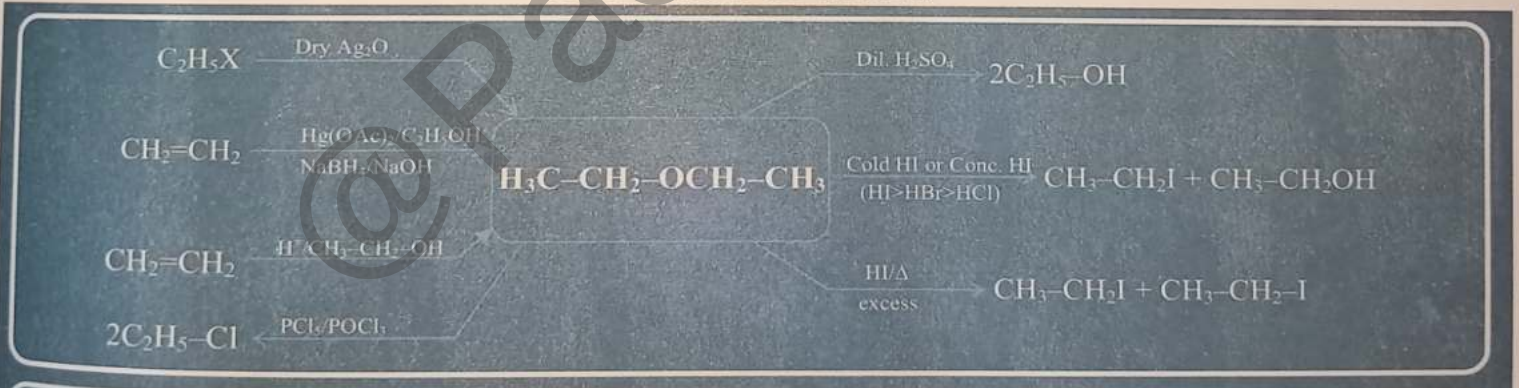
ALCOHOL & ETHER

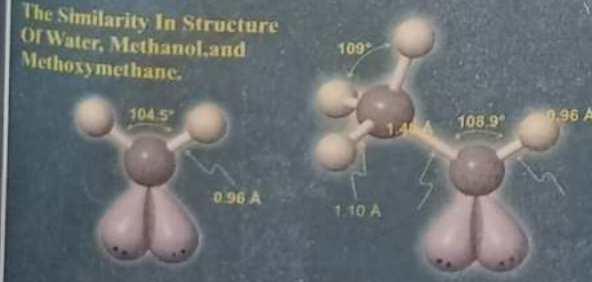


The Fermentation Of The Juice From Crushed Grapes Produces Ethanol In Wine, In This Picture From Bulgaria, Made The Traditional Way.

REACTIONS OF ALCOHOL

REACTIONS OF ETHER





The Similarity In Structure Of Water, Methanol, and Methoxymethane.

The Lowest-Energy Structure Of The Water Hexamer Against A Background Of The Structure Of Ice. The Yellow Lines Depict Hydrogen Bonds.

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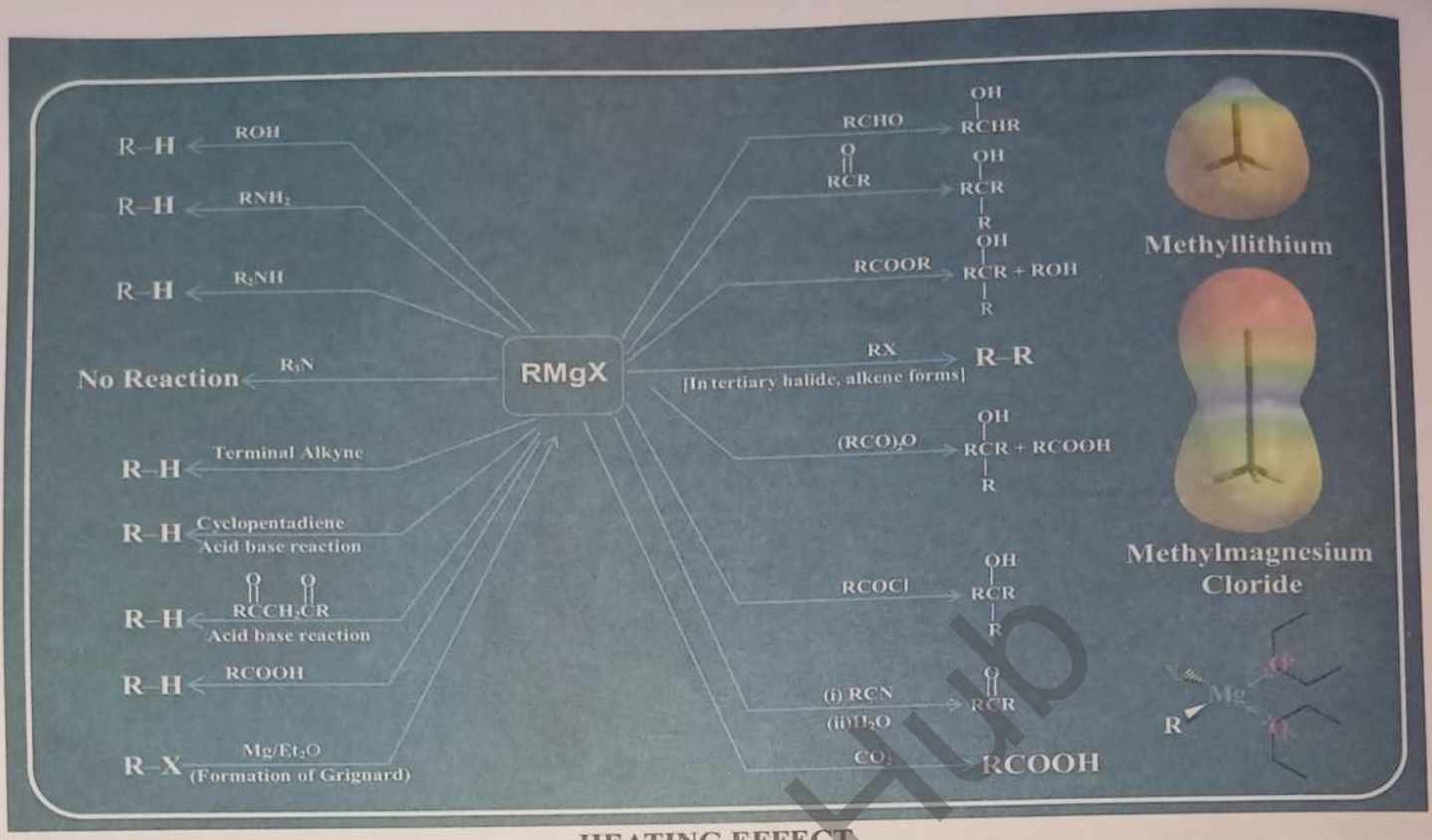
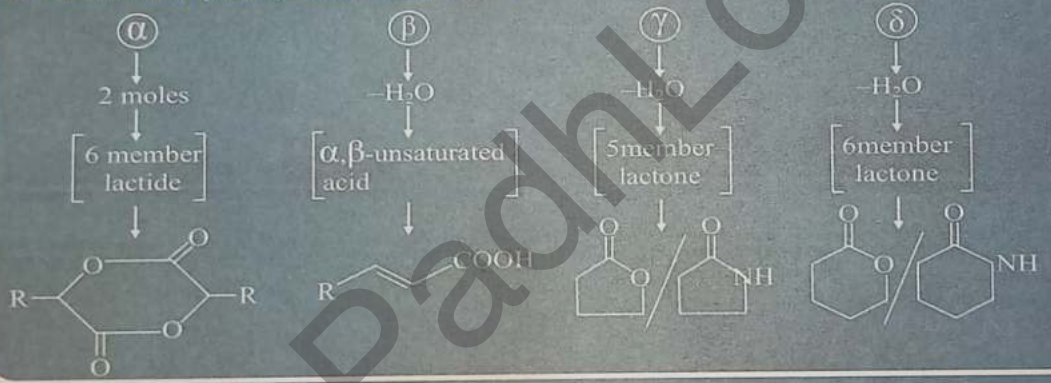
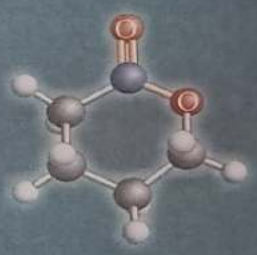


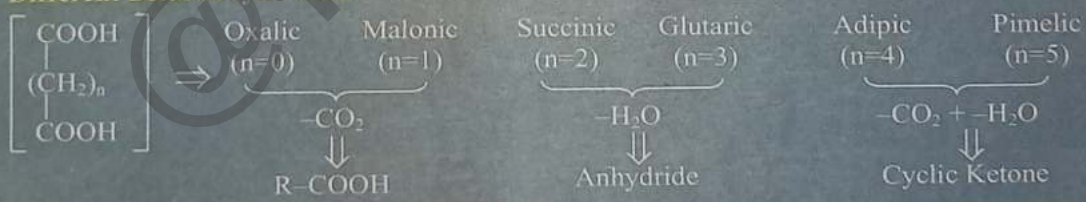
HEATING EFFECT

(a) Different Hydroxy Acid/Amino Acid:





(b) Different Dicarboxylic Acid:



(c) β-γ unsaturated Acid:

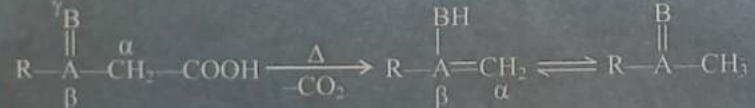


Table 9-1 Boiling Points of Ethers and the Isomeric 1-Alkanols

Ether	Name	Boiling Point (8C)	1-Alkanol	Boiling Point (8C)
CH_3OCH_3	Methoxymethane (Dimethylether)	223.0	CH_3CH_2OH	78.5
$CH_3OCH_2CH_3$	Methoxyethane (Ethyl methyl ether)	10.8	$CH_3CH_2CH_2OH$	82.4
$CH_3CH_2OCH_2CH_3$	Ethoxyethane (Diethylether)	34.5	$CH_3CH_2CH_2CH_2OH$	117.3
$CH_3CH_2CH_2CH_2OCH_2CH_2CH_2CH_3$	1-Butoxybutane (Dibutylether)	142	$CH_3CH_2CH_2CH_2CH_2OH$	194.5

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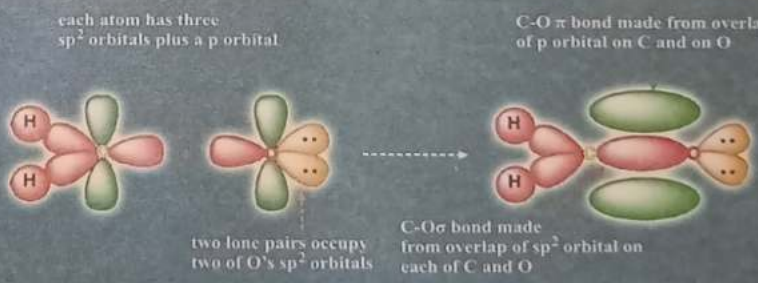
CARBONYL COMPOUND

Cinnamon, obtained from the driedinner bark of Cinnamomum verum trees,is one of the oldest spices known. Inancient times, cinnamon was nearly asvaluable as gold, and was a common giftfor royalty. The principal flavor ingredient in cinnamon is cinnamaldehyde, which is easily synthesized using thealdol condensation of benzaldehydewith acetaldehyde . Most chefs still prefer to usecinnamon sticks or powder made from the tree bark.

OXIDATION OF ALCOHOL

Important Points:



(i) Acid amide gives amine with LAH and B_2H_6

(ii) Imine gives amine with LAH, B_2H_6, SBH

(iii) Acid Halide doesn't react with B_2H_6

(iv) Ester, Acidhalide, Anhydride gives Aldehyde with DIBAL

(v) Cynide with DIBAL followed by water gives aldehyde

(vi) Cynide with $SNCl_2/HCl$ followed by water produces Aldehyde is known as Stephen's reduction

(vii) Acid halide on reaction with hydrogen over lindlar catalyst ($Pd/BaSO_4$) and quinoline gives aldehyde is known as Rosenmund reduction

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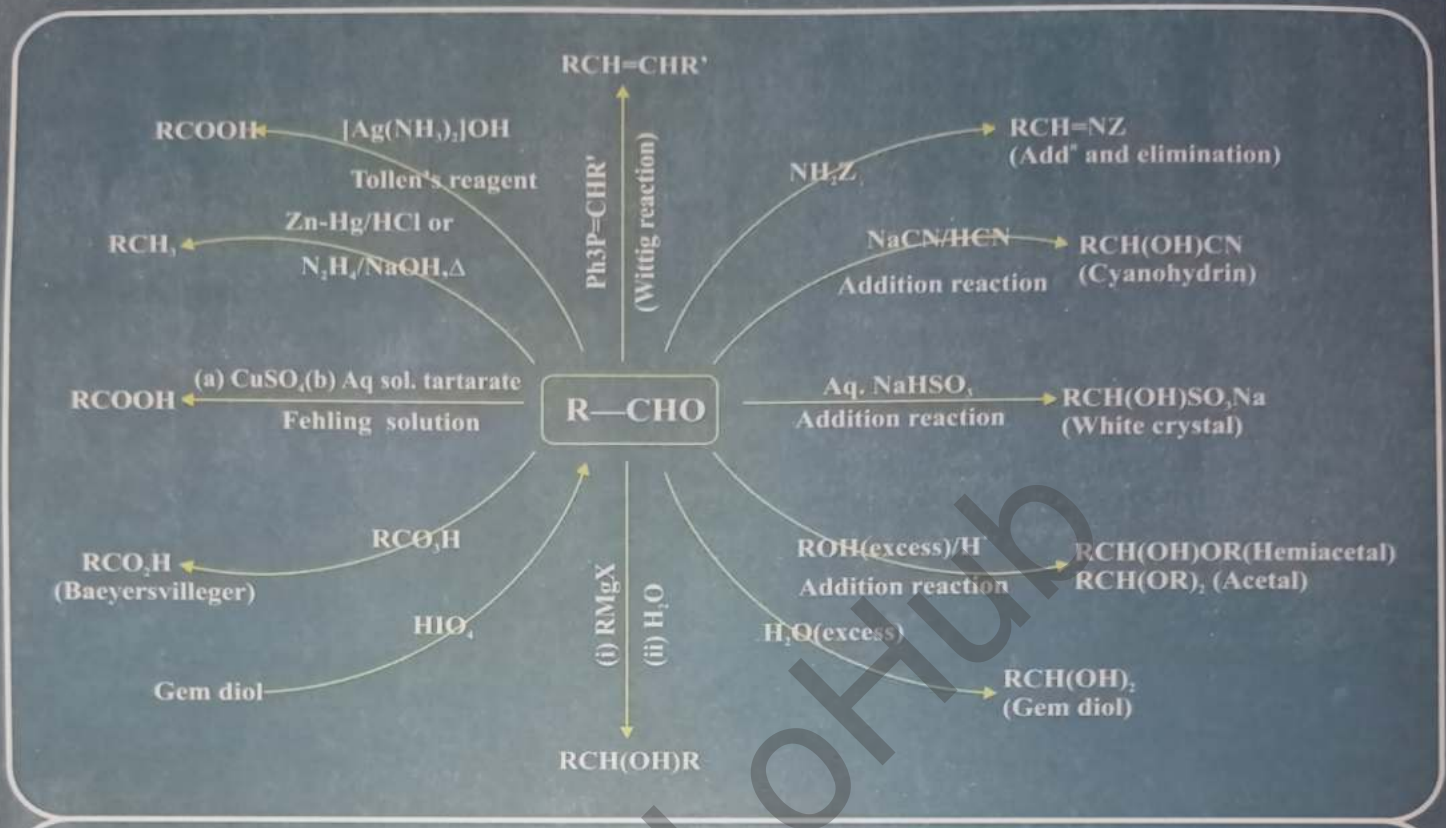
REACTION OF CARBONLY COMPOUND



IMPORTANT POINT :

(i) $NaHSO_3$ test is given by all aldehyde and methyl ketones except acetophenone

(ii) Tollen test is given by all aldehyde, α hydroxyl carbonyl ,hemiacetal and formic acid.

(iii) Fehling test is given by aldehyde(except aromatic), α-hydroxyl carbonyl,hemiacetal and formic acid.

(iv) Schiff test is given by all aldehyde.

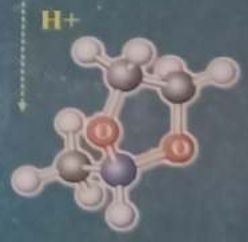
(v) 2,4-DNP test is given by aldehyde and ketones.It is also known as Braddy reagent and Borsch reagent.

(vi) Grignard reagent shows given reaction in sequence :

1. Acid base reaction
2. Nucleophilic addition reaction
3. Substitution reaction

(vii) Migratory aptitude for baeyer villager is H > 3° > 2° > Ph > 1°

(viii) In case of ketone with excess alcohol, ketal and hemiketal are formed.



(ix) In NH_2Z :

- **(a)** If $R\text{-}NH_2$ used then imine will form
- **(b)** If R_2NH used then enamine will form
- **(c)** If NH_2NH_2 used then hydrazone will form
- **(d)** If semicarbazide ($NH_2NHCONH_2$) used then semicarbazone will form
- **(e)** If NH_2OH used then oxime will form

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CARBOXYLIC ACID & AMINES

How many of you can do with-out that morning cup of coffee? The stimulating ingredient Â in coffee is the heterocycle caffeine, 135 mg of it in a normal 8-oz serving. Of course, there is also Roman espresso, which has 100 mg of caffeine in less than an ounce of liquid.

Carboxylic acid derivative

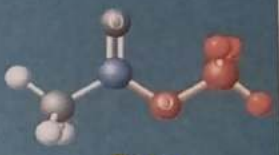
+

Tetrahedral intermediate

Substitution product

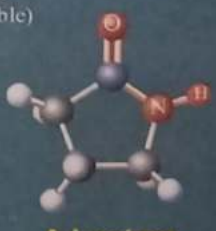
+

Nucleophilic Substitution by Addition-Elimination



Ester

Polyethylene terephthalate is Dacron, used in the Jarvik artificial heart. As a thin film, it is Mylar



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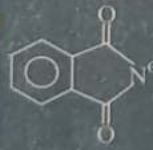

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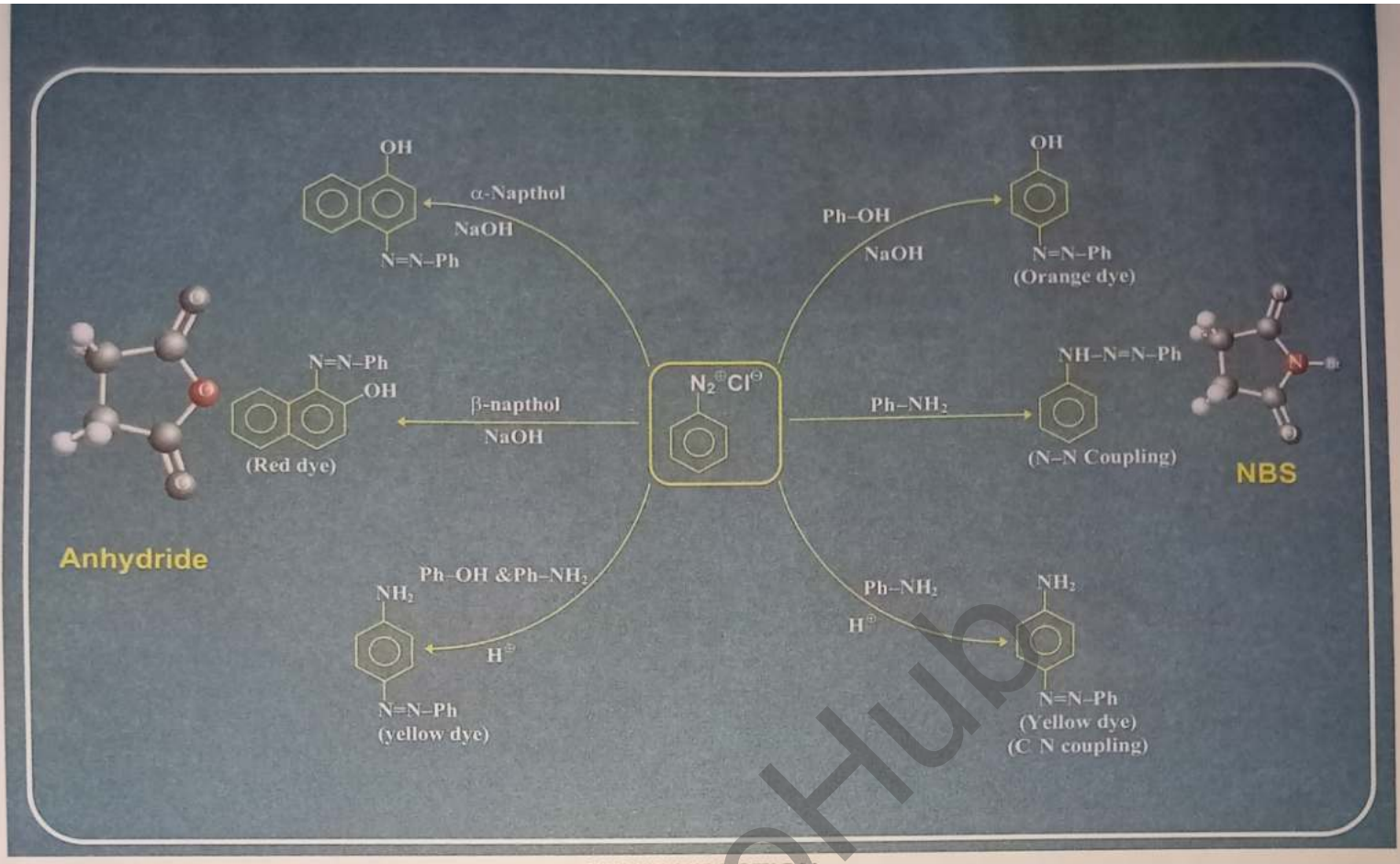


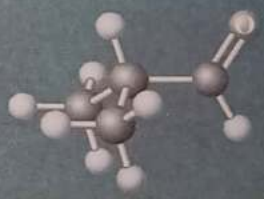
NAME REACTION

1. Aldolcondensation : (requires α-H)

$$R\text{–}CH_2CHO \xrightarrow[NaOH]{Dil} R\text{–}\underset{OH}{CH}\text{–}\underset{CHO}{CH}\text{–}CH_2\text{–}R \xrightarrow{\Delta} R\text{–}\underset{CHO}{C}\text{=}CH\text{–}CH_2\text{–}R$$

2. ClaisenCondensation :

$$R\text{–}CH_2\text{–}\overset{O}{\overset{\|}{C}}\text{–}OR \xrightarrow[(ii)\ H^{\oplus}]{(i)\ RO^{\ominus}} R\text{–}CH_2\text{–}\overset{O}{\overset{\|}{C}}\text{–}\underset{R}{CH}\text{–}COOR$$



2-Methylpropanal

3. Perkin Condensation :

$$Ph\text{–}CHO + CH_3\text{–}\overset{O}{\overset{\|}{C}}\text{–}O\text{–}\overset{O}{\overset{\|}{C}}\text{–}CH_3 \xrightarrow[\Delta]{CH_3COO^{\ominus}} Ph\text{–}CH\text{=}CH\text{–}COOH$$

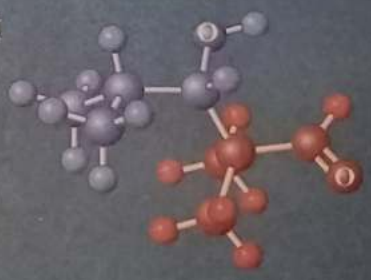
(Cinnamic acid)

Aldol addition | Dil NaOH

4. CannizzaroReaction : (Not require α-H)

$$R\text{–}CHO \xrightarrow[\Delta]{conc.NaOH} R\text{–}COONa + RCH_2OH$$

- Hydride transfer is Rds.
- Redox, disproportional reaction



5. HaloformReaction :

$$R\text{–}\overset{O}{\overset{\|}{C}}\text{–}CH_3 \xrightarrow[4NaOH]{3X_2} R\text{–}COONa + CHX_3$$

- Test is given by $-\overset{O}{\overset{\|}{C}}\text{–}\overset{*}{C}H_3$ | $-\underset{OH}{\overset{*}{C}H}\text{–}CH_3$ | $-\underset{lg}{\overset{*}{C}H}\text{–}CH_3$ | $-\overset{O}{\overset{\|}{C}}\text{–}\overset{*}{C}H_2\text{–}\overset{O}{\overset{\|}{C}}$– groups
- Marked carbon gives haloform

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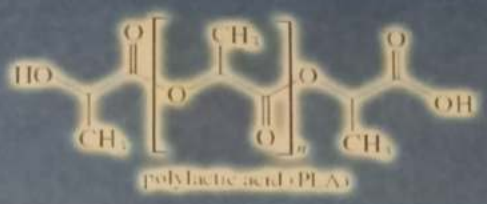
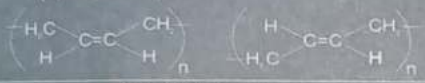
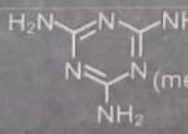
POLYMER



3D printing is an additive process that lays down successive layers of material under computer control. This photo shows a hot printer head sculpting the bones of a human foot by extruding a molten filament of a thermoplastic polymer that hardens in the desired shape. Polylactic acid (PLA), made from plant materials such as cornstarch, is a common filament used in 3D printers. This polyester has a convenient melting point (150â€“160 Â°C) and readily forms fully fused products

S.No.	Name(s)	Monomer	Uses
1.	Polyethylene(low density (LDPE))	$CH_2=CH_2$(ethylene)	film wrap, plastic bags
2.	Polyethylene (high density (HDPE))	$CH_2=CH_2$ (ethylene)	electrical insulation bottles, toys
3.	Polyisoprene (ziglar-natta) Cis-form (natural rubber) (1, 4-addition) Trans-form Gutta-purcha 	$CH_2=CH-C(CH_3)=CH_2$ (isoprene)	requires vulcanization for practical use
4.	Polychloroprene (cis + trans) (Neoprene)	$CH_2=CH-CCl=CH_2$ (chloroprene)	synthetic rubber oil resistant
5.	Nitrile Rubber (Buna-N)	$H_2C=CH-CH=CH_2$ $H_2C=CHCN$	Adhesives hoses
6.	Polystyrene (PS) (Styrene)	$CH_2=CHC_6H_5$ (styrene)	toys, cabinets packaging (foamed)
7.	Polyacrylonitrile (PAN, Orlon, Acrilan)	$CH_2=CHCN$ (acrylonitrile)	rugs, blankets clothing
8.	Polytetrafluoroethylene (PTFE, Teflon)	$CF_2=CF_2$ (tetrafluoroethylene)	non-stick surfaces electrical insulation
9.	SBR styrene butadiene rubber (Buna-S)	$H_2C=CH-CH=CH_2$ $H_2C=CHC_6H_5$	Tires

S.No.	Name(s)	Monomer	Uses
1.	Polyester/Dacron/ Terylene/Mylar	$HO_2C\ C_6H_4\ CO_2H$ (Terephthalic acid) $HO–CH_2CH_2–OH$ (Ethylene glycol)	Fabric, Tyrecord
2.	Glyptal or Alkyds resin	$HO_2C–C_6H_4–CO_2H$ (Phthalic acid) $HO–CH_2CH_2–OH$	Paints and Lacquers
3.	Melamine formaldehyde resin	 (melamine) + HCHO (formaldehyde)	Unbreakable crockery
4.	Polyamide (Nylon 6,6)	$HO_2C–(CH_2)_4–CO_2H$ $H_2N–(CH_2)_6–NH_2$	Parachutes & Clothing
5.	Bakelite	PhOH + HCHO in (excess)	Electrical switch and Bowling Balls
6.	Urea-formaldehyle resin	$H_2N–CO–NH_2$ (Urea) HCHO (Formaldehyde)	

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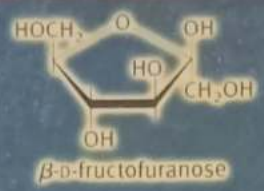
BIOMOLECULE



β-D-fructofuranose

Fructose, also known as fruit sugar, is 50% sweeter than sucrose (table sugar) and is themajor sugar in honey.

1. Carbohydrates Are Naturally Occurring Polyhydroxycarbonyl Compounds That Can Exist As Monomers, Dimers, Oligomers, And Polymers.
2. Monosaccharides Are Called Aldoses If They Are Aldehydes And Ketoses If They Are Ketones. The Chain Length Is Indicated By The Prefix Tri-, Tetr-, Pent-, Hex-, And So Forth.
3. Anomer: The New Stereocenter Formed By Cyclization Of Carbohydrates (Designated □ And □).
4. Mutarotation: Equilibration Between Anomers In Solution Gives Rise To Changes In The Optical Rotation.
5. Reducing Sugars: Sugars Containing Hemiacetal Functions Are Called Reducing Sugars, Because They Readily Reduce Tollens's And Fehling's Solutions.
6. Epimers: Two Diastereomeric Sugars Differing Only In The Configuration At A Single Asymmetric Carbon Atom. The Interconversion Of Epimers Is Called Epimerization.
7. Glycoside: A Cyclic Acetal Form Of Sugar. Glycosides Are Stable To Base, And They Are Non-Reducing Sugars.
8. Starches : A Class Of A-1,4′ Polymers Of Glucose Used For Carbohydrate Storage In Plants And Animals.
9. Amylose: A Linear A-1.4′ Polymer Of D-glucopyranose Used For Carbohydrate Storage In Plants.
10. Amylopectin: A Branched A-1,4′ Polymer Of D-glucopyranose Used For Carbohydrate Storage In Plants. Branching Occurs At A-1,6′ Glycosidic Linkages.
11. Cellulose : A Linear B-1,4′ Polymer Of D-glucopyranose. Cellulose Forms The Cell Walls Of Plants And Is The Majorconstituent Of Wood And Cotton.
12. Glycogen: An Extensively Branched A-1,4′ Polymer Of D-glucopyranose Used For Carbohydrate Storage In Animals. Branching Occurs At A-1,6′ Glycosidic Linkages.
13. Zwitterion (dipolar Ion) : A Structure With An Overall Charge Of Zero But Having A Positively Charged Substituent And A Negatively Charged Substituent. Most Amino Acids Exist In Zwitter Ionic Forms.
14. Isoelectric Point, Pl (isoelectric Ph) : The Ph At Which An Amino Acid (or Protein) Does Not Move Under Electrophoresis. This Is The Ph Where The Average Charge On Its Molecules Is Zero, With Most Of The Molecules In Their Zwitter Ionic Form.
15. Protein : Proteins Are Polypeptides With Molecular Weights Higher Than About 5000 Amu.
16. Primary Structure : The Covalently Bonded Structure Of A Protein; The Sequence Of Amino Acids, Together With Any Disulfide Bridges

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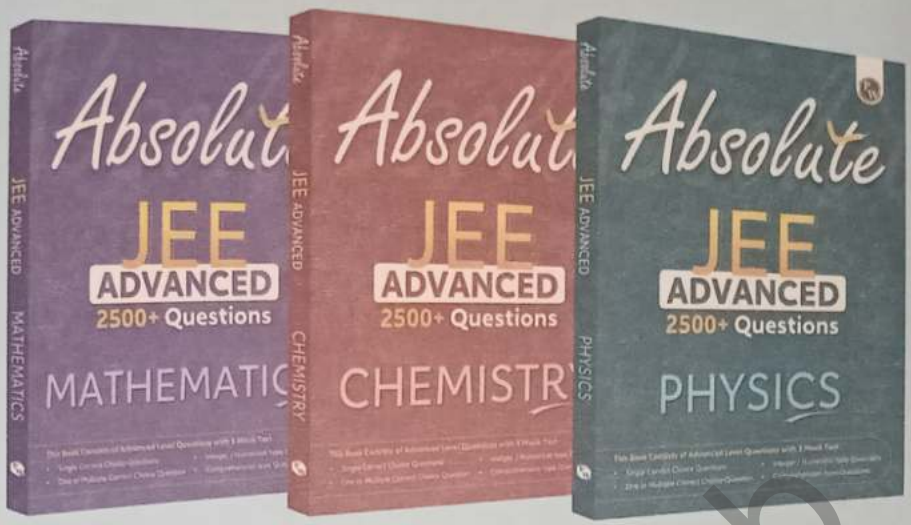

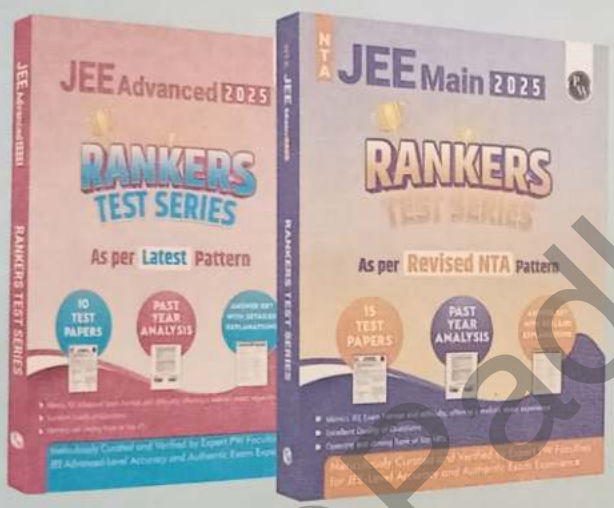

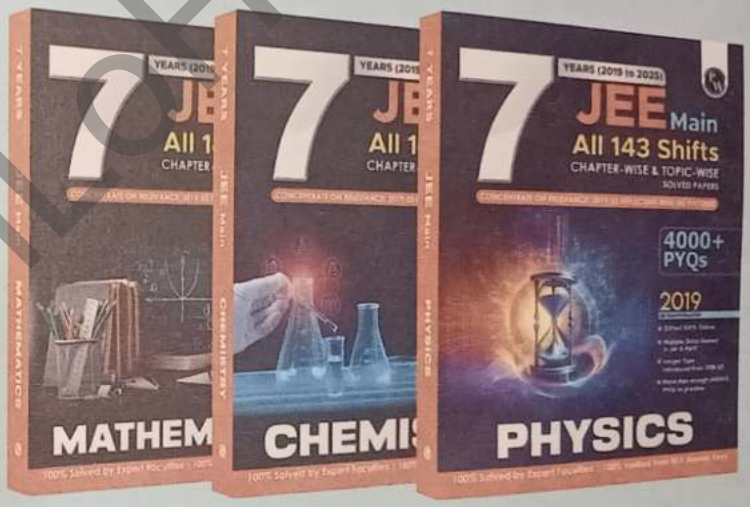

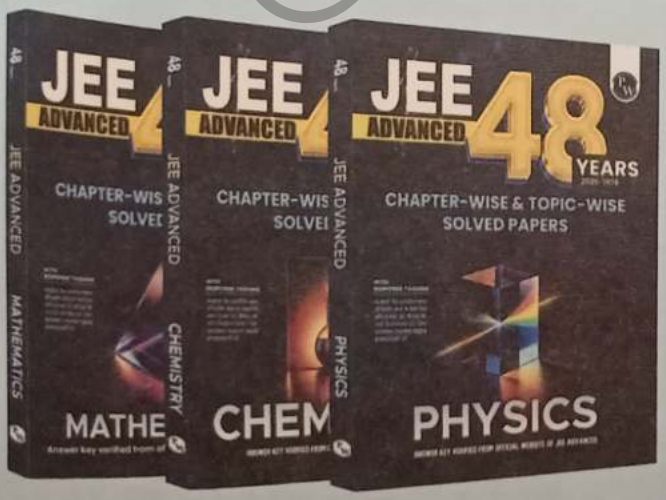

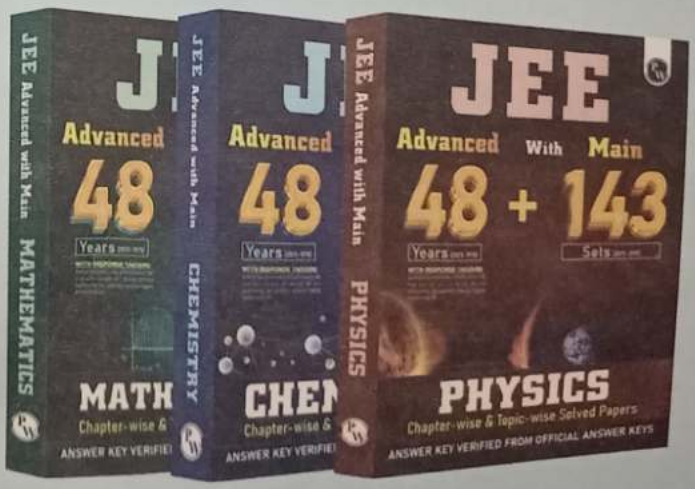

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About The Author

RA Baba

Ex-Director and Head of Organic Chemistry Department, Nucleus, Kota

Popularly known as **RA Baba**, he is a name synonymous with excellence in Organic Chemistry. With over **14 years of teaching experience**, he has mentored countless aspirants, including top JEE Advanced rankers such as **AIR 3, 8, 10, 12, 16, 18, 23, 24, and many others.**

A **two-time IIT-JEE qualifier** himself, he holds a B.Tech degree from **NIT Surathkal**.

He has **authored more than 5 books** for competitive exams, known for their conceptual clarity and student-focused design.

His mentorship extends to **Olympiad achievers** like Mrinal Sharan (currently at Oxford University) and Sushant Padha (IIT Bombay) and many more.

Famous for his **12-hour live Organic Chemistry sessions** on the **JEE Wallah channel**, he is known for teaching difficult topics in a fun and simple way that keeps students interested and motivated.

He believes in the philosophy: **"A student learns better while enjoying."** His intuitive approach and visual tools for reaction mechanisms make complex concepts accessible and enjoyable for learners.

SKC Sir

With **17+ years of teaching experience** in Organic Chemistry, he is a well-known name in **JEE and NEET preparation**. He has guided top NEET rankers including **AIR 1, 2, 3, 5, 6, 9, 11, 15, 20, 21, 25** and many more.

Former **#1 Legend Category Educator** at **Unacademy** (5 years)

Held senior leadership roles such as: **Vice President** of Sarvottam Career Institute Pvt. Ltd., Kota, **Director & HOD** of Chemistrio Institute Pvt. Ltd., Kota, **Former Senior Chemistry Faculty** at Allen Career Institute, Kota.

Author of **4 best-selling books**, known for their simple and effective explanations.

A registered author with top academic publishers: **Pearson India** – for NCERT Mantra and **Universities Press** – for Fundamental Problems in Organic Chemistry.

★★★★★

Majid Husain (Air 3 JEE Adv 2025)

My experience, learning organic chemistry under Rohit Agrawal sir was amazing. The depth of concepts he teaches and how it applies perfectly to JEE problems helps a lot. I can attest to the effectiveness and completeness of his notes. His notesmade the most important part of the preparation for the subject which i followed through out helping me perform well and boost my performance. Apart from this his selected problems were perfect to prepare from scratch and reach JEE advanced level. The problem sets cover all relevant cases and strengthen the concepts. These notes along with the problems are a complete and sufficient course for JEE organic chemistry.

I recommend serious JEE aspirants to use these to learn organic chemistry the way it is intended for JEE advanced.

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